

Certified Bellman Operators for Costly Policy Revision

R38: Lossless Historical Annex

September 24, 2026

This annex preserves, in their entirety, the previously materialized R34 article, technical supplement, response, computation report, and recursive historical annex. Their pages follow in that order. The earlier historical annex contains the retained predecessor manuscripts, proofs, numerical studies, and original stopped-control program. Original PDF and source files also remain unchanged in the repository.

The material is historical evidence, not a claim that all earlier assertions remain current or that earlier protocol-only execution claims have been verified retrospectively. R38's current theorems, numerical declarations, and policy-class qualifications appear in its article and technical supplement. New results come from the separately archived R38 execution.

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CERTIFIED BELLMAN OPERATORS WITH NEURAL PROPOSALS: ALL-RESTART BOUNDS FOR COSTLY POLICY REVISION

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We study policy revision when implementation is costly and actions change future intervention exposure. We construct two-sided Bellman witnesses, unrestricted support problems, and a backward cost-bound propagation that uses operating feasibility at every restart. Their finite rational proof objects bound intervention cost over the full operating-feasible policy set, rather than only a certified local action class. In an action-dependent continuous-state maintenance economy, 42 deployments satisfy their uniform operating tolerances and preserve installed returns. All 18 installed-neural configurations reduce intervention cost relative to the class minimum. The restart propagation tightens 22 global cost intervals without changing the deployments or solving additional unrestricted support problems; twelve spline cases retain exact global optimality. An independent optimizer-free audit checks 294 certificates, and total construction and verification work is reported. We preserve the original stopped-control objective and its separately proved constant-control boundary optimum. Remaining positive global gaps are quantified.

KEYWORDS: Bellman operators, Verified computation, Policy revision, Maintenance, Neural proposals.

1. INTRODUCTION

Replacing an installed decision rule has an economic cost. A maintenance instruction may require retraining operators, changing a contract, or obtaining approval at the states where the instruction is used. Consequently, two policies with similar operating returns need not be interchangeable. This paper studies how to construct a uniformly accurate policy while minimizing the discounted cost of revising an installed rule. The distribution of future implementation costs is endogenous: a revision today changes the states at which later instructions must be changed.

We develop a constructive Bellman-certificate method for this problem. Starting from model primitives, the method builds upper value witnesses, generates a nonempty class of uniformly accurate actions, and solves the intervention-cost problem within that class. A finite proof object specifies the deployed policy and its operating and intervention values. A separate checker verifies whole intervals and isolated boundary points without repeating the optimizing envelope construction. Neural networks and classical approximations supply installed proposals; their fitting errors do not enter a certificate unless separately bounded.

The computational application is a continuous-state maintenance economy with action-dependent transitions. Maintaining or replacing a unit is always inferior to deferring in current reward. Such actions can nevertheless be optimal because they improve future condition. Continuation terms therefore cannot be canceled from the action comparison. Upper witnesses are computed by action-dependent expectation, maximization, and globally checked polygonal compression. The inputs do not include an optimal-value surface, an optimal-action file, or an analytically supplied supersolution. Own-policy values, admissible action sets, and the intervention-cost recursion are also constructed from the primitives and included in the work accounts.

Three results organize the analysis. First, convex piecewise-affine structure permits an explicit construction of upper witnesses with controlled rounding and compression error. Second, a

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uniform local deficit budget produces a nonempty policy class with an all-state, all-restart operating guarantee. Within that class, a second Bellman recursion minimizes intervention cost under the deployed policy’s occupancy. Third, an independently certified incumbent can be retained at zero intervention cost, which is globally minimal even outside the conservative local class. This last result prevents a sufficient local restriction from creating unnecessary revisions when a global own-policy certificate is already available.

The experiments exercise these statements rather than infer them from training loss. Three horizons, two accuracy budgets, and seven proposal rules produce 42 class-based certificates, all checked in exact rational arithmetic. The predeclared mechanism stress incumbent produces strictly positive dynamic-versus-pointwise intervention savings at every horizon and budget. The remaining 36 comparisons have zero savings and are retained. All 12 spline cases qualify for the additional zero-intervention certificate. None of the 18 neural cases meets its raw operating target. These outcomes establish an end-to-end coupled construction and an endogenous-occupancy mechanism; they do not establish neural-specific computational superiority.

A further result advances the original stopped consumption–portfolio problem without changing its primitives. At the economically active boundary restart used previously, we prove a positive derivative over the entire constant-adjustment interval. This identifies the unique global optimum of that scalar control class. The argument combines monotone coupling, killed-process survival derivatives, and rational bounds for the distant-boundary error; it does not assume concavity or extrapolate from five derivative evaluations. The original all-domain current-state target remains stated separately and is not replaced by the auxiliary maintenance-model certificate.

A complementary global-cost result goes beyond the certified action class. We construct a lower operating witness from checked compression defects and solve unrestricted support Bellman problems for a fixed multiplier portfolio. Their values bound the implementation cost of every policy satisfying the original all-state operating constraint. A separately verified deployment supplies the upper bound and an explicit global cost gap. In the coupled application, all 18 installed-neural configurations receive less costly deployments than the conservative class minimum, without weakening the operating target. The remaining global gaps and the twelve zero-cost spline cases are reported alongside these improvements. A backward propagation then transmits future feasible intervention costs through the action-dependent transition law and enforces a necessary current operating inequality with local nonnegative multipliers. The propagated bound tightens 22 of the 42 global intervals while retaining every action and the same deployed policy. This step uses the simultaneous all-restart requirement and is distinct from simply adding a better scalarized candidate. The theory, cost account, and independent proof-object checks are developed together in Section 6. Classical constrained Markov control and its Lagrangian methods provide the underlying comparison principles (Altman, 1999); our result constructs and checks their specific combination with the continuum of restart restrictions.

1.1. *Relation to existing methods*

Bellman comparison and constrained backward induction are classical. We do not claim a new dynamic-programming principle. Fitted value iteration studies approximation and sampling error under a generative model (Munos and Szepesvári, 2008); approximate policy iteration has related performance bounds (Scherrer, 2013). Here fitting is separated from a deterministic, all-state verification layer for a known model. Safe policy improvement with baseline bootstrapping restricts changes when data leave the transition model uncertain (Laroche, Trichelair, and Tachet des Combes, 2019). Our admissible class instead limits a verified operating deficit, and our implementation cost is an economic primitive. Neither information structure subsumes the other.

Numerical economics exploits structure and approximation (Judd, 1998). Adaptive sparse grids address genuinely high-dimensional dynamic models (Brumm and Scheidegger, 2017). Our

dimensional experiment is one-dimensional and does not establish a competing high-dimensional rate. Information-relaxation methods furnish other routes to upper bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014). Their construction and our construction must both be charged; a supplied upper bound is not a free solution input. Verified computation requires both a valid analytic reduction and controlled arithmetic (Rump, 2010). Markov-chain approximation of a diffusion imposes additional consistency requirements (Kushner and Dupuis, 2001); the maintenance economy is not asserted to be a discretization of the stopped diffusion.

The methodological addition is the explicit composition of witness construction, quantified encoding error, a nonempty certified action class, endogenous-cost optimization, and an independently checkable finite representation. The cost recursion alone is standard once the class is known. The constructive witness and class-generation steps are therefore part of the main theorem and computation, not omitted preprocessing. The comparison with uncompressed exact piecewise-affine dynamic programming is intentionally strong: both methods exploit the same affine model structure. No neural advantage, dimension-free complexity, or universal acceleration follows merely from the existence of a regional proof object.

2. ECONOMIC ENVIRONMENT AND OBJECTIVES

2.1. A coupled maintenance economy

There are T decision dates and a terminal date. Condition x lies in $[0, 1]$. Actions $a = 0, 1, 2$ denote defer, maintain, and replace. Current reward is $r(x, a) = x - c_a$, with

$$\beta = 19/20, \quad (c_0, c_1, c_2) = (0, 3/20, 9/20), \quad g(x) = x/2. \quad (1)$$

Two shocks have probabilities $(p_1, p_2) = (3/5, 2/5)$. Their next-state maps are

a	$f_{a1}(x)$	$f_{a2}(x)$
0	$3x/4$	$3x/4 + 1/20$
1	$17x/20 + 1/10$	$17x/20 + 7/50$
2	$x/10 + 4/5$	$x/10 + 17/20$

(2)

All maps preserve the state interval. For a continuation v , write

$$T^a v(x) = x - c_a + \beta \sum_{j=1}^2 p_j v(f_{aj}(x)), \quad Tv = \max_{a \in \{0,1,2\}} T^a v. \quad (3)$$

The optimal operating value satisfies $V_T = g$, $V_t = TV_{t+1}$; a policy π has $J_T^\pi = g$, $J_t^\pi = T^{\pi_t} J_{t+1}^\pi$. Because every intervention has a positive current cost, any maintain or replace decision is justified by continuation value, not by a primitive reward-sign rule.

Let π^0 be an installed policy. Its revision cost from restart (t, x) is

$$C_t^\pi(x) = \mathbb{E}_{t,x}^\pi \sum_{s=t}^{T-1} \beta^{s-t} (1 + X_s) \mathbf{1}\{\pi_s(X_s) \neq \pi_s^0(X_s)\}. \quad (4)$$

One unit is a normalized implementation intervention, not a dollar estimate. For a uniform initial state distribution ν , the reported economic costs are $\int C_0^\pi(x) d\nu(x)$, calculated under each deployed policy. The objective is to minimize this cost subject to

$$\sup_{t \in \{0, \dots, T\}, x \in [0, 1]} \{V_t(x) - J_t^\pi(x)\} \leq \varepsilon. \quad (5)$$

We solve the cost problem in an explicitly constructed inner class and identify an important case in which the minimum is global for (5) itself. Operating-value preservation, $J^\pi \geq J^{\pi^0}$, is a distinct requirement. It is checked separately in the experiment and can alternatively be built into a sufficient class.

2.2. Information and deployment

All solvers have the same known rewards, transition maps, probabilities, terminal payoff, and installed policy. “No precomputed optimum” means that the certificate constructor does not read an optimal-value or optimal-action file. It does not mean model-free or less informed than dynamic programming. An exact reference solve is performed for diagnostics, but its output is not used to build U , the action sets, or the cost-minimizing policy.

Training and verification use different arithmetic objects. Neural and spline fitting use binary64 arithmetic. Saved coefficients are interpreted exactly as rational constants, and their decision rules are compiled into piecewise-constant policies. The guarantee concerns this compiled representation with exact rational evaluation, not an unmodeled floating-point neural forward pass. In particular, every breakpoint has its own stored action; an isolated tie is not discarded because it has probability zero under the initial distribution. The operating guarantee covers every restart, including such points.

3. CONSTRUCTING THE CERTIFICATE INPUTS

A function is represented by rational knots, an affine formula on every open knot interval, and a separate value at every knot. Continuous convex witnesses are a special case; policy values and cost functions can have jumps or exceptional point values. Addition, positive affine composition, integration, maximization, and constrained minimization are performed by exact partition overlays and rational intersections. These operations are finite at every finite horizon, although the number of pieces can grow substantially.

LEMMA 1—Shape-preserving dyadic upper encoding: *Let F be continuous, convex, piecewise affine, and rational on $[0, 1]$. Fix a rational mesh $0 = z_0 < \dots < z_m = 1$. Let H be its chord interpolant through $F(z_i)$. Round $F(0)$ and every chord slope upward to multiples of 2^{-q} , and integrate the rounded slopes successively to obtain G . Then G is convex and*

$$0 \leq G - H \leq 2^{1-q}, \quad F \leq H \leq G. \quad (6)$$

If F is L -Lipschitz and the maximum mesh width is h , then $0 \leq G - F \leq Lh + 2^{1-q}$. For any proposed mesh, its actual global error is determined by the union of the knots of F and G .

PROOF: Convexity orders the exact chord slopes; upward rounding by a common mesh preserves that order. Each rounded slope exceeds its exact value by less than 2^{-q} . The cumulative slope error on an interval of length at most one, plus the rounded starting value, is less than 2^{1-q} . Convexity places the chord above F . For $x \in [z_i, z_{i+1}]$, the Lipschitz bounds from the endpoints imply $H(x) - F(x) \leq Lh$. Finally, $G - F$ is affine between union knots, so its extrema occur at those knots. *Q.E.D.*

The lemma rounds slopes rather than independently rounding ordinates. Independent ordinate rounding need not preserve convexity, which is needed at the next Bellman step. The implementation proposes an adaptive coarsening, rounds retained knot locations to dyadics, recomputes

their exact values, and applies the lemma. It then checks the actual complete error. A failed check increases precision or returns an unresolved result; it is never replaced by a floating tolerance. The uniform-mesh part of the lemma supplies a constructive existence bound independently of the adaptive heuristic.

Put $H_T(\beta) = \sum_{j=0}^{T-1} \beta^j$ and choose

$$\eta = \varepsilon / H_T(\beta), \quad \delta = \eta / 4, \quad e_t = \eta \sum_{j=0}^{T-t-1} \beta^j, \quad e_T = 0. \quad (7)$$

Starting with $U_T = g$, calculate

$$q_t^a = T^a U_{t+1}, \quad F_t = \max_a q_t^a, \quad 0 \leq U_t - F_t \leq \delta. \quad (8)$$

The final step is achieved by Lemma 1, not assumed as an input. Positive affine transitions and affine rewards preserve convexity under expectation and maximization. The construction therefore applies recursively. A Lipschitz bound is also constructive: if U_{t+1} has slope bound L_{t+1} , then F_t has bound $1 + \beta(17/20)L_{t+1}$; the next encoded bound includes its explicit rounding increment.

THEOREM 1—Constructive all-restart class: *For the model (1)–(2), construction (8) produces finite rational upper witnesses. Define*

$$\mathcal{B}_t(x) = \{a : U_t(x) - q_t^a(x) \leq \eta\}. \quad (9)$$

The sets are nonempty at every point. Every measurable selector $\pi_t(x) \in \mathcal{B}_t(x)$ satisfies

$$U_t(x) - e_t \leq J_t^\pi(x) \leq V_t(x) \leq U_t(x), \quad (10)$$

and consequently (5). All certificate inputs are obtained from the primitives by finite exact operations; no optimal or own-policy value is supplied externally.

PROOF: Lemma 1 gives a finite construction at each date. Equation (8) makes U a Bellman supersolution. Backward comparison yields $V_t \leq U_t$. A maximizing action for F_t has deficit at most $\delta < \eta$, proving nonemptiness. If $J_{t+1}^\pi \geq U_{t+1} - e_{t+1}$, positivity and unit mass of the transition probabilities give

$$J_t^\pi \geq q_t^{\pi_t} - \beta e_{t+1} \geq U_t - \eta - \beta e_{t+1} = U_t - e_t.$$

The terminal condition starts the induction. Exact overlays produce $\mathcal{B}_t$ including all equality points. Own-policy values, when requested, follow a separate finite recursion with the policy fixed. *Q.E.D.*

The theorem replaces a verifier for supplied global objects by a construction of those objects. It does not state that the construction is cheaper than every alternative. The next sections report all of its costs and distinguish the size of a stored witness from the potentially larger intermediate envelopes and policy-value partitions.

4. OPTIMAL COSTLY REVISION

With $\mathcal{B}$ fixed, define $D_T = 0$ and

$$D_t(x) = \min_{a \in \mathcal{B}_t(x)} \left\{ (1+x)\mathbf{1}\{a \neq \pi_t^0(x)\} + \beta \sum_j p_j D_{t+1}(f_{aj}(x)) \right\}. \quad (11)$$

All alternatives in this equation are piecewise affine on an exactly computable finite overlay. Minimizing selectors have an exact finite representation with separately checked point values.

THEOREM 2—Endogenous-cost certificate: *A minimizing selector in (11) satisfies (10) and minimizes (4) from every restart among policies selecting actions from $\mathcal{B}$. Randomization and history dependence do not improve this minimum within the fixed class. A certificate consists of U , the deployed policy, its operating values J , and D . It is sufficient to check the Bellman supersolution inequalities, the selected-action deficits, the operating-value equalities, and*

$$D_t \leq c_t^a + \beta P^a D_{t+1} \quad (a \in \mathcal{B}_t), \quad D_t = c_t^{\pi_t} + \beta P^{\pi_t} D_{t+1}, \quad (12)$$

where $c_t^a = (1+x)\mathbf{1}\{a \neq \pi_t^0\}$.

PROOF: The operating claim is Theorem 1. Conditional on the current state and chosen eligible action, backward induction bounds every eligible future intervention cost below by D_{t+1} . Minimization gives the lower bound D_t ; a minimizing selector attains equality. A randomized action gives a convex combination of eligible alternatives. The same conditional argument applies to history-dependent choices. Conditions (12) prove the lower bound and attainment without rerunning the minimizing envelope algorithm. *Q.E.D.*

The occupancy measure in this statement is not frozen at the installed policy. An eligible action that is more expensive now can reduce future exposure to instructions requiring revision. A pointwise guard that keeps the installed action whenever eligible ignores that tradeoff. The experiment isolates this mechanism by comparing both rules on the same $\mathcal{B}$ and evaluating each with its own transition law.

PROPOSITION 1—A global zero-intervention certificate: *Construct the installed value $v_t^0 = T^{\pi_t^0} v_{t+1}^0$, $v_T^0 = g$. If $U_t - v_t^0 \leq \varepsilon$ at every state and restart, then retaining π^0 solves the intervention-cost problem over all policies satisfying (5), with minimum zero. Membership of π^0 in $\mathcal{B}$ is unnecessary.*

PROOF: The upper witness proves $V - v^0 \leq U - v^0 \leq \varepsilon$. Retention has zero intervention cost, and all intervention costs are nonnegative. *Q.E.D.*

This test is performed before recommending a change to a certified incumbent. We preserve the class-only outputs as a diagnostic of conservative over-revision, and add separate zero-cost certificates wherever the test passes. The global claim in Proposition 1 is stronger than class-wise optimality, but applies only when its explicitly computed premise holds.

4.1. Operating preservation and certificate conservatism

A regret guarantee alone does not imply $J^\pi \geq v^0$. A sufficient preservation class intersects (9) with

$$T^a v_{t+1}^0(x) \geq v_t^0(x). \quad (13)$$

When this intersection is nonempty, the same induction yields preservation and the same cost recursion minimizes cost within the intersection. Nonemptiness is an additional condition, not a consequence of (8). In the primary experiment we use (9) and independently evaluate $J^\pi - v^0$ on every open interval and knot. All displayed primary cases satisfy preservation, but this is a checked outcome, not a universal property of the unmodified class.

The local class is an inner approximation to the globally feasible set. A policy can distribute its operating losses across dates more efficiently than the fixed allowance (7), and then be excluded despite satisfying (5). Proposition 1 provides a concrete correction to this conservatism. The full sensitivity comparison below is an envelope over named computed policies; it is not silently promoted to an optimization over every globally feasible policy.

5. GLOBAL BOUNDS BEYOND THE CERTIFIED ACTION CLASS

Class-wise cost optimality and zero-intervention optimality leave a substantive question: how far is a positive-cost repair from the best policy satisfying the original operating constraint? We answer this question by constructing a lower cost bound over the full feasible set, together with a globally feasible upper-cost policy. The construction uses the same primitives and all-restart constraint as Section 2. It neither relaxes the operating tolerance of the deployed policy nor presumes that the local action class exhausts globally feasible policies.

5.1. A lower operating witness from computed defects

Let γ_t select a maximizer of $T^a U_{t+1}$ and let $d_t \geq 0$ be a checked uniform upper bound on $U_t - T^{\gamma_t} U_{t+1}$. Define

$$b_T = 0, \quad b_t = d_t + \beta b_{t+1}, \quad L_t = U_t - b_t. \quad (14)$$

The defect d_t is computed by exact interval-limit and knot comparisons, not inferred from training error. One can replace the maximizing γ by any feasible selector for which the displayed defect is checked.

LEMMA 2—Constructive two-sided operating witnesses: *If $U_T = g$, $U_t \geq T^a U_{t+1}$ for every action, and $U_t - T^{\gamma_t} U_{t+1} \leq d_t$, then*

$$L_t \leq J_t^\gamma \leq V_t \leq U_t \quad (15)$$

from every state and restart. Under the compression rule of Section 3, $d_t \leq \delta$, hence $b_t \leq \delta H_{T-t}(\beta)$.

PROOF: The upper comparison follows by backward induction. If $J_{t+1}^\gamma \geq U_{t+1} - b_{t+1}$, monotonicity and translation by the constant b_{t+1} imply $J_t^\gamma \geq T^{\gamma_t} U_{t+1} - \beta b_{t+1} \geq U_t - d_t - \beta b_{t+1} = L_t$. The terminal values agree. Maximizing the operating return gives the middle inequality. The final bound follows by summing the checked compression defects. *Q.E.D.*

This construction charges both the upper witness and its defect computation. An exact optimal value is unnecessary. Its accuracy can be tightened independently of the operating tolerance at which a policy will be deployed.

5.2. Unrestricted support values and cost lower bounds

Write $k_t(x, a) = (1 + x)\mathbf{1}\{a \neq \pi_t^0(x)\}$ for implementation cost, to distinguish it from the maintenance expenditure c_a . For each nonnegative rational multiplier λ , compute

$$\begin{aligned} W_T^\lambda &= \lambda g, \\ W_t^\lambda(x) &= \max_a \{ \lambda r(x, a) - k_t(x, a) + \beta P^a W_{t+1}^\lambda(x) \}. \end{aligned} \quad (16)$$

Let σ^λ attain the displayed maximum, and independently evaluate its operating and implementation values. Backward induction gives $W_t^\lambda = \lambda J_t^{\sigma^\lambda} - C_t^{\sigma^\lambda}$. The maximization is unrestricted: it does not use $\mathcal{B}$, a fixed-occupancy distribution, or a supplied operating optimum. Whole-cell upper comparisons and equality at the selected action independently certify its value.

Let $\mathcal{F}_\varepsilon$ denote policies satisfying (5). The statements below also hold for randomized and history-dependent policies whenever their continuation from the specified restart satisfies the operating constraint. In particular they bound the all-restart feasible Markov policies studied in the application. For any finite set $\Lambda \subset [0, \infty)$, define

$$B_t(x) = \max \left\{ 0, \max_{\lambda \in \Lambda} [\lambda \{L_t(x) - \varepsilon\} - W_t^\lambda(x)] \right\}. \quad (17)$$

THEOREM 3—Class-free intervention-cost bounds: *For every $\pi \in \mathcal{F}_\varepsilon$,*

$$B_t(x) \leq C_t^\pi(x) \quad \text{for every restart } (t, x). \quad (18)$$

Let $\hat{\pi}$ be any separately verified member of $\mathcal{F}_\varepsilon$, and let ν be the initial distribution. Then, writing $C_{\nu, \varepsilon}^ = \inf_{\pi \in \mathcal{F}_\varepsilon} \int C_0^\pi d\nu$,*

$$\int B_0 d\nu \leq C_{\nu, \varepsilon}^* \leq \int C_0^{\hat{\pi}} d\nu, \quad \int C_0^{\hat{\pi}} d\nu - C_{\nu, \varepsilon}^* \leq \int (C_0^{\hat{\pi}} - B_0) d\nu. \quad (19)$$

A zero right-hand side proves global cost optimality for the original constraint, not only for a certified inner class.

PROOF: Unrestricted support optimality implies $W_t^\lambda \geq \lambda J_t^\pi - C_t^\pi$. Feasibility and Lemma 2 give $J_t^\pi \geq V_t - \varepsilon \geq L_t - \varepsilon$. Multiplication by $\lambda \geq 0$ and rearrangement yield each term of (17) as a lower bound on C_t^π . Nonnegative implementation costs justify the additional zero. Taking the pointwise maximum retains a common lower bound on every feasible policy. Integrating, taking an infimum, and using the feasible $\hat{\pi}$ gives (19). The conditional backward comparison for W^λ also bounds history-dependent and randomized policies. *Q.E.D.*

The argument uses classical weak duality and Bellman comparison; these principles are not claimed as new. The additional construction is a checkable certificate for the original all-state operating constraint: computed two-sided witnesses, unrestricted support values, a separately feasible deployment, and an explicit global cost gap. There is no claim of strong duality, attainment of the lower bound by the finite multiplier set, or global cost optimality merely because a scalarized problem is solved exactly. The lower bound can be loose.

The maximum in (17) may select different multipliers at different states. This is legitimate because every term bounds the same feasible policy's cost. It does not license splicing the actions of different σ^λ : such a splice changes continuation values and must be evaluated and certified again. Our computation selects one complete feasible policy, not a statewise mixture of unchecked recommendations.

5.3. Constructive feasibility and charged work

A finite multiplier portfolio need not rely on accidental feasibility. In the maintenance model, $0 \leq k_t \leq 2$. Put $M_t = 2H_{T-t}(\beta)$.

PROPOSITION 2—A sufficient multiplier without an optimal-value input: *For $\lambda > 0$, an exact maximizing selector in (16) satisfies*

$$V_t - J_t^{\sigma^\lambda} \leq M_t/\lambda, \quad U_t - J_t^{\sigma^\lambda} \leq b_t + M_t/\lambda. \quad (20)$$

Consequently, $\max_t(b_t + M_t/\lambda) \leq \varepsilon$ is a sufficient all-state, all-restart feasibility condition. It uses only the computed defect bound and model primitives.

PROOF: Use an operating-optimal policy as a comparator in the support problem. Its implementation cost is at most M_t , so $\lambda J_t^{\sigma^\lambda} - C_t^{\sigma^\lambda} \geq \lambda V_t - C_t^{\pi^*}$. Since $C_t^{\sigma^\lambda} \geq 0$, this gives the first inequality. Lemma 2 implies $U_t - V_t \leq b_t$, which gives the second. The comparator proves the inequality; the algorithm does not evaluate it to construct its inputs. *Q.E.D.*

For the fixed $T \leq 12$ study below, we build witnesses using accuracy parameter 10^{-4} , so $b_t \leq 1/40000$. The predeclared multiplier 4096 satisfies $1/40000 + 2H_{12}(19/20)/4096 < 1/100$. Thus the portfolio contains a feasible support policy for both declared tolerances independently of the installed proposal. All candidate policies are nevertheless checked directly; a mathematical sufficient bound is not a substitute for checking the serialized object.

For $m = |\Lambda|$, the total work is the sum of witness construction, m unrestricted support solves, their own-policy value evaluations and checks, construction of the lower-bound envelope, and final feasibility checking. If $N_{t,\lambda}$ denotes the actual overlay size for multiplier λ , a conservative addition to (30) is

$$O\left(A(b) \sum_{\lambda \in \Lambda} \sum_t (AD + A^2) \{N_{t,\lambda} + K_t + Z_{t+1,\lambda}\}^2\right) + C_{\text{lower}} + C_{\text{encode}} + C_{\text{check}}. \quad (21)$$

The notation b here is the integer bit-length bound from Section 7, not the defect sequence b_t . All failed feasibility candidates are charged. If the class-minimum candidate is also offered, its complete construction is added rather than treating its inherited certificate as free. No bound on the output sizes, or speedup relative to a classical verified solver, follows from (21) alone. The method buys a stronger economic statement—a bound against every globally feasible policy—at an explicitly recorded computational cost.

6. PROPAGATING THE ALL-RESTART RESTRICTION

The support bound in Section 5 treats each restart separately. Its optimizing comparison policy need not respect the operating restriction at subsequent restarts. Yet an implementable policy in $\mathcal{F}_\varepsilon$ must respect all of them. This section uses that additional restriction to strengthen the cost certificate without solving another unrestricted support problem or changing the deployed policy.

Let $\mathcal{M}$ be a finite set of nonnegative rational numbers containing zero. Write $q_t^{U,a} = T^a U_{t+1}$ and retain the two-sided witnesses L, U and support lower bound B . Starting from $A_T = 0$, calculate backwards

$$z_t^a(x) = k_t(x, a) + \beta P^a A_{t+1}(x), \quad (22)$$

$$m_t^\mu(x) = \min_a \{z_t^a(x) + \mu[L_t(x) - \varepsilon - q_t^{U,a}(x)]\}, \quad (23)$$

$$A_t(x) = \max\{0, B_t(x), m_t^\mu(x) : \mu \in \mathcal{M}\}. \quad (24)$$

Every action remains in the minimization. In particular, we do not impose $q_t^{U,a} \geq L_t - \varepsilon$ separately on the actions available to a randomized controller. That actionwise restriction could exclude a feasible mixture and invalidate a cost lower bound.

THEOREM 4—Restart-propagated global cost certificate: *In the finite-horizon maintenance model, let $L \leq V \leq U$ and B be the verified witnesses of Section 5. Equations (22)–(24) have finite rational piecewise-affine representations. For every all-restart-feasible Markov policy, including a randomized Markov policy,*

$$0 \leq B_t \leq A_t \leq C_t^\pi \quad \text{at every state and date.} \quad (25)$$

The statement also holds for history-dependent policies whose conditional operating continuation is feasible at every history from which it is evaluated. If $\hat{\pi}$ is a separately certified feasible policy, then

$$\int A_0 d\nu \leq C_{\nu,\varepsilon}^* \leq \int C_0^{\hat{\pi}} d\nu, \quad \int C_0^{\hat{\pi}} d\nu - C_{\nu,\varepsilon}^* \leq \int (C_0^{\hat{\pi}} - A_0) d\nu. \quad (26)$$

Relative to (19), the certified excess-cost bound is reduced by exactly $\int (A_0 - B_0) d\nu \geq 0$, with no change in the upper deployment. Enlarging $\mathcal{M}$ cannot decrease A when the other inputs are fixed.

PROOF: At the terminal date implementation cost is zero. Suppose the stated lower bound has been proved at date $t + 1$. Conditional on the current state, let α_a be the action probabilities of a feasible policy. Its continuation cost satisfies

$$C_t^\pi \geq \sum_a \alpha_a \{k_t^a + \beta P^a A_{t+1}\} = \sum_a \alpha_a z_t^a. \quad (27)$$

Its operating continuation is at most the optimal continuation and hence at most U_{t+1} . Feasibility at the current restart and the lower operating witness give

$$L_t - \varepsilon \leq V_t - \varepsilon \leq J_t^\pi \leq \sum_a \alpha_a q_t^{U,a}. \quad (28)$$

For any $\mu \geq 0$, the minimum in (23) is bounded by the same expression averaged with weights α . Equations (27) and (28) imply

$$m_t^\mu \leq \sum_a \alpha_a z_t^a + \mu \left(L_t - \varepsilon - \sum_a \alpha_a q_t^{U,a} \right) \leq \sum_a \alpha_a z_t^a \leq C_t^\pi.$$

The already certified B_t and zero are also lower bounds. Their maximum proves the induction. The same conditional argument applies after every history if its future continuations satisfy the restart restriction. Integrating and comparing with a feasible deployment proves (26) and the exact tightening identity.

Finite positive affine pullbacks, addition, and maxima or minima of finitely many rational affine pieces preserve finite piecewise-affine representability, including exceptional knot values.

They construct every displayed object. Finally, induction shows monotonicity in $\mathcal{M}$: a larger next-date lower bound raises every z_t^a ; a larger multiplier portfolio adds candidates to the outer maximum. *Q.E.D.*

The distinction concerning histories is material. Feasibility only in expectation at time zero does not require each subsequently realized history to meet the same operating tolerance. Equation (27) cannot in general use future all-restart lower bounds for such a history. The theorem therefore does not apply the stronger propagated bound to that different constraint. The Markov all-state problem (5), which is the computational objective of this paper, satisfies precisely the required continuation restriction. The unpropagated support inequality remains valid for any individual feasible restart without this extra continuation premise.

The term $\mu = 0$ already transmits future mandatory intervention costs through the transition law. Positive μ uses the additional necessary operating inequality at the current restart. The resulting construction is a local Lagrangian relaxation coupled backwards across dates. Weak duality and backward comparison are classical; the methodological contribution here is the complete construction and independently checkable composition with two-sided operating witnesses and class-free support bounds. Neither strong duality nor sharpness of the finite multiplier portfolio is a premise. A positive residual gap is not a proof of optimality.

6.1. Construction work and proof objects

An algorithm evaluates (22), constructs the action minima for each local multiplier, and forms the outer maximum. No new unrestricted W^λ solve, optimal operating surface, or externally supplied feasible-policy value is needed. The witnesses and feasible deployment must already have been constructed and charged as in Section 5; their costs do not disappear when the propagation is added.

For clarity, let n_a and n_p be the numbers of actions and shock branches, $m = |\mathcal{M}|$, and S_t the number of cells in the common input overlay at date t , including pulled-back knots of A_{t+1} and U_{t+1} and knots of the installed policy, L_t , and B_t . Each of the m action minima has at most $O(n_a^2 S_t)$ cells before simplification. Overlaying those minima and comparing their $m + 2$ alternatives has at most $O((m + 2)^2 m n_a^2 S_t)$ cells before simplification. These counts yield, for a deliberately elementary implementation, the conservative total bit-work bound

$$O\left(A(b) \sum_t \{n_a n_p + m n_a^2 + (m + 2)^2 m n_a^2\}^2 S_t^2\right) + C_{\text{encode}} + C_{\text{audit}}, \quad (29)$$

where b bounds all intermediate rational bit lengths and $A(b)$ bounds their arithmetic cost. Sorting and repeated exact evaluations are included by the squared expression. This bound is output-sensitive: S_t and b include objects produced by the backward recursion, not small constants assumed in advance. A finite proof object does not imply dimension-independent complexity or a faster operating-value solve.

To verify the object, the checker reconstructs z and each action expression, checks that the supplied m_t^μ lies below every action expression and equals its supplied minimizing selector, and checks that A_t lies above every outer term and equals its supplied source selector. It additionally checks $B_t \leq A_t \leq C_t^\pi$, the terminal value, all hashes linking the inputs, and the reported integrals. These are affine endpoint-limit and isolated-point comparisons, not calls to the optimizing envelope constructor. The selector may choose different bound sources at different states; it is a proof selector, not a new deployment policy.

7. WORK, REPRESENTATION, AND THE ROLE OF PROPOSALS

Let N_t count all partition cells generated in constructing action-continuation functions, intersections, and admissible masks at date t . Let M_t count stored upper-witness pieces, K_t count installed and deployed policy pieces, and Z_t count the operating and cost-value pieces generated by evaluation and optimization. Let b bound actual integer bit lengths, and let $A(b)$ bound the cost of rational arithmetic and comparison, including normalization.

PROPOSITION 3—Charged finite-representation work: *For A actions and D positive affine shock maps per action, a direct overlay implementation constructs the objects in Theorems 1 and 2 with a finite number of rational operations depending on the generated partitions, not on an external state-grid cardinality. A conservative bound is*

$$C_{\text{fit}} + C_{\text{compile}} + O\left(A(b) \sum_t \{(AD + A^2)(N_t + K_t + Z_{t+1})^2 + N_t^2\}\right) + C_{\text{encode}} + C_{\text{check}}. \quad (30)$$

Here N_t^2 includes an elementary adaptive chord check. All quantities are construction outputs, not assumed free witnesses. For the upper-witness subproblem alone, a uniform chord mesh of width $h \leq \delta/(2L_t)$ and slope precision $2^{1-q} \leq \delta/2$ gives $M_t = O(1 + L_t/\delta)$.

PROOF: Each affine composition pulls back finitely many knots. An overlay has at most the sum of its input knot counts, and each action pair contributes at most one intersection on each common affine cell. Sorting, evaluation, and pairwise comparison are bounded by the deliberately conservative squared-count expression. A chord test compares its chord with every intervening source knot; recursive tests are bounded by $O(N_t^2)$. Own-policy evaluation and cost optimization produce their own partitions and are charged through K_t, Z_t . Lemma 1 gives the final mesh statement. Encoding and the separate checker are added rather than hidden in a range-oracle cost. *Q.E.D.*

This is an output-sensitive finite-construction bound, not a polynomial-in-dimension theorem or a guarantee of acceleration. In particular, Z_t can grow rapidly even when M_t is small. Exact intersections and integrals can also have large denominators. The implementation records intermediate and stored piece counts, bit lengths, serialized bytes, and stage times, and rejects an intermediate representation exceeding 200,000 pieces. No cap is reached in the registered cohort. A memory statistic for the whole Python process is reported separately from the portable certificate size; it includes libraries and allocations and is not a bound on deployment memory.

PROPOSITION 4—A quantitative proposal condition: *Suppose a proposal chooses a maximizer of $\hat{q}_t^a$ and satisfies a verified whole-state error $\max_a \|\hat{q}_t^a - q_t^a\|_\infty \leq \kappa_t$. Put $\rho_t = U_t - \max_a q_t^a$. Its selected-action deficit obeys*

$$U_t - q_t^{\pi_t^0} \leq \rho_t + 2\kappa_t \leq \delta + 2\kappa_t. \quad (31)$$

If $\delta + 2\kappa_t \leq \eta$ at every date, the proposal is eligible everywhere and the intervention-cost minimum in (11) is zero. On any region carrying analogous verified local bounds, that region needs no further action-eligibility subdivision for the proposed action.

PROOF: For an exact maximizing action a^* , optimality under $\hat{q}$ gives $q^{a^*} - q^{\pi^0} \leq 2\kappa_t$. Adding ρ_t proves the bound. An eligible installed policy can be followed at every date with zero cost, which is minimal by nonnegativity. The same comparison is pointwise on a region with a valid uniform local error bound. *Q.E.D.*

The proposition identifies a causal role for a verified approximation margin: it can eliminate revisions and action-eligibility refinement. It does not remove the cost of constructing U , computing the approximation bounds, or compiling the proposal. Sampled fitting loss is not a premise of this proposition. The numerical study does not assume that its neural candidates satisfy it. The exact own-policy test in Proposition 1 can be substantially less conservative.

For deployment, a stored piecewise policy answers an exact rational query by knot search and comparison. If an alternative online rule costs c_{on} per query and loading plus offline construction costs $C_{\text{off}} + C_{\text{load}}$, a genuine amortization claim requires $c_{\text{on}} > c_{\text{stored}}$ and

$$Q > (C_{\text{off}} + C_{\text{load}})/(c_{\text{on}} - c_{\text{stored}}). \quad (32)$$

We do not measure this denominator here and therefore make no observed deployment break-even claim. The exact online sign guard from the earlier action-independent model remains a strong control in that model, not a competing solution to the present coupled cost problem.

The next section retains the original constructive-class benchmark and its execution records, including its fitting and timing evidence. Section 9 reports the newly executed unrestricted-cost study separately; inherited timings are not represented as measurements from the new machine.

8. COUPLED COMPUTATIONAL EVIDENCE

8.1. Prospective design and complete accounting

The primitives, horizons $T \in \{4, 8, 12\}$, tolerances $\varepsilon \in \{.01, .05\}$, and proposals were specified in the inherited R31 protocol before the R32 full cohort. This is a referee-informed exploratory study, not an independent preregistered confirmation. We retain the pilot outcomes and amendments rather than describe the research history as prospective from its origin.

Each horizon has an all-defer incumbent, a mechanism stress incumbent, two linear-spline fitted-Q proposals, and three neural fitted-Q proposals. Splines use 33 or 129 equally spaced nodes. Neural proposals use 257 midpoint training states, one hidden ReLU layer of width 16, three outputs, 200 Adam steps per date, learning rate .02, and seeds 31001–31003. Targets use the next fitted continuation, never optimal-value labels. Exact compilation precedes any candidate-specific certificate or reference comparison. One frozen policy is reused across the two tolerances.

The stress incumbent maintains when $x \geq 9/10$ at all nonterminal decision dates and otherwise defers; the final decision always defers. It is intentionally misconfigured and separately labeled. It tests whether a current revision can avoid future exposure to an expensive instruction, not whether a neural model has discovered an institutional practice. All six stress results are reported irrespective of sign.

The reference is uncompressed exact piecewise-affine dynamic programming, not a large brute-force grid. A second accuracy-only control uses the same compressed witnesses, greedily maximizes q_t^a , evaluates its own policy, and checks its error. The two spline controls exploit the natural local basis. Fresh-process neural timing includes interpreter and library initialization, training, and compilation. We separately charge witness construction, installed-value evaluation, action-set construction, cost minimization, final evaluation, serialization, internal verification, and the independent final checker. These single-machine timings are descriptive; neither seeds nor tolerances provide independent machine-speed replications.

All 42 class certificates pass the independent checker, and their deployed operating values weakly exceed their installed values at every restart. Raw neural passes are zero out of 18, whereas all 12 spline cases pass. The separate own-policy witness test also certifies each spline incumbent, yielding 12 additional zero-intervention certificates by Proposition 1. The class-only

TABLE I
COMPLETE COUPLED-MODEL COHORT

Proposal	Cases	Raw passes	Certified	Dynamic savings
Neural	18	0	18	0
Spline	12	12	12	0
All defer	6	0	6	0
Occupancy stress	6	0	6	6

The same frozen proposal is evaluated at two tolerances; these are 21 frozen policy/horizon pairs, not 42 independent models. Raw passes use the exact uncompressed reference only as a diagnostic. The independent upper-witness test separately certifies all 12 spline cases with zero intervention.

spline outputs are preserved to show, rather than conceal, the cost of a conservative sufficient class. No universal post-repair pass statistic is interpreted as neural approximation success.

8.2. End-to-end work and strong classical controls

TABLE II
REPRESENTATION SIZE AND ARITHMETIC IN ONE STATE DIMENSION

T	ϵ	Exact V	Input to compression	Stored U	U bits	Cost pieces
4	0.01	2	2	2	66	11
4	0.05	2	2	2	66	11
8	0.01	11	11	11	67	78
8	0.05	11	11	9	67	78
12	0.01	108	48	27	67	359
12	0.05	108	28	15	67	350

Counts are maxima over dates and, for cost, proposals. Exact roots, knot values, and one-sided limits are included. No state-grid cardinality is used as a complexity proxy. Complete value/cost arithmetic can use more bits than stored witnesses; all stage records and certificates are retained.

TABLE III
MEASURED SOLUTION AND CERTIFICATION COSTS (SECONDS)

T	ϵ	Exact DP	Witness	Greedy	Spline cold	Neural cold
4	0.01	0.003294	0.005446	0.01483	0.0681	2.211
4	0.05	0.003294	0.005648	0.015	0.06876	2.21
8	0.01	0.01027	0.01835	0.04712	0.1801	2.526
8	0.05	0.01027	0.01782	0.0463	0.1655	2.523
12	0.01	0.0883	0.09859	0.2839	0.8417	3.575
12	0.05	0.0883	0.06703	0.2311	0.6392	3.436

Medians across the declared proposal family; single executions are descriptive, not a significance test. Cold totals charge fresh-process neural import/training, compilation, witness construction, own-policy evaluation, admissible sets, cost optimization, encoding, internal verification, and the independent final checker. The exact DP and certified greedy rows solve accuracy-only objectives; neither is represented as a minimum-revision-cost solver.

At horizon 12, the exact value representation reaches 108 pieces. The compressed witness reaches 27 stored pieces at tolerance .01 and 15 at .05; the largest intermediate envelope at

those budgets has 48 and 28 pieces. Stored witnesses use at most 67-bit rational components, while operating and cost representations use up to 256 bits in the cohort. Thus the reduction in stored witness size is real, but it does not eliminate intermediate construction or downstream cost-value complexity. The largest intervention-cost representation has 359 pieces, and the largest uncompressed serialized primary certificate is 370,046 bytes.

The timing comparison does not establish a neural advantage. The natural spline basis is both more accurate before repair and cheaper to fit in this model. The structure-aware exact operating solver is also inexpensive. The proposed cost pipeline solves an additional constrained objective, but that distinction does not justify a claim that it is a faster operating-value solver. It is precisely why the operating-only baseline and the class-minimum-cost output are reported separately. The new positive computational statement is a fully constructed coupled certificate with measured input costs, not an inferred speedup over an artificially enormous state grid.

8.3. Endogenous occupancy and economic sensitivity

TABLE IV
ACTION-DEPENDENT OCCUPANCY CHANGES THE OPTIMAL REVISION PATTERN

T	ε	Regret bound	Pointwise cost	Dynamic cost	Saving
4	0.01	0.00269551	4.21218	4.2099	0.00227628
4	0.05	0.0134775	4.2167	4.20532	0.0113815
8	0.01	0.00148561	9.55894	9.54147	0.0174693
8	0.05	0.00742774	9.59545	9.50629	0.0891575
12	0.01	0.00116031	13.831	13.8072	0.0237425
12	0.05	0.00608421	13.878	13.7581	0.119917

Both policies use the same computed admissible sets. Costs are exact uniform-initial-state integrals under each deployed policy, then displayed as decimals. The stress incumbent is predeclared and intentionally misconfigured; the other 36 cases have zero dynamic-versus-pointwise savings.

The dynamic solution strictly reduces intervention cost relative to the pointwise rule in all six stress cases. Both select only from the same certified action class. Therefore the difference cannot be explained by a more permissive operating target or by supplying the dynamic solver a better upper witness. It arises from the action-dependent continuation of implementation costs. The complete piecewise policies and cost functions document the differing revision pattern. For the remaining candidates, equality of the two cost functions is retained; the mechanism is not asserted to operate for every installed rule.

For economic interpretation, let J_i and C_i be the exactly integrated operating return and intervention cost of a computed deployment. Its net return at shadow price $\lambda \geq 0$ is $J_i - \lambda C_i$. We store all nonnegative intersections and tie points, and report the entire upper envelope over dynamic, pointwise, certified-greedy, exact operating-optimal, and globally eligible raw policies. No large illustrative λ is chosen to manufacture a universal favorable sign. The thresholds are normalized tradeoffs, not calibrated willingness to pay. In particular, a low intervention cost does not mean a policy dominates in operating value. The envelope exposes precisely when the retained operating loss is compensated by reduced implementation cost. The distinction between this finite comparison and the globally constrained problem remains explicit.

TABLE V

COMPLETE SHADOW-PRICE SENSITIVITY FOR THE MECHANISM STRESS CONTROL

T	ε	Envelope winners as λ increases	Switch prices
4	0.01	Exact $\rightarrow$ Dynamic	0.0016623
4	0.05	Exact $\rightarrow$ Dynamic	0.0082732
8	0.01	Exact $\rightarrow$ Dynamic	0.0012955
8	0.05	Exact $\rightarrow$ Dynamic	0.0064472
12	0.01	Exact $\rightarrow$ Greedy $\rightarrow$ Dynamic	1.7362e-05, 0.0013897
12	0.05	Exact $\rightarrow$ Greedy $\rightarrow$ Dynamic	0.00046808, 0.0070918

The envelope is over the displayed computed deployments, not all feasible policies. All pairwise intersections and tie sets are stored as exact fractions. One cost unit is a normalized intervention; no monetary calibration or preferred illustrative price is imposed.

9. A GLOBALLY BOUNDED REVISION PORTFOLIO

9.1. Fixed design and exact verification

The new computation retains the maintenance primitives, all seven installed proposals, horizons 4, 8, 12, and tolerances .01, .05. The inspected staged R33 source specified $\Lambda = \{0, 1, 4, 16, 64, 256, 1024, 4096\}$, the witness parameter 10^{-4} , and the unchanged 200,000-piece cap. This is an exploratory, referee-informed extension, not a new independent random cohort. The three neural seeds and the two spline policies are inherited installed objects. The cost of retraining them is not included in incremental revision work or represented as newly measured. Their original fitting records remain in the computational archive.

For each horizon and installed rule, we construct all eight unrestricted support policies and the common operating witnesses from primitives. We then independently evaluate all candidate operating and implementation values. At each tolerance, the portfolio includes every support policy passing $\sup_{t,x}(U_t - J_t) \leq \varepsilon$, together with a freshly reconstructed class-minimum policy at that tolerance. Selection minimizes the uniformly integrated implementation cost among these complete feasible candidates. The full witness, class, support, own-policy, serialization, and checker costs are recorded. A separately computed exact operating dynamic program is an information-matched diagnostic, not an input to the certificate constructor.

This design produces 168 support certificates, 42 newly constructed class-control certificates, and 42 assembled primal–dual certificates. A read-only audit reloads all 252 objects and disables the maximizing envelope routine while checking them. It verifies all intervals through affine endpoint limits, every isolated knot, all Bellman comparisons and selected-action equalities, the exact support identities, the feasibility envelope, and the cost lower bound. It also recomputes every reported economic scalar from the stored rational functions. The complete cohort passes; all selected policies separately preserve the installed operating values at every restart. No piece cap or unresolved case is suppressed.

The R33 staged sources describe a prior omission of class-construction cost. R34 uses their corrected constructor from its first execution: the class candidate is reconstructed on the same machine, its complete certificate is checked for exact equality to the inherited object, and its full construction is charged. The old staged source archive is preserved; it is not represented as a newly executed R33 publication. The fresh support computation and the later protocol-fixed restart propagation have separate provenance and execution ledgers.

9.2. Economic gain and the remaining global gap

All 18 installed-neural configurations receive a strictly less costly certified deployment than the local-class minimum, at unchanged operating tolerance and with separately verified

TABLE VI
POSITIVE-COST IMPROVEMENTS AND GLOBAL UNCERTAINTY FOR INSTALLED NEURAL RULES

T	ε	Improved	Cost saved	Saved (%)	Global gap
4	0.01	3/3	0.0039–0.0039	2.53–3.35	0.0394–0.0427
4	0.05	3/3	0.0131–0.0131	8.93–12.10	0.0695–0.0873
8	0.01	3/3	0.0140–0.0442	1.36–10.00	0.3019–0.6649
8	0.05	3/3	0.0867–0.4423	17.32–48.08	0.2825–0.4323
12	0.01	3/3	0.0087–0.2729	0.57–9.33	0.7710–1.2740
12	0.05	3/3	0.2143–0.7302	5.74–51.98	0.6333–1.9959

Ranges cover the three fixed installed seeds, not new independent replications. Cost saved compares the selected globally feasible portfolio policy with the freshly reconstructed local-class minimum. The last column is the certified upper bound on excess cost relative to the unrestricted constrained optimum, not relative to that class. All 18 operating-preservation checks pass. Normalized intervention units; uniform initial distribution.

operating preservation. The gain is not attributed to superior neural approximation: all twelve spline configurations have zero-cost feasible incumbents, as already identified by the earlier zero-intervention test. Rather, the positive-cost comparisons demonstrate economically relevant conservatism of the local class. A policy outside that class can distribute its operating losses over time while meeting the original all-restart constraint.

For example, at $T = 12$, $\varepsilon = .05$, and installed seed 31003, the new policy’s normalized intervention cost is approximately .674484, compared with 1.404696 for the class minimum. Its operating-regret certificate remains below .05 and it preserves the installed return. The global lower cost bound is approximately .041222, leaving a certified upper bound of .633262 on the policy’s excess cost over the unrestricted constrained optimum. The approximately .730212 reduction relative to the local class is therefore established, whereas global optimality of this positive-cost deployment is not. Reporting both comparisons prevents a strong improvement over a conservative rule from being mistaken for a sharp global solution.

The lower bound is zero and the upper bound is zero in the twelve spline cases; these are genuine global optima, including relative to randomized or history-dependent feasible policies. For the other cases the complete lower/upper intervals, rather than a binary “success” label, appear in the supplement. Adding multipliers can tighten the lower envelope but cannot worsen it; the finite portfolio supplies no guarantee that this process closes every gap.

9.3. Full work and the role of installed proposals

TABLE VII
CHARGED PORTFOLIO CONSTRUCTION AND FINITE REPRESENTATION

T	Portfolio (s)	Class (s)	Max. pieces	Max. bits	RSS (MiB)
4	0.415	0.042	12	156	22.7
8	1.378	0.145	82	199	26.9
12	8.993	0.885	361	251	51.2

Per-configuration medians across the complete seven-rule, two-budget cohort. Portfolio time includes witness construction, all eight support solves and evaluations, internal checks, exact lower-bound construction, encoding, and a freshly rebuilt class candidate. Class time is included in, not added to, portfolio time. The additional final read-only audit is recorded separately. Maxima cover stored support and own-value representations; they are not bounds on all intermediate overlays. RSS is peak process memory, not deployed-certificate memory. Installed proposal fitting is inherited and excluded from these incremental revision timings.

The displayed per-configuration costs include all eight support solves and their checks, even when a candidate is infeasible, and a fresh class-control construction. They conservatively charge the shared witness and support portfolio again when reporting the second tolerance; unique batch work and wall-clock records are reported separately. The exact operating-control time is not presented as a cost-optimal-revision comparator: solving for the best operating policy and bounding the minimum implementation cost under an operating constraint are different outputs. The class-control timing is the relevant restricted-cost comparator, and the additional work purchases class-free lower bounds. There is no measured neural-specific acceleration or high-dimensional claim.

The experimental price grid consists of computational multipliers used to produce support policies and lower bounds. These are not calibrated monetary shadow prices, nor are they the illustrative external revision-price parameter used in the inherited welfare sensitivity exercise. No universal success count at an arbitrary external price is substituted for the full cost bounds or the retained sensitivity frontier. The normalized unit remains one implementation intervention weighted by $1 + x$ and discounted under the deployed policy’s own occupancy.

10. COMPUTED EFFECT OF THE RESTART RESTRICTION

10.1. *Fixed comparison and provenance*

The restart experiment holds the two-sided operating witnesses, support portfolio, deployed policies, primitives, tolerances, and installed rules fixed. Its only change is to strengthen the lower cost bound by Theorem 4. The local multiplier portfolio is the same eight-element set used for the support calculations. The protocol was committed before implementing or evaluating this propagation. The preceding support results and the inherited maintenance evidence were already known. This is a prospectively specified extension within an exploratory research program, not an independently preregistered confirmation or a new neural training cohort.

All 42 configurations are computed and independently checked. The additional proof object records eight local minima per decision date, their action selectors, the outer source selector, and the complete propagated function. Its canonical hash binds it to the previously certified primal–dual input and its support hashes. In a fresh process, the complete audit rechecks the 252 base objects and then the 42 propagated objects with the optimizing envelope routines disabled. Thus 294 exact certificate objects, including every whole-cell and isolated-point comparison, constitute the final audit.

10.2. *Sharper bounds without policy substitution*

The integrated lower cost bound increases strictly in 22 configurations and is unchanged in 20. None decreases. The upper operating and implementation values are those of exactly the same deployments as before propagation. These improvements therefore measure the value of exploiting the all-restart restriction, not a more favorable policy-selection rule, a looser operating tolerance, or a new neural fit.

The largest absolute increase occurs for NN 31003 at $T = 12$ and $\varepsilon = 0.01$. The certified cost interval changes from $[0.480952, 1.514983]$ to $[0.742636, 1.514983]$; the corresponding excess-cost bound falls from 1.034030 to 0.772346. These displayed interval endpoints are rounded outwards. The largest relative reduction in the excess-cost bound across the cohort is approximately 38.51%. The full table, including all zero changes, is in the supplement.

The twelve zero-cost spline deployments remain global optima. No previously positive global gap becomes zero in this study. Consequently the added information is a strictly tighter global certificate in the indicated configurations, not an assertion that every positive-cost repair is

TABLE VIII
TIGHTENING THE GLOBAL COST INTERVAL WITH THE SAME CERTIFIED DEPLOYMENTS

T	ε	Tightened	Positive gain	Added (s)	Max. pieces
4	0.01	4/7	0.001560–0.007092	0.310	20
4	0.05	1/7	0.016636–0.016637	0.238	13
8	0.01	5/7	0.000014–0.130940	1.748	103
8	0.05	4/7	0.011664–0.042277	1.103	92
12	0.01	4/7	0.008865–0.261685	14.872	471
12	0.05	4/7	0.010245–0.104461	7.909	407

Every row contains the same seven installed rules. Gain is the increase in the integrated cost lower bound, and therefore the decrease in the excess-cost bound; displayed range endpoints are rounded outwards. Added time is the median construction, internal verification, and serialization time, excluding the separately timed read-only final audit. No additional unrestricted support problem is solved. Max. pieces refers to the stored propagated lower bound, not all intermediate overlays. Positive counts are exact, including gains below display precision.

now globally optimal. In particular, the intervention savings over the local action class and the tightening relative to a global feasible-set lower bound answer different questions and are reported separately.

10.3. Cost of stronger certification

The complete propagation batch uses no additional unrestricted support solve. Its measured construction, internal comparison checks, and serialization cost is 178.896 seconds; the additional read-only closure audit costs 90.990 seconds. These are incremental costs on top of all previously charged witness, support, class-control, and feasible-policy computations, not a claim that stronger certification is free. The final combined base-plus-propagation audit takes 154.340 seconds on the recorded machine. Repeated audit passes used during development are retained in the execution ledger rather than hidden in a solver-speed comparison.

The largest stored propagated lower bound has 471 affine pieces, and the stored closure/minimum functions use rational components of up to 716 bits. The 42 compressed closure objects occupy 17.846 MiB. Intermediate action overlays and Python memory can exceed these stored sizes. The computations remain one-dimensional; neither the piece count nor the independence from a chosen state grid establishes a high-dimensional complexity result.

An additional exhaustive two-period check enumerates 108 deterministic contingent plans and 1,512 equal-mixture plans at four rational restart states. The test checks the lower bound for each plan satisfying the conditional restart restriction, including 4 deterministic plans and 20 mixtures. Twelve malformed closure tests must be rejected, including changed isolated points, terminal values, multipliers, selectors, hashes, and reported integrals. These finite tests exercise the implementation; the continuum and randomized-policy guarantees follow from the theorem and the complete affine proof objects, not from sampling.

11. A GLOBAL OPTIMUM AT THE ACTIVE STOPPING BOUNDARY

11.1. The unchanged original economy

The original model has state $(u, x) \in \mathcal{D} = (1.2, 2.8) \times (.5, 2)$, horizon one, and dynamics

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (33)$$

$$dx_s = \{(.02 + .06p_s)x_s - c_s\} ds + .2p_s x_s dZ_s^x, \quad d\langle Z^u, Z^x \rangle_s = -.25 ds. \quad (34)$$

Controls (c, θ, p) lie in $[.05, .8] \times [-.2, .2] \times [-.5, .8]$. At first exit, capped at one, running utility and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (35)$$

$$G(u, x) = -.02(u-2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1-s). \quad (36)$$

Payoffs discount at rate .04. The retained principal target is a current-state actor $\pi(t, u, x)$ with

$$\sup_{(t, u, x) \in [0, 1] \times \overline{\mathcal{D}}} \{V_2(t, u, x) - J_2^\pi(t, u, x)\} \leq .01. \quad (37)$$

The retained whole-domain upper bound is 7.181834580823298. Neither the maintenance model nor the scalar theorem below changes that number. The 47-coordinate current-state derivative/residual bridge remains uninstantiated. We preserve the original target and its derivations rather than silently substitute a smaller policy class.

11.2. From a derivative diagnostic to a complete scalar certificate

Fix the unchanged restart $(0, 61/50, 5/4)$, $k = 2$, $c = 3/4$, $p = 0$, and a constant $\theta \in [-1/5, 1/5]$. Write $J(\theta)$ for the resulting original-model payoff. In the preceding validated calculation the lower-boundary exit probability at zero adjustment lay in $[\text{.68915}, \text{.68916}]$. Curvature changed sign, so a concavity-based search justified at a central restart could not be transferred to this one. We instead prove interval-wide monotonicity.

THEOREM 5—Global optimum in the active-boundary constant class: *For the fixed original-model class above, J is differentiable in the interior and has one-sided derivatives at its endpoints. The following uniform lower bounds hold:*

$$J'(\theta) \geq \begin{cases} 8163/125000, & -1/5 \leq \theta \leq 0, \\ 47251/125000, & 0 \leq \theta \leq 1/5. \end{cases} \quad (38)$$

Consequently the unique global maximizer is $\theta^ = 1/5$. The inherited validated endpoint enclosure $[-3.7609, -3.7607]$ is therefore an enclosure of the optimal payoff in this constant class.*

PROOF: Let $a = 6/5$, $b = 14/5$, $d = 1/50$, and $\sigma = 1/20$. Deterministic wealth is bounded below by $25/49 > 1/2$, so it does not exit. Stopped Itô's formula writes the payoff as its initial settlement plus the discounted occupation integral of $h_\theta = (\partial_t + \mathcal{L}_\theta - .04)g + \ell_2$. On the preference interval, explicit rational bounds give $h_\theta \geq 2$ and $\partial_y h_\theta > 0$. Extend h_θ constantly above b .

For the process killed only at a , the derivative of survival is

$$S'_\theta(t) = 16e^{-16\theta} \Phi\left(\frac{\theta t - 1/50}{(1/20)\sqrt{t}}\right).$$

For fixed t this is log-concave in θ , so its minimum on a compact interval is bounded by the smaller endpoint value. Monotone coupling, with the state dependence of h_θ held fixed, bounds the derivative of the occupation expectation below by $2S'_\theta(t)$. The direct parameter derivative of h is at least $-4/125$ on the negative half and $-54/125$ on the positive half.

On $t \in [1/25, 1/5]$ and $\theta \in [-1/5, 0]$, rational Gaussian and exponential bounds give $S'_\theta(t) \geq 8/25$. On $t \in [1/4, 1]$ and $\theta \in [0, 1/5]$, they give $S'_\theta(t) \geq 352/625$. Also $e^{-.04t} \geq 24/25$. These imply half-line derivative lower bounds .066304 and .379008. The likelihood-score error for paths hitting the distant upper boundary is less than .001 uniformly in θ . Subtracting it proves (38). Strict positivity gives uniqueness. The supplementary proof supplies every constant and the executable rational checks. *Q.E.D.*

The endpoint enclosure is inherited, not a newly run payoff quadrature. What is new is the global optimization certificate connecting it to the entire parameter interval. It improves the active-boundary component while keeping its scope precise: one restart and a constant-control class. A globally verified scalar optimum does not solve the feedback problem (37).

12. CONCLUSION

Uniform operating accuracy and low implementation cost are different objectives. The construction here begins with economic primitives, produces Bellman witnesses with quantified encoding error, and uses their certified action sets to optimize intervention costs under endogenous occupancy. Whole-interval and isolated-point verification make the deployed representation checkable independently of training and of the optimizing envelope routine. An additional own-policy certificate identifies globally cost-minimal zero-intervention solutions without forcing them into a conservative local class.

The coupled maintenance computation establishes these steps end to end and exhibits a genuine dynamic revision mechanism. It also preserves a demanding comparison: classical spline proposals and exact structured dynamic programming are strong alternatives, and the present experiment does not establish a neural speed advantage. The active-boundary theorem converts an earlier local diagnostic into a complete scalar optimum in the original stopped economy. The larger all-domain target remains unchanged. All preceding proofs, unsuccessful experiments, and referee responses are retained in a lossless historical annex, while the current article separates its proved constructions from stronger empirical or high-dimensional claims not established by the evidence.

The unrestricted support certificate further quantifies the distance between a feasible positive-cost revision and the true constrained cost optimum. Its lower bound concerns every feasible policy, while its upper bound comes from an independently checked deployment. This distinction allows substantial cost improvements to be reported together with their remaining global uncertainty, without equating exact scalarized computation with exact constrained optimality.

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SUPPLEMENT TO CERTIFIED BELLMAN OPERATORS: ALL-RESTART BOUNDS FOR COSTLY POLICY REVISION

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1. FINITE REPRESENTATIONS AND VERIFICATION SEMANTICS

The replication uses a finite rational object $f = (z_i, m_i, b_i, v_i)$: $f(x) = m_i x + b_i$ on each open interval (z_i, z_{i+1}) , while $f(z_i) = v_i$. This distinction is necessary even when the original operating value is continuous. Compiled decisions can jump, a cost-minimizing decision can have a special tie value, and an affine transition can send an entire exceptional point to another exceptional point. Integral-zero sets must therefore remain in a certificate for all restart states.

For a common overlay of two such functions, an affine inequality on an open interval holds if and only if it holds in both endpoint limits. The actual endpoint values are checked separately. This is the reason a midpoint may safely identify which affine formula applies on an already constructed cell, but may not by itself verify that formula's inequality. Every range check uses the limits as well as the stored points. A missing cell, a wrong point value, or a changed model fails verification.

An affine composition $f(\alpha x + \gamma)$ with $\alpha > 0$ adds every preimage of a knot that lies in $[0, 1]$. Values at those preimages are inherited exactly. Maximization first overlays all source partitions; on each common affine cell it inserts all pairwise action intersections. The minimum action index breaks an exact tie. Constrained minimization additionally overlays the binary eligibility masks. Thus eligible singleton actions are not inferred from neighboring open cells.

Normalization removes a knot only if both adjacent affine formulas agree and the stored knot value agrees with that same formula. A separate test creates a point spike, normalizes it, composes it with an affine map, and confirms that the spike and its preimage survive. Exact integration ignores singleton masses only when integrating against the continuous initial distribution. The dynamic recursions never ignore them.

2. CONSTRUCTIVE WITNESS DETAILS

2.1. *Existence and implemented adaptive compression*

For a convex Lipschitz function on $[0, 1]$, a chord interpolant on a mesh of width h lies above the function. A conservative Lh error bound follows from the Lipschitz comparisons to both endpoints. Hence a sufficiently fine rational mesh and sufficiently fine dyadic rounding produce a certified upper approximation at every finite backward step. The maximum over finitely many positive affine compositions of convex piecewise-affine functions remains convex and piecewise affine.

The implementation uses a more economical, checked adaptive rule. Initially retain both endpoints. For each candidate chord, find the source knot with the largest chord error. If that error exceeds half the stage compression allowance, retain that knot and recurse on the two children. This yields an unrounded chord cover with error at most half the allowance. Retained locations are rounded to dyadics, and exact function values at the rounded locations are recomputed. Ordered chord slopes, rather than independent point ordinates, are rounded upward; integrating these rounded slopes from an upward-rounded initial value preserves convexity.

Rounding a retained coordinate can change the chord error. The acceptance decision therefore does not rely only on the error before rounding. It evaluates $G - F$ over the union of all source and encoded knots, including endpoints, and requires $0 \leq G - F \leq \delta$. Precisions 24, 32, 48, 64, 96, and 128 bits for coordinates are available, with 16 extra bits for slope and initial-value rounding. A failed precision ladder is an unresolved construction, not a certificate. All registered cases pass at the first coordinate precision. The implemented cap is 200,000 pieces; the existence theorem does not identify that engineering cap with a universal termination bound.

The original local implementation rounded ordinates and ran timing in a shared Python process. The archived attempt and all its results are retained. The final implementation rounds slopes to preserve the induction invariant and times each neural fit in a fresh process. These corrections do not change the economic model, seeds, training recipe, or tolerance. They are documented in the amendment rather than hidden by overwriting the earlier outputs.

2.2. Upper and lower witnesses

Let $\rho_t = U_t - TU_{t+1}$. Exact compression proves $0 \leq \rho_t \leq \delta$. The upper comparison $V_t \leq U_t$ is independent of the proposal. With $e_t = \eta + \beta e_{t+1}$ and $e_T = 0$, an eligible action satisfies

$$T^a(U_{t+1} - e_{t+1}) = q_t^a - \beta e_{t+1} \geq U_t - e_t.$$

The lower witness is therefore the explicitly constructed function $L_t = U_t - e_t$. It is not a supplied policy-value surface. The stronger realized check uses the exactly evaluated J_t^π and reports $\max_{t,x}(U_t - J_t^\pi)$; this can be considerably smaller than the allowance ε .

Exact installed-policy evaluation is a separate recursion. No raw-policy performance statistic is inferred from final repair. The reference diagnostic $\max(V - v^0)$ uses the uncompressed optimum only after the raw policy is frozen. The deployable zero-intervention certificate instead uses $\max(U - v^0)$ and needs no reference optimum. These quantities are recorded separately as `raw_exact_regret` and `raw_regret_upper`.

3. COST OPTIMALITY, PRESERVATION, AND COMPARISON CLASSES

Given nonempty eligible action sets, the intervention-cost recursion is ordinary finite-horizon dynamic programming with state-dependent constraints. Its output is nevertheless an additional economic object beyond the operating-value solve. The exact checker verifies, at every point where action a is eligible, that D_t does not exceed that action's immediate cost plus its continuation. It also verifies equality for the chosen action. This proves minimum cost and attainment from every restart; a uniform initial density is used only to summarize the resulting functions.

The primary computation does not impose one-step installed-value preservation as an action restriction. Instead, it computes the installed and revised values and verifies the complete difference. All 42 primary outputs preserve operating value. It would be incorrect to claim that this empirical fact establishes preservation for all inputs under the deficit-only action class. The optional intersection with $T^a v_{t+1}^0 \geq v_t^0$, together with a nonemptiness condition, provides that theorem when required.

The additional zero-intervention certificates have a different scope of optimality. Once $U - v^0 \leq \varepsilon$ everywhere is checked, zero is a lower bound on the cost of every feasible policy, not merely policies in the local class. Thus retaining a certified incumbent is globally optimal for intervention cost. This does not prove that a nonzero class minimum is globally optimal for the unconstrained-by-class regret problem. The 12 separate zero certificates explicitly expose this distinction.

The pointwise comparator retains an installed action when eligible and otherwise minimizes current implementation cost with action-index tie breaking. It does not optimize future implementation exposure. Its operating and cost functions are recomputed under the policy it actually deploys. The certified greedy comparator maximizes operating continuation against U . The uncompressed reference maximizes operating value exactly. Neither accuracy-only comparator is mislabeled as solving the constrained minimum-cost problem.

The price envelope is an exact comparison over the listed computed policies. For each pair with different costs, the algorithm includes a nonnegative crossing $(J_i - J_j)/(C_i - C_j)$; it checks all open intervals between crossings and every crossing point. A final unbounded interval covers arbitrarily large prices. Identical lines and exact ties remain valid alternatives. This produces a complete finite comparison, without implying an optimization over uncomputed feasible policies or an empirical monetary calibration.

4. COMPLETE ACTIVE-BOUNDARY PROOF

4.1. *Deterministic wealth and the Dynkin integrand*

Throughout this section the primitives, controls, and restart are those in Theorem 5 of the article. With $p = 0$, $c = 3/4$, and $x(0) = 5/4$,

$$x(t) = \frac{75}{2} - \frac{145}{4}e^{t/50}. \quad (1)$$

For $0 \leq t \leq 1$, $e^{t/50} \leq (1 - 1/50)^{-1} = 50/49$, hence $x(t) \geq 25/49 > 1/2$. It is decreasing and starts below two, so no wealth boundary is reached. Let y denote preference. Direct differentiation of the unchanged settlement gives

$$\begin{aligned} h_\theta(t, y) &= \frac{(3/4)^{1-y}}{1-y} - \theta^2 + 8 - .04\theta(y-2) - .00005 + .002 - \frac{.075}{x(t)} \\ &\quad + .0008(y-2)^2 - .004 \log x(t) + .32(1-t), \end{aligned} \quad (2)$$

$$\partial_\theta h_\theta = -2\theta - .04(y-2). \quad (3)$$

For $y \in [6/5, 14/5]$, put $z = y - 1 \in [1/5, 9/5]$ and $L = \log(4/3) < 1/3$. The utility is $-e^{Lz}/z$, which is increasing on this interval because $Lz < 1$. Its lower bound is $-5e^{1/15} \geq -75/14$. Also $|y - 2| \leq 4/5$, $x(t) \geq 1/2$, and $\log x(t) \leq \log(5/4) < 1/4$. Dropping nonnegative terms yields

$$h_\theta \geq 8 - \frac{1}{20000} - \frac{3}{20} - \frac{1}{1000} - \frac{4}{625} - \frac{1}{25} - \frac{75}{14} = \frac{342357}{140000} > 2. \quad (4)$$

The utility derivative is $e^{Lz}(1 - Lz)/z^2 \geq 10/81$. Thus

$$\partial_y h_\theta \geq \frac{10}{81} - \frac{4}{3125} - \frac{1}{125} = \frac{28901}{253125} > 0. \quad (5)$$

Extend $h_\theta(t, y)$ constantly from $b = 14/5$ to larger y . The extension remains bounded, nondecreasing in y , and at least two. Its parameter derivative retains the bounds in (3). Conservative absolute bounds $|h| \leq 20$ and $|\partial_\theta h| \leq 2$ hold for this extension.

4.2. A coupling lower bound for the derivative

Let $Y_t = u_0 + \theta t + \sigma W_t$, killed at $a = 6/5$, where $u_0 = 61/50$ and $\sigma = 1/20$. If drift increases, the coupled process is higher at every date and lower-boundary survival events are nested. For a fixed nondecreasing integrand $h \geq 2$, paths surviving under both drifts have a nonnegative payoff change, while newly surviving paths contribute at least two. Taking a positive difference quotient gives

$$\partial_\theta \mathbb{E}_\theta [h(t, Y_t) \mathbf{1}\{t < \tau_a\}] \geq 2\partial_\theta \Pr(t < \tau_a). \quad (6)$$

The direct dependence of h_θ on θ must be added, not included in this frozen-integrand comparison. It contributes at least $-4/125$ for $\theta \in [-1/5, 0]$ and at least $-54/125$ for $\theta \in [0, 1/5]$, after integrating over a discounted survival time of at most one. Gaussian likelihood scores justify differentiability and dominated passage to the occupation integral for this bounded extension.

The reflection formula with drift, differentiated exactly, yields

$$S'_\theta(t) = 16e^{-16\theta} \Phi\left(\frac{\theta t - 1/50}{(1/20)\sqrt{t}}\right). \quad (7)$$

The function $\log \Phi$ is concave. Indeed, with $r = \varphi/\Phi$, its second derivative is $-r(z)\{z + r(z)\} < 0$; when $z < 0$, the elementary Gaussian tail inequality gives $r(z) > -z$. Equation (7) is therefore log-concave in θ . Its minimum on each of the two half intervals is at least the smaller of its endpoint values.

For $t \in [1/25, 1/5]$, the normal arguments at $\theta = 0$ and $\theta = -1/5$ are bounded below by -2 and $-14/5$, respectively. The latter follows by maximizing $4\sqrt{t} + (2/5)/\sqrt{t}$ on that compact interval; its maximum is $14/5$. The rational inequalities

$$\begin{aligned} 24 < e^{16/5} < 25, \quad \Phi(-2) > 1/50, \quad \Phi(-14/5) > 1/500, \\ \Phi(-4/5) > 1/5, \quad \Phi(6/5) > 22/25. \end{aligned} \quad (8)$$

imply $S'_\theta(t) \geq 8/25$ on the entire negative drift half interval.

For $t \in [1/4, 1]$, the endpoint arguments at $\theta = 0$ and $\theta = 1/5$ are at least $-4/5$ and $6/5$. Hence $S'_\theta(t) \geq 352/625$ on the positive drift half interval. Since $e^{-.04t} \geq 24/25$, the two lower bounds for the derivative of the lower-killed occupation expression are

$$2\frac{4}{25}\frac{24}{25}\frac{8}{25} - \frac{4}{125} = .066304, \quad (9)$$

$$2\frac{3}{4}\frac{24}{25}\frac{352}{625} - \frac{54}{125} = .379008. \quad (10)$$

No derivative sample grid appears in this argument. The intervals cover every admissible constant drift, including the shared point zero.

4.3. Distant-boundary error and rational checks

The upper boundary is distant but not discarded. Under all admissible drifts, a path hitting it must have a Brownian excursion of at least $(b - u_0 - 1/5)/\sigma$. Reflection and a Gaussian tail bound give

$$p_b \leq 2 \exp\left[-\frac{(b - u_0 - 1/5)^2}{2\sigma^2}\right] = 2e^{-380.88} < 2^{-379}. \quad (11)$$

For a path-dependent difference supported on that event, the drift likelihood score has second moment at most $1/\sigma^2$. Cauchy–Schwarz bounds the derivative discrepancy between the actual doubly killed occupation integral and the extended lower-killed one by

$$\frac{20}{\sigma} \sqrt{p_b} + 2p_b < 400 2^{-189} + 2 2^{-379} < 1/1000. \quad (12)$$

Subtracting $1/1000$ from the preceding half-line bounds gives precisely $8163/125000$ and $47251/125000$. A strictly increasing continuous function on the compact drift interval has the unique maximizer $1/5$.

The file `replication/boundary.py` proves the elementary constants in (8) with exact fractions. Exponential bounds use Taylor sums and an explicit geometric tail. A Machin arctangent identity with alternating-series bounds proves a rational bound for π , and $\sqrt{2\pi} < 251/100$ follows. Gaussian lower bounds use monotone right sums of mesh $1/100$ on finite intervals, with every exponential term bounded by a rational Taylor calculation. Truncating the infinite tail only lowers a tail integral. For $\Phi(6/5)$ the integral is from zero to $6/5$, added to $1/2$. Terms are rounded downward to a common dyadic denominator before summation, with the direction of the bound preserved. The code records the resulting exact rational lower bounds and requires them to exceed all displayed thresholds.

The payoff enclosure at $\theta = 1/5$ is inherited from the unchanged validated R30 killed-kernel oracle. Its source path, Git blob hash, and the fact that it is not recomputed are recorded in `results/boundary.json`. The current theorem changes its interpretation from an endpoint evaluation to the optimum of the entire constant class. It does not assert feedback optimality or satisfy the original all-domain current-state target.

5. FULL COHORT AND AUDIT RESULTS

TABLE I: All registered class-based certificates

T	Proposal	ε	Raw regret	Certified bound	Cost saving	Bytes
4	Defer	0.01	1.61894	0.00269551	0	3897
4	Defer	0.05	1.61894	0.0134775	0	3887
4	Stress	0.01	1.61894	0.00269551	0.00227628	4660
4	Stress	0.05	1.61894	0.0134775	0.0113815	4659
4	Spline 33	0.01	4.30982e-16	4.09353e-08	0	3505
4	Spline 33	0.05	4.30982e-16	4.09353e-08	0	3504
4	Spline 129	0.01	4.14599e-16	4.09353e-08	0	3494
4	Spline 129	0.05	4.14599e-16	4.09353e-08	0	3493
4	NN 31001	0.01	0.124593	0.00269551	0	4434
4	NN 31001	0.05	0.124593	0.0134775	0	4425
4	NN 31002	0.01	0.14356	0.00269551	0	4311
4	NN 31002	0.05	0.14356	0.0134775	0	4302
4	NN 31003	0.01	0.143355	0.00269551	0	4440
4	NN 31003	0.05	0.143355	0.0134775	0	4431
8	Defer	0.01	3.35621	0.00148562	0	19038
8	Defer	0.05	3.35621	0.00804234	0	18773
8	Stress	0.01	3.35621	0.00148561	0.0174693	41810

T	Proposal	ε	Raw regret	Certified bound	Cost saving	Bytes
8	Stress	0.05	3.35621	0.00742774	0.0891575	41606
8	Spline 33	0.01	0.0057581	0.00148562	0	15821
8	Spline 33	0.05	0.0057581	0.00575812	0	12143
8	Spline 129	0.01	0.000411466	0.000411565	0	12451
8	Spline 129	0.05	0.000411466	0.000775484	0	12179
8	NN 31001	0.01	0.234831	0.00148562	0	18524
8	NN 31001	0.05	0.234831	0.00804234	0	18250
8	NN 31002	0.01	0.296282	0.00148562	0	18488
8	NN 31002	0.05	0.296282	0.00804234	0	17477
8	NN 31003	0.01	0.224246	0.00148562	0	19178
8	NN 31003	0.05	0.224246	0.00742775	0	18500
12	Defer	0.01	4.75581	0.00116122	0	195094
12	Defer	0.05	4.75581	0.00603201	0	194062
12	Stress	0.01	4.75581	0.00116031	0.0237425	370046
12	Stress	0.05	4.75581	0.00608421	0.119917	359043
12	Spline 33	0.01	0.0057581	0.00116122	0	156289
12	Spline 33	0.05	0.0057581	0.00543908	0	116384
12	Spline 129	0.01	0.000411466	0.000411565	0	91427
12	Spline 129	0.05	0.000411466	0.00105788	0	86102
12	NN 31001	0.01	0.64517	0.00116402	0	184623
12	NN 31001	0.05	0.64517	0.00643136	0	179921
12	NN 31002	0.01	0.611512	0.00115741	0	193825
12	NN 31002	0.05	0.611512	0.00616493	0	173948
12	NN 31003	0.01	0.224246	0.00116122	0	206325
12	NN 31003	0.05	0.224246	0.00603201	0	209874

The tables display rounded summaries only. Every primary certificate stores rational knots, values, policies, Bellman witnesses, and intervention-cost functions. A separate file records the exact interval-wide error and cost residual. The checker does not invoke the envelope optimizer. Its only shared components are the representation, the primitive model, and elementary exact affine operations. This is implementation separation, not a proof assistant or an assertion of independence from all shared arithmetic code.

Eight deliberately malformed inputs are rejected: changed model, absent model, negative tolerance, missing date, missing interval, changed endpoint, false cost, and infeasible action. Tests also retain point spikes and affine preimages and compare the exact two-period Bellman solution with exhaustive adaptive scenario-tree enumeration at four rational restart states. The test count is not substituted for the whole-state inequalities in the certificates.

6. HISTORICAL PRESERVATION AND CURRENT CLAIMS

The document `HISTORY_R32.pdf` concatenates the five R30 submission objects without altering their pages: article, supplement, response, computational record, and the earlier historical annex. The repository retains their original source and result paths. Thus the previous action-independent operating-mode study, inventory comparison, Adam transport and rollback derivations, central-state search, active-boundary quadrature failures, and earlier referee responses are not deleted to simplify the current narrative.

In particular, the unfavorable exhaustive inventory timing ratios, the 26/60 raw passes in the prior operating-mode cohort, and the exact symbolic and online-guard controls remain available. They are historical evidence, not results of the new action-dependent model. The original whole-domain bound remains 7.181834580823298 and its target remains .01. No new result is relabeled as a 47-coordinate derivative bridge, a high-dimensional certificate, or neural-specific acceleration. The constructive coupled certificates and the global active-boundary scalar theorem are additional results with explicitly delimited objects.

7. UNRESTRICTED SUPPORT CERTIFICATES AND THE GLOBAL COST GAP

This appendix supplements Section 5 of the article. The original upper witness and all new support objects use the same rational representation: affine formulas on open cells and independently stored values at their knots. Jumps induced by a compiled installed rule are allowed; measure-zero values are not discarded from any restart guarantee.

7.1. *Verification instructions and proof dependencies*

A support certificate stores the exact model, its multiplier, U , the selector γ , the defect bounds d_t , the installed policy, its support-optimal deployment, W , and that deployment's operating and implementation values. The checker first verifies the model and horizon, legal point and interval actions, and the terminal conditions. It checks $T^a U_{t+1} \leq U_t$ for every action and the displayed defect inequality for γ . The propagated errors and $L = U - b$ are then reconstructed by the checker. No file purporting to contain V is read.

For every support object, the checker verifies that every action-continuation alternative in the support Bellman equation lies below W_t , and that the stored deployment attains equality. It also checks the own-policy Bellman equalities for J, C and the identity $W = \lambda J - C$. These checks imply unrestricted support optimality by backward induction. They do not require a maximizing-envelope computation. The same check proves that no adaptive or randomized action choice can exceed W : conditional expectation and a convex combination preserve the upper comparison.

An assembled certificate stores a feasible deployment, its J, C , its feasibility upper witness, the lower-cost function B , and a selector identifying the attaining lower-bound term on every open cell and knot. It binds the complete ordered multiplier portfolio by SHA-256 hashes of canonical uncompressed JSON. Every support certificate must first pass its own check; hashes alone do not establish mathematical validity. The assembled checker verifies the lower envelope by comparing each candidate term to B and checking equality with the selected term. It verifies $B \leq C$, every restart's operating tolerance, and the exact integrated gap. A separate check compares the selected J to the installed own-policy value to establish operating preservation when it holds.

The implementation of the lower selector accepts integer indices for lower-bound terms, not action labels. Its feasible policy selector accepts only the three legal maintenance actions. Confusing these objects could produce a silently spliced policy, so their validations are distinct. The terminal cost is zero, the terminal operating value is g , and the terminal support value is λg .

7.2. *Feasibility, finite selection, and the scope of the lower bound*

The sufficient-multiplier proposition guarantees that the declared price 4096 has a feasible support selector for every installed rule in the declared model and horizon set. This is an ex ante statement from the bounded implementation cost and witness defect, not an assertion inferred

from the 42 observed passes. The independent audit verifies the rational sufficient bound for each horizon. Every actual serialized candidate is nevertheless tested directly at all dates and states.

The selected primal policy minimizes integrated implementation cost only among the finite set of individually verified complete policies, including the class baseline. This selection is not an optimization over all feasible policies. In contrast, every lower-bound term applies to every feasible policy, so the assembled cost interval bounds the unrestricted constrained optimum. The two scopes must not be exchanged. In particular, the computation does not assume that a support-optimal policy exists for every constrained-optimal operating budget, or that a single common multiplier is a complete dual for the simultaneous continuum of restart constraints.

The pointwise maximum of lower bounds remains a lower bound even when no single policy attains it everywhere. Integration is valid because every admissible policy dominates that pointwise function. Equality of the integrated bounds proves optimality for the stated initial distribution; without a pointwise zero gap it does not by itself prove that the deployment minimizes cost at each individual restart. The supplied all-restart cost-gap envelope is an additional reported quantity, not a consequence of a small integrated gap.

All implementation costs are nonnegative. At zero cost, the integrated gap vanishes for the declared spline cases. Their stronger every-restart zero-cost result follows directly because the retained incumbent has identically zero intervention, not merely zero cost almost everywhere. Isolated action differences in other deployments remain part of every feasibility check even when they do not affect a uniform-distribution integral.

7.3. Costs, unfavorable cases, and preservation of attempts

The eight multiplier problems are all solved and charged, including every candidate that fails an operating tolerance. The common witness is produced from primitives; no precomputed optimal-value file is supplied. The classical class candidate is rebuilt and checked on the same machine using the complete-cost constructor. Every proposed support policy is independently evaluated rather than assigned a cost from its optimization label. The final byte-level audit is an additional cost, reported separately from the checks within construction. Exact operating dynamic programming is run only as a recorded comparator and diagnostic.

The replication stores all 168 support outcomes and all 42 final intervals, not only the neural improvements. Timing records distinguish fitting of the previously installed proposals from revising those fixed proposals now. They do not add incompatible machine timings to obtain a purported speedup. The frozen R32 records and staged R33 source archive remain losslessly preserved. R34 execution records are new and separately identified; no unavailable prior run is reconstructed or asserted as evidence. Compressed encodings and their checksums remain explicitly distinguished from canonical uncompressed certificate bytes.

The checker rejects eight malformed support certificates and eight malformed assembled certificates, including changed endpoint values, missing dates and intervals, false costs or lower bounds, illegal actions, bad multiplier signs, incorrect terminal conditions, and altered support hashes. Four two-period rational restart states are independently compared with all 27 contingent two-step action trees. The final read-only audit disables the optimizing envelope routine, so a hidden repeat of the constructor's optimization is an error rather than a fallback.

7.4. Complete unrestricted-cost cohort

TABLE II: Complete globally bounded portfolio; all 42 declared configurations

T	ε	Installed	λ	Cost lower	Cost upper	Gap	Saved
4	0.01	Defer	64	4.312943	4.937421	0.624478	0.000051
4	0.05	Defer	Class	3.664041	4.936309	1.272268	0.000000
4	0.01	Stress	256	3.628917	4.209701	0.580784	0.000202
4	0.05	Stress	Class	3.000666	4.205319	1.204652	0.000000
4	0.01	Spline 33	0	0.000000	0.000000	0.000000	0.000000
4	0.05	Spline 33	0	0.000000	0.000000	0.000000	0.000000
4	0.01	Spline 129	0	0.000000	0.000000	0.000000	0.000000
4	0.05	Spline 129	0	0.000000	0.000000	0.000000	0.000000
4	0.01	NN 31001	256	0.074641	0.114005	0.039364	0.003946
4	0.05	NN 31001	64	0.025368	0.094890	0.069522	0.013065
4	0.01	NN 31002	256	0.109583	0.152287	0.042704	0.003946
4	0.05	NN 31002	64	0.045915	0.133172	0.087257	0.013065
4	0.01	NN 31003	256	0.083932	0.123969	0.040037	0.003946
4	0.05	NN 31003	64	0.030930	0.104854	0.073924	0.013065
8	0.01	Defer	64	10.203910	10.823575	0.619666	0.001457
8	0.05	Defer	Class	9.445140	10.815918	1.370779	0.000000
8	0.01	Stress	256	8.887172	9.535807	0.648635	0.005663
8	0.05	Stress	64	7.942794	9.491694	1.548900	0.014597
8	0.01	Spline 33	0	0.000000	0.000000	0.000000	0.029666
8	0.05	Spline 33	0	0.000000	0.000000	0.000000	0.000000
8	0.01	Spline 129	0	0.000000	0.000000	0.000000	0.000000
8	0.05	Spline 129	0	0.000000	0.000000	0.000000	0.000000
8	0.01	NN 31001	256	0.205701	0.533681	0.327980	0.030524
8	0.05	NN 31001	64	0.060208	0.415981	0.355774	0.087152
8	0.01	NN 31002	256	0.095675	0.397532	0.301857	0.044178
8	0.05	NN 31002	64	0.000360	0.282891	0.282531	0.086746
8	0.01	NN 31003	1024	0.352314	1.017217	0.664903	0.014023
8	0.05	NN 31003	64	0.045344	0.477618	0.432274	0.442321
12	0.01	Defer	64	15.001167	15.637251	0.636084	0.001873
12	0.05	Defer	Class	14.118948	15.627451	1.508503	0.000000
12	0.01	Stress	256	13.140504	13.797365	0.656861	0.009866
12	0.05	Stress	64	11.957376	13.741126	1.783750	0.016973
12	0.01	Spline 33	0	0.000000	0.000000	0.000000	0.029366
12	0.05	Spline 33	0	0.000000	0.000000	0.000000	0.000880
12	0.01	Spline 129	0	0.000000	0.000000	0.000000	0.000000
12	0.05	Spline 129	0	0.000000	0.000000	0.000000	0.000000
12	0.01	NN 31001	1024	2.620836	3.894877	1.274042	0.065588
12	0.05	NN 31001	256	1.524265	3.520192	1.995927	0.214315
12	0.01	NN 31002	256	1.880895	2.651884	0.770989	0.272947
12	0.05	NN 31002	64	1.033237	2.262773	1.229536	0.452344
12	0.01	NN 31003	1024	0.480953	1.514982	1.034029	0.008744
12	0.05	NN 31003	64	0.041222	0.674484	0.633262	0.730212

Displayed decimals are descriptive rounded summaries of exact rational bounds. The machine-readable fractions, not outward rounding of these printed decimals, define the certificates. “Class” denotes the freshly reconstructed local-class candidate. Saved cost is relative to that class, not a claim that a positive-cost deployment is globally optimal. The 12 spline cases have identically zero cost and exact zero global gap. All selected deployments satisfy their original all-restart operating tolerance and separately preserve installed operating values.

8. RESTART-PROPAGATED LOWER BOUNDS: VERIFIER AND COMPLETE COHORT

8.1. *Precise feasibility class*

For a Markov randomized rule, the conditional expected continuation values $J_t^\pi(x)$ and $C_t^\pi(x)$ include subsequent action randomization as well as the model shocks. Feasibility requires $J_t^\pi(x) \geq V_t(x) - \varepsilon$ for every t, x . It does not require each action in the support of the rule to satisfy this inequality on its own. For a history-dependent extension, impose the analogous conditional expected inequality after every history for which a continuation is being certified. An initial-distribution-only expected constraint defines a different problem and does not justify recursive use of future feasible-continuation lower bounds. These definitions explain exactly the class covered by Theorem 4.

8.2. *Verifiable finite object*

The serialized closure stores the model and version, horizon, tolerance, installed-rule name, the canonical input-certificate SHA-256, the complete ordered support-hash list, local multipliers, every A_t , every m_t^μ , an action selector for each minimum, and the source selector for the outer maximum. The terminal A_T is identically zero. Stored rational knots include zero and one, are strictly ordered, and have separate function values. Every open interval has an affine formula. This representation distinguishes values at ties from their one-sided limits and remains valid when an installed rule induces jumps.

The fresh-process checker first verifies all support and base certificates, rather than treating their hashes as proofs of validity. A hash identifies the exact object; Bellman comparisons establish its validity. It then reconstructs the action expressions in equations (22)–(23). For each local minimum it checks domination by every action expression and equality at its supplied selector. For the outer maximum it checks domination of every source term and equality at the source selector. It verifies $B \leq A \leq C^{\hat{\pi}}$, the terminal condition, horizon and multiplier identities, and both reported integrals. All selector values must be legal integers, including those at isolated knots. Every comparison examines the union of rational knots and both limits of each resulting open affine interval. The optimizing envelope routine is disabled in the constructor module, the shared arithmetic module, and the base-audit module.

These comparisons are sufficient for exact min/max identities: a supplied minimum lying below all alternatives and equaling one alternative is the true minimum, and the dual argument proves a maximum. No optimizer is needed to check these certificates. The checker still uses exact partition overlays, rational arithmetic, and file parsing; these costs are timed. It is an independently executed checker sharing an exact arithmetic library, not a formally verified proof assistant or a fully independent second arithmetic implementation.

8.3. *Adversarial and finite-tree tests*

The mutation suite requires rejection of a negative local multiplier, an omitted multiplier row, an inflated isolated lower-bound point, an inflated isolated minimum point, an illegal minimizing action, a nonintegral outer selector, an incorrect terminal value, a missing date, a changed base hash, a changed support hash, a false integrated gap, and a changed tolerance. Eight support mutations and eight base-summary mutations are also retained. A malformed object is never repaired silently by the checker.

At each of four rational initial states the finite-tree test enumerates the three first actions and all nine shock-contingent second-action pairs. It next enumerates all 378 unordered equal mixtures of the 27 plans. Mixtures sharing the first action use the conditional average second-stage

value after each observed shock; different first actions reveal which branch of the mixture was chosen, and each corresponding continuation is checked separately. This realizes a behavioral randomized rule with no additional remembered private-state constraint. The test rejects no action merely for failing an individual operating test. Only plans or mixtures satisfying the actual conditional operating inequalities enter the lower-bound comparison. The stored test record reports both the total and the eligible counts, so nonvacuity is auditable.

8.4. Complete exact outcomes

TABLE III: All restart-propagated cost intervals

T	ε	Installed	Old lower	New lower	Upper	New gap	Gain
4	0.01	Defer	4.312942	4.312942	4.937421	0.624479	0.000000
4	0.05	Defer	3.664040	3.664040	4.936310	1.272269	0.000000
4	0.01	Stress	3.628916	3.636007	4.209702	0.573694	0.007091
4	0.05	Stress	3.000666	3.017302	4.205319	1.188016	0.016636
4	0.01	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.05	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.01	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.05	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.01	NN 31001	0.074641	0.076202	0.114005	0.037803	0.001560
4	0.05	NN 31001	0.025367	0.025367	0.094890	0.069523	0.000000
4	0.01	NN 31002	0.109582	0.111143	0.152287	0.041143	0.001560
4	0.05	NN 31002	0.045915	0.045915	0.133172	0.087257	0.000000
4	0.01	NN 31003	0.083931	0.085492	0.123970	0.038477	0.001560
4	0.05	NN 31003	0.030930	0.030930	0.104855	0.073925	0.000000
8	0.01	Defer	10.203909	10.203924	10.823576	0.619652	0.000014
8	0.05	Defer	9.445139	9.445139	10.815919	1.370779	0.000000
8	0.01	Stress	8.887171	8.893666	9.535807	0.642140	0.006494
8	0.05	Stress	7.942793	7.954458	9.491694	1.537236	0.011664
8	0.01	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.05	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.01	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.05	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.01	NN 31001	0.205700	0.324516	0.533682	0.209165	0.118815
8	0.05	NN 31001	0.060207	0.102483	0.415982	0.313498	0.042276
8	0.01	NN 31002	0.095675	0.211931	0.397533	0.185601	0.116256
8	0.05	NN 31002	0.000359	0.040296	0.282892	0.242595	0.039936
8	0.01	NN 31003	0.352313	0.483252	1.017217	0.533964	0.130939
8	0.05	NN 31003	0.045343	0.068233	0.477618	0.409385	0.022889
12	0.01	Defer	15.001166	15.001166	15.637251	0.636084	0.000000
12	0.05	Defer	14.118947	14.118947	15.627451	1.508503	0.000000
12	0.01	Stress	13.140504	13.149369	13.797366	0.647996	0.008865
12	0.05	Stress	11.957375	11.967621	13.741127	1.773505	0.010245
12	0.01	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
12	0.05	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
12	0.01	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
12	0.05	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
12	0.01	NN 31001	2.620835	2.711965	3.894878	1.182912	0.091130
12	0.05	NN 31001	1.524265	1.566705	3.520193	1.953488	0.042439
12	0.01	NN 31002	1.880894	2.020437	2.651884	0.631446	0.139543
12	0.05	NN 31002	1.033237	1.137697	2.262773	1.125075	0.104460
12	0.01	NN 31003	0.480952	0.742636	1.514983	0.772346	0.261684
12	0.05	NN 31003	0.041222	0.077511	0.674484	0.596973	0.036289

Lower bounds and gains are rounded downward; upper costs and excess-cost bounds are rounded upward. Exact fractions and all knot values are retained in the machine-readable certificates. The upper deployment is unchanged. A printed zero gain can conceal a positive gain smaller than 10^{-6} ; exact positivity is recorded in the summary.

The lower-bound construction does not change the already verified deployed policy. Its source selectors are proof data, not an action policy obtained by splicing multipliers. This is why the new global cost interval can be compared directly with the previous interval while keeping the upper policy, its operating tolerance, and its occupancy law fixed. All recorded nonzero gaps remain explicit. Raw binary64 neural execution and the full stopped-economy feedback certificate are separate mathematical objects and are not covered by these exact maintenance-model checks.

RESPONSE TO THE R30 REFEREE REPORT: GLOBAL COST BOUNDS

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1. REVISION AND SOURCE OF THE REPORT

This revision responds to the complete R30 report on manuscript commit 3394815. The report is preserved on review commit e9bc144. We read that report together with the intervening R31/R32 construction and numerical records, rather than treating the reviewed R30 manuscript as the latest available science. R34 is based on frozen parent 88015b7, containing the executed R32 package and staged R33 sources and is confined to a new revision branch. The R32 source and executed evidence are inherited work, explicitly identified below; the unrestricted-cost study is freshly executed in R34, and its 252 base certificates are supplemented by 42 new restart-propagated certificates. The full 294-object audit separates inherited evidence from newly executed results.

The substantive addition is a constructive global intervention-cost interval together with a backward propagation that exploits simultaneous feasibility at every restart. Theorem 4 proves the propagated bound for randomized Markov policies and the precisely specified conditional-history extension; it is not only an inner-class optimality statement. A computed lower operating witness, unrestricted scalarized Bellman values, and an independently feasible deployment bound the true minimum implementation cost beyond the conservative local action class. The revised manuscript retains every previous theorem, its original stopped-economy target, the complete boundary proof, all adverse comparisons, and the earlier sensitivity frontier. The historical annex includes the entire R32 article, supplement, response, computation record, and its existing historical annex. No earlier negative experiment has been discarded.

2. MAIN SCIENTIFIC FINDINGS

2.1. *F1: Constructing and charging the witnesses*

The constructive upper-witness algorithm and endogenous-cost computation introduced in R32 are retained in Sections 3 and 4. R34 adds the computed lower witness $L = U - b$ from actual checked compression defects, rather than taking an optimal value as input. Section 5 constructs all support values, their policies, and their operating and implementation values. Their costs are included. The exact operating dynamic program is run separately for comparison, not used as a supplied certificate input.

The staged R33 sources identify an earlier timing omission. R34 uses their complete-cost constructor from the first new execution: the class candidate is rebuilt, checked for exact equality to the inherited object, and charged for witness construction, action masks, cost recursion, evaluation, encoding, and checking. The source archive remains unchanged; we do not claim a new execution of an unavailable prior attempt. Table VII and the complete timing JSON distinguish individual costs, shared work, and the additional final audit.

2.2. *F2: Nontrivial Bellman coupling*

The R32 continuous-state maintenance economy remains the primary computational application. Maintain and replace have lower immediate reward than defer. They can be justified

only by their action-dependent future condition distributions. The unrestricted R34 support problems retain those same transitions, operating returns, and state-dependent revision exposure. No continuation cancellation or primitive reward-sign policy is introduced. The auxiliary action-independent R30 examples are retained as historical controls, not as the principal coupled computation.

2.3. *F3: Strong controls and the value of additional work*

We retain exact structural dynamic programming and both spline controls with the same information. The R34 class comparator is reconstructed from primitives. All 18 installed-neural configurations obtain strictly lower implementation cost outside that conservative class while satisfying their unchanged operating tolerances and preserving their installed operating values. This is an economic improvement over the restricted-cost solution, not evidence that the more expensive portfolio is faster than exact operating dynamic programming. The portfolio delivers an additional output: a lower cost bound against every globally feasible policy.

For the most informative $T = 12$, .05, seed-31003 comparison, cost falls from approximately 1.404696 to .674484, but the unrestricted lower bound is approximately .041222. The reported .633262 global gap makes the remaining economic uncertainty explicit. We do not replace the referee’s computational-advantage question with a false speedup claim. A verified end-to-end advantage over the best classical solver in a difficult higher-dimensional application is not established by this cohort.

2.4. *F4: Endogenous occupancy*

The inherited cost Bellman recursion evaluates future interventions under the revised policy’s own transitions. Its six predeclared stress configurations have positive dynamic-versus-pointwise savings, while the other 36 comparisons have zero savings; all are retained. R34 goes further by optimizing unrestricted support policies under their own transition laws, then independently recomputing their complete implementation costs. Neither policy selection nor the class-free lower bound substitutes the installed occupancy measure for the deployed one.

2.5. *F5: State cardinality, dimension, and geometry*

No trillion-state headline is used. The current contribution is a finite whole-interval representation for a one-dimensional action-dependent economy. Section 7 records the generated representations and integer bit lengths; Table VII adds the support-value geometry and peak process memory. The output-sensitive work bound is not a dimension-independent rate. Earlier 2^{40} statements survive only in their historical context as representation observations, not as a run of a trillion-state dynamic program.

2.6. *F6: The role of neural proposals*

We preserve neural proposals as installed policies whose revision has an economic cost, and keep the exact distinction between trained coefficients, rational compiled policies, and hardware execution. The experiment has three inherited neural seeds, not a newly sampled large cohort. All 18 such cases improve relative to the conservative class, while all twelve spline cases have zero-cost globally feasible incumbents. These findings support an installed-policy revision method that accommodates neural rules, not neural-specific acceleration or superiority over the spline controls. No sampled fitting loss is promoted to a verified whole-state approximation bound.

2.7. *F7: Complexity and a quantitative proposal condition*

The constructive compression bound, charged output-sensitive complexity, and verified proposal-margin condition in Section 7 are retained. Section 5 adds the work of every multiplier solve, its own-policy evaluation, lower-envelope construction, encoding, and checking. The largest declared multiplier has a model-based feasibility guarantee; feasibility is not selected after inspecting successful cases. Certificate size is a measured construction output and can grow with horizon, geometry, and bit length. No favorable unmeasured high-dimensional scaling law is assumed.

2.8. *F8: The unchanged original all-domain target*

The original stopped consumption–portfolio primitives and the exact target remain in Section 11. The retained all-domain bound 7.181834580823298 is not replaced by the maintenance-model bounds, and the missing 47-coordinate current-state bridge is not declared complete. The interval-wide constant-control result is a concrete advance on an explicitly identified restriction of that original economy. R34’s new cost bounds are for the coupled maintenance application, not a disguised certificate for the full stopped feedback problem. The research objective and technical material are preserved without claiming a numerical reduction that has not been computed.

2.9. *F9: Completing the active-boundary scalar problem*

The full R32 global scalar optimization and its proof are retained, rather than returning to the five-point diagnostic criticized in R30. The proof establishes a positive derivative throughout both halves of the constant-adjustment interval, bounds the distant-boundary contribution, and identifies the unique maximizer $\theta = 1/5$ for the declared active-boundary restart and fixed consumption/portfolio controls. The supplement retains all rational details. This is a complete scalar-class result, not an unrestricted feedback-control certificate. Its proof and endpoint enclosure are identified as inherited work; R34 reruns the elementary exact-rational inequality checks, but does not claim a new endpoint quadrature or a new feedback certificate.

2.10. *F10: Economic units and sensitivity*

No universal 60/60 success count at an arbitrary external revision price is reinstated. We retain the entire named-policy sensitivity frontier and normalized intervention units. R34 reports complete cost lower/upper intervals at the two fixed operating tolerances. Its eight multipliers are computational scalarization parameters, not calibrated monetary estimates and not the earlier illustrative external welfare price. The new economic conclusions are cost reductions and certified global gaps in normalized implementation units, not a calibration claim.

2.11. *F11: The theorem-level contribution*

The manuscript explicitly credits classical Bellman comparison and weak duality. The new theorem is not presented as a new duality principle. It composes constructively computed two-sided operating witnesses with unrestricted support certificates, a globally feasible deployment, and a checked global implementation-cost interval for the simultaneous all-state, all-restart operating constraint. This differs from asserting optimality only after a conservative eligible-action set is supplied. It also distinguishes finite policy selection from a lower bound valid for all feasible policies, and does not assume strong duality or a zero gap.

Theorem 4 adds a specific mechanism absent from the separate-restart support inequality: feasible future intervention costs are propagated through the actual action-dependent transition law, while a nonnegative local multiplier enforces a necessary current operating inequality. No action is discarded, so feasible randomized mixtures remain covered. The improvement is verified on the same upper deployment, with 22 strict lower-bound gains and all unchanged outcomes retained. Exact min/max proof selectors allow a checker to establish these identities without invoking optimization. The additional output-sensitive work is charged in equation (29); no new unrestricted support solve is required.

2.12. *F12: A dominant theorem-to-computation chain*

The main chain is now: coupled maintenance primitives; constructive operating witnesses; local-class repair and its limitations; unrestricted support values; a feasible cost-improving deployment; and a lower bound against the original globally feasible set. The 294 independently checked objects, including the additional restart-propagated lower bounds, instantiate this chain in a single unchanged economic model. The stopped-boundary scalar theorem remains a separate, fully proved advancement with an explicit scope. Its results are not counted as evidence for the maintenance algorithm or used to fill a missing high-dimensional scalability result.

The added global-certificate chain now reaches the all-restart restriction itself rather than only supplying a separate-restart relaxation. For the largest absolute gain, the cost interval tightens from $[0.480952, 1.514983]$ to $[0.742636, 1.514983]$ without changing the operating policy. Positive residual gaps remain visible; this is a strengthened verified numerical statement, not a claim that finite scalarization automatically solves every constrained optimum.

3. TECHNICAL COMMENTS AND DISPOSITION

3.1. *T1 and T2: Certificate-input work.*

Upper and lower witnesses, all own-policy values, all support solves, and the newly reconstructed class comparator are constructed and charged. The final checker reads serialized objects and is timed separately. The frozen historical execution and staged source archive remain available; the complete-cost R34 executions and the additional propagation costs have their own ledgers.

3.2. *T3: Deployment amortization.*

The inherited break-even query formula and its assumptions remain in Section 7. No measured break-even is claimed because online-query and loading costs have not been benchmarked under an agreed deployment implementation. The new offline computation is not represented as an online amortization experiment. This requested deployment benchmark remains unmeasured, rather than being claimed complete from a formula alone.

3.3. *T4: Dimension.*

The domain dimension is one throughout the new computation. We report certificate size and process memory without substituting state-cardinality independence for dimension independence.

3.4. *T5: Occupancy.*

Both the inherited class-cost recursion and the new unrestricted support recursion use the revised action’s transition kernel. Complete exact own-policy evaluations check the resulting implementation costs.

3.5. *T6: Proposal quality.*

The verified $\rho + 2\kappa$ deficit bound and its zero-revision implication are preserved. The new global feasibility bound depends on bounded implementation cost and computed witness defects, not on an unverified neural approximation estimate. The observed neural-case improvements do not establish a favorable approximation-rate comparison against splines.

3.6. *T7: Deployment semantics.*

Certificates cover rational compiled policies, including isolated tie points. Floating-point hardware neural execution is not silently included. Changed endpoint and action values are explicitly exercised in adversarial checker tests.

3.7. *T8: Units.*

One cost unit is a normalized implementation intervention weighted by $1 + x$. Multipliers used in the support computation are distinguished from externally calibrated economic prices. No empirical dollar interpretation is introduced.

3.8. *T9: Boundary optimization.*

The complete interval-wide scalar proof and its rational bounds are retained in the article and supplement. We distinguish its unique scalar optimum from the still separate all-domain feedback objective.

3.9. *T10: Preservation and editorial focus.*

All previous technical content remains in the revision or the lossless historical annex. New global-cost theorems and the coupled application are the dominant additions. The preservation map records the unchanged source hashes, deterministic text transformations, and inserted sections. Earlier responses and computations remain accessible without forcing the reader to infer the new contribution from a sequence of response letters.

4. REVIEW PACKAGE AND REPRODUCIBILITY

The root article, supplement, response, computational record, and historical annex are ECTA_R34, SUPP_R34, RESPONSE_R34, COMPUTATION_R34, and HISTORY_R34. Each has an English LaTeX source; the principal documents use the repository’s unmodified official Econometric Society class. The new revision directory contains the fixed protocol, implementation and provenance notes, readable constructors and independent checkers, all 42 outcome rows and 168 multiplier rows, all 294 exact certificates, timing records, mutation tests, source hashes, and publication manifest. The existing review and revision branches are not modified.

5. ADDITIONAL REVIEW-READY CHECK

The complete 42-case propagation table and the 12 corruption tests are in the supplement and execution records. Four-state finite-tree checks include randomized mixtures and explicitly record the eligible count, rather than checking only actionwise feasibility. The current protocol fixes the propagation before its implementation, while identifying the support outcomes that were already known. All newly generated sources, proof objects, tables, and PDFs are committed on the R34 branch; the source report and every preceding branch remain unchanged.

COMPUTATIONAL RECORD FOR REVISION R34

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1. SUBMISSION IDENTITY AND PROVENANCE

R34 is based on frozen parent 88015b77a0c26e7b883156410b684b229fd7e86f. Its isolated branch is revision/econometrica-r34-verified-policy-frontier-2026-09-24. The source R30 report at e9bc144fbb6843d6a3436825a28584efb64fb8f1 reviews manuscript 3394815f4ccbf5917582c8f756b535adf87783cc. R31 and R32 are intervening revisions, not new referee reports. The frozen parent contains an executed R32 publication and a staged R33 source archive. It does not contain an executed R33 publication. We distinguish those objects explicitly.

The repository snapshot was obtained from a successful source-packaging run, 35979134670, at new-branch commit 2a5f133a4f352521a00c8763356c1067b5bb4f90. The R33 source archive was decoded locally and inspected; its mathematical support construction was then executed anew as part of R34. Both the original transfer bytes and the R32 executed evidence remain unchanged. The restart-propagation protocol was committed at 70526abec87eeadd52dd7b30eaf75a78ba70e79c before implementation or evaluation of that addition. At that point the new execution of the support portfolio was already available. The protocol does not relabel known results as prospective observations.

2. NUMERICAL OBJECTS AND PROVENANCE BOUNDARIES

The unchanged model has horizons 4,8,12, tolerances $1/100, 1/20$, and seven installed rules at each horizon. The exact compiled incumbent files are the frozen R32 inputs. No neural or spline fitting is rerun. The support and local multiplier sets are both $\{0, 1, 4, 16, 64, 256, 1024, 4096\}$, the witness accuracy parameter is $1/10000$, and the representation cap is 200,000 pieces. Installed-rule fitting records remain inherited measurements; they are not added to new-machine timings to manufacture a cold-start speedup.

The support stage constructs 168 unrestricted support objects, 42 freshly reconstructed class controls, and 42 primal–dual summaries. Every class control is rebuilt from primitives on the new machine, not treated as a free inherited solution. Its complete rational object must equal the inherited comparator, which supplies a reproducibility check but not its construction. The staged R33 sources contain a description of an earlier timing omission. R34 preserves those source bytes as history; it does not claim to have rerun or reconstructed an unavailable earlier R33 execution. The implemented complete-cost accounting is used from the first R34 support execution.

The new backward propagation constructs another 42 certificates. It does not train a model, alter the feasible deployment, or solve another unrestricted support problem. It builds local action minima and outer bound maxima from the verified primitives and existing witnesses. All local multipliers, all unsuccessful feasibility candidates in the support portfolio, all unchanged propagation outcomes, and all positive remaining gaps are retained.

3. INDEPENDENT VERIFICATION AND TESTS

The final read-only audit reloads 294 serialized objects: 168 support certificates, 42 classical controls, 42 primal–dual summaries, and 42 restart-propagated objects. It disables optimizing envelope routines before checking the complete portfolio and propagated identities. Hashes bind the exact files; all Bellman inequalities, selected equalities, own-policy values, feasible operating envelopes, lower-bound functions, and reported integrals are recomputed with rational arithmetic. Every affine interval is checked through both endpoint limits, and actual isolated knot values are separately checked. The arithmetic library is shared with the constructor; this is a read-only verifier separated from optimization, not a formally verified arithmetic kernel.

Eight malformed support tests, eight malformed primal–dual tests, and twelve malformed propagation tests are required to fail. The support test compares each of four rational restart states with all 27 contingent two-period plans. The additional propagation test enumerates 108 such state–plan combinations and 1,512 equal mixtures, and tests all satisfying the conditional all-restart restriction. Eligible counts are reported, including 4 deterministic plans and 20 mixtures. These finite tests supplement rather than replace the continuum proof and serialized whole-cell comparisons.

The inherited active-boundary rational inequalities are also rerun. They verify the interval-wide positive derivative constants and the upper-exit error bound used in the scalar optimum theorem. The endpoint payoff enclosure remains inherited from the original validated quadrature; it is not described as newly recomputed. No result of the maintenance study changes the original all-domain stopped-economy bound or supplies the missing current-state residual bridge.

4. COMPLETE COMPUTATIONAL ACCOUNTING

Support-stage timings include upper and lower witness construction, all support solves, exact own-policy evaluations, freshly rebuilt class controls, comparison and selection, encoding, and internal checks. Per-configuration totals charge shared portfolios at both tolerances; the summary separately reports unique batch work. The diagnostic exact operating dynamic program is recorded separately and is not an input to the certificate constructor. The first support audit and the final combined audit are separate executed checks, and their command-level wall-clock ledgers are retained.

The new propagation’s construction, internal checks, and encoding total 178.896 seconds. Its read-only additional audit takes 90.990 seconds; the full final base-plus-propagation audit takes 154.340 seconds. These numbers are descriptive measurements on the recorded machine. They do not estimate an average over independent machine runs. All intermediate and output costs remain additional to, not substitutes for, the inherited fitting costs of the already installed rule.

The largest stored propagated lower bound has 471 pieces; closure and local-minimum rationals reach 716 bits. Compressed closure files occupy 17.846 MiB. Raw bytes, compressed bytes, checksums, piece counts, and bit lengths are individually recorded. Stored proof size is not peak process or intermediate-overlay memory. The one-dimensional computation does not establish a dimension-free scalability rate.

5. REPRODUCTION AND PUBLICATION

From the repository root, with Python without the `-O` optimization flag, run:

```
R=revisions/2026-09-24-r34
python "$R/run_study.py"
python "$R/run_closure.py"
python "$R/replication/report_global.py" "$R"
```

```
python "$R/replication/report_restart.py" "$R"
python revisions/2026-09-24-r32/replication/boundary.py \
  --output "$R/results/boundary_recheck.json"
python "$R/assemble.py"
bash "$R/build.sh"
python "$R/publication.py"
```

Only the Python standard library is required for the newly executed exact computations. The inherited fitting procedures have their own recorded dependencies and are not called. The publication workflow reconstructs sources from a checksummed archive, checks the frozen inputs, executes the full computation, builds the article, supplement, response, computation record, and historical annex, and commits the resulting files only to the isolated R34 branch. The source commit, workflow run, publication hashes, and exact certificate hashes are recorded separately. No file attempts to embed the hash of its own future Git commit.

6. PRESERVATION AND INTERPRETATION

The historical annex contains the complete R32 article, supplement, response, computation record, and its existing historical annex, with unchanged pages and file hashes. The current article and supplement also retain the previous source text after input-path changes, with new theory and evidence inserted. The staged R33 transfer archive remains untouched. The preservation map records parent identity, source hashes, all insertions, and inherited document hashes.

All 42 feasible deployments preserve installed operating values. Eighteen installed-neural configurations have strictly lower intervention cost than the conservative class control. The new propagation tightens 22 global intervals without changing any deployment; twelve spline cases retain exact zero global gap. No positive gap closes in the new propagation experiment. These statements distinguish an economic improvement over the restricted class, a sharper global lower bound, and exact global optimality. The original stopped-control objective, adverse comparisons, and complete scalar proof remain intact rather than being removed or relabeled.

Historical Annex to Revision R34

This annex preserves the complete R32 article, supplement, referee response, computational record, and its existing historical annex. These are unchanged historical documents, not new R34 observations. Original sources and results remain on the frozen ancestor and in the inherited executed package. The current manuscript and global-cost evidence are in the separate R34 documents.

CERTIFIED BELLMAN OPERATORS WITH NEURAL PROPOSALS: CONSTRUCTIVE WITNESSES AND COSTLY POLICY REVISION

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We study certified Bellman computation when revising an installed policy is costly and changes future implementation exposure. In a continuous-state maintenance economy with action-dependent transitions, we construct upper witnesses from primitives by exact backward operations and shape-preserving dyadic compression. A nonempty action class guarantees uniform operating accuracy at every restart; a second Bellman recursion minimizes intervention cost within that class. A separate incumbent certificate identifies globally minimal zero-intervention solutions. All 42 registered class configurations pass an independent whole-cell checker, and six predeclared mechanism-stress configurations exhibit strictly positive savings relative to pointwise revision. Twelve spline cases admit zero intervention, whereas none of 18 raw neural cases meets its operating target. Complete construction costs and strong classical controls are reported without inferring neural-specific acceleration. In the unchanged original stopped economy, an interval-wide derivative bound identifies the unique global optimum of an active-boundary constant-control class. The original all-domain feedback target remains distinct from these constructive and computed results.

KEYWORDS: Bellman operators, Certified computation, Policy revision, Maintenance, Neural proposals, Stopped control.

1. INTRODUCTION

Replacing an installed decision rule has an economic cost. A maintenance instruction may require retraining operators, changing a contract, or obtaining approval at the states where the instruction is used. Consequently, two policies with similar operating returns need not be interchangeable. This paper studies how to construct a uniformly accurate policy while minimizing the discounted cost of revising an installed rule. The distribution of future implementation costs is endogenous: a revision today changes the states at which later instructions must be changed.

We develop a constructive Bellman-certificate method for this problem. Starting from model primitives, the method builds upper value witnesses, generates a nonempty class of uniformly accurate actions, and solves the intervention-cost problem within that class. A finite proof object specifies the deployed policy and its operating and intervention values. A separate checker verifies whole intervals and isolated boundary points without repeating the optimizing envelope construction. Neural networks and classical approximations supply installed proposals; their fitting errors do not enter a certificate unless separately bounded.

The computational application is a continuous-state maintenance economy with action-dependent transitions. Maintaining or replacing a unit is always inferior to deferring in current reward. Such actions can nevertheless be optimal because they improve future condition. Continuation terms therefore cannot be canceled from the action comparison. Upper witnesses are computed by action-dependent expectation, maximization, and globally checked polygonal compression. The inputs do not include an optimal-value surface, an optimal-action file, or an analytically supplied supersolution. Own-policy values, admissible action sets, and the intervention-cost recursion are also constructed from the primitives and included in the work accounts.

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Three results organize the analysis. First, convex piecewise-affine structure permits an explicit construction of upper witnesses with controlled rounding and compression error. Second, a uniform local deficit budget produces a nonempty policy class with an all-state, all-restart operating guarantee. Within that class, a second Bellman recursion minimizes intervention cost under the deployed policy’s occupancy. Third, an independently certified incumbent can be retained at zero intervention cost, which is globally minimal even outside the conservative local class. This last result prevents a sufficient local restriction from creating unnecessary revisions when a global own-policy certificate is already available.

The experiments exercise these statements rather than infer them from training loss. Three horizons, two accuracy budgets, and seven proposal rules produce 42 class-based certificates, all checked in exact rational arithmetic. The predeclared mechanism stress incumbent produces strictly positive dynamic-versus-pointwise intervention savings at every horizon and budget. The remaining 36 comparisons have zero savings and are retained. All 12 spline cases qualify for the additional zero-intervention certificate. None of the 18 neural cases meets its raw operating target. These outcomes establish an end-to-end coupled construction and an endogenous-occupancy mechanism; they do not establish neural-specific computational superiority.

A further result advances the original stopped consumption–portfolio problem without changing its primitives. At the economically active boundary restart used previously, we prove a positive derivative over the entire constant-adjustment interval. This identifies the unique global optimum of that scalar control class. The argument combines monotone coupling, killed-process survival derivatives, and rational bounds for the distant-boundary error; it does not assume concavity or extrapolate from five derivative evaluations. The original all-domain current-state target remains stated separately and is not replaced by the auxiliary maintenance-model certificate.

1.1. *Relation to existing methods*

Bellman comparison and constrained backward induction are classical. We do not claim a new dynamic-programming principle. Fitted value iteration studies approximation and sampling error under a generative model (Munos and Szepesvári, 2008); approximate policy iteration has related performance bounds (Scherrer, 2013). Here fitting is separated from a deterministic, all-state verification layer for a known model. Safe policy improvement with baseline bootstrapping restricts changes when data leave the transition model uncertain (Laroche, Trichelair, and Tachet des Combes, 2019). Our admissible class instead limits a verified operating deficit, and our implementation cost is an economic primitive. Neither information structure subsumes the other.

Numerical economics exploits structure and approximation (Judd, 1998). Adaptive sparse grids address genuinely high-dimensional dynamic models (Brumm and Scheidegger, 2017). Our dimensional experiment is one-dimensional and does not establish a competing high-dimensional rate. Information-relaxation methods furnish other routes to upper bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014). Their construction and our construction must both be charged; a supplied upper bound is not a free solution input. Verified computation requires both a valid analytic reduction and controlled arithmetic (Rump, 2010). Markov-chain approximation of a diffusion imposes additional consistency requirements (Kushner and Dupuis, 2001); the maintenance economy is not asserted to be a discretization of the stopped diffusion.

The methodological addition is the explicit composition of witness construction, quantified encoding error, a nonempty certified action class, endogenous-cost optimization, and an independently checkable finite representation. The cost recursion alone is standard once the class is known. The constructive witness and class-generation steps are therefore part of the main theorem and computation, not omitted preprocessing. The comparison with uncompressed exact piecewise-affine dynamic programming is intentionally strong: both methods exploit the

same affine model structure. No neural advantage, dimension-free complexity, or universal acceleration follows merely from the existence of a regional proof object.

2. ECONOMIC ENVIRONMENT AND OBJECTIVES

2.1. A coupled maintenance economy

There are T decision dates and a terminal date. Condition x lies in $[0, 1]$. Actions $a = 0, 1, 2$ denote defer, maintain, and replace. Current reward is $r(x, a) = x - c_a$, with

$$\beta = 19/20, \quad (c_0, c_1, c_2) = (0, 3/20, 9/20), \quad g(x) = x/2. \quad (1)$$

Two shocks have probabilities $(p_1, p_2) = (3/5, 2/5)$. Their next-state maps are

a	$f_{a1}(x)$	$f_{a2}(x)$
0	$3x/4$	$3x/4 + 1/20$
1	$17x/20 + 1/10$	$17x/20 + 7/50$
2	$x/10 + 4/5$	$x/10 + 17/20$

(2)

All maps preserve the state interval. For a continuation v , write

$$T^a v(x) = x - c_a + \beta \sum_{j=1}^2 p_j v(f_{aj}(x)), \quad Tv = \max_{a \in \{0,1,2\}} T^a v. \quad (3)$$

The optimal operating value satisfies $V_T = g$, $V_t = TV_{t+1}$; a policy π has $J_T^\pi = g$, $J_t^\pi = T^{\pi_t} J_{t+1}^\pi$. Because every intervention has a positive current cost, any maintain or replace decision is justified by continuation value, not by a primitive reward-sign rule.

Let π^0 be an installed policy. Its revision cost from restart (t, x) is

$$C_t^\pi(x) = \mathbb{E}_{t,x}^\pi \sum_{s=t}^{T-1} \beta^{s-t} (1 + X_s) \mathbf{1}\{\pi_s(X_s) \neq \pi_s^0(X_s)\}. \quad (4)$$

One unit is a normalized implementation intervention, not a dollar estimate. For a uniform initial state distribution ν , the reported economic costs are $\int C_0^\pi(x) d\nu(x)$, calculated under each deployed policy. The objective is to minimize this cost subject to

$$\sup_{t \in \{0, \dots, T\}, x \in [0,1]} \{V_t(x) - J_t^\pi(x)\} \leq \varepsilon. \quad (5)$$

We solve the cost problem in an explicitly constructed inner class and identify an important case in which the minimum is global for (5) itself. Operating-value preservation, $J^\pi \geq J^{\pi^0}$, is a distinct requirement. It is checked separately in the experiment and can alternatively be built into a sufficient class.

2.2. Information and deployment

All solvers have the same known rewards, transition maps, probabilities, terminal payoff, and installed policy. “No precomputed optimum” means that the certificate constructor does not read an optimal-value or optimal-action file. It does not mean model-free or less informed than dynamic programming. An exact reference solve is performed for diagnostics, but its output is not used to build U , the action sets, or the cost-minimizing policy.

Training and verification use different arithmetic objects. Neural and spline fitting use binary64 arithmetic. Saved coefficients are interpreted exactly as rational constants, and their decision rules are compiled into piecewise-constant policies. The guarantee concerns this compiled representation with exact rational evaluation, not an unmodeled floating-point neural forward pass. In particular, every breakpoint has its own stored action; an isolated tie is not discarded because it has probability zero under the initial distribution. The operating guarantee covers every restart, including such points.

3. CONSTRUCTING THE CERTIFICATE INPUTS

A function is represented by rational knots, an affine formula on every open knot interval, and a separate value at every knot. Continuous convex witnesses are a special case; policy values and cost functions can have jumps or exceptional point values. Addition, positive affine composition, integration, maximization, and constrained minimization are performed by exact partition overlays and rational intersections. These operations are finite at every finite horizon, although the number of pieces can grow substantially.

LEMMA 1—Shape-preserving dyadic upper encoding: *Let F be continuous, convex, piecewise affine, and rational on $[0, 1]$. Fix a rational mesh $0 = z_0 < \dots < z_m = 1$. Let H be its chord interpolant through $F(z_i)$. Round $F(0)$ and every chord slope upward to multiples of 2^{-q} , and integrate the rounded slopes successively to obtain G . Then G is convex and*

$$0 \leq G - H \leq 2^{1-q}, \quad F \leq H \leq G. \quad (6)$$

If F is L -Lipschitz and the maximum mesh width is h , then $0 \leq G - F \leq Lh + 2^{1-q}$. For any proposed mesh, its actual global error is determined by the union of the knots of F and G .

PROOF: Convexity orders the exact chord slopes; upward rounding by a common mesh preserves that order. Each rounded slope exceeds its exact value by less than 2^{-q} . The cumulative slope error on an interval of length at most one, plus the rounded starting value, is less than 2^{1-q} . Convexity places the chord above F . For $x \in [z_i, z_{i+1}]$, the Lipschitz bounds from the endpoints imply $H(x) - F(x) \leq Lh$. Finally, $G - F$ is affine between union knots, so its extrema occur at those knots. *Q.E.D.*

The lemma rounds slopes rather than independently rounding ordinates. Independent ordinate rounding need not preserve convexity, which is needed at the next Bellman step. The implementation proposes an adaptive coarsening, rounds retained knot locations to dyadics, recomputes their exact values, and applies the lemma. It then checks the actual complete error. A failed check increases precision or returns an unresolved result; it is never replaced by a floating tolerance. The uniform-mesh part of the lemma supplies a constructive existence bound independently of the adaptive heuristic.

Put $H_T(\beta) = \sum_{j=0}^{T-1} \beta^j$ and choose

$$\eta = \varepsilon / H_T(\beta), \quad \delta = \eta / 4, \quad e_t = \eta \sum_{j=0}^{T-t-1} \beta^j, \quad e_T = 0. \quad (7)$$

Starting with $U_T = g$, calculate

$$q_t^a = T^a U_{t+1}, \quad F_t = \max_a q_t^a, \quad 0 \leq U_t - F_t \leq \delta. \quad (8)$$

The final step is achieved by Lemma 1, not assumed as an input. Positive affine transitions and affine rewards preserve convexity under expectation and maximization. The construction therefore applies recursively. A Lipschitz bound is also constructive: if U_{t+1} has slope bound L_{t+1} , then F_t has bound $1 + \beta(17/20)L_{t+1}$; the next encoded bound includes its explicit rounding increment.

THEOREM 1—Constructive all-restart class: *For the model (1)–(2), construction (8) produces finite rational upper witnesses. Define*

$$\mathcal{B}_t(x) = \{a : U_t(x) - q_t^a(x) \leq \eta\}. \quad (9)$$

The sets are nonempty at every point. Every measurable selector $\pi_t(x) \in \mathcal{B}_t(x)$ satisfies

$$U_t(x) - e_t \leq J_t^\pi(x) \leq V_t(x) \leq U_t(x), \quad (10)$$

and consequently (5). All certificate inputs are obtained from the primitives by finite exact operations; no optimal or own-policy value is supplied externally.

PROOF: Lemma 1 gives a finite construction at each date. Equation (8) makes U a Bellman supersolution. Backward comparison yields $V_t \leq U_t$. A maximizing action for F_t has deficit at most $\delta < \eta$, proving nonemptiness. If $J_{t+1}^\pi \geq U_{t+1} - e_{t+1}$, positivity and unit mass of the transition probabilities give

$$J_t^\pi \geq q_t^{\pi_t} - \beta e_{t+1} \geq U_t - \eta - \beta e_{t+1} = U_t - e_t.$$

The terminal condition starts the induction. Exact overlays produce $\mathcal{B}_t$ including all equality points. Own-policy values, when requested, follow a separate finite recursion with the policy fixed. *Q.E.D.*

The theorem replaces a verifier for supplied global objects by a construction of those objects. It does not state that the construction is cheaper than every alternative. The next sections report all of its costs and distinguish the size of a stored witness from the potentially larger intermediate envelopes and policy-value partitions.

4. OPTIMAL COSTLY REVISION

With $\mathcal{B}$ fixed, define $D_T = 0$ and

$$D_t(x) = \min_{a \in \mathcal{B}_t(x)} \left\{ (1+x) \mathbf{1}\{a \neq \pi_t^0(x)\} + \beta \sum_j p_j D_{t+1}(f_{aj}(x)) \right\}. \quad (11)$$

All alternatives in this equation are piecewise affine on an exactly computable finite overlay. Minimizing selectors have an exact finite representation with separately checked point values.

THEOREM 2—Endogenous-cost certificate: *A minimizing selector in (11) satisfies (10) and minimizes (4) from every restart among policies selecting actions from $\mathcal{B}$. Randomization and history dependence do not improve this minimum within the fixed class. A certificate consists of U , the deployed policy, its operating values J , and D . It is sufficient to check the Bellman supersolution inequalities, the selected-action deficits, the operating-value equalities, and*

$$D_t \leq c_t^a + \beta P^a D_{t+1} \quad (a \in \mathcal{B}_t), \quad D_t = c_t^{\pi_t} + \beta P^{\pi_t} D_{t+1}, \quad (12)$$

where $c_t^a = (1+x) \mathbf{1}\{a \neq \pi_t^0\}$.

PROOF: The operating claim is Theorem 1. Conditional on the current state and chosen eligible action, backward induction bounds every eligible future intervention cost below by D_{t+1} . Minimization gives the lower bound D_t ; a minimizing selector attains equality. A randomized action gives a convex combination of eligible alternatives. The same conditional argument applies to history-dependent choices. Conditions (12) prove the lower bound and attainment without rerunning the minimizing envelope algorithm. *Q.E.D.*

The occupancy measure in this statement is not frozen at the installed policy. An eligible action that is more expensive now can reduce future exposure to instructions requiring revision. A pointwise guard that keeps the installed action whenever eligible ignores that tradeoff. The experiment isolates this mechanism by comparing both rules on the same $\mathcal{B}$ and evaluating each with its own transition law.

PROPOSITION 1—A global zero-intervention certificate: *Construct the installed value $v_t^0 = T^{\pi_t^0} v_{t+1}^0$, $v_T^0 = g$. If $U_t - v_t^0 \leq \varepsilon$ at every state and restart, then retaining π^0 solves the intervention-cost problem over all policies satisfying (5), with minimum zero. Membership of π^0 in $\mathcal{B}$ is unnecessary.*

PROOF: The upper witness proves $V - v^0 \leq U - v^0 \leq \varepsilon$. Retention has zero intervention cost, and all intervention costs are nonnegative. *Q.E.D.*

This test is performed before recommending a change to a certified incumbent. We preserve the class-only outputs as a diagnostic of conservative over-revision, and add separate zero-cost certificates wherever the test passes. The global claim in Proposition 1 is stronger than class-wise optimality, but applies only when its explicitly computed premise holds.

4.1. Operating preservation and certificate conservatism

A regret guarantee alone does not imply $J^\pi \geq v^0$. A sufficient preservation class intersects (9) with

$$T^a v_{t+1}^0(x) \geq v_t^0(x). \quad (13)$$

When this intersection is nonempty, the same induction yields preservation and the same cost recursion minimizes cost within the intersection. Nonemptiness is an additional condition, not a consequence of (8). In the primary experiment we use (9) and independently evaluate $J^\pi - v^0$ on every open interval and knot. All displayed primary cases satisfy preservation, but this is a checked outcome, not a universal property of the unmodified class.

The local class is an inner approximation to the globally feasible set. A policy can distribute its operating losses across dates more efficiently than the fixed allowance (7), and then be excluded despite satisfying (5). Proposition 1 provides a concrete correction to this conservatism. The full sensitivity comparison below is an envelope over named computed policies; it is not silently promoted to an optimization over every globally feasible policy.

5. WORK, REPRESENTATION, AND THE ROLE OF PROPOSALS

Let N_t count all partition cells generated in constructing action-continuation functions, intersections, and admissible masks at date t . Let M_t count stored upper-witness pieces, K_t count installed and deployed policy pieces, and Z_t count the operating and cost-value pieces generated by evaluation and optimization. Let b bound actual integer bit lengths, and let $A(b)$ bound the cost of rational arithmetic and comparison, including normalization.

PROPOSITION 2—Charged finite-representation work: *For A actions and D positive affine shock maps per action, a direct overlay implementation constructs the objects in Theorems 1 and 2 with a finite number of rational operations depending on the generated partitions, not on an external state-grid cardinality. A conservative bound is*

$$C_{\text{fit}} + C_{\text{compile}} + O\left(A(b) \sum_t \{(AD + A^2)(N_t + K_t + Z_{t+1})^2 + N_t^2\}\right) + C_{\text{encode}} + C_{\text{check}}. \quad (14)$$

Here N_t^2 includes an elementary adaptive chord check. All quantities are construction outputs, not assumed free witnesses. For the upper-witness subproblem alone, a uniform chord mesh of width $h \leq \delta/(2L_t)$ and slope precision $2^{1-q} \leq \delta/2$ gives $M_t = O(1 + L_t/\delta)$.

PROOF: Each affine composition pulls back finitely many knots. An overlay has at most the sum of its input knot counts, and each action pair contributes at most one intersection on each common affine cell. Sorting, evaluation, and pairwise comparison are bounded by the deliberately conservative squared-count expression. A chord test compares its chord with every intervening source knot; recursive tests are bounded by $O(N_t^2)$. Own-policy evaluation and cost optimization produce their own partitions and are charged through K_t, Z_t . Lemma 1 gives the final mesh statement. Encoding and the separate checker are added rather than hidden in a range-oracle cost. *Q.E.D.*

This is an output-sensitive finite-construction bound, not a polynomial-in-dimension theorem or a guarantee of acceleration. In particular, Z_t can grow rapidly even when M_t is small. Exact intersections and integrals can also have large denominators. The implementation records intermediate and stored piece counts, bit lengths, serialized bytes, and stage times, and rejects an intermediate representation exceeding 200,000 pieces. No cap is reached in the registered cohort. A memory statistic for the whole Python process is reported separately from the portable certificate size; it includes libraries and allocations and is not a bound on deployment memory.

PROPOSITION 3—A quantitative proposal condition: *Suppose a proposal chooses a maximizer of $\hat{q}_t^a$ and satisfies a verified whole-state error $\max_a \|\hat{q}_t^a - q_t^a\|_\infty \leq \kappa_t$. Put $\rho_t = U_t - \max_a q_t^a$. Its selected-action deficit obeys*

$$U_t - q_t^{\pi_t^0} \leq \rho_t + 2\kappa_t \leq \delta + 2\kappa_t. \quad (15)$$

If $\delta + 2\kappa_t \leq \eta$ at every date, the proposal is eligible everywhere and the intervention-cost minimum in (11) is zero. On any region carrying analogous verified local bounds, that region needs no further action-eligibility subdivision for the proposed action.

PROOF: For an exact maximizing action a^* , optimality under $\hat{q}$ gives $q^{a^*} - q^{\pi^0} \leq 2\kappa_t$. Adding ρ_t proves the bound. An eligible installed policy can be followed at every date with zero cost, which is minimal by nonnegativity. The same comparison is pointwise on a region with a valid uniform local error bound. *Q.E.D.*

The proposition identifies a causal role for a verified approximation margin: it can eliminate revisions and action-eligibility refinement. It does not remove the cost of constructing U , computing the approximation bounds, or compiling the proposal. Sampled fitting loss is not a premise of this proposition. The numerical study does not assume that its neural candidates satisfy it. The exact own-policy test in Proposition 1 can be substantially less conservative.

For deployment, a stored piecewise policy answers an exact rational query by knot search and comparison. If an alternative online rule costs c_{on} per query and loading plus offline construction costs $C_{\text{off}} + C_{\text{load}}$, a genuine amortization claim requires $c_{\text{on}} > c_{\text{stored}}$ and

$$Q > (C_{\text{off}} + C_{\text{load}})/(c_{\text{on}} - c_{\text{stored}}). \quad (16)$$

We do not measure this denominator here and therefore make no observed deployment break-even claim. The exact online sign guard from the earlier action-independent model remains a strong control in that model, not a competing solution to the present coupled cost problem.

6. COUPLED COMPUTATIONAL EVIDENCE

6.1. Prospective design and complete accounting

The primitives, horizons $T \in \{4, 8, 12\}$, tolerances $\varepsilon \in \{.01, .05\}$, and proposals were specified in the inherited R31 protocol before the R32 full cohort. This is a referee-informed exploratory study, not an independent preregistered confirmation. We retain the pilot outcomes and amendments rather than describe the research history as prospective from its origin.

Each horizon has an all-defer incumbent, a mechanism stress incumbent, two linear-spline fitted-Q proposals, and three neural fitted-Q proposals. Splines use 33 or 129 equally spaced nodes. Neural proposals use 257 midpoint training states, one hidden ReLU layer of width 16, three outputs, 200 Adam steps per date, learning rate .02, and seeds 31001–31003. Targets use the next fitted continuation, never optimal-value labels. Exact compilation precedes any candidate-specific certificate or reference comparison. One frozen policy is reused across the two tolerances.

The stress incumbent maintains when $x \geq 9/10$ at all nonterminal decision dates and otherwise defers; the final decision always defers. It is intentionally misconfigured and separately labeled. It tests whether a current revision can avoid future exposure to an expensive instruction, not whether a neural model has discovered an institutional practice. All six stress results are reported irrespective of sign.

The reference is uncompressed exact piecewise-affine dynamic programming, not a large brute-force grid. A second accuracy-only control uses the same compressed witnesses, greedily maximizes q_t^a , evaluates its own policy, and checks its error. The two spline controls exploit the natural local basis. Fresh-process neural timing includes interpreter and library initialization, training, and compilation. We separately charge witness construction, installed-value evaluation, action-set construction, cost minimization, final evaluation, serialization, internal verification, and the independent final checker. These single-machine timings are descriptive; neither seeds nor tolerances provide independent machine-speed replications.

TABLE I
COMPLETE COUPLED-MODEL COHORT

Proposal	Cases	Raw passes	Certified	Dynamic savings
Neural	18	0	18	0
Spline	12	12	12	0
All defer	6	0	6	0
Occupancy stress	6	0	6	6

The same frozen proposal is evaluated at two tolerances; these are 21 frozen policy/horizon pairs, not 42 independent models. Raw passes use the exact uncompressed reference only as a diagnostic. The independent upper-witness test separately certifies all 12 spline cases with zero intervention.

All 42 class certificates pass the independent checker, and their deployed operating values weakly exceed their installed values at every restart. Raw neural passes are zero out of 18, whereas all 12 spline cases pass. The separate own-policy witness test also certifies each spline incumbent, yielding 12 additional zero-intervention certificates by Proposition 1. The class-only spline outputs are preserved to show, rather than conceal, the cost of a conservative sufficient class. No universal post-repair pass statistic is interpreted as neural approximation success.

6.2. End-to-end work and strong classical controls

TABLE II
REPRESENTATION SIZE AND ARITHMETIC IN ONE STATE DIMENSION

T	ε	Exact V	Input to compression	Stored U	U bits	Cost pieces
4	0.01	2	2	2	66	11
4	0.05	2	2	2	66	11
8	0.01	11	11	11	67	78
8	0.05	11	11	9	67	78
12	0.01	108	48	27	67	359
12	0.05	108	28	15	67	350

Counts are maxima over dates and, for cost, proposals. Exact roots, knot values, and one-sided limits are included. No state-grid cardinality is used as a complexity proxy. Complete value/cost arithmetic can use more bits than stored witnesses; all stage records and certificates are retained.

TABLE III
MEASURED SOLUTION AND CERTIFICATION COSTS (SECONDS)

T	ε	Exact DP	Witness	Greedy	Spline cold	Neural cold
4	0.01	0.003294	0.005446	0.01483	0.0681	2.211
4	0.05	0.003294	0.005648	0.015	0.06876	2.21
8	0.01	0.01027	0.01835	0.04712	0.1801	2.526
8	0.05	0.01027	0.01782	0.0463	0.1655	2.523
12	0.01	0.0883	0.09859	0.2839	0.8417	3.575
12	0.05	0.0883	0.06703	0.2311	0.6392	3.436

Medians across the declared proposal family; single executions are descriptive, not a significance test. Cold totals charge fresh-process neural import/training, compilation, witness construction, own-policy evaluation, admissible sets, cost optimization, encoding, internal verification, and the independent final checker. The exact DP and certified greedy rows solve accuracy-only objectives; neither is represented as a minimum-revision-cost solver.

At horizon 12, the exact value representation reaches 108 pieces. The compressed witness reaches 27 stored pieces at tolerance .01 and 15 at .05; the largest intermediate envelope at those budgets has 48 and 28 pieces. Stored witnesses use at most 67-bit rational components, while operating and cost representations use up to 256 bits in the cohort. Thus the reduction in stored witness size is real, but it does not eliminate intermediate construction or downstream cost-value complexity. The largest intervention-cost representation has 359 pieces, and the largest uncompressed serialized primary certificate is 370,046 bytes.

The timing comparison does not establish a neural advantage. The natural spline basis is both more accurate before repair and cheaper to fit in this model. The structure-aware exact operating solver is also inexpensive. The proposed cost pipeline solves an additional constrained objective,

but that distinction does not justify a claim that it is a faster operating-value solver. It is precisely why the operating-only baseline and the class-minimum-cost output are reported separately. The new positive computational statement is a fully constructed coupled certificate with measured input costs, not an inferred speedup over an artificially enormous state grid.

6.3. Endogenous occupancy and economic sensitivity

TABLE IV
ACTION-DEPENDENT OCCUPANCY CHANGES THE OPTIMAL REVISION PATTERN

T	ε	Regret bound	Pointwise cost	Dynamic cost	Saving
4	0.01	0.00269551	4.21218	4.2099	0.00227628
4	0.05	0.0134775	4.2167	4.20532	0.0113815
8	0.01	0.00148561	9.55894	9.54147	0.0174693
8	0.05	0.00742774	9.59545	9.50629	0.0891575
12	0.01	0.00116031	13.831	13.8072	0.0237425
12	0.05	0.00608421	13.878	13.7581	0.119917

Both policies use the same computed admissible sets. Costs are exact uniform-initial-state integrals under each deployed policy, then displayed as decimals. The stress incumbent is predeclared and intentionally misconfigured; the other 36 cases have zero dynamic-versus-pointwise savings.

The dynamic solution strictly reduces intervention cost relative to the pointwise rule in all six stress cases. Both select only from the same certified action class. Therefore the difference cannot be explained by a more permissive operating target or by supplying the dynamic solver a better upper witness. It arises from the action-dependent continuation of implementation costs. The complete piecewise policies and cost functions document the differing revision pattern. For the remaining candidates, equality of the two cost functions is retained; the mechanism is not asserted to operate for every installed rule.

For economic interpretation, let J_i and C_i be the exactly integrated operating return and intervention cost of a computed deployment. Its net return at shadow price $\lambda \geq 0$ is $J_i - \lambda C_i$. We store all nonnegative intersections and tie points, and report the entire upper envelope over dynamic, pointwise, certified-greedy, exact operating-optimal, and globally eligible raw policies. No large illustrative λ is chosen to manufacture a universal favorable sign. The thresholds are

TABLE V
COMPLETE SHADOW-PRICE SENSITIVITY FOR THE MECHANISM STRESS CONTROL

T	ε	Envelope winners as λ increases	Switch prices
4	0.01	Exact $\rightarrow$ Dynamic	0.0016623
4	0.05	Exact $\rightarrow$ Dynamic	0.0082732
8	0.01	Exact $\rightarrow$ Dynamic	0.0012955
8	0.05	Exact $\rightarrow$ Dynamic	0.0064472
12	0.01	Exact $\rightarrow$ Greedy $\rightarrow$ Dynamic	1.7362e-05, 0.0013897
12	0.05	Exact $\rightarrow$ Greedy $\rightarrow$ Dynamic	0.00046808, 0.0070918

The envelope is over the displayed computed deployments, not all feasible policies. All pairwise intersections and tie sets are stored as exact fractions. One cost unit is a normalized intervention; no monetary calibration or preferred illustrative price is imposed.

normalized tradeoffs, not calibrated willingness to pay. In particular, a low intervention cost

does not mean a policy dominates in operating value. The envelope exposes precisely when the retained operating loss is compensated by reduced implementation cost. The distinction between this finite comparison and the globally constrained problem remains explicit.

7. A GLOBAL OPTIMUM AT THE ACTIVE STOPPING BOUNDARY

7.1. The unchanged original economy

The original model has state $(u, x) \in \mathcal{D} = (1.2, 2.8) \times (.5, 2)$, horizon one, and dynamics

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (17)$$

$$dx_s = \{(.02 + .06p_s)x_s - c_s\} ds + .2p_s x_s dZ_s^x, \quad d\langle Z^u, Z^x \rangle_s = -.25 ds. \quad (18)$$

Controls (c, θ, p) lie in $[.05, .8] \times [-.2, .2] \times [-.5, .8]$. At first exit, capped at one, running utility and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (19)$$

$$G(u, x) = -.02(u-2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1-s). \quad (20)$$

Payoffs discount at rate .04. The retained principal target is a current-state actor $\pi(t, u, x)$ with

$$\sup_{(t,u,x) \in [0,1] \times \overline{\mathcal{D}}} \{V_2(t, u, x) - J_2^\pi(t, u, x)\} \leq .01. \quad (21)$$

The retained whole-domain upper bound is 7.181834580823298. Neither the maintenance model nor the scalar theorem below changes that number. The 47-coordinate current-state derivative/residual bridge remains uninstantiated. We preserve the original target and its derivations rather than silently substitute a smaller policy class.

7.2. From a derivative diagnostic to a complete scalar certificate

Fix the unchanged restart $(0, 61/50, 5/4)$, $k = 2$, $c = 3/4$, $p = 0$, and a constant $\theta \in [-1/5, 1/5]$. Write $J(\theta)$ for the resulting original-model payoff. In the preceding validated calculation the lower-boundary exit probability at zero adjustment lay in $[.68915, .68916]$. Curvature changed sign, so a concavity-based search justified at a central restart could not be transferred to this one. We instead prove interval-wide monotonicity.

THEOREM 3—Global optimum in the active-boundary constant class: *For the fixed original-model class above, J is differentiable in the interior and has one-sided derivatives at its endpoints. The following uniform lower bounds hold:*

$$J'(\theta) \geq \begin{cases} 8163/125000, & -1/5 \leq \theta \leq 0, \\ 47251/125000, & 0 \leq \theta \leq 1/5. \end{cases} \quad (22)$$

Consequently the unique global maximizer is $\theta^* = 1/5$. The inherited validated endpoint enclosure $[-3.7609, -3.7607]$ is therefore an enclosure of the optimal payoff in this constant class.

PROOF: Let $a = 6/5$, $b = 14/5$, $d = 1/50$, and $\sigma = 1/20$. Deterministic wealth is bounded below by $25/49 > 1/2$, so it does not exit. Stopped Itô's formula writes the payoff as its initial settlement plus the discounted occupation integral of $h_\theta = (\partial_t + \mathcal{L}_\theta - .04)g + \ell_2$. On the preference interval, explicit rational bounds give $h_\theta \geq 2$ and $\partial_y h_\theta > 0$. Extend h_θ constantly above b .

For the process killed only at a , the derivative of survival is

$$S'_\theta(t) = 16e^{-16\theta} \Phi\left(\frac{\theta t - 1/50}{(1/20)\sqrt{t}}\right).$$

For fixed t this is log-concave in θ , so its minimum on a compact interval is bounded by the smaller endpoint value. Monotone coupling, with the state dependence of h_θ held fixed, bounds the derivative of the occupation expectation below by $2S'_\theta(t)$. The direct parameter derivative of h is at least $-4/125$ on the negative half and $-54/125$ on the positive half.

On $t \in [1/25, 1/5]$ and $\theta \in [-1/5, 0]$, rational Gaussian and exponential bounds give $S'_\theta(t) \geq 8/25$. On $t \in [1/4, 1]$ and $\theta \in [0, 1/5]$, they give $S'_\theta(t) \geq 352/625$. Also $e^{-.04t} \geq 24/25$. These imply half-line derivative lower bounds .066304 and .379008. The likelihood-score error for paths hitting the distant upper boundary is less than .001 uniformly in θ . Subtracting it proves (22). Strict positivity gives uniqueness. The supplementary proof supplies every constant and the executable rational checks. *Q.E.D.*

The endpoint enclosure is inherited, not a newly run payoff quadrature. What is new is the global optimization certificate connecting it to the entire parameter interval. It improves the active-boundary component while keeping its scope precise: one restart and a constant-control class. A globally verified scalar optimum does not solve the feedback problem (21).

8. CONCLUSION

Uniform operating accuracy and low implementation cost are different objectives. The construction here begins with economic primitives, produces Bellman witnesses with quantified encoding error, and uses their certified action sets to optimize intervention costs under endogenous occupancy. Whole-interval and isolated-point verification make the deployed representation checkable independently of training and of the optimizing envelope routine. An additional own-policy certificate identifies globally cost-minimal zero-intervention solutions without forcing them into a conservative local class.

The coupled maintenance computation establishes these steps end to end and exhibits a genuine dynamic revision mechanism. It also preserves a demanding comparison: classical spline proposals and exact structured dynamic programming are strong alternatives, and the present experiment does not establish a neural speed advantage. The active-boundary theorem converts an earlier local diagnostic into a complete scalar optimum in the original stopped economy. The larger all-domain target remains unchanged. All preceding proofs, unsuccessful experiments, and referee responses are retained in a lossless historical annex, while the current article separates its proved constructions from stronger empirical or high-dimensional claims not established by the evidence.

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SUPPLEMENT TO CERTIFIED BELLMAN OPERATORS: CONSTRUCTIVE WITNESSES AND COSTLY POLICY REVISION

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1. FINITE REPRESENTATIONS AND VERIFICATION SEMANTICS

The replication uses a finite rational object $f = (z_i, m_i, b_i, v_i)$: $f(x) = m_i x + b_i$ on each open interval (z_i, z_{i+1}) , while $f(z_i) = v_i$. This distinction is necessary even when the original operating value is continuous. Compiled decisions can jump, a cost-minimizing decision can have a special tie value, and an affine transition can send an entire exceptional point to another exceptional point. Integral-zero sets must therefore remain in a certificate for all restart states.

For a common overlay of two such functions, an affine inequality on an open interval holds if and only if it holds in both endpoint limits. The actual endpoint values are checked separately. This is the reason a midpoint may safely identify which affine formula applies on an already constructed cell, but may not by itself verify that formula's inequality. Every range check uses the limits as well as the stored points. A missing cell, a wrong point value, or a changed model fails verification.

An affine composition $f(\alpha x + \gamma)$ with $\alpha > 0$ adds every preimage of a knot that lies in $[0, 1]$. Values at those preimages are inherited exactly. Maximization first overlays all source partitions; on each common affine cell it inserts all pairwise action intersections. The minimum action index breaks an exact tie. Constrained minimization additionally overlays the binary eligibility masks. Thus eligible singleton actions are not inferred from neighboring open cells.

Normalization removes a knot only if both adjacent affine formulas agree and the stored knot value agrees with that same formula. A separate test creates a point spike, normalizes it, composes it with an affine map, and confirms that the spike and its preimage survive. Exact integration ignores singleton masses only when integrating against the continuous initial distribution. The dynamic recursions never ignore them.

2. CONSTRUCTIVE WITNESS DETAILS

2.1. *Existence and implemented adaptive compression*

For a convex Lipschitz function on $[0, 1]$, a chord interpolant on a mesh of width h lies above the function. A conservative Lh error bound follows from the Lipschitz comparisons to both endpoints. Hence a sufficiently fine rational mesh and sufficiently fine dyadic rounding produce a certified upper approximation at every finite backward step. The maximum over finitely many positive affine compositions of convex piecewise-affine functions remains convex and piecewise affine.

The implementation uses a more economical, checked adaptive rule. Initially retain both endpoints. For each candidate chord, find the source knot with the largest chord error. If that error exceeds half the stage compression allowance, retain that knot and recurse on the two children. This yields an unrounded chord cover with error at most half the allowance. Retained locations are rounded to dyadics, and exact function values at the rounded locations are recomputed. Ordered chord slopes, rather than independent point ordinates, are rounded upward; integrating these rounded slopes from an upward-rounded initial value preserves convexity.

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Rounding a retained coordinate can change the chord error. The acceptance decision therefore does not rely only on the error before rounding. It evaluates $G - F$ over the union of all source and encoded knots, including endpoints, and requires $0 \leq G - F \leq \delta$. Precisions 24, 32, 48, 64, 96, and 128 bits for coordinates are available, with 16 extra bits for slope and initial-value rounding. A failed precision ladder is an unresolved construction, not a certificate. All registered cases pass at the first coordinate precision. The implemented cap is 200,000 pieces; the existence theorem does not identify that engineering cap with a universal termination bound.

The original local implementation rounded ordinates and ran timing in a shared Python process. The archived attempt and all its results are retained. The final implementation rounds slopes to preserve the induction invariant and times each neural fit in a fresh process. These corrections do not change the economic model, seeds, training recipe, or tolerance. They are documented in the amendment rather than hidden by overwriting the earlier outputs.

2.2. Upper and lower witnesses

Let $\rho_t = U_t - TU_{t+1}$. Exact compression proves $0 \leq \rho_t \leq \delta$. The upper comparison $V_t \leq U_t$ is independent of the proposal. With $e_t = \eta + \beta e_{t+1}$ and $e_T = 0$, an eligible action satisfies

$$T^a(U_{t+1} - e_{t+1}) = q_t^a - \beta e_{t+1} \geq U_t - e_t.$$

The lower witness is therefore the explicitly constructed function $L_t = U_t - e_t$. It is not a supplied policy-value surface. The stronger realized check uses the exactly evaluated J_t^π and reports $\max_{t,x}(U_t - J_t^\pi)$; this can be considerably smaller than the allowance ε .

Exact installed-policy evaluation is a separate recursion. No raw-policy performance statistic is inferred from final repair. The reference diagnostic $\max(V - v^0)$ uses the uncompressed optimum only after the raw policy is frozen. The deployable zero-intervention certificate instead uses $\max(U - v^0)$ and needs no reference optimum. These quantities are recorded separately as `raw_exact_regret` and `raw_regret_upper`.

3. COST OPTIMALITY, PRESERVATION, AND COMPARISON CLASSES

Given nonempty eligible action sets, the intervention-cost recursion is ordinary finite-horizon dynamic programming with state-dependent constraints. Its output is nevertheless an additional economic object beyond the operating-value solve. The exact checker verifies, at every point where action a is eligible, that D_t does not exceed that action's immediate cost plus its continuation. It also verifies equality for the chosen action. This proves minimum cost and attainment from every restart; a uniform initial density is used only to summarize the resulting functions.

The primary computation does not impose one-step installed-value preservation as an action restriction. Instead, it computes the installed and revised values and verifies the complete difference. All 42 primary outputs preserve operating value. It would be incorrect to claim that this empirical fact establishes preservation for all inputs under the deficit-only action class. The optional intersection with $T^a v_{t+1}^0 \geq v_t^0$, together with a nonemptiness condition, provides that theorem when required.

The additional zero-intervention certificates have a different scope of optimality. Once $U - v^0 \leq \varepsilon$ everywhere is checked, zero is a lower bound on the cost of every feasible policy, not merely policies in the local class. Thus retaining a certified incumbent is globally optimal for intervention cost. This does not prove that a nonzero class minimum is globally optimal for the unconstrained-by-class regret problem. The 12 separate zero certificates explicitly expose this distinction.

The pointwise comparator retains an installed action when eligible and otherwise minimizes current implementation cost with action-index tie breaking. It does not optimize future implementation exposure. Its operating and cost functions are recomputed under the policy it actually deploys. The certified greedy comparator maximizes operating continuation against U . The uncompressed reference maximizes operating value exactly. Neither accuracy-only comparator is mislabeled as solving the constrained minimum-cost problem.

The price envelope is an exact comparison over the listed computed policies. For each pair with different costs, the algorithm includes a nonnegative crossing $(J_i - J_j)/(C_i - C_j)$; it checks all open intervals between crossings and every crossing point. A final unbounded interval covers arbitrarily large prices. Identical lines and exact ties remain valid alternatives. This produces a complete finite comparison, without implying an optimization over uncomputed feasible policies or an empirical monetary calibration.

4. COMPLETE ACTIVE-BOUNDARY PROOF

4.1. *Deterministic wealth and the Dynkin integrand*

Throughout this section the primitives, controls, and restart are those in Theorem 3 of the article. With $p = 0$, $c = 3/4$, and $x(0) = 5/4$,

$$x(t) = \frac{75}{2} - \frac{145}{4}e^{t/50}. \quad (1)$$

For $0 \leq t \leq 1$, $e^{t/50} \leq (1 - 1/50)^{-1} = 50/49$, hence $x(t) \geq 25/49 > 1/2$. It is decreasing and starts below two, so no wealth boundary is reached. Let y denote preference. Direct differentiation of the unchanged settlement gives

$$\begin{aligned} h_\theta(t, y) &= \frac{(3/4)^{1-y}}{1-y} - \theta^2 + 8 - .04\theta(y-2) - .00005 + .002 - \frac{.075}{x(t)} \\ &\quad + .0008(y-2)^2 - .004 \log x(t) + .32(1-t), \end{aligned} \quad (2)$$

$$\partial_\theta h_\theta = -2\theta - .04(y-2). \quad (3)$$

For $y \in [6/5, 14/5]$, put $z = y - 1 \in [1/5, 9/5]$ and $L = \log(4/3) < 1/3$. The utility is $-e^{Lz}/z$, which is increasing on this interval because $Lz < 1$. Its lower bound is $-5e^{1/15} \geq -75/14$. Also $|y-2| \leq 4/5$, $x(t) \geq 1/2$, and $\log x(t) \leq \log(5/4) < 1/4$. Dropping nonnegative terms yields

$$h_\theta \geq 8 - \frac{1}{20000} - \frac{3}{20} - \frac{1}{1000} - \frac{4}{625} - \frac{1}{25} - \frac{75}{14} = \frac{342357}{140000} > 2. \quad (4)$$

The utility derivative is $e^{Lz}(1 - Lz)/z^2 \geq 10/81$. Thus

$$\partial_y h_\theta \geq \frac{10}{81} - \frac{4}{3125} - \frac{1}{125} = \frac{28901}{253125} > 0. \quad (5)$$

Extend $h_\theta(t, y)$ constantly from $b = 14/5$ to larger y . The extension remains bounded, nondecreasing in y , and at least two. Its parameter derivative retains the bounds in (3). Conservative absolute bounds $|h| \leq 20$ and $|\partial_\theta h| \leq 2$ hold for this extension.

4.2. A coupling lower bound for the derivative

Let $Y_t = u_0 + \theta t + \sigma W_t$, killed at $a = 6/5$, where $u_0 = 61/50$ and $\sigma = 1/20$. If drift increases, the coupled process is higher at every date and lower-boundary survival events are nested. For a fixed nondecreasing integrand $h \geq 2$, paths surviving under both drifts have a nonnegative payoff change, while newly surviving paths contribute at least two. Taking a positive difference quotient gives

$$\partial_\theta \mathbb{E}_\theta[h(t, Y_t) \mathbf{1}\{t < \tau_a\}] \geq 2\partial_\theta \Pr(t < \tau_a). \quad (6)$$

The direct dependence of h_θ on θ must be added, not included in this frozen-integrand comparison. It contributes at least $-4/125$ for $\theta \in [-1/5, 0]$ and at least $-54/125$ for $\theta \in [0, 1/5]$, after integrating over a discounted survival time of at most one. Gaussian likelihood scores justify differentiability and dominated passage to the occupation integral for this bounded extension.

The reflection formula with drift, differentiated exactly, yields

$$S'_\theta(t) = 16e^{-16\theta} \Phi\left(\frac{\theta t - 1/50}{(1/20)\sqrt{t}}\right). \quad (7)$$

The function $\log \Phi$ is concave. Indeed, with $r = \varphi/\Phi$, its second derivative is $-r(z)\{z + r(z)\} < 0$; when $z < 0$, the elementary Gaussian tail inequality gives $r(z) > -z$. Equation (7) is therefore log-concave in θ . Its minimum on each of the two half intervals is at least the smaller of its endpoint values.

For $t \in [1/25, 1/5]$, the normal arguments at $\theta = 0$ and $\theta = -1/5$ are bounded below by -2 and $-14/5$, respectively. The latter follows by maximizing $4\sqrt{t} + (2/5)/\sqrt{t}$ on that compact interval; its maximum is $14/5$. The rational inequalities

$$\begin{aligned} 24 < e^{16/5} < 25, \quad \Phi(-2) > 1/50, \quad \Phi(-14/5) > 1/500, \\ \Phi(-4/5) > 1/5, \quad \Phi(6/5) > 22/25. \end{aligned} \quad (8)$$

imply $S'_\theta(t) \geq 8/25$ on the entire negative drift half interval.

For $t \in [1/4, 1]$, the endpoint arguments at $\theta = 0$ and $\theta = 1/5$ are at least $-4/5$ and $6/5$. Hence $S'_\theta(t) \geq 352/625$ on the positive drift half interval. Since $e^{-.04t} \geq 24/25$, the two lower bounds for the derivative of the lower-killed occupation expression are

$$2\frac{4}{25}\frac{24}{25}\frac{8}{25} - \frac{4}{125} = .066304, \quad (9)$$

$$2\frac{3}{4}\frac{24}{25}\frac{352}{625} - \frac{54}{125} = .379008. \quad (10)$$

No derivative sample grid appears in this argument. The intervals cover every admissible constant drift, including the shared point zero.

4.3. Distant-boundary error and rational checks

The upper boundary is distant but not discarded. Under all admissible drifts, a path hitting it must have a Brownian excursion of at least $(b - u_0 - 1/5)/\sigma$. Reflection and a Gaussian tail bound give

$$p_b \leq 2 \exp\left[-\frac{(b - u_0 - 1/5)^2}{2\sigma^2}\right] = 2e^{-380.88} < 2^{-379}. \quad (11)$$

For a path-dependent difference supported on that event, the drift likelihood score has second moment at most $1/\sigma^2$. Cauchy–Schwarz bounds the derivative discrepancy between the actual doubly killed occupation integral and the extended lower-killed one by

$$\frac{20}{\sigma} \sqrt{p_b} + 2p_b < 400 2^{-189} + 2 2^{-379} < 1/1000. \quad (12)$$

Subtracting $1/1000$ from the preceding half-line bounds gives precisely $8163/125000$ and $47251/125000$. A strictly increasing continuous function on the compact drift interval has the unique maximizer $1/5$.

The file `replication/boundary.py` proves the elementary constants in (8) with exact fractions. Exponential bounds use Taylor sums and an explicit geometric tail. A Machin arctangent identity with alternating-series bounds proves a rational bound for π , and $\sqrt{2\pi} < 251/100$ follows. Gaussian lower bounds use monotone right sums of mesh $1/100$ on finite intervals, with every exponential term bounded by a rational Taylor calculation. Truncating the infinite tail only lowers a tail integral. For $\Phi(6/5)$ the integral is from zero to $6/5$, added to $1/2$. Terms are rounded downward to a common dyadic denominator before summation, with the direction of the bound preserved. The code records the resulting exact rational lower bounds and requires them to exceed all displayed thresholds.

The payoff enclosure at $\theta = 1/5$ is inherited from the unchanged validated R30 killed-kernel oracle. Its source path, Git blob hash, and the fact that it is not recomputed are recorded in `results/boundary.json`. The current theorem changes its interpretation from an endpoint evaluation to the optimum of the entire constant class. It does not assert feedback optimality or satisfy the original all-domain current-state target.

5. FULL COHORT AND AUDIT RESULTS

TABLE I: All registered class-based certificates

T	Proposal	ε	Raw regret	Certified bound	Cost saving	Bytes
4	Defer	0.01	1.61894	0.00269551	0	3897
4	Defer	0.05	1.61894	0.0134775	0	3887
4	Stress	0.01	1.61894	0.00269551	0.00227628	4660
4	Stress	0.05	1.61894	0.0134775	0.0113815	4659
4	Spline 33	0.01	4.30982e-16	4.09353e-08	0	3505
4	Spline 33	0.05	4.30982e-16	4.09353e-08	0	3504
4	Spline 129	0.01	4.14599e-16	4.09353e-08	0	3494
4	Spline 129	0.05	4.14599e-16	4.09353e-08	0	3493
4	NN 31001	0.01	0.124593	0.00269551	0	4434
4	NN 31001	0.05	0.124593	0.0134775	0	4425
4	NN 31002	0.01	0.14356	0.00269551	0	4311
4	NN 31002	0.05	0.14356	0.0134775	0	4302
4	NN 31003	0.01	0.143355	0.00269551	0	4440
4	NN 31003	0.05	0.143355	0.0134775	0	4431
8	Defer	0.01	3.35621	0.00148562	0	19038
8	Defer	0.05	3.35621	0.00804234	0	18773
8	Stress	0.01	3.35621	0.00148561	0.0174693	41810

T	Proposal	ε	Raw regret	Certified bound	Cost saving	Bytes
8	Stress	0.05	3.35621	0.00742774	0.0891575	41606
8	Spline 33	0.01	0.0057581	0.00148562	0	15821
8	Spline 33	0.05	0.0057581	0.00575812	0	12143
8	Spline 129	0.01	0.000411466	0.000411565	0	12451
8	Spline 129	0.05	0.000411466	0.000775484	0	12179
8	NN 31001	0.01	0.234831	0.00148562	0	18524
8	NN 31001	0.05	0.234831	0.00804234	0	18250
8	NN 31002	0.01	0.296282	0.00148562	0	18488
8	NN 31002	0.05	0.296282	0.00804234	0	17477
8	NN 31003	0.01	0.224246	0.00148562	0	19178
8	NN 31003	0.05	0.224246	0.00742775	0	18500
12	Defer	0.01	4.75581	0.00116122	0	195094
12	Defer	0.05	4.75581	0.00603201	0	194062
12	Stress	0.01	4.75581	0.00116031	0.0237425	370046
12	Stress	0.05	4.75581	0.00608421	0.119917	359043
12	Spline 33	0.01	0.0057581	0.00116122	0	156289
12	Spline 33	0.05	0.0057581	0.00543908	0	116384
12	Spline 129	0.01	0.000411466	0.000411565	0	91427
12	Spline 129	0.05	0.000411466	0.00105788	0	86102
12	NN 31001	0.01	0.64517	0.00116402	0	184623
12	NN 31001	0.05	0.64517	0.00643136	0	179921
12	NN 31002	0.01	0.611512	0.00115741	0	193825
12	NN 31002	0.05	0.611512	0.00616493	0	173948
12	NN 31003	0.01	0.224246	0.00116122	0	206325
12	NN 31003	0.05	0.224246	0.00603201	0	209874

The tables display rounded summaries only. Every primary certificate stores rational knots, values, policies, Bellman witnesses, and intervention-cost functions. A separate file records the exact interval-wide error and cost residual. The checker does not invoke the envelope optimizer. Its only shared components are the representation, the primitive model, and elementary exact affine operations. This is implementation separation, not a proof assistant or an assertion of independence from all shared arithmetic code.

Eight deliberately malformed inputs are rejected: changed model, absent model, negative tolerance, missing date, missing interval, changed endpoint, false cost, and infeasible action. Tests also retain point spikes and affine preimages and compare the exact two-period Bellman solution with exhaustive adaptive scenario-tree enumeration at four rational restart states. The test count is not substituted for the whole-state inequalities in the certificates.

6. HISTORICAL PRESERVATION AND CURRENT CLAIMS

The document `HISTORY_R32.pdf` concatenates the five R30 submission objects without altering their pages: article, supplement, response, computational record, and the earlier historical annex. The repository retains their original source and result paths. Thus the previous action-independent operating-mode study, inventory comparison, Adam transport and rollback derivations, central-state search, active-boundary quadrature failures, and earlier referee responses are not deleted to simplify the current narrative.

In particular, the unfavorable exhaustive inventory timing ratios, the 26/60 raw passes in the prior operating-mode cohort, and the exact symbolic and online-guard controls remain available. They are historical evidence, not results of the new action-dependent model. The original whole-domain bound remains 7.181834580823298 and its target remains .01. No new result is relabeled as a 47-coordinate derivative bridge, a high-dimensional certificate, or neural-specific acceleration. The constructive coupled certificates and the global active-boundary scalar theorem are additional results with explicitly delimited objects.

RESPONSE TO THE R30 REFEREE REPORT

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1. RESPONSE TO THE R30 REPORT

We thank the referee for distinguishing the validity of the earlier verification arguments from the missing end-to-end numerical construction. The revised article replaces the action-independent numerical core with a prospectively specified action-dependent maintenance economy. Upper witnesses, own-policy values, admissible sets, and intervention costs are now constructed from primitives and charged. A new global theorem completes optimization over the unchanged active-boundary constant-control interval. We preserve the original economic target and all historical results. We do not claim neural acceleration or a solved all-domain feedback problem when the computed evidence does not establish them.

The source report is `reviews/2026-09-24-econometrica-r30/referee_report.md`, commit `e9bc144fbb6843d6a3436825a28584efb64fb8f1`. Its reviewed manuscript is commit `3394815f4ccbf5917582c8f756b535adf87783cc`. R32 branches from `6ea1e2fd8ab1928ded4ea98dc74ba6e1e72cf9d0`, retaining the R31 prospective protocol and amendments. This response addresses the report’s individual findings rather than treating a universal final pass rate as closure of every scientific question.

1.1. *R30-F1: supplied witnesses versus a complete construction*

The main article’s constructive theorem starts from rewards, terminal settlement, and action-dependent affine shock maps. Backward expectation and maximization produce F_t ; shape-preserving dyadic compression produces U_t with an exactly checked global error. The lower witness is explicitly $U_t - e_t$. Installed values are computed by exact own-policy recursion, not supplied. The work proposition includes these constructions and downstream cost-value partitions. Tables report witness time, intermediate and stored pieces, bit lengths, complete certificate bytes, and full cold-pipeline timing. This closes the missing input-construction step for the displayed coupled class; it does not establish that every such pipeline beats the strongest classical solver.

1.2. *R30-F2: cancellation of continuation values*

The new maintenance primitives are fixed in the inherited R31 protocol. Maintaining and replacing always have lower current reward than deferring, and have different transition maps. Their usefulness is entirely a continuation effect. The computation constructs the action-dependent Bellman alternatives at every date. No primitive reward-sign identity supplies the optimal policy. The old analytically reducible family is preserved in the historical annex, not used as the new coupled experiment.

1.3. *R30-F3: dominating classical controls*

The comparison now includes uncompressed exact piecewise-affine dynamic programming, an accuracy-only greedy solver using the same constructed upper witnesses, and 33/129-node

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spline fitted-Q proposals. These are more informative controls than a large grid. Their favorable performance is retained. The new contribution demonstrated by the experiment is the coupled construction and minimum-cost repair within its certified class. A neural end-to-end speed advantage is not established. We do not redefine an accuracy-only solver as a minimum-intervention solver, nor omit its much lower operating-solve time.

1.4. *R30-F4: policy-dependent occupancy*

The primary cost recursion uses the transition maps of the deployed action. A predeclared, separately labeled stress incumbent tests future exposure to an expensive installed instruction. In all six stress configurations, the dynamic solution has strictly smaller exact discounted intervention cost than the pointwise comparator using the same admissible sets. Both cost functions are evaluated under their own deployed occupancy. The other 36 cases show equality and are reported. Thus the previously absent mechanism is now exercised, without claiming it is universal or neural-specific.

1.5. *R30-F5: trillion-state rhetoric and dimension*

The new article makes no trillion-state headline comparison. Its geometric table reports state dimension, horizon, exact-value pieces, intermediate envelopes, stored witness pieces, and cost-value pieces. The earlier 2^{40} observation remains unchanged in the historical archive as a representation fact. The current study does not demonstrate high-dimensional scaling. Its output-sensitive construction theorem explicitly retains the possibility of growing partition complexity.

1.6. *R30-F6: the role of neural proposals*

The same frozen neural architecture, fitting recipe, and three seeds are evaluated at every declared horizon and tolerance. Raw neural passes are 0/18; the spline controls pass 12/12 and admit separate zero-intervention certificates. These findings are visible in the abstract, main table, and complete cohort. Neural proposals remain an evaluated installed representation, not a claimed enabling ingredient. The proposal-margin proposition states a verifiable condition under which an approximation eliminates revisions; no sampled training loss is substituted for its premise. Neural-specific superiority remains unestablished by these results.

1.7. *R30-F7: conditional complexity and proposal quality*

The upper-witness construction now has a mesh-and-precision existence bound and a complete output-sensitive work account. The cost of constructing the upper witness, own-policy evaluation, compilation, action sets, cost recursion, serialization, and checking is included. The proposal condition $\rho_t + 2\kappa_t \leq \eta$ quantitatively connects a verified approximation margin to eligibility and zero revision cost. The empirical study reports actual representation geometry rather than treating the favorable univariate case as a dimension-free theorem. No unproved speedup over a classical approximation is inferred.

1.8. *R30-F8: the original all-domain target*

The target remains exactly the original current-state, all-restart .01 operating guarantee. Its retained whole-domain upper bound is still 7.181834580823298, and the 47-coordinate

derivative/residual bridge is not newly instantiated. We have not replaced this target with the maintenance tolerance. We make a substantive advance in the unchanged original economy through the global active-boundary scalar theorem, while keeping that smaller control class explicit. The article's constructive maintenance chain and the original continuous objective are now distinguished, rather than described as two parts of a completed feedback solve.

1.9. *R30-F9: complete optimization at the active boundary*

The five-point derivative diagnostic is replaced by an interval-wide proof for the same restart, controls, and parameter interval. The two drift half intervals have rigorously positive derivative lower bounds .065304 and .378008 after the upper-boundary error is charged. The unique global optimum in that constant class is $\theta = 1/5$. Every Gaussian and exponential constant used in the proof is checked with rational arithmetic. The old validated endpoint payoff interval is identified as inherited; no new quadrature run is invented. This completes the scalar optimization requirement, but not unrestricted feedback optimization.

1.10. *R30-F10: arbitrary implementation prices*

The revised economic tables report the complete piecewise-linear envelope of $J_i - \lambda C_i$ over named computed deployments for every $\lambda \geq 0$. All crossing prices and tie sets are stored as exact fractions. No 60/60 favorable sign at a chosen price is used as a headline. Cost units are normalized interventions, and no monetary calibration is asserted. The earlier illustrative-price experiment remains in the historical annex.

1.11. *R30-F11: the theorem-level distinction*

The article explicitly credits Bellman comparison and constrained backward induction as classical. The stated addition is a constructive, bit-controlled witness pipeline together with its generated action class and a checkable endogenous-cost representation. The cost recursion alone is not claimed as a new Bellman principle. A zero-intervention corollary also gives a global minimum over the true regret-feasible set when the computed incumbent certificate passes, avoiding unnecessary revisions caused by the conservative local class. A lower verified complexity than all established alternatives is not proved; the strong classical numerical comparisons are retained.

1.12. *R30-F12: one principal theorem-to-computation chain*

The article now proceeds from maintenance primitives to constructive witnesses, all-restart admissibility, endogenous minimum-cost revision, and exact economic sensitivity. The active-boundary result is a separate completed optimization theorem in the original economy. Earlier experiments and technical derivations are preserved in a lossless annex rather than interleaved as independent claims of superiority. The dominant chain is complete for the stated one-dimensional coupled class; a frontier-changing high-dimensional application is not claimed from it.

2. TECHNICAL COMMENTS

2.0.0.1. *R30-T1: input complexity.* The charged-work proposition and stage accounting include witness production, raw-value evaluation, compilation, encoding, and checking. The empirical timing table additionally charges the independent checker.

2.0.0.2. *R30-T2: constructing U, L, v^0 .* U is constructed by exact Bellman operations and global compression; $L = U - e$ is explicit; v^0 is an exact policy evaluation. None is loaded from the optimal reference.

2.0.0.3. *R30-T3: online/offline amortization.* The article states the correct break-even formula including construction, loading, and per-query costs, and does not claim a measured break-even threshold because that deployment benchmark was not executed. The old exact online sign guard remains preserved as a strong control in its own action-independent model.

2.0.0.4. *R30-T4: cardinality versus dimension.* Main complexity and evidence are expressed through actual one-dimensional partitions, horizon, bits, and bytes. No state-cardinality observation is used as evidence of dimension independence.

2.0.0.5. *R30-T5: endogenous occupancy.* Six stress configurations show strict dynamic cost savings; 36 equality cases remain. Exact cost Bellman inequalities are independently checked, including isolated eligibility points.

2.0.0.6. *R30-T6: proposal quality.* The verified sup-error/margin proposition connects approximation error to eligibility and zero revision cost. The raw/global own-policy test is reported separately from local-class repair. Unverified training residuals carry no certification force.

2.0.0.7. *R30-T7: hardware execution.* Certificates concern exact compiled policies. Binary64 fitting coefficients are rationalized at compilation; actual floating-point neural forward execution is not certified.

2.0.0.8. *R30-T8: economic units.* The weight $1 + x$ defines normalized state-dependent implementation units. The complete shadow-price envelope is a sensitivity analysis, not an estimated institutional cost schedule.

2.0.0.9. *R30-T9: boundary optimization.* The global monotonicity proof covers all $\theta \in [-1/5, 1/5]$, includes the distant-boundary likelihood-score error, and yields the unique scalar optimum. It does not assume central-state concavity.

2.0.0.10. *R30-T10: historical compression without deletion.* The current article is reorganized around the constructive coupled chain. `HISTORY_R32.pdf` reproduces all five R30 submission documents page-for-page, including their existing history. Earlier source and evidence paths are unchanged.

3. VERIFICATION AND REMAINING DISTINCTIONS

All 42 registered class certificates pass the independent checker. Twelve further zero-intervention certificates pass. Eight malformed-certificate tests are rejected, isolated-point/preimage tests pass, and a two-period exhaustive scenario-tree control agrees at four rational restarts. The scalar proof's rational inequalities all pass. These results support the specific mathematical and computational claims above. They are not a referee acceptance prediction, a proof of neural superiority, a high-dimensional performance study, or a completion of the original all-domain feedback target.

COMPUTATIONAL RECORD FOR REVISION R32

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1. REPRODUCTION AND IMMUTABLE INPUTS

The scientific source is in `revisions/2026-09-24-r32/replication`. The exact maintenance model is defined in `coupled.py`; the registered proposals and frozen-parameter records are generated by `study.py`; the interval-wide scalar proof is checked by `boundary.py`; an independently organized checker and malformed-input tests are in `audit.py`. Tables and a file-hash manifest are generated by `report.py`. All rational calculations use Python’s arbitrary-precision fractions. Neural fitting uses CPU binary64 PyTorch. The recorded initial environment used Python 3.13, NumPy 2.3.5, and PyTorch 2.10.0+cpu. Every execution records its actual Python and platform rather than implying hardware-independent timing.

From the repository root, a complete fresh reproduction is:

```
python -m pip install numpy==2.3.5
python -m pip install torch==2.10.0 \
    --index-url https://download.pytorch.org/whl/cpu
R=revisions/2026-09-24-r32
python "$R/replication/study.py" "$R/results" --horizon 4
python "$R/replication/study.py" "$R/results" --horizon 8
python "$R/replication/study.py" "$R/results" --horizon 12
python "$R/replication/boundary.py" \
    --output "$R/results/boundary.json"
python "$R/replication/audit.py" --folder "$R/results"
python "$R/replication/report.py"
```

Do not use Python’s `-O` flag: the exact construction deliberately refuses execution when assertions are disabled. The independent checker uses explicit exceptions for its substantive comparisons. A failed run is retained as an unresolved record, not silently omitted from a table. The final report generator requires the complete 42-case cohort and all certificates before producing publication tables.

The five publication entry points are `ECTA_R32.tex`, `SUPP_R32.tex`, `RESPONSE_R32.tex`, `COMPUTATION_R32.tex`, and `HISTORY_R32.tex`. The first four use the repository’s unmodified Econometric Society class. Build the article before the supplement, whose external references point to the article. The history PDF includes the five already compiled R30 submission objects without modifying their pages. The build script performs multiple LaTeX passes and records actual exit status. A source manifest hashes the replication, records, and paper sources; the final delivery manifest separately hashes the PDFs.

2. WHAT THE COSTS INCLUDE

Each neural proposal is fitted in a fresh subprocess. Its elapsed time includes interpreter/framework initialization and coefficient serialization; coefficient loading is charged at compilation. Subsequent costs are separated into witness construction, own-policy evaluation, eligible-set construction, dynamic minimum-cost computation, final evaluation, internal verification, and serialization. The independent final checker is timed separately and added in the

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main timing table. Reference solves, pointwise comparisons, sensitivity-envelope generation, and supplementary audit experiments are reported separately, not silently charged as necessary optimizer work.

The class-based pipeline is executed even when the installed proposal eventually qualifies for zero intervention, so that conservative over-revision is visible. The independent audit additionally evaluates the incumbent and writes a distinct zero-intervention certificate when the upper-witness test passes. The incremental own-evaluation and zero-certificate checking time is recorded. A production implementation can perform this test before cost optimization; the paper does not retroactively replace its measured class-pipeline timing by that hypothetical shortcut.

Times are single executions on the recorded machine. The same witness is reused across proposals in the study, but its full measured construction cost is charged to each cold solution for a standalone accounting. The exact dynamic-programming reference and the certified greedy baseline use the same primitives; they optimize operating return rather than the added revision-cost objective. Library memory is included in process peak resident memory. Portable uncompressed certificate bytes and mathematical piece/bit counts are reported separately.

3. FAILURE RETENTION AND IMPLEMENTATION CORRECTIONS

The initial shared-process timing cohort and its source remain under `results/attempt`
`ts/phase-timing-v0`. The final shape-preserving encoder corrects a general issue with independent ordinate rounding; the old tested inequalities are not relabeled as failed, and the old source is not removed. A batch horizon-12 execution interrupted by the local execution limit is retained as an interrupted attempt. It is distinct from a model or certificate failure. The final horizon runs are complete.

No seed, horizon, tolerance, or adverse comparison was dropped. The predeclared stress rule is not included in neural performance counts. The zero-cost corollary was checked for every case, not only the spline successes. The main pipeline and independent checker record 42 primary passes, the additional check records 12 zero-intervention passes, and all eight malformed-certificate mutations are rejected. The rational scalar proof is a separate calculation; its inherited endpoint payoff provenance remains explicit.

4. AUDIT TRAIL AND LIMITATIONS OF VERIFICATION

The checker recomputes whole-cell inequalities using exact affine overlays and point values. It does not invoke the maximizing/minimizing envelope routine or train a model. It shares representation and primitive affine operations with construction, so it is not a fully independent software stack or a machine-checked proof assistant. The supplementary mathematical proofs identify the properties these operations must satisfy. Tiny exhaustive scenario-tree checks and malformed certificates supplement, but do not replace, that proof.

The scientific guarantee is for exact compiled deployment. No hardware floating-point neural execution is covered. Economic sensitivity is over named computed policies with normalized implementation units. The original stopped-economy full-domain target and the neural-specific speedup question remain separate from the certified coupled computation. Those boundaries are part of the record and are not converted into experimental successes by a successful build.

Certified Bellman Operators with Neural Proposals

Historical Preservation Annex, R32

September 24, 2026

This annex reproduces all five R30 submission documents without changing their pages: the article, supplement, response, computational record, and historical annex. The latter retains the previous manuscript lineage. All original source and scientific evidence remain at their existing repository paths. The R31 prospective protocol and amendments are also retained in their original directory. Current claims and their limitations are stated in ECTA_R32 and RESPONSE_R32; historical material is preserved rather than silently rewritten.

CERTIFIED BELLMAN OPERATORS WITH NEURAL PROPOSALS: ADAPTIVE VERIFICATION AND COSTLY POLICY REVISION

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We study Bellman computation with a uniform operating-performance requirement and an explicit cost of revising an installed policy. Regional comparison inequalities certify whole state sets, while a constrained Bellman recursion minimizes discounted intervention cost within a certified policy class. In an operating-mode economy, exact polynomial bounds and rational neural-policy compilation cover a continuous state interval without grid enumeration. Of 60 frozen neural proposals, 26 meet the .01 requirement before repair. Economically weighted revision costs are enclosed relative to the local-class minimum. Exact symbolic, online-guard, polynomial, and vectorized classical controls receive the same model information; neural-specific computational superiority is not inferred. The preceding exhaustive inventory pipeline remains slower than exact dynamic programming. In the original stopped consumption–portfolio economy, a killed-kernel oracle encloses payoff and derivatives at a restart with substantial exit probability and establishes a change in curvature. The original current-state, whole-domain accuracy target remains distinct from these proved and computed results.

KEYWORDS: Bellman operators, Policy revision, Certified computation, Neural proposals, Stopped control.

1. INTRODUCTION

An installed decision rule is not a blank slate. Replacing it can require new operating instructions, contractual approvals, or changes at precisely those states where intervention is expensive. Consequently, two policies with almost identical operating payoffs need not be economically interchangeable. This paper studies Bellman computation when a uniform performance requirement and the cost of policy revision are both explicit objectives. Neural networks provide candidate decisions; the certified object is a compiled policy together with a proof of its performance and intervention cost.

Our starting point is a substantive computational limitation of exhaustive Bellman completion. The preceding inventory experiment evaluated all feasible actions at all states and repaired a candidate whenever its residual allowance was exceeded. The argument was correct, but it repeated the enumeration performed by dynamic programming. On every reported graph, repair plus verification cost more than the independently implemented exact solver. Including new neural fitting raised the displayed cost ratio to between 10.9 and 63.2. Moreover, only 12 of 115 raw candidates met the 0.01 criterion, and none did so in the main cohort with at least 256 states. Those results remain part of the evidence. They motivate a change in the algorithm, not a reinterpretation of a universal post-repair pass rate as approximation success.

We develop two complementary constructions. First, a regional Bellman certificate verifies a whole set of states with one valid range enclosure. Its complexity depends on the construction of the enclosures, the geometry of unresolved regions, and the representation of the candidate, rather than necessarily on the number of grid states. Second, a constrained Bellman recursion minimizes discounted intervention cost within an explicitly certified policy class. This supplies an economic meaning for preservation and separates an optimization claim from a count of unchanged table entries. The admissible class is an inner approximation to the policies satisfying the true regret constraint; we do not identify these two classes.

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The computational example is an eight-period operating-mode economy with an exogenous state process and costly policy revision. This structure permits continuation values to cancel from action comparisons. Exact polynomial range bounds and a rational compilation of a one-hidden-layer ReLU network then certify an entire continuous state interval. The same certificate applies to any midpoint grid, including a grid with 2^{40} states, without materializing that grid. This is a cardinality result in one state dimension, not a high-dimensional approximation theorem. We compare against a structure-aware classical solver using the identical range oracle, polynomial proposals, and vectorized dynamic programming. A good installed proposal can reduce certificate construction, but the classical structural solver remains a strong competitor; neural training is not assigned zero cost.

A separate calculation addresses the economically active stopping boundary of the original consumption–portfolio model. At the new restart, the lower-boundary hitting probability at zero preference adjustment is approximately 0.689, rather than the previous upper bound below $2.153 \cdot 10^{-31}$ at the central restart. We integrate a killed-diffusion kernel with validated Gaussian moments and explicit approximation remainders. First and second derivatives are enclosed, and the curvature changes sign across the tested controls. This is a substantive boundary diagnostic in the unchanged economy. It is not the all-state, all-restart certificate for the original current-state actor.

1.1. *Relation to existing methods*

The comparison argument behind Bellman residual bounds and one-step policy improvement is classical. Fitted value iteration propagates approximation errors under sampling and function-class assumptions (Munos and Szepesvári, 2008); approximate policy iteration and its interpolations have related performance bounds (Scherrer, 2013). A rollout construction improves a base policy by optimizing against its continuation. The exhaustive completion retained in the historical annex belongs to this family. Our general comparison theorem does not claim a new Bellman principle. The additional objects are a checkable regional witness, a complexity statement conditional on its geometry, and an intervention-cost optimization within the resulting admissible action sets.

Safe policy improvement with baseline bootstrapping restricts departures from a behavior policy when a batch dataset leaves transitions uncertain (Laroche, Trichelair, and Tachet des Combes, 2019). Here the transition model is known, the certificate is deterministic, and the intervention weights are economic primitives rather than uncertainty proxies. Neither construction subsumes the other’s information assumptions. The present minimum-cost statement applies only to its displayed certified class. It is not a global minimum-edit theorem for all policies that happen to satisfy the true regret bound.

Numerical economics already uses problem structure, approximation, and error control (Judd, 1998), including adaptive sparse grids for high-dimensional models (Brumm and Scheidegger, 2017). Our one-dimensional region cover is not a substitute for such a dimensionality comparison. Markov-chain approximation of a diffusion requires consistency and convergence conditions (Kushner and Dupuis, 2001); neither the inventory graph nor the operating-mode economy is presented as a verified discretization of the original diffusion. Information-relaxation bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014) and smooth supersolution bounds furnish possible witnesses, but their construction must be charged.

The continuous calculation uses verified numerical integration, rather than Monte Carlo confidence intervals or a decimal quadrature tolerance. Directed and ball arithmetic control different sources of uncertainty only when the analytic hypotheses and truncation errors are also supplied (Rump, 2010, FLINT Project, 2026). Constrained direct search (Kolda, Lewis,

and Torczon, 2003, 2006, Audet and Dennis, 2006) is retained as a certificate wrapper. When a scalar derivative oracle and strict concavity are available, derivative bracketing is the appropriate strong alternative. Neural control approximation (Huré, Pham, Bachouch, and Langrené, 2021, Sirignano and Spiliopoulos, 2018) motivates proposal generation; it does not replace these verification obligations.

2. ECONOMIC OBJECTIVES AND INFORMATION

2.1. The original stopped economy

Time runs from zero to one. Preference u and wealth x satisfy

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (1)$$

$$dx_s = \{(.02 + .06p_s)x_s - c_s\} ds + .2p_s x_s dZ_s^x, \quad d\langle Z^u, Z^x \rangle_s = -.25 ds. \quad (2)$$

The controls lie in $[.05, .8] \times [-.2, .2] \times [-.5, .8]$ for (c, θ, p) . Let τ be the first exit from $\mathcal{D} = (1.2, 2.8) \times (.5, 2)$, capped at one. Running utility and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad k \in [.5, 8], \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1 - s). \quad (4)$$

For an admissible policy adapted to the original filtration,

$$J_k^\pi(t, u, x) = \mathbb{E}_{t,u,x} \left[\int_t^\tau e^{-.04(s-t)} \ell_k(u_s, c_s, \theta_s) ds + e^{-.04(\tau-t)} g(\tau, u_\tau, x_\tau) \right]. \quad (5)$$

Write $V_k = \sup_\pi J_k^\pi$. The original objective is a single current-state actor $\pi(t, u, x)$ satisfying

$$\sup_{(t,u,x) \in [0,1] \times \overline{\mathcal{D}}} \{V_2(t, u, x) - J_2^\pi(t, u, x)\} \leq .01. \quad (6)$$

The best retained whole-domain upper bound is 7.181834580823298. No computation in this revision replaces it by a bound below .01. The finite-state results and the active-boundary scalar derivative calculation have different scopes, identified below. In particular, the financed 47-coordinate actor is not supplied here with a validated full-class derivative/residual bridge. The original model, target, unsuccessful evidence, and historical derivations are retained without alteration.

2.2. The certification and revision problem

For the general discrete-time analysis, let $t = 0, \dots, T - 1$, let the measurable state space be S_t , and let $A_t(s)$ be finite. Rewards and terminal settlement are bounded. A known substochastic kernel P_t^a includes survival; settlement upon absorption is included in the reward. Define

$$(T_t^a v)(s) = r_t(s, a) + \beta \int v(s') P_t(ds' | s, a), \quad T_t v = \max_{a \in A_t(s)} T_t^a v, \quad (7)$$

where $0 < \beta \leq 1$. Assume measurable maximizers, for example a fixed ordering of the finite actions. Let π^0 be an installed policy, possibly obtained by neural approximation. The operating

value is J^π , whereas the revision cost is

$$C_\nu(\pi; \pi^0) = \mathbb{E}_\nu^\pi \sum_{t=0}^{T-1} \beta^t w_t(S_t, \pi_t(S_t)) \mathbf{1}\{\pi_t(S_t) \neq \pi_t^0(S_t)\}, \quad (8)$$

with nonnegative bounded weights and zero cost after absorption. The occupancy measure in this expression is induced by the deployed policy. It is not a uniform count of states. An economically relevant constrained problem is to minimize (8) subject to uniform regret and, when required, operating-payoff preservation. We solve this problem within a certified inner class, and bound the loss from that class in the implemented special case.

The phrase *precomputed-optimum-free* below means that no stored V^* or optimal-action file is read. It does not mean model-free, sampling-only, or less-informed than dynamic programming. All model evaluations, witness construction, interval bounds, compilation, and final checks are charged. An independently implemented reference has independent code, not independent economic information.

3. REGIONAL BELLMAN CERTIFICATES

The following statement separates a proof obligation from the representation used to discharge it.

THEOREM 1—Regional comparison: *Let bounded measurable functions L_t, U_t and a feasible policy π satisfy*

$$L_T \leq g \leq U_T, \quad L_t \leq T_t^{\pi_t} L_{t+1}, \quad U_t \geq T_t U_{t+1}. \quad (9)$$

Then $L_t \leq J_t^\pi \leq V_t^ \leq U_t$ at every state and restart. If $U_t - L_t \leq e_t$, then $V_t^* - J_t^\pi \leq e_t$. It suffices to prove each inequality on every member of a finite covering of each state space, including any exceptional boundaries or tie points.*

PROOF: At T the assertion is the terminal inequality. Positivity of P_t^a preserves order. Thus $J_t^\pi = T_t^{\pi_t} J_{t+1}^\pi \geq T_t^{\pi_t} L_{t+1} \geq L_t$, and $V_t^* = T_t V_{t+1}^* \leq T_t U_{t+1} \leq U_t$. Backward induction proves the assertion simultaneously for every restart. A covering proof is pointwise complete, so it proves the same premises without enumerating the points of a cell. *Q.E.D.*

A regional verifier must bound the complete expression, not its values at selected test points. For example, an interval upper bound on $T_t^a U_{t+1} - U_t$ must cover the whole cell and every feasible action. An implementation may split a cell until its bound is decisive or use an exact finite-state fallback. An unresolved cell prevents certification; it is not counted as a passing cell.

PROPOSITION 1—A certified preservation class: *Suppose $U_T = g$, $U_t \geq T_t U_{t+1}$, $e_T = 0$, and $e_t \geq 0$. Let $v_t^0 = J_t^{\pi^0}$, including $v_T^0 = g$. Define*

$$\mathcal{B}_t(s; e) = \left\{ a \in A_t(s) : \begin{array}{l} T_t^a (U_{t+1} - e_{t+1})(s) \geq U_t(s) - e_t(s), \\ T_t^a v_{t+1}^0(s) \geq v_t^0(s) \end{array} \right\}. \quad (10)$$

If these sets are nonempty, every measurable policy selecting from them satisfies

$$J_t^\pi \geq v_t^0, \quad 0 \leq V_t^* - J_t^\pi \leq e_t \quad (11)$$

for all states and restart dates.

PROOF: The first inequality in (10) makes $U - e$ a subsolution for the selected policy, so Theorem 1 supplies the regret bound. The second inequality and backward induction give $J_t^\pi = T_t^{\pi_t} J_{t+1}^\pi \geq T_t^{\pi_t} v_{t+1}^0 \geq v_t^0$. Q.E.D.

The value v^0 is an own-policy witness, not an optimum. Its exact evaluation may itself be expensive. Its cost and the cost of U are part of the method. A sufficient lower/upper enclosure can replace exact evaluation if it establishes the displayed comparisons conservatively. Empty admissible sets require a different witness or budget; the theorem does not conceal this feasibility condition.

3.1. The exhaustive limit

The original residual-budget algorithm remains a useful fallback on a finite graph. After completing the suffix, it retains a raw action when its local deficit plus the propagated suffix envelope fits in the prescribed allowance; otherwise it chooses a maximizing action against that completed suffix. Its full proof and all previous numerical results are reproduced in the historical annex.

PROPOSITION 2—Zero tolerance: *On a finite exact model, exhaustive backward completion with zero allowance yields $J_t^\pi = V_t^*$ at every state and date. A raw action can be retained at zero tolerance precisely when it is maximizing against the completed suffix. The statement permits retention of any optimal tie; the fixed replacement tie rule does not force a retained tie to be replaced.*

PROOF: At the last decision date, zero allowance permits only a reward-plus-settlement maximizer. Suppose the completed suffix has optimal value. The same zero-deficit condition then permits only a maximizer of the optimal Bellman comparison at the preceding date. Induction gives the result. Q.E.D.

This finite exact limit is not a termination claim for interval subdivision at zero tolerance around irrational indifference points. Nor does a small number of replacements imply a small exhaustive scan. The regional implementation below is what changes the enumeration requirement.

4. THE ECONOMIC VALUE OF PRESERVATION

THEOREM 2—Minimum revision cost in a certified class: *Fix nonempty action sets (10). Put $D_T = 0$ and*

$$D_t(s) = \min_{a \in \mathcal{B}_t(s; e)} \left\{ w_t(s, a) \mathbf{1}\{a \neq \pi_t^0(s)\} + \beta \int D_{t+1}(s') P_t(ds' | s, a) \right\}. \quad (12)$$

A measurable minimizing policy satisfies the uniform regret and preservation bounds of Proposition 1 and minimizes the discounted intervention cost from every restart among policies selecting from $\mathcal{B}(e)$. Randomization and history dependence do not improve the minimum within this class.

PROOF: Every eligible action satisfies the two comparisons in Proposition 1; the chosen policy therefore inherits those guarantees. Conditional on a state and admissible action, the induction hypothesis lower-bounds the remaining intervention cost by D_{t+1} . Taking the minimum

proves that every eligible policy has cost at least D_t . A minimizing selector attains equality. A randomized action forms a convex combination of the displayed alternatives and cannot fall below their minimum. Conditional application of the same induction covers history-dependent policies whose actions remain in the fixed admissible sets. *Q.E.D.*

The conclusion is deliberately about $\mathcal{B}(e)$. A policy excluded by a conservative witness can still meet the true regret requirement. Theorem 2 does not claim otherwise. Likewise, the earlier observed edit-count comparison between two completion rules is not a theorem that one rule changes fewer actions on every input. The new optimization objective supplies the missing preference over policies, rather than strengthening that empirical comparison by assertion.

Budgets can be designed as economic choices. For a finite, prespecified collection $\mathcal{E}$ of state/date-dependent profiles satisfying $0 \leq e_t(s) \leq \varepsilon$, discard profiles with empty admissible sets and solve (12) for each remaining profile. Choosing the profile minimizing $\int D_0 d\nu$ gives the least intervention cost in the union of these certified classes. This follows by taking the minimum of the classwise minima; it is not a polynomial-time claim for unrestricted budget design. Regional weights can encode priority or contractual constraints, and prohibited revisions can be represented by removing actions before optimization. The numerical experiment fixes a common tolerance before execution and reports priority-weighted costs; it does not claim to have optimized unrestricted state-specific budgets.

For a monetary shadow price $\lambda \geq 0$, compare policies using $J_\nu^\pi - \lambda C_\nu(\pi; \pi^0)$. If a more accurate policy π^{ref} has additional intervention cost $\Delta C > 0$ and operating-payoff advantage ΔJ , the preserved policy is preferred precisely when $\lambda > \Delta J / \Delta C$. This is a stated economic tradeoff. It does not value a CPU second in units of utility, and it does not infer economic value from Hamming distance.

5. AN ADAPTIVE CERTIFICATE WITHOUT STATE ENUMERATION

Consider the following operating-mode economy. At each of eight dates an active firm observes $x \in [0, 1]$ and chooses $a \in \{0, 1\}$. Its reward is

$$r(x, a) = x/5 + af(x). \quad (13)$$

Survival is 9/10, discounting is 19/20, and, conditional on survival, the next state has density $2x$, independently of the action. Settlement is zero. Hence $\gamma = 171/200$ and

$$H_n = \sum_{j=0}^{n-1} \gamma^j, \quad H = H_8 = \frac{63065975868557291}{12800000000000000}, \quad \eta = \frac{.01}{H}. \quad (14)$$

An action's local operating loss is $d_a(x) = \max(0, f(x)) - af(x)$. Continuations cancel from action comparisons, so the optimal action is a sign decision on f . Every policy with $d_{\pi_t}(x) \leq \eta$ satisfies regret at most $H_{8-t}\eta \leq .01$ at every state and restart. This exact cancellation is available to the classical comparator as well as to the certificate.

An installed binary policy is represented by constant-action intervals and explicit actions at their endpoints. A one-hidden-layer ReLU score can be compiled into this representation exactly: save its binary64 coefficients as rational constants; split at every activation kink; on each resulting affine piece split again at any score zero. The score's sign is then constant on each open piece. Evaluation at every breakpoint fixes the tie action. This certifies the compiled rule, not a hardware floating-point forward pass.

For a polynomial f on $I = [l, r]$, convert its power coefficients to Bernstein coefficients. Their minimum and maximum enclose the polynomial throughout I . On a raw-action interval let

$\underline{d}, \bar{d}$ bound $(1 - 2a)f$. Retain a if $\bar{d} \leq \eta$; switch if $\underline{d} > \eta$; otherwise bisect. At the prescribed deep-refinement limit, switching is permitted only if $\underline{d} \geq 0$. Every new endpoint is also evaluated exactly. Failure to resolve a cell at the resource cap raises an error rather than a certificate.

PROPOSITION 3—Accuracy, improvement, and a cost enclosure: *Every completed cover produced by this rule has uniform regret at most .01 and weakly improves the raw operating payoff at every restart. Let the initial state have density $2x$, and let the revision priority be $w(x) = 1 + x$. Write F for the union of cells with $\underline{d} > \eta$, and E for the union of all changed cells. Among policies with local loss at most η ,*

$$H \int_F 2x(1 + x) dx \leq C_{\text{local}}^* \leq C(\pi; \pi^0) = H \int_E 2x(1 + x) dx. \quad (15)$$

The same conclusions hold on every midpoint grid with its own transition distribution; the displayed integrals describe the continuous density, not an uncomputed discrete-grid occupancy measure.

PROOF: On a retained cell the raw deficit is at most η . On a changed cell $(1 - 2a)f \geq 0$, so switching makes the decision optimal and weakly increases current reward. Both comparisons hold at the explicitly evaluated endpoints. Because transitions do not depend on actions, backward summation yields the accuracy and improvement claims. Every locally admissible policy must change the raw action throughout F ; all points of E are changed by the reported policy. The weights are nonnegative and absorption contributes the factor H . This proves (15). Endpoints have zero continuous occupancy but are not omitted from the uniform certificate. The pointwise local comparisons also apply to each finite grid, whose transition probabilities sum to one. *Q.E.D.*

PROPOSITION 4—Geometry-dependent work: *Suppose a hierarchical regional verifier visits at most $K2^{s_\ell}$ unresolved cells at level ℓ , terminates by level L , and bounds each visited cell at cost at most c_ℓ . Including a binary split's resolved children, its range-oracle work is*

$$O\left(K \sum_{\ell=0}^L 2^{s_\ell} c_\ell\right). \quad (16)$$

Compilation, witness construction, integration of occupancy weights, serialization, and final verification are additional costs. For a fixed-degree univariate polynomial, simple isolated threshold crossings, and a piecewise constant candidate with finitely many pieces, $s = 0$ applies after a finite initial refinement, giving $O(KL)$ fixed-degree range operations, independently of a later grid cardinality N .

PROOF: Each unresolved cell has two children, so a constant multiple of the sum of unresolved-cell counts bounds all visits. Multiplying by each level's bound cost proves the first assertion. In one dimension, simple crossings have a nonzero derivative in disjoint small neighborhoods. For a fixed polynomial, the Bernstein enclosure differs from the endpoint range by $O(|I|^2)$ on such neighborhoods; its first-order variation is $O(|I|)$. Thus only a bounded number of dyadic cells per crossing can straddle the threshold at a given sufficiently fine level. Away from these neighborhoods a positive margin eventually decides the comparison. Candidate breakpoints add finitely many initial pieces. Summation proves the stated specialization. *Q.E.D.*

Multiple or flat threshold crossings require their own count; the proposition does not silently assume simple crossings in every model. The implemented finite cover is verified directly, whether or not the asymptotic specialization is invoked. In higher dimensions a codimension-one unresolved surface can have $s = d - 1$, so the result is not a dimension-free complexity claim. For a degree- p univariate polynomial the implemented power-to-Bernstein transformation takes $O(p^2)$ rational operations; it is not a free oracle. The one-hidden-layer compilation uses $O(w^2)$ rational operations for width w and at most $O(w)$ sign/kink intervals.

Exact arithmetic has a further cost. If all primitive rational denominators divide a common Q , backward values on a finite horizon have denominators dividing $Q^{2(T-t)+1}$: multiplication by a transition and a discount contributes at most two powers per stage. With bounded rewards, numerator and denominator bit lengths are $O(T \log Q + \log((T+1)R))$, where R bounds the payoff scale. For M separate b -bit primitive denominators the coarse bound $\log Q \leq Mb$ gives $O(TMb + \log((T+1)R))$. Rational normalization and arithmetic on these growing integers must be charged in addition to the $O(TNAD)$ enumeration count. The new records report actual integer bit lengths and serialized certificate bytes.

No reduction in information is implicit. With completely unstructured rewards, an uninspected state-action entry can be changed while preserving all inspected entries and invalidating an all-state guarantee. The positive regional result escapes that ambiguity because a verified structural bound covers the unenumerated entries. It does not infer them from training accuracy.

6. NUMERICAL EVIDENCE

6.1. Design and candidate freezing

The new study fixes three gain functions before execution:

$$f_1(x) = x - 3/8, \quad f_2(x) = 16(x - 1/5)(x - 1/2)(x - 4/5), \quad f_3(x) = 2(x - .33)^2 - .04. \quad (17)$$

Each has 20 neural candidates: widths eight and sixteen, Adam and L-BFGS, and five seeds per architecture/optimizer cell. Training uses 512 fixed midpoint observations of the primitive reward difference, not optimal-value labels. Adam uses 200 steps at rate .02; L-BFGS permits 200 iterations with strong-Wolfe search. Every candidate is saved and hashed before its certification and diagnostic comparison. The entire study is exploratory and informed by the reports. Freezing prevents subsequent seed selection; it does not make the accumulated research history independent or turn the 60 candidates into 60 independent economic models.

All parameters, raw policies, covers, exact metrics, failures, and source hashes are retained. A polynomial least-squares comparator uses the known degree and the same 512 observations. Its compiled dyadic policy is identified separately from an exact polynomial-sign oracle. Constant proposals and a structure-aware direct solver use the same Bernstein bounds. The direct solver at tolerance 10^{-12} is called *near-exact*, not exact. Vectorized full-grid dynamic programming supplies an additional binary64 diagnostic at $N = 2^8, 2^{12}, 2^{20}$; its arithmetic is not promoted to an exact certificate. Deeper rollout and policy iteration cannot improve the sign-based action in this action-independent-transition model: all continuation terms cancel for every lookahead depth. This is an analytic strong-baseline comparison in this family, not a new rollout experiment on the inherited inventory model.

The raw-to-certificate distinction matters. Raw regret is bounded independently by exact polynomial integration and range bounds; it is not inferred from the completed policy. In this model a stationary raw policy with loss $d(x)$ has worst restart-zero regret

$$\sup_x d(x) + (H - 1) \int_0^1 2x d(x) dx. \quad (18)$$

TABLE I
FROZEN NEURAL CANDIDATES: RAW ACCURACY AND EXACT PROOF OBJECTS

Gain	Raw passes	Median cells	Bound calls	Max bits	Certificate KiB
Linear	19/20	4	2–80	203	0.917–29.79
Cubic	0/20	112	83–286	624	47.83–132.6
Quadratic	7/20	77	6–152	415	3.056–73.23

Raw passes require an independently enclosed worst-state, all-restart regret no greater than .01. Counts are not independent economic replications. Certificate size includes exact rational endpoints and proof bounds.

Later restart horizons are shorter. This formula supplies both lower and upper raw-regret bounds. A failed raw candidate is retained even when its final cover certifies. A universal completion pass rate is a consistency check, not the efficacy statistic.

6.2. Work, baselines, and amortization

TABLE II
OBSERVED CONSTRUCTION WORK IN SECONDS

Gain	Fit	Compile	Cert.+check	Total	Structural	Grid DP
Linear	0.04639	0.001201	0.0004445	0.04901	0.002272	0.2178
Cubic	0.06541	0.001625	0.0521	0.1156	0.01557	0.04955
Quadratic	0.06373	0.002	0.02132	0.08557	0.006683	0.03602

Each entry is descriptive single-environment timing. Total is the median of each candidate’s full sum, not a sum of separately rounded medians. Structural is the matched-tolerance Bernstein solver; it is not the exact symbolic sign rule. Grid DP materializes 2^{20} states and uses binary64. All methods have the same model primitives.

The certificate covers the continuum, and therefore also $2^{40} = 1,099,511,627,776$ midpoint states without constructing an array of that size. We did not run a trillion-state dynamic program. The exact integer-index deployment routine compares a queried rational state with certificate breakpoints, including ties. Its guarantee concerns arbitrary-precision rational execution. The proof does not establish the behavior of a hardware neural actor.

An even stronger classical control is available here: the symbolic rule $a^*(x) = \mathbf{1}\{f(x) \geq 0\}$ is exactly optimal by the action-independent continuation identity. It needs no state enumeration or root isolation. Given a raw decision, an online guard that retains it exactly when $(1 - 2a)f(x) \leq \eta$ attains the minimum intervention cost in the local-loss class, with one polynomial comparison per query. Its additional revision cost is zero; our finite cover trades that online comparison for an offline proof object and has the small extra-cost enclosure reported below. We implement and test both controls. Their existence rules out interpreting this family as a demonstrated neural-specific solution advantage. The mathematical contribution is the general certified-class formulation and the explicit proof/representation tradeoff, not the invention of this classical sign rule.

The direct structural comparator exploits the same cancellation of continuation values and remains cheaper than neural fitting plus certification in many comparisons. This is an important control, not an inconvenient baseline to omit. For a genuinely preinstalled reusable proposal, a break-even number of identical certification jobs can be defined by

$$K > \frac{C_{\text{fit}}}{C_{\text{structural}} - (C_{\text{compile}} + C_{\text{certify}} + C_{\text{check}})}, \quad (19)$$

only when the denominator is positive. This is conditional accounting, not evidence that such reuse occurred. The example proves sub-enumerative feasibility and quantifies when a given proposal simplifies a cover; it does not establish a universal neural wall-time advantage.

6.3. Economically weighted policy changes

TABLE III
DISCOUNTED ECONOMIC INTERVENTION METRICS

Gain	Occupancy edits	Priority cost	Operating gain	Extra-cost upper
Linear	0	0	0	$4E - 12$
Cubic	2.382	3.783	0.2736	$2.28E - 10$
Quadratic	0.1387	0.1934	0.001208	$8.4E - 11$

First three numerical columns are medians across the 20 neural candidates. The last is an outward-rounded upper bound on the largest observed upper-minus-lower cost enclosure in the stated local-loss class. All quantities concern the continuous reset density.

The largest cost-enclosure gap is at most $2.28E-10$. All three polynomial proposals meet the raw .01 criterion; their details are in the computational appendix. Among the neural comparisons with positive saved intervention cost, the largest break-even shadow price is below 0.000739. At the illustrative price $\lambda = 1/20$, 60 of 60 exact paired comparisons favor the completed policy over the near-exact structural policy after intervention cost. This is an accounting comparison, not a statistical replication count. We report discounted occupancy changes, priority-weighted changes, and the exact operating-payoff gain over the raw policy. The lower bound in (15) identifies changes that any policy in the local-loss class must make. The difference between upper and lower cost bounds measures conservative extra intervention near unresolved tolerance crossings. Thus numerical minimality is assessed against a stated class and a stated cost, not against unweighted Hamming distance.

The near-exact structural comparator starts from the same installed policy when intervention costs are assessed. A positive net advantage at the illustrative shadow price is a deterministic economic comparison of these two deployments. It is not a statistical treatment effect, a neural-specific theorem, or an estimate of the value of computational time. The threshold $\Delta J/\Delta C$ is reported so that the economic judgment need not be hidden in a chosen price.

6.4. The retained exhaustive benchmark

The old candidate failures, polynomial successes, exact-DP timings, action topologies, and bit-length records are unchanged. The ratios in Table IV are computed from the displayed R29 stage times; their precision should not be read as greater than that of those inputs. The new operating-mode family changes the structural setting, not those adverse measurements. In particular, it does not supply multiple-seed high-dimensional inventory evidence that was absent from R29.

7. AN ECONOMICALLY ACTIVE STOPPING BOUNDARY

Fix the original primitives, $k = 2$, $c = 3/4$, $p = 0$, $x_0 = 5/4$, and constant $\theta \in [-1/5, 1/5]$, but restart preference at $u_0 = 61/50$. Wealth is deterministic and remains above its lower

TABLE IV
UNCHANGED R29 EXHAUSTIVE BENCHMARK AND TOTAL-WORK RATIOS

d/L	States	DP s	Repair+check s	Ratio	Full s	Full/DP
2/4	16	0.001072	0.003879	3.618	–	–
3/4	64	0.00803	0.01485	1.849	–	–
4/4	256	0.03729	0.0654	1.754	2.356	63.19
4/5	625	0.103	0.1841	1.787	3.464	33.63
5/4	1,024	0.2064	0.3054	1.48	4.729	22.91
6/4	4,096	1.103	1.484	1.345	14.81	13.43
7/4	16,384	5.495	8.345	1.519	59.95	10.91

Full includes new fitting, budget repair and verification only. Unmeasured or unavailable costs are not silently added as zero. These ratios use the displayed inherited stage times.

boundary through time one. Put $a = 6/5$, $b = 14/5$, $\sigma = 1/20$, and $d = u_0 - a = 1/50$. The lower-boundary hitting probability is

$$p_a(\theta) = \Phi\left(-\frac{d+\theta}{\sigma}\right) + e^{-2\theta d/\sigma^2} \Phi\left(\frac{-d+\theta}{\sigma}\right). \quad (20)$$

An additional upper-boundary event has probability at most

$$p_b \leq 2 \exp\left[-\frac{(b - u_0 - .2)^2}{2\sigma^2}\right] < 7.709 \cdot 10^{-166}. \quad (21)$$

The lower event is retained exactly; only the distant opposite boundary is bounded as a remainder.

Let $h_\theta = (\partial_t + \mathcal{L}_\theta - .04)g + \ell_2$. Stopped Itô's formula gives

$$J_\theta = g(0, u_0, x_0) + \int_0^1 e^{-.04t} \mathbb{E}_\theta[h_\theta(t, u_t) \mathbf{1}\{t < \tau\}] dt. \quad (22)$$

The half-line killed transition density is

$$p_\theta(t, u_0, y) = \frac{e^{-(y-u_0-\theta t)^2/(2\sigma^2 t)}}{\sigma\sqrt{2\pi t}} \left\{ 1 - e^{-2d(y-a)/(\sigma^2 t)} \right\}, \quad y > a. \quad (23)$$

Its parameter scores are $S_t = (y - u_0 - \theta t)/\sigma^2$ and $S_t^2 = t/\sigma^2$. Consequently the first two integrands are $S_t h_\theta + \partial_\theta h_\theta$ and $(S_t^2 - t/\sigma^2)h_\theta + 2S_t \partial_\theta h_\theta + \partial_\theta^2 h_\theta$. Bounded-domain payoff terms and Gaussian score moments justify differentiation; the supplementary construction gives explicit remainder constants.

The positive curvature near zero and negative curvature near the upper adjustment bound are both separated from zero by their enclosures. Thus the strictly concave central-state scalar objective is not a valid shape assumption near the active stopping boundary. At the five tested controls the first derivative is positive, but five pointwise derivative signs do not prove monotonicity over the entire interval or locate its global maximum. This distinction prevents another form of sample-to-domain extrapolation.

7.1. Verified search and retained mechanisms

The inherited central-state oracle admits a strict-concavity audit. We compare derivative bracketing and the prescribed polling schedule using that identical oracle and explicitly charge construction and the concavity audit. The box-poll theorem remains a generic projected-stationarity

TABLE V
UNCHANGED ORIGINAL ECONOMY AT THE ACTIVE-BOUNDARY RESTART

θ	Exit probability	J_θ	J'_θ	J''_θ
-0.2	[0.99997, 0.99998]	[-7.6652, -7.6650]	[1.7017, 1.7019]	[16.9990, 16.9992]
-0.1	[0.98580, 0.98581]	[-7.3389, -7.3387]	[5.9044, 5.9046]	[82.1152, 82.1154]
0	[0.68915, 0.68916]	[-6.2320, -6.2318]	[15.9905, 15.9907]	[60.6513, 60.6515]
0.1	[0.19903, 0.19904]	[-4.6662, -4.6660]	[12.7887, 12.7889]	[-83.8317, -83.8315]
0.2	[0.04076, 0.04077]	[-3.7609, -3.7607]	[6.0310, 6.0312]	[-45.1336, -45.1334]

All endpoints are rounded outward from validated balls. This is one restart and a fixed constant-control class, not the original whole-domain target. The opposite-boundary and time-truncation errors are included.

TABLE VI
THE SAME VALIDATED CENTRAL-STATE ORACLE UNDER TWO SEARCH PROCEDURES

k	Setup+audit s	Bracket calls	Poll calls	Bracket s	Poll s	Bracket bound	Poll bound
1/2	29.6	2	23	0.02666	0.4102	0	0.005
2	29.19	8	31	0.1074	0.5193	0.00371	0.007763

The bracketing target is .005. The inherited poll meshes have the achieved bound displayed here, not an artificially matched value. Both methods require the same oracle construction and concavity audit.

certificate. Its worst-case call bound is not presented as a prediction of the small scalar experiment, and the new active-boundary calculation does not inherit the central-state concavity assumption.

The historical Adam transport and interval-gated rollback results are reproduced in the historical annex. The transport comparison is an independently implemented recurrence with the same carrier and Adam state (Kingma and Ba, 2015); it is not a cheaper alternative optimizer. The retained nonlinear remainder of roughly 1.39–1.50 explains why the earlier linearized control was inadequate. Rollback verifies safe acceptance and exact restoration, but the inherited paired final payoffs do not establish a positive net welfare improvement. Neither observation is used as evidence for the new regional complexity theorem or the economic intervention-cost result.

8. CONCLUSION

Uniform accuracy and the economics of revising an installed policy are distinct requirements. Regional Bellman inequalities can certify a policy without enumerating every state when validated model structure supports whole-region bounds. A second Bellman recursion assigns preservation an explicit occupancy-weighted objective and minimizes that objective in the certified class. The implemented operating-mode economy supplies exact continuous-domain covers, raw-policy error bounds, a near-minimum intervention-cost enclosure, and strong information-matched classical controls. It does not establish neural exclusivity or eliminate the cost of training and witness construction.

The original stopped economy remains a demanding test of these ideas. The new killed-kernel calculation rigorously instantiates material stopping and reveals nonconcavity absent from the central example. The original current-state, all-domain .01 certificate and the full 47-coordinate derivative/residual bridge remain separate, uncompleted numerical requirements. Retaining them explicitly preserves the scientific target without attributing to a finite model or a scalar calculation a result it has not proved.

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SUPPLEMENT TO CERTIFIED BELLMAN OPERATORS WITH NEURAL PROPOSALS

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1. OBJECTS, ARITHMETIC, AND PRESERVATION

This supplement supplies the analytic construction behind the active-boundary calculation and the detailed numerical records for the regional certificate. The five R29 documents are reproduced, without page alteration, in `HISTORY_R30.pdf`. The original source, exact-rational inventory records, controls, failures, and preceding responses remain in their original directories. Historical statements are to be read with the current response and scope statements, not as an alternative set of current claims.

Three arithmetic objects must remain distinct. Training uses floating point. Compilation interprets the saved binary64 coefficients as exact rational constants and produces an exact piecewise-constant decision rule with explicit tie points. The certificate uses rational polynomial bounds, comparisons, and integrals. An actual floating-point neural forward pass is not certified. The integer-index deployment test covers arbitrary-precision rational comparisons, not unmodeled hardware rounding.

A candidate's saved parameters, compiled intervals, and final cover are linked in the verifier. Every certificate leaf must lie within a compiled constant-action interval and carry the correct raw decision. Every breakpoint is checked against exact network evaluation. Coverage, adjacency, feasible actions, bound coefficients, endpoint actions, and the final tolerance are verified independently of the training loss. Removing an endpoint, altering a bound, making the cover incomplete, supplying a negative tolerance, or querying an invalid grid index is rejected.

2. STOPPED GAUSSIAN MOMENTS AND ANALYTIC ERRORS

Let $a = 6/5$, $b = 14/5$, $u_0 = 61/50$, $\sigma = 1/20$, and $c = 3/4$. Deterministic wealth is

$$x(t) = \left(\frac{5}{4} - \frac{3/4}{.02} \right) e^{.02t} + \frac{3/4}{.02}. \quad (1)$$

It decreases to a value above .51770, so there is no wealth exit in this restricted class. The original preference boundaries and settlement are unchanged. Put $q = y - 2$, so $|q| \leq 4/5$ on $[a, b]$, and $L = \log(4/3)$. The utility term is

$$\frac{c^{1-y}}{1-y} = -\frac{e^{L(1+q)}}{1+q} = \sum_{n=0}^{\infty} c_n q^n, \quad c_n = -\frac{4}{3} \sum_{j=0}^n \frac{L^j}{j!} (-1)^{n-j}. \quad (2)$$

Since $|c_n| \leq e^{2L} = 16/9$, truncation after degree 128 has the uniform bound

$$R_{128} \leq \frac{16}{9} \frac{(4/5)^{129}}{1 - 4/5} < 2.803 \cdot 10^{-12}. \quad (3)$$

All coefficients are enclosed by ball arithmetic. The remainder is an analytic uniform bound, not an interpolation residual.

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The half-line killed kernel equals a difference of two Gaussian densities, with means $m_1 = u_0 + \theta t$ and $m_2 = 2a - u_0 + \theta t$, common variance $v = \sigma^2 t$, and image multiplier $e^{-2\theta(u_0 - a)/\sigma^2}$. For a Gaussian density $\varphi_{m,v}$ define

$$I_n(m, v) = \int_a^b (y - 2)^n \varphi_{m,v}(y) dy. \quad (4)$$

The initial moment is a normal-CDF difference. Integration by parts gives

$$I_1 = (m - 2)I_0 + v\{\varphi_{m,v}(a) - \varphi_{m,v}(b)\}, \quad (5)$$

$$I_n = (m - 2)I_{n-1} + (n - 1)vI_{n-2} + v\{(a - 2)^{n-1}\varphi_{m,v}(a) - (b - 2)^{n-1}\varphi_{m,v}(b)\}, \quad n \geq 2. \quad (6)$$

Their image-weighted difference integrates the utility polynomial and the polynomial components of the Dynkin integrand exactly up to arithmetic enclosure. Moments through degree 130 suffice for two parameter derivatives.

In the fixed $k = 2$ slice,

$$h_\theta(t, y) = \frac{c^{1-y}}{1-y} - \theta^2 + 8 - .04\theta(y - 2) - .00005 + .002 - \frac{c}{10x(t)} \quad (7)$$

$$- .04\{-.02(y - 2)^2 + .1 \log x(t)\} + .32(1 - t),$$

$$\partial_\theta h_\theta = -2\theta - .04(y - 2), \quad \partial_\theta^2 h_\theta = -2. \quad (8)$$

On the domain, the deliberately conservative bounds $|h| \leq M = 20$, $|h'| \leq M_1 = 2$, and $|h''| \leq M_2 = 8$ hold. For example, the utility magnitude is below 5.3, wealth is above .5177, and the remaining displayed terms fit strictly within these bounds.

The time interval $[0, t_0]$, with $t_0 = 10^{-6}$, is bounded rather than numerically integrated. Under the full Gaussian law,

$$\mathbb{E}|S_t| \leq \frac{\sqrt{t}}{\sigma}, \quad \mathbb{E}\left|S_t^2 - \frac{t}{\sigma^2}\right| \leq \frac{\sqrt{2}t}{\sigma^2}. \quad (9)$$

Restricting to survival cannot increase these absolute-moment bounds. Thus the three omitted-time errors are at most

$$E_0^{\text{time}} = Mt_0, \quad (10)$$

$$E_1^{\text{time}} = M_1 t_0 + \frac{2M}{3\sigma} t_0^{3/2}, \quad (11)$$

$$E_2^{\text{time}} = M_2 t_0 + \frac{4M_1}{3\sigma} t_0^{3/2} + \frac{M\sqrt{2}}{2\sigma^2} t_0^2. \quad (12)$$

The uniform utility approximation contributes at most R_{128} , $2R_{128}/(3\sigma)$, and $R_{128}\sqrt{2}/(2\sigma^2)$, respectively.

The difference between the true doubly stopped occupation measure and the lower-killed measure restricted to $[a, b]$ occurs only on a path that has already hit the upper boundary. With the probability bound p_b in the article, Cauchy–Schwarz and the likelihood-score second moments

yield opposite-boundary errors

$$E_0^{\text{opp}} = Mp_b, \quad (13)$$

$$E_1^{\text{opp}} = M\sqrt{p_b}/\sigma + M_1p_b, \quad (14)$$

$$E_2^{\text{opp}} = M\sqrt{2p_b}/\sigma^2 + 2M_1\sqrt{p_b}/\sigma + M_2p_b. \quad (15)$$

This treatment does not discard the economically material lower-boundary event. It only encloses the distant upper-boundary correction. Adding these errors to the ball-integrated polynomial expression produces the reported derivative enclosures.

The remaining integration uses $s = \sqrt{t}$, 200-bit arithmetic, and the fixed breakpoints $s \in \{.001, .01, .05, .1, .2, .4, .8, 1\}$. The complex analytic extension passes holomorphicity checks to square root and logarithm; a ball crossing an unverified branch domain returns nonfinite rather than a false bound. The FLINT integration routine supplies a rigorous integral enclosure; its requested decimal tolerance is not itself treated as the achieved error bound. Final balls are tested for finiteness, reality, and radius below .005. Their actual radii are much smaller. The first attempted nested spatial/time ball integral failed; its source and output remain in `results/stopping/attempts`. The successful moment formulation changes the implementation, not the instance or target controls.

3. EXACT COST ACCOUNTING

For a raw-action interval with local deficit bounds $[l_d, u_d]$, the mandatory-change lower bound includes it only when $l_d > \eta$. The reported upper bound includes every changed interval. Their difference therefore bounds the intervention cost in excess of the minimum local-loss-class cost even when the subdivision stops conservatively near a tolerance crossing. The online exact guard attains the same local-class optimum pointwise without constructing a finite cover. The finite cover provides an offline certificate and an alternative deployment representation, not a lower intervention cost than that guard.

Under the continuous reset density, interval occupancy is $\int_l^r 2x dx$ and priority-weighted occupancy is $\int_l^r 2x(1+x) dx$. Multiplication by H_8 accounts for survival and discounting. Operating gains are integrated from $(a^{\text{new}} - a^0)f(x)$ against that same density. An independently written interval-overlay routine recomputes the policy-change costs and the comparison to the near-exact structural policy. No finite-grid occupancy estimate is substituted for these continuous integrals.

4. DETAILED EVIDENCE

TABLE I: All neural candidates; A = Adam, B = L-BFGS

Gain	Method/width/seed	Raw regret upper	Cells	Calls	Occupancy	Priority	Full s
linear	A/8/1	0.000039	3	3	0	0	0.9121
linear	A/8/2	0.012663	41	77	0.03773	0.0516	0.07108
linear	A/8/3	0.000024	3	3	0	0	0.06399
linear	A/8/4	0.000783	4	4	0	0	0.06383
linear	A/8/5	0.000007	4	4	0	0	0.06447
linear	B/8/1	0.000003	3	3	0	0	0.02075
linear	B/8/2	0.000009	5	5	0	0	0.0221

Gain	Method/width/seed	Raw regret upper	Cells	Calls	Occupancy	Priority	Full s
linear	B/8/3	0.000001	3	3	0	0	0.01558
linear	B/8/4	0.000006	2	2	0	0	0.008753
linear	B/8/5	0.000002	3	3	0	0	0.01174
linear	A/16/1	0.000167	4	4	0	0	0.06609
linear	A/16/2	0.004521	41	77	0.009016	0.01237	0.07411
linear	A/16/3	0.000006	2	2	0	0	0.06553
linear	A/16/4	0.000024	3	3	0	0	0.06516
linear	A/16/5	0.002148	44	80	0.0004114	0.0005666	0.07409
linear	B/16/1	0.000002	6	6	0	0	0.02501
linear	B/16/2	0.000025	6	6	0	0	0.02479
linear	B/16/3	0.000257	6	6	0	0	0.009412
linear	B/16/4	0.000008	5	5	0	0	0.01304
linear	B/16/5	0.000001	7	7	0	0	0.03418
cubic	A/8/1	0.420749	144	286	2.935	4.579	0.127
cubic	A/8/2	0.071483	115	223	0.2479	0.3731	0.1147
cubic	A/8/3	0.384982	110	217	2.387	3.79	0.113
cubic	A/8/4	0.384866	110	217	2.386	3.789	0.1169
cubic	A/8/5	0.420749	144	286	2.935	4.579	0.1844
cubic	B/8/1	0.420749	144	286	2.935	4.579	0.06817
cubic	B/8/2	0.030503	80	152	0.09845	0.1649	0.1278
cubic	B/8/3	0.385395	110	217	2.39	3.795	0.06425
cubic	B/8/4	0.385395	110	217	2.39	3.795	0.06486
cubic	B/8/5	0.385392	110	217	2.39	3.795	0.06136
cubic	A/16/1	0.384537	111	218	2.383	3.785	0.1165
cubic	A/16/2	0.403565	110	217	2.562	4.034	0.1108
cubic	A/16/3	0.384208	112	219	2.38	3.781	0.1208
cubic	A/16/4	0.045918	115	223	0.1029	0.1659	0.1229
cubic	A/16/5	0.018225	47	83	0.04472	0.08034	0.09018
cubic	B/16/1	0.032401	116	224	0.1247	0.1975	0.1729
cubic	B/16/2	0.032752	115	223	0.1425	0.2324	0.09924
cubic	B/16/3	0.383961	112	219	2.378	3.779	0.09589
cubic	B/16/4	0.032828	115	223	0.1442	0.2349	0.1377
cubic	B/16/5	0.029961	117	225	0.09242	0.1585	0.1801
quadratic	A/8/1	0.059549	73	144	0.8961	1.209	0.08238
quadratic	A/8/2	0.001982	6	6	0	0	0.06422
quadratic	A/8/3	0.059549	74	145	0.8961	1.209	0.08016
quadratic	A/8/4	0.058462	77	149	0.5674	0.7913	0.08809
quadratic	A/8/5	0.059549	74	146	0.8961	1.209	0.08306
quadratic	B/8/1	0.059549	73	144	0.8961	1.209	0.02752
quadratic	B/8/2	0.014738	77	149	0.1417	0.1945	0.05122
quadratic	B/8/3	0.010987	78	150	0.07094	0.104	0.0445
quadratic	B/8/4	0.015021	77	149	0.1403	0.1922	0.0612
quadratic	B/8/5	0.203274	74	146	0.9891	1.307	0.02883
quadratic	A/16/1	0.005034	78	150	0.02653	0.03837	0.08954
quadratic	A/16/2	0.059175	78	150	0.8797	1.186	0.08849
quadratic	A/16/3	0.046384	77	149	0.4816	0.6568	0.0917
quadratic	A/16/4	0.004746	78	150	0.01232	0.01552	0.09207
quadratic	A/16/5	0.024038	77	149	0.1023	0.1259	0.08828
quadratic	B/16/1	0.006289	79	151	0.03104	0.04194	0.134
quadratic	B/16/2	0.005631	79	151	0.02927	0.03989	0.1475
quadratic	B/16/3	0.005626	79	151	0.02757	0.0374	0.156
quadratic	B/16/4	0.021356	42	78	0.1371	0.2043	0.05929
quadratic	B/16/5	0.004861	80	152	0.02913	0.04057	0.2042

TABLE II
POLYNOMIAL CONTROLS WITH THE SAME OBSERVATIONS AND KNOWN DEGREE

Gain	Raw regret upper	Fit s	Compile s	Certify s	Cells
linear	0E-12	0.002705	0.001906	0.000161	2
cubic	1E-12	0.0001478	0.01762	0.0008435	4
quadratic	1E-12	0.0002166	0.006903	0.0002222	3

These are exact compiled dyadic policies from fitted polynomial coefficients. An irrational root is not asserted to equal its dyadic approximation.

5. SCOPE OF THE RETAINED CONTINUOUS AND HISTORICAL RESULTS

The scalar comparative statics, the finite inventory completion, the regional operating-mode certificate, and the original continuous all-domain objective have different policy classes and restart sets. The new active-boundary calculation establishes payoff and derivative enclosures for five constant controls at one restart. It neither supplies a global scalar optimizer nor validates all 47 financed coordinates. The original bound 7.181834580823298 is unchanged. The original supersolution/residual theorem and every earlier unsuccessful run remain available in the reproduced R29 documents.

The central-state bracketing comparison uses the inherited validated oracle without changing its error budget or strict-concavity audit. Its target is a projected-gradient upper bound .005; the inherited polling schedule reports its own achieved bound, which need not be identical. Setup and concavity-audit costs are charged before either search. The old direct-search theorem is retained as a sufficient certificate theorem, not as evidence that polling dominates root-based scalar methods. The historical transport identity and restoration tests are preserved in full but no longer serve as independent numerical-superiority claims.

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RESPONSE TO THE R29 ECONOMETRICA REFEREE REPORTS

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1. REVIEW OBJECTS AND SUBSTANTIVE REVISION

We address the independent second-pass report at review commit 3175974aed89dac1595af174a201dcae7bcfae31 and the first R29 report at 3f2724d3db622d26f6b1e439fc01311c4858ee7e. Both concern manuscript commit 788246778893695471015ce4db76e6a61a2c9ca0. This is a new R30 scientific object, not a relabeling of R29’s response to R27.

The revision adds regional certification, a minimum-intervention Bellman problem, exact weighted cost enclosures, a frozen new model/cohort, strong structural classical controls, a genuinely active stopping-boundary oracle, and a matched-oracle search comparison. The original whole-domain .01 target and the 47-dimensional derivative/residual bridge are not falsely marked complete. The entire R29 submission object and earlier history are preserved in HISTORY_R30. The item-by-item dispositions below distinguish new results, corrected interpretation, preserved audit material, and uncompleted numerical requirements.

2. INDEPENDENT SECOND-PASS REPORT

2.1. R29-SP-F1: Regional work and strongest classical baseline

Disposition: New conditional complexity theorem; no universal neural speed advantage. Sections 3–6 replace mandatory state enumeration by whole-region inequalities and a geometry-dependent work bound. The operating-mode example certifies the continuum, but its exact symbolic sign solver and online budget guard exploit the same structure. Their inclusion prevents a claim that neural fitting is necessary or generally fastest. The intervention-cost theorem supplies a separate economic objective. The inherited adverse total-cost ratios are retained.

2.2. R29-SP-F2: Universal completion pass rate

Disposition: Corrected interpretation. The abstract and numerical section no longer treat universal post-completion success as approximation efficacy. The raw-policy error bounds, fitting and proof costs, region counts, policy sizes, weighted intervention costs, and strong structural baselines are reported separately. Passing completed objects are implementation consistency checks.

2.3. R29-SP-F3: Raw accuracy at scale

Disposition: New raw evidence; inherited failures unchanged. The new frozen cohort reports raw continuous-domain regret for all 60 candidates, not only for selected seeds. Raw passes and failures are shown by model, width and optimizer. No new result reclassifies the inherited inventory failures. The operating-mode family is a separate one-dimensional structural example, not a rescue of the old high-dimensional accuracy claim.

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2.4. R29-SP-F4: Neural-specific value

Disposition: Neural exclusivity not established. The title identifies neural proposals and certified Bellman operators as different objects. Polynomial proposals, the exact symbolic sign solver, the online minimum-intervention guard, Bernstein-based structural solving and vectorized DP receive the same model information. Neural fitting is fully charged. The paper does not claim a method-specific speed advantage contradicted by these controls.

2.5. R29-SP-F5: Economic value and minimum intervention

Disposition: New theorem and exact cost enclosure. Section 4 defines discounted, policy-induced occupancy cost and proves a minimum-cost Bellman recursion within explicit certified action sets. Section 5 bounds the implemented cover cost above and below relative to the local-loss class optimum. Priority weights and exact operating gains replace Hamming distance as the economic comparison. The exact online guard is acknowledged to attain that class optimum; global optimality over every true-regret-feasible policy is not claimed.

2.6. R29-SP-F6: Reference-free terminology

Disposition: Corrected terminology and accounting. The current text uses precomputed-optimum-free and independently implemented exact reference. It expressly charges the complete model, own-policy evaluation, witness construction, range bounds, compilation and checking. Absence of an optimal-value file is not presented as a reduction in model information or Bellman work.

2.7. R29-SP-F7: Scalability and seed coverage

Disposition: New regional-cardinality result, not high-dimensional robustness. The new cover applies to a continuum and hence to midpoint grids with cardinality up to 2^{40} without materializing them. The example is explicitly one-dimensional. It uses five seeds, two widths and two optimizers for each of three gain functions. The inherited dimension-five through dimension-seven runs remain single-configuration pipeline demonstrations; no multidimensional neural robustness theorem is inferred.

2.8. R29-SP-F8: Original full-domain target

Disposition: Original target remains numerically unresolved. The original economy, current-state actor requirement, all-state/all-restart .01 target, and bound 7.181834580823298 remain explicit. The new active-boundary derivative computation is a substantive instantiation where stopping is material, but it does not reduce the inherited global bound or certify the unrestricted actor. The response does not mark this numerical requirement closed.

2.9. R29-SP-F9: Economically material stopping

Disposition: New validated active-boundary calculation. Section 7 moves the restart to preference 1.22 while retaining the original model primitives. The lower-boundary hitting probability at zero adjustment is rigorously enclosed near .6891565168. A killed kernel retains that boundary exactly, and validated Gaussian moments enclose payoff and two derivatives for all five predeclared controls. The failed first nested-integration attempt is retained; it is not promoted to a certificate.

2.10. R29-SP-F10: Polling versus verified optimization

Disposition: New derivative-bracketing comparator; 47-dimensional oracle uninstantiated. The inherited central-state polling schedule is compared with derivative bracketing using the identical validated oracle and concavity audit. Construction, audit, search calls and achieved projected-gradient bounds are separated. Bracketing requires fewer oracle calls in these scalar comparisons. The paper retains polling as a generic certificate wrapper and does not extrapolate it to an instantiated 47-dimensional solver.

2.11. R29-SP-F11: Restricted economic comparative static

Disposition: New boundary curvature result; restricted class stated. The active-boundary calculation yields strictly positive second-derivative intervals near zero adjustment and strictly negative ones at .1 and .2. Thus central-state concavity cannot be transferred to the boundary. Consumption and investment remain fixed in this calculation; it is not an unrestricted comparative static or a proof of interval-wide monotonicity from five points.

2.12. R29-SP-F12: Rollback benefit

Disposition: Preserved in audit material; no net-benefit claim. The complete R29 rollback evidence is reproduced in the historical annex. Its acceptance/restoration contract is retained, but the paired final outcomes are not presented as positive welfare or solver-performance evidence. This revision does not claim a new prospective avoided-loss experiment for the inherited production schedule.

2.13. R29-SP-F13: Historical transport identity

Disposition: Preserved as an implementation identity. The original full-recentering argument and numerical records remain in the historical annex. The main text identifies the same carrier, gradient, Jacobian and Adam history and highlights the roughly 1.39–1.50 omitted nonlinear displacement in the earlier control. Equivalence is not called an alternative optimizer or a computational saving.

2.14. R29-SP-F14: Prospective evidence and dependence

Disposition: Frozen new exploratory cohort; no independence claim. The protocol was committed before the new local scientific runs. All seeds, widths, optimizers and model functions were retained. The amendment records the failed nested integral and stronger post-diagnostic classical controls without changing the candidates. The study is explicitly report-informed and exploratory; neither 60 candidates nor repeated certificates are described as independent economic replications.

2.15. R29-SP-F15: Nearest-neighbor novelty

Disposition: Explicit theorem-level comparison. The introduction compares residual propagation, rollout/policy improvement, approximate policy iteration, baseline-bootstrapped safe improvement and verified numerics. Bellman comparison is identified as classical. The stated mathematical additions are the certified-class cost optimization, whole-region proof obligations and their geometry-dependent work accounting. The strong online guard is included rather than obscured by a novelty claim.

2.16. R29-SP-F16: Coherent paper and preservation

Disposition: Restructured main argument; no historical deletion. The main argument now follows regional certification, economically weighted revision, structural computation and active-boundary diagnosis. The complete prior article, supplement, response, computational appendix and historical annex are reproduced in HISTORY_R30. Original source and evidence directories are not overwritten. Transport and rollback no longer carry independent solver-superiority claims.

2.17. R29-SP-F17: Title and certified object

Disposition: Object-specific title. The title is Certified Bellman Operators with Neural Proposals: Adaptive Verification and Costly Policy Revision. It identifies both the approximation stage and the certified object without attributing a representation-independent guarantee or a repaired policy to a raw neural network.

2.18. R29-SP-T1: Zero-tolerance limit

Disposition: Proposition and exact tests. The zero-tolerance proposition states the finite exact zero-tolerance limit and proves optimality by backward induction. Retained optimal ties need not equal the replacement tie rule. Sixteen small exact policy cases are checked. No claim of zero-tolerance interval-subdivision termination at irrational roots is made.

2.19. R29-SP-T2: Total certified-solution ratios

Disposition: Main-table addition. The article displays both (repair+verification)/DP and (fit+repair+verification)/DP for every inherited row with the necessary recorded times. Missing fitting times are not imputed as zero. Ratios are derived from the displayed R29 inputs and not reported with spurious input precision.

2.20. R29-SP-T3: Edit count versus scan cost

Disposition: New algorithm and explicit distinction. The exhaustive algorithm still has full-scan cost regardless of edits. The new implementation certifies complete intervals, counts every range-oracle call and breakpoint, and separately charges compilation and checking. Its work depends on certificate geometry, not automatically on the number of edited actions.

2.21. R29-SP-T4: Weighted edit metrics

Disposition: Exact occupancy and payoff records. The new records include discounted occupancy edits, state-priority-weighted cost with weight $1+x$, exact raw-to-completed operating-payoff gains, independent overlay checks, and break-even intervention shadow prices. The displayed continuous-density integrals are not mislabeled as finite-grid occupancy estimates.

2.22. R29-SP-T5: Stronger rollout and improvement controls

Disposition: Analytic structural comparison; no new inventory rollout panel. In the new exogenous-transition family every continuation cancels from the action comparison, at every lookahead depth and after any policy-evaluation step. Exact symbolic optimization and the online cost-minimizing guard are therefore stronger controls than a shallow rollout. These are implemented. A new deeper-rollout panel for the inherited inventory family is not claimed.

2.23. R29-SP-T6: Optimized exact baseline

Disposition: Vectorized DP and exact structural controls. The new experiment includes full-grid vectorized backward recursion, explicitly labeled binary64 diagnostic arithmetic, and a genuinely exact symbolic structural policy with zero regret. Both use the same primitives as the neural certificate. The old scalar reference and its timings remain unchanged, and no arithmetic certificate is inferred from floating-point DP output.

2.24. R29-SP-T7: Rational bit complexity

Disposition: Asymptotic bound and observed bit sizes. Section 5 gives a common-denominator bound $Q^{2(T-t)+1}$, bit growth $O(T \log Q + \log((T+1)R))$, and the coarse primitive-denominator version. Rational arithmetic and normalization costs are distinguished from enumeration. Each new cover records its maximum integer bit length and serialized size.

2.25. R29-SP-T8: Principled budget choice

Disposition: Explicit economic design problem. Section 4 defines budget selection over a finite prespecified family: solve minimum revision cost in each nonempty certified class, then select the least-cost class. This is exactly optimal over that union, not an efficient solution to unrestricted budget design. The experiment fixes a common budget prospectively and does not claim a numerically optimized state-budget profile.

2.26. R29-SP-T9: Actual stopping boundary

Disposition: Validated boundary experiment. The same response and evidence as R29-SP-F9 apply: preference 1.22, material lower exit, killed-kernel derivatives, explicit tails and time truncation, and all five predeclared controls. The tiny probability bound pertains only to the opposite boundary.

2.27. R29-SP-T10: 47-dimensional bridge

Disposition: Not falsely closed. The full financed/current-state derivative and residual constants are not supplied by the scalar moment oracle. The original all-domain target and this missing instantiation remain explicit in the article, supplement and machine-readable disposition. No readiness test or passing finite certificate is substituted for the missing mathematical/computational object.

2.28. R29-SP-T11: Hardware deployment

Disposition: Exact compiled object; hardware neural rounding not certified. The compiler and integer-index evaluator use exact rational arithmetic and explicit tie points. Tests link every leaf and endpoint to the frozen mathematical network. The manuscript does not claim that arbitrary hardware floating-point evaluation of the network inherits that guarantee. An additional end-to-end rounding enclosure would be needed for that different deployment object.

2.29. R29-SP-T12: Independent implementation and information

Disposition: Terminology corrected. The reference is described as independently implemented, with the same model information. The new structural controls exploit all the same analytic identities and primitives. Software independence and information independence are not conflated.

3. FIRST R29 REPORT: COMPLETE ITEM CROSSWALK

The following crosswalk addresses every first-report finding, technical comment, presentation comment and numbered next-revision request. Each entry adopts the substantive response and qualification in the identified second-pass item; it does not declare an uncompleted requirement resolved.

- 3.0.0.1. *R29-F1*. Universal completion pass rate. See R29-SP-F2. Corrected interpretation.
- 3.0.0.2. *R29-F2*. Regional work and strongest classical baseline. See R29-SP-F1. New conditional complexity theorem; no universal neural speed advantage.
- 3.0.0.3. *R29-F3*. Raw accuracy at scale. See R29-SP-F3. New raw evidence; inherited failures unchanged.
- 3.0.0.4. *R29-F4*. Economic value and minimum intervention. See R29-SP-F5. New theorem and exact cost enclosure.
- 3.0.0.5. *R29-F5*. Neural-specific value. See R29-SP-F4. Neural exclusivity not established.
- 3.0.0.6. *R29-F6*. Scalability and seed coverage. See R29-SP-F7. New regional-cardinality result, not high-dimensional robustness.
- 3.0.0.7. *R29-F7*. Original full-domain target. See R29-SP-F8. Original target remains numerically unresolved.
- 3.0.0.8. *R29-F8*. Polling versus verified optimization. See R29-SP-F10. New derivative-bracketing comparator; 47-dimensional oracle uninstantiated.
- 3.0.0.9. *R29-F9*. Rollback benefit. See R29-SP-F12. Preserved in audit material; no net-benefit claim.
- 3.0.0.10. *R29-F10*. Historical transport identity. See R29-SP-F13. Preserved as an implementation identity.
- 3.0.0.11. *R29-F11*. Nearest-neighbor novelty. See R29-SP-F15. Explicit theorem-level comparison.
- 3.0.0.12. *R29-F12*. Reference-free terminology. See R29-SP-F6. Corrected terminology and accounting.
- 3.0.0.13. *R29-F13*. Prospective evidence and dependence. See R29-SP-F14. Frozen new exploratory cohort; no independence claim.
- 3.0.0.14. *R29-F14*. Scalability and seed coverage. See R29-SP-F7. New regional-cardinality result, not high-dimensional robustness.
- 3.0.0.15. *R29-F15*. Coherent paper and preservation. See R29-SP-F16. Restructured main argument; no historical deletion.

- 3.0.0.16. *R29-F16*. Title and certified object. See R29-SP-F17. Object-specific title.
- 3.0.0.17. *R29-T1*. Zero-tolerance limit. See R29-SP-T1. Proposition and exact tests.
- 3.0.0.18. *R29-T2*. Total certified-solution ratios. See R29-SP-T2. Main-table addition.
- 3.0.0.19. *R29-T3*. Edit count versus scan cost. See R29-SP-T3. New algorithm and explicit distinction.
- 3.0.0.20. *R29-T4*. Weighted edit metrics. See R29-SP-T4. Exact occupancy and payoff records.
- 3.0.0.21. *R29-T5*. Economic value and minimum intervention. See R29-SP-F5. New theorem and exact cost enclosure.
- 3.0.0.22. *R29-T6*. Stronger rollout and improvement controls. See R29-SP-T5. Analytic structural comparison; no new inventory rollout panel.
- 3.0.0.23. *R29-T7*. Independent implementation and information. See R29-SP-T12. Terminology corrected.
- 3.0.0.24. *R29-T8*. Optimized exact baseline. See R29-SP-T6. Vectorized DP and exact structural controls.
- 3.0.0.25. *R29-T9*. Scalability and seed coverage. See R29-SP-F7. New regional-cardinality result, not high-dimensional robustness.
- 3.0.0.26. *R29-T10*. Rational bit complexity. See R29-SP-T7. Asymptotic bound and observed bit sizes.
- 3.0.0.27. *R29-T11*. Polling versus verified optimization. See R29-SP-F10. New derivative-bracketing comparator; 47-dimensional oracle uninstantiated.
- 3.0.0.28. *R29-T12*. Polling versus verified optimization. See R29-SP-F10. New derivative-bracketing comparator; 47-dimensional oracle uninstantiated.
- 3.0.0.29. *R29-T13*. 47-dimensional bridge. See R29-SP-T10. Not falsely closed.
- 3.0.0.30. *R29-T14*. Restricted economic comparative static. See R29-SP-F11. New boundary curvature result; restricted class stated.
- 3.0.0.31. *R29-T15*. Hardware deployment. See R29-SP-T11. Exact compiled object; hardware neural rounding not certified.
- 3.0.0.32. *R29-T16*. Historical transport identity. See R29-SP-F13. Preserved as an implementation identity.
- 3.0.0.33. *R29-T17*. Historical transport identity. See R29-SP-F13. Preserved as an implementation identity.

3.0.0.34. *R29-T18*. Rollback benefit. See R29-SP-F12. Preserved in audit material; no net-benefit claim.

3.0.0.35. *R29-P1*. Total certified-solution ratios. See R29-SP-T2. Main-table addition.

3.0.0.36. *R29-P2*. Hardware deployment. See R29-SP-T11. Exact compiled object; hardware neural rounding not certified.

3.0.0.37. *R29-P3*. Nearest-neighbor novelty. See R29-SP-F15. Explicit theorem-level comparison.

3.0.0.38. *R29-P4*. Coherent paper and preservation. See R29-SP-F16. Restructured main argument; no historical deletion.

3.0.0.39. *R29-N1*. Coherent paper and preservation. See R29-SP-F16. Restructured main argument; no historical deletion.

3.0.0.40. *R29-N2*. Regional work and strongest classical baseline. See R29-SP-F1. New conditional complexity theorem; no universal neural speed advantage.

3.0.0.41. *R29-N3*. Economic value and minimum intervention. See R29-SP-F5. New theorem and exact cost enclosure.

3.0.0.42. *R29-N4*. Stronger rollout and improvement controls. See R29-SP-T5. Analytic structural comparison; no new inventory rollout panel.

3.0.0.43. *R29-N5*. Neural-specific value. See R29-SP-F4. Neural exclusivity not established.

3.0.0.44. *R29-N6*. Prospective evidence and dependence. See R29-SP-F14. Frozen new exploratory cohort; no independence claim.

3.0.0.45. *R29-N7*. Original full-domain target. See R29-SP-F8. Original target remains numerically unresolved.

3.0.0.46. *R29-N8*. 47-dimensional bridge. See R29-SP-T10. Not falsely closed.

3.0.0.47. *R29-N9*. Rollback benefit. See R29-SP-F12. Preserved in audit material; no net-benefit claim.

3.0.0.48. *R29-N10*. Nearest-neighbor novelty. See R29-SP-F15. Explicit theorem-level comparison.

3.0.0.49. *R29-N11*. Optimized exact baseline. See R29-SP-T6. Vectorized DP and exact structural controls.

3.0.0.50. *R29-N12*. Prospective evidence and dependence. See R29-SP-F14. Frozen new exploratory cohort; no independence claim.

COMPUTATIONAL APPENDIX: REVISION R30

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1. READING THE EVIDENCE

All exact fractions, candidate parameters, raw failures, region covers, derivative enclosures, and source identities are stored under `revisions/2026-09-24-r30/results`. Displayed decimal performance bounds are rounded outward. Timings are single-environment observations, not confidence intervals. The article and supplement give model and theorem scopes; this appendix does not extend them. The largest cost-enclosure gap is at most $2.28\text{E-}10$.

TABLE I
FROZEN NEURAL CANDIDATES: RAW ACCURACY AND EXACT PROOF OBJECTS

Gain	Raw passes	Median cells	Bound calls	Max bits	Certificate KiB
Linear	19/20	4	2–80	203	0.917–29.79
Cubic	0/20	112	83–286	624	47.83–132.6
Quadratic	7/20	77	6–152	415	3.056–73.23

Raw passes require an independently enclosed worst-state, all-restart regret no greater than .01. Counts are not independent economic replications. Certificate size includes exact rational endpoints and proof bounds.

TABLE II
OBSERVED CONSTRUCTION WORK IN SECONDS

Gain	Fit	Compile	Cert.+check	Total	Structural	Grid DP
Linear	0.04639	0.001201	0.0004445	0.04901	0.002272	0.2178
Cubic	0.06541	0.001625	0.0521	0.1156	0.01557	0.04955
Quadratic	0.06373	0.002	0.02132	0.08557	0.006683	0.03602

Each entry is descriptive single-environment timing. Total is the median of each candidate’s full sum, not a sum of separately rounded medians. Structural is the matched-tolerance Bernstein solver; it is not the exact symbolic sign rule. Grid DP materializes 2^{20} states and uses binary64. All methods have the same model primitives.

All three polynomial proposals meet the raw .01 criterion; their details are in the computational appendix. Among the neural comparisons with positive saved intervention cost, the largest break-even shadow price is below 0.000739. At the illustrative price $\lambda = 1/20$, 60 of 60 exact paired comparisons favor the completed policy over the near-exact structural policy after intervention cost. This is an accounting comparison, not a statistical replication count.

TABLE III
DISCOUNTED ECONOMIC INTERVENTION METRICS

Gain	Occupancy edits	Priority cost	Operating gain	Extra-cost upper
Linear	0	0	0	$4E - 12$
Cubic	2.382	3.783	0.2736	$2.28E - 10$
Quadratic	0.1387	0.1934	0.001208	$8.4E - 11$

First three numerical columns are medians across the 20 neural candidates. The last is an outward-rounded upper bound on the largest observed upper-minus-lower cost enclosure in the stated local-loss class. All quantities concern the continuous reset density.

TABLE IV
UNCHANGED ORIGINAL ECONOMY AT THE ACTIVE-BOUNDARY RESTART

θ	Exit probability	J_θ	J'_θ	J''_θ
-0.2	[0.99997, 0.99998]	[-7.6652, -7.6650]	[1.7017, 1.7019]	[16.9990, 16.9992]
-0.1	[0.98580, 0.98581]	[-7.3389, -7.3387]	[5.9044, 5.9046]	[82.1152, 82.1154]
0	[0.68915, 0.68916]	[-6.2320, -6.2318]	[15.9905, 15.9907]	[60.6513, 60.6515]
0.1	[0.19903, 0.19904]	[-4.6662, -4.6660]	[12.7887, 12.7889]	[-83.8317, -83.8315]
0.2	[0.04076, 0.04077]	[-3.7609, -3.7607]	[6.0310, 6.0312]	[-45.1336, -45.1334]

All endpoints are rounded outward from validated balls. This is one restart and a fixed constant-control class, not the original whole-domain target. The opposite-boundary and time-truncation errors are included.

TABLE V
THE SAME VALIDATED CENTRAL-STATE ORACLE UNDER TWO SEARCH PROCEDURES

k	Setup+audit s	Bracket calls	Poll calls	Bracket s	Poll s	Bracket bound	Poll bound
1/2	29.6	2	23	0.02666	0.4102	0	0.005
2	29.19	8	31	0.1074	0.5193	0.00371	0.007763

The bracketing target is .005. The inherited poll meshes have the achieved bound displayed here, not an artificially matched value. Both methods require the same oracle construction and concavity audit.

TABLE VI: All neural candidates; A = Adam, B = L-BFGS

Gain	Method/width/seed	Raw regret upper	Cells	Calls	Occupancy	Priority	Full s
linear	A/8/1	0.000039	3	3	0	0	0.9121
linear	A/8/2	0.012663	41	77	0.03773	0.0516	0.07108
linear	A/8/3	0.000024	3	3	0	0	0.06399
linear	A/8/4	0.000783	4	4	0	0	0.06383
linear	A/8/5	0.000007	4	4	0	0	0.06447
linear	B/8/1	0.000003	3	3	0	0	0.02075
linear	B/8/2	0.000009	5	5	0	0	0.0221
linear	B/8/3	0.000001	3	3	0	0	0.01558
linear	B/8/4	0.000006	2	2	0	0	0.008753
linear	B/8/5	0.000002	3	3	0	0	0.01174
linear	A/16/1	0.000167	4	4	0	0	0.06609
linear	A/16/2	0.004521	41	77	0.009016	0.01237	0.07411
linear	A/16/3	0.000006	2	2	0	0	0.06553
linear	A/16/4	0.000024	3	3	0	0	0.06516
linear	A/16/5	0.002148	44	80	0.0004114	0.0005666	0.07409
linear	B/16/1	0.000002	6	6	0	0	0.02501
linear	B/16/2	0.000025	6	6	0	0	0.02479
linear	B/16/3	0.000257	6	6	0	0	0.009412
linear	B/16/4	0.000008	5	5	0	0	0.01304
linear	B/16/5	0.000001	7	7	0	0	0.03418
cubic	A/8/1	0.420749	144	286	2.935	4.579	0.127
cubic	A/8/2	0.071483	115	223	0.2479	0.3731	0.1147
cubic	A/8/3	0.384982	110	217	2.387	3.79	0.113
cubic	A/8/4	0.384866	110	217	2.386	3.789	0.1169
cubic	A/8/5	0.420749	144	286	2.935	4.579	0.1844
cubic	B/8/1	0.420749	144	286	2.935	4.579	0.06817
cubic	B/8/2	0.030503	80	152	0.09845	0.1649	0.1278
cubic	B/8/3	0.385395	110	217	2.39	3.795	0.06425
cubic	B/8/4	0.385395	110	217	2.39	3.795	0.06486
cubic	B/8/5	0.385392	110	217	2.39	3.795	0.06136
cubic	A/16/1	0.384537	111	218	2.383	3.785	0.1165
cubic	A/16/2	0.403565	110	217	2.562	4.034	0.1108
cubic	A/16/3	0.384208	112	219	2.38	3.781	0.1208
cubic	A/16/4	0.045918	115	223	0.1029	0.1659	0.1229
cubic	A/16/5	0.018225	47	83	0.04472	0.08034	0.09018
cubic	B/16/1	0.032401	116	224	0.1247	0.1975	0.1729
cubic	B/16/2	0.032752	115	223	0.1425	0.2324	0.09924
cubic	B/16/3	0.383961	112	219	2.378	3.779	0.09589
cubic	B/16/4	0.032828	115	223	0.1442	0.2349	0.1377
cubic	B/16/5	0.029961	117	225	0.09242	0.1585	0.1801
quadratic	A/8/1	0.059549	73	144	0.8961	1.209	0.08238
quadratic	A/8/2	0.001982	6	6	0	0	0.06422
quadratic	A/8/3	0.059549	74	145	0.8961	1.209	0.08016
quadratic	A/8/4	0.058462	77	149	0.5674	0.7913	0.08809
quadratic	A/8/5	0.059549	74	146	0.8961	1.209	0.08306
quadratic	B/8/1	0.059549	73	144	0.8961	1.209	0.02752
quadratic	B/8/2	0.014738	77	149	0.1417	0.1945	0.05122
quadratic	B/8/3	0.010987	78	150	0.07094	0.104	0.0445
quadratic	B/8/4	0.015021	77	149	0.1403	0.1922	0.0612
quadratic	B/8/5	0.203274	74	146	0.9891	1.307	0.02883
quadratic	A/16/1	0.005034	78	150	0.02653	0.03837	0.08954
quadratic	A/16/2	0.059175	78	150	0.8797	1.186	0.08849
quadratic	A/16/3	0.046384	77	149	0.4816	0.6568	0.0917
quadratic	A/16/4	0.004746	78	150	0.01232	0.01552	0.09207

Gain	Method/width/seed	Raw regret upper	Cells	Calls	Occupancy	Priority	Full s
quadratic	A/16/5	0.024038	77	149	0.1023	0.1259	0.08828
quadratic	B/16/1	0.006289	79	151	0.03104	0.04194	0.134
quadratic	B/16/2	0.005631	79	151	0.02927	0.03989	0.1475
quadratic	B/16/3	0.005626	79	151	0.02757	0.0374	0.156
quadratic	B/16/4	0.021356	42	78	0.1371	0.2043	0.05929
quadratic	B/16/5	0.004861	80	152	0.02913	0.04057	0.2042

TABLE VII
POLYNOMIAL CONTROLS WITH THE SAME OBSERVATIONS AND KNOWN DEGREE

Gain	Raw regret upper	Fit s	Compile s	Certify s	Cells
linear	0E-12	0.002705	0.001906	0.000161	2
cubic	1E-12	0.0001478	0.01762	0.0008435	4
quadratic	1E-12	0.0002166	0.006903	0.0002222	3

These are exact compiled dyadic policies from fitted polynomial coefficients. An irrational root is not asserted to equal its dyadic approximation.

TABLE VIII
UNCHANGED R29 EXHAUSTIVE BENCHMARK AND TOTAL-WORK RATIOS

d/L	States	DP s	Repair+check s	Ratio	Full s	Full/DP
2/4	16	0.001072	0.003879	3.618	–	–
3/4	64	0.00803	0.01485	1.849	–	–
4/4	256	0.03729	0.0654	1.754	2.356	63.19
4/5	625	0.103	0.1841	1.787	3.464	33.63
5/4	1,024	0.2064	0.3054	1.48	4.729	22.91
6/4	4,096	1.103	1.484	1.345	14.81	13.43
7/4	16,384	5.495	8.345	1.519	59.95	10.91

Full includes new fitting, budget repair and verification only. Unmeasured or unavailable costs are not silently added as zero. These ratios use the displayed inherited stage times.

Certified Bellman Operators with Neural Proposals

Historical Preservation Annex, R30

September 24, 2026

This annex reproduces the complete R29 article, supplement, response, computational appendix and historical annex without changing their pages. The R29 historical annex retains the R26 documents and earlier embedded material. Original source and evidence remain unchanged in their directories. Current scientific interpretation is given in `ECTA_R30` and `RESPONSE_R30`; historical text is preserved for audit rather than silently rewritten.

NEURAL BELLMAN OPERATORS

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We study Bellman solution procedures that separate neural proposals, economically meaningful improvement, and independent certification. A backward residual-budget operator preserves candidate actions whenever the completed continuation leaves sufficient accuracy allowance. We prove all-state, all-restart accuracy and pointwise payoff improvement without an optimal-value reference. Exact-rational experiments audit 115 frozen proposals and 286 completions, including trivial-policy controls and graphs with up to 16,384 states. For continuous control, precision-aware coordinate polling certifies projected stationarity at active box constraints. A likelihood-ratio oracle instantiates its assumptions in a stopped consumption–portfolio economy with costly preference adjustment. Independently implemented historical transport and paired rollback experiments identify the roles of parameter geometry and deployment checks. Classical comparators, raw failures, and verification costs remain explicit. The original whole-domain accuracy target is retained separately from the proved finite-state and restricted-class results.

KEYWORDS: Bellman operators, Stochastic control, Certified computation, Preference adjustment, Neural approximation.

1. INTRODUCTION

Numerical dynamic programming must answer two different questions. Can an approximator discover a useful policy, and can the computation establish that the policy is economically accurate? A small training loss, a favorable simulated return, and a smooth optimizer trajectory do not answer the second question when stopping, financing restrictions, and active controls are present. Conversely, a rigorous certificate for a known policy does not demonstrate that an algorithm can discover the solution of an unknown Bellman problem. This paper joins discovery and certification through explicitly specified operators, rather than identifying either component with the other.

Our organizing economic problem is a consumption–portfolio economy in which the agent can alter a stochastic preference state at a quadratic cost. Consumption, investment, preference adjustment, and the continuation region interact. We retain the original objective: a current-state policy whose regret is at most .01 uniformly over the economic state domain and all restart times. The revision adds three constructive results without changing that objective. First, a backward Bellman completion operator turns a learned finite-state candidate into a policy with a prescribed uniform regret bound. The operator uses its own policy values and local action comparisons, not an optimal-value oracle. Second, a precision-aware direct-search result provides projected-gradient stationarity at box constraints. An explicit likelihood-ratio bridge makes its smoothness and oracle requirements numerically checkable on a slice of the original stopped economy. Third, an exact recentering identity isolates finite-path neural Jacobian and optimizer-moment transport, including the nonlinear output displacement omitted by a linearized control.

The computations are designed to test these statements separately. A stopped, coupled inventory problem supplies unknown-solution Bellman discovery and exact policy evaluation. A constant-control slice of the original economy supplies active-boundary and interior economic optima. Matched production-schedule experiments supply actual deployment decisions, including optimizer-state restoration. The inherited stochastic manufactured example remains a

verification stress test, and the inherited full-domain experiment remains the test of the original objective. We do not substitute the former for the latter.

This distinction is quantitatively consequential. All 34 inventory candidates meet .01 after the same completion rule, whereas only 12 meet it before completion. Yet exact tabular backward induction is also available at these sizes and is charged as a direct baseline. The matched central-state frontier does not identify neural superiority. The best retained whole-domain bound for the original economy is 7.181834580823298, approximately 718.18 times its stated tolerance. Our new accuracy results concern precisely the problems and policy classes for which their hypotheses are instantiated. They are neither a numerical proof of that original target nor a statistical claim about optimizer superiority.

1.1. *Relation to the literature and the incremental contribution*

Classical computational economics emphasizes approximation design, economic feasibility, and error control; see Judd (1998). Markov-chain approximation for continuous stochastic control provides a systematic discretization theory under its own consistency and convergence conditions (Kushner and Dupuis, 2001). Adaptive sparse grids offer a major classical response to high-dimensional economic state spaces (Brumm and Scheidegger, 2017). Our finite inventory graph is an independently specified economic model, not a consistency-verified discretization of the original diffusion. Its exact finite-state accuracy therefore makes no assertion about a continuous-domain discretization error or a sparse-grid performance comparison.

Fitted value iteration propagates approximation error through Bellman recursion; finite-sample bounds quantify that propagation under distributional and function-class conditions (Munos and Szepesvári, 2008). Deep stochastic-control algorithms study approximation and generalization in finite-horizon problems (Huré, Pham, Bachouch, and Langrené, 2021). Neural PDE methods provide mesh-free residual approximation (Sirignano and Spiliopoulos, 2018). These works motivate the discovery stage, but neither a sampled residual nor a convergence statement with an optimization-error term is the same object as a fixed-candidate, all-state arithmetic certificate. Our inventory implementation performs fitted Bellman recursion without a supplied value witness, freezes its actions, and then constructs a deterministic residual envelope and a reference-free corrective sweep. Bellman residual bounds and policy improvement are classical principles. The contribution here is their explicit tolerance-preserving composition, exact arithmetic implementation, and separation from both learned policy generation and a subsequent independent optimal reference.

Information-relaxation duality bounds stochastic dynamic programs by relaxing nonanticipativity and charging penalties (Brown, Smith, and Sun, 2010, Brown and Smith, 2014). Our stopped supersolution argument is complementary: it bounds a policy and an optimal value using a smooth witness, a boundary mismatch, and generator residuals. Neither form of upper bound makes a candidate feasible automatically. The financed policy construction is evaluated only after admissibility is checked in the original information filtration. The historical dual library is a shared input with a separately identified construction cost, not a free neural training oracle.

Constrained derivative-free optimization is well developed. Mesh adaptive direct search treats general constraints and nonsmooth stationarity under appropriate assumptions (Audet and Dennis, 2006); direct-search sufficient-decrease arguments give worst-case iteration bounds (Vicente, 2010). We do not claim to introduce constrained direct search. Our box result exposes the cost of an interval oracle at the necessary precision and gives a short-face projected-gradient bound: geometrically unavailable full steps are treated through projection, while computationally unresolved evaluations cannot establish stationarity. The resulting theorem is instantiated below with validated derivatives and payoff enclosures, rather than inferred from a software test.

Adam uses a history of coordinatewise moments (Kingma and Ba, 2015). Limited-memory quasi-Newton methods are serious classical controls, not optional weak baselines (Byrd, Lu, Nocedal, and Zhu, 1995). Function-space kernel descriptions of neural learning are central to the neural tangent kernel literature (Jacot, Gabriel, and Hongler, 2018); our identity instead concerns a finite-width network, a realized finite step, and the complete historical Adam state. It is an exact chain-rule comparison, not a fixed-kernel or infinite-width limit. The independent recentered implementation retains both the network carrier and the nonlinear output remainder, so its agreement with neural Adam identifies a mechanism without establishing a cheaper nonneural solver.

Finally, rigorous floating-point verification requires explicit control of rounding, truncation, and domain coverage (Rump, 2010). We distinguish three arithmetic objects: exact rational finite-state policies; mathematical real networks with saved binary64 coefficients interpreted exactly; and hardware floating-point deployment. Only the first two are certified here. A hardware execution theorem would require an additional rounding enclosure at deployment.

2. THE ECONOMIC PROBLEM AND THE CERTIFICATE TARGET

Time runs from 0 to $T = 1$. Before stopping, preference u and wealth x evolve according to

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (1)$$

$$dx_s = \{(.02 + .06p_s)x_s - c_s\} ds + .2p_s x_s dZ_s^x, \quad (2)$$

where $d\langle Z^u, Z^x \rangle_s = -.25 ds$. Controls satisfy

$$c \in [.05, .8], \quad \theta \in [-.2, .2], \quad p \in [-.5, .8]. \quad (3)$$

The continuation domain is $\mathcal{D} = (1.2, 2.8) \times (.5, 2)$. Let τ be first exit from $\mathcal{D}$, capped at 1. The running payoff and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad k \in [.5, 8], \quad (4)$$

$$G(u, x) = -.02(u-2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1-s). \quad (5)$$

At $s = 1$, $g = G$. An admissible progressively measurable policy, adapted to the original Brownian filtration, has payoff

$$J_k^\pi(t, u, x) = \mathbb{E}_{t,u,x} \left[\int_t^\tau e^{-.04(s-t)} \ell_k(u_s, c_s, \theta_s) ds + e^{-.04(\tau-t)} g(\tau, u_\tau, x_\tau) \right]. \quad (6)$$

Write $V_k = \sup_{\pi \in \mathcal{A}} J_k^\pi$. The unrestricted value includes admissible path-dependent policies; the target implementation is a single current-state actor $\pi(t, u, x)$. The original full-domain numerical experiment fixes $k = 2$ and asks for

$$\sup_{(t,u,x) \in [0,1] \times \overline{\mathcal{D}}} \{V_2(t, u, x) - J_2^\pi(t, u, x)\} \leq .01. \quad (7)$$

The boundary values use immediate settlement. We retain (7) throughout. A certificate at $(0, 2, 1.25)$, a certificate on a small restart rectangle, or an optimum within a constant-control class is not this statement.

The adjustment cost has a direct economic role. Positive θ changes future utility curvature, but its flow cost rises with k . Stopping adds a substantial loss before the horizon. Consequently

a policy can have accurate typical-state behavior and still have large worst-restart regret near financially or psychologically fragile states. This is the reason to report a uniform criterion, rather than a mean simulation score alone.

2.1. Stopped residual verification

For a smooth witness v , define

$$\begin{aligned} \mathcal{R}^a v = & v_t + \theta v_u + \{(.02 + .06p)x - c\}v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux} \\ & + \ell_k(u, c, \theta) - .04v. \end{aligned} \quad (8)$$

No differentiability of V_k is required in the next statement.

PROPOSITION 1—Uniform policy certificate: *Suppose $v \in C^{1,2}$ has the boundedness needed for stopped Itô's formula on the closed cylinder. Suppose the policy is admissible, $|v - g| \leq b$ on the parabolic boundary, and, throughout the continuation cylinder,*

$$|\mathcal{R}^\pi v| \leq e, \quad \sup_{a \in \mathcal{A}} \mathcal{R}^a v - \mathcal{R}^\pi v \leq q, \quad (9)$$

where $b, e, q \geq 0$. Then

$$0 \leq V_k(t, u, x) - J_k^\pi(t, u, x) \leq 2b + (2e + q) \frac{1 - e^{-.041-t}}{.04}. \quad (10)$$

State-dependent and slabwise versions improve this bound when signed residual information is available. The all-restart suffix envelope used in the retained full-state calculation is proved in Appendix E. A numerical grid of test points alone cannot establish any of these hypotheses; the intervals must cover each whole cell and each feasible control range.

2.2. Feasible memory and the financed comparison

The retained financed family uses sixteen time slabs. Its network has 355 parameters and 48 raw outputs. A common raw utility intercept is absorbed by the financing root, leaving 47 economic quotient coordinates. This quotient removes a redundancy; it does not turn every neural parameter displacement into an independent economic direction.

The construction can retain the adapted internal state

$$dY_s = .02Y_s ds - .3Y_s dZ_s^x, \quad (11)$$

with expert wealth processes $X_i = F_i(s, Y_s)$ and portfolio fractions $p_i = -1.5Y_s F_{i,y}/F_i$. Non-negative initial mixture weights yield aggregate wealth and wealth-weighted investment, subject to the inherited feasibility checks. This policy is Markov in an augmented state containing Y and the fixed mixture weights, not necessarily in (t, u, x) alone. Since the augmentation is generated from the original observations and carries no independent randomness, it does not enlarge the original admissible filtration. The unrestricted augmented and original control problems can therefore represent the same admissible policies, while the particular finite financed library remains a restricted class.

A central financed certificate is useful evidence about proposal generation and deployment. It cannot establish (7) for a current-state actor. We preserve that distinction in the production tables, the historical comparison, and the response to the referee.

3. CERTIFIED BELLMAN AND DEPLOYMENT OPERATORS

We distinguish a proposal operator, a completion operator, and a deployment operator. The first may be neural and approximate. The second uses a Bellman error criterion on a specified state model. The third compares economic payoffs and restores the entire optimizer state after rejection. Their conclusions are different: approximation quality, uniform policy regret, and monotone deployed payoff, respectively.

3.1. Reference-free Bellman completion

Consider a finite-horizon Markov decision problem on finite state sets S_t , with nonempty feasible actions $A_t(s)$, discount $\beta \in (0, 1]$, known transition probabilities, bounded rewards, and a fixed terminal payoff. A stopped state has only its settlement action and zero continuation. Let

$$(T_t v)(s) = \max_{a \in A_t(s)} \left\{ r_t(s, a) + \beta \sum_{s'} P_t(s' | s, a) v(s') \right\}. \quad (12)$$

For a deterministic policy π , compute its own value exactly by backward evaluation, $v_t^\pi = T_t^\pi v_{t+1}^\pi$. Neither this evaluation nor a comparison of its local actions requires V^* . Define the nonnegative Bellman deficit

$$\delta_t^\pi(s) = (T_t v_{t+1}^\pi)(s) - v_t^\pi(s), \quad (13)$$

and its state-sensitive envelope

$$B_T^\pi = 0, \quad B_t^\pi(s) = \delta_t^\pi(s) + \beta \max_{a \in A_t(s)} \sum_{s'} P_t(s' | s, a) B_{t+1}^\pi(s'). \quad (14)$$

Both quantities are zero at a stopped state.

THEOREM 1—Residual envelope and uniform completion: *For every deterministic policy, $0 \leq V_t^*(s) - v_t^\pi(s) \leq B_t^\pi(s)$ at every state and restart time. Fix $\varepsilon > 0$ and*

$$\eta = \varepsilon \left(\sum_{j=0}^{T-1} \beta^j \right)^{-1}. \quad (15)$$

Given any raw policy π^0 , construct a completed policy backward. At time t , evaluate each action against the already completed continuation $v_{t+1}^{\bar{\pi}}$. Retain $\pi_t^0(s)$ if its deficit from the largest such action value is at most η ; otherwise select a maximizing feasible action. Then

$$0 \leq V_t^*(s) - v_t^{\bar{\pi}}(s) \leq \eta \sum_{j=0}^{T-t-1} \beta^j \leq \varepsilon \quad (16)$$

for all states and restarts. The algorithm uses no optimal-value reference. With at most N states, A actions, and D successor outcomes, each evaluation, envelope, and completion sweep uses $O(TNAD)$ arithmetic operations. Exact-integer bit complexity additionally depends on the bit lengths of rewards, probabilities, discount factors, and accumulated values.

The proof is a backward comparison argument, given in Appendix A. The terminal stopping convention is essential: a settled state cannot be improved by continuing the chain. In the implementation, exact rational deficits, rather than their printed decimal representations, determine both action replacement and the tolerance test.

The completed policy is the compiled action table, with the changed states recorded. It is not generally the raw neural network evaluated in floating-point arithmetic. In particular, the theorem must not be described as a certificate for an unchanged neural policy when completion replaces an action. The neural representation supplies the candidate value recursion, and completion preserves its decision wherever the suffix-conditioned Bellman deficit is sufficiently small. This construction permits an exact accuracy guarantee without assuming that neural optimization finds a global minimum.

The operation count also reveals the principal computational qualification. At the state sizes studied here, exact backward induction has the same enumeration order. Completion is not a new way to evade tabular complexity. The experiment tests whether discovery and certification can be combined without reference leakage; the classical reference and its cost remain visible.

3.2. Synchronous economic deployment

Let S contain the policy parameters and all optimizer state, including moment estimates, step counters, and the rule's current hyperparameters. Suppose a verifier returns $[L(\pi), U(\pi)]$ containing the exact economic payoff of a feasible policy at the declared restart. A proposal generated from S is deployed only if

$$L(\pi_{\text{candidate}}) > U(\pi_{\text{incumbent}}) + \zeta, \quad \zeta \geq 0. \quad (17)$$

Otherwise the saved S is restored. A verifier failure gives no usable interval and is not an acceptance.

PROPOSITION 2—Monotone deployment: *Every accepted proposal improves exact payoff by more than ζ . Rejection followed by exact state restoration leaves both the deployed policy and the next proposal's initial optimizer state equal to their saved values. If $\zeta > 0$ and payoffs are bounded above by M , at most $\lfloor (M - J_0)/\zeta \rfloor$ accepted proposals can occur.*

For $\zeta = 0$, monotonicity does not imply a finite number of acceptances, convergence of Adam, or discovery of a global optimum. Nor is (17) equivalent to comparing two regret upper bounds. A tighter upper bound can result from a better witness or finer verifier while the candidate's actual payoff falls.

An interval pair can overlap even when the candidate is genuinely better. If a true improvement is $\gamma > 0$ and each interval has width at most w , separation is guaranteed when $2w < \gamma$. Refining widths $w_j = w_0 2^{-j}$ therefore resolves a fixed positive margin after finitely many successful verifier levels. Zero margin, a finite precision floor, or an exhausted resource budget leaves the comparison unresolved; it is not evidence of economic indifference.

3.3. Box-constrained stationarity with explicit oracle work

Let $C = \prod_{i=1}^n [\ell_i, u_i]$ and let $F : C \rightarrow \mathbb{R}$ be differentiable with L -Lipschitz gradient on the relevant feasible segments. For $\tau > 0$, define the ascent projected-gradient mapping

$$G_\tau(x) = \frac{\Pi_C(x + \tau \nabla F(x)) - x}{\tau}. \quad (18)$$

Its zero set is the first-order box-stationary set. The gradient need not be zero at a constrained optimum.

At mesh $h > 0$, a coordinate poll evaluates every feasible full step $x \pm he_i$. The interval oracle must enclose F with width at most κh^2 at the incumbent and at every feasible trial used in a complete unsuccessful poll. A step is accepted if its lower endpoint exceeds the incumbent's upper endpoint by σh^2 , where $\sigma > 0$. A full step outside C is recorded as a geometric short face with its distance to the boundary. A feasible step whose verification is unresolved is not a short face.

THEOREM 2—Constrained precision-aware polling: *Suppose $F \leq M$ on the feasible search region and the stated width is obtained whenever required, at work at most $W(\kappa h^2)$ per call. Starting from x_0 , repeated accepted coordinate steps followed by complete polls reach an unsuccessful complete poll using work at most*

$$(2n+1) \left[1 + \left\lfloor \frac{M - F(x_0)}{\sigma h^2} \right\rfloor \right] W(\kappa h^2). \quad (19)$$

At its final point x ,

$$\|G_\tau(x)\|_2 \leq \sqrt{n} h \max \left\{ \sigma + 2\kappa + \frac{L}{2}, \frac{1}{\tau} \right\}. \quad (20)$$

If any required feasible evaluation fails to reach the requested width, the conclusion is oracle unresolved, *not* (20).

The maximum in (20) is the boundary term missing from an interior positive-spanning argument. When a full step exists in the sign of a gradient component, unsuccessful comparison bounds that component by $(\sigma + 2\kappa + L/2)h$. When it does not, the distance to the corresponding face is less than h , so projection bounds the component of G_τ by h/τ . This proves constrained stationarity without treating an infeasible direction as a failed economic comparison.

Equation (19), not a bare $O(\varepsilon^{-2})$ poll count, is the complexity statement. Choosing h proportional to a desired projected-gradient tolerance gives an iteration factor of order ε^{-2} but requires width of order ε^2 . A rapidly growing W can dominate all proposal work. The theorem does not establish arbitrary precision for the original 47-dimensional stopped checker. Our implementation below verifies the needed widths at four declared meshes on an original-economy slice, and reports actual setup, derivative-audit, and oracle work. It does not extrapolate a fixed truncation order to arbitrarily small meshes.

3.4. Exact recentered historical transport

Let $f(\theta) \in \mathbb{R}^{48}$ denote the raw network outputs, and let $Q \in \mathbb{R}^{48 \times 47}$ have orthonormal columns spanning the economically relevant quotient. Locally the proposal loss depends on raw outputs only through the quotient after the financing intercept has been eliminated. Write g_r for its quotient gradient and $B_r = Q^\top Df(\theta_r)$. For minimization, Adam's first moment is

$$\hat{m}_n = \sum_{r=1}^n a_{nr} B_r^\top g_r, \quad a_{nr} = \frac{(1 - \beta_1) \beta_1^{n-r}}{1 - \beta_1^n}, \quad (21)$$

with the corresponding coordinatewise second moment and denominator D_n . Thus

$$f(\theta_{n+1}) - f(\theta_n) = -\alpha \sum_{r=1}^n a_{nr} Df(\theta_n) D_n^{-1} B_r^\top g_r + R_n, \quad (22)$$

where R_n is the exact nonlinear Taylor remainder. If Df is H -Lipschitz on the step segment, $\|R_n\| \leq H\|\theta_{n+1} - \theta_n\|^2/2$. The exact identity does not require this bound to be small.

PROPOSITION 3—Recentered path equivalence: *Initialize two implementations with identical carrier parameters, Adam moments, counters, and hyperparameters. In the second implementation evaluate the economic gradient at the carrier’s realized raw output, pull it back using $B_n^\top$, apply the same parameter-space Adam recurrence, and recenter the next economic coordinates on the full realized output $f(\theta_{n+1})$. In exact arithmetic the two implementations have identical parameters, moments, raw outputs, and proposal policies at every step, including the entire nonlinear remainder in (22).*

This proposition is an induction using the chain rule. It does not say that a 47-dimensional optimizer with no neural carrier reproduces 355-dimensional Adam, nor does it imply a chart-independent adaptive optimizer. The implementation uses two independently coded recurrences and measures their floating-point discrepancy. Exact real equality is a mathematical result; the measured discrepancy is a consistency check, not a substitute for its assumptions.

The previous moving control used a capped linearized output update, omitted R_n , and let the economic quotient drift from the carrier. Its poor performance consequently confounded transport, clipping, nonlinear displacement, and misalignment. The new control removes those confounds by construction. Agreement identifies what the neural trajectory is doing; it cannot by itself establish that the neural representation is necessary or cheaper.

4. STATE-DEPENDENT RESIDUAL BUDGETS

The uniform local tolerance in Theorem 1 reserves the same amount for every possible future stage. Once a continuation has been completed, however, its actual residual envelope is known. We can allocate only the amount needed by that continuation, and test the raw action against the remaining budget. This distinction concerns preservation of candidate decisions, not a change in the information available to the algorithm.

Let $e_t : S_t \rightarrow [0, \infty)$ be a prescribed family of error budgets, with $e_T = 0$ and zero budget at settled states. Assume that, at every nonsettled state,

$$e_t(s) \geq \beta \max_{a \in A_t(s)} \sum_{s'} P_t(s' | s, a) e_{t+1}(s'). \quad (23)$$

A common tolerance $e_t(s) = \varepsilon$ before settlement satisfies this condition when $\beta \leq 1$. During the backward construction, let v_{t+1} and B_{t+1} be the exact value and residual envelope of the already completed suffix. Set

$$q_t(s, a) = r_t(s, a) + \beta P_{t,s,a} v_{t+1}, \quad (24)$$

$$c_t(s) = \beta \max_{a \in A_t(s)} P_{t,s,a} B_{t+1}, \quad a_t^{\text{rem}}(s) = e_t(s) - c_t(s). \quad (25)$$

Retain the raw action $a^0 = \pi_t^0(s)$ exactly when

$$\max_a q_t(s, a) - q_t(s, a^0) \leq a_t^{\text{rem}}(s). \quad (26)$$

Otherwise choose a maximizing feasible action, using a fixed tie rule. For the selected action a , set $v_t(s) = q_t(s, a)$ and

$$B_t(s) = \max_b q_t(s, b) - q_t(s, a) + c_t(s). \quad (27)$$

At a settled state, apply its settlement payoff and set $B_t = 0$; the terminal value is the stated terminal payoff and $B_T = 0$.

THEOREM 3—Budgeted completion and policy preservation: *Under the finite-state assumptions of Theorem 1 and (23), the backward construction is well defined. At every state and restart,*

$$v_t^{\pi^0}(s) \leq v_t^{\bar{\pi}}(s) \leq V_t^*(s), \quad 0 \leq V_t^*(s) - v_t^{\bar{\pi}}(s) \leq B_t(s) \leq e_t(s). \quad (28)$$

The local remaining allowance is nonnegative. Conditional on the already completed suffix and on using envelope (27), the rule retains the raw action whenever that envelope can certify it within $e_t(s)$. No optimal value or optimal action reference is needed. Its arithmetic enumeration order is $O(TNAD)$; exact bit complexity and storage are charged separately.

PROOF: The assertion holds at the terminal time and at settlement. Suppose it holds at $t + 1$. Since $0 \leq B_{t+1} \leq e_{t+1}$ and transition probabilities are nonnegative, (23) gives $0 \leq c_t(s) \leq e_t(s)$. Thus the allowance in (25) is nonnegative. If the raw action passes (26), adding c_t proves $B_t \leq e_t$. If it fails, a maximizing feasible action exists by finiteness; its local deficit is zero, so $B_t = c_t \leq e_t$.

For the accuracy statement, write $d_{t+1} = V_{t+1}^* - v_{t+1}^{\bar{\pi}}$. By the induction hypothesis $0 \leq d_{t+1} \leq B_{t+1}$. The Bellman equation and the elementary inequality $\max_a(x_a + y_a) \leq \max_a x_a + \max_a y_a$ yield

$$V_t^* - v_t^{\bar{\pi}} = \max_a \{q_t(s, a) + \beta P_{t,s,a} d_{t+1}\} - q_t(s, \bar{\pi}_t(s)) \leq B_t(s).$$

The loss is nonnegative by optimality of V^* . For preservation, assume $v_{t+1}^{\bar{\pi}} \geq v_{t+1}^{\pi^0}$. The selected action has $q_t(s, \bar{\pi}_t(s)) \geq q_t(s, \pi_t^0(s))$, because it is either the raw action or a maximizer. Positivity of the transition law then gives

$$v_t^{\bar{\pi}}(s) \geq r_t(s, \pi_t^0(s)) + \beta P_{t,s,\pi_t^0(s)} v_{t+1}^{\pi^0} = v_t^{\pi^0}(s).$$

This closes the induction. Finally, once v_{t+1} and B_{t+1} are fixed, c_t does not depend on the selected current action. The raw action satisfies $B_t \leq e_t$ if and only if (26) holds. Computing the action values and propagating the envelope each enumerates at most $TNAD$ successor contributions. None of these steps calls an optimal-reference solver. *Q.E.D.*

The preservation inequality also holds for the uniform-threshold operator, by the same induction. It is a pointwise economic comparison on the finite model, stronger than observing that a printed regret bound decreased. It does not carry over automatically to a continuous-state economy or to a floating-point neural implementation.

The theorem is locally permissive, not globally minimum-edit. Changing a later action changes both its value and its envelope, which can change earlier decisions. Consequently we make no claim that the budget rule changes fewer actions on every input, nor that either rule yields the cheapest certified policy. The two rules are compared on exactly the same raw tables. A fixed common tolerance has the advantage of certifying every restart, rather than just one initial distribution. More general budgets satisfying (23) can incorporate state-specific accuracy priorities.

This construction uses the full model, feasible-action enumeration, and exact suffix evaluation. These are the same information resources available to exact backward induction. Classical dynamic programming supplies the Bellman comparison argument; the methodological addition

here is the explicit candidate-preservation rule, its state-dependent certificate budget, and the joint proof of local permissiveness, whole-state accuracy, and payoff monotonicity. It is not an invention of policy iteration or an unconditional neural acceleration theorem.

4.1. Relation to established numerical methods

The oracle and conclusion must be matched before comparing methods. Fitted value iteration propagates approximation and sampling errors through a value recursion; Munos and Szepesvári (2008) give finite-sample bounds under their approximation and distributional assumptions. Our finite graph instead permits exact model evaluation after a policy is frozen. Its deterministic, all-state certificate is obtained by enumeration; it cannot be advertised as a better statistical rate or as a solution of the dimensionality problem addressed by approximation methods.

Constrained direct search predates the present work. Kolda, Lewis, and Torczon (2003) organize direct-search methods and their convergence mechanisms; Kolda, Lewis, and Torczon (2006) relate a stationarity measure to the step-length control in generating-set search under linear constraints. The contribution of Theorem 2 is the explicit interval-width requirement, the short-face alternative for a box, the work factor $W(\kappa h^2)$, and the unresolved-oracle outcome. The theorem is an instantiated certificate argument, not a claim to have originated constrained derivative-free optimization.

The verified-numeric literature, represented by Rump (2010), supplies the reason for using directed enclosures rather than tolerances on floating-point residuals. The stopped slice contributes a concrete analytic oracle: change of path measure controls the stopping discrepancy, while polynomial moments, Gaussian tails, and directed rounding enclose the objective and its first two derivatives. Neither neural stochastic-control approximation (Huré, Pham, Bachouch, and Langrené, 2021) nor a small training residual alone supplies those particular deterministic enclosures. Conversely, the slice has one constant preference-control parameter and does not establish a general high-dimensional oracle.

The comparative economic benchmark remains the unchanged consumption–portfolio model. Dual bounds and information relaxation (Brown, Smith, and Sun, 2010, Brown and Smith, 2014), classical constrained quasi-Newton optimization (Byrd, Lu, Nocedal, and Zhu, 1995), and projection and sparse-grid methods (Judd, 1998, Brumm and Scheidegger, 2017) are substantive comparators, not incidental implementation choices. The numerical contribution is identified by theorem, oracle, coupling rule, and executed application separately. A general claim of neural superiority would require a favorable accuracy–total-work comparison; the present revision reports that comparison rather than assuming its outcome.

5. AN INSTANTIATED CERTIFICATE IN THE ORIGINAL STOPPED ECONOMY

Theorem 2 becomes a computational result only after the payoff oracle, smoothness constant, feasible geometry, and precision are supplied. We do so without changing the primitives in Section 2. Fix the restart $(0, 2, 5/4)$, consumption $c = 3/4$, investment $p = 0$, and constant adjustment $\theta \in [-1/5, 1/5]$. We examine $k = 1/2$ and $k = 2$. This is an admissible one-dimensional class inside the original economy, not a replacement for all current-state policies.

Wealth is deterministic:

$$x_s = \left(\frac{5}{4} - \frac{3/4}{.02} \right) e^{.02s} + \frac{3/4}{.02}. \quad (29)$$

Validated endpoints give $x_1 \in [.5177014240300721, .517701424030129]$, so wealth does not exit. Preference is $u_s = 2 + \theta s + .05W_s$. Since $|\theta| \leq .2$, a preference exit implies

TABLE I
ORIGINAL-ECONOMY SLICE: INSTANTIATED BOUNDARY AND INTERIOR CERTIFICATES

k	Final θ	L	m	Class regret upper	Total seconds
1/2	1/5	1.615	0.9622	0	38.94
2	37/200	3.086	2.432	6.157×10^{-6}	39.5

The objective and stopping rule are those of the original economy; consumption, investment, restart, and constant-adjustment class are fixed as stated. m is a validated lower bound on $-F''$. Total time includes construction, derivative auditing, polls, and the final optimum check. It is not a full-domain policy certificate.

$\sup_{s \leq 1} |.05W_s| \geq .6$. Reflection and a Gaussian Chernoff bound give

$$p_{\text{exit}} \leq 4e^{-72} < 2.153 \times 10^{-31}. \quad (30)$$

This is an upper bound for every θ in the box, not an observed exit frequency.

The stopped objective is compared with an unstopped extension that clips preference to $[1.2, 2.8]$ inside utility and settlement. Their pathwise payoffs coincide off the exit event and have absolute value at most $H = 14.4$. A likelihood-ratio derivative on the full path, with score $W_1/.05$, bounds the payoff, first-derivative, and second-derivative differences. It avoids differentiation of a random first-exit time. The bounds, including the explicit θ -dependence of the adjustment cost, are proved in Appendix C.

We then integrate Gaussian moments analytically. On $|Z| \leq 10$, the flow expansion about $u = 2$ has $|u - 2| \leq .7$. A degree-100 power series and a degree-20 discount expansion yield rigorous geometric and exponential remainders. Gaussian tail moments control the complement, including the derivative score factors. MPFR directed arithmetic encloses every coefficient and polynomial evaluation; saved binary64 endpoints are rounded outward. The script records separate flow, terminal, stopping, and arithmetic enclosures rather than reporting one unexplained tolerance.

Eighty intervals of width $1/200$ cover the complete θ box. They certify strict concavity. For $k = 1/2$, $-F'' \geq .9622197$ and $|F'''| \leq 1.6146843$; for $k = 2$, $-F'' \geq 2.4326157$ and $|F'''| \leq 3.0850803$. The oracle is executed at meshes $1/25, 1/50, 1/100, 1/200$, with $\sigma = 1/100$ and $\kappa = 1/100000$. Every required width is checked, including at the active face. The fixed expansion orders are claimed only at these meshes.

For $k = 1/2$, the derivative remains positive at $\theta = 1/5$. Strict concavity therefore certifies the upper-face global optimum within the class. Its class regret is zero, although its ordinary gradient is not zero. For $k = 2$, a validated sign bracket locates a unique interior optimum near .183. The final poll is $\theta = 37/200$, with class regret at most 6.157×10^{-6} from $F'(\theta)^2/(2m)$. Thus a higher adjustment cost changes the economic solution from a binding upper control to an interior action. The result concerns actual stopped welfare, not the differentiable proposal loss.

The instantiation resolves the boundary exclusion and demonstrates a stopped-gradient bridge on this class. It does not instantiate the financed 47-dimensional gradient bridge, and it does not show that fixed c and p are optimal among unrestricted controls. Those are distinct mathematical statements.

6. UNKNOWN-SOLUTION BELLMAN DISCOVERY AND COMPLETION

6.1. A coupled stopped inventory economy

There are d goods and eight decision periods, with discount $99/100$. Inventory states are $x \in \{0, \dots, L - 1\}^d$. A feasible order purchases zero units or one or two units of one good,

never exceeding capacity before demand. Demand is equiprobable over $d + 2$ vectors: zero, the all-ones vector, and each vector with one component equal to two and all other components equal to one. These common demand realizations couple the inventories.

For order a , let $z = x + a$, sales $s = \min(z, D)$, remaining stock $y = (z - D)_+$, and shortage $h = (D - z)_+$. The period reward is

$$r(x, a, D) = \sum_j s_j - \frac{3}{5} \sum_j a_j - \frac{1}{10} \mathbf{1}_{\{a \neq 0\}} - \frac{1}{50} \sum_j y_j^2 - \frac{2}{5} \sum_j h_j^2 - \frac{1}{20d} \left(\sum_j h_j \right)^2. \quad (31)$$

The last term makes aggregate shortage costly beyond the sum of itemwise shortages. The all-zero state stops immediately and pays $-2 - d/5$. Other terminal states pay $\sum_j x_j/5 - \sum_j x_j^2/50$. The stopping state cannot order its way back into the continuation economy. Hence both common shocks and the ability to avoid joint stockout matter for a policy's worst-state value.

The model is specified by rational primitives. It has no engineered value function or action oracle. We study $(d, L) = (2, 4), (3, 4), (4, 4)$, with an additional $(4, 5)$ resolution and capacity control. These yield 16, 64, 256, and 625 states. Changing L changes the stated finite economic model; it is not an estimate of a continuous-state discretization rate.

6.2. Equal information and reference ordering

Raw policies are obtained by backward fitted Bellman recursion. At each stage a candidate constructs action values using its own preceding value approximation and takes the maximum over feasible actions. The targets are not taken from an optimal reference. Neural candidates use tanh networks of width 16 or 32 and one or two hidden layers, with seeds 27301 and 27302. Each fitted stage uses 400 Adam steps at .01, followed by at most 150 L-BFGS-B objective-gradient calls. The last accepted iterate is restored if the call cap is reached. Polynomial candidates use total degrees two and three and least squares on exactly the same states and candidate-specific Bellman targets. All candidates have the same primitives, transition information, terminal payoff, and stopping convention.

There are eight neural configurations and two polynomial configurations at each $L = 4$ model. The $d = 4, L = 5$ control uses the two width-32, depth-two neural seeds and both polynomial degrees, giving 34 retained cases in total. The period-zero action is compiled but its value approximation is not fitted, since no predecessor would use it. This explicit implementation reduction gives seven fitted stages rather than eight and changes no period-zero action target.

For each case, the complete raw action table is saved and hashed before certification. Exact own-policy evaluation produces (13) and (14). Completion uses a freshly constructed model and only the frozen action table, with the same $\varepsilon = 1/100$ for every case. It saves the changed actions, their original suffix-conditioned deficits, the completed policy, and a second reference-free certificate. All candidates and both certificates for a model are frozen before a separately coded scalar-integer optimal recursion is invoked. That recursion does not call the vectorized action-value or maximization routines used by the certificate. It verifies the policy gaps afterward; it supplies no labels, features, stopping decisions, or hyperparameters to training.

This is a designed experiment, not a preregistered clinical trial. The base R27 model was frozen before the present executions, but its completion amendment was explicitly written after prior diagnostic work. R28 inherits that post-diagnostic algorithmic choice and applies it to every new case without selection. The present source commit, run environment, all weights, all policies, and their freeze-order hashes are recorded. No claim of confirmatory independence from the earlier model-design process is made.

TABLE II
UNKNOWN-SOLUTION INVENTORY: ALL CASES, ALL STATES, ALL RESTARTS

d	L	States	Cases	Raw passes	Completed passes	Worst completed bound
2	4	16	10	8	10	2.089×10^{-4}
3	4	64	10	4	10	0
4	4	256	10	0	10	0
4	5	625	4	0	4	3.962×10^{-4}

A pass means the exact-rational reference-free bound is at most .01. Counts are computational coverage, not independent statistical replications. The complete 34-case table, raw bounds, exact gaps, changed-action counts, tolerance profile, architectures, and timings appear in the supplement.

TABLE III
INVENTORY COMPUTATION: MEASURED WORK AND DIRECT-REFERENCE COST

d, L	Neural train	Poly train	Raw check	Completion	Completed check	Exact DP
2,4	1.687	6.819×10^{-4}	0.001872	0.001072	0.00179	0.001284
3,4	1.778	9.284×10^{-4}	0.007657	0.004351	0.007219	0.007796
4,4	2.06	0.002249	0.03527	0.01918	0.03299	0.04531
4,5	4.134	0.003528	0.08468	0.04699	0.0806	0.1218

Seconds; candidate-stage columns are within-model medians, with neural and polynomial generation separated. Exact DP is the independently executed scalar optimal recursion and excludes candidate training. Construction, compilation, independent gap evaluation, total wall time, model-array bytes, action-table bytes, and process high-water memory are reported separately in the supplement. Shared reference costs are shown both unamortized and per case; no reference cost is silently assigned zero.

6.3. All-state accuracy, correction burden, and classical work

All 34 completed candidates satisfy the uniform target. The largest completed envelope is 3.962×10^{-4} , while 12 raw candidates already pass. We retain the raw failures rather than treating completion as evidence that neural training alone attains the tolerance. The independent optimal recursion confirms every saved envelope at every state and restart. The gap between mean fitting accuracy and a worst-restart guarantee becomes visible as the state model grows: local action mistakes at financially fragile inventory states can propagate backward, and completion corrects those mistakes against the completed suffix rather than against an unavailable optimal continuation.

Two distinctions govern the interpretation. First, completed policies can contain nonneural corrections. The supplement reports the number and fraction of nonstopped decisions changed, so the burden of correction is observable. Second, exact tabular backward induction is extremely competitive at these sizes. It is shown as a standalone reference solve in the cost table, not hidden inside the certificate’s timing. The experiment establishes unknown-solution discovery plus certified completion; it does not establish that neural approximation beats tabulation, polynomials, or adaptive sparse grids in a high-dimensional limit.

7. MECHANISM AND PRODUCTION-SCHEDULE EXPERIMENTS

7.1. Recentered historical transport

The new mechanism study uses seeds 27101–27103, proposal quadrature orders (8, 16) and (12, 24), 200 steps, and learning rate .08. One implementation uses the neural optimizer directly;

TABLE IV
INDEPENDENT IMPLEMENTATIONS OF THE EXACT RECENTERED IDENTITY

Seed	Proposal orders	Parameter difference	Output difference	Retained remainder
27101	(8,16)	2.466×10^{-12}	2.463×10^{-12}	1.451
27101	(12,24)	2.107×10^{-12}	2.123×10^{-12}	1.451
27102	(8,16)	5.045×10^{-13}	4.992×10^{-13}	1.392
27102	(12,24)	5.165×10^{-13}	5.125×10^{-13}	1.392
27103	(8,16)	1.072×10^{-12}	1.066×10^{-12}	1.503
27103	(12,24)	8.398×10^{-13}	8.648×10^{-13}	1.503

Maxima are over 200 steps. Carrier/economic mismatch is zero by construction and capped-step count is zero. The remainder is measured in quotient output norm. All first- and second-moment discrepancies and effective step distributions are retained. Representative policies from seed 27101 at both quadrature orders are separately checked by the stopped MPFR verifier.

the other independently computes the quotient gradient, Jacobian pullback, first and second moments, and the full parameter displacement. Economic coordinates are recentered on realized carrier outputs after every step. There is no output cap, and no nonlinear remainder is dropped.

The maximum parameter discrepancy across the six paths is 2.466×10^{-12} . This numerical agreement supports consistency of the implementations with Proposition 3. It does not certify equality of floating-point executions, and it does not identify a representation-free method: the carrier, its Jacobian, and all historical parameter moments remain present. In the earlier capped control, high nominal rates could produce the same effective trajectory because every step hit the cap. The new comparison removes that tuning confound rather than reinterpreting the old nominal-rate grid as uncapped rate identification.

7.2. Production schedules, rejected proposals, and matched controls

The production study uses seeds 27201 and 27202 and the fixed (8, 16) proposal rule. Neural Adam uses the inherited selected rate .08; direct quotient Adam uses .32. Direct L-BFGS-B provides a classical comparator. Each method has four blocks of at most 100 objective-gradient calls, with otherwise identical gated and ungated arms. Adam retains its moments across accepted blocks. L-BFGS-B is restarted in both arms at each block, so restart treatment is not a confound. Both arms pay for the same initial and block-end verifications; the ungated arm simply deploys the candidate regardless of the interval comparison.

The gate uses strict payoff separation with no injected extreme learning rate. A rejected block restores the network or quotient parameters, all Adam moments and counters where applicable, and all saved optimizer settings. Saved, candidate, restored, and deployed state hashes are retained, together with the candidate’s actual payoff interval, incumbent interval, strict margin, and interval width. These are causal online trajectories, not replayed selections from a completed path.

There are 6 actual rejections in the gated trajectories, all with exact restoration of the saved optimizer state. The direct L-BFGS-B paths exercise late-stage rejection under a production schedule; the Adam paths need not reject in order for the safety proposition to hold. Importantly, interval overlap can cause rejection of an economically indistinguishable or slightly better proposal. We therefore do not call these counts a measured frequency of harmful proposals. The paired intervals in the supplement determine which improvements, if any, are strictly certified between final gated and ungated policies. Absent strict separation, a lower printed regret upper bound is not proof that rollback improves payoff.

TABLE V
FIXED-SCHEDULE CENTRAL-STATE DEPLOYMENT: ALL TWELVE TRAJECTORIES

Seed	Method/arm	Calls	Rejections	Final regret upper	Total seconds
27201	Neural Adam / U	400	0	0.001926	88.54
27201	Neural Adam / G	400	0	0.001926	88.33
27201	Direct Adam / U	400	0	0.001937	88.53
27201	Direct Adam / G	400	0	0.001937	88.68
27201	Direct L-BFGS-B / U	114	0	0.001924	87.69
27201	Direct L-BFGS-B / G	136	3	0.001924	87.71
27202	Neural Adam / U	400	0	0.001926	88.52
27202	Neural Adam / G	400	0	0.001926	88.82
27202	Direct Adam / U	400	0	0.001937	88.44
27202	Direct Adam / G	400	0	0.001937	88.31
27202	Direct L-BFGS-B / U	107	0	0.001924	87.82
27202	Direct L-BFGS-B / G	115	3	0.001924	87.64

The payoff is at the central restart in the financed class. Total seconds are measured same-process, warm-infrastructure wall times, including initial and checkpoint verification and file bookkeeping. They are not cold from-scratch times; historical dual construction is separately disclosed. A rejected interval comparison need not mean the candidate truly loses payoff.

The results do not displace direct L-BFGS-B from the strongest tested central-state frontier. They establish that the same certificate gate is implementable for neural and direct methods, and that full-state rollback can be exercised without an adversarial rate injection. An algorithmic efficiency claim would require the gains from avoided losses to exceed verification and restoration costs over a broader population of economically relevant problems. That claim is not inferred from an implementation test.

8. CONTINUOUS VERIFICATION, ORIGINAL-DOMAIN PROGRESS, AND WORK ACCOUNTING

8.1. Retained stochastic verification benchmark

The inherited three-dimensional state–time benchmark has two independent Brownian sources, stopping, and an unspanned state. It is intentionally manufactured: its value witness and optimal actions are known and used in training. We retain it to test whole-cell verification, not as evidence of unknown-solution discovery. Its details and exact Hamiltonian-square identity are reproduced in the supplement; the full earlier derivations remain in the historical annex.

For the two fixed neural policies, the R26 second-order interval verifier gives bounds .0008343368 and .0007884142 on 32,768 cells. The same-cell first-order bounds are .00137310 and .00133294. Thus the second-order enclosure genuinely sharpens verification for unchanged policies. The exact rational degree-eight comparator gives $1.2978908373252045 \times 10^{-7}$ on 64 one-dimensional cells. That comparator exploits known optimal structure and must not be confused with the information-matched polynomial discovery candidates in Section 6.

Neither this three-dimensional success nor the finite-state dimension study proves scalable continuous-state interval-Hessian verification. Mixed derivatives grow quadratically in dimension, rectangular covers can grow exponentially, and interval dependency can dominate. Our new finite-state study quantifies actual enumeration, architecture, tolerance, and storage at 16–625 states; it is not relabeled as a high-dimensional continuous verifier result.

8.2. What changed in the original full-domain bound

The original history is decomposed as follows:

Object	Regret upper	Identified change
Coarse retained object	7.2414624431	Original retained actor and witness, coarse verification.
Fine unchanged object	7.2004087464	Same actor and witness; finer verification only.
Fine revised object	7.1818345808	Changed actor/witness candidate, common finer verification.

The first decrease, about .041054, is verifier tightening. The second, about .018574 or .258% of the unchanged fine bound, is a joint actor/witness certificate improvement. It is not a certified increase in policy payoff. The historical artifact explicitly records that no such payoff comparison was proved. Moreover, the first continuation block used 82 gradient calls and 48 accepted optimizer iterations, whereas the next three blocks used one call each, accepted no optimizer iteration, and left the candidate hashes unchanged. This stagnation is an empirical observation, not the conclusion of Theorem 2 for the full-domain class.

The new original-economy slice establishes a different, correctly restricted guarantee. We do not splice its small regret into the full-state ledger or replace the target in (7). The full-domain bound and its unresolved payoff-improvement comparison remain visible in both the article and the referee disposition table.

8.3. Complete cost categories and the measured boundary

For a deployed method, the relevant cost is

$$C_{\text{total}} = C_{\text{construction}} + C_{\text{tuning}} + C_{\text{proposal}} + C_{\text{geometry}} + C_{\text{verification}} + C_{\text{completion}} + C_{\text{rollback}} + C_{\text{reference}} + C_{\text{bookkeeping}}. \quad (32)$$

The inventory experiment separately measures model construction, fitted-stage generation, action compilation, own-policy certification, fresh-model completion, completed-policy certification, independent reference construction and solution, and final reference evaluation. The slice includes the power-series oracle construction and whole-box derivative audit. Neither uses a hidden optimal-value witness in candidate generation.

The central-state experiment intentionally reuses the frozen historical dual library and process-level validated quadrature cache. Its run time therefore excludes cold reconstruction of those inputs. The library’s original construction record reports 415.788436934 seconds for 20 freshly generated nodes, including policy and dual generation and verification, with maximum resident memory 509,348 KiB on its original machine. We retain that record and its input hashes, identify warm reuse explicitly, and do not add seconds from different machines into a purported new measured total. A deployment amortizing this library over K solves must additionally charge its reconstruction cost divided by K ; $K = 1$ is not free. The new scripts also record the fixed rates as inherited tuning choices rather than claiming zero-cost tuning from scratch.

Peak process memory is a high-water measure that includes the numerical runtime; it is not the sum of certificate-array bytes and model weights. Both object sizes and process memory are reported. No cell count, tuning run, or checkpoint is counted as an independent replication of a statistical neural effect. The complete ledger permits method comparisons at a common declared information and infrastructure boundary and makes the remaining cold-pipeline measurement explicit.

9. CANDIDATE ATTRIBUTION, ROBUSTNESS, AND STATE-SPACE SCALING

The new cohort holds the 34 R28 raw policies fixed and adds 81 candidates under a committed protocol. On each original model we add ten random feasible tables, a one-period greedy policy, a base-stock rule, and two-period truncated dynamic programming. The width-32, depth-two seed panel is expanded to ten seeds at $(d, L) = (4, 4)$ and five at $(4, 5)$. At dimensions five, six, and seven we run six fixed proposal rules each. No failed candidate is replaced. The seven models contain between 16 and 16,384 states per time slice; the largest is 64 times the original 256-state four-dimensional graph. This is a finite-graph scaling study, not a claim to solve a high-dimensional continuous problem.

Every new raw action table and training record is hashed before its own completion. All reference-free variants are frozen before the independent full-horizon optimal reference is constructed for that model. A runtime audit rejects optimal-reference file reads, and the candidate/completion context disables the independent optimal solver. The exact reference is generated by a separate scalar implementation. The program checks every state and restart against it, not only a start-state average. The process had access to prior R27/R28 outcomes, so these additions and the residual-budget rule are exploratory, despite their fixed cohort and enforced program dependencies. The missing original R27 raw-run artifacts have not been reconstructed; R28 and R29 are separately identified executions.

Table VI distinguishes true raw regret, a reference-free raw residual bound, and the guarantees after repair. There are 115 raw policies and 286 completed variants, including the 14-policy tolerance panel at .001, .01, and .05. All completed variants satisfy their exact rational tolerance. A failed raw residual test is not, by itself, proof that raw regret exceeds the tolerance; the independent reference determines the latter count. The budget rule makes fewer changes than the fixed rule in 60 main-cohort cases, more in 0, and the same number in 55. These are paired deterministic comparisons, not independent statistical replications.

TABLE VI
EXACT ALL-STATE/ALL-RESTART COVERAGE AT TOLERANCE .01

d/L	N	Cases	Raw bound	Raw regret	Fixed	Budget
2/4	16	23	8	8	23	23
3/4	64	23	4	4	23	23
4/4	256	31	0	0	31	31
4/5	625	20	0	0	20	20
5/4	1024	6	0	0	6	6
6/4	4096	6	0	0	6	6
7/4	16384	6	0	0	6	6

The control experiment makes attribution observable. Random, myopic, base-stock, and two-period policies receive exactly the same two completion operators as neural and polynomial proposals. Detailed corrections by time, inventory total, minimum inventory, and number of empty coordinates are retained, together with suffix-conditioned deficit quantiles. Thus a small correction fraction can be distinguished from corrections concentrated near default or at early restart times. Raw-to-completed payoff monotonicity follows from Theorem 3, not from a reduction in a certificate alone.

Completion cost need not be proportional to the number of corrected actions: even a policy requiring no changes incurs the model-wide action scan. A small neural correction fraction therefore does not establish a computational advantage. Table VII retains exact dynamic programming

visibly and charges both repair and verification. All times within that table are measured on the same R29 execution environment. A dash means that no new neural fitting run was specified for that model; inherited R28 fitting times are not combined with R29 hardware timings. Full candidate-specific ledgers are in the computational appendix. In the largest graph, the specified neural proposal requires 634 replacements out of 131064 nonsettled state–time decisions under the budget rule, compared with 114341 for the myopic control. Its raw worst regret is 1.549, compared with 3.659 for that control. This is a substantial candidate-preservation difference in the executed model. It is not a total-work advantage: the neural fitting stage costs 51.6 seconds, whereas the independent full optimal solve costs 5.495 seconds in the same execution.

TABLE VII
SAME-EXECUTION STAGE COSTS (SECONDS; MEDIAN OVER THE SPECIFIED POLICIES)

d/L	N	Exact DP	New neural	Fixed repair	Budget repair	Verify
2/4	16	0.001072	–	0.001992	0.002268	0.001611
3/4	64	0.00803	–	0.007419	0.008217	0.006628
4/4	256	0.03729	2.291	0.02754	0.03296	0.03244
4/5	625	0.103	3.28	0.08989	0.1022	0.08187
5/4	1024	0.2064	4.424	0.1365	0.1498	0.1556
6/4	4096	1.103	13.33	0.6438	0.6971	0.7865
7/4	16384	5.495	51.6	3.282	3.677	4.668

The logical work count is $T \sum_s |A(s)|D$ successor contributions for one feasible-action sweep. The vector implementation also records the larger dense count $TNAD$, model-array bytes, compiled-policy bytes, integer bit lengths, CPU time, wall time, and process high-water resident memory. High-water memory includes libraries, previous cases, and live models; it is not an isolated per-candidate peak. The standalone component sum charges the full model and independent reference to each policy without amortization. Interpreter startup and file serialization are included in the outer cohort measurement but not silently allocated into individual method timings. Reusing an inherited candidate does not make its historical training free; those rows are explicitly marked as replay-only rather than new standalone fitting measurements.

The original economy remains distinct. The two rigorously enclosed preference-adjustment slices, six independently implemented transport paths, and twelve paired production trajectories are reproduced from the frozen R28 evidence. They instantiate boundary-aware polling, the stopped likelihood-ratio bridge for the stated scalar class, and full-state rollback. They do not establish the financed 47-dimensional derivative bridge or change the original current-state, whole-domain bound 7.181834580823298 against .01. Nor does the present cost ledger recover the unmeasured construction time of inherited central dual infrastructure. These are identified mathematical and measurement requirements, not replaced by auxiliary success.

10. CONCLUSION

A Bellman method must identify its approximation object, control class, restart set, verification oracle, and total computational work. The residual-budget theorem constructs a uniformly accurate finite-state policy while preserving every candidate action that its suffix envelope can certify. The constrained-poll theorem and stopped slice give a separate continuous-control guarantee, including boundary stationarity and oracle-precision costs. The exact transport identity and paired deployment experiment identify mechanisms without attributing generic repair to neural fitting. These contributions retain the original economic problem and its .01 target.

They do not turn a finite graph into a continuous-domain certificate, and they do not declare the original 7.181834580823298 bound closed. The complete computational and historical appendices retain all unsuccessful candidates, detailed derivations, and prior evidence.

APPENDIX A: PROOFS OF THE CERTIFICATE AND COMPLETION RESULTS

PROOF OF PROPOSITION 1: Apply stopped Itô's formula to $e^{-.04(s-t)}v(s, u_s, x_s)$, first on compactly localized cylinders and then pass to the stopping time using the stated boundedness. For any admissible a ,

$$J^a - v(t, u, x) = \mathbb{E} \left[e^{-.04(\tau-t)}(g - v)(\tau, u_\tau, x_\tau) + \int_t^\tau e^{-.04(s-t)} \mathcal{R}^a v(s, u_s, x_s) ds \right]. \quad (33)$$

Since $\mathcal{R}^a v \leq e + q$, $J^a \leq v + b + (e + q) \frac{1 - e^{-.041-t}}{.04}$. For the candidate, $J^\pi \geq v - b - e \frac{1 - e^{-.041-t}}{.04}$. Take the supremum in the first inequality and subtract the second. Admissibility gives the nonnegative lower bound on regret. This argument uses a smooth test witness, not a differentiable optimal value. Q.E.D.

PROOF OF THEOREM 1: The terminal gap is zero. Suppose $0 \leq V_{t+1}^* - v_{t+1}^\pi \leq B_{t+1}^\pi$. Monotonicity of the Bellman operator gives the lower bound and

$$V_t^* - v_t^\pi = T_t V_{t+1}^* - T_t v_{t+1}^\pi + \delta_t^\pi \quad (34)$$

$$\leq \beta \max_a P_t^a B_{t+1}^\pi + \delta_t^\pi = B_t^\pi. \quad (35)$$

The inequality follows by comparing each action's continuation difference before taking the maximum. At a stopped state there is no continuation, so the gap and the envelope are zero.

For completion, proceed backward after fixing the completed suffix. If the raw action is retained, its deficit against that suffix is at most η by the test. If it is replaced by a maximizing action, the deficit is zero. Thus $\delta_t^\pi \leq \eta$ everywhere. Induction in the previous bound yields $B_t^\pi \leq \eta \sum_{j=0}^{T-t-1} \beta^j$, proving the assertion. This argument does not assume that the raw policy is neural, near optimal, or generated independently of the model. Reference-free means that the implemented construction never reads V^* ; the proof may of course compare to the mathematically defined optimal value.

Each backward sweep enumerates at most TNA action values, each with at most D successors. A maximization and a retained-action test add linear overhead. When primitives are rational, a common denominator can be propagated backward and all comparisons reduced to integer comparisons. Their bit cost is not a unit-cost real operation and must be counted separately for growing horizons or large denominators. Q.E.D.

A.1. Exact arithmetic convention in the inventory implementation

Let $R = 100d$, $D = d + 2$, and let r_{saz}^{num} be the integer reward numerator. Terminal values have denominator $q_T = R$. Define $q_t = 100Dq_{t+1}$. If the next value numerator is n_{t+1} , the action numerator is

$$Q_{t,s,a}^{\text{num}} = 100 \frac{q_{t+1}}{R} \sum_{z=1}^D r_{saz}^{\text{num}} + 99 \sum_{z=1}^D n_{t+1}(s'(s, a, z)). \quad (36)$$

The stopped action is instead assigned its settlement numerator at denominator q_t . Equation (36) exactly equals the rational expectation and discount recursion. Python arbitrary-size integers

carry all accumulated numerators. Binary64 training is allowed to choose a poor action; it is not allowed to change a certificate comparison. Exported decimal bounds are rounded upward for display, whereas the decision uses the exact rational fraction. The independent optimal recursion uses a scalar outcome loop and separately selects feasible maximizers. Every saved policy gap is then compared with its saved statewise envelope using common integer denominators.

PROOF OF PROPOSITION 2: Containment gives $J(\pi_{\text{candidate}}) \geq L(\pi_{\text{candidate}}) > U(\pi_{\text{incumbent}}) + \zeta \geq J(\pi_{\text{incumbent}}) + \zeta$. A rejected candidate is not deployed. Restoring a byte-equivalent serialization of every deterministic optimizer state component makes the next state identical to the saved state; restoring only policy parameters would not establish this fact for an adaptive optimizer. Summing the strict improvements and using $J \leq M$ gives the finite-acceptance bound when $\zeta > 0$. *Q.E.D.*

APPENDIX B: CONSTRAINED POLLING: GEOMETRY, PRECISION, AND WORK

PROOF OF THEOREM 2: Write $w = \kappa h^2$. Failure of the strict acceptance test at a feasible trial y means $L(y) \leq U(x) + \sigma h^2$. Since intervals contain the true values and each has width at most w ,

$$F(y) - F(x) \leq (\sigma + 2\kappa)h^2. \quad (37)$$

For $y = x + she_i \in C$, $s \in \{-1, 1\}$, smoothness gives

$$F(y) - F(x) \geq sh \partial_i F(x) - \frac{L}{2}h^2. \quad (38)$$

Thus $s\partial_i F(x) \leq Ch$, where $C = \sigma + 2\kappa + L/2$.

For each i with nonzero derivative, take $s = \text{sgn } \partial_i F(x)$. If the full signed step is feasible, the preceding inequality bounds $|\partial_i F(x)|$, and scalar projection implies $|(G_\tau(x))_i| \leq Ch$. If it is infeasible, the distance to the face in that sign is less than h , and the same scalar projection gives $|(G_\tau(x))_i| < h/\tau$. A zero-width coordinate has projected component zero. Squaring and summing establishes (20).

Every accepted step improves true payoff by more than σh^2 , by the same containment argument as Proposition 2. Therefore the number of accepted steps is bounded by $\lfloor (M - F(x_0))/(\sigma h^2) \rfloor$. Each accepted or final unsuccessful poll requires no more than one incumbent evaluation and $2n$ trial evaluations. Multiplying by the stated per-evaluation work gives (19). Computational failure at a feasible trial destroys the failed-poll inequality for that direction; it cannot be classified as a geometric short face. This proves the final qualification as well. *Q.E.D.*

For the constant-control slice, $M = 0$ is a valid upper payoff bound: flow utility and adjustment costs are nonpositive, while terminal wealth is below one and any earlier boundary settlement includes a nonpositive early-exit penalty. A lower endpoint for the initial payoff supplies a conservative $M - F(x_0)$. At the four executed meshes the code checks every required width, and a payoff call evaluates a degree-100 interval polynomial after construction. Its arithmetic cost is linear in the degree; setup, coefficient construction, and derivative auditing are separate. Measured wall time is an observation, not a proved worst-case CPU-time oracle. The large sufficient-decrease work bound is not presented as a sharp predictor of the observed call count.

For completeness, the sign of the settlement payoff in this upper-bound argument can be checked without a numerical approximation. Wealth in the slice is

$$x_s = \frac{75}{2} - \frac{145}{4}e^{s/50}, \quad x_{1/2} < \frac{75}{2} - \frac{145}{4} \left(1 + \frac{1}{100}\right) = \frac{71}{80} < 1.$$

The strict inequality uses $e^z > 1 + z$ for $z > 0$. Wealth is decreasing. If $s \leq 1/2$, settlement is at most $.1 \log(5/4) - 4 < 0$. If $s \geq 1/2$, wealth is below one, the quadratic preference term is nonpositive, and the early-exit charge is nonpositive; settlement is again nonpositive. The same conclusion holds at the terminal time. Running utility and the adjustment-cost term are nonpositive throughout the stopped domain. Thus the complete stopped payoff is nonpositive pathwise, which justifies $M = 0$ in the finite-work bound independently of the computed payoff intervals.

The unconstrained positive-spanning result is recovered when all poll directions are feasible. At a binding upper face an increasing objective can have a nonzero ordinary gradient but a zero projected-gradient mapping. This is precisely the $k = 1/2$ case in the original slice. In general, neither projected stationarity nor a complete unsuccessful poll implies global optimality without further structure. The slice adds verified strong concavity; the finite-state completion result instead uses Bellman comparison and does not require parameter-space convexity.

APPENDIX C: THE STOPPED LIKELIHOOD-RATIO BRIDGE AND ANALYTIC ORACLE

C.1. Stopping and explicit parameter dependence

In the slice, x_s decreases monotonically and its interval endpoint remains above .5. Thus only preference can cause early exit. The drift moves preference by at most .2 over the horizon, leaving a Brownian excursion of at least .6 to reach either boundary from two. The reflection bound $\Pr(\sup_{s \leq 1} |W_s| \geq 12) \leq 4 \Pr(W_1 \geq 12)$ and $\Pr(W_1 \geq 12) \leq e^{-72}$ give (30).

Let H_θ be the stopped payoff as a functional of the preference path, and $\tilde{H}_\theta$ its full-horizon clipped extension. The path functionals agree when no exit occurs. Both are bounded in absolute value by $H = 14.4$ for the stated parameters: clipped flow utility is at most $5(4/3)^{1/5} < 6$ in magnitude, adjustment cost is at most .04 when $k \leq 2$, and the boundary/terminal settlement is bounded by the early-exit charge eight plus the bounded quadratic and logarithmic terms. At a fixed preference path, the only explicit θ dependence is the quadratic cost. The difference $D_\theta = H_\theta - \tilde{H}_\theta$ therefore satisfies

$$|D_\theta| \leq 2H \mathbf{1}_A, \quad |\partial_\theta D_\theta| \leq k(.2) \mathbf{1}_A, \quad |\partial_{\theta\theta} D_\theta| \leq k \mathbf{1}_A, \quad (39)$$

where A is the exit event for that path.

Changing the constant drift of the full preference path yields the likelihood-ratio score $S = W_1/\sigma$, $\sigma = .05$, under the current drift measure. The second score is $S^2 - 1/\sigma^2$. Bounded path functionals, their displayed explicit derivatives, and Gaussian exponential moments justify differentiating the expectation twice. Consequently,

$$|\mathbb{E} D_\theta| \leq 2Hp, \quad (40)$$

$$|\partial_\theta \mathbb{E} D_\theta| \leq 2H \sqrt{p}/\sigma + k(.2)p, \quad (41)$$

$$|\partial_{\theta\theta} \mathbb{E} D_\theta| \leq 2H \sqrt{2p}/\sigma^2 + 2k(.2) \sqrt{p}/\sigma + kp, \quad (42)$$

where $p = 4e^{-72}$. The square roots follow from Cauchy–Schwarz, $\mathbb{E} S^2 = 1/\sigma^2$, and $\mathbb{E}(S^2 - 1/\sigma^2)^2 = 2/\sigma^4$. The argument differentiates a change of path measure, not the hitting-time map. The explicit cost derivatives cannot be dropped from these bounds.

C.2. Power series and Gaussian moments

Set $a = \log(4/3)$ and $v = u - 2$. With $c = 3/4$, untruncated flow utility is

$$f(v) = -\frac{4}{3} \frac{e^{av}}{1+v} = -\frac{4}{3} \sum_{n=0}^{\infty} c_n v^n, \quad c_0 = 1, \quad c_n = \frac{a^n}{n!} - c_{n-1}. \quad (43)$$

The absolute coefficient of f is bounded by $(4/3)e^a = 16/9 < 2$. On $|Z| \leq 10$, $v = \theta s + \sigma\sqrt{s}Z$ satisfies $|v| \leq .7$, so a degree- N truncation has error at most

$$d_N = \frac{2(.7)^{N+1}}{1-.7}. \quad (44)$$

For $N = 100$ this is approximately 1.509423×10^{-15} . There is no singularity in this central region. Clipping handles the Gaussian tails without evaluating $1/(1+v)$ beyond its valid economic interval.

The full Gaussian polynomial moments are

$$\mathbb{E}(\theta s + \sigma\sqrt{s}Z)^n = \sum_{\substack{0 \leq j \leq n \\ j \text{ even}}} \binom{n}{j} \theta^{n-j} \sigma^j (j-1)!! s^{n-j/2}. \quad (45)$$

The convention $(-1)!! = 1$ is used. Each time moment $I_m = \int_0^1 e^{-\rho s} s^m ds$, $\rho = .04$, is enclosed by

$$I_m = \sum_{\ell=0}^K \frac{(-\rho)^\ell}{\ell!(m+\ell+1)} + r_{m,K}, \quad |r_{m,K}| \leq \frac{e^\rho \rho^{K+1}}{(K+1)!}, \quad K = 20. \quad (46)$$

The code propagates this remainder into every coefficient, including the running adjustment cost. The untruncated terminal expectation is the exact quadratic $-.02(\theta^2 + \sigma^2)$ plus $.1 \log x_1$, discounted at ρ .

C.3. Tail control for values and derivatives

Let T_j bound $\mathbb{E}[|Z|^j \mathbf{1}_{\{|Z|>10\}}]$. Dropping the normal density's factor $(2\pi)^{-1/2} < 1$ gives the rigorous recursions

$$T_0 = \frac{2e^{-50}}{10}, \quad T_1 = 2e^{-50}, \quad T_j = 2 \cdot 10^{j-1} e^{-50} + (j-1)T_{j-2} \quad (j \geq 2). \quad (47)$$

Define

$$P_r = 2 \sum_{n=0}^{100} \sum_{j=0}^n \binom{n}{j} (.2)^{n-j} \sigma^j T_{j+r}. \quad (48)$$

This controls the polynomial tail weighted by $|Z|^r$ uniformly over time and the drift box. The clipped utility has magnitude below six. For a normal expectation at time s , drift differentiation gives factors $\sqrt{s}Z/\sigma$ and $s(Z^2 - 1)/\sigma^2$. Hence the flow errors of orders zero, one, and two are bounded respectively by

$$E_0^{\text{flow}} = d_N + 6T_0 + P_0, \quad (49)$$

$$E_1^{\text{flow}} = (d_N + 6T_1 + P_1)/\sigma, \quad (50)$$

$$E_2^{\text{flow}} = \{2d_N + 6(T_2 + T_0) + P_2 + P_0\}/\sigma^2. \quad (51)$$

We used $\mathbb{E}|Z| \leq 1$ and $\mathbb{E}|Z^2 - 1| \leq 2$ on the central part and bounded the discount integral by one. These are remainder bounds for expectations and their derivatives, not pointwise derivative bounds for a clipped function.

For the terminal quadratic, the clipped and untruncated versions coincide in the central region. On the tail their difference is at most $.0144 + .0001Z^2$, because $|\theta| \leq .2$ and the clipping radius is $.8$. Thus the terminal errors are

$$E_0^{\text{term}} = .0144T_0 + .0001T_2, \quad (52)$$

$$E_1^{\text{term}} = (.0144T_1 + .0001T_3)/\sigma, \quad (53)$$

$$E_2^{\text{term}} = \{.0144(T_2 + T_0) + .0001(T_4 + T_2)\}/\sigma^2. \quad (54)$$

Adding the likelihood-ratio stopping bounds gives the total analytic errors. Directed interval arithmetic adds coefficient and evaluation rounding. The first two derivatives of the resulting polynomial are evaluated on each entire drift cell; their analytic error budgets are then added. This is the source of the validated L and strong-concavity constant m used in the main text.

C.4. From derivatives to class regret

If $F'' \leq -m < 0$ on the whole interval, then for any feasible y ,

$$F(y) - F(x) \leq F'(x)(y - x) - \frac{m}{2}(y - x)^2 \leq \frac{|F'(x)|^2}{2m}. \quad (55)$$

The code uses an upper enclosure for $|F'(x)|$ and a lower enclosure for m . At the upper face, a nonnegative lower bound on F' and strict concavity show that F is increasing on the entire box, giving exact class optimality. The analogous statement holds at the lower face. A sign-changing interval bracket for F' identifies the unique interior optimum. These arguments certify optimality within the fixed consumption/investment/constant-adjustment class; they do not compare that class to arbitrary adapted controls.

APPENDIX D: HISTORICAL TRANSPORT AND THE FINANCING-GRADIENT FORMULA

PROOF OF PROPOSITION 3: Suppose the two implementations agree through step n . They then evaluate the same raw carrier output and the same economic quotient. By the chain rule their parameter gradients are identical, $\nabla_\theta L = B_n^\top g_n$. Their first and second moments, bias corrections, denominators, and parameter displacement are consequently identical. Evaluating the full network at the new parameters gives identical raw outputs and includes the entire nonlinear displacement. Exact recentering supplies the induction hypothesis for the next step. A linearized output update would not supply it unless its Taylor remainder vanished. Expanding the first-moment recurrence gives the historical sum in (22); Taylor's integral remainder gives its stated bound when the Jacobian is Lipschitz. *Q.E.D.*

The inherited financing calculation is retained because it identifies what remains to be instantiated for the general stopped gradient. Let $P(y, a) = B$ be the exact financing equation and P_N its proposal approximation. Suppose $P_a, P_{N,a} \geq d_* > 0$ on the relevant root segment and

$|P - P_N| \leq e_P$. Then the root error is at most $\delta = e_P/d_*$. The reduced gradient of an objective H is

$$H_y - H_a \frac{P_y}{P_a}. \quad (56)$$

At the exact and approximate roots write $A = H_y$, $B_1 = H_a$, $C_1 = P_y$, and $D_1 = P_a$. Let E_A, E_B, E_C, E_D enclose both approximation errors and the root-shift effects over distance δ , and let M_B, M_C bound the absolute values of both exact and approximate B_1, C_1 . The reduced-gradient discrepancy is at most

$$E_A + \frac{E_B M_C + M_B E_C}{d_*} + \frac{M_B M_C E_D}{d_*^2}. \quad (57)$$

Indeed, add and subtract the mixed products in $B_1 C_1 / D_1 - B_{1,N} C_{1,N} / D_{1,N}$, and use $|D_1^{-1} - D_{1,N}^{-1}| \leq E_D / d_*^2$. All four error terms and both magnitude bounds are needed. Finite payoff secants do not supply these derivative enclosures. R28 instantiates a different exact stopped-gradient bridge on the one-dimensional slice, not the constants in (57) along the financed 47-dimensional optimizer path.

APPENDIX E: ALL-RESTART DISCOUNTED SUFFIX ENVELOPES

Let a nonnegative residual envelope equal a_i on time slab $[t_i, t_{i+1}]$, with $\Delta_i = t_{i+1} - t_i$. The discounted suffix at a slab boundary obeys

$$C_i = a_i \frac{1 - e^{-\rho \Delta_i}}{\rho} + e^{-\rho \Delta_i} C_{i+1}, \quad C_m = 0. \quad (58)$$

At a restart t in that slab, put $r = t_{i+1} - t$. The suffix is

$$H_i(r) = \frac{a_i}{\rho} + \left(C_{i+1} - \frac{a_i}{\rho} \right) e^{-\rho r}, \quad 0 \leq r \leq \Delta_i. \quad (59)$$

Its derivative has a constant sign unless it is identically zero. The maximum within the slab is therefore attained at an endpoint. Taking the maximum over the slab endpoints proves the all-restart envelope used by the postprocessor. When a separate uniform boundary mismatch is present, its contribution must be added with the coefficient justified by the underlying stopped verification inequality; it is not absorbed into an unexplained legacy output field. The postprocessed all-restart number is consequently a theorem-backed use of the same slab bounds, not an empirical maximum over sampled restart times.

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SUPPLEMENT TO NEURAL BELLMAN OPERATORS: REVISION R29

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1. REVIEW OBJECT AND PRESERVATION

The article, this supplement, the computational appendix, the point-by-point response, and the historical annex form one revision object. The base evidence commit is `cd1191f278948942f42c6507b4566108f2aa2f6d`; the latest addressed report is R27, dated September 24, 2026, at `d6f6f7fa2b0ef40e42206dc64406f004cc83c71e`. R26 and the independent R25 second pass are addressed individually in the response. No earlier source, result, or review branch is overwritten. The old root index is retained in the source audit. The original R27 raw-run files were absent from its reviewed tree; a later re-execution cannot supply an earlier freeze event.

2. EXACT ARITHMETIC AND INDEPENDENT CHECKS

The inventory implementation uses arbitrary-precision Python integers, NumPy object arrays, and exact Fraction comparisons. At (d, L) , rewards have denominator $100d$. Backward denominator recursion multiplies by $100D$ per stage, where $D = d + 2$ is the number of equiprobable demand outcomes. Hence action deficits, envelope comparisons, and tolerance acceptance use integers after exact scaling. Printed decimals are explanatory only; bounds are rounded outward by the table renderer. The independently implemented scalar reference enumerates primitive transitions and rewards, rather than calling the vectorized action-value or completion routines.

For each model, candidate and completion files are sealed before reference creation. The recorded dependency audit prohibits reference files and the optimal solver during generation. This establishes the program’s information path, not blinding of the investigator to earlier results. Each policy’s feasibility, settlement convention, own-value recursion, residual envelope, and every reference gap are checked. Uniform and budgeted completion both improve the raw policy pointwise on this graph. Hash and event-order checks are supplemented by unit tests of infeasible actions, settlement, exact boundary comparisons, and forbidden reference reads.

3. CONTROLS AND CORRECTION BURDEN

A myopic action maximizes the exact one-period reward. The base-stock policy orders toward $\lceil (L - 1)/2 \rceil$, choosing the lowest-index most deficient good and at most two units. The two-period comparator uses the exact terminal payoff at the endpoint of a truncated two-step horizon; it does not read the full-horizon reference. Random policies are sampled independently from feasible actions at each nonsettled state and time. All receive the same repair rules and budgets. The fixed cohort, including the pre-execution seed-index clarification, is preserved in the protocol files.

TABLE I: Proposal quality and median correction burden; percentages exclude settled states

d/L	Proposal	Cases	Raw regret max	Fixed %	Budget %	Repair s
2/4	neural	8	0	0	0	0.001778
2/4	polynomial	2	0.1391	5.833	5.833	0.001871
2/4	random	10	5.054	63.33	63.33	0.002297
2/4	myopic	1	3.849	81.67	81.67	0.002414
2/4	base_stock	1	1.935	48.33	48.33	0.002281
2/4	tabular2	1	0.8426	26.67	26.67	0.002075
3/4	neural	8	0.01418	0.2976	0.2976	0.006931
3/4	polynomial	2	0	0	0	0.006899
3/4	random	10	5.141	69.25	69.15	0.009537
3/4	myopic	1	3.59	87.9	87.9	0.009774
3/4	base_stock	1	1.339	39.48	39.48	0.008217
3/4	tabular2	1	0.7878	18.45	18.45	0.00757
4/4	neural	16	0.3666	2.868	2.255	0.03275
4/4	polynomial	2	0.3666	1.863	1.225	0.03208
4/4	random	10	5.736	73.43	73.21	0.04108
4/4	myopic	1	3.273	86.42	86.42	0.04022
4/4	base_stock	1	1.245	36.13	35.54	0.03699
4/4	tabular2	1	0.8514	17.5	17.45	0.03238
4/5	neural	5	0.3664	2.244	1.643	0.08003
4/5	polynomial	2	0.1925	1.763	1.102	0.08268
4/5	random	10	6.933	77.9	77.7	0.1064
4/5	myopic	1	4.057	83.79	83.79	0.1005
4/5	base_stock	1	1.884	39.28	39.04	0.09759
4/5	tabular2	1	1.491	25.56	25.56	0.08222
5/4	neural	1	0.2399	0.2933	0.2933	0.1362
5/4	polynomial	1	0.2399	0.4643	0.4032	0.1456
5/4	random	1	7.066	75.77	75.6	0.1838
5/4	myopic	1	3.552	86.46	86.46	0.1783
5/4	base_stock	1	2.544	30.36	30.36	0.1526
5/4	tabular2	1	2.72	17.69	17.69	0.1471
6/4	neural	1	0.8042	0.8364	0.8364	0.6143
6/4	polynomial	1	0.865	1.004	1.004	0.6497
6/4	random	1	7.541	77.65	77.65	0.8053
6/4	myopic	1	3.543	86.88	86.88	0.8977
6/4	base_stock	1	4.752	25.4	25.4	0.7107
6/4	tabular2	1	5.346	14.04	14.04	0.6834
7/4	neural	1	1.549	0.4998	0.4837	3.356
7/4	polynomial	1	1.809	1.227	1.211	3.542
7/4	random	1	10.41	78.46	78.46	4.315
7/4	myopic	1	3.659	87.25	87.24	4.151
7/4	base_stock	1	7.072	22.16	22.14	3.812
7/4	tabular2	1	8.011	10.46	10.46	3.411

4. TOLERANCE SENSITIVITY

The same 14 policies are evaluated at all three tolerances; these repeated audits are not independent replications. Changes depend on the completed suffix, so the table is descriptive rather than a global minimum-edit theorem.

TABLE II: Identical raw candidates under fixed and state-dependent budgets

d/L	Policy	ε	Fixed changes	Budget changes	Fixed bound	Budget bound
2/4	N32/2/301	1/1000	0	0	0	0
2/4	N32/2/301	1/100	0	0	0	0
2/4	N32/2/301	1/20	0	0	0	0
2/4	Myopic	1/1000	98	98	0	0
2/4	Myopic	1/100	98	98	0	0
2/4	Myopic	1/20	98	98	0	0
3/4	N32/2/301	1/1000	1	1	0	0
3/4	N32/2/301	1/100	1	1	0	0
3/4	N32/2/301	1/20	1	0	0	0.01355
3/4	Myopic	1/1000	443	443	0	0
3/4	Myopic	1/100	443	443	0	0
3/4	Myopic	1/20	443	440	0	0.01355
4/4	Myopic	1/1000	1763	1763	0	0
4/4	Myopic	1/100	1763	1763	0	0
4/4	Myopic	1/20	1763	1759	0	0.0325
4/4	N32/2/303	1/1000	61	61	0	0
4/4	N32/2/303	1/100	61	45	0	0.009483
4/4	N32/2/303	1/20	60	32	0.002989	0.0458
4/5	Myopic	1/1000	4183	4183	0	0
4/5	Myopic	1/100	4183	4183	0	0
4/5	Myopic	1/20	4183	4111	0	0.03431
4/5	N32/2/303	1/1000	118	113	0	3.962×10^{-4}
4/5	N32/2/303	1/100	113	82	3.962×10^{-4}	0.009483
4/5	N32/2/303	1/20	95	35	0.006091	0.04628
5/4	N32/2/303	1/1000	24	24	0	0
5/4	N32/2/303	1/100	24	24	0	0
5/4	N32/2/303	1/20	24	13	0	0.04974
5/4	Myopic	1/1000	7076	7076	0	0
5/4	Myopic	1/100	7076	7076	0	0
5/4	Myopic	1/20	7076	7061	0	0.04995
6/4	N32/2/303	1/1000	274	274	0	0
6/4	N32/2/303	1/100	274	274	0	0
6/4	N32/2/303	1/20	274	231	0	0.04879
6/4	Myopic	1/1000	28468	28462	0	5.167×10^{-4}
6/4	Myopic	1/100	28462	28462	5.167×10^{-4}	5.167×10^{-4}
6/4	Myopic	1/20	28462	28456	5.167×10^{-4}	0.01098
7/4	N32/2/303	1/1000	655	655	0	0
7/4	N32/2/303	1/100	655	634	0	0.009704
7/4	N32/2/303	1/20	655	537	0	0.02474
7/4	Myopic	1/1000	114348	114348	0	0
7/4	Myopic	1/100	114348	114341	0	0.004821
7/4	Myopic	1/20	114341	114341	0.004821	0.004821

5. MEMORY, BIT GROWTH, AND ENUMERATION WORK

TABLE III: Storage and arithmetic coverage; RSS is a process high-water mark in KiB

d/L	Model B	Policy B	Feasible terms	Dense slots	Bits	RSS KiB
2/4	6112	1024	1632	2560	80	322444
3/4	40672	4096	11880	17920	84	323596

d/L	Model B	Policy B	Feasible terms	Dense slots	Bits	RSS KiB
4/4	244448	16384	73296	110592	86	343612
4/5	596105	40000	197568	270000	87	349856
5/4	1371856	65536	415128	630784	89	365856
6/4	7328752	262144	2227392	3407872	91	465000
7/4	37733696	1048576	11500488	17694720	93	968908

6. ORIGINAL-ECONOMY ORACLE AND PRODUCTION EVIDENCE

The article gives the complete likelihood-ratio proof, analytic remainder construction, projected-poll theorem, and transport identity. The following tables retain the executed precision levels and matched production comparison. Class regret is relative to the fixed scalar preference-adjustment class, not the original unrestricted policy problem. A successful pointwise interval gate is a payoff comparison, whereas a smaller original full-domain bound is only a certificate improvement. The original financed dual-construction time remains unmeasured; all recorded central timing totals state their conditional scope.

TABLE IV: All executed scalar polling meshes and precision-dependent work

k	h	θ	Calls	$\ G_1\ $ bound	Seconds	Call bound
1/2	1/25	1/5	17	0.04001	0.3453	257166
1/2	1/50	1/5	2	0.02001	0.07249	975450
1/2	1/100	1/5	2	0.01001	0.07256	3901800
1/2	1/200	1/5	2	0.005001	0.07264	15607200
2	1/25	1/5	17	0.06211	0.3399	257166
2	1/50	9/50	5	0.03106	0.1249	997506
2	1/100	9/50	3	0.01553	0.08933	3988965
2	1/200	37/200	6	0.007763	0.1434	15955851

TABLE V: Analytic derivative remainder budgets (orders zero, one, and two)

k	Order	Flow	Terminal	Stopping	Total
1/2	0	1.51×10^{-15}	9.451×10^{-25}	6.198×10^{-30}	1.51×10^{-15}
1/2	1	3.019×10^{-14}	1.898×10^{-22}	2.673×10^{-13}	2.974×10^{-13}
1/2	2	1.208×10^{-12}	3.872×10^{-20}	7.56×10^{-12}	8.768×10^{-12}
2	0	1.51×10^{-15}	9.451×10^{-25}	6.198×10^{-30}	1.51×10^{-15}
2	1	3.019×10^{-14}	1.898×10^{-22}	2.673×10^{-13}	2.974×10^{-13}
2	2	1.208×10^{-12}	3.872×10^{-20}	7.566×10^{-12}	8.773×10^{-12}

TABLE VI: Full-path displacement and moment diagnostics

Seed	Order	Min step	Median	95th pct.	m discrepancy	v discrepancy
27101	8/16	0.0101	0.03226	3.328	1.148×10^{-18}	4.765×10^{-22}
27101	12/24	0.0101	0.03233	3.328	1.789×10^{-18}	3.706×10^{-22}
27102	8/16	0.01073	0.03079	2.493	2.386×10^{-18}	3.177×10^{-22}

Seed	Order	Min step	Median	95th pct.	m discrepancy	v discrepancy
27102	12/24	0.01073	0.03074	2.493	1.952×10^{-18}	5.294×10^{-22}
27103	8/16	0.02044	0.0489	3.154	7.387×10^{-18}	5.294×10^{-22}
27103	12/24	0.02038	0.04887	3.153	5.34×10^{-18}	4.236×10^{-22}

TABLE VII: Paired final-payoff interval: gated minus ungated

Seed	Method	Lower	Upper	Proved positive	Rejections
27201	Neural Adam	-3.9377×10^{-7}	3.9377×10^{-7}	no	0
27201	Direct Adam	-3.9377×10^{-7}	3.9377×10^{-7}	no	0
27201	Direct L-BFGS-B	-3.9397×10^{-7}	3.9357×10^{-7}	no	3
27202	Neural Adam	-3.9377×10^{-7}	3.9377×10^{-7}	no	0
27202	Direct Adam	-3.9377×10^{-7}	3.9377×10^{-7}	no	0
27202	Direct L-BFGS-B	-3.9365×10^{-7}	3.9389×10^{-7}	no	3

7. INTERPRETATION OF EXACT-COMMIT VALIDATION

The read-only validation workflow checks the final source commit from a clean checkout, verifies frozen scientific hashes, rebuilds the PDFs, and records the exact SHA, run identifier, and artifact identity in its output. The record is external to the commit it identifies, avoiding a self-referential hash assertion. Compilation, tests, and workflow success do not prove uninstantiated hypotheses of the original continuous-control theorems.

RESPONSE TO THE REFEREES: NEURAL BELLMAN OPERATORS, R29

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Peking University

1. OVERVIEW

We thank the referee for identifying both the absent R27 submission object and the scientific attribution problem. This revision supplies a materialized article, supplement, complete computational ledger, historical annex, code, frozen evidence, and executable validation. It retains the title, original economy, original .01 whole-domain target, and every earlier source and result. We respond through proofs, explicit oracles, independently evaluated controls, and complete disclosure of outcomes. We do not treat a response entry, successful compilation, or an auxiliary-model certificate as proof that the original all-state target has been achieved.

The R27 report is the latest report addressed. The R26 findings and the independent R25 second pass are cross-referenced individually below. The original R27 raw-run artifacts were not present at the reviewed commit; the later R28 execution is not represented as recovery of that missing record. The R29 extensions are exploratory relative to all prior outcomes, with fixed cohorts and programmatically enforced reference isolation.

2. R27

2.1. R27-F1 — Materialized referee object

The revision now contains a complete article, essential supplement, computational appendix, response, and historical annex, with source, frozen results, tests, and reproduction instructions. The previous design-only snapshot is not described as a completed manuscript. The current index identifies R29 and the exact reports addressed.

Evidence: Article; SUPP_R29; COMPUTATION_R29; HISTORY_R29; REVISION_MANIFEST.json

Disposition: Correction implemented with the specified evidence.

2.2. R27-F2 — Missing original raw R27 artifacts

The absent R27 raw-run files are not silently reconstructed or assigned retrospective hashes. The existing R28 run is identified by its source and evidence commits. All 34 of its raw policy files are preserved, and the new 81 candidates have their own training records, hashes, dependency audit, and freeze events before reference generation. This supplies a new auditable record, not proof of the missing historical chronology.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json; R28 frozen inventory records

Disposition: New evidence supplied; missing historical R27 record not retroactively established.

2.3. R27-F3 — Executable exact-head validation

The new validation workflow runs on every push to the new revision branch without a commit-message gate. It checks the frozen data and source identities from a clean checkout, rebuilds every current PDF, and retains the checked SHA and run identity in an external artifact. The separate expensive scientific-generation workflow is not confused with final-head validation.

Evidence: test_revision.py; source_audit/PRESERVATION.json; exact-commit validation artifact

Disposition: Correction implemented with the specified evidence.

2.4. R27-F4 — Generic repair versus neural contribution

Both repair rules are applied to neural, polynomial, ten-seed random-feasible, myopic, base-stock, and two-period tabular proposals. Raw regret, raw residual bounds, correction counts, suffix deficits, topology, and full stage costs are reported separately. The theorem is representation independent. Any neural benefit must be visible in the paired raw quality or burden comparison, rather than inferred from a completed certificate.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; SUPP_R29: attribution table

Disposition: Correction implemented with the specified evidence.

2.5. R27-F5 — Post-diagnostic status

The R27 completion amendment remains identified as post-diagnostic. R28 and R29 are later executions, and the new state-dependent budget rule and expanded controls are exploratory relative to prior outcomes. A committed cohort and programmatic reference isolation are useful reproducibility controls, but they are not described as investigator blinding or retroactive preregistration.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

2.6. R27-F6 — Small unknown-solution benchmark

The unchanged original four models are retained and the fixed cohort adds dimensions five, six, and seven, reaching 16,384 states per slice. The study measures both approximation and exact-enumeration costs. These larger finite graphs are a scaling diagnostic, not evidence that neural approximation is necessary or that continuous high-dimensional control has been solved.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.7. R27-F7 — Tabular certification complexity

The paper states feasible and dense state-action-successor counts, $O(\text{TNAD})$ arithmetic enumeration, exact-integer bit growth, and model/policy storage. Exact dynamic programming remains visible in the main scaling table. A complete action-table audit is not claimed to evade the curse of dimensionality.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

2.8. R27-F8 — *Raw four-dimensional failure*

The raw neural policies, including the previously reported loss near 0.36656, are preserved. The complete ledger distinguishes raw loss, raw bound, each completed bound, and edited actions for every policy. Fitting, completion, verification, and exact reference costs are separate. The raw failure is neither removed nor relabeled as neural success after repair.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

2.9. R27-F9 — *Constrained stationarity*

The article states the projected-gradient mapping, a precision-aware finite-work theorem on a closed box, the short-face alternative, and explicit unresolved-oracle semantics. Its assumptions are instantiated in two original-economy slices at four meshes with measured oracle calls, validated derivative bounds, and boundary or interior optima.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

2.10. R27-F10 — *Scalar slice versus original policy class*

The fixed consumption, zero risky investment, constant preference-adjustment interval, initial state, and two cost coefficients are stated explicitly. The class regret certificate is not substituted for the financed 47-dimensional class or for all states and restarts of the original economy. The economic comparative-static content is retained without changing the original target.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Correction implemented with the specified evidence.

2.11. R27-F11 — *Stopped derivative bridge*

A change-of-path-measure argument gives value, first-derivative, and second-derivative stopping-error bounds for the scalar class, including the explicit parameter dependence of the quadratic cost. Analytic Gaussian-tail and polynomial remainders are enclosed with directed arithmetic. The general financed 47-dimensional derivative-error constants are still not instantiated; finite secants are not presented as that bridge.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: financing-gradient formula

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.12. R27-F12 — *Historical-transport isolation*

The six executed paths independently implement direct neural differentiation and quotient-gradient pullback with the same carrier, Adam state, and full nonlinear recentering. Every nonlinear remainder is retained and there is no unmatched displacement cap. Parameter, output, moment, and displacement diagnostics are included. This is a path-equivalence experiment, not a cheaper carrier-free optimizer.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

2.13. R27-F13 — *Production rollback*

The fixed production-schedule study contains twelve gated and ungated trajectories, matched by seed and optimizer, including direct L-BFGS-B. Every checkpoint interval, strict comparison, rejection/restoration record, gradient count, and component timing is retained. The paired final-payoff interval determines whether an improvement is actually proved; restoration correctness alone is not called solver superiority.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

2.14. R27-F14 — *Original whole-domain target*

The original current-state, all-state, all-restart tolerance remains 0.01. The inherited bound is 7.181834580823298. The revision develops constructive finite-state and scalar stopped-control guarantees but does not claim that they close this original quantitative requirement. Neither the title nor the original economy is replaced by the auxiliary model.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

2.15. R27-F15 — *Stale review lineage*

The index now names R29, the R27 report and exact review head, and the R26 and R25 second-pass crosswalks. The previous index is archived byte-for-byte. The missing review files on the R28 base are carried into the new branch as their original Git blobs rather than rewritten summaries.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

2.16. R27-F16 — *Literature and novelty*

The positioning distinguishes fitted-value-iteration error analysis, constrained generating-set search, verified arithmetic, information relaxation, neural stochastic control, and computational economics. The additions are identified as an explicit residual-budget preservation operator, an interval-oracle boundary theorem with stated work, a stopped scalar oracle, and independently executed mechanism controls. The paper does not claim to invent Bellman comparison, policy iteration, direct search, or interval arithmetic.

Evidence: Article: literature positioning; paper/references.tex

Disposition: Correction implemented with the specified evidence.

2.17. R27-F17 — *Robustness beyond two seeds*

The width-32, depth-two panel has ten seeds at the four-dimensional L=4 model and five at L=5; the original architecture controls remain. Ten random policies per original model, larger dimensions, and a paired three-tolerance panel are added under fixed rules. This is structured computational coverage, not a complete robustness theorem over all charts, optimizers, or economies.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.18. R27-F18 — *Certified deployment object*

The deployed finite-state object is the exact compiled action table after any replacements. The raw neural network remains the proposal object. An unchanged neural execution certificate would need separate floating-point execution analysis, which is not asserted. Each raw and completed policy has its own hash and edit record.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; per-policy action tables

Disposition: Correction implemented with the specified evidence.

2.19. R27-T1 — *Residual identity and performance bound*

The article defines the policy value, nonnegative local Bellman deficit, and state-sensitive residual envelope. A backward proof covers terminal payoff, settlement, feasible actions, time dependence, and every restart. The new budget theorem additionally proves pointwise improvement over the original candidate and nonnegative remaining allowances.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls

Disposition: Correction implemented with the specified evidence.

2.20. R27-T2 — *Information available to completion*

Completion uses the full transition law, exact rewards, exact completed-suffix values, all feasible actions, and exact arithmetic. All of these are stated as powerful model resources also available to exact dynamic programming. Absence of an optimal-value file is not equated with a weaker computational oracle.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; main scaling table

Disposition: Correction implemented with the specified evidence.

2.21. R27-T3 — *Trivial-policy controls*

Random-feasible, myopic, base-stock, and two-period tabular policies are executed under both identical completion rules. The results retain all seeds and failures and report correction burden and costs alongside neural and polynomial proposals.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: Correction implemented with the specified evidence.

2.22. R27-T4 — *Correction topology*

Every changed action records time, state index, original and replacement action, exact suffix deficit, and local allowance. Aggregations distinguish total inventory, minimum inventory, and number of empty coordinates. Restart-specific worst losses and near-default edit counts are reported in the computational appendix.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: topology

Disposition: Correction implemented with the specified evidence.

2.23. R27-T5 — *Exact arithmetic cost*

The implementation records arbitrary-precision integer and Fraction backends, logical and dense successor counts, value/denominator bit lengths, wall time, CPU time, model arrays, policy bytes, and scoped process high-water memory. It does not call cumulative RSS an isolated per-case peak. Inherited dual setup time is still marked unmeasured.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.24. R27-T6 — *Remaining-horizon tolerance allocation*

The original fixed eta rule is retained unchanged as a control. The new theorem propagates the already completed suffix envelope and allocates the remaining state-dependent budget, with all-state/all-restart accuracy and policy-preservation proofs. Both rules are evaluated at 0.001, 0.01, and 0.05 on the same tolerance cohort.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; SUPP_R29: tolerance panel

Disposition: Correction implemented with the specified evidence.

2.25. R27-T7 — *Separate numerical stages*

The paper separates candidate generation and compilation, raw audit, completion, completed certification, exact reference, and gap comparison. Full model and reference work are charged to non-amortized component sums. Startup and serialization remain visible at cohort level and are not assigned artificial per-method estimates.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

2.26. R27-T8 — *Programmatic reference isolation*

The runtime audit rejects optimal-reference reads during candidate generation and completion, and the independent full-horizon solver is disabled in that context. Candidate, variant, and model-freeze hashes precede reference-creation events. Validation checks these events and file identities. This does not establish recovery of the missing R27 chronology.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json; controls.py; test_revision.py

Disposition: Correction implemented with the specified evidence.

2.27. R27-T9 — *Strongest classical baseline*

Exact scalar backward induction is in the article scaling table, with model construction and solve times retained. Direct L-BFGS-B remains in the continuous production and historical comparisons. Neither is hidden in an appendix to imply a neural advantage.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

2.28. R27-T10 — *Finite versus continuous certification*

The finite-state theorem enumerates a graph; the original continuous problem requires domain coverage, derivative or residual enclosures, and a stopped-control comparison. The paper states these as separate complexity regimes. The largest finite graph is not claimed to establish continuous-domain verifier scaling.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Correction implemented with the specified evidence.

2.29. R27-N1 — *Complete referee copy*

The revision now contains a complete article, essential supplement, computational appendix, response, and historical annex, with source, frozen results, tests, and reproduction instructions. The previous design-only snapshot is not described as a completed manuscript. The current index identifies R29 and the exact reports addressed.

Evidence: Article; SUPP_R29; COMPUTATION_R29; HISTORY_R29; REVISION_MANIFEST.json

Disposition: Correction implemented with the specified evidence.

2.30. R27-N2 — *Immutable raw evidence*

The absent R27 raw-run files are not silently reconstructed or assigned retrospective hashes. The existing R28 run is identified by its source and evidence commits. All 34 of its raw policy files are preserved, and the new 81 candidates have their own training records, hashes, dependency audit, and freeze events before reference generation. This supplies a new auditable record, not proof of the missing historical chronology.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json; R28 frozen inventory records

Disposition: New evidence supplied; missing historical R27 record not retroactively established.

2.31. R27-N3 — *R26 and R25 crosswalk*

The index now names R29, the R27 report and exact review head, and the R26 and R25 second-pass crosswalks. The previous index is archived byte-for-byte. The missing review files on the R28 base are carried into the new branch as their original Git blobs rather than rewritten summaries.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

2.32. R27-N4 — *Attribution-proof completion*

Both repair rules are applied to neural, polynomial, ten-seed random-feasible, myopic, base-stock, and two-period tabular proposals. Raw regret, raw residual bounds, correction counts, suffix deficits, topology, and full stage costs are reported separately. The theorem is representation independent. Any neural benefit must be visible in the paired raw quality or burden comparison, rather than inferred from a completed certificate.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; SUPP_R29: attribution table

Disposition: Correction implemented with the specified evidence.

2.33. R27-N5 — *Complete completion proof*

The article defines the policy value, nonnegative local Bellman deficit, and state-sensitive residual envelope. A backward proof covers terminal payoff, settlement, feasible actions, time dependence, and every restart. The new budget theorem additionally proves pointwise improvement over the original candidate and nonnegative remaining allowances.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls

Disposition: Correction implemented with the specified evidence.

2.34. R27-N6 — *Executed constrained stationarity*

The article states the projected-gradient mapping, a precision-aware finite-work theorem on a closed box, the short-face alternative, and explicit unresolved-oracle semantics. Its assumptions are instantiated in two original-economy slices at four meshes with measured oracle calls, validated derivative bounds, and boundary or interior optima.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

2.35. R27-N7 — *Executed faithful transport*

The six executed paths independently implement direct neural differentiation and quotient-gradient pullback with the same carrier, Adam state, and full nonlinear recentering. Every nonlinear remainder is retained and there is no unmatched displacement cap. Parameter, output, moment, and displacement diagnostics are included. This is a path-equivalence experiment, not a cheaper carrier-free optimizer.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

2.36. R27-N8 — *Paired production rollback*

The fixed production-schedule study contains twelve gated and ungated trajectories, matched by seed and optimizer, including direct L-BFGS-B. Every checkpoint interval, strict comparison, rejection/restoration record, gradient count, and component timing is retained. The paired final-payoff interval determines whether an improvement is actually proved; restoration correctness alone is not called solver superiority.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

2.37. R27-N9 — *Standalone cost ledger*

The paper separates candidate generation and compilation, raw audit, completion, completed certification, exact reference, and gap comparison. Full model and reference work are charged to non-amortized component sums. Startup and serialization remain visible at cohort level and are not assigned artificial per-method estimates. The earlier central dual-construction cost is not recoverable from its timing records and remains unmeasured.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.38. R27-N10 — *Scaling evidence*

The unchanged original four models are retained and the fixed cohort adds dimensions five, six, and seven, reaching 16,384 states per slice. The study measures both approximation and exact-enumeration costs. These larger finite graphs are a scaling diagnostic, not evidence that neural approximation is necessary or that continuous high-dimensional control has been solved.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.39. R27-N11 — *Primary-source positioning*

The positioning distinguishes fitted-value-iteration error analysis, constrained generating-set search, verified arithmetic, information relaxation, neural stochastic control, and computational economics. The additions are identified as an explicit residual-budget preservation operator, an interval-oracle boundary theorem with stated work, a stopped scalar oracle, and independently executed mechanism controls. The paper does not claim to invent Bellman comparison, policy iteration, direct search, or interval arithmetic.

Evidence: Article: literature positioning; paper/references.tex

Disposition: Correction implemented with the specified evidence.

2.40. R27-N12 — *Current lineage*

The index now names R29, the R27 report and exact review head, and the R26 and R25 second-pass crosswalks. The previous index is archived byte-for-byte. The missing review files on the R28 base are carried into the new branch as their original Git blobs rather than rewritten summaries.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

2.41. R27-N13 — *Exact-head CI*

The new validation workflow runs on every push to the new revision branch without a commit-message gate. It checks the frozen data and source identities from a clean checkout, rebuilds every current PDF, and retains the checked SHA and run identity in an external artifact. The separate expensive scientific-generation workflow is not confused with final-head validation.

Evidence: test_revision.py; source_audit/PRESERVATION.json; exact-commit validation artifact

Disposition: Correction implemented with the specified evidence.

3. R26

3.1. R26-F1 — *Latest-review mismatch*

The response identifies the actual R27 report as latest and includes a complete item-by-item R26 and R25 second-pass matrix. The original reports are preserved, not replaced by the stale R24 citation. The revision index and manifest now agree.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

3.2. R26-F2 — Original 718-fold gap

The original 0.01 target is unchanged, and the retained all-domain bound remains 7.181834580823298. The new finite-state and scalar-class proofs are separate results. No table, abstract, or test flag reports the original target as closed.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

3.3. R26-F3 — No demonstrated neural advantage

The strongest exact tabular and direct L-BFGS-B comparators remain in the main text. Raw neural success, repair burden, and total costs are reported separately. The neural representation is not credited with a generic exact-model repair outcome, and numerical superiority is not inferred from feasibility.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

3.4. R26-F4 — Conditional precision complexity

The full work expression includes the interval-oracle factor $W(\kappa h^2)$. The derivative and payoff oracle is instantiated at the four executed mesh levels of the stopped scalar class. The theorem does not assert arbitrary precision of the 47-dimensional checker or a bare epsilon-inverse-square total-work result.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.5. R26-F5 — Boundary exclusion

The projected-gradient theorem includes feasible full coordinate steps and geometric short-face cases at closed-box boundaries. Oracle failure is distinct from an infeasible step. Both a binding upper-face optimum and an interior optimum are executed in the original-economy scalar class.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.6. R26-F6 — Stopped gradient

The scalar likelihood-ratio bridge is proved and instantiated with directed analytic remainder budgets, including explicit cost derivatives. The financing-gradient formula retains all approximation and root-shift terms. Its general 47-dimensional constants remain uninstantiated.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: financing-gradient formula

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.7. R26-F7 — *Confounded historical transport*

The new independent control recenters to the full nonlinear carrier output after each step, retains the complete remainder, and removes the unmatched output cap. Six paths report parameter, quotient, and moment agreement. The old capped/drifted runs remain archived and are not reinterpreted as faithful controls.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

3.8. R26-F8 — *Post-diagnostic stress versus production*

The inherited rate-20.48 stress experiment remains historical engineering evidence. The later production study fixes rates and paired gated/ungated rules before that study, retains all checkpoints, and reports interval comparisons of final payoffs. It is not relabeled as the original R25 held-out test.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

3.9. R26-F9 — *Known-solution manufactured example*

The manufactured problem is retained as a verifier stress test with its known-witness information stated. The unknown-solution inventory cohort trains only on model primitives and its own approximate suffix values, then freezes policies before a separately generated exact reference.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

3.10. R26-F10 — *High-dimensional verification scaling*

The new finite graphs reach dimension seven and 16,384 states and record enumeration, memory, and bit growth. This broadens finite-state coverage. It does not supply a high-dimensional continuous Hessian/interval-cover complexity theorem, which remains a distinct requirement.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.11. R26-F11 — *From-scratch cost*

Model construction, proposal generation, compilation, completion, certification, and independent reference are separately recorded for the new inventory executions. Non-amortized totals and resource scope are explicit. The old central dual-construction timing is not invented or inferred from later checker times.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.12. R26-F12 — *Parameterization robustness*

The expanded seed and architecture cohorts, trivial controls, state sizes, and tolerance panels are all retained. The central production study also compares neural and direct optimizers under matched gates. This is wider deterministic coverage, not proof of robustness to every parameter chart or training rule.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.13. R26-F13 — *Sparse literature*

The article retains the expanded primary bibliography and adds direct-search stationarity comparisons. Each numerical ingredient is attributed to its established literature, and the contribution statement identifies the added oracle, budget rule, coupling result, and application rather than a generic claim of invention.

Evidence: Article: literature positioning; paper/references.tex

Disposition: Correction implemented with the specified evidence.

3.14. R26-F14 — *Bound-improvement decomposition*

The same-cover warm-start and retained-actor bounds, the separate cover refinement, and the all-restart suffix proof remain in the article and historical annex. A tighter bound is not asserted to prove higher original-domain payoff. Only a strict payoff interval comparison or the finite-model preservation theorem supplies that different conclusion.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json; Article: all-restart suffix envelope

Disposition: Correction implemented with the specified evidence.

3.15. R26-T1 — *Full oracle work*

The work theorem displays $W(\kappa h^2)$ and reports actual oracle work at executed precisions. No headline substitutes an iteration count for total verified work.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.16. R26-T2 — *Oracle failure semantics*

A feasible point with unresolved verification cannot complete a stationarity poll. The algorithm and theorem distinguish this event from an infeasible full step at a nearby face.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.17. R26-T3 — *Exact coefficients versus execution*

Rational policy-table certificates and directed coefficient enclosures are distinguished from floating-point network execution. The repaired finite-state table is the deployed object.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls

Disposition: Correction implemented with the specified evidence.

3.18. R26-T4 — *Structurally advantaged comparator*

The manufactured reference remains labeled as known-solution and its direct comparator as structurally informed. The independent unknown-solution experiment is separate; neither universal neural inferiority nor superiority is inferred from the manufactured case.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; historical annex

Disposition: Correction implemented with the specified evidence.

3.19. R26-T5 — *Tests versus theorem hypotheses*

Validation checks executable identities, hashes, fixed cohorts, and recorded numerical enclosures. The manuscript explicitly states that passing these checks does not establish hypotheses that have not been analytically or numerically instantiated.

Evidence: test_revision.py; source_audit/PRESERVATION.json; exact-commit validation artifact; Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.20. R26-T6 — *Trajectory distributions*

The transport ledger retains every step and reports displacement distribution, nonlinear remainder, moment mismatch, and carrier alignment. The production ledger retains every candidate interval and deployment decision.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

3.21. R26-T7 — *Distinct guarantees*

The paper distinguishes monotone deployed payoff, constrained stationarity, finite-state epsilon optimality, and Adam convergence. Only the corresponding proved guarantee is asserted for each operator.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

3.22. R26-T8 — *Economic content*

The unchanged stopped economy links costly preference adjustment to a binding boundary at low adjustment cost and an interior solution at the higher cost. This is a verified comparative-static result within the stated class; broader optimal-policy implications are not inferred.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.23. R26-N1 — *R25 second-pass disposition*

The response identifies the actual R27 report as latest and includes a complete item-by-item R26 and R25 second-pass matrix. The original reports are preserved, not replaced by the stale R24 citation. The revision index and manifest now agree.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

3.24. R26-N2 — *Original quantitative target*

The original 0.01 target is unchanged, and the retained all-domain bound remains 7.181834580823298. The new finite-state and scalar-class proofs are separate results. No table, abstract, or test flag reports the original target as closed.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

3.25. R26-N3 — *Unknown-solution stochastic control*

The manufactured problem is retained as a verifier stress test with its known-witness information stated. The unknown-solution inventory cohort trains only on model primitives and its own approximate suffix values, then freezes policies before a separately generated exact reference.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

3.26. R26-N4 — *Instantiate meaningful original-economy control*

The full work expression includes the interval-oracle factor $W(\kappa h^2)$. The derivative and payoff oracle is instantiated at the four executed mesh levels of the stopped scalar class. The theorem does not assert arbitrary precision of the 47-dimensional checker or a bare epsilon-inverse-square total-work result.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.27. R26-N5 — *Boundary-aware proof*

The projected-gradient theorem includes feasible full coordinate steps and geometric short-face cases at closed-box boundaries. Oracle failure is distinct from an infeasible step. Both a binding upper-face optimum and an interior optimum are executed in the original-economy scalar class.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.28. R26-N6 — *Faithful mechanism isolation*

The new independent control recenters to the full nonlinear carrier output after each step, retains the complete remainder, and removes the unmatched output cap. Six paths report parameter, quotient, and moment agreement. The old capped/drifted runs remain archived and are not reinterpreted as faithful controls.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

3.29. R26-N7 — *Production rollback regime*

The inherited rate-20.48 stress experiment remains historical engineering evidence. The later production study fixes rates and paired gated/ungated rules before that study, retains all checkpoints, and reports interval comparisons of final payoffs. It is not relabeled as the original R25 held-out test.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

3.30. R26-N8 — *Dimension and tolerance scaling*

The new finite graphs reach dimension seven and 16,384 states and record enumeration, memory, and bit growth. This broadens finite-state coverage. It does not supply a high-dimensional continuous Hessian/interval-cover complexity theorem, which remains a distinct requirement.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.31. R26-N9 — *All infrastructure cost*

Model construction, proposal generation, compilation, completion, certification, and independent reference are separately recorded for the new inventory executions. Non-amortized totals and resource scope are explicit. The old central dual-construction timing is not invented or inferred from later checker times.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.32. R26-N10 — *Primary literature*

The article retains the expanded primary bibliography and adds direct-search stationarity comparisons. Each numerical ingredient is attributed to its established literature, and the contribution statement identifies the added oracle, budget rule, coupling result, and application rather than a generic claim of invention.

Evidence: Article: literature positioning; paper/references.tex

Disposition: Correction implemented with the specified evidence.

3.33. R26-N11 — *Bound decomposition*

The same-cover warm-start and retained-actor bounds, the separate cover refinement, and the all-restart suffix proof remain in the article and historical annex. A tighter bound is not asserted to prove higher original-domain payoff. Only a strict payoff interval comparison or the finite-model preservation theorem supplies that different conclusion.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json; Article: all-restart suffix envelope

Disposition: Correction implemented with the specified evidence.

3.34. R26-N12 — *Title and claim alignment*

The title Neural Bellman Operators is retained, as are the original economic problem and quantitative target. The abstract and theorem statements identify actual neural-proposal, certified-completion, and continuous-control contributions, rather than replacing the project by a weaker problem or asserting an unsupported neural speedup.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

4. R25-SP

4.1. R25-SP-F1 — *Integrated manuscript*

The revision now contains a complete article, essential supplement, computational appendix, response, and historical annex, with source, frozen results, tests, and reproduction instructions. The previous design-only snapshot is not described as a completed manuscript. The current index identifies R29 and the exact reports addressed.

Evidence: Article; SUPP_R29; COMPUTATION_R29; HISTORY_R29; REVISION_MANIFEST.json

Disposition: Correction implemented with the specified evidence.

4.2. R25-SP-F2 — *Original whole-domain accuracy*

The original current-state, all-state, all-restart tolerance remains 0.01. The inherited bound is 7.181834580823298. The revision develops constructive finite-state and scalar stopped-control guarantees but does not claim that they close this original quantitative requirement. Neither the title nor the original economy is replaced by the auxiliary model.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

4.3. R25-SP-F3 — *Held-out frontier*

The strongest exact tabular and direct L-BFGS-B comparators remain in the main text. Raw neural success, repair burden, and total costs are reported separately. The neural representation is not credited with a generic exact-model repair outcome, and numerical superiority is not inferred from feasibility.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

4.4. R25-SP-F4 — *Faithful moving transport*

The new independent control recenters to the full nonlinear carrier output after each step, retains the complete remainder, and removes the unmatched output cap. Six paths report parameter, quotient, and moment agreement. The old capped/drifted runs remain archived and are not reinterpreted as faithful controls.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

4.5. R25-SP-F5 — *Cap-driven nominal learning rates*

The old nominal-rate tuning is retained as a result for the capped algorithm, not clean identification of uncapped learning-rate effects. The new recentered transport experiment has no unmatched cap and reports the complete effective output-step distribution, nonlinear remainder, and moment discrepancies. It does not retroactively change the earlier tuning outcome.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.6. R25-SP-F6 — *Known-witness stochastic benchmark*

The manufactured problem is retained as a verifier stress test with its known-witness information stated. The unknown-solution inventory cohort trains only on model primitives and its own approximate suffix values, then freezes policies before a separately generated exact reference.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

4.7. R25-SP-F7 — *Stopped objective gradient*

A change-of-path-measure argument gives value, first-derivative, and second-derivative stopping-error bounds for the scalar class, including the explicit parameter dependence of the quadratic cost. Analytic Gaussian-tail and polynomial remainders are enclosed with directed arithmetic. The general financed 47-dimensional derivative-error constants are still not instantiated; finite secants are not presented as that bridge.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: financing-gradient formula

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

4.8. R25-SP-F8 — *Rollback exercised*

The fixed production-schedule study contains twelve gated and ungated trajectories, matched by seed and optimizer, including direct L-BFGS-B. Every checkpoint interval, strict comparison, rejection/restoration record, gradient count, and component timing is retained. The paired final-payoff interval determines whether an improvement is actually proved; restoration correctness alone is not called solver superiority.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.9. R25-SP-F9 — *Certificate versus payoff improvement*

The same-cover warm-start and retained-actor bounds, the separate cover refinement, and the all-restart suffix proof remain in the article and historical annex. A tighter bound is not asserted to prove higher original-domain payoff. Only a strict payoff interval comparison or the finite-model preservation theorem supplies that different conclusion.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json; Article: all-restart suffix envelope

Disposition: Correction implemented with the specified evidence.

4.10. R25-SP-F10 — Dominant verification and full costs

Model construction, proposal generation, compilation, completion, certification, and independent reference are separately recorded for the new inventory executions. Non-amortized totals and resource scope are explicit. The old central dual-construction timing is not invented or inferred from later checker times.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

4.11. R25-SP-F11 — Robustness coverage

The width-32, depth-two panel has ten seeds at the four-dimensional L=4 model and five at L=5; the original architecture controls remain. Ten random policies per original model, larger dimensions, and a paired three-tolerance panel are added under fixed rules. This is structured computational coverage, not a complete robustness theorem over all charts, optimizers, or economies.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

4.12. R25-SP-F12 — Performance and convergence theorem

The article now supplies complete all-state/all-restart finite-state performance theorems, a projected-stationarity theorem with explicit precision-dependent work, and a rigorously instantiated scalar original-economy oracle. These are constructive guarantees. They do not constitute an epsilon-accurate convergence theorem for the original full continuous-state neural iteration.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

4.13. R25-SP-T1 — Current revision index

The index now names R29, the R27 report and exact review head, and the R26 and R25 second-pass crosswalks. The previous index is archived byte-for-byte. The missing review files on the R28 base are carried into the new branch as their original Git blobs rather than rewritten summaries.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

4.14. R25-SP-T2 — Preserve chronology

The R27 completion amendment remains identified as post-diagnostic. R28 and R29 are later executions, and the new state-dependent budget rule and expanded controls are exploratory relative to prior outcomes. A committed cohort and programmatic reference isolation are useful reproducibility controls, but they are not described as investigator blinding or retroactive preregistration.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

4.15. *R25-SP-T3 — Effective output-step statistics*

The transport ledger retains every step and reports displacement distribution, nonlinear remainder, moment mismatch, and carrier alignment. The production ledger retains every candidate interval and deployment decision.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.16. *R25-SP-T4 — Carrier recentering*

The six executed paths independently implement direct neural differentiation and quotient-gradient pullback with the same carrier, Adam state, and full nonlinear recentering. Every nonlinear remainder is retained and there is no unmatched displacement cap. Parameter, output, moment, and displacement diagnostics are included. This is a path-equivalence experiment, not a cheaper carrier-free optimizer.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

4.17. *R25-SP-T5 — Known-structure comparator*

The manufactured reference remains labeled as known-solution and its direct comparator as structurally informed. The independent unknown-solution experiment is separate; neither universal neural inferiority nor superiority is inferred from the manufactured case.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; historical annex

Disposition: Correction implemented with the specified evidence.

4.18. *R25-SP-T6 — All-restart suffix proof*

The same-cover warm-start and retained-actor bounds, the separate cover refinement, and the all-restart suffix proof remain in the article and historical annex. A tighter bound is not asserted to prove higher original-domain payoff. Only a strict payoff interval comparison or the finite-model preservation theorem supplies that different conclusion.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json; Article: all-restart suffix envelope

Disposition: Correction implemented with the specified evidence.

4.19. *R25-SP-T7 — Deployment versus regret-bound gate*

The paper distinguishes monotone deployed payoff, constrained stationarity, finite-state epsilon optimality, and Adam convergence. Only the corresponding proved guarantee is asserted for each operator.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.20. *R25-SP-T8 — No pseudo-replication*

Seeds, architectures, dimensions, repeated tolerance checks, time slabs, and verification cells are reported as computational coverage. They are not pooled as independent observations of one statistical treatment effect. Paired repairs and gated/ungated paths retain their dependence explicitly.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: Correction implemented with the specified evidence.

4.21. *R25-SP-N1 — Materialize and seal referee object*

The revision now contains a complete article, essential supplement, computational appendix, response, and historical annex, with source, frozen results, tests, and reproduction instructions. The previous design-only snapshot is not described as a completed manuscript. The current index identifies R29 and the exact reports addressed.

Evidence: Article; SUPP_R29; COMPUTATION_R29; HISTORY_R29; REVISION_MANIFEST.json

Disposition: Correction implemented with the specified evidence.

4.22. *R25-SP-N2 — Original orders-of-magnitude progress*

The original current-state, all-state, all-restart tolerance remains 0.01. The inherited bound is 7.181834580823298. The revision develops constructive finite-state and scalar stopped-control guarantees but does not claim that they close this original quantitative requirement. Neither the title nor the original economy is replaced by the auxiliary model.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

4.23. *R25-SP-N3 — Faithful neural geometry*

The six executed paths independently implement direct neural differentiation and quotient-gradient pullback with the same carrier, Adam state, and full nonlinear recentering. Every nonlinear remainder is retained and there is no unmatched displacement cap. Parameter, output, moment, and displacement diagnostics are included. This is a path-equivalence experiment, not a cheaper carrier-free optimizer.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

4.24. *R25-SP-N4 — Proposal-gradient bridge*

A change-of-path-measure argument gives value, first-derivative, and second-derivative stopping-error bounds for the scalar class, including the explicit parameter dependence of the quadratic cost. Analytic Gaussian-tail and polynomial remainders are enclosed with directed arithmetic. The general financed 47-dimensional derivative-error constants are still not instantiated; finite secants are not presented as that bridge.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: financing-gradient formula

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

4.25. *R25-SP-N5 — Unknown-solution stochastic benchmark*

The manufactured problem is retained as a verifier stress test with its known-witness information stated. The unknown-solution inventory cohort trains only on model primitives and its own approximate suffix values, then freezes policies before a separately generated exact reference.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

4.26. *R25-SP-N6 — Natural production rejection*

The fixed production-schedule study contains twelve gated and ungated trajectories, matched by seed and optimizer, including direct L-BFGS-B. Every checkpoint interval, strict comparison, rejection/restoration record, gradient count, and component timing is retained. The paired final-payoff interval determines whether an improvement is actually proved; restoration correctness alone is not called solver superiority.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.27. *R25-SP-N7 — Broader robustness and scaling*

The width-32, depth-two panel has ten seeds at the four-dimensional $L=4$ model and five at $L=5$; the original architecture controls remain. Ten random policies per original model, larger dimensions, and a paired three-tolerance panel are added under fixed rules. This is structured computational coverage, not a complete robustness theorem over all charts, optimizers, or economies.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

COMPUTATIONAL APPENDIX TO NEURAL BELLMAN OPERATORS: R29

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1. COMPLETE FIXED-COHORT RECORD

Every candidate appears below. N denotes width/depth/seed minus 27000; P denotes polynomial degree; R denotes random seed minus 29000. Raw regret is computed only after the candidate has been frozen. Raw and completed bounds are reference-free exact residual certificates. The complete statewise values, action tables, restart losses, and correction topology are retained as machine-readable files.

TABLE I: All 115 candidates at .01: raw performance, completion bounds, and action changes

d/L	Policy	Raw regret	Raw bound	Fixed bound	Budget bound	Fixed edits	Budget edits
2/4	N16/1/301	0	0	0	0	0	0
2/4	N16/1/302	0	0	0	0	0	0
2/4	N16/2/301	0	0	0	0	0	0
2/4	N16/2/302	0	0	0	0	0	0
2/4	N32/1/301	0	0	0	0	0	0
2/4	N32/1/302	0	0	0	0	0	0
2/4	N32/2/301	0	0	0	0	0	0
2/4	N32/2/302	0	0	0	0	0	0
2/4	P2	0.1391	0.1571	2.089×10^{-4}	2.089×10^{-4}	7	7
2/4	P3	0.1391	0.1571	0	0	7	7
2/4	R1	4.767	7.419	0	0	84	84
2/4	R2	5.054	8.931	2.089×10^{-4}	2.089×10^{-4}	84	84
2/4	R3	4.598	7.377	0	0	76	76
2/4	R4	4.214	7.046	0	0	75	75
2/4	R5	4.083	6.029	0	0	76	76
2/4	R6	4.134	8.743	0	0	78	78
2/4	R7	4.899	8.06	0	0	71	71
2/4	R8	3.964	6.25	0	0	73	73
2/4	R9	5.019	7.77	2.089×10^{-4}	2.089×10^{-4}	76	76
2/4	R10	4.293	6.747	0	0	76	76
2/4	Myopic	3.849	4.472	0	0	98	98
2/4	Base stock	1.935	2.017	2.089×10^{-4}	2.089×10^{-4}	58	58
2/4	Two-period	0.8426	1.111	2.089×10^{-4}	2.089×10^{-4}	32	32
3/4	N16/1/301	0.01418	0.01418	0	0	3	3
3/4	N16/1/302	0.01418	0.01418	0	0	3	3
3/4	N16/2/301	0	0	0	0	0	0
3/4	N16/2/302	0	0	0	0	0	0
3/4	N32/1/301	0.01355	0.01355	0	0	2	2
3/4	N32/1/302	0.01418	0.01418	0	0	1	1
3/4	N32/2/301	0.01355	0.01355	0	0	1	1
3/4	N32/2/302	0.01418	0.01418	0	0	3	3
3/4	P2	0	0	0	0	0	0
3/4	P3	0	0	0	0	0	0

d/L	Policy	Raw regret	Raw bound	Fixed bound	Budget bound	Fixed edits	Budget edits
3/4	R1	5.048	9.497	0	0	350	350
3/4	R2	4.797	9.901	0	0.006601	351	350
3/4	R3	4.472	8.613	0	0	344	344
3/4	R4	4.523	8.488	0	0	351	351
3/4	R5	4.496	9.133	0	0.006601	337	336
3/4	R6	5.141	11.62	0	0.006601	362	361
3/4	R7	4.578	10.21	0	0.006601	342	341
3/4	R8	4.761	8.865	0	0	351	351
3/4	R9	4.781	7.755	0	0.006601	348	347
3/4	R10	4.987	10.16	0	0	343	343
3/4	Myopic	3.59	5.165	0	0	443	443
3/4	Base stock	1.339	1.374	0	0	199	199
3/4	Two-period	0.7878	0.8156	0	0	93	93
4/4	N16/1/301	0.3666	0.4915	0	0.009483	61	47
4/4	N16/1/302	0.3666	0.4914	0	0.006551	59	46
4/4	N16/2/301	0.3666	0.4915	0	0.009483	55	43
4/4	N16/2/302	0.3666	0.4915	0	0.009483	56	46
4/4	N32/1/301	0.3666	0.4914	0	0.006551	61	48
4/4	N32/1/302	0.3666	0.4914	0	0.006551	59	47
4/4	N32/2/301	0.3666	0.4915	0	0.009483	61	48
4/4	N32/2/302	0.3666	0.4915	0	0.009483	58	44
4/4	P2	0.03934	0.03934	0	0.006551	17	4
4/4	P3	0.3666	0.4914	0	0.006551	59	46
4/4	R1	5.736	15.08	0	0.009483	1515	1510
4/4	R2	5.253	12.99	0	0.009483	1516	1509
4/4	R3	4.998	11.95	0	0.006551	1493	1491
4/4	R4	4.437	10.65	0	0.009483	1512	1506
4/4	R5	4.593	12.53	0	0.009483	1471	1465
4/4	R6	4.969	12.96	0	0.006551	1494	1490
4/4	R7	5.271	14.35	0	0.006551	1502	1496
4/4	R8	4.816	11.17	0	0.006551	1460	1456
4/4	R9	5.539	15.8	0	0.009483	1484	1480
4/4	R10	5.266	13.27	0	0.006551	1505	1500
4/4	Myopic	3.273	5.858	0	0	1763	1763
4/4	Base stock	1.245	1.29	0	0.006551	737	725
4/4	Two-period	0.8514	0.8695	0	0.002989	357	356
4/4	N32/2/303	0.3666	0.4915	0	0.009483	61	45
4/4	N32/2/304	0.3666	0.4915	0	0.009483	58	45
4/4	N32/2/305	0.3666	0.4915	0	0.009483	58	46
4/4	N32/2/306	0.3666	0.4915	0	0.009483	58	45
4/4	N32/2/307	0.3666	0.4915	0	0.009483	59	47
4/4	N32/2/308	0.3666	0.4915	0	0.009483	57	47
4/4	N32/2/309	0.3666	0.4915	0	0.009483	56	46
4/4	N32/2/310	0.3666	0.4915	0	0.009483	60	45
4/5	N32/2/301	0.3664	0.4913	3.962×10^{-4}	0.006804	112	87
4/5	N32/2/302	0.3664	0.4913	3.962×10^{-4}	0.006804	120	92
4/5	P2	0.1925	0.2019	3.962×10^{-4}	0.006804	92	62
4/5	P3	0.1029	0.1108	3.962×10^{-4}	0.006551	84	48
4/5	R1	5.827	15.86	1.856×10^{-4}	0.009483	3878	3865
4/5	R2	6.422	14.98	3.962×10^{-4}	0.006551	3867	3855
4/5	R3	6.098	13.37	3.962×10^{-4}	0.009483	3907	3896
4/5	R4	6.299	15.81	3.962×10^{-4}	0.006804	3910	3899
4/5	R5	5.839	14.05	0	0.009483	3877	3864
4/5	R6	6.337	19.6	3.962×10^{-4}	0.009483	3903	3893
4/5	R7	6.933	17.09	3.962×10^{-4}	0.009483	3901	3885
4/5	R8	5.76	12.84	3.962×10^{-4}	0.006551	3895	3885

d/L	Policy	Raw regret	Raw bound	Fixed bound	Budget bound	Fixed edits	Budget edits
4/5	R9	5.84	12.78	3.962×10^{-4}	0.006551	3883	3873
4/5	R10	5.867	13.72	3.962×10^{-4}	0.006551	3850	3837
4/5	Myopic	4.057	7.221	0	0	4183	4183
4/5	Base stock	1.884	2.231	0	0.006551	1961	1949
4/5	Two-period	1.491	1.563	1.856×10^{-4}	1.856×10^{-4}	1276	1276
4/5	N32/2/303	0.3664	0.4913	3.962×10^{-4}	0.009483	113	82
4/5	N32/2/304	0.3664	0.4913	3.962×10^{-4}	0.006804	112	82
4/5	N32/2/305	0.3664	0.4913	3.962×10^{-4}	0.006804	107	79
5/4	N32/2/303	0.2399	0.2399	0	0	24	24
5/4	P2	0.2399	0.24	0	0.006237	38	33
5/4	R1	7.066	21.65	0	0.009065	6201	6187
5/4	Myopic	3.552	6.384	0	0	7076	7076
5/4	Base stock	2.544	5.244	0	0	2485	2485
5/4	Two-period	2.72	5.302	0	0	1448	1448
6/4	N32/2/303	0.8042	0.9011	0	0	274	274
6/4	P2	0.865	0.9819	0	0	329	329
6/4	R1	7.541	25.26	0	0	25437	25437
6/4	Myopic	3.543	6.752	5.167×10^{-4}	5.167×10^{-4}	28462	28462
6/4	Base stock	4.752	11.09	0	0	8321	8321
6/4	Two-period	5.346	11.56	0	0	4600	4600
7/4	N32/2/303	1.549	2.148	0	0.009704	655	634
7/4	P2	1.809	2.245	0	0.009704	1608	1587
7/4	R1	10.41	34.27	0	0.009704	102839	102836
7/4	Myopic	3.659	7.056	0	0.004821	114348	114341
7/4	Base stock	7.072	17.73	0	0.009704	29044	29023
7/4	Two-period	8.011	18.54	0	0	13713	13713

2. STAGE-BY-STAGE COST LEDGER

An asterisk marks inherited-policy replay: its R28 fitting work is not repeated or included in the R29 generation entry. The component sum charges the full reference and model per policy and includes the raw audit and selected budget repair, not all sensitivity variants. It excludes startup and serialization, which remain in the cohort resource record. CPU and wall time are different measurements. Full model-construction, reference, policy compilation, comparison, and amortization fields remain in each JSON record.

TABLE II: Seconds by candidate; totals are non-amortized measured component sums

d/L	Policy	Generate	Raw audit	Repair	Verify	Wall sum	CPU sum
2/4	N16/1/301	$2.003 \times 10^{-4}*$	0.001766	0.001802	0.001595	0.01101	0.01101
2/4	N16/1/302	$1.801 \times 10^{-4}*$	0.001609	0.001757	0.001577	0.01067	0.01067
2/4	N16/2/301	$1.587 \times 10^{-4}*$	0.001615	0.001792	0.00159	0.01074	0.01074
2/4	N16/2/302	$1.723 \times 10^{-4}*$	0.00162	0.001765	0.001631	0.01075	0.01075
2/4	N32/1/301	$2.06 \times 10^{-4}*$	0.001635	0.001788	0.001598	0.01081	0.01081
2/4	N32/1/302	$1.765 \times 10^{-4}*$	0.001639	0.001774	0.001592	0.01072	0.01072
2/4	N32/2/301	$1.777 \times 10^{-4}*$	0.001622	0.001769	0.001587	0.01073	0.01073
2/4	N32/2/302	$2.057 \times 10^{-4}*$	0.001628	0.001781	0.001592	0.01077	0.01077
2/4	P2	$2.055 \times 10^{-4}*$	0.001714	0.001912	0.00165	0.01131	0.01131
2/4	P3	$1.771 \times 10^{-4}*$	0.00169	0.001831	0.001599	0.011	0.01101
2/4	R1	0.001038	0.001753	0.002347	0.001622	0.01252	0.01252
2/4	R2	9.675×10^{-4}	0.001756	0.002434	0.001661	0.01265	0.01265
2/4	R3	9.7×10^{-4}	0.001757	0.002277	0.001661	0.0124	0.01239

d/L	Policy	Generate	Raw audit	Repair	Verify	Wall sum	CPU sum
2/4	R4	9.524×10^{-4}	0.001758	0.002275	0.001611	0.01236	0.01236
2/4	R5	9.437×10^{-4}	0.001762	0.002582	0.001607	0.01266	0.01266
2/4	R6	9.45×10^{-4}	0.001762	0.002296	0.001625	0.01238	0.01238
2/4	R7	9.332×10^{-4}	0.001748	0.002278	0.001621	0.01236	0.01236
2/4	R8	0.001315	0.001766	0.002268	0.001601	0.01266	0.01267
2/4	R9	9.197×10^{-4}	0.001753	0.002381	0.00167	0.01255	0.01255
2/4	R10	9.247×10^{-4}	0.001765	0.002298	0.00161	0.01237	0.01237
2/4	Myopic	1.379×10^{-4}	0.001764	0.002414	0.001619	0.01172	0.01172
2/4	Base stock	1.102×10^{-4}	0.00174	0.002281	0.001659	0.01164	0.01164
2/4	Two-period	0.001391	0.001695	0.002075	0.001661	0.01261	0.01261
3/4	N16/1/301	$1.802 \times 10^{-4*}$	0.006947	0.006917	0.006565	0.04626	0.04626
3/4	N16/1/302	$1.965 \times 10^{-4*}$	0.006898	0.006945	0.006553	0.04606	0.04605
3/4	N16/2/301	$2.246 \times 10^{-4*}$	0.006596	0.006839	0.006633	0.04487	0.04488
3/4	N16/2/302	$2.454 \times 10^{-4*}$	0.006564	0.006857	0.006572	0.04459	0.04459
3/4	N32/1/301	$2.14 \times 10^{-4*}$	0.006878	0.007034	0.006655	0.04562	0.04562
3/4	N32/1/302	$2.185 \times 10^{-4*}$	0.007027	0.006949	0.006585	0.04559	0.0456
3/4	N32/2/301	$2.269 \times 10^{-4*}$	0.006908	0.006873	0.006613	0.04508	0.04508
3/4	N32/2/302	$2.173 \times 10^{-4*}$	0.006891	0.007057	0.00659	0.04568	0.04569
3/4	P2	$2.094 \times 10^{-4*}$	0.006564	0.006926	0.006578	0.0444	0.0444
3/4	P3	$2.051 \times 10^{-4*}$	0.006563	0.006873	0.0066	0.04444	0.04444
3/4	R1	0.005023	0.007098	0.009017	0.006628	0.05295	0.05295
3/4	R2	0.003786	0.00717	0.009542	0.006903	0.05299	0.05299
3/4	R3	0.003681	0.007228	0.009065	0.006584	0.05199	0.05199
3/4	R4	0.003651	0.007186	0.009127	0.006589	0.05181	0.05182
3/4	R5	0.003627	0.007222	0.009531	0.007022	0.05325	0.05325
3/4	R6	0.003655	0.007226	0.009722	0.006994	0.05353	0.05353
3/4	R7	0.003656	0.007214	0.009573	0.007014	0.05303	0.05303
3/4	R8	0.003547	0.007185	0.01152	0.006827	0.05411	0.05411
3/4	R9	0.003593	0.007149	0.009597	0.006941	0.05281	0.0528
3/4	R10	0.00369	0.007123	0.009283	0.00674	0.05193	0.05193
3/4	Myopic	4.535×10^{-4}	0.007152	0.009774	0.006689	0.04906	0.04907
3/4	Base stock	3.048×10^{-4}	0.007136	0.008217	0.006709	0.04739	0.04738
3/4	Two-period	0.005767	0.007036	0.00757	0.006601	0.05188	0.05189
4/4	N16/1/301	$3.17 \times 10^{-4*}$	0.03234	0.03227	0.03212	0.2151	0.2151
4/4	N16/1/302	$4.228 \times 10^{-4*}$	0.03262	0.03276	0.03244	0.2151	0.2151
4/4	N16/2/301	$3.897 \times 10^{-4*}$	0.03269	0.0326	0.03448	0.2176	0.2176
4/4	N16/2/302	$3.767 \times 10^{-4*}$	0.03321	0.03212	0.03251	0.2154	0.2153
4/4	N32/1/301	$3.823 \times 10^{-4*}$	0.03202	0.03207	0.03229	0.2154	0.2155
4/4	N32/1/302	$3.996 \times 10^{-4*}$	0.03195	0.03203	0.03287	0.2174	0.2174
4/4	N32/2/301	$3.904 \times 10^{-4*}$	0.03256	0.03286	0.03248	0.2162	0.2162
4/4	N32/2/302	$4.133 \times 10^{-4*}$	0.03241	0.03224	0.03235	0.2153	0.2153
4/4	P2	$3.62 \times 10^{-4*}$	0.03227	0.03183	0.03223	0.2144	0.2144
4/4	P3	$3.901 \times 10^{-4*}$	0.03235	0.03232	0.03208	0.2151	0.215
4/4	R1	0.01417	0.03351	0.04214	0.03317	0.2432	0.2432
4/4	R2	0.01416	0.03319	0.04088	0.03237	0.2386	0.2386
4/4	R3	0.01403	0.03343	0.04042	0.03232	0.2378	0.2378
4/4	R4	0.01429	0.03423	0.04139	0.03257	0.2418	0.2418
4/4	R5	0.01401	0.03407	0.04163	0.03272	0.2399	0.2399
4/4	R6	0.01406	0.03378	0.04128	0.03237	0.2397	0.2397
4/4	R7	0.01402	0.03434	0.0406	0.03235	0.2394	0.2394
4/4	R8	0.01433	0.03366	0.04032	0.03193	0.2376	0.2375
4/4	R9	0.01427	0.03376	0.04192	0.03186	0.2408	0.2408
4/4	R10	0.01425	0.03361	0.04061	0.03255	0.2392	0.2392
4/4	Myopic	0.001593	0.03313	0.04022	0.03134	0.2215	0.2215
4/4	Base stock	8.257×10^{-4}	0.03338	0.03699	0.03288	0.2221	0.2221
4/4	Two-period	0.02705	0.0323	0.03238	0.03106	0.2394	0.2394

d/L	Policy	Generate	Raw audit	Repair	Verify	Wall sum	CPU sum
4/4	N32/2/303	2.296	0.0326	0.03257	0.03271	2.511	2.511
4/4	N32/2/304	2.287	0.03332	0.03274	0.03235	2.504	2.504
4/4	N32/2/305	2.287	0.03347	0.03296	0.03241	2.504	2.504
4/4	N32/2/306	2.291	0.03377	0.03332	0.03318	2.508	2.508
4/4	N32/2/307	2.29	0.03319	0.0328	0.03296	2.506	2.506
4/4	N32/2/308	2.292	0.03383	0.03309	0.03268	2.508	2.508
4/4	N32/2/309	2.288	0.03354	0.0328	0.03278	2.504	2.504
4/4	N32/2/310	2.293	0.03466	0.03394	0.03386	2.513	2.513
4/5	N32/2/301	$6.162 \times 10^{-4*}$	0.08153	0.07898	0.0785	0.5774	0.5404
4/5	N32/2/302	$5.888 \times 10^{-4*}$	0.08061	0.08003	0.08133	0.5788	0.5418
4/5	P2	$6.194 \times 10^{-4*}$	0.0817	0.08207	0.082	0.5833	0.5463
4/5	P3	$6.596 \times 10^{-4*}$	0.0832	0.08328	0.08367	0.5881	0.5512
4/5	R1	0.035	0.08778	0.1058	0.08232	0.6499	0.6129
4/5	R2	0.03424	0.0876	0.1112	0.08576	0.6608	0.6237
4/5	R3	0.03448	0.08777	0.1084	0.08241	0.6519	0.6149
4/5	R4	0.03464	0.08494	0.1072	0.08275	0.6506	0.6137
4/5	R5	0.03473	0.08517	0.1041	0.08124	0.6448	0.6078
4/5	R6	0.03387	0.0858	0.1071	0.08278	0.6488	0.6119
4/5	R7	0.03431	0.08791	0.1058	0.08107	0.6486	0.6116
4/5	R8	0.03498	0.08563	0.1069	0.08291	0.6508	0.6138
4/5	R9	0.03437	0.08678	0.1039	0.08424	0.65	0.613
4/5	R10	0.03419	0.08665	0.106	0.08284	0.6527	0.6156
4/5	Myopic	0.004207	0.08421	0.1005	0.07823	0.6022	0.5653
4/5	Base stock	0.001781	0.08706	0.09759	0.08174	0.6061	0.5691
4/5	Two-period	0.06739	0.08129	0.08222	0.07834	0.6406	0.6036
4/5	N32/2/303	3.278	0.08119	0.08017	0.08021	3.856	3.819
4/5	N32/2/304	3.28	0.08277	0.07986	0.08126	3.864	3.827
4/5	N32/2/305	3.3	0.08139	0.08086	0.08053	3.882	3.845
5/4	N32/2/303	4.424	0.1562	0.1362	0.1488	5.439	5.438
5/4	P2	0.009263	0.16	0.1456	0.1579	1.065	1.065
5/4	R1	0.05625	0.1695	0.1838	0.1585	1.176	1.176
5/4	Myopic	0.007529	0.1646	0.1783	0.1504	1.089	1.089
5/4	Base stock	0.002879	0.1646	0.1526	0.1534	1.062	1.062
5/4	Two-period	0.1317	0.1599	0.1471	0.1604	1.18	1.18
6/4	N32/2/303	13.33	0.7603	0.6143	0.7353	18.46	18.45
6/4	P2	0.04243	0.8219	0.6497	0.7728	5.301	5.301
6/4	R1	0.2291	0.8564	0.8053	0.784	5.825	5.825
6/4	Myopic	0.03442	0.8714	0.8977	0.8448	5.867	5.867
6/4	Base stock	0.01109	0.867	0.7107	0.7889	5.48	5.48
6/4	Two-period	0.7142	0.8352	0.6834	0.799	6.15	6.149
7/4	N32/2/303	51.6	4.137	3.356	4.438	79.84	79.83
7/4	P2	0.2275	4.766	3.542	4.659	29.81	29.81
7/4	R1	0.8988	5.056	4.315	4.697	31.92	31.91
7/4	Myopic	0.1622	5.249	4.151	4.677	31	31
7/4	Base stock	0.04446	5.209	3.812	4.946	31.08	31.08
7/4	Two-period	4.105	5.068	3.411	4.531	33.8	33.8

TABLE III: Deficits of raw actions against completed suffixes under the budget rule

d/L	Policy	Median	90th pct.	99th pct.	Maximum	Bits
2/4	N16/1/301	0	0	0	0	80
2/4	N16/1/302	0	0	0	0	80

d/L	Policy	Median	90th pct.	99th pct.	Maximum	Bits
2/4	N16/2/301	0	0	0	0	80
2/4	N16/2/302	0	0	0	0	80
2/4	N32/1/301	0	0	0	0	80
2/4	N32/1/302	0	0	0	0	80
2/4	N32/2/301	0	0	0	0	80
2/4	N32/2/302	0	0	0	0	80
2/4	P2	0	0	0.1119	0.1211	80
2/4	P3	0	0	0.1119	0.1211	80
2/4	R1	0.4343	1.192	2.131	2.662	80
2/4	R2	0.2907	1.439	2.192	2.728	80
2/4	R3	0.2376	1.378	2.871	2.971	80
2/4	R4	0.2114	1.518	2.617	3.148	80
2/4	R5	0.2518	1.301	2.391	2.905	80
2/4	R6	0.1725	1.341	1.996	3.148	80
2/4	R7	0.1633	1.294	2.617	2.905	80
2/4	R8	0.1517	1.274	3.102	3.148	80
2/4	R9	0.168	1.549	2.859	3.148	80
2/4	R10	0.2359	1.378	2.628	2.971	80
2/4	Myopic	0.5401	1.631	2.681	2.971	80
2/4	Base stock	1.045×10^{-4}	0.5148	1.097	1.196	80
2/4	Two-period	0	0.2114	0.4312	0.4433	80
3/4	N16/1/301	0	0	0	0.01418	84
3/4	N16/1/302	0	0	0	0.01418	84
3/4	N16/2/301	0	0	0	0	84
3/4	N16/2/302	0	0	0	0	84
3/4	N32/1/301	0	0	0	0.01355	84
3/4	N32/1/302	0	0	0	0.01418	84
3/4	N32/2/301	0	0	0	0.01355	84
3/4	N32/2/302	0	0	0	0.01418	84
3/4	P2	0	0	0	0	84
3/4	P3	0	0	0	0	84
3/4	R1	0.4902	1.254	1.805	1.997	84
3/4	R2	0.5278	1.396	1.982	2.036	84
3/4	R3	0.5257	1.208	1.952	2.036	84
3/4	R4	0.5869	1.301	1.931	2.056	84
3/4	R5	0.4777	1.16	1.843	2.036	84
3/4	R6	0.5278	1.273	1.807	2.056	84
3/4	R7	0.4736	1.158	1.843	2.017	84
3/4	R8	0.5426	1.271	1.884	2.056	84
3/4	R9	0.4433	1.307	1.843	2.056	84
3/4	R10	0.4778	1.328	1.983	2.017	84
3/4	Myopic	0.7182	1.203	1.57	1.845	84
3/4	Base stock	0	0.5577	0.766	0.8601	84
3/4	Two-period	0	0.326	0.5627	0.62	84
4/4	N16/1/301	0	0	0.1632	0.2253	86
4/4	N16/1/302	0	0	0.1632	0.2253	86
4/4	N16/2/301	0	0	0.1632	0.2253	86
4/4	N16/2/302	0	0	0.1632	0.2253	86
4/4	N32/1/301	0	0	0.1632	0.2253	86
4/4	N32/1/302	0	0	0.1632	0.2253	86
4/4	N32/2/301	0	0	0.1632	0.2253	86
4/4	N32/2/302	0	0	0.1632	0.2253	86
4/4	P2	0	0	0	0.03934	86
4/4	P3	0	0	0.1632	0.2253	86
4/4	R1	0.6107	1.392	1.73	2.306	86
4/4	R2	0.6168	1.371	1.717	1.86	86

d/L	Policy	Median	90th pct.	99th pct.	Maximum	Bits
4/4	R3	0.6243	1.368	1.73	2.306	86
4/4	R4	0.6215	1.378	1.726	2.281	86
4/4	R5	0.6111	1.306	1.716	2.281	86
4/4	R6	0.6111	1.31	1.717	2.281	86
4/4	R7	0.6259	1.379	1.73	2.306	86
4/4	R8	0.5938	1.308	1.728	2.281	86
4/4	R9	0.6215	1.326	1.733	2.306	86
4/4	R10	0.5956	1.318	1.73	2.306	86
4/4	Myopic	0.7088	1.164	1.502	1.611	86
4/4	Base stock	0	0.5887	0.7004	0.7239	86
4/4	Two-period	0	0.4476	0.692	0.6978	86
4/4	N32/2/303	0	0	0.1632	0.2253	86
4/4	N32/2/304	0	0	0.1632	0.2253	86
4/4	N32/2/305	0	0	0.1632	0.2253	86
4/4	N32/2/306	0	0	0.1632	0.2253	86
4/4	N32/2/307	0	0	0.1632	0.2253	86
4/4	N32/2/308	0	0	0.1632	0.2253	86
4/4	N32/2/309	0	0	0.1632	0.2253	86
4/4	N32/2/310	0	0	0.1632	0.2253	86
4/5	N32/2/301	0	0	0.04569	0.225	87
4/5	N32/2/302	0	0	0.04569	0.225	87
4/5	P2	0	0	0.02476	0.1577	87
4/5	P3	0	0	0.006551	0.1029	87
4/5	R1	0.5853	1.342	1.802	2.306	87
4/5	R2	0.592	1.342	1.864	2.306	87
4/5	R3	0.6193	1.347	1.864	2.281	87
4/5	R4	0.5852	1.35	1.848	2.306	87
4/5	R5	0.592	1.342	1.864	2.306	87
4/5	R6	0.6215	1.352	1.864	2.281	87
4/5	R7	0.6215	1.347	1.864	2.281	87
4/5	R8	0.5958	1.351	1.864	2.306	87
4/5	R9	0.592	1.357	1.864	2.306	87
4/5	R10	0.5796	1.321	1.848	2.281	87
4/5	Myopic	0.6812	1.129	1.473	1.611	87
4/5	Base stock	0	0.6134	0.9805	1.143	87
4/5	Two-period	0	0.5771	0.9805	1.143	87
4/5	N32/2/303	0	0	0.03951	0.225	87
4/5	N32/2/304	0	0	0.04426	0.225	87
4/5	N32/2/305	0	0	0.03951	0.225	87
5/4	N32/2/303	0	0	0	0.2398	89
5/4	P2	0	0	0	0.2398	89
5/4	R1	0.7081	1.468	1.816	2.547	89
5/4	Myopic	0.7604	1.014	1.474	1.856	89
5/4	Base stock	0	0.5978	0.7643	0.8984	89
5/4	Two-period	0	0.5978	0.7643	1.16	89
6/4	N32/2/303	0	0	0	0.7549	91
6/4	P2	0	0	0.0178	0.7549	91
6/4	R1	0.7964	1.544	2.24	3.164	91
6/4	Myopic	0.833	0.9146	1.542	2.089	91
6/4	Base stock	0	0.531	0.8061	1.425	91
6/4	Two-period	0	0.5293	0.8061	2.217	91
7/4	N32/2/303	0	0	0	1.23	93
7/4	P2	0	0	0.5536	1.23	93
7/4	R1	0.8443	1.596	2.685	4.162	93
7/4	Myopic	0.8503	0.9031	1.557	2.315	93
7/4	Base stock	0	0.4119	0.8073	1.926	93

d/L	Policy	Median	90th pct.	99th pct.	Maximum	Bits
7/4	Two-period	0	0.3865	0.8125	3.235	93

TABLE IV: Corrections by time and inventory boundary; worst losses over all candidates in a model

d/L	Time	Edits	Min inventory 0	Raw restart loss	Budget restart loss
2/4	0	121	56	5.054	4.78×10^{-5}
2/4	1	135	57	4.485	5.699×10^{-5}
2/4	2	125	58	4.134	6.396×10^{-5}
2/4	3	123	57	3.905	1.034×10^{-4}
2/4	4	125	53	3.434	2.089×10^{-4}
2/4	5	123	48	3.042	0
2/4	6	115	51	2.296	0
2/4	7	104	49	1.677	0
3/4	0	531	305	5.141	1.065×10^{-4}
3/4	1	542	311	4.96	1.416×10^{-4}
3/4	2	540	306	4.776	2.009×10^{-4}
3/4	3	538	311	4.732	2.841×10^{-4}
3/4	4	541	315	3.937	4.099×10^{-4}
3/4	5	522	307	3.726	0.001035
3/4	6	516	306	3.013	0.001307
3/4	7	492	292	2.017	0.006601
4/4	0	2478	1732	5.736	0.002999
4/4	1	2531	1805	5.539	7.755×10^{-5}
4/4	2	2483	1763	5.285	2.566×10^{-4}
4/4	3	2241	1507	5.194	3.834×10^{-4}
4/4	4	2231	1514	4.516	0.001808
4/4	5	2272	1515	3.918	0.006259
4/4	6	2099	1513	3.323	0.009483
4/4	7	2197	1510	2.306	0
4/5	0	6068	3465	6.462	2.001×10^{-4}
4/5	1	5977	3379	6.933	2.327×10^{-4}
4/5	2	5936	3381	5.975	9.397×10^{-4}
4/5	3	5892	3308	6.299	0.001218
4/5	4	5883	3337	5.165	0.006349
4/5	5	5920	3354	4.331	0.005074
4/5	6	5455	3303	3.514	0.009483
4/5	7	5561	3324	2.306	0
5/4	0	2305	1422	7.066	9.741×10^{-6}
5/4	1	2290	1431	6.407	3.449×10^{-5}
5/4	2	2265	1411	5.962	9.435×10^{-5}
5/4	3	2275	1412	5.28	1.572×10^{-4}
5/4	4	2271	1411	4.638	3.627×10^{-4}
5/4	5	2245	1394	4.114	0.001283
5/4	6	1798	1381	3.076	0.009065
5/4	7	1804	1399	2.547	0
6/4	0	8874	6144	7.541	7.463×10^{-8}
6/4	1	8812	6087	7.363	5.196×10^{-7}
6/4	2	8781	6070	6.573	2.43×10^{-6}
6/4	3	8840	6103	5.786	9.088×10^{-6}
6/4	4	8743	6117	5.144	2.155×10^{-5}
6/4	5	8745	6126	4.346	4.748×10^{-5}
6/4	6	7343	6028	3.392	1.279×10^{-4}

d/L	Time	Edits	Min inventory 0	Raw restart loss	Budget restart loss
6/4	7	7285	6015	2.796	5.167×10^{-4}
7/4	0	34200	26094	10.41	3.438×10^{-7}
7/4	1	34230	26124	9.569	3.126×10^{-6}
7/4	2	33712	25577	8.717	2.558×10^{-5}
7/4	3	33751	25589	7.72	1.938×10^{-4}
7/4	4	33816	25661	6.182	0.001292
7/4	5	33674	25733	4.863	0.006405
7/4	6	29333	25486	3.927	0.009704
7/4	7	29418	25487	3.032	0

3. INHERITED R28 DETAILED EXECUTION LEDGERS

These are the original R28 hardware measurements, not cross-hardware speedup comparisons. Their underlying records remain byte-for-byte unchanged.

TABLE V: All original R28 inventory candidates

d, L	Policy	Raw bound	Raw loss	Complete bound	Complete loss	Edits
2,4	N16/1/301	0	0	0	0	0
2,4	N16/1/302	0	0	0	0	0
2,4	N16/2/301	0	0	0	0	0
2,4	N16/2/302	0	0	0	0	0
2,4	N32/1/301	0	0	0	0	0
2,4	N32/1/302	0	0	0	0	0
2,4	N32/2/301	0	0	0	0	0
2,4	N32/2/302	0	0	0	0	0
2,4	P2	0.1571	0.1391	2.089×10^{-4}	2.089×10^{-4}	7
2,4	P3	0.1571	0.1391	0	0	7
3,4	N16/1/301	0.01418	0.01418	0	0	3
3,4	N16/1/302	0.01418	0.01418	0	0	3
3,4	N16/2/301	0	0	0	0	0
3,4	N16/2/302	0	0	0	0	0
3,4	N32/1/301	0.01355	0.01355	0	0	2
3,4	N32/1/302	0.01418	0.01418	0	0	1
3,4	N32/2/301	0.01355	0.01355	0	0	1
3,4	N32/2/302	0.01418	0.01418	0	0	3
3,4	P2	0	0	0	0	0
3,4	P3	0	0	0	0	0
4,4	N16/1/301	0.4915	0.3666	0	0	61
4,4	N16/1/302	0.4914	0.3666	0	0	59
4,4	N16/2/301	0.4915	0.3666	0	0	55
4,4	N16/2/302	0.4915	0.3666	0	0	56
4,4	N32/1/301	0.4914	0.3666	0	0	61
4,4	N32/1/302	0.4914	0.3666	0	0	59
4,4	N32/2/301	0.4915	0.3666	0	0	61
4,4	N32/2/302	0.4915	0.3666	0	0	58
4,4	P2	0.03934	0.03934	0	0	17
4,4	P3	0.4914	0.3666	0	0	59
4,5	N32/2/301	0.4913	0.3664	3.962×10^{-4}	3.962×10^{-4}	112
4,5	N32/2/302	0.4913	0.3664	3.962×10^{-4}	3.962×10^{-4}	120
4,5	P2	0.2019	0.1925	3.962×10^{-4}	3.962×10^{-4}	92
4,5	P3	0.1108	0.1029	3.962×10^{-4}	3.962×10^{-4}	84

TABLE VI: Original R28 work and correction ledger

d, L	Policy	Build	Compile	Ref.-free	Standalone	% edits	99th loss
2,4	N16/1/301	6.704×10^{-4}	6.738×10^{-4}	1.505	1.508	0	0
2,4	N16/1/302	4.819×10^{-4}	5.66×10^{-4}	1.497	1.501	0	0
2,4	N16/2/301	4.684×10^{-4}	5.76×10^{-4}	1.862	1.865	0	0
2,4	N16/2/302	4.557×10^{-4}	6.148×10^{-4}	1.863	1.866	0	0
2,4	N32/1/301	4.304×10^{-4}	5.766×10^{-4}	1.526	1.529	0	0
2,4	N32/1/302	4.384×10^{-4}	5.761×10^{-4}	1.53	1.534	0	0
2,4	N32/2/301	4.63×10^{-4}	5.88×10^{-4}	1.982	1.985	0	0
2,4	N32/2/302	4.552×10^{-4}	6.102×10^{-4}	2.092	2.095	0	0
2,4	P2	4.006×10^{-4}	2.652×10^{-4}	0.007922	0.01115	5.833	0.1212
2,4	P3	3.829×10^{-4}	2.462×10^{-4}	0.007613	0.0108	5.833	0.1212
3,4	N16/1/301	0.001102	8.04×10^{-4}	1.555	1.569	0.5952	0.005613
3,4	N16/1/302	0.001103	7.941×10^{-4}	1.567	1.581	0.5952	0.002806
3,4	N16/2/301	0.001148	8.265×10^{-4}	1.977	1.991	0	0
3,4	N16/2/302	0.001173	7.869×10^{-4}	1.989	2.003	0	0
3,4	N32/1/301	0.001166	8.163×10^{-4}	1.628	1.642	0.3968	1.051×10^{-4}
3,4	N32/1/302	0.001148	7.93×10^{-4}	1.63	1.644	0.1984	0.002806
3,4	N32/2/301	0.001194	8.188×10^{-4}	2.174	2.188	0.1984	0
3,4	N32/2/302	0.001213	8.081×10^{-4}	2.175	2.189	0.5952	0.008419
3,4	P2	0.001105	4.383×10^{-4}	0.02348	0.03774	0	0
3,4	P3	0.001157	4.629×10^{-4}	0.02406	0.03833	0	0
4,4	N16/1/301	0.006404	0.001782	1.78	1.855	2.99	0.2052
4,4	N16/1/302	0.005622	0.001828	1.782	1.857	2.892	0.2052
4,4	N16/2/301	0.005445	0.00179	2.368	2.442	2.696	0.2052
4,4	N16/2/302	0.005581	0.00176	2.369	2.444	2.745	0.2052
4,4	N32/1/301	0.005494	0.001831	1.958	2.033	2.99	0.2052
4,4	N32/1/302	0.005471	0.001807	1.95	2.025	2.892	0.2052
4,4	N32/2/301	0.005458	0.001779	2.974	3.048	2.99	0.2052
4,4	N32/2/302	0.005549	0.001773	2.963	3.038	2.843	0.2052
4,4	P2	0.005477	0.00141	0.1017	0.1762	0.8333	0.00649
4,4	P3	0.005537	0.001415	0.1036	0.1782	2.892	0.2052
4,5	N32/2/301	0.02482	0.003514	4.392	4.586	2.244	0.1706
4,5	N32/2/302	0.01684	0.003527	4.386	4.578	2.404	0.1706
4,5	P2	0.01272	0.003051	0.2448	0.4368	1.843	0.05718
4,5	P3	0.01273	0.003075	0.2451	0.4385	1.683	0.01822

TABLE VII: Original R28 restart losses

d, L	Time	Raw worst	Complete worst
2,4	0	0.01797	4.78×10^{-5}
2,4	1	0.019	5.699×10^{-5}
2,4	2	0.02281	6.396×10^{-5}
2,4	3	0.02997	1.034×10^{-4}
2,4	4	0.03888	2.089×10^{-4}
2,4	5	0.1391	0
2,4	6	0.0728	0
2,4	7	0	0
3,4	0	5.014×10^{-4}	0
3,4	1	6.384×10^{-4}	0
3,4	2	6.754×10^{-4}	0
3,4	3	0.001651	0

d, L	Time	Raw worst	Complete worst
3,4	4	0.002779	0
3,4	5	0.00842	0
3,4	6	0.01418	0
3,4	7	0	0
4,4	0	0.3666	0
4,4	1	0.2068	0
4,4	2	0.04572	0
4,4	3	0.001242	0
4,4	4	0.007523	0
4,4	5	0.0456	0
4,4	6	0.0325	0
4,4	7	0	0
4,5	0	0.3664	1.856×10^{-4}
4,5	1	0.2067	6.536×10^{-5}
4,5	2	0.04569	3.962×10^{-4}
4,5	3	0.08895	0
4,5	4	0.1925	0
4,5	5	0.05719	0
4,5	6	0.1156	0
4,5	7	0	0

TABLE VIII: Recentered transport timing

Seed	Order	Setup	Neural	Transport	Diagnostics	Total
27101	8/16	0.003616	0.5799	0.8522	0.06444	36.62
27101	12/24	0.002385	0.8794	1.157	0.06613	37.18
27102	8/16	0.002306	0.578	0.8489	0.06433	1.507
27102	12/24	0.002181	0.8853	1.165	0.06696	2.135
27103	8/16	0.002029	0.5797	0.851	0.06446	1.511
27103	12/24	0.002136	0.8814	1.158	0.06652	2.124

TABLE IX: Paired production timing

Seed	Method	Setup	Generation	Checker	Bookkeeping	Total
27201	Neural Adam U	0.001439	1.174	87.31	0.00562	88.54
27201	Neural Adam G	0.00157	1.173	87.1	0.005491	88.33
27201	Direct Adam U	0.001669	1.047	87.43	0.00343	88.53
27201	Direct Adam G	0.00168	1.057	87.58	0.003551	88.68
27201	Direct L-BFGS-B U	0.001572	0.298	87.34	0.003031	87.69
27201	Direct L-BFGS-B G	0.001619	0.354	87.31	0.005319	87.71
27202	Neural Adam U	0.001469	1.163	87.3	0.005566	88.52
27202	Neural Adam G	0.00149	1.163	87.6	0.005498	88.82
27202	Direct Adam U	0.001584	1.045	87.35	0.003501	88.44
27202	Direct Adam G	0.001564	1.047	87.22	0.003501	88.31
27202	Direct L-BFGS-B U	0.00157	0.2793	87.5	0.002768	87.82
27202	Direct L-BFGS-B G	0.001563	0.3027	87.29	0.005414	87.64

TABLE X: Every production checkpoint decision

Seed	Method	Block	Lower	Upper	Margin	Width	Decision
27201	Neural Adam U	1	-1.28723	-1.287229	0.03313	3.938×10^{-7}	D
27201	Neural Adam U	2	-1.287225	-1.287224	4.549×10^{-6}	3.938×10^{-7}	D
27201	Neural Adam U	3	-1.287223	-1.287222	1.913×10^{-6}	3.938×10^{-7}	D
27201	Neural Adam U	4	-1.287222	-1.287221	6.623×10^{-7}	3.938×10^{-7}	D
27201	Neural Adam G	1	-1.28723	-1.287229	0.03313	3.938×10^{-7}	A
27201	Neural Adam G	2	-1.287225	-1.287224	4.549×10^{-6}	3.938×10^{-7}	A
27201	Neural Adam G	3	-1.287223	-1.287222	1.913×10^{-6}	3.938×10^{-7}	A
27201	Neural Adam G	4	-1.287222	-1.287221	6.623×10^{-7}	3.938×10^{-7}	A
27201	Direct Adam U	1	-1.287282	-1.28728	0.03308	3.938×10^{-7}	D
27201	Direct Adam U	2	-1.287251	-1.287249	3.056×10^{-5}	3.938×10^{-7}	D
27201	Direct Adam U	3	-1.28724	-1.287238	1.082×10^{-5}	3.938×10^{-7}	D
27201	Direct Adam U	4	-1.287234	-1.287232	5.481×10^{-6}	3.938×10^{-7}	D
27201	Direct Adam G	1	-1.287282	-1.28728	0.03308	3.938×10^{-7}	A
27201	Direct Adam G	2	-1.287251	-1.287249	3.056×10^{-5}	3.938×10^{-7}	A
27201	Direct Adam G	3	-1.28724	-1.287238	1.082×10^{-5}	3.938×10^{-7}	A
27201	Direct Adam G	4	-1.287234	-1.287232	5.481×10^{-6}	3.938×10^{-7}	A
27201	Direct L-BFGS-B U	1	-1.28722	-1.287219	0.03314	3.938×10^{-7}	D
27201	Direct L-BFGS-B U	2	-1.28722	-1.287219	-3.936×10^{-7}	3.938×10^{-7}	D
27201	Direct L-BFGS-B U	3	-1.28722	-1.287219	-3.938×10^{-7}	3.938×10^{-7}	D
27201	Direct L-BFGS-B U	4	-1.28722	-1.287219	-3.938×10^{-7}	3.938×10^{-7}	D
27201	Direct L-BFGS-B G	1	-1.28722	-1.287219	0.03314	3.938×10^{-7}	A
27201	Direct L-BFGS-B G	2	-1.28722	-1.287219	-3.936×10^{-7}	3.938×10^{-7}	R
27201	Direct L-BFGS-B G	3	-1.28722	-1.287219	-3.936×10^{-7}	3.938×10^{-7}	R
27201	Direct L-BFGS-B G	4	-1.28722	-1.287219	-3.936×10^{-7}	3.938×10^{-7}	R
27202	Neural Adam U	1	-1.28723	-1.287229	0.03274	3.938×10^{-7}	D
27202	Neural Adam U	2	-1.287225	-1.287224	4.779×10^{-6}	3.938×10^{-7}	D
27202	Neural Adam U	3	-1.287223	-1.287222	1.253×10^{-6}	3.938×10^{-7}	D
27202	Neural Adam U	4	-1.287222	-1.287221	5.336×10^{-7}	3.938×10^{-7}	D
27202	Neural Adam G	1	-1.28723	-1.287229	0.03274	3.938×10^{-7}	A
27202	Neural Adam G	2	-1.287225	-1.287224	4.779×10^{-6}	3.938×10^{-7}	A
27202	Neural Adam G	3	-1.287223	-1.287222	1.253×10^{-6}	3.938×10^{-7}	A
27202	Neural Adam G	4	-1.287222	-1.287221	5.336×10^{-7}	3.938×10^{-7}	A
27202	Direct Adam U	1	-1.287282	-1.28728	0.03269	3.938×10^{-7}	D
27202	Direct Adam U	2	-1.287251	-1.287249	3.057×10^{-5}	3.938×10^{-7}	D
27202	Direct Adam U	3	-1.287239	-1.287238	1.078×10^{-5}	3.938×10^{-7}	D
27202	Direct Adam U	4	-1.287234	-1.287232	5.455×10^{-6}	3.938×10^{-7}	D
27202	Direct Adam G	1	-1.287282	-1.28728	0.03269	3.938×10^{-7}	A
27202	Direct Adam G	2	-1.287251	-1.287249	3.057×10^{-5}	3.938×10^{-7}	A
27202	Direct Adam G	3	-1.287239	-1.287238	1.078×10^{-5}	3.938×10^{-7}	A
27202	Direct Adam G	4	-1.287234	-1.287232	5.455×10^{-6}	3.938×10^{-7}	A
27202	Direct L-BFGS-B U	1	-1.28722	-1.287219	0.03275	3.938×10^{-7}	D
27202	Direct L-BFGS-B U	2	-1.28722	-1.287219	-3.939×10^{-7}	3.938×10^{-7}	D
27202	Direct L-BFGS-B U	3	-1.28722	-1.287219	-3.938×10^{-7}	3.938×10^{-7}	D
27202	Direct L-BFGS-B U	4	-1.28722	-1.287219	-3.938×10^{-7}	3.938×10^{-7}	D
27202	Direct L-BFGS-B G	1	-1.28722	-1.287219	0.03275	3.938×10^{-7}	A
27202	Direct L-BFGS-B G	2	-1.28722	-1.287219	-3.939×10^{-7}	3.938×10^{-7}	R
27202	Direct L-BFGS-B G	3	-1.28722	-1.287219	-3.939×10^{-7}	3.938×10^{-7}	R
27202	Direct L-BFGS-B G	4	-1.28722	-1.287219	-3.939×10^{-7}	3.938×10^{-7}	R

Neural Bellman Operators

Historical Preservation Annex, R29

September 24, 2026

This annex reproduces the complete last materialized R26 article, supplement, and response without changing their pages. The R26 supplement itself retains the complete reviewed R24 article and supplement. The current article incorporates the later R28 scientific sources and adds R29 results; all intermediate source and evidence directories are retained in the repository. Historical claims are to be read with the current point-by-point response, not as a replacement for its updated conclusions.

NEURAL BELLMAN OPERATORS

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We study certified computation in a stopped consumption–portfolio economy with costly preference adjustment. In the executed financed policy class, direct L-BFGS-B remains more accurate and cheaper than the neural alternatives. We develop a synchronous policy-improvement operator with complete optimizer-state rollback, a finite-work quotient-polling theorem under explicit oracle and smoothness conditions, and second-order interval verification for current-state neural policies. Expanded, disjoint calibration brackets all six first-order configurations; a moving-transport control retains historical Jacobians and Adam moments. Eighteen new online decisions include five actual rejections. In a separate stopped economy with unspanned risk and an independently known value, unchanged neural policies obtain whole-state–time regret certificates below .001 on 32,768 cells, rather than 262,144. Exact rational polynomial verification provides a stronger classical reference. The original economy and its uniform .01 objective are retained: its present bound is 7.181835, and neither neural efficiency dominance nor a continuous stopped-gradient certificate is asserted.

KEYWORDS: Stochastic control, Bellman operators, Preference adjustment, Validated numerics, Policy improvement, Computational economics.

1. INTRODUCTION

Economic policy computation requires more than a small optimization loss. A proposed policy must be feasible in the economy being studied, its payoff must refer to the intended stopping rule, and its error must hold on the state set to which the conclusion applies. These requirements become distinct when a neural parameterization, numerical financing, and a continuation-value approximation are used together. Our objective is a Neural Bellman Operator for the stopped consumption–portfolio economy defined below, with a uniform regret tolerance of .01 on its original state–time domain. This objective is unchanged throughout the revision.

The present contribution is a set of constructive steps toward that objective, with the evidence attached to each step. First, we couple proposal generation to certified policy improvement. A rejected proposal restores not only a deployed policy but also the optimizer’s parameters, first and second moments, and iteration counter. This changes the next proposal. The distinction matters: certifying a completed trajectory and subsequently selecting its best checkpoint does not implement such an operator. Second, we give a derivative-free quotient-polling theorem with explicit accuracy and work requirements for the payoff oracle. It provides a route from verified improvement to a stationarity statement without treating an unverified proposal gradient as an economic gradient. Its smoothness and vanishing-width hypotheses are stated separately from the properties established for the current stopped-payoff checker.

Third, we strengthen whole-domain verification. In a stopped investment economy with an unspanned productivity shock, an exact Hamiltonian completion converts policy regret into an integral of squared action errors. Centered second-order interval bounds certify two fixed neural policies on the entire state–time box, including arbitrary restart times. They meet a .001 tolerance on 32,768 cells. The same policies previously required 262,144 cells with the inherited first-order verifier. A separate exact rational Bernstein calculation certifies the direct polynomial policy without the floating derivative-coefficient shortcut identified in our audit. These are improvements in verification of fixed policies, not claims of better training.

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The classical comparison is essential to the economic interpretation. The expanded calibration study stored in the repository before this revision uses disjoint tuning and evaluation seeds and includes a moving historical-transport control. All six first-order selections lie strictly inside the expanded rate grid with deterioration at a larger rate. Nevertheless, direct L-BFGS-B has both a lower regret bound and strictly higher certified payoff than neural Adam in every matched central-state comparison. The manufactured stochastic benchmark likewise favors a classical approximant. These results rule out a neural-efficiency claim for the tested designs; they do not alter the policy class, economic model, or target that the paper seeks to solve.

The original current-state experiment has a complete-domain bound of 7.181834580823298, compared with the previously reviewed 7.241462443133396. Neither number establishes the .01 objective. Moreover, an improved verification bound is not, by itself, evidence of improved policy payoff. The stopped-gradient bridge also remains uninstantiated for the original proposal objective. We therefore maintain separate records for a proved identity, a certified numerical result, an optimization diagnostic, and a research objective. The new stochastic result is not substituted for the original economy.

Our framework draws on verification and approximation methods for continuous-time control (Kushner and Dupuis, 2001), the distinction between feasible policies and bounds (Brown, Smith, and Sun, 2010), and the numerical treatment of economic dynamic models (Judd, 1998, Brumm and Scheidegger, 2017). The proofs and complete numerical records from earlier revisions remain available without deletion. The companion historical annex reproduces all 93 pages of the reviewed R24 article and all 58 pages of its technical supplement; their source files are retained at their original paths. This article presents the current argument, while the new supplement supplies the complete calibration, checkpoint, verification, and provenance records.

2. THE STOPPED ECONOMY AND THE NUMERICAL OBJECTIVE

The horizon is $T = 1$. The state is $(u, x) \in D = (1.2, 2.8) \times (.5, 2)$, where u indexes preferences and x is wealth. Controls $a = (c, \theta, p)$ take values in

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

Under the original Brownian filtration,

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (1)$$

$$dx_s = ((.02 + .06p_s)x_s - c_s) ds + .2p_s x_s dZ_s^x, \quad d\langle Z^u, Z^x \rangle_s = -.25 ds. \quad (2)$$

The adjustment price is $k \in [.5, 8]$; the central comparison uses $k = 2$. Let τ be the first exit from D , capped at 1. Running utility, terminal preference, and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u-2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1-s). \quad (4)$$

For an admissible progressively measurable control,

$$J_k^a(t, u, x) = \mathbb{E}_{t,u,x} \left[\int_t^\tau e^{-.04(s-t)} \ell_k(u_s, c_s, \theta_s) ds + e^{-.04(\tau-t)} g(\tau, u_\tau, x_\tau) \right],$$

$$V_k = \sup_{a \in \mathcal{A}} J_k^a. \quad (5)$$

Admissibility includes a well-defined stopped state process. The bounded state and control ranges keep payoffs bounded; neural feedbacks used in the current-state calculations are smooth before stopping. We do not assume that the unknown value is classically differentiable.

For a smooth test function v , the controlled discounted residual is

$$\begin{aligned} \mathcal{R}_k^a v = & v_t + \theta v_u + ((.02 + .06p)x - c)v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux} \\ & + \ell_k(u, c, \theta) - .04v. \end{aligned} \quad (6)$$

The numerical objective concerns a single current-state actor $a = \pi(t, u, x)$ and a uniform bound for all starting times and all states in the original domain. An initial-state financed expert library and an auxiliary economy are distinct objects, even when their numerical bounds are smaller.

PROPOSITION 1—Stopped residual certificate: *Suppose a smooth candidate v and an admissible policy π satisfy $|v - g| \leq b$ on the stopping boundary, $|\mathcal{R}_k^\pi v| \leq e$ throughout the state–time domain, and*

$$\sup_{a \in A} \mathcal{R}_k^a v - \mathcal{R}_k^\pi v \leq q.$$

Then, at every starting state,

$$0 \leq V_k - J_k^\pi \leq 2b + (2e + q) \frac{1 - e^{-.04(1-t)}}{.04}. \quad (7)$$

The same argument applies to separate upper and lower test functions with their respective one-sided residual and boundary inequalities, and to time-dependent error envelopes integrated up to 1.

The statement requires inequalities over a complete cover, not losses at training nodes. The inherited full-state certificate uses the stronger slabwise, sign-separated implementation, rather than replacing it by the constant bound in (7).

2.1. State information in the financed implementation

The central experiments use sixteen time slabs and a 1–16–16–3 tanh generator with 355 parameters. Its 48 node outputs determine a financed family with 47 economically distinct coordinates: a common intercept displacement is offset by the financing root. The decoder bounds consumption slopes and preference drift and reconstructs wealth through exact conditional pricing. This construction supplies admissible candidates; it is not a numerical identity between a general current-state policy and a financed library.

For expert i , write its conditional wealth as $X_i = F_i(s, Y_s)$, where

$$dY_s = .02Y_s ds - .3Y_s dZ_s^x, \quad p_i = -\frac{1.5Y_s F_{i,y}(s, Y_s)}{F_i(s, Y_s)}.$$

With fixed initialization weights $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, aggregate consumption and preference adjustment are weighted sums and portfolio weight is wealth-weighted. On the financed reachable set, $x = \sum_i \lambda_i X_i$ and the aggregate dynamics satisfy (2). The deployed implementation is Markov in (t, u, x, Y, λ) , not generally in (t, u, x) alone.

A causal auxiliary state with no additional randomness does not enlarge the class of progressively measurable controls under the original filtration. Conversely, an original admissible

control can ignore the auxiliary state. Thus unrestricted optimal values agree when the augmented admissible class is precisely this causal class. The equality says nothing about optimality of the restricted expert library. Off-manifold admissible extensions do not provide numerical accuracy at arbitrary memory states. The retained R24 regional bound below .004281 concerns $t = 0$, $u \in [1.98, 2.02]$, $x \in [1.24, 1.27]$, with initialized memory and nine experts. Its economic scope is kept separate from the whole-domain current-state objective.

3. CERTIFIED POLICY IMPROVEMENT AND NUMERICAL GEOMETRY

Let $y \in \mathbb{R}^{47}$ parameterize the financed quotient, and let $F(y)$ denote the exact stopped payoff at a specified initial state. This section distinguishes the proposal objective, the certified payoff oracle, and the state of the optimizer. No identity between their gradients is presumed.

3.1. Synchronous improvement with complete rollback

An optimizer state s contains economic coordinates, neural or chart-carrier parameters, first and second moments, and the iteration counter. Let $\Phi_h(s)$ be a block of proposal updates under a declared resource budget. The proposed policy receives a certified payoff enclosure $I(y) = [L(y), U(y)]$. If the checker fails, it returns no enclosure and the proposal is not accepted. The improvement operator is

$$\mathcal{T}(s) = \begin{cases} \Phi_h(s), & L(y(\Phi_h(s))) > U(y(s)), \\ s, & \text{otherwise.} \end{cases} \quad (8)$$

On rejection the entire saved state is restored. The block size and next learning rate are controlled by an external, recorded schedule; a rejected candidate's moment state is not used by the next block.

PROPOSITION 2—Verified improvement: *If every accepted enclosure contains the exact payoff of the corresponding admissible policy, every accepted update in (8) strictly increases that payoff, and the deployed sequence is nondecreasing. If acceptance instead requires $L(y') > U(y) + \sigma h^2$ with $\sigma > 0$, $h > 0$, and $F \leq M$, then there are at most $\lfloor (M - F(y_0))/(\sigma h^2) \rfloor$ accepted updates at that fixed scale.*

This is a safety and finite-improvement statement, not a convergence claim for Adam. If intervals overlap, failure to certify an improvement is not evidence that the candidate is worse. Restoring the optimizer is consequential only when a rejection occurs; the earlier R25 execution accepted all 24 blocks. Section 5.2 therefore executes a declared stress schedule that exercises the rejection branch, retains full states, and separately continues rejected states as diagnostic shadow runs.

3.2. Precision-adaptive payoff decisions

The absence of a stopped-gradient certificate does not prevent exact-payoff comparisons when sufficiently narrow payoff enclosures are available. Let the width of each policy enclosure at level j be at most $w_j = w_0 2^{-j}$. For a prescribed improvement threshold $s \geq 0$, compare

$$[L(y') - U(y), U(y') - L(y)]$$

with s . Accept if its lower endpoint exceeds s , reject the threshold claim if its upper endpoint is below s , and otherwise refine or return *undecided* at the resource cap.

LEMMA 1—Finite separation of a nonzero payoff margin: *Assume the enclosures above are valid at every level. If $\gamma = |F(y') - F(y) - s| > 0$, the threshold decision is certified whenever $2w_j < \gamma$. It suffices to use at most*

$$\max\{0, \lceil \log_2(2w_0/\gamma) \rceil + 1\}$$

refinements. At zero margin no finite separation is guaranteed.

The current original-economy checker supplies valid intervals at its executed settings. We have not proved that all its components shrink to zero along every relevant policy trajectory. That additional oracle property is a hypothesis of Lemma 1, not an empirical conclusion from two quadrature rules. In particular, rigorous secants are not pointwise derivative enclosures.

3.3. A quotient-polling theorem with explicit verification work

A finite-work result can be obtained without differentiating the proposal quadrature. Consider a direct poll in the economically identified coordinates. Let $\mathcal{D}$ be a finite set of unit directions with positive cosine measure

$$\mu = \inf_{g \neq 0} \max_{d \in \mathcal{D}} \frac{g'd}{\|g\|} > 0.$$

For the coordinate poll $\mathcal{D} = \{\pm e_i : 1 \leq i \leq 47\}$, $\mu = 47^{-1/2}$. At scale h , accept a feasible trial $y + hd$ only if its certified lower payoff exceeds the incumbent upper payoff by σh^2 . A complete unsuccessful poll tests every direction.

ASSUMPTION 1—Poll regularity and oracle accuracy: *On all poll segments under consideration, F is continuously differentiable with L -Lipschitz gradient and is bounded above by M . Every prescribed poll point and segment is feasible and inside this regularity region; accepted iterates remain in it. The payoff oracle returns a valid interval of width at most κh^2 at each point, at cost at most $W(\kappa h^2)$. Here $\sigma > 0$ and $\kappa \geq 0$ are fixed.*

THEOREM 1—Accuracy–work bound for a certified quotient poll: *Under Assumption 1, a complete unsuccessful poll satisfies*

$$\|\nabla F(y)\| \leq \frac{\sigma + 2\kappa + L/2}{\mu} h. \quad (9)$$

Starting at y_0 and continuing fixed-scale polls until the first unsuccessful one requires no more than

$$(|\mathcal{D}| + 1) \left(1 + \left\lfloor \frac{M - F(y_0)}{\sigma h^2} \right\rfloor \right) \quad (10)$$

payoff-oracle evaluations, with their total verification work bounded by (10) times $W(\kappa h^2)$. Taking $h = \mu\varepsilon/(\sigma + 2\kappa + L/2)$ yields an ε -stationarity bound with $O(\varepsilon^{-2})$ polls, apart from the explicitly retained oracle cost.

The estimate is a conditional numerical-method theorem, not an assertion that the original stopped objective already meets Assumption 1. It does not establish a global optimum, a rate for Adam, or a dimensional advantage of neural coordinates. A missing feasible direction at a boundary is not counted as an unsuccessful complete poll. In this revision the theorem is

checked on an exact quadratic example; a 47-dimensional certified poll was not run on the original economy.

The poll itself depends on quotient policies and certified payoffs, not on an equivalent parameter chart of the neural generator. This is an exact form of chart independence for its acceptance decisions. Chart independence of this component must not be confused with invariance of Adam or L-BFGS-B in neural weights. A complete implementation may use neural proposals first and the quotient poll as a safeguard, but its evaluation cost must include every verification required by (10).

3.4. Historical neural transport and an executed control

For a raw output map $f : \mathbb{R}^{355} \rightarrow \mathbb{R}^{48}$, let $J_n = Df(\vartheta_n)$ and $g_n = \nabla_y L(f(\vartheta_n))$. With bias-corrected Adam first moment,

$$\hat{m}_n = \sum_{r=1}^n a_{nr} J'_r g_r, \quad a_{nr} = \frac{(1 - \beta_1) \beta_1^{n-r}}{1 - \beta_1^n},$$

and $D_n = \text{diag}(\sqrt{\hat{v}_n} + \epsilon)$, Taylor's theorem gives

$$f(\vartheta_{n+1}) - f(\vartheta_n) = -\eta_n \sum_{r=1}^n a_{nr} J_n D_n^{-1} J'_r g_r + \epsilon_n, \quad (11)$$

$$\|\epsilon_n\| \leq \frac{H}{2} \eta_n^2 \|D_n^{-1} \hat{m}_n\|^2, \quad (12)$$

provided Df is H -Lipschitz on the update segment. Current $J_n D_n^{-1} J'_n$ alone omits the historical transport. The retained static tangent, diagonal, and whitening charts do not reproduce (11).

The additional direct arm uses quotient coordinates $z \in \mathbb{R}^{47}$ and a separate 355-parameter chart carrier ϑ . At each step it computes $B_n = Q' J_n$, pulls the direct payoff gradient back through B'_n , updates parameter-space Adam moments, and pushes the resulting parameter displacement forward through B_n . Its economic displacement is

$$\delta z_n = B_n \delta \vartheta_n - \eta_n 10^{-6} \nabla_z L, \quad \|\delta z_n\| \leq .5, \quad (13)$$

where a common cap factor rescales the economic and carrier displacements. The carrier is updated in parameter space, while z is updated linearly. The regularizer and cap were fixed before evaluation. The method retains historical pullbacks, adaptive second moments, and the current pushforward, but intentionally omits the finite-step nonlinear output remainder in (11). It is therefore a moving-transport control, not an exact neural-trajectory replica. Its 355 carrier coordinates and two 355-dimensional moment arrays are real costs; calling it a memory-free 47-variable method would be incorrect.

3.5. What financing-gradient control requires

Let $P(y, a) = B_0$ be the exact financing equation, P_N its numerical approximation, and $a(y), a_N(y)$ their roots on a common bracket. Let $H(y, a)$ and $H_N(y, a)$ denote exact and numerical payoff functions before root substitution. Suppose $P_a, P_{N,a} \geq d_* > 0$ and $|P - P_N| \leq e_P$, so $|a - a_N| \leq \delta = e_P / d_*$. At the respective roots, let

$$\|H_{N,y} - H_y\| \leq E_A, \quad |H_{N,a} - H_a| \leq E_B, \quad \|P_{N,y} - P_y\| \leq E_C, \quad |P_{N,a} - P_a| \leq E_D.$$

Each E includes fixed-offset approximation error and the applicable offset-Lipschitz allowance times δ . The derivative factor bounds are explicitly two-sided in the numerical/exact distinction:

$$\max\{|H_a|, |H_{N,a}|\} \leq M_B, \quad \max\{\|P_y\|, \|P_{N,y}\|\} \leq M_C. \quad (14)$$

PROPOSITION 3—Reduced-gradient perturbation: *Under these assumptions and differentiability of both implicit roots,*

$$\|\nabla_y H_N(y, a_N(y)) - \nabla_y H(y, a(y))\| \leq E_A + \frac{E_B M_C + M_B E_C}{d_*} + \frac{M_B M_C E_D}{d_*^2}. \quad (15)$$

The bounded proposal extension clips preference before running-utility evaluation. Its integrability and derivative implementation are not an identification of its gradient with that of (5). Neither this revision nor the imported R25 experiment supplies verified values for every term in (15) for the original stopped payoff along the executed optimizer paths. We instead report certified payoff decisions and finite secants as such. The theorem removes an ambiguity in the previous derivative-factor statement; it does not manufacture the missing constants.

4. WHOLE-DOMAIN VERIFICATION IN AN UNSPANNED-RISK STOPPED ECONOMY

To test stochastic propagation, stopping, and current-state restart without the original financed decoder, consider a separate investment economy on

$$(t, x, z) \in [0, 1] \times [.5, 2] \times [-1, 1], \quad a \in [0, 2.2], \quad p \in [0, 1],$$

with independent Brownian motions W, B :

$$dX = (.02X + a + .04pX) dt + .25pX dW, \quad dZ = .3 dB. \quad (16)$$

The B shock is not spanned by the risky wealth exposure. Let stopping occur at first exit from the interior box or time 1. Set $A(t) = .8 + .2t$ and supply the smooth witness

$$v(t, x, z) = A(t)(\log x + .1xz - .1z^2). \quad (17)$$

At every stopping face, settlement equals v . Running reward is $b(t, x, z) - a^2/2$, where the model's transfer is

$$b = .04v - v_t - .02xv_x - \frac{.3^2}{2}v_{zz} - \frac{v_x^2}{2} + \frac{.04^2v_x^2}{2(.25)^2v_{xx}}. \quad (18)$$

All these coefficients are bounded on the closed box, and $v_{xx} = -A/x^2 < 0$. This is a manufactured model: the transfer is chosen to make the value independently known. The experiment validates policy approximation and continuous verification in a stochastic stopped setting; it is not an exercise in discovering an unknown value function or calibrating an empirical economy.

THEOREM 2—Exact stochastic value and restart regret: *For the model (16)–(18), the optimal current-state actions are*

$$a^*(t, x, z) = A(t)(1/x + .1z), \quad p^*(t, x, z) = .64(1 + .1xz). \quad (19)$$

They lie in the declared action bounds, and the unrestricted stopped value is v . For every admissible current-state policy $\pi = (a, p)$,

$$v(t, x, z) - J^\pi(t, x, z) = \mathbb{E}_{t,x,z} \int_t^\tau e^{-.04(s-t)} \frac{(a - a^*)^2 + .0625A(s)(p - p^*)^2}{2} ds. \quad (20)$$

Consequently, a uniform upper bound $\bar{d}$ on the integrand gives

$$0 \leq v - J^\pi \leq \bar{d} \frac{1 - e^{-.04(1-t)}}{.04} \quad (21)$$

at every original state of this auxiliary economy and at every restart time, without stored initialization weights or a financing root.

The state–time statement is stronger than a list of simulated starting states. It follows from a complete cover and a stopped verification identity. The tabulated start times below are evaluations of the same uniform formula; no interpolation between those times is used in the proof.

4.1. Centered second-order action certificates

Let $e_j = \pi_j - \pi_j^*$ be an action error on a rectangular cell with center m and half-widths r_i . Interval propagation supplies point enclosures of $e_j(m)$ and $\partial_i e_j(m)$, and whole-cell enclosures of $\partial_{ik} e_j$. Denote valid upper absolute bounds by E_j , G_{ij} , and H_{ikj} .

THEOREM 3—Second-order cell enclosure: *If e_j is twice continuously differentiable on the cell, then throughout it*

$$|e_j| \leq E_j + \sum_i G_{ij} r_i + \frac{1}{2} \sum_i H_{iij} r_i^2 + \sum_{i < k} H_{ikj} r_i r_k. \quad (22)$$

The minimum of this bound and a valid first-order mean-value enclosure remains valid. Substituting such componentwise error bounds into the nonnegative quadratic integrand in (20), maximizing over a complete cover, and applying (21) yields a whole-state–time certificate.

For each tanh or sigmoid layer, the implementation propagates

$$D(\phi \circ q) = \phi'(q)Dq, \quad D^2(\phi \circ q) = \phi'(q)D^2q + \phi''(q)Dq Dq'.$$

All affine operations and transcendental enclosures use the inherited directed interval arithmetic. Saved binary64 network weights are interpreted as exact real coefficients; displayed economic decoder constants are exact rationals. The certificate therefore concerns the real-valued policy represented by those coefficients, not an unspecified hardware rounding path at deployment. The control scales are exact 2.2 and 1. A separate floating-point comparison with automatic differentiation tests the implementation of the chain rule; it does not replace the interval proof.

Both neural policies are the unchanged R25 3–12–12–2 policies. They were fitted using the known actions on a $5 \times 9 \times 7$ tensor rule. R26 does not retrain them. It verifies nested covers of sizes 512, 4096, and 32768, retaining the unsuccessful coarse certificates. Every endpoint and center of these covers is dyadic. The code saves every cell, its action-error bounds, its quadratic deficit bound, and a file digest. No sampled cell is substituted for an unverified remainder of the domain.

4.2. Exact rational verification of the classical reference

The retained direct policy approximates $1/x$ by a degree-eight polynomial $P(\xi)$, $\xi = (x - 1.25)/.75$, uses $a = A(P(\xi) + .1z)$ with projection onto $[0, 2.2]$, and uses the exact portfolio expression in (19). Projection cannot increase distance to the feasible optimum. The polynomial coefficients are the unchanged stored binary64 coefficients, interpreted exactly as rational numbers.

The inherited direct verifier formed derivative coefficients with the ordinary floating multiplication $i c_i$ before treating them as point intervals. That operation does not itself enclose the exact product of the stored coefficient and i . We do not retroactively label the old derivative-coefficient computation as a directed enclosure. Instead we remove the derivative calculation from the current certificate.

LEMMA 2—Exact residual-polynomial certificate: *Let $R(\xi) = ((5 + 3\xi)/4)P(\xi) - 1$. On a cell $[a, b] \subset [-1, 1]$, express $R(a + (b - a)s) = \sum_{i=0}^n \beta_i \binom{n}{i} s^i (1 - s)^{n-i}$ with exact rational coefficients. Then*

$$\sup_{\xi \in [a, b]} |P(\xi) - 1/x| \leq \frac{\max_i |\beta_i|}{(5 + 3a)/4}. \quad (23)$$

With E the largest bound over a complete cover, the direct policy's regret is at most $\frac{1}{2}E^2(1 - e^{-.04(1-t)})/.04$ for every starting state and time.

The computation uses 64 one-dimensional cells. The t, z dependence is handled analytically through $A \leq 1$, and portfolio error cancels exactly. The resulting bound is $1.2978908373252045 \times 10^{-7}$. Its small value is not a claim about superiority of a general classical Bellman solver; it is the correct strong baseline for this structured, known-value problem. In particular, it prevents a verifier improvement from being reported as neural-policy superiority.

5. NUMERICAL EVIDENCE AND ECONOMIC SCOPE

The evidence has two distinct chronologies. We import the completed R25 experiment at commit `c78d934` without changing its scientific files. Its protocols precede its execution. The present R26 protocol was recorded locally after examining those results and before the new stress tests and fixed-policy verification. It is not described as a remotely preregistered or independently held-out study. The new seeds 26101, 26102 are diagnostic stress seeds; the fixed stochastic policies retain their original seeds 25201, 25202.

5.1. Expanded calibration and held-out comparison

The imported experiment expands the learning-rate grid to

$$\{.005, .02, .08, .32, 1.28, 5.12, 20.48\}.$$

At seeds 25001, 25002, every arm receives 200 objective/gradient calls per rate. Selection uses the mean certified endpoint payoff lower bound; infeasible endpoints are ineligible, and the fixed tie rule selects the smaller rate. The selected rates are .08 for neural and whitening Adam and .32 for direct, diagonal, static tangent, and moving-transport Adam. All six selections have a worse larger-rate trial and are strictly below the grid ceiling. This repairs the earlier endpoint-censoring

TABLE I
ORIGINAL ECONOMY, CENTRAL STATE: IMPORTED HELD-OUT COMPARISON

Method	Regret upper	Calls	Generation seconds	Checking seconds
Neural Adam	0.0019264	400.0	1.144	13.836
Direct Adam	0.0019367	400.0	1.016	13.887
Diagonal Adam	0.0019329	400.0	1.039	13.961
Static tangent Adam	0.0020835	400.0	1.023	13.923
Whitening Adam	0.0179096	400.0	1.041	13.958
Moving transport Adam	0.0019275	400.0	1.687	13.897
Neural L-BFGS-B	0.0096083	243.0	0.671	13.840
Direct L-BFGS-B	0.0019235	123.0	0.307	13.891

The state is $(t, u, x) = (0, 2, 1.25)$, $k = 2$, proposal orders $(8, 16)$. Regret is the worst certified deployed bound over two evaluation seeds; work columns are seed means. These are financed policies, not whole-domain current-state actors. Checking sums scheduled checkpoint verification; it is not a measured online latency. Carrier construction and common preprocessing, and any common dual construction, are not included in the generation column. The full run records remain available.

problem for this finite design, not a proof that all schedules or all hyperparameters have been optimized.

Evaluation uses seeds 25101, 25102 and proposal orders $(8, 16)$ and $(12, 24)$, without retuning. Each first-order arm has a 400-call cap. L-BFGS-B has the same cap, with its declared line search and tolerances; early termination is retained rather than padded into fictitious calls. The 84 tuning trajectories, 32 held-out trajectories, six earlier online trajectories, and separate reference studies have different inferential purposes. Their sum is bookkeeping, not a replication count for one effect.

Direct L-BFGS-B remains the strongest tested central-state solver. This is not merely a difference in common upper bounds: each matched interval for neural Adam payoff minus direct L-BFGS-B payoff is strictly negative. At the coarse rule the two intervals are approximately $[-2.3223, -1.5593] \times 10^{-6}$ and $[-3.2620, -2.4991] \times 10^{-6}$. The fine-rule comparisons have the same sign. These are certified differences, not significance tests across a population of seeds.

Moving transport substantially closes the gap between the static tangent arm and neural Adam, but does not beat the direct L-BFGS-B frontier. Because its carrier incurs differentiation and storage, its generation time is charged rather than presented as costless quotient preconditioning. The retained R24 equivalent-chart and neural L-BFGS-B failures are preserved. Neither a static-chart failure nor this moving-control result establishes an irreducible neural advantage.

5.2. Actual online rejection and recovery

The new stress schedule has three blocks: 200 calls at the selected base rate, 50 calls at 20.48, and 100 recovery calls. The recovery rate is halved after a rejected stress block. This deliberately aggressive block tests the mechanism, not the solver frontier. Neural, direct, and moving-transport arms use the original central economy and the unchanged stopped checker. Every proposal, incumbent, decision, and optimizer snapshot is retained.

Across six trajectories, 13 proposals are accepted and five rejected. Four rejected aggressive candidates produce separate shadow continuations from the rejected optimizer state. All four shadow final states differ from the actual rolled-back trajectories. The moving arm accepts both aggressive proposals, but one later recovery proposal is rejected. Thus rejection is not asserted for a branch never entered. Exact equality of the canonical before/after optimizer-state

TABLE II
R26 SYNCHRONOUS STRESS EXECUTION IN THE ORIGINAL CENTRAL ECONOMY

Seed	Arm	A/R	Regret upper	Gen.	Check	Coupled
26101	Neural	2/1	0.0019257	1.057	8.176	9.269
26101	Direct	2/1	0.0019471	0.941	8.056	9.012
26101	Moving transport	2/1	0.0019339	1.646	8.001	9.674
26102	Neural	2/1	0.0019266	1.058	7.896	8.989
26102	Direct	2/1	0.0019469	0.943	7.994	8.952
26102	Moving transport	3/0	0.0019310	1.543	7.862	9.433

A/R denotes accepted/rejected blocks. Times are seconds from the new local serial execution. Check includes the initial incumbent certificate. Coupled time is the measured interleaved run from initial evaluation through the final decision; it includes bookkeeping but excludes initial model construction, inherited common dual construction, and the separately timed shadow continuation. All arms use 350 block gradient calls; these stress runs are not the held-out comparison in Table I.

TABLE III
FIXED-POLICY STOCHASTIC CERTIFICATES ON THE SAME CELL COVERS

Seed	Cells	First order	New certificate	Check seconds
25201	512	0.15838620	0.12137299	0.145
25201	4,096	0.00939394	0.00423980	1.123
25201	32,768	0.00137310	0.00083434	8.848
25202	512	0.20476383	0.17534400	0.144
25202	4,096	0.01128653	0.00515876	1.109
25202	32,768	0.00133294	0.00078841	8.846

Both methods certify the same stored neural policy over $[0, 1] \times [.5, 2] \times [-1, 1]$ in the manufactured, unspanned-risk stopped economy. The new certificate takes the minimum of two valid componentwise error bounds. Times include both same-cover calculations and cell-file writing in the local execution. They are not a hardware-normalized speedup relative to the earlier CI run.

serializations is checked at every rejection, including moments and step counters. No shadow certificate is used to replace the production incumbent.

These executions establish synchronous certification and actual state restoration. They do not establish an Adam convergence rate or show that the stress schedule improves production efficiency. The monotonicity theorem applies to the executed acceptance decisions; the finite-work polling theorem concerns the separately defined poll and its stated hypotheses.

5.3. Whole-state-time reference verification

Table III compares first-order and second-order error enclosures on identical cells for fixed neural policies in the auxiliary stochastic economy. Each attempted cover is retained, including those above the target.

The target .001 is attained at 32,768 cells for both policies, whereas the retained first-order stopping rule required 262,144. The reduction is eightfold in cell count at the same declared tolerance, not an eightfold measured runtime gain. The final bounds, .000834337 and .000788415, are valid at all starting times; the earlier finer certificates may of course be numerically tighter. The policies have not changed. The exact rational direct bound, approximately 1.298×10^{-7} , remains much stronger. A successful stochastic neural certificate therefore demonstrates nontrivial verified approximation and restart coverage, not a neural accuracy/work victory.

5.4. *Original full-state progress and stopped-payoff sensitivity*

The imported continuation experiment retains the original economy, current-state actor, full domain, and .01 target. It uses 85 gradient calls across four blocks and delivers an all-start-time bound of 7.181834580823298. Relative to 7.241462443133396, the bound decreases by about .059628, or .823%. The remaining bound is approximately 718 times the target. No separately certified actor-payoff improvement is recorded. Neither a local financed bound nor a manufactured-model certificate is used to change this conclusion.

The imported secant experiment certifies five original stopped-payoff policies along a raw-coordinate interpolation between neural Adam and direct L-BFGS-B outputs. Of four adjacent secant intervals, only the first is strictly positive, approximately $[5.332, 8.383] \times 10^{-6}$. The remaining three contain zero. These comparisons rigorously concern finite policy differences; they do not bound every continuous derivative, prove an optimizer ordering robust to all surrogate errors, or instantiate Proposition 3. They provide an actionable comparison primitive while the full stopped-gradient bridge remains a separate mathematical and computational task.

5.5. *Reproducibility and preservation*

The new source contains a protocol, all three attempted covers for each fixed neural policy, exact rational Bernstein coefficients on all 64 cells, every online state snapshot, shadow records, machine-readable publication tables, and checks for the completed Hamiltonian identity, derivative propagation, exact polynomial conversion, complete cover, acceptance/rollback invariants, and historical preservation. The single root revision index is updated, with its old bytes archived. No historical scientific source or result is overwritten. The new response identifies the disposition of all twelve blocking findings, all six technical findings, and all eight requested next steps.

6. CONCLUSION

A Neural Bellman Operator for the original stopped preference-adjustment economy remains the organizing objective. This revision adds an executed certification-coupled proposal mechanism, a conditional finite-work quotient-polling theorem, a moving historical-transport comparison, and sharper whole-domain verification in a stopped unspanned-risk economy. The experiments distinguish a real optimizer rollback from checkpoint replay and a verifier gain from a policy gain. They retain a strong direct benchmark and the adverse neural outcomes rather than selecting away the relevant comparison.

The next successful original-economy calculation must supply a current-state actor with a complete-domain certificate near the declared .01 tolerance and a separate payoff-improvement statement when such a statement is made. A proof that the original checker supplies arbitrarily narrow policy intervals, or instantiated stopped-gradient constants, would connect the verified-improvement component more tightly to the optimization analysis. The present results do not claim those conclusions. They provide reproducible methods, proofs, and evidence with which to pursue them, while preserving the original economic problem and every prior derivation for further scrutiny.

APPENDIX A: PROOFS

A.1. *Stopped verification*

PROOF OF PROPOSITION 1: Apply stopped Itô's formula to $e^{-.04(s-t)}v(s, u_s, x_s)$, localizing first if needed. On the bounded stopped domain the stochastic integral has zero expectation

under the stated admissibility and smoothness conditions. Thus

$$J_k^a - v(t, u, x) = \mathbb{E} \int_t^\tau e^{-.04(s-t)} \mathcal{R}_k^a v(s, u_s, x_s) ds + \mathbb{E}[e^{-.04(\tau-t)}(g - v)(\tau, u_\tau, x_\tau)].$$

For every a , $\mathcal{R}_k^a v \leq e + q$, giving $J_k^a \leq v + b + (e + q)C_t$, where $C_t = (1 - e^{-.04(1-t)})/.04$. For π , $\mathcal{R}_k^\pi v \geq -e$, giving $J_k^\pi \geq v - b - eC_t$. Take the supremum over a and subtract. One-sided test functions follow from the same identity; deterministic time envelopes may be integrated to 1 because $\tau \leq 1$. *Q.E.D.*

A.2. Improvement, oracle precision, and polling

PROOF OF PROPOSITION 2: Validity of the two enclosures gives $F(y') \geq L(y') > U(y) \geq F(y)$ on acceptance. A rejection restores the original policy and optimizer state. With forcing term σh^2 , each accepted payoff gain is strictly larger than that quantity. The sum of these gains is bounded by $M - F(y_0)$, yielding the stated conservative integer bound. No assertion about gradient convergence is needed. *Q.E.D.*

PROOF OF LEMMA 1: For a valid enclosure of width at most w_j , its lower endpoint is no less than the true value minus w_j , and its upper endpoint is no greater than that value plus w_j . Consequently the lower and upper difference endpoints differ from the true payoff difference by at most $2w_j$ in the relevant direction. If $2w_j < \gamma$, the whole difference interval lies strictly on the correct side of the threshold. The displayed refinement count ensures this strict inequality. If the margin is zero, no positive finite width need separate it. *Q.E.D.*

PROOF OF THEOREM 1: At an unsuccessful poll, $L(y + hd) \leq U(y) + \sigma h^2$ for every $d \in \mathcal{D}$. Interval widths bounded by κh^2 imply

$$F(y + hd) - F(y) \leq (\sigma + 2\kappa)h^2.$$

The Lipschitz-gradient inequality on the feasible segment gives

$$F(y + hd) - F(y) \geq h \nabla F(y)' d - \frac{L}{2} h^2.$$

Combining these inequalities, maximizing over d , and using the cosine measure yields (9). Each successful poll increases payoff by more than σh^2 . The payoff upper bound therefore allows at most the number of successes in Proposition 2 before an unsuccessful complete poll. At most $|\mathcal{D}| + 1$ oracle evaluations per poll, including a conservative incumbent reevaluation, gives (10). Multiplying by the uniform oracle cost gives the work bound. The chosen h gives the stationarity tolerance. All feasibility, regularity, and oracle requirements are precisely those in Assumption 1; failure of any requirement does not imply stationarity. *Q.E.D.*

A.3. Transport and financing perturbations

For completeness, the first-moment recursion $m_n = \beta_1 m_{n-1} + (1 - \beta_1) J'_n g_n$ with zero initialization expands to the coefficients a_{nr} after bias correction. Taylor expansion of f along $-\eta_n D_n^{-1} \hat{m}_n$ yields (11); integrating the Lipschitz Jacobian difference along the segment bounds the remainder by $H \|\delta \vartheta_n\|^2 / 2$. The executed carrier uses the corresponding sequence of pull-backs and pushforwards but advances economic coordinates linearly, as stated in (13).

PROOF OF PROPOSITION 3: The mean-value inequality and the lower bound on P_a imply $|a - a_N| \leq e_P/d_*$. Implicit differentiation gives $a_y = -P_y/P_a$ and $a_{N,y} = -P_{N,y}/P_{N,a}$. Hence the two reduced gradients are $H_y - H_a P_y/P_a$ and $H_{N,y} - H_{N,a} P_{N,y}/P_{N,a}$. Add and subtract products one factor at a time. The payoff-derivative and price-derivative differences contribute $(E_B M_C + M_B E_C)/d_*$; the reciprocal difference contributes at most $M_B M_C E_D/d_*^2$, since both denominators are at least d_* . The direct derivative difference contributes E_A . The root-shift allowances are already included in the E terms, so no offset error is counted twice. *Q.E.D.*

A.4. The stochastic reference and its interval certificates

PROOF OF THEOREM 2: The controlled residual of v in the reference economy is

$$v_t + (.02x + a + .04px)v_x + \frac{.25^2 p^2 x^2}{2} v_{xx} + \frac{.3^2}{2} v_{zz} + b - a^2/2 - .04v.$$

Complete the square in a and p , using $v_x = A(1/x + .1z)$ and $v_{xx} = -A/x^2$. With (18), the result is exactly

$$-\frac{1}{2}\{(a - a^*)^2 + .0625A(p - p^*)^2\}.$$

On the closed box, $.32 \leq a^* \leq 2.1$ and $.512 \leq p^* \leq .768$, so the unconstrained maximizers are admissible. The optimal controls are smooth on a neighborhood of the stopped domain. Stopped Itô's formula applies, with settlement equal to v on every stopping face. The residual is nonpositive for every admissible policy and zero for the displayed optimizer. Thus $V = v$ and the same formula with an arbitrary policy gives (20). Bounding the nonnegative integrand and using $\tau \leq 1$ proves (21) for every starting point and time. *Q.E.D.*

PROOF OF THEOREM 3: Write a point as $m + h$, with $|h_i| \leq r_i$. The second-order integral Taylor formula gives

$$e_j(m + h) = e_j(m) + De_j(m)h + \int_0^1 (1-s)h'D^2e_j(m + sh)h ds.$$

The integral weight is $1/2$. Absolute values and the symmetric Hessian expansion yield the diagonal half-weights and the single mixed terms in (22). Whole-cell interval Hessians cover every point on the segment. Both first- and second-order bounds are individually valid, so their componentwise minimum is valid. The Hamiltonian deficit is nondecreasing in the nonnegative squared-error bounds and in the positive coefficient A ; a complete cover and (21) then give the uniform result. *Q.E.D.*

PROOF OF LEMMA 2: The Bernstein basis is nonnegative on $[0, 1]$ and sums to one. Hence the polynomial lies between the smallest and largest exact coefficients β_i . The denominator $x = (5 + 3\xi)/4$ is positive and bounded below by $(5 + 3a)/4$ on the cell, proving (23). The exact action error before projection is $A(P - 1/x)$; projection onto the feasible interval cannot increase its distance from a^* . As $A \leq 1$ and the portfolio action is exact, the deficit is bounded by $E^2/2$. Apply Theorem 2. The conversion itself uses the identity that a local power coefficient q_k contributes $q_k \binom{i}{k} / \binom{n}{k}$ to Bernstein coefficient i , for $k \leq i$; every operation in this conversion is rational. *Q.E.D.*

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NEURAL BELLMAN OPERATORS: TECHNICAL SUPPLEMENT AND PRESERVED HISTORY

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This supplement records calibration, held-out comparisons, every new online decision, whole-domain restart bounds, arithmetic conventions, source lineage, and reproduction instructions. It then reproduces the complete reviewed R24 article and technical supplement without removing historical derivations or adverse evidence. The appended documents retain their original revision labels and page numbers; R26’s new conclusions are stated in the current article and response.

1. READING THE EVIDENCE

Three numerical objects must be kept distinct. The original full-state actor uses (t, u, x) alone in the stopped preference-adjustment economy and retains target .01. Its imported bound is 7.181834580823298, with no separately certified payoff improvement. The financed central-state and regional constructions use the original economy but a restricted implementation that may store a factor and initialization weights. The manufactured unspanned-risk economy has its own state (t, x, z) , target .001, known value witness, and current-state actors. A small number in the third object is not evidence of solving the first.

The latest report addressed is `reviews/2026-09-23-econometrica-r24/reference_report.md`, at review head `0eb0400ae311333fb77826ef9e2f067718926ae8`. It reviewed manuscript commit `14ce582e188f437cc8d99f310edd4f06515936a4`. During this revision we found the completed R25 scientific experiment, frozen at `c78d934c0b3b8d346a40c7ddcaf415e167462265`. Its results are imported and attributed as pre-existing evidence. R26’s new executions are the stress/rollback study and fixed-policy second-order and exact rational verification. They are not retrospectively labeled R25 results.

2. EXPANDED RATE CALIBRATION

Table I gives every rate-level mean from the two declared tuning seeds. There are 42 rate–representation configurations and 84 underlying trajectories. Selection is by mean certified endpoint payoff lower bound, subject to feasibility at both seeds. All raw per-seed files are retained under the R25 results directory. Larger-rate deterioration brackets the selected point within this finite rate family; it does not optimize every schedule, architecture, or resource allocation.

TABLE I: All expanded-grid tuning means

Representation	Rate	Mean payoff lower	Feasible	Selection
Neural	0.005	-1.28750818	Yes	–
Neural	0.02	-1.28726134	Yes	–
Neural	0.08	-1.28722814	Yes	Selected

Representation	Rate	Mean payoff lower	Feasible	Selection
Neural	0.32	-1.29126462	Yes	—
Neural	1.28	-1.29493757	Yes	—
Neural	5.12	-1.29516817	Yes	—
Neural	20.5	-1.29870289	Yes	—
Direct	0.005	-1.30441066	Yes	—
Direct	0.02	-1.29001771	Yes	—
Direct	0.08	-1.28752144	Yes	—
Direct	0.32	-1.28725009	Yes	Selected
Direct	1.28	-1.28730312	Yes	—
Direct	5.12	-1.28973026	Yes	—
Direct	20.5	-1.29830178	Yes	—
Diagonal	0.005	-1.30254589	Yes	—
Diagonal	0.02	-1.28942345	Yes	—
Diagonal	0.08	-1.28745977	Yes	—
Diagonal	0.32	-1.28724261	Yes	Selected
Diagonal	1.28	-1.28728777	Yes	—
Diagonal	5.12	-1.29091535	Yes	—
Diagonal	20.5	-1.29136187	Yes	—
Static tangent	0.005	-1.29338548	Yes	—
Static tangent	0.02	-1.28809121	Yes	—
Static tangent	0.08	-1.28736038	Yes	—
Static tangent	0.32	-1.28733298	Yes	Selected
Static tangent	1.28	-1.29490638	Yes	—
Static tangent	5.12	-1.29490573	Yes	—
Static tangent	20.5	-1.29498455	Yes	—
Whitening	0.005	-1.31592177	Yes	—
Whitening	0.02	-1.30735092	Yes	—
Whitening	0.08	-1.30550875	Yes	Selected
Whitening	0.32	-1.30651228	Yes	—
Whitening	1.28	-1.30637072	Yes	—
Whitening	5.12	-1.30863433	Yes	—
Whitening	20.5	-1.32439113	Yes	—
Moving transport	0.005	-1.28751271	Yes	—
Moving transport	0.02	-1.28729186	Yes	—
Moving transport	0.08	-1.28724663	Yes	—
Moving transport	0.32	-1.28723401	Yes	Selected
Moving transport	1.28	-1.28723602	Yes	—
Moving transport	5.12	-1.28723602	Yes	—
Moving transport	20.5	-1.28723602	Yes	—

3. FINE-RULE HELD-OUT COMPARISON

The fine rule uses the same held-out seeds and selected configurations as the coarse rule, without retuning. Its stronger quadrature is a robustness diagnostic for optimization outcomes, not a rigorous derivative-error enclosure. Every method continues to use the same independent stopped-payoff checker.

Both neural-minus-direct L-BFGS-B payoff difference intervals remain negative at the fine rule. These enclosures are computed from matched policy intervals, with outward-rounded subtraction. A solver’s smaller regret upper bound alone would not establish a strict payoff

TABLE II
HELD-OUT CENTRAL-STATE COMPARISON AT PROPOSAL ORDERS (12, 24)

Method	Regret upper	Calls	Generation seconds	Checking seconds
Neural Adam	0.0019264	400.0	2.113	13.920
Direct Adam	0.0019367	400.0	1.942	13.975
Diagonal Adam	0.0019328	400.0	1.931	13.898
Static tangent Adam	0.0020835	400.0	1.962	13.796
Whitening Adam	0.0179100	400.0	1.900	13.845
Moving transport Adam	0.0019278	400.0	2.639	13.826
Neural L-BFGS-B	0.0096082	242.0	1.218	13.821
Direct L-BFGS-B	0.0019235	104.0	0.499	13.734

Worst regret over two evaluation seeds and mean work. The state is $(0, 2, 1.25)$ in the original economy, $k = 2$, under the financed implementation. These are not all-start-time certificates or hardware-normalized timings for the R26 local execution.

ordering; the paired intervals do. The current presentation retains direct L-BFGS-B as the strongest tested central-state method.

4. EVERY SYNCHRONOUS STRESS DECISION

Each trajectory begins from a valid certified incumbent. The first block uses 200 calls at the base rate, the stress block uses 50 calls at 20.48, and the final block uses 100 calls at the base rate or half that rate after a rejected stress proposal. Acceptance requires the candidate lower payoff to exceed the incumbent upper payoff. The proposed policy's regret in the next table is not substituted for the incumbent's regret when the proposal is rejected.

TABLE III: All R26 online decisions, including rejected proposals

Seed	Arm	Block	Rate	Candidate regret	Decision
26101	Neural	1	0.08	0.001927	Accept
26101	Neural	2	20.5	0.009641	Restore
26101	Neural	3	0.04	0.001926	Accept
26101	Direct	1	0.32	0.001954	Accept
26101	Direct	2	20.5	0.039585	Restore
26101	Direct	3	0.16	0.001947	Accept
26101	Moving transport	1	0.32	0.001937	Accept
26101	Moving transport	2	20.5	0.001934	Accept
26101	Moving transport	3	0.32	0.001942	Restore
26102	Neural	1	0.08	0.001927	Accept
26102	Neural	2	20.5	0.051789	Restore
26102	Neural	3	0.04	0.001927	Accept
26102	Direct	1	0.32	0.001954	Accept
26102	Direct	2	20.5	0.035360	Restore
26102	Direct	3	0.16	0.001947	Accept
26102	Moving transport	1	0.32	0.001939	Accept
26102	Moving transport	2	20.5	0.001937	Accept
26102	Moving transport	3	0.32	0.001931	Accept

There are 13 accepts and five restores. For every restore, the before and after canonical state files are equal, and their SHA-256 digests agree. The saved state includes parameters, moments, and the optimizer counter; the moving arm additionally includes its independent economic coordinates and carrier. A common external rate schedule is not confused with a rejected optimizer’s moments. The restore decision is taken before the next block is generated.

The four shadow continuations start from the rejected aggressive candidate, use the same recovery call count and rate as their actual counterpart, and never influence the production incumbent. Their final regret bounds are approximately .009641149 and .051788673 for the two neural seeds, and .039585071 and .035359719 for the direct seeds. Every shadow trajectory differs from the actual restored trajectory. Their times and state files are preserved separately. The moving arm accepts both aggressive blocks, so no aggressive-block shadow is reported for it; one later moving proposal is rejected and restored. We do not fabricate a shadow where the declared condition did not occur.

5. WHOLE-DOMAIN RESTART CERTIFICATES

The fixed neural actors in the auxiliary stochastic economy use only (t, x, z) . The certificate’s proof integrates a uniform nonnegative Hamiltonian-deficit upper bound from any restart time to 1. It neither stores initialization weights nor assumes the restart lies on a financed reachable manifold. The following numbers are evaluations of that all-time bound, not separate Monte Carlo estimates.

TABLE IV
AUXILIARY-ECONOMY UNIFORM REGRET BOUNDS AT SELECTED RESTART TIMES

Seed	$t = 0$	$t = .25$	$t = .5$	$t = .75$	$t = 1$
25201	0.000834337	0.000628871	0.000421340	0.000211723	0.000000000
25202	0.000788414	0.000594257	0.000398149	0.000200070	0.000000000

The neural certificate saves all $512 + 4096 + 32768$ cells attempted per policy. The finest new cover has $16 \times 64 \times 32$ cells. The inherited first-order stopping rule required $32 \times 128 \times 64$ cells. The minimum of the first-order and second-order action-error enclosures is used componentwise before the nonnegative quadratic deficit is formed. Taking the minimum of two valid upper bounds does not require a probabilistic selection correction. Point derivatives, however, may not substitute for interval Hessians over the complete cell.

6. ARITHMETIC AND EXACT POLYNOMIAL CONVERSION

The interval backend is the inherited rational-Taylor binary64 implementation. Rational constants are first enclosed; elementary operations and affine transformations are outward rounded. Tanh and sigmoid bounds use the backend’s exponential enclosure. Positive derivative ranges may be intersected with their analytic ranges, $0 \leq \tanh' \leq 1$ and $0 \leq \text{sigmoid}' \leq 1/4$. The second derivative is then propagated by the exact chain rule. All stored weights are fixed binary64 numbers interpreted as exact reals.

For the direct polynomial policy, write $P(\xi) = \sum_{j=0}^8 c_j \xi^j$, with c_j the exact rational value of the stored float. On a rational cell $\xi = a + (b - a)s$, local power coefficients of the residual $R = xP - 1$ are

$$q_k = \sum_{j=k}^n R_j \binom{j}{k} a^{j-k} (b - a)^k.$$

Its Bernstein coefficients are

$$\beta_i = \sum_{k=0}^i q_k \frac{\binom{i}{k}}{\binom{n}{k}}.$$

Every coefficient and cell endpoint is stored as an exact rational string in `results/jets/direct_exact.json`. The basis convex-hull bound, the positive lower bound on x , nonexpansiveness of action projection, and exact portfolio cancellation give the certificate in the article. This avoids the inherited ordinary floating multiplication in derivative coefficients. We preserve that old implementation and its output as historical material rather than changing a past calculation silently.

7. ORACLE PRECISION AND THE ORIGINAL STOPPED GRADIENT

An interval decision for $F(y') - F(y)$ is a statement about a finite payoff difference. Dividing by a finite step gives a secant enclosure, not a derivative enclosure. To enclose a directional derivative one additionally needs a rigorously controlled local remainder, such as a verified Hessian bound along the relevant policy segment. No such complete stopped-payoff derivative remainder is supplied for the original economy here.

The article’s reduced-gradient perturbation result explicitly bounds both numerical and exact derivative factors. Its root conditioning, price error, derivative errors, and offset-Lipschitz allowances must all be instantiated before it bounds the original executed economic gradient. Its proof is not an excuse to omit these quantities. Likewise, the payoff-polling theorem requires sufficiently narrow valid payoff intervals, smoothness on feasible poll segments, and an upper payoff bound. The actual online gate establishes safe accepted improvements with the available checker, while the theorem’s shrinking-oracle hypotheses remain additional requirements for stationarity.

The exact quadratic regression test verifies the logic of the finite-poll estimate in a setting where the oracle conditions hold exactly. It is labeled as a mathematical implementation test, not a new original-economy optimizer experiment. The stopped reference has a stronger independent exact identity, but it is a separate model.

8. WORK ACCOUNTING

For a batch of m policies, a complete cost account has the form

$$T_{\text{dual}} + \sum_{i=1}^m (T_{\text{setup},i} + T_{\text{proposal},i} + T_{\text{check},i} + T_{\text{bookkeeping},i}).$$

The dual cost is amortized only if the same validated dual is reused legitimately. The imported held-out table reports proposal and scheduled-checkpoint times, not this entire expression. Carrier construction and shared preprocessing have their own records. The new online table measures actual interleaving of generation and checking from initial incumbent evaluation; it excludes construction of the inherited common dual and initial model object. The shadow continuation is separately timed and excluded from the production online latency. These distinctions prevent a sub-two-second proposal from being presented as an end-to-end certified solve.

R25 and R26 were executed in different recorded environments. In particular, the eightfold cell-count reduction is not reported as an eightfold wall-clock speedup. The local same-cover

table includes both first-order and second-order calculations and file writing. Timings are observations of these runs, not a benchmark against an external implementation.

9. SOURCE PRESERVATION AND REPRODUCTION

The R24 source archive contains 5,760 files. Their SHA-256 values are recorded before the new edits. The old root revision index is preserved verbatim under the new source-audit directory; all historical scientific files retain their contents. The imported R25 artifact contains 781 non-cache files, whose unchanged bytes are separately checked. The root-tree and commit identities of the frozen remote base are recorded in the delivery metadata. A local artifact or branch is not represented as remotely published unless a remote write has actually succeeded.

From a complete checkout containing the frozen R25 evidence and this revision, the new computation is reproduced from the repository root with:

```
python revisions/2026-09-23-r26/replication/online_stress.py
python revisions/2026-09-23-r26/replication/jet_certificate.py
python revisions/2026-09-23-r26/replication/publication.py
python revisions/2026-09-23-r26/replication/test_revision.py
```

Set `PYTHONDONTWRITEBYTECODE=1` to avoid local cache files. The original R25 protocols and scripts reproduce the imported calibration, full-state continuation, stochastic training, and secants separately; the new fixed-policy verification does not retrain those actors. A fresh reproduction should preserve the frozen evidence directory or run in a separate working tree. Runtime fields and ZIP container metadata need not be bitwise identical, while the mathematical policy inputs and certificate inequalities must agree within the stated arithmetic conventions.

The new checks cover the symbolic Hamiltonian completion, interval-jet chain rule against a separately computed floating derivative diagnostic, exact Bernstein identities, complete cell covers and file digests, restart bounds, all acceptance inequalities, complete rejected-state restoration, shadow isolation, original-model scope, interior tuning selection, and preservation. An initial test located the imported manifest relative to the repository root rather than the R25 directory; that test-path error and its correction are recorded. It was not a changed scientific result. The initial all-runs online command encountered an execution timeout after completing the first seed; the second seed was then executed arm by arm with the same parameters, preserving completed records. No new rate or policy was selected in response to those outcomes.

HISTORICAL ANNEX I: REVIEWED R24 ARTICLE

The following 93 pages are the complete article at reviewed manuscript commit 14ce582e188f437cc8d99f310edd4f06515936a4. They are included as historical development, not silently relabeled as new R26 results. The new manuscript's statements control where an earlier interpretation is corrected. Every original page is retained.

NEURAL BELLMAN OPERATORS

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We develop Neural Bellman Operators for a stopped consumption–portfolio economy with costly preference adjustment. A financed expert controller admits an explicit augmented-state Bellman formulation with the original unrestricted value. We derive momentum-correct neural output transport and a conditional financing-gradient perturbation theorem. A prospectively fixed 110-trajectory study uses disjoint tuning, direct preconditioners, two quadrature rules, certified checkpoints, a four-cell cover and an analytic reference. Tuned neural Adam attains a central-state regret bound below .001926; direct L-BFGS-B remains more accurate and cheaper. Nine neural experts certify an expanded initial-state region below .004281. The regional controller stores initial weights and a common factor, rather than using the original state alone. All earlier derivations and adverse outcomes remain. The original full-domain .01 target is unchanged and is not established by the regional evidence.

KEYWORDS: Stochastic control, Neural Bellman operators, Preference adjustment, Validated numerics, Policy verification, Computational economics.

1. INTRODUCTION

A numerical method for dynamic economics must specify its policy’s information, the optimization mechanism that generates it, and the economic error certified for deployment. These are separate questions. A policy can use information stored at initialization while remaining admissible in the original economy. Parameterizations can represent identical policies but induce different optimization paths. A sharper upper comparison need not improve the policy being compared. We develop Neural Bellman Operators with these distinctions explicit.

The application is the stopped consumption–portfolio economy with costly preference adjustment in Section 2. We retain its continuous actions, correlated shocks, settlement, complete state–time domain and uniform regret objective of .01. The current-state actor and financed expert construction are different implementations. The former maps (t, u, x) to actions. The latter uses initial interpolation weights and an observable common pricing factor to reconstruct financed accounts. Its regional certificates concern an augmented-state feedback policy, not a feedback map on the original state alone.

Our first result supplies the dynamic-programming formulation of this implementation. The decoder has a finite-dimensional Markov realization with constant initial weights and one common factor. Conditional prices reconstruct expert wealth without independently simulated subaccounts. We give the generator, boundary condition and Bellman semigroup. Under the original full Brownian information structure, unrestricted augmentation preserves economic value: memory supplies no additional future information. This equality does not imply optimality of a restricted expert library or accuracy throughout the complete original domain.

Our second result identifies the optimization mechanism after finite-node class equality. The financed family has a 47-dimensional quotient chart, while the neural generator has 355 parameters. Its Jacobian induces a local output metric. With momentum, Adam transports historical gradients through historical Jacobians; its update is not generally the current output gradient multiplied by $JD^{-1}J'$. We prove the correct identity and Taylor remainder and state when a simpler metric interpretation is justified. Experiments include explicit diagonal, tangent-

metric and whitening charts for direct optimization, alongside direct and neural Adam and L-BFGS-B.

The numerical study uses disjoint tuning and a prospectively fixed design. Each first-order method has equal tuning-call allocation but separately measured time. Evaluation crosses two new seeds, two proposal quadrature rules and five certification checkpoints. All configurations, failures and carry-forward decisions remain. A four-cell cover uses nine genuinely distinct experts per arm. A deterministic reference with an analytic optimum separates optimization error from approximation, temporal quadrature and arithmetic uncertainty.

A third analytical result propagates price-equation and derivative errors into the reduced payoff gradient. Root conditioning and derivative accuracy enter separately. A small price residual alone cannot provide the required derivative bounds. Executed two-rule differences at frozen policies are useful diagnostics, not rigorous errors for the gradient of the original stopped payoff.

The measured comparison in Section 7 includes direct-method advantages. Tuned neural Adam reaches a central-state bound below .001926, while direct L-BFGS-B is more accurate and cheaper. Nine neural experts certify the expanded four-cell region below .004281. A transferred-rate reference case and neural L-BFGS-B failures prevent a universal neural-superiority interpretation. These results strengthen the method's mathematical definition and empirical identification without replacing the original objective by an easier one.

The current complete-domain current-state bound remains 7.241462443133396, and no separate payoff improvement for that jointly changed actor is established. The retained residual-correction construction does not require exact replication; successful policy-sensitive execution of that construction remains a further numerical requirement. A local certificate or an analytic reference does not discharge it.

The background includes numerical economics and diffusion approximation in Judd (1998), Kushner and Dupuis (2001), Barles and Souganidis (1991), state coverage in Brumm and Scheidegger (2017), unrestricted upper comparisons in Brown, Smith, and Sun (2010), Brown and Smith (2011), and neural control constructions in Han and E (2016), Han, Jentzen, and E (2018). Our contribution is the explicit relation between policy information, financed output geometry and certified economic performance. Earlier derivations, experiments and adverse outcomes are retained in historical appendices and supplements, rather than deleted or represented as observations from this new study.

2. THE STOPPED PREFERENCE-ADJUSTMENT ECONOMY

Time is continuous, with horizon $T = 1$. Preferences u and wealth x evolve as

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (1)$$

$$dx_s = \{(.02 + .06p_s)x_s - c_s\} ds + .2p_s x_s dZ_s^x, \quad (2)$$

where $d\langle Z^u, Z^x \rangle_s = -.25 ds$. The state domain is

$$D = (1.2, 2.8) \times (.5, 2), \quad A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

An admissible control $a = (c, \theta, p)$ is progressively measurable, takes values in A , and is used up to the first exit from D or time one, whichever occurs first. The controls in the original optimization are not restricted to deterministic time functions or to neural networks.

For adjustment price $k \in [.5, 8]$, the running payoff and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2} \theta^2, \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1 - s). \quad (4)$$

Let $\tau = 1 \wedge \inf\{s \geq t : (u_s, x_s) \notin D\}$ and $\rho = .04$. The criterion is

$$J_k(t, z; a) = \mathbb{E}_{t,z} \left[\int_t^\tau e^{-\rho(s-t)} \ell_k(u_s, c_s, \theta_s) ds + e^{-\rho(\tau-t)} g(\tau, u_\tau, x_\tau) \right]. \quad (5)$$

Here $V_k = \sup_{a \in \mathcal{A}} J_k(\cdot; a)$. The liquidation fee makes a stopping-trace error economically meaningful even when the training distribution assigns little mass near an exit. It is therefore included explicitly in verification rather than approximated by an average boundary loss.

For a smooth test function v , define the discounted residual

$$\begin{aligned} \mathcal{R}_a v = & v_t + \theta v_u + \{(.02 + .06p)x - c\}v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux} \\ & + \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2 - .04v. \end{aligned} \quad (6)$$

All constants in the certificate denote the displayed exact decimals. Training uses floating-point approximations of them, but the certification program encloses the exact constants and treats stored binary64 network weights as exact real numbers.

The central initial condition is $(t, u, x) = (0, 2, 1.25)$. We also study the nondegenerate rectangle

$$K = [1.98, 2.02] \times [1.24, 1.26]. \quad (7)$$

The claim on K is uniform over its interior, not a statement about 72 isolated initial conditions. Its proof is given in Section Y.13, and its tolerance differs from the sharper central-state tolerance.

3. CONSTRUCTIVE NEURAL VERIFICATION

We first state the verification result for a class of stopped controlled diffusions. Let Z take values in a bounded domain, with drift b_a , covariance matrix Σ_a , running reward f_a , constant discount $\rho \geq 0$, and settlement g on the parabolic boundary. Write

$$\mathcal{R}_a v = v_t + b_a \cdot Dv + \frac{1}{2} \Sigma_a : D^2 v + f_a - \rho v.$$

The assumptions below concern the stochastic model and the computed witness. They do not assume regularity of the unknown optimal value.

ASSUMPTION 1: *For every admissible control, the stopped diffusion and its payoff are well defined. The controlled coefficients and payoffs are bounded on the stopped domain. The candidate feedback π is admissible. The witness v is continuously differentiable in time and twice continuously differentiable in state on a neighborhood of the closed domain, and its derivatives are bounded there. Discounted Itô's formula and optional stopping hold for these bounded stopped quantities.*

The compact economy in Section 2 with the stored smooth, bounded neural feedback satisfies these requirements. Boundedness also permits localization of the stochastic integral before taking expectations.

THEOREM 1—A complete residual and improvement certificate: *Under Assumption 1, suppose verified nonnegative functions $e(s), q(s)$ and a number $b \geq 0$ satisfy, at every state and time,*

$$|\mathcal{R}_\pi v| \leq e(s), \quad 0 \leq \sup_{a \in A} \mathcal{R}_a v - \mathcal{R}_\pi v \leq q(s), \quad |v - g| \leq b \text{ on the stopping boundary.} \quad (8)$$

Then, for every initial state z and time t ,

$$0 \leq V(t, z) - J(t, z; \pi) \leq 2b + \int_t^T e^{-\rho(s-t)} \{2e(s) + q(s)\} ds. \quad (9)$$

The same conclusion applies to any admissible implementation whose action lies in a verified output-error neighborhood of the stored feedback, provided that the residual and improvement inequalities have been enclosed over that neighborhood.

The theorem is deliberately an a posteriori statement. It is useful because every item in (8) has a finite implementation, not because its stochastic comparison identity is new. A training history can propose the witness and policy, but cannot replace any of the inequalities. The proof in Appendix O separates the policy-evaluation and upper-comparison errors; each stopping-trace discrepancy is paid once.

3.1. A finite implementation

Fix a rational rectangular cover of the entire state–time domain, including every stopping face. On each cell, interval forward differentiation gives the value, first derivatives, and relevant second derivatives of the stored network. An action oracle bounds the maximum over the continuous action set. Interval evaluation of the deployed policy gives its residual and the difference from this global maximum. Taking the largest bounds in each time slab produces e and q . Facewise interval evaluation produces b . Finally, integration of these piecewise-constant bounds yields (9).

A cell is accepted only if all operations have finite, valid enclosures and every potential action branch is covered. No failed or inconclusive cell can be removed from the domain. Refinement replaces a cell by a complete cover of that cell, while keeping previous results in the execution record. The method either returns the certified inequality for the specified stored object or records that a requested accuracy has not been established at the available resolution. This is a stopping criterion for an accuracy claim, not a criterion based on validation loss.

In the economy, the action maximization separates into three scalar problems conditional on the interval-enclosed derivatives. Consumption is strictly concave, and its maximizer is

$$c^* = \text{proj}_{[.05, .8]}(v_x^{-1/u}) \quad \text{when } v_x > 0, \quad c^* = .8 \quad \text{when } v_x \leq 0.$$

The preference action is $\theta^* = \text{proj}_{[-.2, .2]}(v_u/k)$. The portfolio objective is

$$h_1 p + \frac{1}{2} h_2 p^2, \quad h_1 = .06 x v_x - .0025 x v_{ux}, \quad h_2 = .04 x^2 v_{xx}.$$

A negative curvature gives a clipped vertex; nonnegative curvature gives an endpoint maximum. When an interval does not determine the sign, the calculation encloses the whole feasible portfolio interval. Thus uncertainty enlarges the bound instead of excluding an action. These formulas preserve the original continuous control opportunities.

3.2. Transport across model coefficients

The affine dependence on k used later is not a general argument for perturbing dynamics. For the latter we use residual transport. Consider another model with the same state domain, action set, and settlement, but possibly different coefficients and discount. Let its discount be at least $\underline{\rho} \geq 0$, and let $\tilde{\mathcal{R}}_a$ denote its residual.

PROPOSITION 1—Coefficient transport: *Suppose the original checks (8) hold, and interval coefficient bounds verify*

$$\sup_{a \in A} |\tilde{\mathcal{R}}_a v - \mathcal{R}_a v| \leq d(s) \quad (10)$$

throughout the common domain. If Assumption 1 also holds for the new model, its policy regret is at most

$$2b + \int_t^T e^{-\underline{\rho}(s-t)} \{2e(s) + q(s) + 2d(s)\} ds. \quad (11)$$

If the settlement changes by at most b_{par} , replace b by $b + b_{\text{par}}$.

For example, the modulus in (10) can be evaluated directly from

$$|(\tilde{b}_a - b_a) \cdot Dv| + \frac{1}{2} |(\tilde{\Sigma}_a - \Sigma_a) : D^2 v| + |\tilde{f}_a - f_a| + |\tilde{\rho} - \rho| |v|.$$

It includes the change in the covariance matrix rather than treating volatility perturbations as drift changes. The proposition does not couple paths or equate stopping times across economies. Each model is verified against its own stopped dynamics on the common domain. The executed coefficient box is reported in Section Y.20.

4. THE BELLMAN PROBLEM WITH CONTROLLER MEMORY

Write $z = (u, x)$ for the original economic state and $a = (c, \theta, p)$ for an action. The original admissible class uses progressively measurable controls under the completed Brownian filtration of the economy. A feedback map on (t, u, x) is a particular implementation in that class, not its definition. This distinction matters for the financed expert library.

4.1. A finite-dimensional realization

Let $Y_0 = 1$ and

$$dY_t = .02Y_t dt - .3Y_t dW_t^x.$$

For expert i , freeze the sixteen-slab consumption coefficients, the deterministic slabwise preference drift θ_i , and the exact-price-defined reserve R_i . Its conditional-price wealth is

$$F_i(t, y) = R_i e^{-.02(1-t)} + \int_t^1 e^{-.02(s-t)} \mathbb{E}^Q[c_i(s, Y_s) | Y_t = y] ds, \quad (12)$$

$$p_i(t, y) = -1.5 y F_{i,y}(t, y) / F_i(t, y).$$

The common-factor dynamics under Q have log mean increment $.065(s - t)$ and variance $.09(s - t)$; under the original measure the log mean increment is $-.025(s - t)$. These are the

same conditional-price and hedge identities as in the retained financed-decoder theorem. Frozen initial weights $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, are chosen from the initial cell. Define

$$\begin{aligned} c^\lambda(t, y) &= \sum_i \lambda_i c_i(t, y), & \theta^\lambda(t) &= \sum_i \lambda_i \theta_i(t), \\ p^\lambda(t, x, y) &= \frac{\sum_i \lambda_i p_i(t, y) F_i(t, y)}{x}. \end{aligned} \quad (13)$$

The initialized reachable manifold is

$$x = \sum_i \lambda_i F_i(t, y), \quad u = \sum_i \lambda_i u_i(t), \quad (14)$$

where all virtual u_i use the same preference shock. Their differences are deterministic conditional on initialization, since the expert drifts are deterministic. Virtual experts may be continued for this coupling after their individual exit times; the deployed aggregate is stopped at the original aggregate exit time and receives the original settlement.

PROPOSITION 2—Markov realization of the financed library: *Under the retained financing and feasibility hypotheses, the aggregate controller is Markov in (t, u, x, Y, λ) , with the expert library frozen. Initialized on (14), it preserves that manifold until original aggregate stopping and obeys the original economic equations and continuous action constraints. Independent stochastic subaccount coordinates are unnecessary: each wealth account is reconstructed as $F_i(t, Y_t)$. This is not, in general, a Markov policy on (t, u, x) alone.*

Two histories with the same original state can have different (y, λ) and hence different controls. Outside the reachable manifold one may specify any admissible extension; the reported certificates concern initialized reachable states and do not certify an arbitrary off-manifold extension. Cell selection is made at initialization and stored, rather than repeatedly switched during a path.

4.2. Generator, value, and dynamic programming

For a smooth function $v(t, u, x, y, \lambda)$, with λ fixed, the augmented generator excluding the time derivative is

$$\begin{aligned} \mathcal{L}_{\text{aug}}^a v &= \theta v_u + \{(.02 + .06p)x - c\}v_x + .02yv_y + .00125v_{uu} + .02p^2x^2v_{xx} + .045y^2v_{yy} \\ &\quad - .0025pxv_{ux} + .00375yv_{uy} - .06pxyv_{xy}. \end{aligned} \quad (15)$$

The mixed derivatives use the original correlation $-.25$; the preference-factor covariance is positive. Settlement is $g(t, u, x)$, independent of memory. The unrestricted augmented Bellman equation is

$$v_t + \sup_{a \in A} \{\mathcal{L}_{\text{aug}}^a v + \ell_k(u, c, \theta)\} - .04v = 0, \quad v = g \text{ on the original stopping boundary.} \quad (16)$$

The semigroup below is the primary definition. This differential equation is interpreted classically only when sufficient regularity holds; writing it does not assume differentiability of the unknown value.

ASSUMPTION 2—Information and restart: *The original and augmented problems have the same completed Brownian filtration and action constraints. Frozen memory is deterministic at a restart, future Brownian increments have their original law, and no auxiliary variable reveals future shocks. The stopped controlled equations have well-posedness, admissible concatenation, and dynamic-programming properties. Payoffs are integrable and the stopping payoff is unchanged.*

THEOREM 2—Augmentation preserves unrestricted economic value: *Under Assumption 2, the unrestricted augmented value equals $V_k(t, u, x)$ for every deterministic admissible memory initialization. It is independent of y and λ . For $h \geq 0$ the augmented dynamic-programming operator is*

$$(\mathcal{T}_h w)(s) = \sup_a \mathbb{E}_s \left[\int_t^{(t+h) \wedge \tau} e^{-0.04(r-t)} \ell_k(u_r, c_r, \theta_r) dr + \mathbf{1}_{\{\tau \leq t+h\}} e^{-0.04(\tau-t)} g(\tau, u_\tau, x_\tau) + \mathbf{1}_{\{\tau > t+h\}} e^{-0.04h} w(S_{t+h}) \right], \quad (17)$$

where S is the augmented state. It preserves pointwise order and satisfies the dynamic-programming semigroup identity on admissible continuation functions. Its unrestricted value is the original optimum, while (13) has a restricted policy value below that optimum.

The theorem uses the original full-information class. It does not say that adding an unobserved factor to a partially observed problem leaves value unchanged. Nor does it turn a local certificate for an augmented policy into a full-domain certificate for the current-state actor.

4.3. The computational operator and acceptance rule

The numerical construction approximates policy values associated with this Bellman problem. A proposal optimizer is an implementation component, not the exact Bellman semigroup. At checkpoint n , let $[L_n, U_n]$ enclose candidate payoff and $[L_I, U_I]$ enclose incumbent payoff. The deployment update is

$$\mathcal{A}(I, n) = \begin{cases} n, & L_n > U_I, \\ I, & \text{otherwise.} \end{cases} \quad (18)$$

Every accepted replacement has strictly greater actual payoff at the certified initialization. With a fixed unrestricted comparator, true regret falls by the same amount. This comparison needs no dual; a dual is needed to give a numerical upper bound on regret.

The experiment archives snapshots prospectively during optimization and certifies them serially afterward. The proposal optimizer continues its trajectory independently of the replayed deployment decision. Thus the gate neither restores optimizer moments nor supplies a demonstrated convergence mechanism. The historically retained rollback experiment is a different implementation. Cumulative generation-plus-checking curves are accounting reconstructions, not directly timed online deployments.

5. FINANCED OUTPUT GEOMETRY AND PROPOSAL ACCURACY

Let $f : \mathbb{R}^p \rightarrow \mathbb{R}^{48}$ be the raw-output map of the neural generator evaluated at the sixteen slab nodes. Here $p = 355$. The continuous financed policy depends on these outputs through bounded

TABLE I
COMPLETE NUMERICAL-METHOD SPECIFICATION

Component	Definition in the executed study
Policy information	(t, u, x, Y, λ) with frozen expert coefficients; original-state actor retained separately.
Generator	1–16–16–3 tanh network or an invertible direct output chart.
Proposal objective	Sixteen slabs; Gauss–Legendre/Gauss–Hermite orders (8, 16) or (12, 24); finite-rule implicit financing root.
Tuning	Two disjoint seeds; three rates per first-order arm; maximize mean certified lower payoff at 200 calls.
Deployment	Frozen dyadic coefficients; exact price-defined reserve, conditional-price wealth and hedge.
Acceptance	Serial replay of independently certified strict payoff comparisons; no proposal rollback.
Payoff checker	Unchanged directed stopped-model checker, including financing, tails, stopping and continuous actions.
Comparator	Inherited non-neural dual library; unnecessary for pairwise payoff improvement.
Cost	Preprocessing, generation, diagnostics, checking, tuning and upper-library reuse separated.

slope and drift decoders and the budget offset. The common raw-intercept direction is a gauge: shifting every intercept is undone by financing. Let $Q \in \mathbb{R}^{48 \times 47}$ have orthonormal columns perpendicular to that direction. The retained quotient and interpolation results establish equality of the attainable financed finite-node families, not good conditioning of a trained neural map.

5.1. The actual output update with momentum

For a differentiable loss $L(y)$, write $J_n = Df(\vartheta_n)$ and $g_n = \nabla_y L(f(\vartheta_n))$. Adam’s bias-corrected first moment is

$$\hat{m}_n = \sum_{r=1}^n a_{nr} J'_r g_r, \quad a_{nr} = \frac{(1 - \beta_1) \beta_1^{n-r}}{1 - \beta_1^n}.$$

Let D_n be diagonal with entries $\sqrt{\hat{v}_n} + \epsilon$, including the numerical stabilizer.

THEOREM 3—Output transport of an adaptive parameter update: *Suppose Df is Lipschitz with constant H on the segment between ϑ_n and ϑ_{n+1} . For Adam with learning rate η_n ,*

$$f(\vartheta_{n+1}) - f(\vartheta_n) = -\eta_n \sum_{r=1}^n a_{nr} J_n D_n^{-1} J'_r g_r + \varepsilon_n, \quad (19)$$

$$\|\varepsilon_n\| \leq \frac{1}{2} H \eta_n^2 \|D_n^{-1} \hat{m}_n\|^2. \quad (20)$$

The matrix $J_n D_n^{-1} J'_n$ is positive semidefinite and has rank at most 48. It describes a current-gradient step only without momentum, or after an additional approximation replacing historical gradients and Jacobians. A small parameter-gradient norm bounds the quotient output gradient only when the relevant singular values of the output Jacobian are bounded away from zero.

A neural–direct Adam difference can therefore arise from the current metric, historical gradient transport, or finite-step curvature. Reporting only the adaptive accumulator omits the Jacobian;

reporting only JJ' omits adaptive scaling and momentum. The executed diagnostic records actual Jacobian singular values, the spectrum of $JD_n^{-1}J'$, actual output displacement and its linearization remainder at declared checkpoints. These floating-point spectra are diagnostics, not directed rank certificates.

PROPOSITION 3—Equivalent parameter charts: *For invertible S , define $\tilde{f}(\varphi) = f(S\varphi)$. At $\varphi = S^{-1}\vartheta$, outputs agree and $D\tilde{f} = JS$. A gradient step in φ induces first-order output metric $JSS'J'$. Where J has full row rank, the minimum-norm right-inverse step $\delta\vartheta = -\eta J'(JJ')^{-1}d$ produces $\delta y = -\eta d + O(\|\delta\vartheta\|^2)$ for any specified output direction d , provided the Jacobian is locally Lipschitz.*

The right-inverse statement is a first-order invariance, not a claim that the poorly conditioned trained networks meet a usable rank bound. It is not the executed optimizer. The diagnostic rescaling uses powers of two in the parameter chart $\vartheta = S\varphi$ with identical initial outputs; it does not change tanh arguments and call the resulting function unchanged.

5.2. Explicit direct controls for geometry

At the matched initial network compute

$$G_0 = Q'J_0J_0'Q = W \operatorname{diag}(\gamma_j)W', \quad \tilde{\gamma}_j = \max\{\gamma_j, 10^{-5}\gamma_{\max}, 10^{-12}\}.$$

The direct raw outputs are $y = y_0 + QBz$. We compare $B = I$, the square root of the regularized diagonal metric, $B = W \operatorname{diag}(\tilde{\gamma}_j^{1/2})W'$, and $B = W \operatorname{diag}(\tilde{\gamma}_j^{-1/2})W'$. Normalize $\|QB\|_F^2 = 47$. These are the direct, diagonal, tangent and whitening charts. Each is invertible on the same quotient and none enlarges the policy class.

The tangent chart freezes a local approximation to neural geometry, not Adam's moving momentum-dependent trajectory. Whitening tests the reverse scaling. Jacobian and factorization costs are charged as preprocessing; the ordinary direct arm need not incur them in a standalone implementation. The original 47-coordinate gauge control alone is not a test of these anisotropic geometries.

5.3. Financing and derivative error

Let y collect independent raw coordinates, a be the scalar financing offset, and $P(y, a) = B_0$ the exact price equation. Let P_N be its approximation, with roots $a(y)$ and $a_N(y)$ in a common bracket. Let $F(y, a)$ and $F_N(y, a)$ be payoff objectives. These may describe an integrable proposal extension; identification with the stopped economic payoff requires an additional argument.

THEOREM 4—Reduced-gradient perturbation: *On a common root bracket suppose $P_a, P_{N,a} \geq d_* > 0$, $|P - P_N| \leq e_P$, and the required first derivatives exist and are Lipschitz in the offset. Then $|a - a_N| \leq \delta = e_P/d_*$. At the respective roots suppose*

$$\|F_{N,y} - F_y\| \leq E_A, \quad |F_{N,a} - F_a| \leq E_B, \quad \|P_{N,y} - P_y\| \leq E_C, \quad |P_{N,a} - P_a| \leq E_D,$$

where each E includes its fixed-offset approximation error and its offset-Lipschitz allowance times δ . If both versions satisfy $|F_a| \leq M_B$ and $\|P_y\| \leq M_C$, then

$$\|\nabla_y F_N(y, a_N(y)) - \nabla_y F(y, a(y))\| \leq E_A + \frac{E_B M_C + M_B E_C}{d_*} + \frac{M_B M_C E_D}{d_*^2}. \quad (21)$$

Existence of an exact-price correction root does not establish the accuracy of a delivered finite-rule root. Finite-difference agreement for a discrete objective verifies its implementation derivative, not distance to the derivative of the continuous stopped payoff.

The proposal extends preference beyond the stopping interval by clipping to $[1.2, 2.8]$ before running-utility evaluation. This supplies a bounded, integrable Gaussian-tail extension instead of extending CRRA through preference one. The original payoff checker and stopping model are unchanged. The two proposal rules agree with the inherited objective at regression-tested unclipped nodes. Whether any checkpoint node is clipped is recorded. Two-rule differences are evaluated at the same raw policy, not different optimized iterates. They do not instantiate the uniform derivative hypotheses of Theorem 4.

6. PROSPECTIVE NUMERICAL DESIGN

The protocol was committed before new scientific execution, after both R23 reports and the complete preceding evidence were examined. This is a prospective response to known methodological questions, not independent rediscovery of the old qualitative pattern. Tuning, evaluation and multicell construction are distinct stages; every current-run result is retained.

6.1. *Tuning and held-out checkpoints*

At $(u, x) = (2, 1.25)$, seeds 24001 and 24002 form the tuning set. Each of five Adam arms tests .005, .02, .08 for 200 objective/gradient calls with proposal orders (8, 16). Select the rate maximizing mean final certified lower payoff over the two seeds. Infeasible candidates are ineligible and ties within 10^{-10} use the smaller rate. Each arm receives six trials and 1, 200 calls. Actual generation, preprocessing and checking times are retained; equal calls are not equal work.

Evaluation seeds 24101, 24102 are crossed with orders (8, 16), (12, 24) and the five selected Adam methods, neural L-BFGS-B and direct L-BFGS-B. L-BFGS-B has call cap 400, maximum line searches 40, function tolerance 10^{-13} and gradient tolerance 10^{-8} . No hyperparameter is selected on these outcomes.

Checkpoints are fixed at calls 25, 50, 100, 200, 400. For line-search methods use the latest accepted iterate available by the call, not an unaccepted trial or future iterate. Carry a terminated endpoint forward while reporting actual calls. Snapshots are archived during optimization and certified serially afterward by the independent checker. The schedule is prospective, but cumulative generation-plus-checking curves reconstruct costs rather than time an online rollback algorithm. Repeated converged endpoints are distinguished from changed proposals that fail the strict gate.

Initial checking, preprocessing, tuning and diagnostic-only calls are separately recorded. The inherited dual construction cost is 415.789 seconds. Adding it to each fresh solve supplies an unamortized accounting scenario, not a controlled same-machine timing experiment. For M solves reusing the same library, charge $415.789/M$ per solve. Reuse is not assumed after arbitrary changes in economic primitives. All rates selected in this study were .08, the upper edge of the declared grid; this finite calibration family does not identify an optimum outside its range.

6.2. *A larger four-cell cover*

The grid is

$$u \in \{1.98, 2, 2.02\}, \quad x \in \{1.24, 1.255, 1.27\}.$$

It gives nine distinct initial states and four cells, extending the preceding wealth span by 50 percent. Lexicographic vertex i uses seed $24300 + i$. Neural, direct and tangent Adam use their selected rates; direct L-BFGS-B is included without retuning. All receive the 400-call cap and (8, 16) rule. Every endpoint is checked, including unsuccessful proposals.

PROPOSITION 4—Multicell regret certificate: *Suppose every deployed vertex policy in a cell satisfies the financed-decoder conditions with stopped payoff lower bound L_i . Suppose the inherited upper $\bar{V}$ is convex on the cell and the retained common-shock concavity and stopping hypotheses hold with nonsynchronized stopping allowance ζ . Then the initialized augmented mixture satisfies*

$$V(0, z) - J(0, z; \pi^\lambda) \leq \max_{i \text{ in cell}} \{\bar{V}(z_i) - L_i\} + \zeta \quad (22)$$

throughout the cell. The largest bound certifies the complete union. A failed vertex invalidates its cells unless an independently certified incumbent is deployed.

The preference range and drift bounds are unchanged, giving the retained allowance $\zeta = 16p_E$, $p_E = 2 \exp\{-.58^2/(2 \cdot .0025)\}$. The convex upper is evaluated at new vertices, not interpolated from the old four values. Cell bounds use actual deployed certificates. This is a substantive low-dimensional multicell execution, not removal of worst-case exponential library growth in dimension. The new result certifies regret; it does not infer uniform mixture payoff improvement merely from improved vertices.

6.3. A known-solution reference and failure diagnostics

The reference economy has deterministic wealth, horizon one, zero interest and discounting, fixed preference two, no risky asset or preference adjustment, running utility $-1/c$, terminal payoff $\log R$, and initial wealth 1.25. Controls satisfy $.05 \leq c \leq .8$ and $R = 1.25 - \int_0^1 c_t dt > 0$. It is a separately named benchmark, not the original stochastic economy.

PROPOSITION 5—Reference optimum and error separation: *The unique optimum, up to equality almost everywhere, is constant consumption*

$$c_* = \frac{\sqrt{6} - 1}{2}, \quad R_* = c_*^2, \quad V_* = -1/c_* + 2 \log c_*.$$

It is interior to the control bounds and $R_ > .5$. Every feasible sixteen-slab policy has exact payoff*

$$J(c) = -\frac{1}{16} \sum_{j=1}^{16} \frac{1}{c_j} + \log \left(1.25 - \frac{1}{16} \sum_{j=1}^{16} c_j \right).$$

Both continuous parameter families contain the optimum. Class-approximation and temporal quadrature errors are zero; directed arithmetic encloses the residual optimization gap and payoff uncertainty.

The decoder $c_j = .05 + .70\sigma(y_{j0})$ contains c_* and guarantees reserve above .5. There is no financing gauge, so direct charts retain all 48 raw coordinates. The other 32 outputs are inactive in payoff. Seeds 24501, 24502 use transferred selected rates and the same checkpoints, without reference-specific calibration.

The adverse R23 neural L-BFGS-B run at seed 23203 is diagnosed separately. Preserve its original source, endpoint and stopping message. Regenerate it, change only an equivalent powers-of-two parameter chart with layer-group scales (1/4, 1, 4), restart with tighter tolerance, and restart direct optimization from the failed raw outputs. Record Jacobian spectra, gradients, call histories, raw-output and financed-coefficient distances to the frozen direct solution, and certificates. Restart caps are additional budgets, not hidden equal-total-work comparisons. These diagnostics explain a known failure and never enter held-out selection.

7. NUMERICAL RESULTS

The study contains 110 trajectories and 286 prescribed checkpoint records, separated into tuning, evaluation, multicell construction, an analytic reference and post-hoc diagnostics. These are not pooled as independent replications of a single claim. The supplement and machine-readable ledgers retain every configuration, candidate, termination and gate decision.

7.1. Calibrated performance and the direct frontier

Table II charges every tuning trial. Every arm selects .08, the upper edge of the finite grid. Table III reports final deployed policies, not favorable endpoints selected after evaluation. Equal calls do not imply equal work. At the coarse rule, neural Adam’s worst bound is .001925713,

TABLE II
DISJOINT TUNING, WITH ALL TRIALS CHARGED

Arm	Rate	Calls	Gen. s	Prep. s	Check s	Mean lower payoff
Neural–Adam	0.08	1200	3.944	0.000	12.420	-1.287225
Direct–Adam	0.08	1200	3.555	0.000	12.387	-1.287524
Diagonal–Adam	0.08	1200	3.570	0.018	12.474	-1.287466
Tangent–Adam	0.08	1200	3.547	0.018	12.425	-1.287377
Whitened–Adam	0.08	1200	3.535	0.018	12.375	-1.304690

Two seeds, three rates, 200 calls per trial. Costs include every trial, not only the selected rate. Every arm selected the upper grid endpoint .08; optimal calibration beyond this declared finite grid is not established. Shared preprocessing is charged conservatively as a standalone cost.

versus .002039228 for direct Adam and .002008735 for diagonal Adam. The frozen tangent chart does not reproduce the neural endpoint, and inverse-metric whitening is worse. These findings are consistent with roles for adaptive geometry, transported momentum and nonlinear steps, but do not separately identify those causal contributions.

Direct L-BFGS-B remains the strongest tested solver: its deployed bound is below .001923443 with mean 111.5 calls and .321 generation seconds, versus 400 calls and 1.303 seconds for neural Adam. Neural L-BFGS-B retains an adverse held-out endpoint near .009609. The dual-free intervals in Table IV identify payoff ordering without confounding it with a common unrestricted upper. The old shared-rate comparison remains historical; the present finite tuning envelope governs the new interpretation.

7.2. Quadrature, geometry and the acceptance rule

The largest ordinary raw-gradient difference between the two rules is 4.409180×10^{-7} , and the largest financing-offset difference is 9.345210×10^{-6} . The largest relative difference

TABLE III
HELD-OUT CENTRAL STATE: ACCURACY AND COMPUTATION

Rule	Arm	Regret upper	Calls	Gen. s	Check s	Cert. cost s
8/16	Neural–Adam	0.001925713	400.0	1.303	10.379	13.744
8/16	Direct–Adam	0.002039228	400.0	1.181	10.413	13.656
8/16	Diagonal–Adam	0.002008735	400.0	1.179	10.330	13.574
8/16	Tangent–Adam	0.002157232	400.0	1.175	10.333	13.573
8/16	Whitened–Adam	0.01874203	400.0	1.175	10.377	13.617
8/16	Neural–L-BFGS-B	0.009608327	251.0	0.792	10.478	13.333
8/16	Direct–L-BFGS-B	0.001923443	111.5	0.321	10.400	12.784
12/24	Neural–Adam	0.001925705	400.0	1.997	10.370	14.427
12/24	Direct–Adam	0.002039189	400.0	1.859	10.352	14.272
12/24	Diagonal–Adam	0.002008701	400.0	1.853	10.338	14.253
12/24	Tangent–Adam	0.002157182	400.0	1.847	10.377	14.286
12/24	Whitened–Adam	0.01874227	400.0	1.878	10.480	14.420
12/24	Neural–L-BFGS-B	0.009608181	243.5	1.183	10.479	13.722
12/24	Direct–L-BFGS-B	0.001923667	112.0	0.520	10.457	13.037

Regret is the worse of two seeds at $(t,u,x)=(0,2,1.25)$; calls and times are means. Policies are final deployed checkpoints. Checking includes all five scheduled checks, including converged repeats. Certified cost adds preprocessing and initial checking. Diagnostic-only calls and tuning are separate.

TABLE IV
NEURAL ADAM MINUS COMPARATOR: DUAL-FREE PAYOFF INTERVALS

Rule	Comparator	Lower	Upper
8/16	Direct–Adam	0.0001125274	0.0001138956
8/16	Diagonal–Adam	0.00007642732	0.00008377674
8/16	Tangent–Adam	0.0001587773	0.0002319002
8/16	Whitened–Adam	0.01560812	0.01681708
8/16	Neural–L-BFGS-B	-2.561433×10^{-6}	0.007683369
8/16	Direct–L-BFGS-B	-2.798121×10^{-6}	-1.515012×10^{-6}
12/24	Direct–Adam	0.0001125009	0.0001138648
12/24	Diagonal–Adam	0.00007639466	0.00008375511
12/24	Tangent–Adam	0.0001587550	0.0002318586
12/24	Whitened–Adam	0.01560848	0.01681733
12/24	Neural–L-BFGS-B	-2.572662×10^{-6}	0.007683235
12/24	Direct–L-BFGS-B	-2.419365×10^{-6}	-1.474925×10^{-6}

Each row encloses both evaluation-seed payoff differences using independent delivered-policy intervals. Positive lower endpoints prove a payoff advantage in both matched problems; negative upper endpoints prove the reverse. This is not a confidence interval or an unrestricted optimality comparison.

is 22.99217, at a nearly stationary fine-rule direct L-BFGS-B checkpoint. Small absolute differences therefore do not justify interpreting a surrogate stopping tolerance as a continuous-gradient tolerance. The principal endpoint ordering survives this finite refinement comparison, but no uniform stopped-gradient error bound follows. No held-out checkpoint node was clipped. Actual Jacobian and adaptive-metric spectra and transport remainders are retained as ordinary diagnostics, not rigorous rank certificates.

The 140 held-out replayed decisions contain 121 accepted replacements and 19 rejections: seven repeat the preceding candidate exactly and twelve concern changed candidates. The

optimizer was not rolled back after rejection. These observations establish the recorded deployment effect, not a contribution to proposal convergence. Curves retain every scheduled checkpoint, including carried-forward converged endpoints, against actual calls, generation time and reconstructed generation-plus-checking cost.

7.3. Expanded state coverage and known-solution calibration

Table V reports all four cells through their union bound. Nine distinct policies per method are trained and checked. The neural union bound .004280526 and direct L-BFGS-B bound .004274863 both meet .01 throughout the expanded initialization region under the augmented-state realization. They are not sampled-point, high-dimensional or all-start-time results. In

TABLE V
FOUR CELLS AND NINE DISTINCT EXPERTS PER METHOD

Arm	Region regret upper	Calls	Gen. s	Check s
Neural-Adam	0.004280526	3600	11.755	18.686
Direct-Adam	0.004398790	3600	10.553	18.638
Tangent-Adam	0.004508256	3600	10.548	18.659
Direct-L-BFGS-B	0.004274863	1250	3.592	18.694

Region $[1.98, 2.02] \times [1.24, 1.27]$ at time zero. Every cell is certified using its deployed expert records and the retained concavity/exit theorem. The controller has the explicit augmented state. Generation/checking are totals across nine genuinely distinct experts; initial checks and preprocessing remain in the ledger.

the analytic reference, class and temporal quadrature errors are zero, so the remaining gap is optimization error plus directed evaluation uncertainty. Direct L-BFGS-B reaches a worst gap below 3.393×10^{-13} with a payoff enclosure width around 10^{-14} . Neural Adam has one gap below 3.621×10^{-8} and another near .002772, worse in the latter case than the direct first-order baselines. That adverse transferred-rate outcome is retained; it precludes universal neural superiority.

TABLE VI
ANALYTIC REFERENCE: SOLVER GAP AND ARITHMETIC WIDTH

Arm	Gap upper	Payoff width	Calls	Gen. s	Check s
Neural-Adam	0.002771190	9.769963×10^{-15}	400.0	0.321	0.066
Direct-Adam	0.00005492341	1.021406×10^{-14}	400.0	0.198	0.066
Diagonal-Adam	0.00006450277	1.021406×10^{-14}	400.0	0.197	0.066
Tangent-Adam	0.00001955686	1.021406×10^{-14}	400.0	0.197	0.067
Whitened-Adam	0.5646302	1.332268×10^{-14}	400.0	0.196	0.069
Direct-L-BFGS-B	3.392842×10^{-13}	1.021406×10^{-14}	14.0	0.007	0.067

Worst gap and width across two seeds, mean calls and times. This separate deterministic economy has zero class and temporal quadrature errors. Rates transfer from original-economy tuning without reference-specific calibration. The common consumption intercept is retained; the other two output channels are inactive.

7.4. Failure mechanism, verification slack and the unchanged objective

The regenerated seed-23203 bound .010503068 reproduces the old failure. An equivalent layer-scaled chart reduces it to .002554827 within its 400-call diagnostic budget. Tighter tolerance and a direct restart from the already saturated raw outputs leave the bound near .010503067. Removing network redundancy only after reaching the adverse output is not sufficient; decoder saturation and the preceding trajectory also matter. Raw distances are not economic payoff distances. Restart budgets are explicitly additional, not equal-total-work comparisons. The

TABLE VII
POST-HOC SEED-23203 FAILURE DIAGNOSIS

Case	Regret upper	Calls	Raw distance	Financed distance
regenerated	0.01050307	92	299.963	2.691
scaled	0.002554827	400	186.416	7.409
restart tighter	0.01050307	43	296.812	2.753
direct restart	0.01050307	4	299.963	2.691

Distances are to the unchanged frozen R23 direct solution. Tighter and direct restarts use additional budgets. Regenerated and chart-scaled trajectories start at identical outputs; these diagnostics are excluded from tuning and held-out selection.

largest held-out policy enclosure width is 3.814276×10^{-7} . For exact policy payoff J and unrestricted upper $\bar{V}$,

$$\bar{V} - L = (\bar{V} - V) + (V - J) + (J - L), \quad 0 \leq J - L \leq U - L.$$

The first two terms are not separately identified in the original economy. Tight policy intervals alone do not identify dual slack. The reference separates these components in its own economy, while the original-economy pairwise intervals eliminate the dual. Preprocessing, generation, initial and checkpoint checking, scientific instrumentation, tuning and upper-library allocations are separately reported. The historical 415.789-second dual cost is an accounting scenario, not a same-machine comparison; the reference/failure recovery stages also used a separately provisioned host.

The original-state all-start-time bound remains 7.241462443133396 against the unchanged .01 objective. A new versioned record corrects the stale name of the older undiscounted field without changing historical bytes. This revision does not establish a new original-state full-domain accuracy certificate, a separate payoff increase for that actor, a successful nonreplicable stochastic execution, or rigorous numerical stopped-gradient errors. The new analytical and empirical results specify their actual domains while retaining the stronger objectives and their historical derivations.

8. CONCLUSION

Neural Bellman Operators require explicit policy information, optimization geometry and economic verification. The financed library has a Markov realization with controller memory and the original unrestricted value under the original information structure. Its local accuracy remains separate from the full-state current-state target. Neural parameterization acts through geometry and historical gradient transport, not a larger finite-node policy family.

Disjoint tuning, explicit direct charts, quadrature refinement, all certified checkpoints, a genuine multicell cover and an analytic reference strengthen the method's empirical identification.

Direct-method advantages and neural failures are scientific findings, not configurations to omit. The original .01 full-domain objective still requires stronger policy-sensitive verification and a separately established actor payoff improvement. The state realization, derivative theorem and executed diagnostics supply constructive ingredients without substituting an easier economic problem.

APPENDIX A: PROOFS OF THE STATE AND GEOMETRY RESULTS

A.1. Proof of Proposition 2

The conditional-price identity gives $X_i(t) = F_i(t, Y_t)$ and

$$dX_i = \{(.02 + .06p_i)X_i - c_i\}dt + .2p_iX_i dW^x.$$

Multiply by constant λ_i and sum. On (14), this is the original wealth equation with (13). All virtual preference processes share the same shock and $\sum_i \lambda_i = 1$, so their weighted sum solves $du = \theta^\lambda dt + .05dW^u$. Pathwise uniqueness on each slab and continuity of prices at slab dates preserve both equalities until aggregate stopping. Consumption and drift are convex combinations within their admissible intervals. Portfolio exposure is a wealth-weighted convex combination because $F_i > 0$ and $x = \sum_i \lambda_i F_i$. The global bounds of Theorem 5 give admissibility. Frozen coefficients, constant weights, and current (t, Y_t) determine prices and actions. Preference differences are initial differences plus integrated deterministic drift differences, requiring no additional stochastic coordinate. *Q.E.D.*

A.2. Proof of Theorem 2

Fix deterministic initial memory. Every augmented progressively measurable control is an original progressively measurable control: both use the same filtration, and memory is constructed causally from it. The economic coordinates have identical dynamics and payoff under that control. Hence the augmented supremum is at most $V_k(t, u, x)$. Conversely, every original admissible control is available by ignoring memory, giving the reverse inequality. This compares admissible classes and does not require that a neural family attain the supremum.

Covariances follow from diffusion coefficients and correlations. In particular,

$$d\langle u, Y \rangle_t = .05(-.3Y_t)(-.25)dt = .00375Y_t dt, \quad d\langle x, Y \rangle_t = -.06p_t x_t Y_t dt.$$

Also $d\langle Y \rangle_t = .09Y_t^2 dt$. The factor 1/2 applies to diagonal Hessian terms, not mixed terms written once. There are no derivatives in constant weights. Itô's formula gives (15).

Conditioning at an intermediate time settles paths already stopped and uses admissible concatenation on surviving paths, giving (17). If $w \leq \tilde{w}$, each admissible control has weakly larger continuation payoff under $\tilde{w}$; taking suprema preserves order. Conditioning and concatenating twice gives the semigroup identity. A classical first-order time expansion, when justified, gives (16). Memory derivatives of the unrestricted value vanish, reducing it to the original Bellman equation. The value comparison itself needs no differentiability. *Q.E.D.*

A.3. Proof of Theorem 3

The chain rule gives $\nabla_\vartheta L = J'_n g_n$. Iterate $m_n = \beta_1 m_{n-1} + (1 - \beta_1)J'_n g_n$ from $m_0 = 0$ and divide by $1 - \beta_1^n$ to obtain the historical expansion. Adam uses $\delta\vartheta_n = -\eta_n D_n^{-1} \hat{m}_n$. The integral remainder is

$$f(\vartheta_n + \delta\vartheta_n) - f(\vartheta_n) - J_n \delta\vartheta_n = \int_0^1 \{Df(\vartheta_n + s\delta\vartheta_n) - J_n\} \delta\vartheta_n ds.$$

Its norm is at most $H\|\delta\vartheta_n\|^2/2$. Substitute the update to obtain the two displayed conclusions. For any v , $v'JD_n^{-1}J'v = \|D_n^{-1/2}J'v\|^2 \geq 0$; rank cannot exceed the output dimension. On an output-gradient subspace with smallest singular value $s_* > 0$ for J' , $\|J'g\| \geq s_*\|g\|$. Without that lower bound, small parameter gradient can coexist with large output gradient. Historical J_r cannot generally be replaced by J_n ; even equality of Jacobians leaves an average of historical gradients, not necessarily the current gradient. *Q.E.D.*

A.4. Proof of Proposition 3

The chain rule yields $D\tilde{f} = JS$. A gradient step is $\delta\varphi = -\eta S'J'g$, so $J\delta\vartheta = JS\delta\varphi = -\eta JSS'J'g$. Local Lipschitz continuity of the Jacobian bounds the nonlinear remainder by a constant times the squared parameter step. If J has full row rank, JJ' is invertible and the stated minimum-norm right inverse satisfies $J\delta\vartheta = -\eta d$ exactly in the linearization. Other full-rank parameter charts recover that prescribed first-order output direction, although step norm and nonlinear remainder can differ. *Q.E.D.*

A.5. Proof of Theorem 4

At the approximate root, $|P(y, a_N) - B_0| = |P(y, a_N) - P_N(y, a_N)| \leq e_P$. Monotonicity and the mean-value theorem imply $|a_N - a| \leq e_P/d_*$. A fixed-offset derivative error e_X and offset-Lipschitz constant H_X imply a root-evaluated error $E_X = e_X + H_X\delta$ by the triangle inequality. Implicit differentiation gives

$$\nabla_y F(y, a(y)) = F_y - F_a P_y / P_a,$$

and similarly for the numerical functions. Abbreviate derivatives by A, B, C, D and numerical versions by A_N, B_N, C_N, D_N . Add and subtract BC_N/D_N and BC/D_N to obtain

$$\|B_N C_N / D_N - BC / D\| \leq \frac{|B_N - B| \|C_N\|}{d_*} + \frac{|B| \|C_N - C\|}{d_*} + \frac{|B| \|C\| \|D_N - D\|}{d_*^2}.$$

Adding the A error proves (21). Every derivative bound is a hypothesis, not an inference from an ordinary two-rule difference or a payoff-value error bound. *Q.E.D.*

A.6. Proof of Proposition 4

The retained common-shock mixture argument gives $J(0, z; \pi^\lambda) \geq \sum_i \lambda_i L_i - \zeta$ under its concavity, integrability and stopping hypotheses. Convexity of the explicit upper gives $V(0, z) \leq \bar{V}(z) \leq \sum_i \lambda_i \bar{V}(z_i)$. Subtract and bound the convex combination of vertex gaps by its largest term. Maximizing over a complete finite cover proves the union bound. At a shared face a deterministic initialization rule chooses a cell; either neighboring certified construction supplies its corresponding bound. This is not repeated switching of controllers during a path. The result is a regret bound. Improved vertex payoffs alone do not prove uniform mixture payoff improvement, because the old and new mixtures can have different Jensen gains; the retained separate mixture-gain theorem addresses that stronger comparison. *Q.E.D.*

A.7. Proof of Proposition 5

The function $c \mapsto -1/c$ is strictly concave for $c > 0$. At a fixed average $\bar{c}$, Jensen's inequality gives $\int_0^1 -1/c_t dt \leq -1/\bar{c}$, with equality only for constant consumption almost everywhere.

Terminal wealth is $1.25 - \bar{c}$. Thus maximize $h(c) = -1/c + \log(1.25 - c)$, whose second derivative is $-2/c^3 - 1/(1.25 - c)^2 < 0$. Its first-order condition gives $c^2 + c = 1.25$ and positive root $(\sqrt{6} - 1)/2$. Direct substitution verifies interior control bounds and reserve above .5. Exact integration on each slab gives the stated finite-slab payoff. Constant output bias represents the optimum in both continuous parameter families, so there is no policy-class loss. The optimum and candidate expressions require rational operations, a square root and a logarithm, enclosed by the directed backend. Their interval difference encloses the optimization gap rather than estimating it statistically. *Q.E.D.*

APPENDIX B: PRESERVED R23 MAIN-TEXT DEVELOPMENT

The following unchanged source components retain all preceding arguments and outcomes. Their revision-specific claims are historical; the R24 main text defines the current objects and evidence. The original foundations are already reproduced in the main text.

APPENDIX C: INTRODUCTION

A numerical solution of a dynamic economic model has to deliver a policy, improve its economic payoff, and control its loss relative to the original optimum. A neural representation can change the geometry of policy generation without changing the attainable policies. A value witness can sharpen a comparison without improving the policy it accompanies. Confusing these effects obscures both the economic accuracy of a solver and the source of a computational advantage.

We develop Neural Bellman Operators that distinguish these effects while retaining a common stochastic control problem. The application is a stopped consumption–portfolio economy with costly preference adjustment, correlated shocks, continuous controls, and a time-dependent liquidation settlement. Its unrestricted objective, full state–time domain, and accuracy target of .01 are unchanged. The operator couples a candidate-generation map to policy-specific payoff acceptance and a separate unrestricted value comparison. The former determines whether a proposed policy should replace its incumbent; the latter determines the policy’s certified economic error.

Our first contribution is a precise analysis of the financed policy coordinates. The budget equation removes one common-intercept direction. We establish its unique solution, its implicit derivative, finite price-root brackets that include unbounded Gaussian tails, and an explicit inverse-conditioning bound. The derivative is an oblique projection, not an arbitrary normalization. We also prove that the neural and direct parameterizations can represent the same finite-node raw-output arrays. Consequently, a comparison between them identifies a difference in parameterization and optimization geometry, not an unmeasured difference in the size of the delivered policy class. A directed full-rank certificate verifies a concrete interpolation construction.

Our second contribution is a continuum payoff comparison for independently initialized expert policies. Verified pair moments control the old mixture’s Jensen benefit when both its consumption slopes and preference drifts differ across corners. The resulting theorem proves policy improvement and a true-regret reduction at every initial state of a rectangle. Correlated shocks do not require an independence approximation: their contribution is bounded on the joint time–probability measure. The unrestricted upper comparison is used for regret but is unnecessary for the mathematical ordering of two feasible policy payoffs.

The experiments separate parameterization from optimizer and include a control that removes the redundant budget coordinate. The new held-out design contains 72 configurations: three

independently initialized four-expert systems, three coordinate representations, and two optimizers. Each configuration has the same 400-gradient-call cap. The four primary neural/direct arms start from exactly the same delivered policy within each block. The 47-coordinate direct control is mapped to that policy, with the numerical discrepancy recorded and its incumbent separately checked whenever it is not identical. All configurations and all prescribed comparisons are reported.

In this design every neural–Adam system has regional regret below 0.003022 and a uniform initial-to-final payoff gain of at least 0.026360. Its uniform payoff advantage over each of the two direct–Adam coordinate systems is at least 0.003044. Removing the redundant intercept does not remove this conditional advantage. Direct L-BFGS-B nevertheless reaches a tighter regional bound with substantially fewer objective calls, and one neural L-BFGS-B system remains above .01. The evidence identifies a reproducible optimizer-conditional effect; it does not establish an unconditional neural accuracy–cost frontier advantage.

The full-state experiment is a distinct numerical object. Its actor and witness take the current (t, u, x) as inputs and are checked on every cell of the original domain against the original continuous action set. A recorded warm-start refinement tightens its all-start-time certificate from 7.2783190635 to 7.241462444. This is an improvement in a certified comparison, not a separately proved increase in that actor’s payoff. We retain the full-domain .01 objective rather than substitute the successful regional experiment for it. A constructive residual-correction cone, valid without exact financial replication, makes the requirements for further global improvement explicit.

The paper also separates mathematical deployment from numerical implementation. Regional policies are compiled to slab coefficients; their conditional prices and hedges still require integration. The selected consumption streams are replicable, but the preference shock has an unspanned component: exact replication of those streams does not make every payoff in the joint filtration replicable. The independent dual upper is non-neural and its construction cost is charged. A wealth counterfactual is a sufficient re-budgeted increment for fixed learned shapes, not a minimum compensating variation for an unchanged policy map.

The numerical dynamic-programming and verification background includes Judd (1998), Kushner and Dupuis (2001), and Barles and Souganidis (1991); adaptive sparse-grid comparisons are developed by Brumm and Scheidegger (2017). Dual comparisons relate to Brown, Smith, and Sun (2010), Brown and Smith (2011); neural control and high-dimensional equation methods include Han and E (2016), Han, Jentzen, and E (2018). Our results combine financed-coordinate identification, nonsynchronized continuum comparison, and executed acceptance with an explicit accounting of their distinct roles. Earlier theory, numerical failures, and high-dimensional comparisons remain in the appendices and supplement with their original scopes. They are not relabeled as evidence from the new held-out study.

APPENDIX D: STOCHASTIC NEURAL POLICIES WITH SEPARATE PAYOFF ACCEPTANCE

The economy and notation are as specified above. In this section $r = .02$, the risky asset has excess drift .06 and volatility .2, and $Y_0 = 1$ with

$$dY_t = .02Y_t dt - .3Y_t dW_t^x.$$

The equivalent martingale measure satisfies $dW_t^{x,Q} = dW_t^x + .3 dt$. Thus $\log Y_t$ has mean $-.025t$ under the physical measure and $.065t$ under the martingale measure, in both cases with variance $.09t$.

D.1. Neural generation and exact deployment semantics

A 1–16–16–3 network with two tanh hidden layers takes only the normalized time-slab coordinate and produces coefficients for sixteen equal time slabs. Current preference and the traded factor are not network inputs; factor dependence enters the fixed decoder. The budget offset is differentiated inside the proposal objective. For slab j , the deployed policy is

$$c_j(y) = .5 + .3 \sigma(b_j - s_j \log y), \quad .1 \leq s_j \leq 6.5, \quad 0 \leq \theta_j \leq .2, \quad (23)$$

where $\sigma(a) = (1 + e^{-a})^{-1}$. The stored dyadic coefficients, not an unspecified real-number neural forward pass, define the delivered policy. The network is compiled once; the contingent-claim decoder below then acts on the observable factor. Each expert has 355 neural parameters. Four experts have 1,420 parameters, apart from the decoder. The network is trained from random weights without inherited policy or value labels. Its accuracy is established by checking its delivered policy, not by assuming convergence of Adam.

For a given initial state (u_i, x_i) , let

$$R_i = e^r \left(x_i - \int_0^1 e^{-rt} \mathbb{E}^Q[c_i(t, Y_t)] dt \right). \quad (24)$$

The directed price check verifies $.5 < R_i < .5001$. In particular, R_i is an exact price-defined reserve enclosed by an interval; it is not an approximately balanced budget. Define

$$F_i(t, y) = R_i e^{-r(1-t)} + \int_t^1 e^{-r(s-t)} \mathbb{E}^Q[c_i(s, Y_s) | Y_t = y] ds, \quad (25)$$

$$X_t^i = F_i(t, Y_t), \quad p_i(t, y) = -1.5 y \partial_y F_i(t, y) / F_i(t, y). \quad (26)$$

Consumption and preference drift may jump at the deterministic slab dates; wealth is continuous. The price is smooth inside each slab and has matching values at its endpoints.

THEOREM 5—Self-financing stochastic decoder: *Every policy satisfying (23)–(24) is a progressively measurable admissible policy of the original economy. Its wealth satisfies*

$$dX_t = ((.02 + .06p_t)X_t - c_t) dt + .2p_t X_t dW_t^x.$$

For all traded-factor realizations, before preference stopping,

$$.5 < X_t < 1.282253 < 2, \quad 0 \leq p_t < .706888 < .8.$$

There is no wealth-boundary exit. The preference process and its stopping rule remain those of the original model.

The proof uses the conditional-price equation, not a simulated or time-discretized budget. It also checks the portfolio bound globally in $y > 0$. Thus the nonzero portfolio is an actual admissible control, not a diagnostic evaluated only on sampled paths.

D.2. Strict acceptance of policy values

Let $[L_i^n, U_i^n]$ enclose the stopped expected payoff of expert i at retained work level n . A proposed expert is accepted only if

$$L_i^{n+1} > U_i^n. \quad (27)$$

The checker completes before further proposals. Rejection restores both the network and Adam state to the incumbent. The original experiment retains every proposal, certificate, decision, network, compiled coefficients, and cost. It contains no learned critic in the acceptance test. Therefore (27) proves a strict increase in that expert's actual payoff at its specified initial state. It does not assert global optimization of the neural parameters.

The optimal comparator is separate. Let $\bar{V}_k(z)$ be the inherited unrestricted dual upper bound on the original stopped economy. A policy can be accurate because $\bar{V}_k(z) - L_k^\pi(z)$ is small, but merely reducing $\bar{V}_k - L_k^\pi$ is never substituted for (27). The source identities and costs of the dual library are retained separately from neural generation.

D.3. A policy on a continuum of initial states

Order the four vertices $z_i = (u_i, x_i)$ of K lexicographically. For $z \in K$, let $\lambda_i(z)$ be its bilinear interpolation weights. On the same Brownian paths, run the four subaccounts and set

$$\begin{aligned} u_t &= \sum_i \lambda_i u_t^i, & X_t &= \sum_i \lambda_i X_t^i, & c_t &= \sum_i \lambda_i c_t^i, \\ \theta_t &= \sum_i \lambda_i \theta_t^i, & p_t &= \frac{\sum_i \lambda_i p_t^i X_t^i}{X_t}. \end{aligned} \quad (28)$$

Initial wealth is exactly $x = \sum_i \lambda_i x_i$. The wealth-weighted portfolio remains in the original admissible interval. This is an adapted policy with stored initial-state context and observable subaccounts, not a claim that a single map of current (t, u, x) alone represents every expert history.

THEOREM 6—Uniform optimality from verified expert payoffs: *Let $p_E = 2 \exp\{-.58^2/(2 \cdot .0025)\}$. For any fixed $k \in [.5, 8]$, the mixture (28) satisfies*

$$J_k^{\text{mix}}(z) \geq \sum_i \lambda_i(z) L_{i,k} - 16p_E, \quad z \in K. \quad (29)$$

If $\bar{V}_k$ is convex in z on K , then

$$0 \leq V_k^*(z) - J_k^{\text{mix}}(z) \leq \max_i \{\bar{V}_k(z_i) - L_{i,k}\} + 16p_E. \quad (30)$$

The explicit dual upper used here is convex: its state dependence is affine, the support function of an interval slope, and a nonnegative exponential localization term. Convexity of the unknown value function is not assumed.

The theorem uses joint concavity of $f(u, c) = c^{1-u}/(1-u)$ on $[1.2, 2.8] \times [.5, .8]$, concavity of settlement, and the common Brownian coupling. Stopping is dealt with by the explicit event correction, rather than by an invalid assertion that arbitrary stopped payoffs are globally concave.

D.4. Why expert improvement does not suffice for mixture improvement

Even if all four experts improve, the old mixture might have a larger Jensen benefit than the new mixture. Ordering the four corners alone therefore does not prove a policy improvement at every interior state. We address this additional identification problem directly.

THEOREM 7—Policy-sensitive continuum improvement: *Suppose Γ_0 verifies*

$$J_2^{\text{mix},0}(z) \leq \sum_i \lambda_i(z) U_i^0 + \Gamma_0, \quad z \in K.$$

Then the final mixture has uniform payoff improvement at least

$$d = \min_i (L_i^f - U_i^0) - \Gamma_0 - 16p_E. \quad (31)$$

When $d > 0$ and (30) gives final regret at most B , true regret contracts pointwise by

$$\frac{V_2^*(z) - J_2^{\text{mix},f}(z)}{V_2^*(z) - J_2^{\text{mix},0}(z)} \leq \frac{B}{B + d}, \quad z \in K. \quad (32)$$

For each executed initial four-expert system, the common slopes and preference drifts allow the directed Hessian construction in the appendix to compute Γ_0 . This construction yields $d \geq 0.01762319$ and $B/(B + d) \leq 0.14110003$ for all five seeds.

We distinguish the sixty strict expert transitions, the four work-level accuracy bounds on K , and the proved initial-to-final mixture improvement. We do not assert that every consecutive mixed policy improves throughout K : the small later expert gains do not by themselves establish that stronger statement.

D.5. A fixed-policy price-sensitivity theorem

For a fixed policy, neither the state law nor the stopping time changes with k . Write

$$B_i^{\text{full}} = \frac{1}{2} \int_0^1 e^{-.04t} (\theta_t^i)^2 dt.$$

Then $B_i^{\text{stop}} \in [B_i^{\text{full}} - .02C_{\rho}p_E, B_i^{\text{full}}]$ and

$$J_{i,k} = J_{i,2} - (k - 2)B_i^{\text{stop}}. \quad (33)$$

Use this interval identity with the fixed final neural coefficients; do not retrain or interpolate neural solutions in k . The unrestricted dual upper is bounded between its verified price nodes by convex chords. On each price segment, an affine payoff lower bound subtracted from the affine dual chord is affine at each vertex. Its maximum is therefore bounded by its endpoints. Applying Theorem 6 at those endpoints proves, without a sampled price grid,

$$\sup_{z \in K, k \in [1.5, 2.5]} \{V_k^*(z) - J_k^{\text{mix},f}(z)\} \leq 0.00332694 < .005 \quad (34)$$

for every seed. This is a post-training sensitivity theorem about the fixed policies, not an additional confirmatory training experiment.

APPENDIX E: MATCHED POLICY EXPERIMENTS AND NONSYNCHRONIZED CONTINUUM COMPARISONS

We use the stochastic policy, exact self-financing decoder, and rectangle $K = [1.98, 2.02] \times [1.24, 1.26]$ defined in Section D. An expert's policy is specified by the sixteen triples (b_j, s_j, θ_j) and its price-defined reserve, not by an optimizer termination message. Unless stated otherwise, $k = 2$ and the starting time is zero.

E.1. Crossing parameterization and optimizer

For each ensemble and vertex, a randomly initialized 1–16–16–3 tanh network produces a 16×3 array of raw outputs at the slab nodes. We initialize the direct parameterization to that exact binary64 array. Both Adam and L-BFGS-B start from the same network weights in the neural arm and the same array in the direct arm. The delivered initial consumption intercepts, slopes, and preference drifts coincide exactly within each block. The dimensions are 355 neural parameters and 48 direct coefficients. Vertex v in ensemble s uses seed $s + v$, for $s \in \{22000, 22100, 22200\}$; corners are therefore not synchronized.

Each configuration has a cap of 400 function/gradient evaluations. Adam uses learning rate .005. L-BFGS-B uses at most 40 line-search evaluations per iteration, function tolerance 10^{-13} , and gradient tolerance 10^{-8} . At budget exhaustion we restore the last accepted optimizer iterate, not an unaccepted line-search trial. We report gradient calls separately from accepted optimizer iterations, wall time, and checker calls. An initial-policy check is shared within each block and is charged when a configuration is considered as a standalone method. Final proposals are delivered only when their certified lower payoff exceeds the incumbent's certified upper payoff. Failed feasibility checks leave the incumbent in place and remain in the results.

This design varies the parameterization, including its induced geometry and sharing, within a fixed optimizer. It does not identify a causal effect of hidden layers independently of every change in optimization geometry. Nor are Adam iterations equated with L-BFGS-B iterations. The earlier neural–Adam versus direct–L-BFGS-B comparison remains a useful end-to-end baseline, but is not a representation ablation.

E.2. Pair-moment control of the mixture's Jensen benefit

Let $i = 1, \dots, 4$ index experts and let $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, denote the fixed bilinear weights of an initial state in K . Define unstopped auxiliary preference means

$$m_i(t) = u_i + \int_0^t \theta_i(s) ds, \quad c_i(t) = .5 + .3\sigma\{b_i(t) - s_i(t) \log Y_t\}.$$

All auxiliary experts are driven by the same Brownian motions. They define the mixture up to the mixture's own stopping time; an auxiliary expert's exit does not silently liquidate all subaccounts. On the common no-exit event, auxiliary and individually stopped payoffs agree.

Write $I_j = [j/16, (j+1)/16]$, $M_j = \int_{I_j} e^{-.04t} dt$, and let D_j bound the range of $m_i(t)$ on I_j , obtained from both slab endpoints. Let $T_j = \max_i \theta_{ij} - \min_i \theta_{ij}$. The directed pair-moment inputs are

$$Q_{ik,j} \geq \int_{I_j} e^{-.04t} \mathbb{E}[(c_i(t) - c_k(t))^2] dt, \quad V_j^c = \frac{3}{8} \max_{i < k} Q_{ik,j}, \quad V_j^u = \frac{M_j D_j^2}{4}. \quad (35)$$

The expectation is under the physical measure. The factor and preference shocks need not be independent.

Let H_{uu}, H_{uc}, H_{cc} bound the absolute Hessian of the running utility on $[1.78, 2.42] \times [.5, .8]$. Let D_T be the range of $m_i(1)$, D_R the range of the enclosed reserves, and $R_- > .5$ a common lower reserve bound. Define

$$\Gamma = \sum_j \left\{ \frac{1}{2} H_{uu} V_j^u + H_{uc} \sqrt{V_j^u V_j^c} + \frac{1}{2} H_{cc} V_j^c + \frac{1}{4} M_j T_j^2 \right\} + 6p_4 \sum_j M_j$$

$$+ e^{-.04} \left\{ .005 D_T^2 + \frac{.0125 D_R^2}{R_-^2} \right\} + 16p_E, \quad (36)$$

where $p_4 = 2\phi(4)/4$ and $p_E = 2 \exp\{-.58^2/(2 \cdot .0025)\}$. These are analytic tail bounds, not estimated failure probabilities.

THEOREM 8—Continuum comparison without synchronized initialization: *Suppose the expert coefficients satisfy the decoder's admissibility conditions and each expert payoff is enclosed by $[L_i, U_i]$. For arbitrary, possibly unequal, slopes and preference drifts, the corresponding mixture payoff satisfies*

$$\sum_i \lambda_i L_i - 16p_E \leq J_\lambda \leq \sum_i \lambda_i U_i + \Gamma.$$

Let superscripts 0 and 1 denote old and proposed expert systems. If

$$d = \min_i (L_i^1 - U_i^0) - \Gamma^0 - 16p_E > 0, \quad (37)$$

then $J_\lambda^1 - J_\lambda^0 \geq d$ throughout K . If W is the convex unrestricted dual upper comparison used here, then

$$V - J_\lambda^1 \leq B := \max_i \{W(u_i, x_i) - L_i^1\} + 16p_E.$$

Consequently $(V - J_\lambda^1)/(V - J_\lambda^0) \leq B/(B + d)$ whenever $d > 0$. Convexity is required of the dual comparison W , not asserted for the original value function in all state coordinates.

The theorem also compares two different trained parameterizations: use the direct final mixture as the old system and the neural final mixture as the proposed system. No optimal-value upper is needed for that payoff-ordering comparison. The older range-based Jensen certificate remains available as a separate valid upper bound; it is not discarded when pair moments are sharper.

E.3. A constructive global comparison beyond exact replication

For a controlled diffusion with generator $\mathcal{L}^a$, running reward f^a , discount ρ , and stopped payoff g , set

$$R^a v = v_t + \mathcal{L}^a v - \rho v + f^a, \quad D^a \psi = \psi_t + \mathcal{L}^a \psi - \rho \psi.$$

These definitions do not require complete markets or an analytic hedge. Let v be a smooth candidate witness and π an admissible current-state policy. Choose nonnegative correction functions ψ_ℓ and nonnegative coefficients α_ℓ, β_ℓ .

PROPOSITION 6—Verified residual-correction cone: *Suppose a finite covering of the state-time and action domains supplies valid interval upper bounds for $R^a v$ and $D^a \psi_\ell$, and lower bounds for $R^\pi v$. If the finite linear inequalities certify*

$$\sup_a R^a \left(v + \sum_\ell \alpha_\ell \psi_\ell \right) \leq 0, \quad R^\pi \left(v - \sum_\ell \beta_\ell \psi_\ell \right) \geq 0,$$

with the corresponding upper stopping trace and policy-accessible lower stopping trace, then

$$J^\pi \geq v - \sum_\ell \beta_\ell \psi_\ell, \quad V \leq v + \sum_\ell \alpha_\ell \psi_\ell.$$

A floating-point linear-programming solution is a proposal only; its inequalities must be checked with directed arithmetic. Including the constant correction 1 and the time correction $C_\rho(T - t)$ provides a feasible fallback whenever finite global residual and trace bounds are already available. Nonreplicable controls alter the generator and admissible set, not this verification argument.

We implement the time-slab member of this correction cone, retaining the independently enclosed complete-domain residuals. For slab errors $e_j = e_j^+ + e_j^-$, define $F(T) = 0$ and

$$F(t_j) = C_\rho(h_j)e_j + e^{-\rho h_j} F(t_{j+1}). \quad (38)$$

On each slab $F(t) = e_j/\rho + [F(t_{j+1}) - e_j/\rho]e^{-\rho(t_{j+1}-t)}$ is monotone or constant. Thus $\max_j F(t_j)$ certifies every starting time. The broader state-dependent cone is a constructive extension, not a reported incomplete-market experiment.

E.4. Cover and deployment complexity

At a fixed starting time, an affine, self-financing mixture on a d -simplex uses $d + 1$ active experts. A regular grid with n subdivisions per coordinate has $(n + 1)^d$ distinct vertices and $d!n^d$ simplices. Sorting fractional coordinates locates the active simplex in $O(d \log d)$ operations. Shared vertex storage avoids duplicating a complete policy for every simplex, but does not eliminate the worst-case dimension dependence.

PROPOSITION 7—Conditional cover complexity: *Suppose the state mixture preserves admissibility, the running and terminal criteria are concave on a common event, and the complement correction is certified separately. Then the vertex-payoff and convex-upper argument extends to each simplex, with its own complement allowance. With m time slabs, q_t time nodes and q_z Gaussian nodes, direct conditional-price evaluation costs $O((d + 1)m q_t q_z)$ per query; compiled active-policy storage is $O((d + 1)m)$ and a complete vertex library costs $O((n + 1)^d m)$.*

For a fixed current-state neural policy and witness, suppose a compact-domain interval extension overestimates its certificate functional by at most Ch^α on a box of diameter h , and the desired inequality has a strict uniform margin $\eta > 0$. Then a dyadic adaptive cover terminates after at most $O((D(C/\eta)^{1/\alpha})^d)$ boxes, up to geometric constants, on a domain of diameter D . Neither a positive margin nor a low effective dimension follows merely from using a neural network.

The simplex statement concerns state mixtures at a common starting time; it does not justify mixing policies initialized at different times. The second statement applies directly to a current-state residual checker and therefore can include time among its coordinates. The actual full-state execution below covers every cell of the original three-dimensional state-time domain. Its residual margin for the .01 target is not positive.

APPENDIX F: FINANCED-COORDINATE IDENTIFICATION AND EXACT-PRICE CONDITIONING

Let μ be a finite positive pricing measure, with mass $M > 0$. In the present economy it integrates discounted risk-neutral consumption over the remaining horizon. For a time-slab intercept vector b and slope vector s , write $\ell(t, z) = b_j - s_j \log Y_t(z)$ on slab j . Put $c_- = .5$, $\Delta c = .3$, and

$$P(a) = c_- M + \Delta c \int \sigma(\ell + a) d\mu, \quad B = x - e^{-rT} R, \quad \nu = \frac{B - c_- M}{\Delta c M}.$$

Here R is a target terminal reserve, $r = .02$, and the numerical application has $T = 1$. Assume ℓ is finite μ -almost everywhere and $0 < \nu < 1$. The exact-price family solves $P(a) = B$. The delivered finite-precision policies are instead priced independently; their reserve is defined by that exact price and enclosed by the checker. The two constructions must not be silently equated.

THEOREM 9—The budget quotient and a continuous-price bracket: *The price equation has a unique finite root a^* . A common intercept shift $b \mapsto b + h\mathbf{1}$ changes that root to $a^* - h$ and leaves the financed stream unchanged. For slabs of positive pricing measure, define*

$$D_j = \Delta c \int_{I_j} \sigma'(\ell + a^*) d\mu, \quad D = \sum_j D_j > 0, \quad \omega_j = D_j/D.$$

Then the financed intercepts $b^f = b + a^\mathbf{1}$ satisfy*

$$\frac{\partial a^*}{\partial b_j} = -\omega_j, \quad \frac{\partial b^f}{\partial b} = I - \mathbf{1}\omega^\top, \quad \frac{\partial a^*}{\partial B} = D^{-1}. \quad (39)$$

The matrix in (39) is a rank- $(m-1)$ projection whose null space is the common-intercept direction. Fixing one raw intercept to zero is a complete coordinate chart for the same financed family, with $3m-1$ rather than $3m$ direct coordinates.

Suppose a measurable set G has $\mu(G) \geq (1-p)M$, $\ell \in [\ell_-, \ell_+]$ on G , and $0 \leq p < \min(\nu, 1-\nu)$. With $\text{logit}(q) = \log(q/(1-q))$, the root is enclosed by

$$a_- = \text{logit}\left(\frac{\nu-p}{1-p}\right) - \ell_+, \quad a_+ = \text{logit}\left(\frac{\nu}{1-p}\right) - \ell_-. \quad (40)$$

If Z bounds $|\ell + a|$ on G for $a \in [a_-, a_+]$, then

$$P'(a) \geq d_* := \Delta c M(1-p) \frac{e^{-Z}}{(1+e^{-Z})^2} > 0. \quad (41)$$

For any $\hat{a}$ in that bracket, an exact-price residual bound $|P(\hat{a}) - B| \leq \varepsilon$ implies $|\hat{a} - a^| \leq \varepsilon/d_*$. A proposal quadrature residual must include a separately certified quadrature error before it can be used as ε .*

For a finite positive quadrature rule, $p = 0$ and the extreme node logits give the bracket directly. For the continuous Gaussian pricing measure, the checker uses $G = \{|Z_0| \leq 8\}$ in the standard-normal integration coordinate and the directed Mills bound $p \leq 2\phi(8)/8$. Under that measure $\log Y_t = .065t + .3\sqrt{t}Z_0$; all time slabs are covered when obtaining ℓ_- and ℓ_+ . The unbounded tail is therefore included rather than treated as a finite support assumption. The derivative lower bound can be conservative. Its role is to establish a valid conditioning enclosure, not to claim that every observed root is well conditioned.

The proposal program centers intercepts before evaluating prices and retains centered coordinates when forming the delivered logits. This avoids reintroducing a large canceled common offset into the price calculation. It uses the finite-rule bracket, rejects nonfinite or saturated arithmetic, and applies the analytic first derivative after bisection. The independent stopped-policy checker remains unchanged. Mathematical translation invariance does not imply bitwise invariance under arbitrary floating-point shifts.

PROPOSITION 8—Equality of the finite-node policy classes: *At m distinct scalar time nodes, a $1-m-m-3$ tanh network with unrestricted finite real weights can interpolate any prescribed $m \times 3$ raw-output array. Thus the neural, direct, and direct-quotient representations used here generate the same finite-node financed family, for finite logits and the interior sigmoid action ranges. They do not thereby have the same optimization geometry or the same finite-precision trajectory. Nor does interpolation at fixed time nodes establish approximation of a current-state policy on a higher-dimensional domain.*

The proof constructs an invertible hidden-feature matrix and solves only for the final affine layer. For the actual sixteen nodes, the replication record supplies a directed MPFR nonsingularity certificate for rational first- and second-layer parameters. This identifies what the crossed experiment can attribute to parameterization. It does not assert that Adam or L-BFGS-B finds the interpolating weights, or that this construction is itself a competitive training method.

PROPOSITION 9—Fixed-shape re-budgeting: *Consider two exact-price policies with the same finite intercept contrasts, slopes, deterministic preference drifts, and terminal reserve R , but with initial wealth $x' < x''$ in the attainable budget interval. Suppose both price-defined wealth processes stay strictly inside the wealth boundaries until preference exit or the horizon. Then their consumption streams and wealth processes are pathwise ordered under common shocks, their stopping times coincide, and $J(x'') \geq J(x')$. The policy map changes through its financing offset. This result therefore concerns a re-budgeted family, not the compensating variation of a fixed complete policy map.*

For the target reserve .50001 and consumption in $[\cdot 5, \cdot 8]$, the conditional-price bounds establish the stated wealth nonexit condition over the one-year horizon. The earlier .002 counterfactual is supported by its separately computed payoff intervals; Proposition 9 does not replace those calculations or identify a minimum increment. Small differences between the proposed reserve and the independently enclosed delivered reserve remain in the numerical checks.

APPENDIX G: CERTIFIED POLICY COMPARISONS AND CURRENT-STATE VERIFICATION

G.1. A held-out crossed design with a budget-coordinate control

The new experiment fixes ensembles 23000, 23100, and 23200, the four vertices of $K = [1.98, 2.02] \times [1.24, 1.26]$, sixteen time slabs, and a cap of 400 joint objective/gradient evaluations per configuration before execution. Corner v uses seed ensemble $+v$, so the four corners are independently initialized. The neural architecture is $1-16-16-3$ with tanh hidden layers and 355 trainable coordinates. The direct and direct-quotient versions use 48 and 47 coordinates. Adam uses step size .005; L-BFGS-B uses the same objective and a maximum of 40 line-search steps, with function and gradient tolerances 10^{-13} and 10^{-8} . On budget exhaustion the last accepted L-BFGS-B iterate is restored.

All four primary neural/direct arms have exactly matched delivered initial policies. The quotient arm fixes the sixteenth raw intercept to zero and subtracts it from the other fifteen. This changes coordinates but not the financed family. Its delivered initial discrepancy is bounded and separately recorded; nonidentical binary64 incumbents are priced separately rather than assigned the neural incumbent's interval. No configuration is removed. A proposed final policy replaces the initial incumbent only when its independently certified payoff lower bound exceeds the incumbent's payoff upper bound. Unsuccessful proposals would leave the incumbent in place.

The primary computation uses positive quadrature only to generate proposals. Regret and acceptance use the inherited continuous-time stopped-policy checker, including its quadrature remainder, Gaussian tail, settlement, and unrestricted dual comparison. The nonsynchronized Jensen theorem then certifies every reported regional statement. This is not inference from four pointwise successes alone. These continuum properties are a posteriori checks of the frozen policies; the comparisons to be reported were specified prospectively.

TABLE G.I
HELD-OUT MATCHED-INITIAL-POLICY SYSTEMS, INCLUDING THE DIRECT QUOTIENT

Ensemble	Configuration	Regret upper	Gain lower	Gradients	Gen. sec.
23000	Neural–Adam	0.00302112	0.02641307	1600	5.994
23000	Neural–L-BFGS-B	0.00283890	0.02657559	1600	5.806
23000	Direct–Adam	0.01206722	0.01880295	1600	4.736
23000	Direct–L-BFGS-B	0.00283871	0.02657561	405	1.176
23000	Quotient–Adam	0.01208949	0.01878120	1600	4.909
23000	Quotient–L-BFGS-B	0.00283872	0.02657561	496	1.480
23100	Neural–Adam	0.00300570	0.02658891	1600	5.644
23100	Neural–L-BFGS-B	0.00283873	0.02673407	1600	5.349
23100	Direct–Adam	0.01196019	0.01889670	1600	4.574
23100	Direct–L-BFGS-B	0.00283872	0.02673412	452	1.276
23100	Quotient–Adam	0.01198223	0.01887538	1600	4.748
23100	Quotient–L-BFGS-B	0.00283871	0.02673412	428	1.240
23200	Neural–Adam	0.00301599	0.02636096	1600	5.676
23200	Neural–L-BFGS-B	0.01050307	0.01856495	1300	4.345
23200	Direct–Adam	0.01215505	0.01873897	1600	4.561
23200	Direct–L-BFGS-B	0.00283871	0.02651360	362	1.029
23200	Quotient–Adam	0.01217718	0.01871731	1600	4.757
23200	Quotient–L-BFGS-B	0.00283871	0.02651360	498	1.430

Every bound covers K at $t = 0, k = 2$. Gain is relative to the independently initialized incumbent mixture. Each system contains four experts. Generation excludes checking and the shared dual. All prescribed configurations are retained.

All 72 final proposals are feasible and strictly improve their respective incumbent at the vertex where they were trained. All eighteen delivered four-expert systems have a positive certified uniform initial-to-final payoff gain. The neural–Adam systems have uniform regret below 0.003022 and payoff gain at least 0.026360. Direct–Adam and quotient–Adam remain above .01. The additional coordinate control therefore rules out a common-intercept redundancy as the sole explanation of the observed Adam comparison.

The correct interpretation remains conditional on optimizer and budget. Direct L-BFGS-B obtains regional regret below .002839 for every ensemble, with fewer objective/gradient evaluations than the corresponding neural L-BFGS-B run. One neural L-BFGS-B ensemble has regret upper .01050307 despite a valid financing check and strict policy improvement. That observation is retained in Table G.I. Fixing the financing root does not eliminate nonlinear optimization sensitivity.

The comparison in Table G.II is between actual feasible-policy payoffs, not between two regret envelopes. The general Jensen bound is applied to each direct incumbent mixture, with its own heterogeneous slopes and drifts. The smallest uniform neural–Adam payoff advantage exceeds .003044 against both direct coordinate systems. Together with Proposition 8, this establishes an optimizer-conditional parameterization effect within the same finite-node policy class. It is

TABLE G.II
UNIFORM ACTUAL-PAYOFF ADVANTAGE OF NEURAL-ADAM

Ensemble	Comparator	Vertex gain lower	Jensen upper	Uniform gain lower
23000	Direct-Adam	0.00751990	0.00436704	0.00315287
23000	Quotient-Adam	0.00751639	0.00436521	0.00315119
23100	Direct-Adam	0.00748368	0.00434849	0.00313519
23100	Quotient-Adam	0.00748020	0.00434739	0.00313282
23200	Direct-Adam	0.00743883	0.00439387	0.00304496
23200	Quotient-Adam	0.00743540	0.00439114	0.00304427

The uniform bound subtracts the comparator mixture's Jensen benefit and the stopping correction. It orders actual payoffs throughout K ; it does not compare only regret envelopes.

neither a larger expressive class nor unconditional neural superiority. The earlier independently initialized crossed study gives the same qualitative Adam ordering; its full 48-configuration ledger, including a rejected neural L-BFGS-B proposal and separately labeled repairs, remains in the historical appendix and supplement.

G.2. Costs, numerical checks, and proof-critical arithmetic

The held-out study uses 21,541 objective evaluations and the same number of gradient evaluations, 68.729 seconds of policy generation, and 188.034 seconds of final checking. Its complete elapsed time is 290.946 seconds, including initial-policy checks and overhead. The peak process resident set is 428,544 KiB. These are single-thread measurements in the recorded environment, not cross-hardware speed claims. The inherited non-neural unrestricted dual library cost of 415.789 seconds is additional and common to the methods. Both an unamortized cost and an explicitly shared-library interpretation are appropriate; neither sets that construction cost to zero. The supplement reports generation, checking, optimizer calls, parameter counts, environment, and the precise meaning of the extra-call proxy for every vertex.

The largest observed proposal-rule root residual is 4.45×10^{-16} after outward display rounding. This is a numerical diagnostic, not a bound on exact economic prices. The unchanged directed checker certifies the delivered reserves and payoffs. In a separate proof-instantiation pass, all 72 frozen policies receive continuous Gaussian price brackets under Theorem 9. The smallest verified endpoint margin exceeds 2.05×10^{-10} . The weakest bracket-wide derivative lower bound is only 5.45×10^{-23} ; it is valid but conservative and is not presented as useful local conditioning. No policies are modified by this pass.

A separate MPFR rank computation encloses the determinant of the sixteen-node hidden-feature matrix in $[5.48546568232, 5.48546568245] \times 10^{-6}$, away from zero. This supports the explicit interpolation construction, not the behavior of either optimizer. The finite-difference, shift, and nonfinite-input regressions are code tests rather than substitutes for the proofs. The theorem-to-function-to-result map in the supplement distinguishes these roles.

G.3. The original current-state operator and active acceptance

The full-state numerical object remains distinct from the regional experiments. In the recorded current-state refinement, the actor and witness consume (t, u, x) , jointly have 2,500 parameters, and begin from the retained full-state checkpoint. Four hundred gradient evaluations produce 267 accepted L-BFGS-B iterations. A complete MPFR cover of 1,024 cells, with no skipped cells, and the discounted suffix construction tighten the all-start-time original-economy upper bound

from 7.2783190635 to 7.241462444. This is a genuine continuous-domain witness improvement. It is not a proof that the jointly changed actor has improved in payoff, and it is not the unchanged .01 accuracy target. The successful regional systems cannot be substituted into that sentence.

The residual-correction cone of Proposition 6 supplies a constructive comparison for general generators, including generators without exact replication. Its computed coefficients remain limited by the residuals they have to dominate. Thus the mathematical extension beyond the selected hedgeable consumption streams does not imply an executed sharp solution of the unspanned full-state problem. The inherited high-dimensional warm-start comparisons retain their separate models, finite-query scopes, training costs, and the stronger accelerated baseline at the strictest high-dimensional target.

An active-rejection study is retained because the new 72-configuration design has no rejected final proposals. In that study four of eight prescribed proposals are rejected, and exact model, optimizer, and random-state restoration is checked. Repeating an unchanged deterministic block reproduces rejection, exposing stagnation rather than hiding it. A separately identified supervisor changes the proposal schedule outside the restored state and yields six accepted and two rejected blocks. These are targeted robustness tests, not a convergence theorem for Adam or L-BFGS-B. They demonstrate a needed distinction between safe rollback and productive proposal generation.

G.4. Deployment, scaling, and economic scope

The regional implementation retains four financed subaccounts and fixed initial-state mixture weights. Networks are compiled to 192 binary64 coefficients, occupying 1,536 coefficient bytes per four-expert system; conditional prices and their derivatives still require integration. The retained four-expert timing diagnostic has at most 8,192 quadrature nodes per query and median latency about .160 milliseconds. Tests with 8, 16, and 32 duplicated experts measure implementation scaling only. They neither train a larger cover nor certify the accuracy of an online quadrature rule. Proposition 7 states the conditional domain/tolerance complexity, including its dimension dependence; it does not remove a high-dimensional covering cost by assertion.

The earlier fixed-policy price transport remains a sensitivity result on its stated k interval, not current-state generalization. The .002 wealth increment is a *sufficient re-budgeted wealth increment*: learned slopes and preference-drift shapes are retained while the financing offset is changed and checked again. Proposition 9 explains the order of the corresponding exact-price family. Neither that proposition nor the numerical counterfactual estimates the minimum compensating variation of an unchanged policy map.

APPENDIX H: CONCLUSION

Neural Bellman Operators require an economic comparison, not merely a numerical stopping rule for training. The new results identify the financed policy coordinates, remove a redundant budget direction, and prove continuum payoff ordering without synchronized initialization. The held-out experiment shows that the neural–Adam payoff advantage survives both optimizer matching and removal of that redundant direct coordinate. Its interpretation is a parameterization effect within a fixed optimizer and work budget, not greater finite-node expressiveness or overall numerical dominance.

The stronger direct L-BFGS-B results and the adverse neural L-BFGS-B outcome remain visible. Active rejection, exact restoration, and supervisory recovery are supported by their executed records. The exact-price bracket, rank certificate, and nonsynchronized mixture theorem provide separate analytic and executable checks. All depend on stated mathematical hypotheses; agreement between arithmetic backends is not an independent economic proof.

The original full-domain .01 objective remains the target for the current-state neural operator. The complete-domain witness has improved, but the available bound does not yet establish that accuracy or a strict payoff improvement for the jointly refined actor. The general residual-correction cone identifies a route that does not rely on the selected replicable consumption family. We retain that objective, the original economy, and the earlier substantive development while reporting exactly which economic comparisons the present calculations establish.

APPENDIX I: FINANCED-COORDINATE PROOFS

I.1. Existence, identification, and the price derivative

For every finite ℓ , the logistic function is continuous, strictly increasing, and takes values strictly between zero and one. Its bound by one permits dominated convergence under the finite measure μ . Hence P is continuous and strictly increasing, with limits c_-M and $(c_- + \Delta c)M$ at minus and plus infinity. The interior-budget condition gives existence and uniqueness of a finite root. Replacing (b, a) by $(b + h\mathbf{1}, a - h)$ leaves every logit and price unchanged, proving the common-shift identity.

The derivative satisfies $0 < \sigma'(y) \leq 1/4$. Differentiation under the integral is therefore justified for the offset and each intercept coordinate. Positive measure of a slab and finiteness of its logits give $D_j > 0$. The implicit function theorem gives $\partial a^*/\partial b_j = -D_j/D$ and $\partial a^*/\partial B = 1/D$. Differentiating $b^f = b + a^*\mathbf{1}$ proves (39). Since $\omega^\top \mathbf{1} = 1$,

$$(I - \mathbf{1}\omega^\top)^2 = I - \mathbf{1}\omega^\top, \quad (I - \mathbf{1}\omega^\top)\mathbf{1} = 0.$$

If this projection annihilates z , then $z = \mathbf{1}(\omega^\top z)$, so its null space is exactly one-dimensional and its rank is $m - 1$. It need not be an orthogonal projection. Each common-shift equivalence class has exactly one representative with final intercept zero. Slopes and preference drifts are unchanged, giving $3m - 1$ direct coordinates. The implementation uses raw intercepts scaled by .3; applying the same common-shift chart before that scaling gives the same quotient. $\square$

I.2. The unbounded-tail bracket and inverse conditioning

Normalize μ to a probability measure. On G , monotonicity bounds $\sigma(\ell + a)$ by its values at ℓ_- and ℓ_+ . On the complement it lies in $[0, 1]$. Writing the possibly larger actual mass of G explicitly and then using its lower bound gives

$$(1 - p)\sigma(\ell_- + a) \leq \frac{P(a) - c_-M}{\Delta cM} \leq (1 - p)\sigma(\ell_+ + a) + p.$$

The upper expression equals ν at a_- and the lower equals ν at a_+ . Monotonicity of P proves (40). This argument uses a bound on the complement's measure; it does not assign the Gaussian law compact support.

For $|y| \leq Z$, the logistic derivative is at least $e^{-Z}/(1 + e^{-Z})^2$, because it is even and decreases with $|y|$. Restrict the derivative integral to G and use its mass lower bound to obtain (41). The mean value theorem between $\hat{a}$ and a^* gives the residual-to-root bound. If P_q is a quadrature price, the triangle inequality supplies the valid residual bound $|P(\hat{a}) - B| \leq |P_q(\hat{a}) - B| + |P(\hat{a}) - P_q(\hat{a})|$. Omitting the second term is not justified by a small first term.

For the Gaussian implementation, t and $\sqrt{t}$ are interval-enclosed on every slab, and $Z_0 \in [-8, 8]$ gives the compact logit range under the pricing measure. The Mills inequality supplies the omitted relative mass. Directed logarithms and exponentials enclose both endpoints of (40). The executable check adds a small outward endpoint allowance and verifies strictly positive

price margins at both ends. It computes the derivative bound using $e^{-Z}/(1 + e^{-Z})^2$, avoiding subtraction of a rounded logistic value from one. Each of the 72 frozen delivered policies is checked. The resulting finite existence and conditioning enclosures do not alter those policies or replace their original financing certificates. $\square$

APPENDIX J: FINITE-NODE CLASS EQUIVALENCE AND ITS CERTIFICATE

Order the distinct nodes as $t_1 < \dots < t_m$. Select $a_1 < t_1$ and $a_j \in (t_{j-1}, t_j)$ for $j \geq 2$. Define the first hidden-feature matrix $H_{ij}(K) = \tanh(K(t_i - a_j))$. As $K \rightarrow \infty$, it tends to a matrix whose first column consists of ones and whose successive columns switch from minus one to plus one at successive rows. Subtracting successive rows shows that this limiting matrix is nonsingular. By continuity of the determinant, $H(K)$ is nonsingular for some finite K .

Use a diagonal second-layer weight matrix ϵI and zero biases. Its feature matrix is $G(\epsilon) = \tanh(\epsilon H(K))$, with $\tanh$ applied elementwise. Since $G(\epsilon)/\epsilon \rightarrow H(K)$ as $\epsilon \downarrow 0$, it is nonsingular for sufficiently small positive finite ϵ . For any raw-output array $Y \in \mathbb{R}^{m \times 3}$, the final affine layer with zero bias and coefficient matrix $G(\epsilon)^{-1}Y$ interpolates Y exactly in real arithmetic. Every network also produces such an array, so the two finite-node output sets coincide. The common decoder and Theorem 9 then give the financed-family and quotient statements. Bounds attained only by infinite sigmoid logits belong to the closure, not to the asserted finite-logit interior.

The numerical certificate fixes the actual sixteen nodes $t_j = (2j + 1)/16 - 1$, thresholds $a_j = 2j/16 - 1$, $j = 0, \dots, 15$, $K = 80$, and $\epsilon = 1/4$. MPFR interval evaluation encloses the entire exact feature matrix. Directed Gaussian elimination has no pivot interval containing zero and encloses its determinant in the strictly positive interval reported in the supplement. Elimination is an algebraic enclosure of that fixed matrix, not a rank judgment based on a singular-value tolerance. It proves existence of the exact final linear solve; rounding a computed solve would require its own output-error analysis. $\square$

APPENDIX K: RE-BUDGETING AND PRESERVATION OF ECONOMIC COMPARISONS

Higher initial wealth raises B . Theorem 9 therefore increases the exact financing offset. With the contrasts and slopes unchanged, logistic consumption is ordered at every time and factor realization. Under common shocks, the deterministic preference drifts give identical preference paths. The price-defined wealth difference before stopping is the positive conditional price of the remaining consumption difference, since both policies have the same terminal reserve. Their wealth processes are therefore ordered as well.

The maintained nonexit condition rules out wealth stopping for either policy. They stop at the same preference exit or terminal date. Running utility has derivative $c^{-u} > 0$ with respect to consumption on the original domain; effort cost is identical. At the common stopping time the settlement's wealth derivative is $.1/x > 0$, while its preference and liquidation components agree. Comparing these terms pathwise and taking discounted expectations proves Proposition 9.

For the specified reserve and consumption range, the conditional price at remaining horizon $h \in [0, 1]$ lies between $Re^{-.02h} + .5C_{.02}(h)$ and $Re^{-.02h} + .8C_{.02}(h)$. The lower bound is minimized at $h = 0$ and exceeds .5; the upper is maximized at $h = 1$ and is below 2. This proves wealth nonexit for the exact-price family. In the experiment the reserve is independently enclosed rather than set equal to its floating-point target. Its original feasibility and payoff enclosures remain authoritative; this ideal-family monotonicity result is not used to bypass them. $\square$

APPENDIX L: PROOFS FOR NONSYNCHRONIZED POLICY COMPARISON

L.1. Weighted variances and the Jensen upper

For weights λ summing to one, the exact identity

$$\sum_i \lambda_i \|z_i - \bar{z}\|^2 = \sum_{i < k} \lambda_i \lambda_k \|z_i - z_k\|^2, \quad \bar{z} = \sum_i \lambda_i z_i, \quad (42)$$

follows by expansion. In particular, $\sum_{i < k} \lambda_i \lambda_k = (1 - \sum_i \lambda_i^2)/2 \leq 3/8$ for four experts. For scalar points with range D , the weighted variance is at most $D^2/4$; translate the range to $[0, D]$, use $z^2 \leq Dz$, and maximize $D\bar{z} - \bar{z}^2$.

On $A_t = \{.05W_t^u \leq .2\}$ the auxiliary preferences lie in $[1.78, 2.42]$, since their means lie in $[1.98, 2.22]$. The same interval contains their weighted average. Consumption and its average lie in $[\cdot 5, \cdot 8]$. Taylor's formula with integral remainder, expanded about the weighted average, eliminates the weighted linear term. Taking absolute Hessian bounds gives

$$f(\bar{u}, \bar{c}) - \sum_i \lambda_i f(u_i, c_i) \leq \frac{1}{2} H_{uu} \text{Var}_\lambda(u) + H_{uc} \sqrt{\text{Var}_\lambda(u) \text{Var}_\lambda(c)} + \frac{1}{2} H_{cc} \text{Var}_\lambda(c).$$

The common preference shock cancels in every pair difference, so its weighted variance equals the variance of the deterministic means and is at most $D_j^2/4$ on slab j . Identity (42) and the directed pair integrals imply

$$\int_{I_j} e^{-.04t} \mathbb{E}[\text{Var}_\lambda(c(t))] dt \leq V_j^c.$$

The expectation may be restricted to A_t without increasing this nonnegative upper bound. Cauchy–Schwarz, on the joint time-probability measure $e^{-.04t} dt dP$, bounds the cross term by $\sqrt{V_j^u V_j^c}$. Correlation of the Brownian motions is immaterial to this argument; independence has not been assumed.

On the common no-exit event, all preferences lie in the original domain. The joint utility is concave there by the determinant calculation in the retained proof, and its range is contained in $[-6, 0]$. Thus the nonnegative Jensen gap on A_t^c is at most six. The two-sided Mills inequality gives $P(A_t^c) \leq 2\phi(4)/4$ for every $t \leq 1$. The effort contribution is exactly $k \text{Var}_\lambda(\theta)/2$; for $k = 2$ it is bounded by $T_j^2/4$. These observations prove the running part of (36).

At the terminal date the exact subaccount wealth is its reserve R_i . The quadratic preference term has Jensen gap $.02 \text{Var}_\lambda(m_i(1)) \leq .005 D_T^2$. Since $-(\log x)'' = 1/x^2 \leq 1/R_-^2$, the log-wealth term contributes at most $.0125 D_R^2/R_-^2$. Both are discounted by $e^{-.04}$.

The event $E = \{\sup_{t \leq 1} |.05W_t^u| < .58\}$ prevents preference exit for every expert and the mixture; the financial decoder prevents wealth exit. The reflected Gaussian tail bound gives $P(E^c) \leq p_E$. On E , stopped and full-horizon expressions agree. On its complement, both the mixture payoff and a convex combination of expert payoffs lie in $[-15, 1]$, so replacing their difference costs at most $16p_E$. This proves the upper bound with (36). Joint concavity on E and the same complement allowance prove the lower bound. Subtract the old upper from the new lower to obtain (37). Finally, $V \leq W$ and convexity of W give the regret upper. If $r_1 = V - J^1$ and $r_0 = V - J^0$, then $r_0 \geq r_1 + d$ and $0 \leq r_1 \leq B$, whence $r_1/r_0 \leq B/(B + d)$. $\square$

L.2. Certification of the pair moments

With $v = \sqrt{t}$ and z standard normal under the physical measure,

$$\log Y_t = -.025v^2 + .3vz.$$

For fixed slab coefficients the integrand of a pair moment is

$$2ve^{-.04v^2} \phi(z) \left[.3\sigma\{b_i - s_i(-.025v^2 + .3vz)\} - .3\sigma\{b_k - s_k(-.025v^2 + .3vz)\} \right]^2.$$

The checker uses rationally isolated Gauss nodes on directed cells in v and z , with $|z| \leq 8$. The Gaussian tail omitted from a pair moment is at most $.092\phi(8)/8$; using this bound separately on every time slab is conservative.

We justify the analytic constant used by the inherited quadrature-error routine for this new integrand. In simultaneous complex discs of radii $1/16$ in v and $1/2$ in z , centered at real $v \in [0, 1]$ and $z \in [-8, 8]$, the imaginary part of the logistic argument is bounded by

$$6.5 \left\{ .0252 \frac{17}{16} \frac{1}{16} + .3 \left(\frac{17}{16} \frac{1}{2} + \frac{17}{2} \frac{1}{16} \right) \right\} < 2.094 < \frac{9}{4} < \frac{3\pi}{4}.$$

Consequently the logistic function has no pole in these discs and its modulus is at most $\sqrt{2} < 1.5$. The complex consumption modulus is below $.95$, the Gaussian density modulus is below one, and $|2ve^{-.04v^2}| < 3$. The pair integrand therefore has modulus less than 11 , hence less than the deliberately conservative Cauchy constant 100 . The machine check obtains the still sharper product upper bound 3.468414 ; the proof uses 100 throughout.

All cell endpoints, node-isolation errors, tensor-product quadrature remainders, and Gaussian tails are added outward. The global quadrature remainder is applied to each slab separately, never divided by the number of slabs. The resulting six pair integrals enter (35); numerical similarity of policies is not used as a substitute for a pair bound. This construction establishes an arithmetic certificate conditional on the stated analytic proof. It is not proof-assistant verification or independent validation of the economic model.

L.3. A simpler range certificate, retained as a cross-check

An alternative bound needs no pair integrations. Put $L = 97/40$. The Gaussian representation yields $P(|\log Y_t| > L) \leq 2\phi(8)/8$ for $t \leq 1$. On $|\log Y_t| \leq L$, the maximum range of affine logits is attained at an endpoint of $[-L, L]$:

$$A_j = \max_{\ell \in \{-L, L\}} \left\{ \max_i (b_{ij} - s_{ij}\ell) - \min_i (b_{ij} - s_{ij}\ell) \right\}.$$

The maximal difference of two logistic values whose arguments differ by at most A_j is $\tanh(A_j/4)$, attained at symmetric arguments. Hence consumption range is at most $C_j = .3 \tanh(A_j/4)$. The running Jensen upper on slab j is

$$M_j \{ H_{uu} D_j^2 + 2H_{uc} D_j C_j + H_{cc} C_j^2 + kT_j^2 \} / 8,$$

with the additional six-unit allowance for the union of preference and factor tails. The same terminal and stopping corrections apply. This proves the independently executable range formula without requiring equal slopes or drifts. The range and pair-moment bounds are both retained; either can be used, or their smaller certified upper can be selected.

APPENDIX M: PROOFS FOR GLOBAL VERIFICATION AND COVER COMPLEXITY

M.1. Residual corrections

For $w^+ = v + \sum_{\ell} \alpha_{\ell} \psi_{\ell}$, linearity of D^a gives $R^a w^+ = R^a v + \sum_{\ell} \alpha_{\ell} D^a \psi_{\ell}$. Nonnegative coefficients permit interval upper bounds to be summed without reversing any inequality. On a

state box B and an action cell C , sufficient linear constraints are

$$\overline{R^a v}(B, C) + \sum_{\ell} \alpha_{\ell} \overline{D^a \psi_{\ell}}(B, C) \leq 0.$$

For $w^- = v - \sum_{\ell} \beta_{\ell} \psi_{\ell}$, sufficient constraints are

$$\underline{R^{\pi} v}(B) - \sum_{\ell} \beta_{\ell} \overline{D^{\pi} \psi_{\ell}}(B) \geq 0.$$

A complete action cover or an exact continuous-action oracle is required; sampled maximizing actions do not suffice. Trace inequalities must be checked on their specified stopping faces. Discounted Itô's formula, localization, and the integrability conditions already stated for stopped verification yield $J^a \leq w^+$ for every admissible a and $J^{\pi} \geq w^-$. Taking the supremum over a proves Proposition 6. The argument is unchanged for a vector diffusion with nonreplicable shocks, provided the corresponding generator, admissibility, and integrability conditions hold.

For a feasible fallback, $D^a 1 = -\rho$ and $D^a C_{\rho}(T - t) = -1$. Nonnegative constants dominate bounded stopping-trace discrepancies, and nonnegative multiples of C_{ρ} dominate finite positive and negative residual errors. For time-slab corrections $\psi_j(t) = \int_t^T e^{-\rho(s-t)} 1_{I_j}(s) ds$, one has $D^a \psi_j = -1_{I_j}$ away from slab endpoints. Values match at the endpoints, so piecewise discounted Itô integration applies. This proves the time-slab correction program and recurrence (38). The envelope's monotonicity within a slab proves the all-start-time maximum rule. $\square$

M.2. Expert covers and adaptive verification

A simplex represents each point by nonnegative barycentric weights on $d + 1$ vertices. Under the stated affine-mixture conditions, weighted subaccounts preserve the state law and admissibility. Concavity and the separately bounded common-event complement transfer vertex payoffs; convexity of the upper comparison transfers its upper values. No statement about arbitrary nonlinear state mixtures is implied. A regular d -dimensional grid has $(n + 1)^d$ distinct vertices; the standard ordering of fractional coordinates divides each of its n^d cells into $d!$ simplices. Active price evaluation requires one set of at most $m_{q_t} q_z$ integrand evaluations per active expert, proving the stated cost and storage bounds. The exact conditional-price object need not possess a closed-form elementary integral.

For the adaptive interval statement, subdivide until $Ch^{\alpha} < \eta$. The strict margin then separates every interval upper from the failure threshold. A dyadic cover reaches such a diameter after finitely many levels, with a number of boxes of the indicated order. The assertion is conditional on uniform consistency and a positive margin; it does not prove that the full-domain error objective has been attained. Nor does reducing the number of active experts to $d + 1$ eliminate the number of distinct vertices needed for a global library. $\square$

M.3. Root invariance and supervisory recovery

For a fixed quadrature rule, the financed budget function $P(a)$ is continuous and strictly increasing in the common offset a whenever finite logistic coefficients are used. Its limits are $.5 \int_0^1 e^{-.02t} dt$ and $.8 \int_0^1 e^{-.02t} dt$. The chosen budget lies strictly between them. A finite bracket therefore exists. Shifting every raw intercept by h and the root by $-h$ leaves the exact delivered consumption stream unchanged. Centering the intercepts is a numerical choice of this redundant common-offset coordinate, not a change of the policy family. Geometric bracket expansion

remedies a failed fixed bracket, but the independent financing check remains necessary because the proposal uses finite quadrature and finite arithmetic.

Exact rollback restores the proposal state. If the next proposal is a deterministic function of that state, it repeats after rejection. A supervisory state outside the rolled-back proposal state is therefore necessary for deterministic recovery by step-size modification. In the executed recovery rule, the state-hash comparison precedes the learning-rate change. The observed recovery is a finite execution result; no convergence rate or universal acceptance guarantee follows from it.

APPENDIX N: PROOFS FOR THE STOCHASTIC POLICY CONSTRUCTION

N.1. Self-financing and admissibility

For each slab, the martingale-measure backward equation for (25) is

$$F_t + .11yF_y + .045y^2F_{yy} - .02F + c = 0, \quad F(1, y) = R.$$

Applying Itô's formula under the physical measure gives

$$dF = (.02F - c - .09yF_y)dt - .3yF_y dW^x.$$

Substitution of $p = -1.5yF_y/F$ yields exactly the original wealth equation. Continuity of the conditional price at the finitely many slab boundaries eliminates any jump or unfinanced reset. Formula (24) makes $F(0, 1) = x$ exactly.

Let $a(y) = c(y) - .01$. Since $a \in [.49, .79]$,

$$F(t, y) = .5 + (R - .5)e^{-.02(1-t)} + \int_t^1 e^{-.02(s-t)} \mathbb{E}^Q[a(s, Y_s) | Y_t = y] ds.$$

This implies $.5 < F < X_{\max}$, where

$$X_{\max} = .5 + .79 \frac{1 - e^{-.02}}{.02} + .0001 < 1.282253.$$

For $q = \sigma(b - s \log y)$, elementary maximization over $q \in [0, 1]$ gives

$$\frac{|yc_y|}{c - .01} = s \frac{.3q(1-q)}{.49 + .3q} \leq s\kappa, \quad \kappa = \frac{\sqrt{.79} - \sqrt{.49}}{\sqrt{.79} + \sqrt{.49}}.$$

The conditional factor is multiplicative in y , so differentiation under its bounded integrand gives

$$|yF_y| \leq 6.5\kappa(F - .5), \quad F_y \leq 0.$$

Consequently $0 \leq p \leq 1.5(6.5)\kappa(1 - .5/X_{\max}) < .706888$. The deterministic preference drift and consumption satisfy the original action constraints. These arguments establish Theorem 5 for every $y > 0$, not merely inside a quadrature truncation.

N.2. Concavity, common stopping event, and continuum accuracy

Write $r_u = u - 1 \in [.2, 1.8]$ and $a = -\log c \in [-\log .8, \log 2]$. The diagonal entries of the Hessian of $f = -e^{r_u a}/r_u$ are negative. Its determinant has the sign of

$$D(r_u, a) = r_u^2 a^2 - 2r_u(r_u + 1)a + 2(r_u + 1).$$

Since $a < .7$, D decreases in a . The polynomial $D(r_u, .7)$ is concave in r_u , so its minimum is at an endpoint. At $r_u = 1.8$ it is $.1316 > 0$ and at $r_u = .2$ it is larger. Hence f is jointly concave on the full stated rectangle. Both $-k\theta^2/2$ and $G(u, x) = -.02(u-2)^2 + .1 \log x$ are concave for $k \geq 0$.

All expert preference drifts are in $[0, .2]$. On

$$E = \{\sup_{t \leq 1} |.05W_t^u| < .58\},$$

all four expert preference processes, and their convex mixture, stay inside $(1.2, 2.8)$. There is no wealth exit by Theorem 5. The exponential maximal inequality gives $\Pr(E^c) \leq p_E$. On E , running and settlement concavity give the required pathwise mixture lower bound up to the common terminal time. For these particular admissible policies, the original stopped payoff lies in $[-15, 1]$ for all $k \in [.5, 8]$: running utility has magnitude below six, the adjustment penalty is at most $.16$, and the remaining stopping settlement fits in this conservative range. The error on E^c is at most $16p_E$. This proves (29). Convexity of the explicit dual bound gives

$$\bar{V}_k(z) \leq \sum_i \lambda_i \bar{V}_k(z_i),$$

and proves (30). The argument uses neither concavity nor a numerical approximation of V^* .

N.3. An upper bound for the initial mixture's Jensen gain

The stored initial experts have identical slope and preference-drift vectors within each seed; this is checked exactly from their serialized coefficients. The four consumption intercepts differ only by budget offsets. Thus, in slab j ,

$$d_{c,j} := .3 \tanh\{(\max_i b_{ij} - \min_i b_{ij})/4\}$$

bounds the full-factor range across the four consumptions. The preference range is $d_u = .04$ at every time. No loss in this bound is hidden by testing a few factor values.

On the event $|.05W_t^u| \leq .2$, all four preferences belong to $[1.78, 2.42]$. Directed interval evaluation on a 64-by-64 cover of this rectangle times $[.5, .8]$ gives

$$|f_{uu}| \leq 4.436941, \quad |f_{uc}| \leq 3.709523, \quad |f_{cc}| \leq 25.902278.$$

These are enclosing Hessian evaluations over boxes, not derivative samples. For any four nonnegative weights summing to one, weighted variance of a quantity in a range of width d is at most $d^2/4$. Taylor's theorem about the weighted mean and Cauchy-Schwarz therefore bound the flow Jensen gap by

$$\gamma_j = \frac{H_{uu}d_u^2 + 2H_{uc}d_ud_{c,j} + H_{cc}d_{c,j}^2}{8}.$$

The complement of the marginal event has probability at most $2\phi(4)/4 = e^{-8}/(2\sqrt{2\pi})$, uniformly for $t \leq 1$. Its flow Jensen gap is at most six on E . The initial effort Jensen gap is zero because all preference drifts coincide. If d_R bounds the range of exact terminal reserves, the terminal Jensen gap is at most

$$e^{-.04} \left(.005d_u^2 + \frac{.0125}{.5^2} d_R^2 \right).$$

Integrate γ_j using the exact discounted slab masses, add six times the marginal-tail bound integrated over the horizon, add the terminal term, and add $16p_E$ for E^c . The resulting directed upper endpoint is Γ_0 . Each reserve range is formed from its entire enclosing interval. This proves the premise of Theorem 7. Subtract the old upper bound from the final lower bound to obtain (31). If final regret is $e(z) \leq B$, old regret is at least $e(z) + d$. Since $e/(e + d)$ increases in e , (32) follows. In particular, a zero final regret creates no division problem because $d > 0$ bounds the old regret away from zero.

N.4. Conditional Gaussian evaluation and analytic error bounds

The independent payoff checker does not import the neural optimizer or simulate paths. Put $t = v^2$ and represent $\log Y = -.025t + .3vz$, with z standard normal. Conditional on z , preference minus one has mean

$$r_0 = u_0 - 1 + \int_0^t \theta_s ds + .0125vz$$

and independent Gaussian variance $.00234375t$. The sign of the conditional mean reflects the original negative correlation and the convention $z = -W_t^x/\sqrt{t}$.

On $|z|, |z_2| \leq 8$, the checker has $r_0 \geq .88$, actual $u - 1 \geq .49$, actual $u - 1 \leq 1.71$, and $|\log c| < .7$. Outside the original preference interval, define a bounded utility extension by clipping preference to $[1.2, 2.8]$. The extension is an integration device only: it is never a change to the stopped economic model, whose difference is bounded below. Expand conditional utility to degree ten in the independent Gaussian increment. Its first omitted even moment gives the uniform remainder

$$10(10395) \frac{(.00234375)^6}{(.49)^{13}} < 1.836 \times 10^{-7}.$$

For completeness, the derivative bound follows from

$$\frac{e^{ar}}{r} = \int_{-a}^{\infty} e^{-ry} dy, \quad r > 0,$$

by bounding its twelfth derivative divided by $12!$. The integral on $[0, \infty)$ equals r^{-13} , and the integral on $[-a, 0]$ is bounded by $e^{ra}a^{13}/13!$. The conservative multiplier ten bounds their sum on the stated ranges.

Let $T_0 = 2\phi(8)/8$ and $T_{2j} = 28^{2j-1}\phi(8) + (2j-1)T_{2j-2}$. These bound the omitted even Gaussian tail moments. The conditional-polynomial correction is

$$\sum_{j=0}^5 \frac{10(.00234375)^j}{.88^{2j+1}} T_{2j},$$

and $12T_0$ bounds the two omitted bounded-utility tails. The price integral and its derivative have separate normal-tail bounds $.8T_0$ and $1.95T_0$. This calculation never integrates the singular unextended CRRA expression over an unrestricted Gaussian preference distribution.

Each transformed time slab is subdivided to width at most $1/40$ in v . The normal interval $[-8, 8]$ has 96 cells. Both integrals use degree-15 exact, eight-point Gauss rules, with rationally isolated nodes and outward weights. The inherited analytic rule bounds its remainder on a complex disc of radius $1/16$ in v and $1/2$ in z . Here is an explicit justification for its common

modulus bound $M = 100$. On these discs the imaginary logistic arguments are respectively bounded by

$$6.5\{(2(.065) + 2.4)/16 + .065/256\} < 1.03, \quad 6.5(.3)/2 = .975 < \pi/3.$$

Hence the sigmoid has no pole, its real part lies between zero and one, $\Re c \geq .5$, $|c| \leq .8$, and $|\log c| < .8$. Also $\Re r_0 > .8$, $|r_0| < 1.4$, and the complex independent variance has modulus below .0027. The conditional Taylor polynomial is bounded by

$$\sum_{j=0}^5 \frac{10(2j-1)!!(.0027)^j}{.8^{2j+1}} < 12.554.$$

The coefficient estimate uses $e^{2(1.4)(.8)} < 10$. Multiplication by $|2v| \leq 17/8$, the discounted density bounds, and the Gaussian density modulus below .5 leaves a bound below fifteen. The price and price-derivative integrands are smaller. Thus $M = 100$ is conservative for each analytic integrand. The quadrature remainder is charged separately from arithmetic enclosure width and from Gaussian truncation. The two arithmetic implementations may choose different outward time subdivisions; they need not have identical node counts.

Finally, the full-horizon extended-flow calculation differs from the original stopped payoff only after preference exit. With

$$M_4 = .22^4 + 6(.22)^2(.0025) + 3(.0025)^2,$$

the correction $20p_E + .02\sqrt{M_4 p_E}$ bounds the bounded running and stopping terms and the unbounded quadratic terminal-preference term by Cauchy–Schwarz. Terminal log wealth uses the enclosed exact reserve. Each reported payoff interval includes all these errors. Independent MPFR arithmetic checks the same analytic derivation, rather than proving an unrelated theorem.

N.5. Price transport and welfare transfer

For fixed policies τ is independent of k . The adjustment integral is affine in k and the omitted cost after stopping is at most $.02C_\rho p_E$, proving (33). The dual-node chords are upper bounds because the unrestricted value is a supremum of affine functions of k . Apply the appropriate sign of $k - 2$ to the payoff-cost interval. On every dual-node segment the resulting vertex upper-minus-lower bound is affine; its endpoint maximum bounds the whole segment. State mixing then proves (34).

For compensation, apply Theorem 6 to expert initial wealths $x_i + .002$, with the same weights for the original z . The actual initial wealth is $x + .002$. Rechecked reserves and slopes retain Theorem 5. Compare the resulting policy lower bound with the convex dual upper at the original, uncompensated z . The smallest directed vertex margin, less $16p_E$, is $0.00004475 > 0$. This proves the sufficient initial-wealth compensation without interpreting a utility gap as an unfinanced consumption gift.

APPENDIX O: PROOFS OF THE VERIFICATION AND TRANSPORT RESULTS

PROOF OF THEOREM 1: For any admissible control a , stopped Itô's formula gives

$$J(t, z; a) - v(t, z) = \mathbb{E}_{t,z} \left[\int_t^\tau e^{-\rho(s-t)} \mathcal{R}_{a_s} v(s, Z_s) ds + e^{-\rho(\tau-t)} (g - v)(\tau, Z_\tau) \right].$$

One may first stop the local martingale at a bounded localization sequence. Bounded derivatives and coefficients on the stopped compact domain give integrability, and dominated convergence then yields the displayed identity. For the candidate π , the absolute value of the right side is at most $b + \int_t^T e^{-\rho(s-t)} e(s) ds$. For an arbitrary original control, the residual is at most $e(s) + q(s)$, because its action is covered by the continuous maximum in (8). Hence

$$V(t, z) - v(t, z) \leq b + \int_t^T e^{-\rho(s-t)} \{e(s) + q(s)\} ds.$$

Adding the lower-policy error proves (9). Nonnegativity follows from admissibility of π . If an implemented action belongs to an enclosed neighborhood, the same residual and action-gap checks hold for it by interval inclusion, so the proof is unchanged. No statement about the convergence of the process that generated v or π is used. *Q.E.D.*

PROOF OF PROPOSITION 1: In the perturbed model, the candidate residual is at least $-e - d$, while the residual of any original admissible action is at most $e + q + d$. Applying the stopped identity separately in that model bounds the lower-policy and upper-value errors by the corresponding integrals. Their sum is $2e + q + 2d$, not an extra improvement-gap penalty on top of those errors. Nonnegative integrands permit replacement of the perturbed discount by ρ . The stopping-trace error is at most $b + b_{\text{par}}$ if the settlement differs by at most b_{par} . This argument neither equates the two stopping times nor assumes a pathwise coupling of different diffusions. *Q.E.D.*

APPENDIX P: PROOF OF THE JOINT STATE-PRICE CONSTRUCTION

LEMMA 1—Concavity of the covered policy functional: *For each stored policy satisfying (67), there is a concave function $F_j(z; k)$ of $z \in K$, affine in k , such that $J_k(0, z; \pi_{j,x}) \geq F_j(z; k)$. Its values at the four corners have the independently computed lower endpoints used in the state-price calculation.*

PROOF: Put $r = u - 1$ and $L = -\log c$. The determinant expression in Section Y.13 has the sign of

$$r^2 L^2 - 2r(r+1)L + 2(r+1) = r^2 L^2 + 2(r+1)(1 - rL) > 0,$$

while $f_{cc} < 0$. Therefore $D^2 f$ is negative definite over the stated region. In the covered Gaussian integral, the preference input is affine in initial preference and consumption is affine in initial wealth. Their image remains in the region just verified. Integrating this concave composition with positive Gaussian and time weights preserves concavity.

The full-horizon preference settlement is the concave quadratic $-.02\{(u - 2 + \int \theta)^2 + .0025\}$. By the budget shift the terminal wealth, and hence its logarithmic settlement, is independent of initial wealth. The deterministic expenditure is independent of the initial state. Subtract a uniform normal-tail and stopping allowance from this covered integral plus its terminal settlement and expenditure. The supplement proves this allowance for all $z \in K$ and $k \in [.5, 8]$. The resulting exact function is F_j , which retains concavity and affine dependence on price. Its coefficient of $-k$ is

$$B_j^{\text{full}} = \frac{1}{2} \int_0^1 e^{-\rho s} \theta_{j,s}^2 ds.$$

At each corner the interval quadrature encloses the exact covered integral and terminal term. Its lower endpoint after the uniform corrections is no larger than F_j there. The finite quadrature

rule is not assumed to reproduce the exact functional, and sampled corner values are not used without their enclosure errors. *Q.E.D.*

PROOF OF PROPOSITION 18: Let the four corners of K be z_v , and write $z = \sum_v \lambda_v(z) z_v$ using the nonnegative bilinear corner weights, which sum to one. Convexity of the verified upper object and concavity of the lower functional imply

$$V_{k_i}(0, z) \leq U_i(z) \leq \sum_v \lambda_v(z) U_i(z_v), \quad J_k(0, z; \pi_{j,x}) \geq F_j(z; k) \geq \sum_v \lambda_v(z) F_j(z_v; k).$$

For k between k_i and k_{i+1} , fixed-policy affinity in price and a common admissible policy set give convexity of V_k . Applying the value chord at the two endpoints yields the chord between the two verified upper corner combinations.

Suppose a policy lower corner endpoint is L_{jv} at k_j and its expenditure belongs to $[B_j^-, B_j^+]$. Its sign-aware lower line is

$$L_{jv} - (k - k_j)B_j^+ \quad (k \geq k_j), \quad L_{jv} - (k - k_j)B_j^- \quad (k \leq k_j).$$

Each price cell between consecutive nodes has a fixed sign for every policy. The upper chord minus this lower line is therefore an affine function $g_{jv}(k)$. For every state, the regret of policy j is at most $\sum_v \lambda_v(z) g_{jv}(k) \leq \max_v g_{jv}(k)$. Selecting a policy minimizing this last bound proves the uniform state guarantee

$$R(k) = \min_j \max_v g_{jv}(k).$$

Take all pairwise intersections of the finitely many affine functions and the price-cell endpoints. Between successive such points, their ordering does not change. The maximum for each policy and then the minimum across policies select a single affine function on that interval. Its maximum occurs at an endpoint. Exact rational evaluation at this finite set thus bounds the supremum over all real prices. A corner line is discarded only when another corner line from the same policy dominates it at both price-cell endpoints, which proves domination on the entire cell. The program records those reductions and checks the remaining intersections without a sampled-grid extrapolation. *Q.E.D.*

PROOF OF PROPOSITION 19: The independent primal and dual constructions in the supplement supply, respectively, valid original-policy lower endpoints and upper functions (66). Exact action and terminal-budget checks hold for every central policy and every shifted policy at the extremes of the initial-state rectangle. Affinity of the shift extends these feasibility checks to its interior. Lemma 1 supplies the concave lower functional with a common tail and stopping allowance. All quantities in Proposition 18 are therefore present.

The execution uses the 18 prices

$$.5, .875, 1.25, 1.625, 2, 2.5, 3, 3.5, 4, 4.25, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8.$$

The stored state–price record checks 211 candidate breakpoints and returns the exact fraction displayed following (69); upward decimal conversion gives (68). The central-state calculation uses the exact stopped-policy expenditure intervals and checks 187 candidate breakpoints with a separate all-intersections implementation. Its exact worst-case fraction is

$$\frac{1600901779824582944359922627717}{160587876576305697877131733762048},$$

whose upward display is 0.009969007711. These are computer-assisted conclusions conditional on the explicitly stated arithmetic model and analytic enclosure lemmas, not consequences of agreement between two floating-point approximations. *Q.E.D.*

APPENDIX Q: ECONOMIC RESOLUTION AND THE COMMON-ENDPOINT STRESS BOUND

For $p < q$, every policy loses $(q - p)B(\pi) \geq 0$, and $B(\pi) \leq .02C_\rho(1)$. Taking suprema preserves monotonicity and gives $0 \leq V_p - V_q \leq .02C_\rho(1)(q - p)$. Combining this fact with the independently verified intervals at p and q gives (81). On any segment where the upper and lower envelope pieces are affine, both strict separation and the conservative relative-width inequality reduce to affine inequalities in q . Their roots, included with the exact envelope breakpoints, produce the resolution regions without a grid test.

Finally, if every nodal upper endpoint is enlarged by δ and every nodal lower endpoint is decreased by δ , every upper chord is enlarged by δ and every lower policy line is decreased by δ . Their maximum/minimum envelope gap increases by at most 2δ . This proves the recorded endpoint stress test and also its scope: it tests sensitivity to valid endpoint inflation, not the validity of upstream node proofs.

APPENDIX R: PROOFS FOR STOPPING ACCESSIBILITY AND CERTIFIED ITERATION

PROOF OF PROPOSITION 14: Fix $x_0 \in [1.5, 2)$ and use $c = .8$, $\theta = p = 0$. Wealth decreases, stays below two, and satisfies $x_s \geq x_0 - .8s \geq .7$ for $s \leq 1$. Thus it does not hit either wealth face. Preferences equal $2 + .05W_s$. On the event $H = \{\sup_{s \leq 1} |.05W_s| \leq .2\}$ there is no preference exit, $u \in [1.8, 2.2]$, and running utility is at least -1.5 . To see the last inequality, set $r = u - 1 \in [.8, 1.2]$. The function $-(5/4)^r/r$ is increasing on this interval since $r \log(5/4) < 1$. Its value at $.8$ exceeds -1.5 , using $(5/4)^4 < (6/5)^5$. Settlement on H is at least $-.05$, because $-.02(.2)^2 + .1 \log(.7) > -.05$. Discounting negative terms cannot worsen these lower bounds. The pathwise discounted payoff on H is therefore at least -1.55 .

Until the original stopping time, $u \in [1.2, 2.8]$. The same elementary monotonicity bounds running utility by -6 from below, and $G \geq -.1$. Even on H^c , the stopped payoff is at least $-6 - 8 - .1 = -14.1$. The Brownian maximal inequality gives $\Pr(H^c) \leq 2e^{-8}$. Hence

$$J_2(0, 2, x_0; .8, 0, 0) \geq -1.55 - 12.55(2e^{-8}).$$

Subtracting $g(0, 2, 2) = .1 \log 2 - 8$ proves (49), uniformly as $x_0 \uparrow 2$. Directed MPFR evaluation encloses its right-hand side in the interval recorded in the replication; its lower endpoint exceeds 6.37.

Finally, a smooth witness equal to g on all stopping faces and with $\sup_a \mathcal{R}_a v \leq e_+(s)$ gives $V(0, z) \leq v(0, z) + \int_0^1 e^{-.04s} e_+(s) ds$ by the stopped Itô identity. Take $z = (2, x)$, let $x \uparrow 2$, and use continuity of v and the established strict gap. The integrated uniform positive residual must exceed 6.37. A sequence with vanishing such residuals is excluded, without making any assertion that the economic control problem lacks a solution. *Q.E.D.*

PROOF OF THEOREM 11: Discounted Itô's formula, localization of the stochastic integral, and the bounded stopped-domain hypotheses give, for every admissible control a ,

$$J(t, z; a) - v(t, z) = \mathbb{E}_{t,z}^a \left[\int_t^\tau e^{-\rho(s-t)} \mathcal{R}_a v(s, Z_s) ds + e^{-\rho(\tau-t)} (g - v)(\tau, Z_\tau) \right].$$

The positive residual bound and the facewise upper discrepancy bounds imply

$$V - v \leq \int_t^T e^{-\rho(s-t)} e_+(s) ds + \sum_i \beta_i^+ \omega_i^*.$$

Apply the same identity to π with the opposite sign to obtain

$$v - J^\pi \leq \int_t^T e^{-\rho(s-t)} e_-(s) ds + \sum_i \beta_i^- \omega_i^\pi.$$

Adding proves the assertion. No common stopping time across two distinct controlled paths is assumed. For the proposed smooth hitting-probability bound w_i , optional stopping yields $\mathbb{E}[e^{-\rho(\tau-t)} w_i(\tau, Z_\tau)] \leq w_i(t, z)$; domination of the indicator on the stopping boundary then gives the stated probability bound. *Q.E.D.*

PROOF OF PROPOSITION 15: Write $Y = 2 - x > 0$ and $p = Yq$, where $q = (.15 + .65 \tanh N_p)/1.5$ is bounded. Until another exit,

$$dY = (c - .04)ds + Y\{.02 - .06xq\}ds - .2xqY dZ_s^x.$$

The bounded coefficients define a strictly positive stochastic exponential Φ . Variation of constants gives

$$Y_s = \Phi_s \left(Y_t + \int_t^s \Phi_r^{-1} (c_r - .04) dr \right) > 0,$$

since $c \geq .05$. The upper wealth face is therefore inaccessible from the interior. On the remaining state faces $B = 0$; at terminal time $1 - t = 0$. On the upper wealth face $B \geq 0$ and softplus is positive, so $v - g \geq 0$. Thus every upper-comparison trace discrepancy is zero, and the only potentially positive policy discrepancy has zero hitting probability. The same calculation applies to admissible bounded clipped head perturbations before the distance multiplication. *Q.E.D.*

PROOF OF THEOREM 12: Let $d_n = \epsilon_n/(1 - q)$. The resolvent of $I - qP_{\pi_n}$ is nonnegative and has row sums at most $(1 - q)^{-1}$, so $\|v_n - V^{\pi_n}\|_\infty \leq d_n$. Put $a_n = 2qd_n + \eta_n$. The contraction inequalities and approximate greedy inequality give

$$TV^{\pi_n} - T_{\pi_{n+1}} V^{\pi_n} \leq a_n \mathbf{1}.$$

Since $TV^{\pi_n} \geq V^{\pi_n}$, the same nonnegative resolvent gives $V^{\pi_{n+1}} \geq V^{\pi_n} - a_n \mathbf{1}/(1 - q)$. Therefore

$$\begin{aligned} V^* - V^{\pi_{n+1}} &= TV^* - T_{\pi_{n+1}} V^{\pi_{n+1}} \\ &\leq qE_n \mathbf{1} + a_n \mathbf{1} + qP_{\pi_{n+1}} (V^{\pi_n} - V^{\pi_{n+1}}) \\ &\leq \{qE_n + a_n/(1 - q)\} \mathbf{1}. \end{aligned}$$

Taking the supremum norm proves (54). The rate follows by summing the resulting geometric convolution. The comparison with a continuous-time problem is the triangle inequality after separately bounding the two transfer terms; it does not follow from the finite Markov contraction alone. *Q.E.D.*

APPENDIX S: PROOF OF THE STATE-GLOBAL INVENTORY CERTIFICATE

PROOF OF THEOREM 16: The scalar backward Riccati solutions exist on the finite horizon, remain at least .5, and are bounded above. Indeed, in remaining time their drift is $2a_i p - p^2 + q_i$, positive at .5 and negative above its positive equilibrium. The neural gains also remain at least .5 by construction. Let $d_i = \hat{p}_i - p_i^*$ and let R_i be the neural Riccati residual. Then

$$d'_i = (\hat{p}_i + p_i^* - 2a_i)d_i + R_i, \quad d_i(1) = 0.$$

The coefficient is at least $\kappa_i = 1 - 2a_i$. Integrating backward therefore yields $|d_i(t)| \leq \varepsilon_i(1 - e^{-\kappa_i(1-t)})/\kappa_i \leq D$.

The quadratic optimal value has matrix $P^* = p_\perp^*(I - M) + p_m^*M$ and a scalar noise term. Completing the square in its generator and applying Itô's formula gives the exact identity

$$\mathcal{J}(0, x; \hat{a}) - \mathcal{V}(0, x) = \mathbb{E} \int_0^1 |\{\hat{P}(s) - P^*(s)\}X_s|^2 ds.$$

The two projectors are orthogonal, so the integrand is bounded by $D^2|X_s|^2$. Both eigenvalues of $A - \hat{P}$ are at most $-.4$. Itô's formula for $|X_s|^2$ then gives

$$\frac{d}{ds} \mathbb{E}|X_s|^2 \leq -.8\mathbb{E}|X_s|^2 + .01d.$$

Gronwall's inequality and $|x|^2/d \leq 1$ imply $d^{-1}\mathbb{E}|X_s|^2 \leq e^{-.8s} + (.01/.8)(1 - e^{-.8s})$. Integration proves (80). The bounded deterministic gains give finite second moments and justify localization and removal of the stopping sequence in both Itô identities. No bounded-state truncation or simulated tail probability is used. *Q.E.D.*

APPENDIX T: PROOF OF THE COMPENSATION BOUND

PROOF OF PROPOSITION 21: Holding the specified paths fixed, the derivative of consumption utility with respect to the top-up is

$$\frac{\partial}{\partial \lambda} \frac{\{(1 + \lambda)c\}^{1-u}}{1-u} = c^{1-u}(1 + \lambda)^{-u} \geq (1 + \lambda)^{-2.8}.$$

The inequality uses $c \leq 1$, $u > 1$, $u \leq 2.8$, and $\lambda \geq 0$. Integrating from zero to λ , then over the discounted alive-time measure, bounds the welfare gain below by $A\{1 - (1 + \lambda)^{-1.8}\}/1.8$. Solving for a gain of ϵ proves (82). For the time-control library, wealth feasibility excludes a wealth exit, and the retained drift bounds and Brownian maximal inequality give the displayed preference-exit probability. Surviving to terminal time alone yields the stated lower bound on discounted alive time. This argument requires no extrapolation of the certified optimal value to an altered wealth state. *Q.E.D.*

APPENDIX U: PROOFS FOR THE CERTIFICATE-ALIGNED ALGORITHM

PROOF OF THEOREM 13: For a fixed slab and sign, write $A = \max\{0, \max_i a_i\}$ and $B = \max\{0, \max_i b_i\}$. The definition of d gives $b_i \leq \max\{a_i, 0\} + d \leq A + d$, hence $B \leq A + d$. For $K = M + 1$ logits including zero, factor $e^{A/\tau}$ out of their exponential sum. The remaining sum lies in $[1, K]$. It follows that

$$A \leq \tau \log(1 + \sum_i e^{a_i/\tau}) \leq A + \tau \log K.$$

Multiply by the positive discounted slab mass, sum over the two signs and all slabs, and apply Theorem 11. This proves (57); taking $a = b$ and $d = 0$ proves (58). A nonzero independently certified trace budget can be added to both policy-loss bounds without changing the argument.

For the implementation, every stored logit is interpreted as its exact dyadic value. The maximum used to stabilize the logarithm is also an exact stored value. Directed subtraction, exponentiation, pairwise summation, logarithm, multiplication, and discounted-mass evaluation enclose the smooth objective. Directed subtraction of each nonnegative proposal logit from its retained certified endpoint encloses the correction. Its construction is valid regardless of the source of the difference between the two computational graphs. An incomplete cover would invalidate the initial use of Theorem 11; the implementation rejects that case rather than increasing a numerical tolerance. *Q.E.D.*

PROOF OF PROPOSITION 16: The initial accepted certificate is valid by independent verification. At each subsequent checkpoint, rejection leaves the incumbent unchanged, and acceptance replaces it by a valid strictly smaller bound and its associated policy. Induction proves the first assertion.

For the second assertion, consider a deterministic unit-horizon problem with discount $\rho \geq 0$, zero terminal reward, and two actions with constant rewards 0 and -1 . Put $C_\rho(h) = \int_0^h e^{-\rho s} ds$ and $v_a(t) = aC_\rho(1-t)$. Then $v'_a - \rho v_a = -a$, so the optimal residual is $-a$ and the residual of an action with reward r is $r - a$. The signed certificate is

$$C_\rho(1)\{(-a)_+ + (a-r)_+\}.$$

The optimal action $r = 0$ paired with $a = -10$ has certificate $10C_\rho(1)$ and true regret zero. The worse action $r = -1$ paired with $a = -1$ has certificate $C_\rho(1)$ and true regret $C_\rho(1) > 0$. Both witnesses match terminal reward and both certificates are valid. Strict certificate acceptance therefore does not imply increasing policy values. This is an analytical counterexample to the inference, not a simulated observation about the original economy. *Q.E.D.*

PROOF OF PROPOSITION 17: Let $A_n = \|V_n^* - v_n\|_\infty$ and $P_n = \|V_n^\pi - v_n\|_\infty$. The evaluation and improvement inequalities imply

$$\|v_n - T_n v_{n+1}\|_\infty \leq \epsilon_n + \eta_n.$$

The Lipschitz inequalities for T_n and $T_n^{\pi_n}$ give

$$A_n \leq qA_{n+1} + \epsilon_n + \eta_n, \quad P_n \leq qP_{n+1} + \epsilon_n,$$

with $A_N, P_N \leq b$. Backward iteration and the triangle inequality give (59). Finally, $C_\rho(h) \sum_{n=0}^{N-1} e^{-\rho nh} = C_\rho(Nh)$, including its continuous extension at $\rho = 0$. The bound concerns this finite-horizon Bellman problem. Any use for a different continuous-time problem still requires a transfer theorem or a direct continuous-time certificate. *Q.E.D.*

PROOF OF PROPOSITION 20: The upper half of the verification argument gives

$$V_2(0, 2, x) \leq v(0, 2, x) + \int_0^1 e^{-.04s} e_+(s) ds$$

for every interior x . The upper trace inequalities remove any adverse settlement discrepancy. Let $x \uparrow 2$, use continuity of the fixed witness and Proposition 14, and subtract $g(0, 2, 2)$. The result is $\Delta \leq h_v + \int e^{-.04s} e_+(s) ds$. Since the integral is nonnegative, (70) follows. No policy payoff enters this lower bound; it is a statement about the residual needed to support that particular witness. *Q.E.D.*

APPENDIX V: PROOF FOR NONLINEAR VECTOR-STATE CONTROL

PROOF OF THEOREM 14: Let D be the cyclic first-difference matrix, so $(Dz)_i = z_i - z_{i+1}$ and $\|D\|_2 \leq 2$. Differentiating (73) gives

$$\nabla^2 \Phi(z) = \text{diag}(q_i + \frac{1}{2} \text{sech}^2 z_i) + D^\top \text{diag}(\frac{3}{10} \text{sech}^2 (Dz)_i) D.$$

All terms are positive semidefinite. Since $1 \leq q_i \leq 6/5$ and $\text{sech}^2 \leq 1$, $\nabla^2 \Phi(z) \succeq (6/5 + 1/2 + (3/10)4)I = (29/10)I$ globally.

Stack the H propagated states. Their derivative with respect to U is a block lower-triangular convolution operator B with coefficients $1, 3/5, (3/5)^2, \dots$. If S is the finite shift operator, then $B = \sum_{j=0}^{H-1} (3/5)^j S^j \otimes I_d$, so $\|B\|_2 \leq \sum_j (3/5)^j \leq 5/2$. Therefore

$$\nabla_U^2 F_x(U) = 4I + B^\top \text{diag}(\nabla^2 \Phi(z_{h+1})) B,$$

and its spectrum lies in $[4, 4 + (25/4)(29/10)] = [4, 177/8]$. This also establishes coercivity, existence, and uniqueness of the minimizer for every x .

Let $m = 4$, $L = 177/8$, and $\alpha = 2/(m + L) = 16/209$. With $g_k = \nabla F_x(U_k)$, the fundamental theorem of calculus yields

$$g_{k+1} = \left(I - \alpha \int_0^1 \nabla^2 F_x(U_k - t\alpha g_k) dt \right) g_k.$$

The integral is symmetric with spectrum in $[m, L]$. Consequently $\|g_{k+1}\|_2 \leq (L - m)/(L + m) \|g_k\|_2 = (145/209) \|g_k\|_2$. Strong convexity, followed by minimization of its quadratic lower tangent, gives $F_x(U) - \inf F_x \leq \|\nabla F_x(U)\|_2^2 / (2m) = \|\nabla F_x(U)\|_2^2 / 8$.

It remains to bound the initial gradient uniformly in the state cube and in the neural weights. The output constraint gives $|u_{0,hi}| \leq 1/4$, and $|b_{hi}| \leq .03$. State propagation therefore implies

$$|z_{hi}| \leq 1 + \frac{.25 + .03}{1 - .6} = 1.7.$$

Each coordinate of $\nabla \Phi$ has absolute value at most $(6/5)(1.7) + 1/2 + 2(3/10) = 3.14$. The backward adjoint is a geometrically weighted sum of such gradients. Hence every coordinate of $\nabla_U F_x(U_0)$ has absolute value at most $4(1/4) + 3.14/(1 - .6) = 8.85$. There are Hd coordinates, so $\|g_0\|_2^2 \leq Hd(8.85)^2$. Combining the inequalities proves (76) and solving it for k gives the work allocation.

Forward propagation, cyclic neighbor differences, and backward adjoints each require $O(Hd)$ operations; no dense Hessian need be formed. Directed verification of a stored plan replaces its unknown exact gradient norm by an upper enclosure. It certifies that stored plan at that stored initial vector even when its floating-point optimization trajectory differs from the exact-real iteration. It does not, by itself, certify the floating-point neural/correction map simultaneously at every state. Q.E.D.

APPENDIX W: PROOF OF BUDGET-FEASIBLE COMPENSATION

PROOF OF PROPOSITION 22: Couple the original and compensated policies with the same preference Brownian motion. Their preference processes and adjustment costs coincide. Their portfolios are both zero. The wealth difference before stopping is

$$z(s) = \Delta x e^{.02s} - \frac{d_c}{.02} (e^{.02s} - 1) = \Delta x \frac{e^{.02} - e^{.02s}}{e^{.02} - 1} \geq 0,$$

with $z(1) = 0$. The verified wealth and consumption conditions make both policies admissible and exclude wealth stopping. Thus they have the same preference or terminal stopping time. At preference stopping, the wealth-dependent settlement increment is $.1 \log((X_s + z(s))/X_s) \geq 0$; at time one it is zero.

For any preference $u \in [1.2, 2.8]$, marginal utility is c^{-u} . Along the compensated consumption segment, $c \leq .8 < 1$, so $c^{-u} \geq .8^{-1.2}$. The mean value theorem therefore bounds the instantaneous utility increment below by $d_c .8^{-1.2}$. The survival probability at every time is at least $1 - p_e$. Integrating with discount and adding the nonnegative settlement change proves (83). The library's uniform policy-loss inequality then gives the concluding comparison with the original optimal value. The proof neither redefines that value nor asserts that the sufficient compensation is the minimum possible one. *Q.E.D.*

APPENDIX X: PROOFS FOR POLICY-SENSITIVE CONTINUATION AND LEARNED WORK

X.1. Separated acceptance and regret contraction

PROOF OF PROPOSITION 11: The enclosure inequalities give $J_{\pi'} - J_{\pi} \geq L_{\pi'} - U_{\pi}$. Moreover, $\bar{V} - L_{\pi} \geq V - J_{\pi}$. Under (44), subtracting the resulting gain inequality from $V - J_{\pi}$ proves the contraction. Under the last hypotheses, write $g = V - J_{\pi'} \geq 0$ and $a = J_{\pi'} - J_{\pi} \geq d$. Then $g/(g + a) \leq g/(g + d) \leq B/(B + d)$. Since $d > 0$, the denominator is positive even when the new policy is optimal. These arguments apply separately at each state and therefore uniformly wherever the hypotheses are uniform. *Q.E.D.*

X.2. A difference-sensitive state modulus

PROOF OF THEOREM 10: First consider the unstopped reference functional used by the independent evaluator. At time s , couple the two preference inputs by the same standard normal variable Z :

$$U_s^i = u + m_i(s) + .05\sqrt{s}Z, \quad m_i(s) = \int_0^s \theta_v^i dv, \quad i \in \{a, b\}.$$

For the reference functional, clip the preference input to $[1.2, 2.8]$ in the running utility. Retain the analytically integrated terminal quadratic. This clipping does not change the original running payoff before a preference exit. The stopping allowance stated below controls the difference from the original stopped functional.

Write $f(u, c) = c^{1-u}/(1-u)$ and $r = u - 1$. Direct differentiation gives

$$\begin{aligned} |f_{uu}| &= \frac{e^{-r \log c}}{r^3} \{(r \log c + 1)^2 + 1\}, \\ |f_{uc}| &= (-\log c)e^{-u \log c}, \quad |f_{cc}| = ue^{-(u+1) \log c}. \end{aligned}$$

On the larger clipped rectangle, $|f_u| = e^{-r \log c}(1 + r \log c)/r^2$ and $|f_c| = e^{-u \log c}$; $1 + r \log c$ is positive there. The code evaluates enclosing intervals for these formulas, not derivatives estimated from network outputs.

On $|Z| \leq 4$, all four coupled inputs, including those at z_* , are in $[1.78, 2.42] \times [.7, .8]$. This follows from $u \in [1.98, 2.02]$, $0 \leq m_i(s) \leq .2$, and $.05\sqrt{s}|Z| \leq .2$. No clipping occurs on this event. The difference between the two policies' preference derivatives is bounded by $H_{uu}d_m + H_{uc}d_c$; the corresponding consumption derivative difference is bounded by $H_{uc}d_m + H_{cc}d_c$. Integrate along the straight segment from z_* to z . Its preference displacement is at most .02, and

its common consumption displacement is at most $.01/Q_r$. The resulting bound on the change of the payoff difference is the central term inside the brackets of (47).

On the complementary event, clipping is 1-Lipschitz. For each of the two initial states the difference of the two clipped running utilities has absolute value at most $A_u d_m + A_c d_c$. Each clipped running utility is negative and has absolute value less than 6: on this rectangle, $e^{-r \log c}/r$ decreases in r , and its largest value is at $r = .2, c = .7$, where it is less than 6. The oscillation of the difference between two initial states is consequently at most D_4 . Mills' inequality gives

$$\Pr(|Z| > 4) \leq 2 \frac{e^{-8}}{4\sqrt{2\pi}} = P_4.$$

This is a one-time normal-tail bound inside the time integral, not a bound on the maximum of a Brownian path. Integrating the two running-payoff bounds with the discount gives the first term in (47). Preference-adjustment expenditure is independent of the initial state and cancels from the state difference.

The common wealth shift implies

$$X_1^i = e^{.02} \left\{ x - \int_0^1 e^{-.02s} \left(c_s^i + \frac{x - 1.25}{Q_r} \right) ds \right\} = e^{.02} \left\{ 1.25 - \int_0^1 e^{-.02s} c_s^i ds \right\}.$$

Thus each policy's terminal wealth is independent of x , even when the stored dyadic controls do not satisfy the ideal budget identity exactly. Their log-settlement difference has zero initial-state variation. The difference of their expected terminal preference settlements has initial-preference derivative $-.04(m_b(1) - m_a(1))$. Its discounted variation is at most $.04e^{-\rho}(.02)d_T$.

Finally, both policies have strictly inward wealth drift and terminal wealth above $.5$, verified over K . Their only possible premature exit is in preference. For $0 \leq \theta \leq .2$, the distance of the deterministic preference path from either boundary is at least $.58$ on K . The exponential Brownian maximal bound therefore gives

$$p_e = 2e^{-.58^2/(2(.05)^2)}.$$

The independent original-payoff argument bounds the discrepancy between the clipped reference functional and the stopped value by

$$\epsilon_{\text{stop}} = 15p_e + .02\sqrt{\{.22^4 + 6(.22)^2(.0025) + 3(.0025)^2\}p_e}.$$

The second term uses the fourth moment and Cauchy–Schwarz for the terminal quadratic, rather than treating that unbounded random variable as uniformly bounded. There are two policies at two initial states, so the transfer calculation pays four such allowances. Adding these terms proves (48). The central quadrature enclosures are then combined with the directed modulus by interval subtraction and addition. No simulation approximation to an exit time enters this argument. Q.E.D.

X.3. Four-corner optimality for the new actors

The exact lower functional in Lemma 1 depends on feasibility, the budget-preserving shift, and the original utility, not on the algorithm that generated the controls. Each compiled neural policy passes those same conditions. Its four independently evaluated lower endpoints are bounds for the same concave covered functional after uniform quadrature, tail, and stopping allowances. The two inherited dual upper functions are convex in initial state and are valid at their respective

prices. Convexity of V in k makes their chord a valid optimal upper comparison at $k = 2$. The argument of Proposition 18 therefore bounds regret over all of K by the largest of the four new corner gaps. This application does not interpolate raw Monte Carlo values and does not infer concavity of an arbitrarily fitted neural critic.

X.4. A constructive continuation trace

PROOF OF PROPOSITION 13: Fix the stored no-transfer time controls. The wealth solution is deterministic, affine in initial wealth, and continuous in that initial condition. For every $x \in [1.25, 2)$, its drift satisfies $.02X_s - c_s < 0$ and its terminal wealth exceeds .5 by the directed budget check. Thus it does not hit a wealth boundary. Preference exit depends on u , the Brownian motion, and the same fixed θ , but not on initial wealth. The stopping time is therefore common as $x \uparrow 2$. On this range the running payoff is bounded before stopping, and the wealth component of settlement is bounded and continuous. Dominated convergence proves existence of the interior limit and identifies its payoff by inserting the limiting wealth path into the same preference-stopped expectation.

The independent quadrature evaluates the unclipped Gaussian representation with its normal-tail and stopping allowances. Those allowances continue to hold for this fixed control context and the limiting wealth path: consumption and preference ranges are unchanged, wealth remains in $[\cdot.5, 2]$, and the sole premature exit is in preference. The evaluator's no-transfer input adapter checks these hypotheses; it does not invoke the usual K -dependent wealth shift at $x = 2$.

For uniformity in u , consider the covered reference payoff minus $g(0, u, 2)$. Discounting leaves a positive terminal-minus-initial quadratic curvature of $.04(1 - e^{-.04})$; those quadratic terms do not cancel. The running utility supplies a larger negative curvature. Indeed, on the covered region $r = u - 1 \leq 1.8$, $c \leq .8$, so

$$-f_{uu} = \frac{e^{-r \log c}}{r^3} \{(r \log c + 1)^2 + 1\} \geq \frac{125}{729}.$$

The covered normal mass is at least $1 - 2e^{-32}$, and $e^{-.04s} \geq e^{-.04}$. Thus the second derivative of the covered payoff minus initial settlement is at most

$$-\frac{125}{729}e^{-.04}(1 - 2e^{-32}) + .04(1 - e^{-.04}) < 0.$$

Expenditure and terminal log wealth are independent of u . Subtracting the common tail, quadrature, and stopping allowances preserves this strict concavity. Its minimum on $[1.98, 2.02]$ is bounded below by the smaller endpoint enclosure. Directed evaluation of both endpoints for all five fixed policies gives the stated minimum; the additional center calculation is a diagnostic rather than the basis of uniformity.

The process defined by immediate settlement when initialized exactly at $x = 2$ is a different initial-value convention. The proposition concerns only the interior limit and makes no change to that boundary convention. Q.E.D.

X.5. Budget-feasible neural compensation

PROOF OF PROPOSITION 12: Let $Q_r(s) = \int_0^s e^{-.02v} dv$. For the paired original and compensated plans the difference in wealth is

$$\tilde{X}_s - X_s = e^{.02s} \delta \left(1 - \frac{Q_r(s)}{Q_r(1)} \right) \geq 0, \quad \tilde{X}_1 - X_1 = 0.$$

The independently checked consumption ranges lie inside $[\cdot 7, \cdot 8]$ even after both the K shift and the compensating increment. Both wealth paths are strictly decreasing and remain above $\cdot 5$; the compensated initial wealth is below 2. Neither can exit in wealth. Their preference paths, hence preference exit times, coincide. Logarithmic wealth settlement cannot fall at that time. Preference expenditure and preference settlement are unchanged.

Before stopping, $f_c = c^{-u} \geq (\cdot 8)^{-1.2}$. Hence the running utility gain is at least $(\delta/Q_r)(\cdot 8)^{-1.2}$ per unit of discounted time. Since the probability of a premature preference exit is at most p_e , the expected discounted time before stopping is at least $C_\rho(1)(1 - p_e)$. The directed lower bound on the gain is larger than $\cdot 0119058$, whereas every new rectangular regret upper bound is at most 0.011816625 . Subtracting regret proves the last inequality in the proposition. This construction changes the feasible consumption policy with wealth; it is not a claim about the minimum wealth increment for a fixed policy. *Q.E.D.*

X.6. Learned-gradient work

PROOF OF THEOREM 15: For fixed x , let $g = \nabla F_x(U)$ and $\eta = 2/(m + L)$. The integral Hessian identity gives

$$\nabla F_x(U - \eta g) = \left[I - \eta \int_0^1 \nabla^2 F_x(U - s\eta g) ds \right] g.$$

The integral matrix is symmetric with spectrum in $[m, L]$. The operator norm of the bracket is at most $(L - m)/(L + m) = 145/209$. Iteration and the strong-convexity bound $F_x(U) - F_x^* \leq \|\nabla F_x(U)\|^2/(2m)$ prove (77). In particular the initial gradient, not merely bounded network outputs, determines this guarantee.

For the state extension, write the stacked inventory trajectory as $AU + Sx$ plus its fixed offset. The model's geometric propagation gives $\|A\|_2 \leq 5/2$ and $\|S\|_2 \leq 3/4$. The nonlinear inventory potential has Hessian norm at most $29/10$; the quadratic control cost has no state cross derivative. Consequently

$$\|D_x \nabla_U F_x(U)\|_2 \leq (5/2)(29/10)(3/4) = 87/16.$$

For any covering center x_i ,

$$\begin{aligned} \|\nabla F_x(f_\theta(x))\|_2 &\leq \|\nabla F_{x_i}(f_\theta(x_i))\|_2 + L\|f_\theta(x) - f_\theta(x_i)\|_2 + (87/16)\|x - x_i\|_2 \\ &\leq G_\theta(x_i) + (LK_\theta + 87/16)h. \end{aligned}$$

A tanh activation is 1-Lipschitz. Multiplying enclosing matrix-norm bounds, including the output scale, gives a valid K_θ ; Frobenius norms can replace spectral norms conservatively. This statement requires a genuine cover of the asserted set. A finite query maximum instead corresponds to $h = 0$ on that finite set alone. *Q.E.D.*

APPENDIX Y: DATED HISTORICAL THEORY AND NUMERICAL DEVELOPMENT

The following material is retained from R14–R22. Each outcome retains its original policy, model, domain, and work budget. Historical descriptions of then-current evidence do not replace the present abstract and results. In particular, the R21 end-to-end baseline is not the later crossed representation experiment, and a re-budgeted wealth increment is not an unchanged-map welfare estimate.

*Y.1. Crossed evidence, active rejection, and the current-state witness**Y.1.1. All registered configurations*

Table Y.I reports all twelve four-expert configurations, comprising 48 independently checked final proposals. The initial policy is matched within every vertex block; corner initialization is independent across the four vertices. A rejected proposal is represented by its retained incumbent, not silently omitted. Bounds in tables are rounded outward.

TABLE Y.I
MATCHED-INITIAL-POLICY CROSSED EXPERIMENT: COMPLETE FOUR-EXPERT SYSTEMS

Ensemble	Configuration	Regret upper	Gain lower	Gradients	Gen. sec.
22000	Neural–Adam	0.00300464	0.02611105	1600	6.075
22000	Neural–L-BFGS-B	0.03770374	-0.00408098	1272	4.052
22000	Direct–Adam	0.01203417	0.01857673	1600	4.695
22000	Direct–L-BFGS-B	0.00283871	0.02626800	430	1.261
22100	Neural–Adam	0.00300355	0.02664950	1600	5.195
22100	Neural–L-BFGS-B	0.00700843	0.02675515	1126	3.585
22100	Direct–Adam	0.01209862	0.01903333	1600	4.691
22100	Direct–L-BFGS-B	0.00283871	0.02675520	434	1.277
22200	Neural–Adam	0.00301293	0.02666651	1600	5.163
22200	Neural–L-BFGS-B	0.00283910	0.02677308	1492	4.719
22200	Direct–Adam	0.01192986	0.01897113	1600	4.651
22200	Direct–L-BFGS-B	0.00283871	0.02677312	433	1.260

Uniform bounds concern K at $t = 0, k = 2$. Gain is relative to the initial mixture. A negative gain lower bound makes no positive-improvement claim. Generation excludes checking; full costs are in the supplement.

Every neural–Adam ensemble has uniform regret below 0.003013 and a certified initial-to-final mixture payoff gain of at least 0.026111. The maximum upper bound on the true regret ratio is 0.1031964. These conclusions hold throughout K despite unequal initial slopes and preference drifts. The pair-moment theorem, not corner improvement alone, establishes them. The more conservative range-based calculation also certifies positive improvement for these three ensembles and remains in the replication record.

Table Y.II applies the same theorem to the final neural–Adam and direct–Adam mixtures. The neural parameterization has a strictly positive uniform payoff advantage in each ensemble under the common 400-gradient-call budget. Direct Adam does not attain a .01 uniform bound at this work level. However, direct L-BFGS-B attains approximately .00283871 in all three ensembles, with fewer objective calls than the neural configurations. The experiment therefore establishes an optimizer-conditional representation advantage, not overall neural numerical superiority. Neural L-BFGS-B also displays substantially greater between-ensemble variability than direct L-BFGS-B.

TABLE Y.II
NEURAL-ADAM VERSUS DIRECT-ADAM: UNIFORM TRUE-PAYOFF COMPARISON

Ensemble	Vertex gain lower	Direct Jensen upper	Uniform gain lower
22000	0.00747909	0.00436669	0.00311240
22100	0.00749716	0.00437404	0.00312312
22200	0.00748655	0.00435908	0.00312747

The cost ledger distinguishes generation from checking and identifies the shared initial-policy check. Wall times are measurements in the recorded environment, not hardware-independent complexity constants. Objective calls include the complete proposal calculation; internal budget-root iterations are not additional gradient calls. The original dual upper library is still necessary for regret, remains non-neural, and retains its separately reported construction cost. No such upper is needed for the strict neural-Adam versus direct-Adam payoff comparison.

Y.1.2. A numerical financing failure and its repair

In the retained local preflight of the registered experiment, the neural L-BFGS-B proposal for ensemble 22000 at (1.98, 1.26) terminates numerically but fails the independent reserve check. Its reserve enclosure is approximately $[.4774000812, .4774000935]$, below the required .5. The fixed proposal root bracket $[-10, 10]$ does not contain the budget root for those raw coefficients. The correct decision is rejection. The frozen failing parameters, actor, and check are retained as a regression fixture; any corresponding rejection in the reference execution also retains its incumbent in Table Y.I. Optimizer convergence is not a financing theorem.

A subsequent, separately specified diagnostic centers the raw intercepts, expands the budget-root bracket until it encloses the quadrature budget, and performs 80 bisections before implicit differentiation. The independent checker is unchanged. A common intercept shift of 1,000 leaves the delivered policy invariant to within 1.084×10^{-13} in the numerical diagnostic. Re-budgeting the frozen failed network restores a valid reserve but does not make that network a competitive optimum. Rerunning the four configurations in the selected block gives the complete diagnostic results in the supplement. They are not substituted for the prospective primary outcomes. In particular, the repaired neural L-BFGS-B run remains less accurate than direct L-BFGS-B.

Y.1.3. The gate is exercised, and rollback alone is not recovery

The prospective stress test uses seed 22300, Adam learning rate .5, and eight successive blocks of 25 gradient evaluations at the first vertex. It accepts 4 blocks and rejects 4. Each rejection restores network parameters, optimizer state, and the Python, NumPy, and PyTorch random-number states exactly. This validates the restoration path but also exposes a limitation: with the same deterministic proposal rule, rejected candidates can repeat after rollback. Repeated rejections are not independent adverse draws.

The follow-up recovery test uses the root-safe objective and a separate supervisory controller. After a rejected block, exact restoration is checked first; only then does the controller halve the learning rate. The controller itself is not rolled back. This execution accepts 6 blocks and rejects 2, with accepted improvement after the first rejection. All proposal, incumbent, learning-rate, and state-hash records are retained. Explicit no-op and state-corruption tests are listed as fault-injection regressions, not empirical optimizer outcomes. These tests demonstrate an active gate and one recovery mechanism; they do not establish broad convergence or architecture robustness.

Y.1.4. Current-state execution on the full domain

The global experiment does not use the four-expert decoder. It refines the accessible-face architecture whose actor and critic both take (t, u, x) , starting from the recorded R17 seed17107 checkpoint. Joint L-BFGS-B uses 400 gradient evaluations on the complete $4 \times 16 \times 16$ differentiable interval-surrogate cover. A separate MPFR checker evaluates every original-domain cell, with no skipped or failed cells and with the original continuous action comparison. The current-state dependence of the actor is also checked by diagnostic derivatives; those pointwise derivatives are not offered as the coverage proof.

The initial full-domain bound is 7.2783190635 and the new bound is 7.241462444. The discounted suffix calculation in (38) extends that bound to every starting time, rather than using an unnecessarily undiscounted sum. The run is an explicitly charged warm-start refinement, not a new from-scratch success. Because the witness and actor change together and no separate sharp policy-payoff interval is obtained for this full-state actor, the bound decrease is not claimed to prove actual policy improvement. The original .01 full-domain objective remains unmet; regional bounds in Tables Y.I–Y.II do not replace it.

Y.1.5. Deployment and economic interpretation

The online calculation for a regional mixture is not merely a network forward pass. The network is compiled to slab coefficients; conditional prices and their factor derivatives require integration. A separate ordinary-float timing diagnostic evaluates the price and hedge with eight time nodes and sixteen Gaussian nodes per remaining slab. Four experts use at most 8,192 quadrature nodes per query and 1,536 bytes for their coefficient arrays, excluding reserves, indexing, and language-runtime overhead. The measured median time is 0.160 milliseconds. Duplication to 8, 16, and 32 experts tests computational scaling only; it does not enlarge the certified economic domain. These timings are not rigorous online approximation-error certificates.

We use *sufficient re-budgeted wealth increment* for the earlier .002 result. The counterfactual increases initial wealth, retains learned slopes and preference-drift shapes, recomputes the consumption budget offset, and re-verifies the financed stream. It does not keep the entire original policy map fixed and is not a minimum compensating variation or an estimated structural welfare parameter. The earlier price-transport, wealth, and synchronized-initialization calculations remain a posteriori properties of frozen policies; only the previously registered training design has confirmatory status.

Y.2. Earlier regional results, retained with their original scopes

The next results are the frozen R20/R21 analyses. Their end-to-end baseline is not the new crossed representation experiment. The precise re-budgeted interpretation above governs current welfare terminology.

Y.3. Executed neural accuracy and matched local comparisons

The confirmatory design fixes seeds 20100–20104, the four vertices of K , Adam step size .005, and retained budgets 0, 100, 400, 1000. The exploratory seed 20000 is excluded. The independent checker encloses physical-measure utility, preference costs, settlement, martingale-measure consumption prices, Gaussian and stopping tails, and the global control bounds. Eighty neural expert checkpoints were originally checked; all twenty final experts were also checked by a separate MPFR arithmetic implementation. R21 replays the frozen-policy checks and uses the

TABLE Y.III

FOUR-EXPERT NEURAL POLICIES: BOUNDS UNIFORM ON K , AT $t = 0, k = 2$. FINAL BOUNDS USE THE HULL OF BOTH ARITHMETIC IMPLEMENTATIONS.

Seed	0 updates	100	400	1,000	Gain over initial
20100	0.03819183	0.00363672	0.00299180	0.00290848	0.01840213
20101	0.03760120	0.00417052	0.00301777	0.00290261	0.01785789
20102	0.03732944	0.00384199	0.00301067	0.00289514	0.01762319
20103	0.03755641	0.00385873	0.00300813	0.00289022	0.01783956
20104	0.03778334	0.00364572	0.00299527	0.00289318	0.01803023

hull of the available enclosures for the reported final bounds. The shared mathematics is not thereby an independent formal proof.

The initial policies miss .01. Every seed reaches both .01 and .005 at the first retained noninitial budget of 100 updates. All five final bounds are below .003. None reaches the additional .002 reporting threshold. The last column is an actual-payoff improvement bound, obtained from Theorem 7; it is not the change in an optimality upper bound. The final arithmetic hull gives bounds between approximately 0.00289021 and 0.00290848. Table entries for error bounds are rounded upward, and gain lower bounds downward.

Y.3.1. The neural and classical cost frontier

The conventional stochastic comparator directly optimizes sixteen triples of coefficients in (23), with no hidden neural layers. It uses L-BFGS-B, the same conditional-price decoder, the same four initial states, and precisely the same payoff and dual checks. The retained budgets are 0, 10, 50, 200, with convergence recorded when earlier. This is a matched *local* continuous-time accuracy comparison, not a surrogate comparison on a different discrete economy.

The direct stochastic method reaches .005 at ten retained iterations for each expert and has final K -uniform bound 0.00283871. At the first .005 hit, neural seed 20100 uses 1.107 seconds for proposal generation and 15.036 seconds for the accumulated binary interval checks, or 16.143 seconds in total after outward rounding. The direct method uses .153 and 15.360 seconds, respectively, or 15.512 seconds. All five seed-specific ledgers, target hits and misses, and final wall times appear in the supplement. Different optimizer iterations are not equated: the comparison is made at the common checked tolerance.

These are recorded costs conditional on access to the common optimal upper. A cold-start ledger additionally charges the full 415.789 seconds of the inherited twenty-node dual-library construction, or explicitly declares its reuse. The final twenty-expert MPFR audit and R21 replays are additional publication-validation costs, not silently free checks. Recorded neural peak process memory is retained. The R20 direct-stochastic run did not record peak RSS; its 192 scalar coefficients must not be substituted for process memory. Consequently, the data support a local accuracy frontier and an explicit cost ledger, but neither a complete global MC/SL frontier nor a neural memory or speed advantage.

Y.3.2. A policy-sensitive stochastic improvement

Three separately executed deterministic SLSQP transcriptions, with 16, 32, and 64 time slabs, receive their own supporting affine payoff upper bounds on K . These are upper bounds for the named policies, not for the optimal value. The neural mixtures' lower bounds exceed each such affine upper by at least 0.00455889 throughout K . This comparison orders actual stopped payoffs without changing the evaluator to favor the neural policy. It establishes an improvement over these three deterministic policies; the sharper and faster direct stochastic comparator remains visible and precludes a blanket neural-superiority claim.

Y.3.3. Budget-feasible welfare compensation for the learned policies

For every final neural expert, retain the learned slopes and preference drifts and raise only its consumption intercept by a common budget offset. Finance the resulting consumption stream from initial wealth $x_i + .002$, and verify the exact residual reserve again using (24). The twenty compensated-policy lower bounds exceed the original, uncompensated optimal-value upper bounds by at least 0.00004475 after continuum transfer. Thus each learned mixture, provided with an additional .002 of initial wealth, achieves at least $V_2^*(u, x)$ for every $(u, x) \in K$. The a sufficient re-budgeted wealth increment is .16 percent at reference wealth 1.25, and less than .162 percent throughout K . It is neither the minimum compensating variation nor an external unfinanced transfer. Unlike the historical .0078 result for a deterministic library, this statement concerns the new learned stochastic policies.

Y.3.4. Relation to the historical whole-domain experiments

The R18 complete-domain feedback results remain 7.278–7.344 against the unchanged .01 target. Their final policy-side residual component is zero for all ten accessibility-aware seeds; the remaining bound is the positive comparison component. The all-seed actor–critic factorial, signed decompositions and trace-floor trajectories are retained below and in the supplement. A smaller historical bound cannot identify actor progress.

The new evaluator does not apply another interval wrapper to those failed actors or critics. It evaluates newly trained stochastic consumption policies and uses a separate localized unrestricted dual upper. Its success on K is therefore compatible with the historical fixed-critic floor above 6.27. It does not prove that the historical critic has learned the correct continuation trace, or supply a sharp supersolution on the whole cylinder. The retained one-sided continuation targets provide constructive information for that separate global problem.

The nonlinear high-dimensional control material is likewise preserved with its actual attribution. The initialization-independent strongly-convex theorem is not evidence that training is necessary. The subsequent learned-gradient theorem and 32-query matched-tolerance experiment do identify a limited reduction in online gradient work, including 31 of 32 paired queries at dimension 128 and tolerance .01. They do not establish a same-hardware amortized speedup, performance for all cube states from those 32 queries, or a broad stochastic high-dimensional advantage.

Y.4. State-feedback realization and the representation ablation

The same-family comparison is a test of the neural parameterization itself, not merely another external baseline. To interpret that test correctly, we specify both the information observed by the network and the state feedback implemented by its decoder.

The network input is only the normalized time-slab coordinate. Neither the current preference nor the traded factor is a network input. The factor enters the prescribed logistic consumption rule; its self-financing portfolio is supplied by conditional pricing. Within a seed, the random-number generator is reset before each of the four experts, and their initial parameter dictionaries are exactly identical. Different seeds give five different initial dictionaries. The four objectives and budget offsets differ. Thus there are five independent seed replicates, not twenty independently initialized experts. The exact equality of the initial slopes and effort vectors used in the Jensen proof is consistent with this design.

PROPOSITION 10—Restricted Markov realization: *Fix an expert or fixed initial-state mixture, and denote its consumption, reserve, and conditional price by c_λ , R_λ , and F_λ . Put $A_r(t) =$*

$(1 - e^{-r(1-t)})/r$. For every $t < 1$, the map $y \mapsto F_\lambda(t, y)$ is continuously differentiable and strictly decreasing from

$$H_\lambda(t) = R_\lambda e^{-r(1-t)} + .8A_r(t) \quad \text{to} \quad L_\lambda(t) = R_\lambda e^{-r(1-t)} + .5A_r(t).$$

It has an inverse $y_\lambda(t, x)$ on this open wealth interval. Before stopping, the deployed policy consequently has the equivalent feedback representation

$$\hat{c}_\lambda(t, x) = c_\lambda(t, y_\lambda(t, x)), \quad \hat{p}_\lambda(t, x) = -1.5 y_\lambda(t, x) \frac{\partial_y F_\lambda(t, y_\lambda(t, x))}{x}, \quad \hat{\theta}_\lambda(t) = \theta_\lambda(t).$$

This is a restricted wealth-feedback policy indexed by its fixed initial-state context. It is not a single trained network on all current states (t, u, x) .

PROOF: Every slab slope is strictly positive. For any positive future interval, logistic consumption is strictly decreasing in its factor argument and has limits .8 and .5. Conditional factors multiply the present factor by a strictly positive lognormal random variable. Differentiation under the bounded integrand gives $\partial_y F_i < 0$ for $t < 1$; the derivative is continuous at every positive y . Dominated convergence gives the two displayed limits. Finite convex combinations preserve strict monotonicity and these endpoint formulas, since the mixture reserve is $R_\lambda = \sum_i \lambda_i R_i$. The inverse-function theorem applies. Substitution of $X_t = F_\lambda(t, Y_t)$ proves equality of the feedback and subaccount implementations along every pre-stopping path. At $t = 1$ the interval collapses to the reserve and no inverse is required. The portfolio formula agrees with the wealth-weighted mixture in (28). Q.E.D.

The proposition identifies real feedback in the delivered controls without attributing it to an unobserved neural input. It does not execute a full-state neural Bellman solve, remove the restricted policy class, or add reactions to current preference shocks. The unrestricted value in the optimality bounds is nevertheless the original economy's value, not the optimum of that restricted class.

TABLE Y.IV

REPRESENTATION ABLATION AT THE FINAL RETAINED WORK LEVEL. BOTH METHODS USE THE SAME STOCHASTIC LOGISTIC FAMILY AND UNRESTRICTED UPPER. THE FINAL COLUMN IS A CERTIFIED PAYOFF ORDERING, NOT MERELY A COMPARISON OF REGRET BOUNDS.

u	x	Best neural bound	Direct bound	Direct payoff gain $\geq$
1.98	1.24	0.00261425	0.00258714	0.00002670
1.98	1.26	0.00289022	0.00283871	0.00005111
2.02	1.24	0.00268964	0.00266465	0.00002459
2.02	1.26	0.00260024	0.00255442	0.00004542

The direct parameterization has a smaller final certified regret at every vertex than every neural seed. A further interval subtraction, using the hulls of the independently recomputed payoff intervals, establishes a strict *actual payoff* advantage at those vertices; the last column reports its lower bound. This is adverse evidence for the hidden-layer parameterization in the tested regime. The vertex ordering is not asserted for every interior mixture: different mixtures may have different Jensen gains. The direct mixture does have the sharper proved uniform regret bound shown earlier.

TABLE Y.V

UNPOOLED FOUR-EXPERT COST AT THE FINAL RETAINED WORK LEVEL (SECONDS). THE FINAL-CHECK COLUMN IS A SUBSET OF ALL CHECKPOINT CHECKING, NOT AN ADDITIONAL CHARGE. THE SHARED DUAL AND MPFR COSTS REMAIN SEPARATE.

Method	Generation	All checking	Final checking
Seed 20100	10.631	30.002	7.509
Seed 20101	10.730	30.197	7.610
Seed 20102	10.788	30.245	7.597
Seed 20103	10.627	30.174	7.577
Seed 20104	10.638	30.242	7.560
Direct	1.143	30.597	7.598

Pooling the common verifier cost would obscure the generation result. The direct final four-expert solve takes approximately 1.14 seconds of generation, against 10.63–10.79 seconds for a neural seed. Both are charged for all retained checkpoints. Neural training is the reported generation work, not an omitted extra offline stage. The shared dual construction, final MPFR pass, and R21 frozen-policy replay are separately charged as explained above. Missing direct process peak memory remains unreported, rather than being imputed from parameter storage.

The full-state target is unchanged. The best inherited complete-domain feedback certificate remains 7.278319, and the earlier R19 bound over K is .011816625 despite its sub-.01 reference-state result. The new sub-.003 uniform result concerns the explicitly defined stochastic mixture and the new proof, not a revision of either historical observation. The current experiment therefore establishes useful neural-generated policy accuracy and policy improvement on K , but does not establish a neural representation advantage or global full-state NBO accuracy.

Y.5. Retained solver theory and experimental development

The following sections preserve the substantive R14–R19 mathematical and numerical development. Their experiment-specific bounds concern the objects named in those sections; they are not relabeled as R21 training. In particular, deterministic price libraries, historical full-state feedback networks, and the new stochastic neural mixtures remain separate policies. The new uniform results above do not revise the old observations by deletion.

Y.6. Policy-value-separated neural updates

For a policy π and initial state z , write $L_\pi(z) \leq J_\pi(z) \leq U_\pi(z)$ for independently established payoff enclosures. Write $\bar{V}(z) \geq V(z)$ for an optimal upper comparison. The two upper bounds have different meanings: U_π evaluates one policy, whereas $\bar{V}$ ranges over every admissible policy. A better $\bar{V}$ can sharpen a certificate without improving π .

PROPOSITION 11—Separated acceptance and observable progress: *Let the preceding bounds hold on a set K . A proposal π' satisfying*

$$\inf_{z \in K} \{L_{\pi'}(z) - U_\pi(z)\} > 0 \quad (43)$$

strictly improves the true payoff at every $z \in K$. More strongly, if, for some $0 < \kappa < 1$,

$$L_{\pi'} - U_\pi \geq \kappa(\bar{V} - L_\pi) \quad \text{on } K, \quad (44)$$

then $V - J_{\pi'} \leq (1 - \kappa)(V - J_{\pi})$ on K . If instead a verified gain $J_{\pi'} - J_{\pi} \geq d > 0$ and final regret bound $V - J_{\pi'} \leq B$ hold uniformly, then

$$\frac{V - J_{\pi'}}{V - J_{\pi}} \leq \frac{B}{B + d}. \quad (45)$$

All inequalities compare economic payoffs. An increase in L_{π} alone does not imply improvement: both the policy and the width of its evaluation interval can change. Equation (44) is a checkable sufficient progress condition, not an assumption that Adam always produces such a proposal. Its numerical use below is an a posteriori certificate for retained transitions, not an a priori optimization rate.

The evaluator need not be neural. For example, a smooth fixed-policy evaluation function q obeys the stopped identity

$$J_{\pi}(z) = q(z) + \mathbb{E}_z^{\pi} \int_0^{\tau} e^{-\rho s} \mathcal{R}_{\pi} q(s, Z_s) ds + \mathbb{E}_z^{\pi} [e^{-\rho \tau} (g - q)(\tau, Z_{\tau})].$$

Separate upper and lower enclosures of these two terms yield L_{π}, U_{π} . A distinct supersolution, relaxation, or monotone comparison can produce $\bar{V}$. These roles cannot be interchanged merely because both calculations involve a value function.

Y.6.1. The executed neural policy block

The new actor has two hidden tanh layers of width 16 and 338 trainable parameters. It maps the midpoints of 16 time slabs to two outputs N_c, N_{θ} . Let

$$w_j = \int_{j/16}^{(j+1)/16} e^{-\rho s} ds, \quad Q_r = \sum_j w_j, \quad b = 1.25 - .5e^{-\rho \cdot 0.25} - 10^{-7}.$$

At the reference wealth its mathematical controls are

$$c_j = \frac{b}{Q_r} + .006 \left\{ \tanh N_c(t_j) - \frac{\sum_i w_i \tanh N_c(t_i)}{Q_r} \right\}, \quad (46)$$

$$\theta_j = .1 + .1 \tanh N_{\theta}(t_j), \quad p_j = 0.$$

The exact budget equation implies terminal wealth $.5 + 10^{-7}e^{-\rho \cdot 0.25}$. For an initial wealth $x \in [1.24, 1.26]$, add $(x - 1.25)/Q_r$ to every consumption slab. Terminal wealth is then unchanged. The implementation serializes the compiled binary64 controls, checks their actual budget and action bounds by directed arithmetic, and treats these exact dyadic controls as the delivered policy. It does not assume that floating-point network compilation is exact.

The initial-state map is held fixed after initialization. Thus this experiment is a neural-generated time-control subclass of the original adapted policy set, not the state-feedback architecture of the complete-cover study. The upper bound still concerns the unrestricted original problem. Keeping these two facts together permits a genuine optimality certificate for the delivered neural policy without asserting a whole-domain feedback result.

Seeds 19100–19104, budgets 0, 50, 200, 800, learning rate .003, and the payoff gate were declared before execution. Full-batch proposal quadrature uses 16 time nodes and 40 Gaussian nodes per slab. It evaluates the original CRRA expectation, adjustment cost, and terminal

reward; it does not use an inherited policy or an optimal-value label. This ordinary floating-point objective generates proposals only. At each retained checkpoint, the independent rational-Gauss evaluator, with Cauchy remainders, normal-tail bounds, and an original-stopping correction, establishes the policy-value interval. The final reproduction executes this value gate before the next Adam update; rejected proposals would restore the incumbent network and optimizer state. Every prescribed proposal is retained. All fifteen noninitial proposals pass the strict gate at the reference state.

Y.6.2. Transferring policy improvement to initial states

Separate Lipschitz bounds for two values can overwhelm a small but economically meaningful improvement. We instead control their difference under common shocks. Let a, b denote two feasible slab policies with the same wealth-transfer rule. On a common refinement of their time partitions define

$$d_c = \sup_s |c_s^b - c_s^a|, \quad d_m = \sup_s \left| \int_0^s (\theta_v^b - \theta_v^a) dv \right|, \quad d_T = \left| \int_0^1 (\theta_v^b - \theta_v^a) dv \right|.$$

The differences are computed from the exact stored controls. No bound on the unknown optimal policy is required.

Let H_{uu}, H_{uc}, H_{cc} bound the absolute utility Hessian entries on $[1.78, 2.42] \times [.7, .8]$, and let A_u, A_c bound its first derivatives on $[1.2, 2.8] \times [.7, .8]$. Put

$$\begin{aligned} P_4 &= \frac{e^{-8}}{2\sqrt{2\pi}}, \quad D_4 = \min\{12, 2(A_u d_m + A_c d_c)\}, \\ \Omega(a, b) &= C_\rho(1) \left[.02(H_{uu}d_m + H_{uc}d_c) + \frac{.01}{Q_r}(H_{uc}d_m + H_{cc}d_c) + D_4 P_4 \right] \\ &\quad + .04e^{-\rho}(.02)d_T + 4\epsilon_{\text{stop}}. \end{aligned} \tag{47}$$

Here ϵ_{stop} is the uniform fixed-policy stopping allowance from the original-payoff evaluator. The same allowance is valid for both policies and all initial states in K .

THEOREM 10—Coupled-payoff initial-state transfer: *Suppose both policies satisfy the checked consumption, preference, zero-portfolio, and wealth-feasibility conditions above. For $z_* = (2, 1.25)$ and $z \in K$,*

$$|\{J_b(z) - J_a(z)\} - \{J_b(z_*) - J_a(z_*)\}| \leq \Omega(a, b). \tag{48}$$

Consequently, a directed central value-difference interval $[d_-, d_+]$ gives the uniform interval $[d_- - \Omega, d_+ + \Omega]$ for the true policy gain on K .

The proof uses a central normal event, a Lipschitz tail allowance, and exact cancellation of the initial-wealth dependence of terminal wealth. The constants are evaluated with directed arithmetic, not fitted to observed state samples. The four corner payoff evaluations used for optimality certification are an additional calculation; they are not used as empirical evidence for the policy-improvement modulus.

Y.7. New policy-sensitive evidence in the original economy

Table Y.VI reports every prescribed neural trajectory. At the reference state, the initial bounds are approximately .0392–.0398, and all final bounds are below .01. Unlike the decrease in the old complete-cover certificates, every accepted decrease is accompanied by a disjoint fixed-policy payoff interval in the improving direction. The common optimal upper comparator is consulted only after actor generation; it supplies no training labels.

TABLE Y.VI
NEW NEURAL POLICIES: POLICY-SENSITIVE PROGRESS AND OPTIMAL COMPARISON

Seed	B_0	B_{50}	B_{200}	B_{800}	$\inf_K \Delta J$	B_K
19100	0.039437	0.011268	0.009965	0.009764	0.016679	0.011806
19101	0.039587	0.011453	0.009996	0.009766	0.016759	0.011808
19102	0.039755	0.014144	0.010018	0.009775	0.016830	0.011817
19103	0.039190	0.012475	0.010012	0.009757	0.016513	0.011799
19104	0.039502	0.012358	0.009973	0.009765	0.016758	0.011807

B_k is the original-economy regret upper bound at $(0, 2, 1.25)$, adjustment price 2, after k Adam updates. The last two columns are uniform over $K = [1.98, 2.02] \times [1.24, 1.26]$, at time zero. All upper bounds are rounded upward and gain lower bounds downward. These policies are neural-generated time controls with an initial-wealth budget map; they are not the full-state feedback benchmark.

The coupled-payoff theorem certifies initial-to-final payoff improvement of at least 0.016513 throughout K , for every seed. The four-vertex argument of Proposition 18, now applied to the newly generated neural policies at $k = 2$, gives the uniform bound 0.011816625. The covered payoff is concave in the initial state, the dual upper comparison is convex, and the quadrature, tail, and stopping allowances are uniform. The numerical check does not infer a continuum conclusion from sampled values.

These two independently verified quantities also give a policy-learning statement. Equation (45) bounds the final-to-initial true-regret ratio by a number below .42 for every seed and every initial state in K . At the reference state, the direct initial-to-final test (44) passes with $\kappa = .75$. Each of the fifteen successive checkpoint transitions also passes that test with $\kappa = .02$. The rates are a posteriori statements about these actual policies. They neither claim convergence of every random initialization nor make a prediction about an unexecuted number of Adam updates.

Y.7.1. A common accuracy frontier and a policy comparison

Table Y.VII compares freshly generated SLSQP time transcriptions at 16, 32, and 64 slabs with the neural generators at first attainment of the same reference-state accuracy. The original stopped payoff evaluator and the certified optimal upper comparator are common. The neural method is not faster in this experiment. The comparison establishes actual neural policy improvement and sharp reference-state accuracy, not a speed advantage over direct optimization in a small time-control class.

A further use of Theorem 10 compares each final neural policy directly with the fresh 16-slab SLSQP policy. Uniformly on K , the latter’s payoff exceeds the neural payoff by at most 0.000206240. This bound does not suppose that the SLSQP policy is optimal. It supplies a trusted, policy-specific comparison on a nontrivial state set, rather than comparing descriptive action ranges or two loose regret bounds.

TABLE Y.VII

MATCHED REFERENCE-STATE ACCURACY, WITH A SHARED INDEPENDENTLY CERTIFIED UPPER COMPARATOR

Generator	Updates	Gen. s	Eval. s	Sum s	Bound
19100	200	0.307	8.144	8.451	0.009965
19101	200	0.309	7.916	8.225	0.009996
19102	800	1.215	10.534	11.749	0.009775
19103	800	1.210	10.517	11.727	0.009757
19104	200	0.310	7.892	8.203	0.009973
SLSQP-16	–	0.021	2.635	2.656	0.009722
SLSQP-32	–	0.032	5.266	5.298	0.009704
SLSQP-64	–	0.083	10.495	10.578	0.009699

Neural evaluation cost includes all retained checkpoint checks through first attainment of .01; SLSQP has one final check. Generation includes neural compilation. Costs are conditional on the shared fixed upper comparator. Its historical fitting and verification costs are reported separately in the ledger, not represented as newly executed or silently free. All candidates target the original stopped payoff. SLSQP time transcription is not a Markov-chain or semi-Lagrangian solver.

The six retained Markov-chain and semi-Lagrangian feedback candidates remain separate objects. Their bounds do not become sharp merely because a time transcription reaches the reference-state target. The supplement reports the positive and negative contributions for all six. At the finest grids the negative policy-side term becomes positive while the positive witness term changes comparatively little. This identifies the component responsible for the reported bound's deterioration; it does not by itself identify whether interpolation, witness fitting, or true policy quality is the ultimate cause.

Y.7.2. Welfare compensation for the neural policies

The new neural certificate has its own budget-feasible economic interpretation. Increase initial wealth by $\delta = .0092$ and add δ/Q_r to every consumption slab of the neural policy. Preference adjustment and portfolio controls are unchanged. Directed checks over all five policies and all initial wealths in K establish the original consumption cap, identical terminal wealth, and no wealth exit. Intermediate wealth is weakly higher, so the preference-exit time is common and the settlement contribution cannot fall.

PROPOSITION 12—Sufficient neural-policy wealth compensation: *For every reported final neural policy and $(u, x) \in K$, the preceding feasible adjustment satisfies*

$$J_k(0, u, x + \delta; \tilde{\pi}) - J_k(0, u, x; \pi) \geq \frac{\delta}{Q_r} (.8)^{-1.2} C_\rho(1)(1 - p_e) > .0119058,$$

where $k = 2$ and $p_e = 2 \exp\{-.58^2/(2(.05)^2)\}$. Hence $J_k(0, u, x + \delta; \tilde{\pi}) \geq V_k(0, u, x)$.

The increment is 0.736 percent of reference wealth 1.25. It is sufficient compensation with an explicitly changed feasible consumption policy, not the exact minimum compensating variation of an unchanged policy. It applies to the new neural-generated policies on K , not to the old full-domain feedback certificate or solely to the non-neural price library.

Y.8. Continuation targets and attribution

A fixed-policy continuation value gives a constructive target even when the optimal value is unknown. Fix one newly trained time-control vector, without the initial-wealth transfer used on

K. For $x \in [1.25, 2)$, its deterministic wealth remains above .5, and its wealth drift is strictly negative. The upper face is therefore inaccessible before the preference exit or terminal date. Let q_π denote its true stopped policy value with this fixed control context.

PROPOSITION 13—Policy-induced continuation trace: *The one-sided limit $q_\pi(0, u, 2-)$ exists for $u \in [1.98, 2.02]$. The original-utility Gaussian evaluator, with the same uniform stopping allowance and no wealth transfer, encloses this limit. Its certified excess over $g(0, u, 2)$ is at least 6.708771 uniformly on that preference segment for all five new policies.*

The proof does not redefine the absorbing boundary value. A process initialized exactly on the stopping boundary has immediate payoff g ; the displayed object is the limit from interior initial wealth. This distinction is precisely why forcing a critic to match settlement on this face is inconsistent. The true q_π has the fixed-policy Feynman–Kac characterization on each time slab. It is not an optimal upper witness, and the numerical execution here certifies its continuation trace rather than a global Bellman residual for a learned approximation to q_π .

A continuation-aware critic update can now impose a verified inequality against a feasible-policy continuation target, rather than merely omitting an equality constraint. For an upper witness w , the appropriate necessary requirement is $w(0, u, 2-) \geq q_\pi(0, u, 2-)$, up to an explicitly allocated comparison error. The target is generated from the new policy and original payoff; no optimal-value labels are supplied. Joint control of this trace and the interior positive Bellman residual remains necessary.

The latter qualification is tested directly. In a predeclared intervention, only the final additive bias of each of five old critics is increased until directed arithmetic proves the earlier feasible-policy trace requirement at $(0, 2, 2)$. All actors remain fixed. Every one of the 1,024 boxes is recomputed. Table Y.VIII shows that the resulting total bounds are about 22.7–23.4. A scalar lift removes the particular pointwise obstruction but worsens the global certificate. It is neither selected as an improved witness nor used to claim full-domain accuracy.

TABLE Y.VIII
A CONTINUATION-TRACE CONSTRAINT IS NECESSARY, NOT SUFFICIENT

Seed	Bias lift	Trace excess	B^+	B^-
17100	8.69086	6.372265	16.464523	6.295713
17101	9.28629	6.372265	16.469820	6.673984
17102	8.66455	6.372265	16.468794	6.199523
17103	9.30010	6.372265	16.470158	6.888356
17104	8.70903	6.372265	16.453049	6.300897

Only the final critic bias changes; actors and all other weights remain fixed. Each lifted critic satisfies the directed point-trace constraint. The recomputed global bound worsens, so the intervention is not described as a successful global witness solve.

Y.8.1. Attribution in every retained feedback seed

Table Y.IX supplies the complete actor–critic interchange, including the signed policy component, for all ten old feedback seeds. Pairing the initial actor with the final critic reproduces the final certificate in every row. The final negative component is zero; the entire remaining bound is the positive optimal-comparison component. Thus the earlier result establishes witness improvement but does not identify actor improvement. The new fixed-policy comparisons in Section Y.7 address a different, policy-sensitive question with newly generated actors.

TABLE Y.IX
FULL-DOMAIN ACTOR–CRITIC INTERCHANGE: ALL TEN FROZEN FEEDBACK RUNS

Seed	A_0, C_0	A_1, C_0	A_0, C_1	A_1, C_1	Final B^-
17100	26.300514	23.718686	7.339624	7.339624	0.0
17101	26.499830	24.195287	7.281385	7.281385	0.0
17102	26.326374	23.626229	7.343312	7.343312	0.0
17103	26.511217	24.396419	7.280303	7.280303	0.0
17104	26.367488	23.701299	7.336433	7.336433	0.0
17105	26.468295	24.307436	7.280018	7.280018	0.0
17106	26.352908	23.639651	7.343665	7.343665	0.0
17107	26.666728	24.502208	7.278319	7.278319	0.0
17108	26.339971	23.757239	7.338341	7.338341	0.0
17109	26.527472	24.463132	7.279140	7.279140	0.0

Entries are complete time-zero regret bounds on the original spatial domain, not policy values. The final positive component equals the total in every row; the final negative component is zero. This table audits the earlier full-state experiment and is not relabeled as new training.

The supplement also reports verified upper-face trace excess, the implied fixed-witness floor, and the positive component at all 30 accessible checkpoints. The matched all-face experiment is retained unchanged. Its width-dependent ranking is evidence about finite-budget performance, not a contradiction of the approximation-space consistency theorem and not evidence for a uniform empirical advantage of the accessibility-aware architecture.

Y.9. Stopping accessibility and the neural approximation space

A stopping contract need not be the limit of continuation value at every geometrical boundary point. This distinction matters for a degenerate controlled diffusion: an investor can eliminate portfolio risk and move away from a wealth boundary. It therefore cannot be decided by assigning a larger penalty to every boundary observation in a neural loss.

PROPOSITION 14—Trace consistency in the unchanged economy: *For the economy in Section 2, at $k = 2$,*

$$\liminf_{x \uparrow 2} V_2(0, 2, x) - g(0, 2, 2) \geq 6.45 - .1 \log 2 - 25.1e^{-8} > 6.37. \quad (49)$$

Consequently, a sequence of witnesses continuous on the closed cylinder, each equal to g on its entire stopping boundary, cannot have its uniform positive Bellman residual converge to zero.

The proposition concerns the consistency of an approximation space, not the existence of a solution to the economic decision problem. The original utility, adapted controls, horizon, and liquidation fee are unchanged. Its implication is constructive: enforce the settlement where the proposed policy can stop, and use the correct one-sided comparison where it cannot. In particular, imposing the upper-wealth settlement as a hard equality would exclude the relevant continuation limit. The proof uses the elementary feasible policy $(c, \theta, p) = (.8, 0, 0)$ and does not use an estimated optimal value.

Y9.1. A signed, facewise comparison

Partition the stopping boundary into measurable faces F_i , assigning intersections to one face. Let

$$\omega_i^*(t, z) \geq \sup_a \mathbb{E}_{t,z}[e^{-\rho(\tau-t)} 1_{F_i}(\tau, Z_\tau)], \quad \omega_i^\pi(t, z) \geq \mathbb{E}_{t,z}^\pi[e^{-\rho(\tau-t)} 1_{F_i}(\tau, Z_\tau)].$$

These are upper bounds on discounted hitting probabilities, not probabilities estimated by simulation. A nonnegative smooth function whose discounted generator is nonpositive and whose stopping trace dominates the face indicator supplies such a bound by optional stopping. A separate invariance argument can give the exact value zero for an inaccessible face.

THEOREM 11—Accessibility-weighted neural verification: *Under Assumption 1, suppose complete enclosures establish*

$$\sup_a \mathcal{R}_a v \leq e_+(s), \quad -\mathcal{R}_\pi v \leq e_-(s), \quad e_+, e_- \geq 0,$$

and $(g - v)_+ \leq \beta_i^+$, $(v - g)_+ \leq \beta_i^-$ on F_i . For an interior initial state,

$$\begin{aligned} 0 \leq V(t, z) - J(t, z; \pi) &\leq \int_t^T e^{-\rho(s-t)} \{e_+(s) + e_-(s)\} ds \\ &\quad + \sum_i \beta_i^+ \omega_i^*(t, z) + \sum_i \beta_i^- \omega_i^\pi(t, z). \end{aligned} \quad (50)$$

The upper comparison retains the entire original action set. Inaccessibility for the candidate policy does not remove an action from that comparison.

This theorem retains the stopped verification identity underlying Theorem 1. The new distinction is which sign of a trace discrepancy must be paid, and under whose stopping distribution it is paid. Replacing both signs by a common supremum discards precisely the boundary information identified by Proposition 14. The inequality is still an a posteriori certificate; convergence of a candidate-generation procedure is a separate statement.

Y9.2. An executable accessibility-preserving architecture

Let N_c, N_θ, N_p, N_v be smooth neural outputs. We use

$$\begin{aligned} c &= .425 + .375 \tanh N_c, & \theta &= .2 \tanh N_\theta, \\ p &= \frac{2-x}{1.5} \{ .15 + .65 \tanh N_p \}, \end{aligned} \quad (51)$$

$$v = G + (1-t) \{-8 + B(u, x) \text{softplus}(N_v)\}, \quad (52)$$

$$B(u, x) = (1 - e^{-12(u-1.2)})(1 - e^{-12(2.8-u)})(1 - e^{-4(x-.5)}).$$

Here $\text{softplus}(y) = \log(1 + e^y)$. Consumption and preference controls obey the original bounds. The portfolio head also obeys the original bounds throughout the state domain. Its boundary factor is part of the policy, not a replacement of the economy's portfolio opportunity.

PROPOSITION 15—Exact elimination of the trace budget: *For an interior initial state, the policy in (51) cannot hit $x = 2$ before another stopping event. The witness in (52) equals g on the two preference faces, the lower wealth face, and the terminal face, and is at least g on the upper wealth face. Hence every trace contribution in (50) is zero.*

The statement remains true for bounded admissible perturbations of the action heads, clipped to their stated intervals *before* multiplication of the portfolio head by $(2 - x)/1.5$. An unrestricted additive error in the final portfolio output is different: it can destroy inaccessibility and is not covered by this claim. The replication records this implementation contract explicitly. The certificate describes a real mathematical feedback with stored dyadic weights; it is not a claim about an undocumented floating-point activation routine or an SDE simulator.

The architecture does not by itself make the residual small. Full-domain derivative enclosures must still validate both terms in (50). Reporting a zero trace budget without the remaining residual budget would therefore be incomplete. Nor do we force the numerical continuation witness to equal the known immediate-settlement value at an initially stopped upper-boundary state: at that state every policy has value g and regret zero directly.

Y.10. An auxiliary finite-Bellman policy-iteration bound

The following result is an auxiliary statement about a stationary finite Bellman problem, not the algorithm executed in the nonlinear experiments. The executed certificate-aligned algorithm is defined in Section Y.11. It is useful to distinguish three sources of approximation: policy evaluation, policy improvement, and approximation of a continuous-time transition problem. We state a finite Bellman result that makes the first two requirements quantitative. It does not assume that a particular stochastic-gradient optimizer attains them.

Consider a finite discounted Markov problem with rewards r_a , stochastic or substochastic transition matrices P_a , and contraction modulus $q \in (0, 1)$. Absorbing settlement is incorporated in the reward. Write $T_a v = r_a + q P_a v$ and $T v = \max_a T_a v$. A neural evaluation v_n of policy π_n and an improved policy π_{n+1} are accepted with recorded defects

$$\|v_n - T_{\pi_n} v_n\|_\infty \leq \epsilon_n, \quad 0 \leq T v_n - T_{\pi_{n+1}} v_n \leq \eta_n \mathbf{1}. \quad (53)$$

On a continuous state space the same inequalities require a complete enclosure, not a finite collection of collocation residuals. On a finite state space they can be checked by enumeration. They give a stopping and refinement criterion for a learned representation without identifying its training loss with its Bellman defect.

THEOREM 12—A quantitative certified policy-iteration recurrence: *Let $E_n = \|V^* - V^{\pi_n}\|_\infty$ for the finite Bellman problem. If (53) holds, then*

$$E_{n+1} \leq q E_n + \frac{2q\epsilon_n}{(1-q)^2} + \frac{\eta_n}{1-q}. \quad (54)$$

Thus geometrically decreasing certified defects imply convergence of policy values, with the explicit convolution bound obtained by iterating (54). If both defects are bounded by $C r^n$, $r \in (0, 1)$, the bound decreases at rate $O(\max\{q, r\}^n)$ when $q \neq r$, and $O(nq^n)$ when $q = r$.

The contraction and policy-improvement ingredients are classical. The operational content is the use of independently checked evaluation and continuous-action defects as acceptance criteria. A neural optimizer that does not achieve those defects has not met the hypothesis. In

particular, the full-horizon experiments below are not assigned a convergence guarantee by virtue of being called policy iteration.

A time or state discretization requires an additional comparison. If a numerical optimal value differs from continuous-time value by at most ζ_h , and the numerical value of the transferred policy differs from its continuous-time payoff by at most ξ_h , its continuous-time loss is at most $E_n + \zeta_h + \xi_h$. Neither term is assumed zero. An alternative is to bypass both transfers and certify the actual continuous-time feedback with Theorem 11; that is the procedure used for the classical candidates below. At an inward stopping face, imposing an inconsistent Dirichlet approximation does not supply the required ζ_h . Proposition 14 explains why this qualification is material rather than technical.

For a fixed positive discount rate, $q = e^{-\rho h}$ approaches one as h decreases. The factors in (54) must therefore be included in any accuracy allocation. The theorem is not a dimension-free complexity result. The original nonlinear economy still uses a full rectangular certificate; the structured extension in Section Y.17 avoids a state tensor cover for a different, explicitly proved reason.

Y.11. Complete-cover envelope proposals

The complete-cover benchmark has three explicit objects: a policy π_θ , a smooth witness v_ϕ , and a complete-cover certificate. Its target is the original continuous-time policy loss, not the optimal value of a discretized training problem. The architecture in Section Y.9 supplies admissibility and a zero signed trace budget. The following construction addresses the remaining residuals.

Partition time into $0 = t_0 < \dots < t_N = T$ and cover the spatial domain on slab j by M_j closed boxes. Write

$$m_j = \int_{t_j}^{t_{j+1}} e^{-\rho s} ds, \quad b_{ji}^+ \geq \sup_{(s,z) \in Q_{ji}, a} \mathcal{R}_a v_\phi(s, z), \quad b_{ji}^- \geq \sup_{(s,z) \in Q_{ji}} (-\mathcal{R}_{\pi_\theta} v_\phi)(s, z).$$

Each $b_{ji}^\pm$ is nonnegative; the optimal comparison ranges over the original continuous action set. Define

$$B(b) = \sum_j m_j \sum_{\sigma \in \{+, -\}} \max\{0, \max_i b_{ji}^\sigma\}, \quad (55)$$

$$S_\tau(a) = \sum_j m_j \sum_{\sigma \in \{+, -\}} \tau \log \left(1 + \sum_{i=1}^{M_j} \exp(a_{ji}^\sigma / \tau) \right), \quad \tau > 0. \quad (56)$$

The extra unit in each logarithm is the zero logit. It is necessary because only the positive part of each signed residual is charged. All bounds in this section are for time zero; replacing m_j by the appropriate discounted remaining-slab masses gives the corresponding bound at another initial time.

THEOREM 13—Trainable envelope and certified policy loss: *Suppose the hypotheses of Theorem 11 hold, the signed trace budget is zero, and the boxes cover the entire state–time domain. For any finite proposal logits a_{ji}^σ , let*

$$d_j^\sigma \geq \max_i (b_{ji}^\sigma - \max\{a_{ji}^\sigma, 0\})_+, \quad D = \sum_j m_j (d_j^+ + d_j^-).$$

Then every interior initial state satisfies

$$0 \leq V(0, z) - J(0, z; \pi_\theta) \leq B(b) \leq S_\tau(a) + D. \quad (57)$$

If the certified endpoints themselves are used as logits, then

$$B(b) \leq S_\tau(b) \leq B(b) + 2\tau \sum_j m_j \log(M_j + 1). \quad (58)$$

A directed upper enclosure of $S_\tau(a) + D$, computed from the exact stored proposal logits and the certified endpoints, is therefore a sufficient, checkable stopping criterion for the policy-loss target.

The theorem applies to the implemented differentiable interval-envelope loss even when its ordinary floating-point arithmetic has not itself been proved enclosing. The correction D covers any endpoint discrepancy between that proposal graph and the independently evaluated certificate; it is not assumed to be a rounding error of a particular library routine. In the new audit, the logits are recomputed from the stored neural weights and then interpreted as exact dyadic numbers for the directed soft-maximum calculation. Neither differentiation through an ordinary floating-point interval graph nor a small sampled loss is called a certificate.

The smoothing allowance is explicit. With four time slabs, 256 spatial boxes per slab, $\rho = .04$, $T = 1$, and $\tau = .15$, its upper bound is 1.631868. This is a worst-case allowance, not an observed policy loss. Refining the cover while keeping τ fixed need not remove this allowance. A sufficient joint refinement rule is $\tau \log(1 + \max_j M_j) \rightarrow 0$, together with consistent residual enclosures. The theorem does not assert that a nonconvex optimizer reaches a requested envelope level.

Y.11.1. The executed algorithm

The complete-cover neural algorithm uses the following rule. Initialize the actor and critic, form the fixed cover, and minimize $S_\tau(a(\theta, \phi))$ by joint Adam updates of their parameters. At each predeclared checkpoint, freeze and serialize both networks. Independently recompute every signed residual box with directed arithmetic and the original action maximization. Reject an incomplete, nonfinite, or failed enclosure. For a complete object, compare its bound B with the incumbent bound, retain the proposal regardless of the comparison, and replace the incumbent only under a strict decrease. Stop at the declared work budget or when the complete bound meets the stated target. The implementation is `guided_neural.py`; the separate checker is `accessibility_certificate.py`.

The reported original-economy run uses checkpoints 0, 100, 400, temperature .15, learning rate .003, gradient clipping at 100, and two hidden tanh layers of width 16 or 32. There is no point sampling in its envelope objective: all 1,024 boxes enter every proposal update. The ten seeds and paired all-face ablations use matched initialization logic, optimization budget, cover, and arithmetic. The acceptance decision uses the complete MPFR bound, not the smooth proposal loss. The new R18 audit applies Theorem 13 to these already executed, predeclared R17 trajectories; it is not represented as a newly randomized training study.

PROPOSITION 16—What certificate acceptance guarantees: *The incumbent bound of the preceding algorithm is nonincreasing and remains a valid upper bound for its associated incumbent policy. Neither this property nor a decrease in raw proposal bounds implies that the true values of successive policies increase.*

The distinction is substantive. A better witness can certify the same policy more sharply. Conversely, a worse policy can have a smaller bound if it is paired with a much better witness. Accordingly, the numerical record reports raw proposals, incumbent decisions, and actor–critic interchanges separately. A claim of true policy improvement would require an additional policy-value comparison, not just the incumbent rule.

Y.11.2. A finite-horizon error allocation without a stationary resolvent

The auxiliary stationary recurrence in Section Y.10 has a different role. Its amplification factors become large when $q = e^{-\rho h} \uparrow 1$. For a finite horizon, a backward error allocation is more appropriate when a time-discrete implementation is analyzed directly. Let T_n and $T_n^{\pi_n}$ be monotone Bellman operators with Lipschitz modulus q , and let v_N approximate the terminal reward g .

PROPOSITION 17—Finite-horizon evaluation and improvement defects: *If*

$$\|v_N - g\|_\infty \leq b, \quad \|v_n - T_n^{\pi_n} v_{n+1}\|_\infty \leq \epsilon_n, \quad 0 \leq T_n v_{n+1} - T_n^{\pi_n} v_{n+1} \leq \eta_n \mathbf{1},$$

then the nonstationary policy $\pi = (\pi_0, \dots, \pi_{N-1})$ satisfies

$$\|V_0^* - V_0^\pi\|_\infty \leq 2q^N b + \sum_{n=0}^{N-1} q^n (2\epsilon_n + \eta_n). \quad (59)$$

For $q = e^{-\rho h}$, $Nh = T$, and $\epsilon_n \leq \bar{\epsilon} C_\rho(h)$, $\eta_n \leq \bar{\eta} C_\rho(h)$, the accumulated defect term is at most $(2\bar{\epsilon} + \bar{\eta}) C_\rho(T)$.

The finite-horizon statement is representation independent. Its 80 exact-rational finite-MDP regression cases do not constitute execution of a continuous-time neural policy-iteration scheme. The original-economy algorithm instead applies Theorem 13 directly to the continuous-time policy, avoiding an unproved discretization-transfer bound. Both finite-Bellman results are retained, with their different hypotheses and numerical roles made explicit.

Y.12. A sharp economic comparison construction

A neural certificate should be compared with independently identified bounds for the same economy. We therefore retain the affine-preference market-deflator construction and reimplement its numerical evaluation. The lower policy and the upper comparison object are identified separately from the neural feedback throughout.

Y.12.1. Feasible policies and the original stopping contract

A library policy consists of 16 exact stored consumption constants $c_j \in [.7, .8]$, preference actions $\theta_j \in [0, .2]$, and $p_j = 0$. These controls are held on their time intervals until the original first exit. Wealth is deterministic and satisfies

$$x_1 = e^{.02} \left(x_0 - \int_0^1 e^{-.02s} c_s ds \right).$$

The program proves $x_1 > .5$ and the complete action bounds before evaluating a payoff. It does not round a budget residual to zero or assume that a nearly feasible policy is feasible.

The previous implementation evaluated a polynomial representation with Gaussian moments. The new evaluator instead applies an interval-certified Gauss rule directly to the original CRRA expression on a covered Gaussian domain. After $t = v^2$, it integrates over v and a standard-normal coordinate z . Rationally isolated Gauss nodes, interval weights, and analytic complex-neighborhood remainder bounds enclose the integrals. The unbounded normal tail and the discrepancy from stopping are accounted for analytically. The use of an auxiliary bounded full-horizon extension in this proof does not change the objective (5): it is connected to that objective by an explicit stopping correction, less than 3.87×10^{-18} on K for these policies.

At $k = 4.25$, the second implementation encloses the policy payoff in

$$[-1.315469674845588, -1.315469674834266], \quad (60)$$

with the stored unrounded interval width approximately 1.13×10^{-11} . This is a policy-payoff width, not an optimality gap. The latter still requires an upper bound on V_k .

Y.12.2. The affine-preference dual

Let $dY_s = .02Y_s ds - .3Y_s dZ_s^x$. Its product with wealth satisfies

$$d(Y_s x_s) = (.04Y_s x_s - Y_s c_s) ds + \text{a local martingale.}$$

The cancellation holds for every original adapted portfolio control. Define

$$\Psi(u, y) = \max_{.05 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad S_0(u, y) = \Psi(u, y) + .01y - .004 \log(.5), \quad \beta = \partial_u \Psi.$$

For a deterministic reference $\bar{\theta} \in [0, .2]$, set $m(t) = 2 + \int_0^t \bar{\theta}_s ds$ and let Q be the affine tangent to $\phi(u) = -.02(u-2)^2$ at $m(1)$. On $u \in [1.5, 2.5]$, $y \in [.2, 12]$, the supporting inequality is

$$S_0(u, y) \leq S_0(m, y) + \beta(m, y)(u - m), \quad 2 \leq m \leq 2.2. \quad (61)$$

It is justified by analytic monotonicity and complete interval covers, not by sampled tangency or presumed global concavity. Its new primitive checks have strictly positive margins; the supplement gives the proof and the recorded numerical margins.

Put $\hat{y} = \text{proj}_{[.2, 12]} y$ and replace the source by the right side of (61) at $\hat{y}$. The resulting full-line auxiliary problem $A_k(t, u, y)$ retains adapted preference actions, quadratic adjustment cost, and terminal payoff Q . Its source and terminal value are affine in the initial preference. The following positive localization potentials reconnect it to the original first-exit problem:

$$Z_u(h, u) = 8h \{ e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h} \}, \quad (62)$$

$$Z_y(h, y) = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (63)$$

For $y_0 \in [1.2, 2.4]$ the resulting upper bound is

$$V_k(0, 2, 1.25) \leq .75y_0 + .1 \log(.5) + A_k(0, 2, y_0) + Z_u(1, 2) + Z_y(1, y_0). \quad (64)$$

The auxiliary value has a computable upper bound consisting of one-Gaussian source expectations, a terminal tangent, a correlated-shock correction, a reference-action gap, and a conditional-variance allowance. The latter is necessary because the adapted marginal value of adjustment is random; replacing it by its unconditional mean without a correction would not be

valid. The supplement states and proves the complete formula, including the factors .00375 and .09/(24k) in the covariance and variance terms.

In the second implementation, the source uses separate analytic formulas for its capped and interior consumption branches. Interval automatic differentiation supplies Hessian enclosures, and a rational-root Gauss construction encloses the weighted moments. This differs from the inherited manually differentiated source and polynomial moment integrator. The analytic dual theorem is shared; its mathematical validity is not established merely by agreement between programs. Its finite supporting-plane checks and all decisive numerical quantities are recomputed by the new implementation.

Y.13. Joint initial-state and price certification

The state-domain extension is obtained without assuming a Lipschitz modulus for the unknown optimal value. Two structural properties suffice: a convex upper comparison function and a concave feasible-policy lower function.

Y.13.1. Extending the dual in the initial state

For the fixed dual witness, define

$$b_0 = \int_0^1 e^{-\rho s} \mathbb{E} \beta(m(s), \hat{Y}_s) ds + e^{-\rho} Q'. \quad (65)$$

An interval for b_0 is obtained in the same independent execution as the dual. Since the coefficient of the initial u in the auxiliary criterion is independent of the chosen adapted preference action,

$$A_k(0, u, y_0) = A_k(0, 2, y_0) + (u - 2)b_0.$$

Let U_0 be a certified upper endpoint in (64). It follows that

$$\begin{aligned} V_k(0, u, x) \leq & U_0 + y_0(x - 1.25) + (u - 2)b_0 \\ & + 8e^{-18} \{e^{120(u-2)} + e^{-120(u-2)} - 2\}, \end{aligned} \quad (66)$$

for the verified initial rectangle. Interval multiplication is sign-aware when $u - 2 < 0$. With an interval for b_0 , its upper support function is convex in u . Thus the computable right side of (66) is convex in (u, x) . This is a property of the upper comparison object, not an assertion that V_k itself is convex in wealth.

Y.13.2. A budget-preserving policy map

Write $Q_r = \int_0^1 e^{-.02s} ds$. For each stored policy, define

$$c_j(x) = c_j + \frac{x - 1.25}{Q_r}, \quad \theta_j(x) = \theta_j, \quad p_j(x) = 0. \quad (67)$$

Its terminal wealth is identical to that of the central-state policy. All shifted consumption values remain in $[.7, .8]$ on K ; this is checked for every library member. The policy depends on the initial state and is thereafter a time control until exit. It is not described as an approximate neural feedback surface.

The utility $f(c, u) = c^{1-u}/(1-u)$ is jointly concave over the region needed by the Gaussian cover. With $r = u - 1$ its Hessian satisfies

$$\begin{aligned} f_{cc} &= -uc^{-u-1}, \\ f_{cu} &= -(\log c)c^{-u}, \\ f_{uu} &= -\frac{c^{-r}}{r^3} \{(r \log c + 1)^2 + 1\}, \\ \det D^2 f &= \frac{c^{-2u}}{r^3} \{r^2(\log c)^2 + 2r(r+1)\log c + 2(r+1)\}. \end{aligned}$$

On $u \in [1.2, 2.8]$, $c \in [.7, .8]$, one has $r(-\log c) < 1$, so the determinant is positive. The covered expected payoff, with the exact budget shift, is therefore concave in the initial state. A single uniform tail and stopping correction turns this covered functional into a lower bound for the original stopped policy throughout K .

PROPOSITION 18—Four vertices and a continuum of prices: *Suppose, at each price node k_i , a convex upper function $U_i(z)$ bounds $V_{k_i}(0, z)$ on a rectangle K . For each policy j , suppose a concave lower function $F_j(z; k)$ bounds its original payoff on K , is affine in k , and has a verified expenditure slope interval. Then corner values alone produce a uniform regret certificate on $K \times [k_1, k_N]$ by the following finite construction. Between adjacent price nodes use the chord of U_i and U_{i+1} . For each policy transfer its four lower corner values affinely in price, using the appropriate expenditure endpoint. At each price select the policy minimizing the largest of its four upper-minus-lower corner gaps. The supremum of the resulting bound is attained at an endpoint or an intersection of the finitely many affine corner-gap functions.*

The optimal value is convex in k because the policy set, dynamics, and stopping contract are common and fixed-policy payoffs have the form $A(\pi) - kB(\pi)$. This justifies the upper price chord, not interpolation of V as though it were affine. The expenditure used for the concave covered lower functional is its full-horizon deterministic expenditure, with the stopping discrepancy already paid uniformly at the largest price. For the separate central-state exact-policy envelope, the program instead bounds the stopped expenditure. Confusing these two slopes would invalidate the transfer; they are separately recomputed and labeled in the evidence.

PROPOSITION 19—Executed state–price certificate: *For the unchanged economy (5), the 18 stored policies with the state map (67) and the price selector in Proposition 18 satisfy*

$$\sup_{(u,x) \in K} \sup_{k \in [.5, 8]} \{V_k(0, u, x) - J_k(0, u, x; \pi_{k,x})\} \leq 0.012095431554. \quad (68)$$

The central-state selector based on the independently evaluated stopped policies satisfies

$$\sup_{k \in [.5, 8]} \{V_k(0, 2, 1.25) - J_k(0, 2, 1.25; \pi_k)\} \leq 0.009969007711. \quad (69)$$

The execution evaluates all 72 policy–state corners, uses all 18 independently verified upper witnesses, and checks 211 exact candidate breakpoints for the joint state–price construction. No unsampled state requires an empirical smoothness extrapolation. The exact rational worst-case joint bound is

$$\frac{2187960974729572475442514300375}{180891518006739738096917284913152}.$$

TABLE Y.X
ALL PRESCRIBED ORIGINAL-ECONOMY NEURAL TRAJECTORIES

Seed	Width	B_0	B_{100}	B_{400}	All-face B_{400}	Seconds
17100	16	26.300514	10.104267	7.339624	7.345561	104.1
17101	32	26.499830	7.547734	7.281385	7.263538	193.4
17102	16	26.326374	10.237435	7.343312	7.351242	96.1
17103	32	26.511217	7.535185	7.280303	7.263378	143.4
17104	16	26.367488	10.591643	7.336433	7.345015	71.3
17105	32	26.468295	7.555938	7.280018	7.261518	114.4
17106	16	26.352908	9.701133	7.343665	7.349999	51.2
17107	32	26.666728	7.604406	7.278319	7.262408	97.2
17108	16	26.339971	9.745279	7.338341	7.345879	48.6
17109	32	26.527472	7.553444	7.279140	7.261511	79.0

All entries use the full original state–time cylinder and continuous action upper comparison. B_j denotes a raw proposal certificate, not an incumbent replacement. Seconds include all three certificate checks and the complete accessible training trajectory. Both architectures use identical declared budgets. Printed values are rounded; exact directed bounds, signed decompositions, and all 60 objects are retained.

The displayed decimal in (68) is rounded upward. The computational record contains the complete selected-policy regions and every corner interval. Appendix P proves the interpolation and finite-breakpoint steps.

Y.14. Certificate-aligned computation on the original economy

All experiments in this section use the original economy, its full continuous action comparison, $k = 2$, and the complete state–time domain. They are not price-library or inventory results. The immutable R17 scientific object contains the prescribed trajectories and all raw certificates. The new R18 calculations audit their objective implications, reproduce two full-domain certificates, and reconstruct every displayed aggregate from the retained records.

Y.14.1. Raw trajectories, acceptance, and the training objective

Table Y.X reports every seed. All ten accessibility-aware raw sequences decrease strictly at both transitions; all paired all-face sequences also decrease. Thus the improvement is not created by selecting the incumbent or hiding a rejected proposal. All 40 post-initialization proposals are accepted by the strict rule. The historical sampled-training experiment is different: all ten of its 800-to-2,400-update certificates deteriorate, although sampled residual mean squared error falls in nine of ten cases. The new training loss reverses that particular anti-alignment in the observed experiment; it is not a proof that every longer Adam run must improve the certificate.

The complete-cover objective audit covers 60 neural objects and 61,440 cells. It verifies all 589 files in the immutable scientific manifest, checks network/certificate identities and complete cell coverage, and re-evaluates the proposal logits. Directed evaluation of the soft maximum and its endpoint correction verifies Theorem 13 for every object. The largest integrated correction D is below 1.20×10^{-13} ; the remaining differences between a smooth objective and a box maximum are not an arithmetic precision effect. The audit is deliberately separate from the training experiment and retains the exact recomputed logits.

The final accessibility-aware bounds are between 7.2783190 and 7.3436651, and none reaches .01. The original target is not relaxed. The result establishes a reproducible improvement in the certified objective, not sharp accuracy of the original nonlinear neural policy. Training,

verification, and checkpoint selection costs are included separately in the retained resource records. At the final checkpoint, the observed mean total trajectory cost is 99.87 seconds; timings describe the recorded hardware and concurrent workload, not a matched-accuracy speed ranking.

Y.14.2. A matched boundary ablation and a fixed-witness floor

At width 16, the accessibility-aware final bound is smaller than its paired all-face bound in all five pairs. At width 32, the ranking reverses in all five pairs. The paired mean difference (accessibility-aware minus all-face) is approximately .005049. This is not evidence of a uniform finite-budget advantage for the corrected trace class. The structural result in Proposition 14 remains an asymptotic consistency statement, while the actual finite-budget comparison depends on the learned witness and optimization trajectory.

For seed 17100, refining a fixed final accessibility-aware object from 1,024 to 8,192 and 65,536 cells changes its bound from 7.339625 to 7.280222 and 7.251560. The corresponding all-face bounds are 7.345561, 7.267777, and 7.236151. These are verifier-refinement curves around fixed objects, not new policy training. They quantify enclosure dependence without suggesting that a sufficiently fine cover would reach .01.

PROPOSITION 20—A computable fixed-witness obstruction: *Let v satisfy the upper-comparison trace conditions and let $h_v = v(0, 2, 2) - g(0, 2, 2)$ be its upper-wealth trace excess. Write $\Delta = 6.45 - .1 \log 2 - 25.1e^{-8}$. Any nonnegative, time-dependent bound on its positive optimal residual must satisfy*

$$\int_0^1 e^{-.04s} e_+(s) ds \geq (\Delta - h_v)_+. \quad (70)$$

This is a lower bound on the certificate component of this fixed witness, not a lower bound on the regret of its policy.

The final accessible critics have verified trace excesses only about .052–.097. Their lower floors from (70) range from 6.274882 to 6.320490. Consequently, refining their covers or increasing arithmetic precision cannot close the original accuracy gap. A materially different continuation witness is required. The all-face class has the stronger structural obstruction $h_v = 0$.

The actor–critic interchange sharpens this diagnosis. Pairing the initial actor with the final critic for seed 17100 gives exactly the same retained bound, 7.339624178645005, as the final actor/final critic pair. Pairing the final actor with the initial critic gives 23.718685694842687. Thus the certificate descent cannot be attributed to a demonstrated increase in the actor’s true payoff. Both facts are retained rather than reinterpreted as policy improvement.

Y.14.3. Unrestricted candidate generation and independent witnesses

The new controlled Markov-chain and semi-Lagrangian candidates do not multiply their interior portfolio choices by the neural wealth-distance factor. Each solves over the original bounded action grid, including zero portfolio. The upper-wealth grid column represents the interior continuation trace: zero-portfolio continuation is computed and compared with immediate settlement for nonzero portfolio at that trace. The retained policies select zero portfolio on this column. The certified controls are the original-action-clipped, exact-real bilinear state interpolants, constant within each time interval. Their interpolation consequently has bounded $p/(2 - x)$ and cannot hit the upper face before another stopping event; no interior distance multiplier is imposed.

TABLE Y.XI
ORIGINAL-ACTION CLASSICAL CANDIDATES WITH METHOD-SPECIFIC WITNESSES

Method	Nodes/axis	Time steps	Bound	Generate	Witness	Verify
MC	9	64	7.339815	0.34	62.14	9.41
MC	17	128	7.314510	7.60	62.41	9.71
MC	25	256	7.763463	124.01	39.84	6.71
SL	9	64	7.342250	0.35	67.46	8.60
SL	17	128	7.314559	8.52	45.80	7.58
SL	25	256	7.774123	74.71	39.22	5.34

Times are seconds. MC: controlled Markov chain; SL: semi-Lagrangian. Interior actions use no neural distance multiplier. Every transferred policy has a separate witness and a complete continuous-time check; all six fail the unchanged .01 target. These bounds do not rank true policy values. Discrete central values remain diagnostics.

Each candidate receives a separately trained witness and a full continuous-time residual check against the original continuous action set. Hence neither the discrete central value nor a common inaccurate neural witness is used as its accuracy certificate. Table Y.XI gives all six resulting bounds and separates candidate generation, witness construction, and verification. The method-specific witnesses remove the earlier common-witness confound but do not produce a sharp matched-accuracy frontier. In particular, the finest-grid bounds increase because their policy residual contribution is no longer zero. The numerical values of the discrete schemes remain diagnostics, not certified continuous-time values.

Y.14.4. Control changes and numerical scope

The supplement also reports whole-domain ranges for all six classical policies and whole-price/time ranges for all twenty fresh-library policies. For the latter, consumption lies in $[-.7618225, .7721747]$, preference adjustment in $[.0031022, .2]$, and portfolio is exactly zero. These library ranges hold for every policy selected over $k \in [.5, 8]$, not only the displayed price nodes.

Complete-box bounds on the original neural controls are reported in the supplement for all ten final actors. For seed 17100, the final ranges are approximately $[-.530998, .749371]$ for consumption, $[-.093966, .113275]$ for preference adjustment, and $[0, .087417]$ for portfolio. The corresponding uniform bounds on the change from initialization are .218738, .019170, and .100661. These are conservative ranges of mathematical control functions over the full state–time domain, not simulated confidence intervals or policy-optimality claims.

The certificate interprets stored weights as exact dyadic numbers and activations as mathematical tanh/softplus. The allowed head perturbation is 2^{-24} and precedes the wealth-distance factor. Neither the proposal graph’s ordinary floating-point activations, a deployed inference engine, nor an SDE simulator is certified by this contract. Independent arithmetic reduces rounding risk but does not independently establish every shared analytical premise.

Y.15. Nonlinear multi-product control with a state-uniform certificate

A high-dimensional test should require the policy to depend on the high-dimensional state. We therefore distinguish the following nonlinear planning problem from the two-mode quadratic inventory model retained in Section Y.17. Here there is no Riccati reduction or supplied optimal solution.

There are d heterogeneous products and $H = 12$ dates. Initial inventory is $x \in [-1, 1]^d$. The deterministic dynamics and total cost are

$$z_0 = x, \quad z_{h+1} = \frac{3}{5}z_h + u_h - b_h, \quad b_{hi} = \frac{((h+i) \bmod 7) - 3}{100}, \quad (71)$$

$$F_x(U) = \sum_{h=0}^{H-1} \{2\|u_h\|_2^2 + \Phi(z_{h+1})\}, \quad U = (u_0, \dots, u_{H-1}) \in \mathbb{R}^{Hd}, \quad (72)$$

$$\Phi(z) = \sum_{i=0}^{d-1} \left\{ \frac{q_i z_i^2}{2} + \frac{1}{2} \log \cosh z_i \right\} + \frac{3}{10} \sum_{i=0}^{d-1} \log \cosh(z_i - z_{i+1}), \quad q_i = 1 + \frac{i}{5(d-1)}, \quad (73)$$

where indices in the coupling term are cyclic. Production/adjustment costs are quadratic, while individual and cross-product inventory penalties are nonquadratic. Every coordinate enters the nonlinear coupling; heterogeneity prevents the two-invariant-mode decomposition of the quadratic comparison model. Its Bellman equation is

$$W_h(z) = \inf_{u \in \mathbb{R}^d} \{2\|u\|_2^2 + \Phi(\frac{3}{5}z + u - b_h) + W_{h+1}(\frac{3}{5}z + u - b_h)\}, \quad W_H = 0.$$

The computation instead uses the equivalent deterministic control-plan formulation, without treating either formulation's unknown value as training data.

A two-hidden-layer neural map takes all d coordinates of x and produces all Hd controls:

$$U_0(x) = \frac{1}{4} \tanh N_\theta(x).$$

It is trained on the original cost, not optimal labels, using 400 Adam updates, batches of 32 uniformly drawn initial states, learning rate .003, width $\min\{64, 2d\}$, and three predeclared seeds. A correction stage applies

$$U_{k+1}(x) = U_k(x) - \frac{16}{209} \nabla_U F_x(U_k(x)). \quad (74)$$

The complete solver is the neural initialization together with the declared number of corrections. The correction, not a change of model or an optimal-value label, supplies its quantitative convergence mechanism.

THEOREM 14—Uniform nonlinear control accuracy and work: *For every $d \geq 2$ and every x , the cost in (72) satisfies*

$$4I \preceq \nabla_U^2 F_x(U) \preceq \frac{177}{8} I, \quad U \in \mathbb{R}^{Hd}. \quad (75)$$

Let $r = 145/209$. For every $x \in [-1, 1]^d$ and every bounded neural initialization above, the exact-real correction obeys

$$0 \leq F_x(U_k(x)) - \inf_U F_x(U) \leq \frac{\|\nabla F_x(U_k(x))\|_2^2}{8} \leq \frac{Hd(8.85)^2}{8} r^{2k}. \quad (76)$$

A gradient evaluation requires $O(Hd)$ arithmetic operations using forward state propagation and backward adjoints. Thus, after initialization, a sufficient number of correction steps for total

TABLE Y.XII
STATE-UNIFORM TOTAL-COST LOSS FOR NONLINEAR CONTROL

Corrections	$d = 8$	$d = 32$	$d = 128$
0	939.87	3759.48	15037.9
8	2.70779	10.8312	43.3247
16	0.00780122	0.0312049	0.12482
24	2.24755×10^{-5}	8.99022×10^{-5}	0.000359609
32	6.47527×10^{-8}	2.59011×10^{-7}	1.03604×10^{-6}

The domain is every initial $x \in [-1, 1]^d$. Bounds concern the exact-real neural initialization and rational-step gradient correction, not an uncertified floating-point inference graph. Entries are total cost losses, not per-coordinate losses. The theorem holds for every bounded initialization, including an untrained one.

loss at most ε is

$$k \geq \max \left\{ 0, \left\lceil \frac{\log(Hd(8.85)^2/(8\varepsilon))}{2\log(209/145)} \right\rceil \right\}.$$

No tensor cover of the initial-state cube is required.

The work statement counts exact-real arithmetic operations, not bit complexity or the cost of floating-point certification. The result uses strong convexity and the sparse coupled dynamics of this application. It is not a theorem for a general stochastic high-dimensional HJB equation. The factor d in total loss is visible, and no per-coordinate normalization is used to hide it.

Y.15.1. Uniform and pointwise numerical objects

Table Y.XII reports the directed evaluation of the state-uniform bound for every specified dimension and correction budget. The $d = 128$ total bound is below .000360 after 24 corrections and below 1.037×10^{-6} after 32. These inequalities hold for the exact-real neural map and correction at every initial state in the cube, regardless of which of the three stored neural seeds is used.

Separately, the actual stored floating-point plans are checked by MPFR at two fixed initial vectors: an alternating corner and a predeclared pseudorandom vector. The checker evaluates the original cost and KKT gradient using exact rational model constants, and applies the first inequality in (76). It does not assume that the floating-point optimizer has converged. All 90 neural query/budget objects and 66 classical query/budget objects are retained across dimensions. At $d = 128$ and 32 corrections, the six retained neural query-plan bounds range from 8.28771×10^{-15} to 1.62604×10^{-14} . Those sharper values concern those six exact stored plans, not every state in the cube. The new R18 execution rechecks all 18 final neural query plans across the three dimensions.

Y.15.2. Matched certificate comparisons

The comparison methods are zero-start gradient descent with the same step, strongly-convex accelerated gradient, and L-BFGS. All use the same original model and the same independent gradient-based cost-loss inequality. Table Y.XIII reports the final $d = 128$ query results and separates neural training, online solution, and verification time. The supplement includes the corresponding tables at $d = 8$ and 32 and every retained correction budget.

The data do not establish a neural speed advantage: strong classical methods attain comparable or smaller certified errors without the training cost. The state-uniform theorem likewise holds for

TABLE Y.XIII
MATCHED CERTIFICATE COMPARISONS AT 128 COORDINATES

Method	Largest query bound	Train	Online	Verify
Neural + gradient	1.63×10^{-14}	12.709	0.078	0.678
Zero + gradient	2.06×10^{-14}	0.000	0.078	0.555
Accelerated gradient	2.6×10^{-16}	0.000	0.071	0.648
L-BFGS	3.13×10^{-14}	0.000	0.096	0.699

All query plans use the same MPFR KKT-gradient certificate. Gradient methods use 32 corrections; L-BFGS uses its retained stopping result. Neural entries summarize six query/seed plans; classical entries summarize two queries. Times are descriptive means in seconds; neural training is a one-time per-network cost, not a per-query charge. Query bounds do not replace the state-uniform bound in Table Y.XII.

any initialization in the specified bounded class, including an untrained one. Neural initialization and its amortized cost are therefore reported as components of the solver, not credited with the contraction supplied by the correction. This application establishes an accurately certified, full-state-dependent nonlinear control-plan solver; it does not replace the original stochastic economy or close that economy's remaining neural accuracy gap.

Y.16. Learned quality and matched online work

The nonlinear inventory problem has strong convexity $m = 4$, Hessian upper bound $L = 177/8$, step $2/(m + L) = 16/209$, and gradient contraction factor $r = 145/209$. The existing state-uniform theorem holds for every bounded initialization, including an untrained one. Training is therefore irrelevant to that particular guarantee. To identify a contribution from learning, the initial approximation quality and the required online work must be measured separately.

THEOREM 15—Learned-gradient correction bound: *For any exact initial plan $f_\theta(x)$, let $G_\theta(x) \geq \|\nabla F_x(f_\theta(x))\|_2$. After j exact-real gradient corrections,*

$$F_x(U_j) - F_x^* \leq \frac{G_\theta(x)^2}{8} \left(\frac{145}{209} \right)^{2j}. \quad (77)$$

On a finite query set, the largest independently enclosed initial gradient gives a uniform bound on that set. More generally, suppose a set S is covered by radius- h Euclidean balls around verified queries and f_θ is K_θ -Lipschitz. Then

$$\sup_{x \in S} \|\nabla F_x(f_\theta(x))\|_2 \leq \max_i G_\theta(x_i) + \left(\frac{177}{8} K_\theta + \frac{87}{16} \right) h$$

holds, and its right-hand side is a valid uniform choice for G_θ in (77).

The last term follows from the state-to-control-gradient cross derivative of this particular model. A directed product of matrix-norm bounds gives a valid K_θ for the stored tanh network. It may be conservative. The theorem does not claim that 32 queries cover a 128-dimensional cube, and the existing initialization-agnostic cube bound is not replaced by an unverified learned-quality estimate.

Y.16.1. A new, predeclared finite query distribution

For each dimension $d = 8, 32, 128$, 32 uniform-cube draws are fixed with seed $19200 + d$ before their evaluation. All three stored neural initializers are compared with zero-start gradient descent and zero-start acceleration at target total losses $10^{-2}, 10^{-3}, 10^{-4}, 10^{-6}$. Each delivered floating-point plan is serialized and independently checked by its directed gradient certificate. Ordinary floating-point stopping diagnostics can request a check but cannot accept a plan. The maximum budget is 64 corrections, and no failed query is removed.

For a fair work count, acceleration returns its checked search point. Thus it does not pay an artificial second gradient call merely because its internal extrapolated and unextrapolated points differ. Every method pays for one gradient per correction and one final gradient evaluation. Neural inference, corrections, final directed checking, and the recorded offline training cost are identified separately.

TABLE Y.XIV
MATCHED DIRECTED TOLERANCE ON A NEW FINITE QUERY DISTRIBUTION

d	Target	Neural GD	Zero GD	Accelerated	Wins / 96
8	10^{-2}	1.05	3.44	3.84	96
8	10^{-3}	2.35	5.19	5.12	96
8	10^{-4}	4.27	7.16	6.97	96
8	10^{-6}	8.56	11.56	10.84	96
32	10^{-2}	1.86	4.72	4.94	96
32	10^{-3}	3.31	6.47	6.06	96
32	10^{-4}	5.36	8.66	8.00	96
32	10^{-6}	9.84	13.00	12.00	96
128	10^{-2}	5.00	6.00	5.97	93
128	10^{-3}	7.00	8.00	7.00	0
128	10^{-4}	9.06	10.00	9.00	0
128	10^{-6}	14.00	14.00	13.00	0

Mean correction counts; each method uses one gradient evaluation per correction and one final gradient evaluation. Neural means pool three frozen initializers and 32 predeclared queries. Classical means use the same 32 queries. A win means strictly fewer gradient evaluations than acceleration for the paired query; it is not a wall-time claim. Acceleration returns its independently checked search point. The set of 32 queries is not a cover of the state cube.

The learned map reduces online gradient work substantially at 8 and 32 coordinates. At 128 coordinates and tolerance .01, every neural seed uses five mean corrections, compared with six for zero-start gradient descent and approximately 5.97 for acceleration; it uses fewer gradients than acceleration on 31 of 32 paired queries per seed. At 10^{-3} the neural and accelerated counts coincide, and at 10^{-6} acceleration needs fewer corrections. These outcomes identify a limited, tolerance-dependent contribution from learning rather than an initialization-independent guarantee misattributed to training.

The finite-query initial-gradient theorem is evaluated as a second, separate work statement. The supplement reports the proved uniform correction budgets on each fixed query set. They can be more conservative than the actual checked stopping counts. Stored floating-point plans receive their own certificates; exact-real contraction is not used as an undocumented bound on a floating-point deployment graph.

Y.16.2. Amortization and the cost of completeness

For a declared query distribution and target, let T_{train} be offline cost and let $\bar{t}_0, \bar{t}_\theta$ be measured mean per-query times. When $\bar{t}_0 > \bar{t}_\theta$, the accounting break-even count is

$$Q_{\text{break}} = \left\lfloor \frac{T_{\text{train}}}{\bar{t}_0 - \bar{t}_\theta} \right\rfloor + 1.$$

No finite break-even is reported when the measured saving is nonpositive. The supplement gives both solver-only and final-check-inclusive counts, all three seeds, and every target. The training time comes from the stored training execution; query times are five-repeat medians in the new execution. These are explicit transferred-cost estimates, not a matched-hardware retraining experiment, statistical confidence intervals, or certified speedups. In particular, no end-to-end acceleration advantage is asserted for the 128-dimensional experiment.

The complete-cover feedback algorithm has a different scaling object. With M boxes, state dimension d , depth ℓ , and hidden width w , storing the value, time derivative, state gradient, and state Hessian requires $s_d = 2 + d + d(d+1)/2$ components. A direct dense-jet implementation costs on the order of

$$M\{s_d(dw + \ell w^2) + C_A\}$$

scalar operations, where C_A is the per-box cost of certified action maximization. Direct back-propagation stores on the order of $M\ell w s_d$ intermediate quantities. Streaming verification can replace M by its chunk size in working memory, but does not remove the total box work. For a tensor cover, $M = N_t \prod_i N_i$; interval dependency and action subdivision can increase the required counts. These costs are not suppressed in a dimension-free complexity claim. In contrast, one correction in the sparse nonlinear inventory model costs $O(Hd)$ after neural inference.

Y.17. Retained quadratic inventory validation

For historical validation of the dynamic certificate, retain the quadratic inventory-adjustment economy below. Its numerical gain problem reduces to two scalar Riccati equations; the genuinely vector-state nonlinear experiment is Section Y.15. With $M = \mathbf{1}\mathbf{1}'/d$,

$$dX_s = (AX_s + a_s)ds + .1dW_s, \quad A = -.2I + .3M, \quad (78)$$

and minimize

$$\mathcal{J}(t, x; a) = \mathbb{E} \left[\int_t^1 \{X'_s(I + .5M)X_s + |a_s|^2\} ds + .5|X_1|^2 \right]. \quad (79)$$

Controls are square-integrable adapted adjustments. Mean inventory affects drift and aggregate holding cost; idiosyncratic inventory also has its own adjustment dynamics and shocks. This is a coupled d -state dynamic problem, not an action maximization at frozen derivatives. It is a structured, uncalibrated application, and the structure is available to classical solvers as well.

The orthogonal and mean modes have $(a_\perp, q_\perp) = (-.2, 1)$ and $(a_m, q_m) = (.1, 1.5)$. Their Riccati gains satisfy

$$(p_i^*)' + 2a_i p_i^* - (p_i^*)^2 + q_i = 0, \quad p_i^*(1) = .5.$$

We train two neural gain outputs $\hat{p}_i(t) = .5 + (1 - t) \text{softplus}(N_i(t))$ against these *differential residuals*, not against exact Riccati solutions. The feedback is

$$\hat{a}(t, x) = -\hat{P}(t)x, \quad \hat{P} = \hat{p}_\perp(I - M) + \hat{p}_m M.$$

The network changes its gain with time and controls all d state coordinates. One learned pair is reused across the six dimensions; this reuse is not counted as six independent training runs.

THEOREM 16—State-global quadratic neural policy certificate: *Suppose interval differentiation on a complete time cover bounds the two neural Riccati residuals by ε_i . Put $\kappa_\perp = 1.4$, $\kappa_m = .8$, and*

$$D = \max_i \varepsilon_i \frac{1 - e^{-\kappa_i}}{\kappa_i}, \quad C = \frac{1 - e^{-.8}}{.8}.$$

For every $d \geq 1$ and every $|x|^2/d \leq 1$,

$$0 \leq \frac{\mathcal{J}(0, x; \hat{a}) - \mathcal{V}(0, x)}{d} \leq D^2 \left\{ C + \frac{.01}{.8}(1 - C) \right\}. \quad (80)$$

The bound concerns the continuous-time controlled diffusion on all of $\mathbb{R}^d$, not a finite collection of initial states.

The proof uses a quadratic performance-difference identity and a closed-loop Lyapunov moment bound. It replaces state covering by algebra valid on the entire state space. The certificate’s time-enclosure work depends on network size and the number of time cells, rather than a tensor grid in d . Applying the feedback requires a mean and coordinatewise operations, hence $O(d)$ work. This advantage comes from the model’s two invariant modes; it is not a generic escape from dimensional dependence for nonlinear Bellman equations.

Ten fixed seeds, two widths, and both predeclared checkpoints are retained. Every final checkpoint has per-coordinate certified loss at most 0.000095606; the complete range and costs appear in Table Y.XV. The certificate scales to total loss by multiplication by d . The supplement also reports the earlier checkpoint and fixed-network time-cover refinements separately. A structure-aware classical calculation solves only two scalar Riccati equations. Its low cost is reported rather than hidden: this application establishes accurate, reproducibly generated neural feedback with full dynamic certification, not superiority over the classical solution of this specially structured problem.

Y.17.1. A rigorously validated classical comparison for the quadratic model

The preceding quadratic model has only two independent gain equations. We retain it as a transparent validation case, not as the nonlinear high-dimensional benchmark. Its classical comparison is now certified rather than merely an RK45 diagnostic.

For each scalar mode, with coefficients $(a, q) = (-.2, 1)$ or $(.1, 1.5)$ and terminal gain $p(0) = .5$ in remaining time h , let $\kappa = \sqrt{a^2 + q}$ and $y = (.5 - a)/\kappa$. The exact gain is

$$p^*(h) = a + \kappa \frac{y + \tanh(\kappa h)}{1 + y \tanh(\kappa h)}.$$

It is increasing on $[0, 1]$. Directed endpoint values therefore enclose its full range on each time cell. The implemented policy stores a dyadic midpoint gain on each cell; MPFR bounds its

TABLE Y.XV
CERTIFIED NEURAL POLICY LOSS PER INVENTORY COORDINATE

Seed	Width	250 updates	1,000 updates	Train (s)	Verify (s)
16200	8	0.00117290	0.00009561	5.28	1.18
16201	16	0.00045470	0.00007661	7.05	2.85
16202	8	0.00051798	0.00006084	5.40	1.27
16203	16	0.00011929	0.00003574	7.44	3.26
16204	8	0.00179439	0.00008973	5.09	1.09
16205	16	0.00054105	0.00002530	6.28	3.05
16206	8	0.00036825	0.00004136	5.59	1.36
16207	16	0.00023854	0.00009053	6.34	3.34
16208	8	0.00099196	0.00005105	5.88	1.60
16209	16	0.00036174	0.00007427	5.08	2.29

Each checkpoint uses a complete 512-cell time cover and a state-global quadratic proof. Bounds hold for every $|x_0|^2/d \leq 1$. Multiply the reported bound by d for total loss at $d = 4, 8, 16, 32, 64, 128$. One gain pair is reused across dimensions. Timings cover one training trajectory and its two checks, not six duplicated runs.

TABLE Y.XVI
VALIDATED CLASSICAL POLICIES FOR THE TWO-MODE QUADRATIC MODEL

Time cells	Per-coordinate bound	Total bound, $d = 128$	Seconds
32	0.000304119	0.0389272	0.064
128	1.91901×10^{-5}	0.00245634	0.127
512	1.20221×10^{-6}	0.000153883	0.380
2048	7.51824×10^{-8}	9.62334×10^{-6}	0.929

Complete time-cell range enclosures use the closed-form scalar Riccati solutions and MPFR-directed arithmetic. The policies are dyadic piecewise-constant gains. Both generation and validation are included in seconds. This is the structured quadratic comparison, not the nonlinear application.

uniform gain error δ . All stored gains are strictly above .5, so the moment estimate used in the quadratic performance-difference identity gives per-coordinate loss at most

$$\delta^2 \left\{ C_{.8}(1) + \frac{.01}{.8} (1 - C_{.8}(1)) \right\}.$$

Total loss is this quantity multiplied by d . Table Y.XVI reports both units and includes the full generation-and-verification cost. In particular, the 128-cell classical policy has a $d = 128$ total bound below .002457, and the 2,048-cell policy has a total bound below 9.624×10^{-6} . This is the appropriate rigorous comparison for a problem whose exact structure is so favorable to classical methods.

Y.18. Historical sampled-training and restricted-baseline experiments

This section retains the R16 experiment without substituting later results for its observations. The original nonlinear economy is unchanged in all experiments in this section. The architecture is (51)–(52). Seeds 16100–16109 use two hidden tanh layers, with width 16 for even seeds and width 32 for odd seeds. Every trajectory retains the 800- and 2,400-update checkpoints. Each actor update follows three critic updates; Adam learning rates are .0002 and .0005, respectively, with batch size 128, double precision, and one Torch thread. A quarter of each sampling mixture

targets preference and lower-wealth boundary layers. No optimal-value labels are used. These settings were fixed before the canonical remote execution; no best-seed substitution is made.

For every checkpoint, MPFR interval differentiation encloses the complete $[0, 1] \times [1.2, 2.8] \times [.5, 2]$ cylinder on $4 \times 16 \times 16$ cells. The continuous consumption, adjustment, and portfolio maxima use the original action set. The certificate allows a componentwise head perturbation of 2^{-24} , clipped to the head bounds before the portfolio-distance multiplication. This location of the perturbation matters: an unrestricted additive error in the final portfolio could invalidate the boundary-inaccessibility argument. The proof concerns mathematical real-valued networks with stored dyadic weights under this explicit output contract. It does not certify a particular machine-library tanh implementation or an SDE simulation stack.

TABLE Y.XVII
FULL-HORIZON NEURAL CHECKPOINTS IN THE ORIGINAL ECONOMY

Seed	Width	800 updates	2,400 updates	Train (s)	Verify (s)
16100	16	30.19468	54.00405	165.70	28.36
16101	32	41.45535	96.87386	180.11	79.46
16102	16	28.84397	53.60464	156.26	29.22
16103	32	42.60784	93.28879	162.35	59.47
16104	16	29.69505	68.44466	138.88	23.06
16105	32	48.37134	100.68512	139.71	65.74
16106	16	30.03684	59.19474	129.16	27.60
16107	32	45.64637	102.35557	143.36	62.72
16108	16	28.54732	49.67500	132.95	25.88
16109	32	44.11682	95.77234	74.49	23.67

Each bound is uniform over the original interior state rectangle at time zero, $k = 2$, and uses 1,024 cells with the original continuous action maximum. The trace contribution is exactly zero. Training time covers both checkpoints in one trajectory; verification time sums their two audits. No checkpoint meets 10^{-2} . Resource measurements are descriptive, not matched-accuracy efficiency comparisons.

Table Y.XVII retains every prescribed checkpoint. The exact trace budget is zero, but the positive optimal residual and negative policy residual still produce large bounds. An increase in optimization budget uniformly loosens these certificates. Since the bounds also include interval dependency, this observation does not by itself prove that the realized policy payoff deteriorates. The refinement experiment holds the first seed’s final network fixed and doubles the cover in each coordinate; the complete cell records separately retain the signed budget and the more conservative unsigned residual/action budget. The hard-all-face ablation is retained as well. Its finite-resolution certificate is not evidence that the excluded approximation space becomes consistent as residual error tends to zero.

Y.18.1. State-space candidate generators

We implement a controlled Markov-chain approximation with nonnegative transition weights and a semi-Lagrangian weak Euler approximation with correlated shocks. Each uses the same utility, discount, action bounds, state domain, and first-exit settlement. Refinement uses 9, 17, and 25 nodes per state coordinate, 64, 128, and 256 time steps, and progressively richer bounded control-head grids. The Markov-chain implementation uses wide diffusion jumps and a checked nonnegative stay probability; the semi-Lagrangian method uses four symmetric shock directions. Approximate trajectories apply the liquidation contract at their first straight-segment exit. This

discretization is a candidate-generation device, not an asserted exact solution of the stopped diffusion.

For verification, each numerical policy is extended by bilinear interpolation in state and piecewise-constant interpolation in time. Its portfolio head is multiplied by $(2 - x)/1.5$, exactly as in the neural candidate class. These are feasible policies in the unchanged economy, although the candidate class is narrower than the original collection of all admissible controls. The upper comparison still uses every original control. We enclose the interpolated policies with the same complete verifier and the fixed seed 16100 final critic. Thus their discrete value outputs are never used as continuous-time upper or lower bounds.

TABLE Y.XVIII
HISTORICAL RESTRICTED STATE-SPACE CANDIDATES WITH A COMMON WITNESS

Method	Nodes/axis	Time steps	Regret bound	Generate (s)	Verify (s)
Markov chain	9	64	48.98350	0.12	3.84
Markov chain	17	128	48.95819	3.41	3.73
Markov chain	25	256	49.45469	42.90	3.74
Semi-Lagrangian	9	64	48.93352	0.08	3.67
Semi-Lagrangian	17	128	49.47658	1.06	3.69
Semi-Lagrangian	25	256	49.69852	25.90	3.74

All six policies are continuous bilinear state interpolants of bounded heads, piecewise constant in time, with the portfolio-distance factor. The common witness is the predeclared seed 16100 final critic, not a fitted value from either baseline. Generation discretizes the model; the separate certificate evaluates the continuous-time policy and unchanged stopping contract. None meets 10^{-2} , so this is not a matched-accuracy ranking.

The common witness standardizes the verification contract but is inaccurate; these loose bounds are not a comparison of solver accuracy. None of these six candidates attains the 10^{-2} target under that certificate. The experiment supplies two distinct state-space constructions, three refinements, and actual continuous-time policy verification; it does not establish a matched- 10^{-2} efficiency frontier or an optimal implementation of either classical method. In particular, a loose common upper witness cannot rank realized policy quality merely by the size of a certified upper bound. Section Y.14 reports the later method-specific witnesses. Their remaining error is not an omitted term that can be declared negligible.

Y.19. Arithmetic independence and fresh price-continuum construction

We distinguish two complete calculations. The first re-evaluates all eighteen historical policy/dual nodes with a new arithmetic implementation. Every basic operation and every exponential, logarithm, square root, and tanh endpoint is evaluated using 128-bit MPFR directed rounding, followed by directed conversion to binary64 endpoints. Stored network weights and policy constants are interpreted as exact dyadic inputs, while model decimals are enclosed as exact rationals. Invalid domains and division through zero fail closed. The kernel passes 450 exact-rational regression checks, but these checks are diagnostics rather than a formal proof of the implementation.

The full-node MPFR calculation recomputes stopped policy payoffs, dual source integrals, expenditure intervals, and the final exact rational envelope. It does not read historical certified endpoints as inputs. It does share the R14 analytic decomposition, rational Gauss-node isolation, derivative remainder lemmas, and stopping inequalities. Thus it is an independent established-arithmetic cross-check of the decisive calculation, not an independent discovery of

every mathematical premise. Its complete price-uniform bound is 0.009969007708; full lower and upper intervals, not three pointwise comparisons, are retained at every node.

The second calculation starts from the economic model alone. It initializes the price set at $1/2$ and 8 , generates a 16-slab feasible time-control policy at each node by deterministic SLSQP initialization, and fits the corresponding dual from deterministic starts. It then selects the worst exact current envelope breakpoint, rounded to a spacing of $1/1024$; a repeated node triggers bisection of the widest price interval. The declared stopping rule is a proved uniform gap below .01 or thirty freshly generated nodes. No inherited policy, dual pilot, or historically selected intermediate node is read. Optimizer attempts, unsuccessful intermediate envelopes, and resource use are retained.

TABLE Y.XIX
COMPLETE PRICE-CONTINUUM CALCULATIONS ON THE ORIGINAL ECONOMY

Calculation	Nodes	Uniform regret	Wall time (s)	Peak RSS (MiB)
Historical policies; MPFR	18	0.009969007708	1065.89	276.3
Fresh model-to-library	20	0.009981645276	415.79	497.5

Both cover $(t, u, x) = (0, 2, 1.25)$ and every $k \in [.5, 8]$. Historical inputs are frozen policies and dual pilots, not inherited endpoints; MPFR recomputes every node. The fresh calculation starts from model constants and deterministic initializations, with no historical policy or pilot. Its arithmetic is the retained rational–Taylor implementation; the MPFR label does not apply to the fresh library. Peak MPFR RSS is the largest worker, not summed concurrent memory.

The fresh pipeline reaches 0.009981645276 with 20 generated nodes. This result belongs to the time-control/dual method, not to the neural feedback. The fresh calculation uses the retained rational–Taylor interval kernel; the independent MPFR label belongs only to the separately re-evaluated eighteen-node object. Fresh generation closes a reproducibility distinction without removing the dual relaxation’s accuracy floor. The original 16/32/64-slab experiment and its failure to attain .005 are retained below. Neither extra price nodes nor quadrature refinement alone supplies a convergence law to arbitrarily small error.

The historical R16 replication command executes that revision’s neural trajectories and checks, both state-space solvers, the fresh price library, the historical-node MPFR audit, the structured inventory experiment, and the normalization calculations. Source identities, input hashes, complete cell records, failures, and hardware/software metadata accompany the results. Timings in different concurrent experiment processes are descriptive; they are not used to infer solver superiority from unequal accuracy or unequal processor contention.

Y.19.0.1. Historical evidence retained. The following original-economy experiments, bounds, comparisons, and welfare-resolution calculations are the R14 research record. Their sources and results are unchanged. They are not additional R16 training runs or endpoints of the fresh R16 library.

Y.20. Retained R14: executed neural verification on the original economy

The retained R14 neural experiment uses a state-feedback actor and critic with two hidden layers of 48 tanh units. Inputs are $(2t - 1, (u - 2)/.8, (x - 1.25)/.75)$. The actor’s output transformation is

$$c = .425 + .375 \tanh a_1, \quad \theta = .2 \tanh a_2, \quad p = .15 + .65 \tanh a_3.$$

TABLE Y.XX
WHOLE-DOMAIN NEURAL CERTIFICATES ON THE UNCHANGED ECONOMY, $k = 2$

Updates	Cells	$2b$	$\int e^{-\rho s}(2e + q)ds$	Regret upper	Seconds
100	2048	14.9778	8.8713	23.8491	4.40
300	2048	14.2970	9.0431	23.3400	4.75
800	256	14.2484	25.9010	40.1494	0.42
800	2048	14.1073	16.1408	30.2480	4.41
800	16384	14.0463	9.4470	23.4932	25.09

Upper bounds are rounded upward. Time is the measured verification stage, excluding training. Every row covers the complete domain with zero skipped and zero failed cells. The last three rows refine the cover for the same 800-update snapshot. None attains the .01 target.

TABLE Y.XXI
STOPPING-TRACE ERRORS FOR THE FINEST NEURAL COVER

Stopping face	Cells	Trace error upper
$u = 1.2$	512	4.63971373
$u = 2.8$	512	7.02049755
$x = .5$	512	5.17780198
$x = 2$	512	7.02310795

Each entry is a uniform bound over the complete face and time interval. The comparison theorem pays the largest face bound twice. The error is an evaluation of the stored critic against $G - 8(1 - t)$, not a change in that settlement.

The critic is $v(t, u, x) = G(u, x) + (1 - t)N(t, u, x)$, so its terminal trace is exact. The first-exit trace is not imposed by this architecture and must be checked.

All weights are freshly generated from seed 1401. Each actor-improvement step is preceded by three critic residual updates, with batch size 128. Adam learning rates are .0005 for the critic and .0002 for the actor, and the boundary-loss coefficient is .025. Snapshots at 100, 300, and 800 updates are retained. The experiment uses the 48-by-48 tanh architecture of the retained comparisons, but its continuous economic residual and boundary objective are a new adapter; it is not the same finite-step training objective as the external benchmark. The optimizer and sampled objective are proposal mechanisms only.

The certification program uses an independent interval implementation for network evaluation and differentiation. It checks all cells of $[0, 1] \times [1.2, 2.8] \times [.5, 2]$, all four first-exit faces, and the terminal face. A componentwise output neighborhood of radius 2^{-24} is included in the action ranges. This proves robustness for any admissible action within that neighborhood of the mathematical real-tanh network. It does not assert an undocumented error bound for a machine-library tanh routine or for a discretized diffusion simulator.

Table Y.XX separates the stopping trace from the integrated policy residual and action gap. The finest cover has 16,384 interior cells and 512 cells on each stopping face. It gives the time-zero bound 23.493172463 and an all-initial-times bound below 23.656877772. The terminal trace is zero. The 800-update sampled residual mean square is approximately .0079446, but this does not control either the worst-case residual or the first-exit trace. Moreover, the 800-update snapshot is not uniformly better than the 300-update snapshot at the same coarse cover. Both outcomes remain in the record.

The coefficient-transport calculation uses the same stored snapshot and cover, with

$$r \in [.019, .021], \quad \mu \in [.058, .062], \quad \sigma_u \in [.049, .051], \quad \sigma_x \in [.195, .205],$$

TABLE Y.XXII
INDEPENDENT CENTRAL-STATE CERTIFICATES FOR ALL RETAINED PRICES

k	Policy lower	Optimum upper	Regret upper	Policy width
0.5	-1.27266301	-1.26281817	0.009844825	1.13e-11
0.875	-1.27897331	-1.26925551	0.009717794	1.13e-11
1.25	-1.28479324	-1.27510918	0.009684054	1.13e-11
1.625	-1.29013398	-1.28045672	0.009677256	1.13e-11
2	-1.29501788	-1.28533017	0.009687701	1.13e-11
2.5	-1.30083961	-1.29112854	0.009711056	1.13e-11
3	-1.30590538	-1.29616492	0.009740453	1.13e-11
3.5	-1.31024866	-1.30047467	0.009773977	1.13e-11
4	-1.31389776	-1.30408950	0.009808244	1.13e-11
4.25	-1.31546968	-1.30564683	0.009822839	1.13e-11
4.5	-1.31688907	-1.30705332	0.009835742	1.13e-11
5	-1.31935183	-1.30949330	0.009858515	1.13e-11
5.5	-1.32141556	-1.31153758	0.009877966	1.13e-11
6	-1.32317048	-1.31327568	0.009894785	1.13e-11
6.5	-1.32468139	-1.31477191	0.009909467	1.14e-11
7	-1.32599609	-1.31607368	0.009922399	1.14e-11
7.5	-1.32715060	-1.31721673	0.009933858	1.14e-11
8	-1.32817262	-1.31822851	0.009944109	1.14e-11

Every upper endpoint covers all original adapted controls; every lower endpoint belongs to the identified feasible policy. The last column is the width of that policy's payoff enclosure, not its regret. Regret uses unrounded endpoints. All 18 cases are included, and no inherited generation time is counted as zero.

$$\rho \in [.039, .041], \quad k \in [1.9, 2.1],$$

and correlation fixed at $-.25$. Here μ denotes the risky-wealth drift coefficient multiplying p . The action bounds, state domain, and settlement are unchanged. Proposition 1 gives a time-zero regret bound below 23.530407814 uniformly over this model box. This demonstrates an executed transport mechanism beyond affine objective continuation. It does not make the base neural certificate accurate: the same boundary and approximation errors remain visible.

The finest computation records separate cellwise evaluation, action, and transport quantities. A diagnostic comparison of the interval neural derivatives with automatic differentiation checks 140 derivative components at 20 states and 60 actor components, with no discrepancy outside the numerical tolerance. This diagnostic detects implementation disagreement; it is not substituted for the full-cell interval proof.

Y.21. Retained R14: independent numerical evidence and robustness

The sharp economic comparison uses a second implementation that imports none of the inherited primal or dual certifiers. Its basic arithmetic encloses exact rational constants and exact stored binary64 inputs. Addition, multiplication, division, and reductions are rounded outward. Exponentials and logarithms are bounded by rational series with analytic remainders; tanh is obtained from the enclosed exponential. The Gauss nodes are isolated by exact polynomial signs and the weights are enclosed by their defining formula. The supplement states the floating-point model and the analytic remainder bounds, so these assertions are not based solely on rerunning a production function.

The independent primal evaluator uses 262,144 integrand evaluations per 16-interval policy. The fine dual uses 262,144 source boxes per price. Complete supporting-plane covers certify

TABLE Y.XXIII
SEPARATE QUANTITIES IN THE INDEPENDENT $k = 4.25$ CALCULATION

Quantity	Lower	Upper
Source integral	-2.38605836997	-2.38604817428
Covariance	-0.00017879144	-0.00016944433
Reference-control gap	0	0.00003256016
Conditional variance	0.00003697298	0.00003697299
Localization	2.4594E-7	2.4595E-7
Primal payoff	-1.315469674846	-1.315469674834

Intervals are displayed outward. The source-remainder enclosure, Gaussian tail, terminal tangent, reference cost, and arithmetic allowances are present in the machine-readable record. The conditional-variance allowance is positive and is not removed when quadrature is refined.

TABLE Y.XXIV
CONTINUUM CERTIFICATES UNDER INDEPENDENT VERIFICATION AND PERTURBATION

Certificate	Uniform regret upper
R12 retained implementation, original inputs	0.009994024847
R14 independent implementation, finer cover	0.009969007711
R14 perturbed witnesses and enlarged allowances	0.009970977450
R14 joint state–cost rectangle	0.012095431554

The first three rows refer to the central initial state. The fourth covers the additional initial-state rectangle K and is not compared with the central-state .01 tolerance as though the domains were identical. The perturbation experiment changes proof witnesses and allowances, not the economic primitives.

a negative second-derivative upper bound below $-.06717$ on the first preference region and a strict derivative separation on the remaining region. They also certify a source floor above -6.98956 , a source ceiling below zero, and an absolute preference coefficient below $.97119$. These independently computed margins imply the analytic bounds $S_0 \geq -7.1$ and $|\beta| \leq 1.1$ used for stopping continuation. The complete old policy and dual inputs are retained and identified by their content hashes.

Exact price postprocessing is independently implemented by enumerating affine intersections with rational arithmetic, rather than importing the original hull routine. Its input expenditures are newly computed from the stored preference actions with an explicit stopping correction. Thus the upstream policy values, upper values, expenditure intervals, and downstream envelope all have a new execution path. Reproducibility of this path remains distinct from independence of its analytic premises.

Y.21.1. Stress tests for the entire library

The robustness experiment changes every dual pilot: y_0 is increased by 10^{-6} , and its reference preference drifts receive alternating perturbations of $\pm 10^{-6}$ with inward projection into the admissible reference interval. Simultaneously, the proof uses the weaker source floor -7.11 , the larger coefficient bound 1.11 , twice the Gaussian-tail allowance, and 1.1 times the conditional-variance allowance. The resulting continuation floor remains strictly above -8 . All 18 upper certificates are recomputed, while the original economy and feasible lower policies remain unchanged.

The unperturbed independent bound leaves approximately 3.10×10^{-5} below $.01$, compared with approximately 5.98×10^{-6} for the retained R12 bound. The perturbed calculation remains

TABLE Y.XXV
FRESH GENERATION AND INDEPENDENT VERIFICATION AS POLICY RICHNESS INCREASES

Slabs	Generation	Dual fit	Primal audit	Dual audit	Regret upper
16	0.235	0.284	0.737	7.408	0.009688058
32	0.364	0.434	0.926	6.815	0.009669548
64	0.847	0.379	1.791	5.748	0.009665074

All stage times are seconds. Each case starts without inherited policy or dual inputs, and all three optimizations report success. End-to-end measured case times are 8.67, 8.54, and 8.77 seconds. The experiment changes policy richness, unlike a price-node refinement experiment. All cases pass .01; none passes .005, .0025, or .001.

below .01. Neither statement claims a bound below .009. An additional exact sensitivity calculation shows how a common enlargement of every upper and lower endpoint changes the final envelope: enlarging each side by δ increases the guaranteed gap by at most 2δ . It is a sensitivity result conditional on valid node intervals, not a replacement for the independent primitive audit.

A third, materially different decomposition integrates the capped and interior conjugate using closed truncated-lognormal moments. At prices .5, 4.25, and 8, a separate program computes the source integral, initial affine coefficient, and covariance term at 40 and 70 decimal digits. Every computed quantity lies inside the independently certified interval. The agreement provides a useful cross-check of algebra and decomposition; these point-valued high-precision quadratures are not themselves rigorous enclosures, and their convergence is not used as a proof of the interval certificate.

Y.22. Retained R14: accuracy, computational work, and comparators

A certification time measured from frozen proof objects is not the cost of generating those objects. The 18-node independent audit takes approximately 103.08 seconds of measured wall time in the retained run, but starts from existing policies and dual pilots. The 72-corner state audit takes approximately 29.71 seconds. These are verification costs. The generation costs of inherited and interrupted historical attempts are not reconstructed from their absence and are not assigned zero.

The fresh neural run records all 800 actor updates and 2,400 critic updates, the fixed seed, every retained checkpoint, and resource usage. Its recorded execution after program entry is approximately 13.08 seconds, with interpreter and import time separately recorded. It uses one Torch CPU thread and no GPU. The execution environment reports an AMD EPYC 9V74 processor and five exposed logical CPUs; this identifies the environment, not exclusive ownership of a physical machine. Stage times recorded during overlapping local experiments are descriptive clocks, not controlled hardware-efficiency comparisons.

Y.22.1. A newly generated policy-richness frontier

The same original economy at $k = 2$ is also solved within deterministic time-control classes of 16, 32, and 64 intervals. Each problem starts from a fixed feasible constant-consumption construction and preference action .05. Constrained sequential quadratic optimization uses analytic gradients to propose a policy. A separately initialized scalar-deflator and reference-drift fit proposes its upper witness. The fresh policy is then checked by the independent stopped evaluator and the fresh upper witness by the independent dual. The fitting objective is not counted as a certificate.

The frontier is informative because its approximation object is held distinct from its upper relaxation. Policy refinement improves the lower bound, but the independently fitted affine upper

bound is nearly unchanged. A fixed specialized relaxation can therefore limit the attainable regret even when quadrature error is very small. This calculation is not evidence of convergence to arbitrary tolerance, and it does not establish convergence of the neural optimizer.

The classical comparator is a freshly executed direct time-control optimization. It is a serious feasible-policy and upper-bound calculation on the same original economy, but it is not two distinct state-space HJB methods and it does not furnish a matched-accuracy advantage for the neural method. A comparative claim requiring monotone PDE, semi-Lagrangian, or controlled Markov-chain baselines at the same certified accuracy remains outside the completed experiment. Likewise, the preserved Riccati-reference study used structured gain outputs; its errors cannot be relabeled as errors of the present 48-by-48 tanh actor and residual-training pipeline.

Y.22.2. Retained comparisons and high-dimensional scope

The historical SOC-MartNet holdouts, method-specific tuning studies, and clipped-feedback comparator are preserved without suppressing unfavorable panels. The clipped-feedback comparator's lower mean cost and the absence of a robust ordering in several higher-dimensional seed panels limit what can be inferred about candidate-generation quality. They do not invalidate the stopped comparison theorem, just as the theorem does not establish empirical superiority.

The 128-dimensional experiment in the retained record certifies a rank-one-coupled Hamiltonian action oracle at frozen jets. Its nonconvex coupling can be reduced to a scalar aggregate search with strongly convex conditional subproblems. This is an action-optimization result. It is not a certified solution of a 128-dimensional dynamic economic model. The actual neural state-domain calculation in this revision concerns the two-state economy in Section 2; no action-dimension result is used to inflate that state dimension.

Y.23. Retained R14: what the certified counterfactuals identify

For any policy, raising k subtracts $(q - p)B(\pi)$, where

$$0 \leq B(\pi) = \mathbb{E} \int_0^\tau e^{-\rho s} \theta_s^2 / 2 \, ds \leq .02 \frac{1 - e^{-\rho}}{\rho}.$$

Consequently V_k is nonincreasing and convex. Monotonicity determines a weak sign without numerical computation, but not a strictly positive or economically well-resolved welfare loss.

Let $[L(k), U(k)]$ be the independently certified central-state value envelope. For $p < q$, a valid loss interval is

$$[\max\{0, L(p) - U(q)\}, \min\{U(p) - L(q), .02C_\rho(1)(q - p)\}], \quad C_\rho(h) = \frac{1 - e^{-\rho h}}{\rho}. \quad (81)$$

A positive lower endpoint establishes strict loss. We report a separate relative-resolution criterion: the width of this interval is no greater than its positive lower endpoint. Neither criterion follows automatically from a .01 bound on policy regret.

The contrast between $2 \rightarrow 8$ and $4 \rightarrow 4.1$ is substantive. The first comparison is strictly positive and relatively resolved by the current intervals. The second has a small absolute upper bound but no certified positive lower bound from interval separation. A policy may therefore be sufficiently accurate for an absolute deployment tolerance while the current value bounds remain too wide for a local comparative-static statement.

The policy library also has a direct economic interpretation. Its selector chooses a stored control vector on each exact price region; within that region no new optimization is required. The

TABLE Y.XXVI
CERTIFIED WELFARE LOSSES WHEN THE ADJUSTMENT PRICE RISES

p	q	Lower loss	Upper loss	Strict	Relative
0.5	2	0.01266717	0.02940793	Yes	No
2	8	0.02321063	0.04284245	Yes	Yes
4	4.1	0E-8	0.00196053	No	No
4	8	0.00433075	0.02408312	Yes	No
6	8	0E-8	0.01489694	No	No

“Strict” means a strictly positive lower endpoint. “Relative” means interval width no greater than that endpoint. A “No” in the strict column does not reverse the economic monotonicity theorem; it means that strict loss is not separated from zero by these intervals.

TABLE Y.XXVII
PRICE REGIONS IN WHICH WELFARE LOSSES ARE SEPARATED

Anchor p	Strictly positive loss	Relative width at most one
0.5	(1.093292, 8.0]	(2.617755, 8.0]
2	(2.886123, 8.0]	(6.450558, 8.0]
4	(6.207877, 8.0]	–
6	–	–

Decimal endpoints are descriptive approximations of exact rational thresholds, not proof endpoints. The exact fractions and open-endpoint convention are in the price-audit file. A dash means no such region up to price eight is established by this predicate. The relative criterion here uses the untruncated difference of value intervals and is therefore conservative.

supplement lists the selected regions, their widths, consumption ranges, and average preference adjustments. These are properties of feasible chosen policies. They are not claims that the optimal policy itself is piecewise constant in the price, or that policy expenditures equal derivatives of the unknown optimum.

The tolerance .01 is an absolute loss criterion in the utility units of (5). It is not a scale-free welfare measure. The relative and strict-loss diagnostics in (81), rather than the target alone, determine which of the reported economic comparisons are resolved. No empirical calibration or external welfare normalization is imposed to make this declared numerical target appear economically universal.

Y.23.1. A consumption-top-up interpretation

An absolute utility tolerance is not an economic unit. We therefore define a compensation experiment before reporting it. Keep the policy’s preference, wealth, adjustment cost, and stopping paths fixed, and provide an externally financed proportional consumption top-up $\lambda \geq 0$. Only the consumption entering instantaneous utility changes, from c_s to $(1 + \lambda)c_s$. This is an in-kind welfare comparison; it is not a claim that the extra consumption is financeable under the unchanged budget or the original consumption cap.

PROPOSITION 21—A scale-normalized compensation bound: *For a feasible time-control policy with $c \leq 1$, $u \in [1.2, 2.8]$ until stopping, suppose $V - J^\pi \leq \epsilon$ and $\mathbb{E} \int_0^\tau e^{-.04s} ds \geq A > 0$. If $1 - 1.8\epsilon/A > 0$, a top-up no larger than*

$$\bar{\lambda} = (1 - 1.8\epsilon/A)^{-1/1.8} - 1 \quad (82)$$

is sufficient to compensate the certified policy loss in the specified welfare experiment.

For the fresh central-state price library, the rigorous original stopping calculation supplies

$$A \geq \frac{1 - e^{-.04}}{.04} \left\{ 1 - 2 \exp\left(-\frac{.58^2}{2(.05)^2}\right) \right\}.$$

The resulting uniform top-up bound is 1.033008 percent. It is not assigned to a neural policy. Under a common positive rescaling of the entire welfare criterion, both the loss and the marginal utility gain rescale together, leaving this compensation experiment unchanged. This does not supply an empirical calibration or a wealth-equivalent claim. The exact definition prevents a numerical threshold from being presented as economically meaningful merely because it is slightly below .01.

Y.24. Budget-feasible welfare compensation in the original economy

An absolute utility tolerance is not self-interpreting. The consumption-top-up experiment in the preceding section specifies an external financing convention, but it is not a budget-feasible wealth equivalent. We now give a sufficient initial-wealth compensation with an explicit feasible policy adjustment under the original budget and stopping rules. The object being compensated is the sharp deterministic time-control library, not the original neural policies.

Let $\pi = (c, \theta, 0)$ be the selected library policy at $(u_0, x_0) = (2, 1.25)$ and an arbitrary $k \in [.5, 8]$. Add initial wealth Δx and a constant consumption increment

$$d_c = \frac{\Delta x}{C_{.02}(1)}, \quad C_{.02}(1) = \frac{1 - e^{-.02}}{.02},$$

while retaining the preference control and zero portfolio. Before preference stopping, the wealth difference solves $\dot{z} = .02z - d_c$, $z(0) = \Delta x$, and has $z(1) = 0$ and $z(s) \geq 0$. No extra payoff, transfer after time zero, or modified consumption cap is introduced.

PROPOSITION 22—Sufficient wealth compensation: *Suppose every selected original and compensated policy avoids wealth stopping before time one, satisfies $c + d_c \leq .8$, and has preference-exit probability at most p_e . Then*

$$J_k(0, 2, 1.25 + \Delta x; \pi^+) - J_k(0, 2, 1.25; \pi) \geq d_c \cdot 8^{-1.2} C_{.04}(1)(1 - p_e). \quad (83)$$

Consequently, if the right side exceeds the original library's uniform regret bound ε_{lib} , the compensated feasible policy satisfies $J_k(0, 2, 1.25 + \Delta x; \pi^+) \geq V_k(0, 2, 1.25)$ for every library-covered price.

The comparison is sufficient compensation with an explicit policy change. It is neither the exact minimum wealth equivalent of the unchanged feedback nor a calibrated empirical welfare estimate.

Directed calculations check every one of the 20 freshly generated library policies. The preference-exit bound is below 1.207×10^{-29} . The analytic sufficient wealth increment is at most .007713143; the implementation chooses the simple rational increment $39/5000 = .0078$, or .624% of initial wealth. Its consumption increment is enclosed by $[\text{.007878259998266659}, \text{.007878259998266711}]$. All compensated consumption paths satisfy the original .8 cap, all terminal wealth values remain strictly above .5, and the global wealth upper bound is below $1.257958 < 2$.

The resulting uniform utility gain is at least .010094048517141695, greater than the original uniform regret upper bound .009981645275150275, with a verified slack exceeding .00011240324. An additional replay through the original stopped-payoff evaluator is retained separately. Thus the normalization is tied to a feasible economic experiment rather than to the numerical size of .01 alone. Its coverage remains the original library's central initial preference and complete price interval, not the full-state neural experiment.

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HISTORICAL ANNEX II: REVIEWED R24 TECHNICAL SUPPLEMENT

The following 58 pages are the complete technical supplement of the same reviewed object. Its full derivation chain, prior experiments, and adverse outcomes remain available together with the unchanged source files. These pages retain their original typography and numbering.

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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Complete crossed-configuration ledgers, proof dependencies, root and rejection diagnostics, deployment costs, and retained historical technical material accompany the main paper. Prospective executions, post-observation repairs, analytic certification, and arithmetic diagnostics are identified separately.

1. CURRENT REPLICATION OBJECT

R24 contains thirty tuning trajectories, twenty-eight held-out trajectories, thirty-six multicell trajectories, twelve reference trajectories and four post-hoc failure trajectories. All 110 are retained. There are 286 prescribed checkpoint records, of which 140 belong to the central held-out experiment. These categories are not pooled to inflate the number of independent replications.

1.1. *Chronology and the operational repair*

Protocol commit `b54afa8c9107d70094a1d84be4a3c0279916530c` precedes the scientific execution. Primary run `35832563885` completed all tuning, held-out and multicell computations. It then stopped while strict JSON encoded an unused financing-root diagnostic as infinity in the reference economy, which has no financing equation. The exact metadata patch encodes the unused nonfinite value as null; it changes no objective, seed, rate, checkpoint, policy or certificate.

The full primary archive is retained at `archive/primary_execution.zip`, SHA-256 `cf8da93098d0e32c6d6bd219e57aedc84a9b31d1adbee941a004a85d2a44cde8`. Recovery run `35833755321` restored this archive, did not repeat or select among primary trajectories, and completed only the reference and failure stages. Every primary result was checked byte-identical. Failed execution, source identity and correction are preserved. Reference/failure stages used a separately provisioned machine; cross-stage times are not controlled same-machine comparisons.

Every pre-existing file from the reviewed branch is checked against its Git blob identifier in the publication audit. All new evidence and manuscripts are confined to R24 paths. The preceding main-text and supplement components remain in the new manuscripts with explicit historical notices; their revision-specific observations do not override the R24 domains.

1.2. *Analytical hypotheses and numerical instantiations*

2. COMPLETE COST ACCOUNTING

Generator-only curves exclude scientific instrumentation. Certified-solve cost includes preprocessing, initial checking and every checkpoint check. Instrumentation, tuning and the historical upper library are additional and separately listed. Shared initial policies and initial geometry can be reused, but standalone columns conservatively charge each method. Their sum is not presented as the measured wall time of the shared study. The dual reuse allocation is valid only when the same unrestricted library remains applicable to the economy.

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TABLE I
DEPENDENCY MAP FOR THE CURRENT CLAIMS

Claim	Analytical assumptions	Executed evidence
State realization	Common observed factor; constant weights; conditional-price identity; positive accounts; common shocks.	Frozen expert coefficients and the unchanged financing/feasibility checker.
Augmented value	Original full Brownian filtration; causal memory; identical actions and stopping; admissible concatenation.	Analytical theorem, not a numerical full-state accuracy claim.
Regional regret	Joint concavity on the actual range; shared shocks; convex explicit upper; uniform nonsynchronized exit allowance.	Nine actual experts per arm, every cell, directed vertex certificates.
Payoff replacement	New lower bound strictly exceeds old upper bound at identical initialization.	Every replayed acceptance and rejection.
Output geometry	Differentiability and local Jacobian Lipschitz continuity for the Taylor remainder.	Actual Jacobian and adaptive-metric spectra and transport remainders; ordinary diagnostics.
Gradient error	Common price bracket; positive price derivative; derivative approximation and offset-Lipschitz bounds.	Conditional theorem; ordinary quadrature differences do not instantiate a rigorous stopped-gradient bound.
Reference gap	Deterministic reference; exact constant optimum in both continuous parameter classes.	Directed reference/candidate intervals and exact slab integration.

TABLE II
STANDALONE COSTS, INSTRUMENTATION AND UPPER-LIBRARY ALLOCATION

Rule	Arm	Prep.	Initial	Diag.	Cert.	+ dual	Tune
8/16	Neural-Adam	0.000	2.063	0.127	13.744	429.533	16.364
8/16	Direct-Adam	0.000	2.063	0.111	13.656	429.445	15.942
8/16	Diagonal-Adam	0.002	2.063	0.111	13.574	429.363	16.062
8/16	Tangent-Adam	0.002	2.063	0.112	13.573	429.362	15.991
8/16	Whitened-Adam	0.002	2.063	0.111	13.617	429.406	15.927
8/16	Neural-L-BFGS-B	0.000	2.063	0.110	13.333	429.122	0.000
8/16	Direct-L-BFGS-B	0.000	2.063	0.105	12.784	428.573	0.000
12/24	Neural-Adam	0.000	2.060	0.125	14.427	430.216	16.364
12/24	Direct-Adam	0.000	2.060	0.111	14.272	430.061	15.942
12/24	Diagonal-Adam	0.002	2.060	0.110	14.253	430.042	16.062
12/24	Tangent-Adam	0.002	2.060	0.110	14.286	430.075	15.991
12/24	Whitened-Adam	0.002	2.060	0.113	14.420	430.209	15.927
12/24	Neural-L-BFGS-B	0.000	2.060	0.111	13.722	429.511	0.000
12/24	Direct-L-BFGS-B	0.000	2.060	0.106	13.037	428.826	0.000

Seconds. Diag. is additional optional scientific instrumentation. Cert. includes generation, preprocessing, initial and checkpoint checking. + dual adds the unamortized historical 415.789-second construction; divide that charge by M for M valid reuses. Tune charges all tuning trials and is additional. The historical dual charge is not a same-machine timing comparison.

3. CERTIFIED CHECKPOINT CURVES

Each curve includes caps 25, 50, 100, 200 and 400. For line-search methods, the most recent accepted iterate available by the cap is used; terminated endpoints retain actual calls. The

ordinate's maximum across two seeds is an envelope of two certified problems, not a confidence interval. Snapshots were archived prospectively and checked serially afterward. The cumulative certification axis reconstructs the specified checking schedule, not an online rollback experiment.

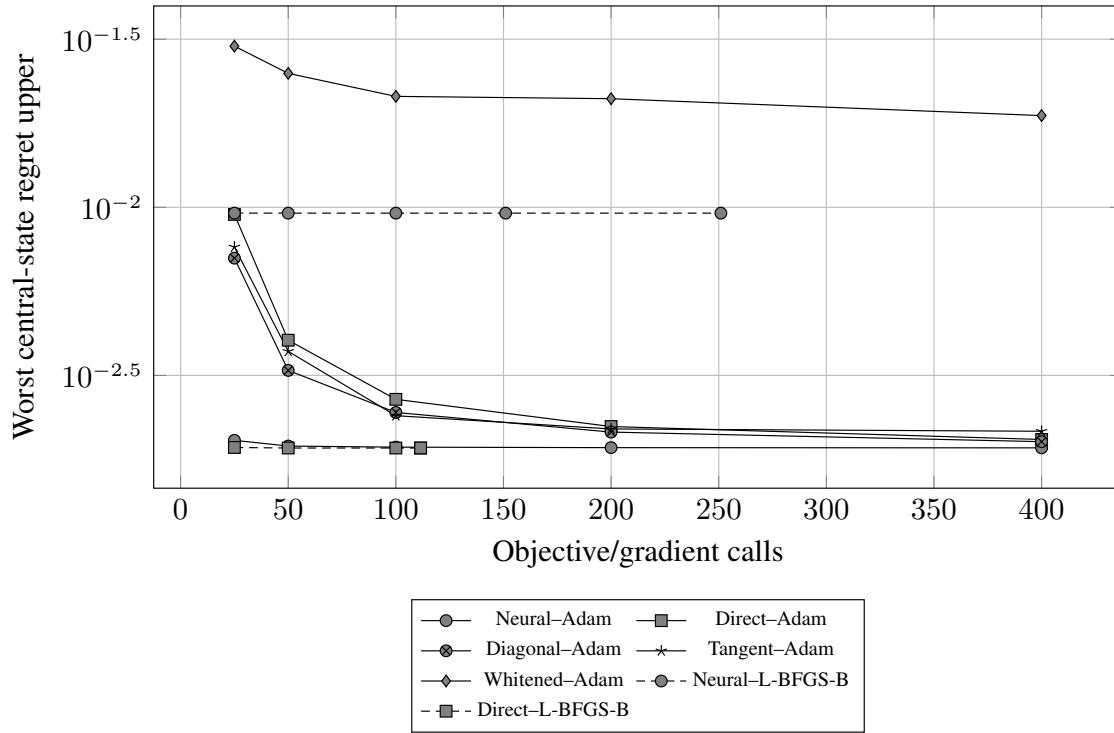


FIGURE 1.—All five prospective checkpoints, rule 8/16. Ordinate: worse regret bound across two seeds; abscissa: mean actual calls or cost. Lines connect recorded checkpoints, not extra observations. Certification is serial replay, so cumulative cost is reconstructed rather than directly timed online wall time.

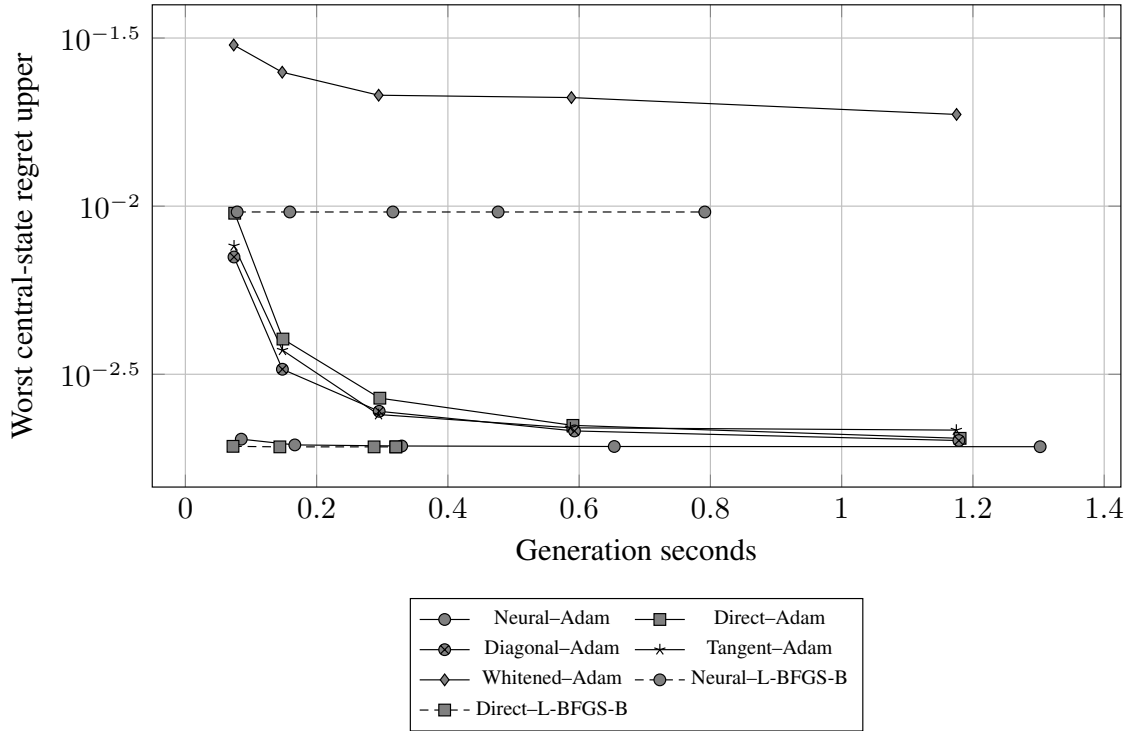


FIGURE 2.—All five prospective checkpoints, rule 8/16. Ordinate: worse regret bound across two seeds; abscissa: mean actual calls or cost. Lines connect recorded checkpoints, not extra observations. Certification is serial replay, so cumulative cost is reconstructed rather than directly timed online wall time.

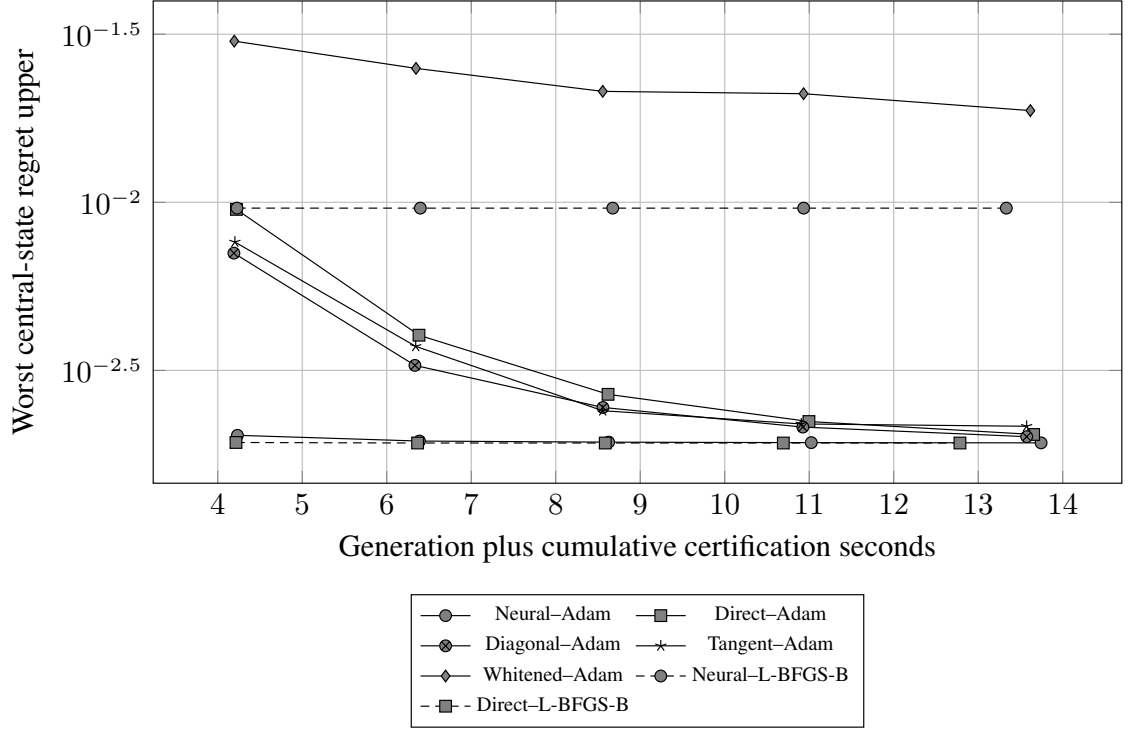


FIGURE 3.—All five prospective checkpoints, rule 8/16. Ordinate: worse regret bound across two seeds; abscissa: mean actual calls or cost. Lines connect recorded checkpoints, not extra observations. Certification is serial replay, so cumulative cost is reconstructed rather than directly timed online wall time.

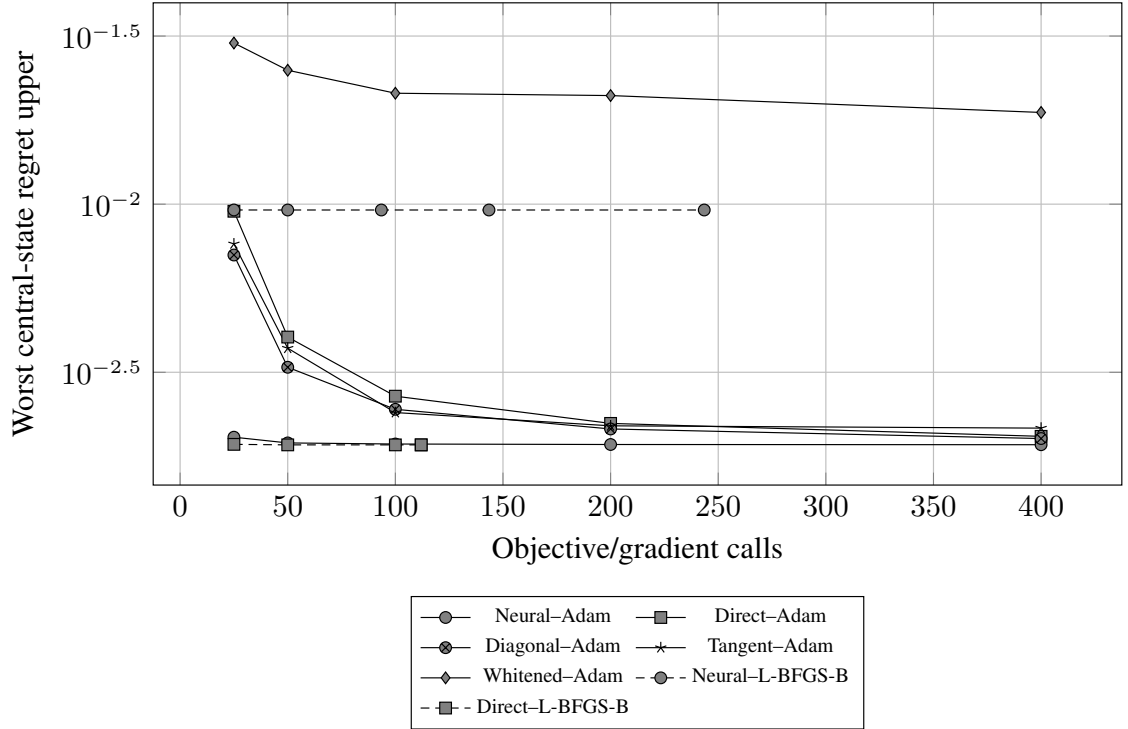


FIGURE 4.—All five prospective checkpoints, rule 12/24. Ordinate: worse regret bound across two seeds; abscissa: mean actual calls or cost. Lines connect recorded checkpoints, not extra observations. Certification is serial replay, so cumulative cost is reconstructed rather than directly timed online wall time.

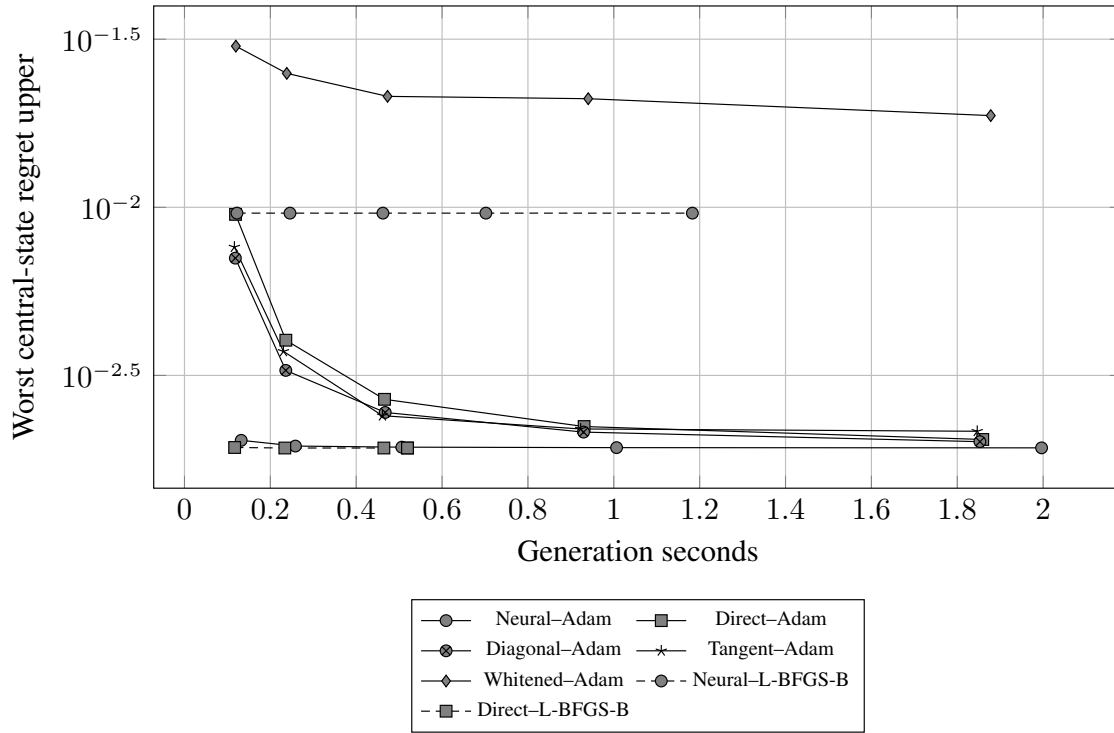


FIGURE 5.—All five prospective checkpoints, rule 12/24. Ordinate: worse regret bound across two seeds; abscissa: mean actual calls or cost. Lines connect recorded checkpoints, not extra observations. Certification is serial replay, so cumulative cost is reconstructed rather than directly timed online wall time.

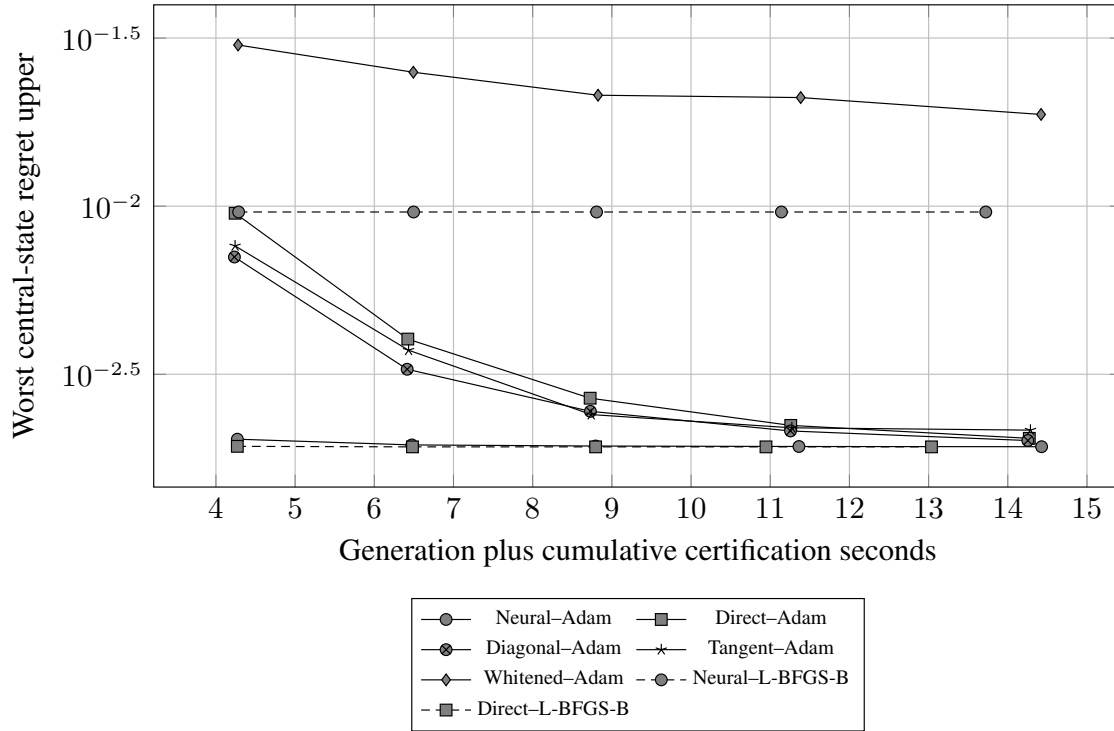


FIGURE 6.—All five prospective checkpoints, rule 12/24. Ordinate: worse regret bound across two seeds; abscissa: mean actual calls or cost. Lines connect recorded checkpoints, not extra observations. Certification is serial replay, so cumulative cost is reconstructed rather than directly timed online wall time.

4. COMPLETE CONFIGURATION LEDGER

Dataset	Arm/rate	Seed	Calls	Gen. s	Check s	Candidate bound
tuning	Diagonal-Adam/0.005	24001	200	0.590	2.082	0.01691209
tuning	Diagonal-Adam/0.02	24001	200	0.612	2.113	0.003987860
tuning	Diagonal-Adam/0.08	24001	200	0.599	2.089	0.002148714
tuning	Direct-Adam/0.005	24001	200	0.593	2.053	0.01915781
tuning	Direct-Adam/0.02	24001	200	0.586	2.057	0.004725880
tuning	Direct-Adam/0.08	24001	200	0.605	2.065	0.002225349
tuning	Neural-Adam/0.005	24001	200	0.659	2.049	0.002210394
tuning	Neural-Adam/0.02	24001	200	0.654	2.079	0.001964497
tuning	Neural-Adam/0.08	24001	200	0.658	2.067	0.001927726
tuning	Tangent-Adam/0.005	24001	200	0.587	2.066	0.007097344
tuning	Tangent-Adam/0.02	24001	200	0.590	2.092	0.002566933
tuning	Tangent-Adam/0.08	24001	200	0.592	2.075	0.002049251
tuning	Whitened-Adam/0.005	24001	200	0.587	2.053	0.03177225
tuning	Whitened-Adam/0.02	24001	200	0.588	2.059	0.02071565
tuning	Whitened-Adam/0.08	24001	200	0.590	2.065	0.01775094
tuning	Diagonal-Adam/0.005	24002	200	0.589	2.060	0.01820277
tuning	Diagonal-Adam/0.02	24002	200	0.588	2.063	0.004390295
tuning	Diagonal-Adam/0.08	24002	200	0.593	2.067	0.002188804
tuning	Direct-Adam/0.005	24002	200	0.591	2.059	0.01960521
tuning	Direct-Adam/0.02	24002	200	0.589	2.088	0.004771570
tuning	Direct-Adam/0.08	24002	200	0.592	2.066	0.002228170
tuning	Neural-Adam/0.005	24002	200	0.657	2.073	0.002236912
tuning	Neural-Adam/0.02	24002	200	0.655	2.073	0.001970312
tuning	Neural-Adam/0.08	24002	200	0.661	2.078	0.001928848
tuning	Tangent-Adam/0.005	24002	200	0.588	2.057	0.009563505
tuning	Tangent-Adam/0.02	24002	200	0.592	2.075	0.003215310
tuning	Tangent-Adam/0.08	24002	200	0.598	2.060	0.002110697
tuning	Whitened-Adam/0.005	24002	200	0.587	2.050	0.02985379
tuning	Whitened-Adam/0.02	24002	200	0.589	2.073	0.02241046
tuning	Whitened-Adam/0.08	24002	200	0.594	2.075	0.02103428
heldout	Diagonal-Adam	24101	400	1.855	10.365	0.002008701
heldout	Direct-Adam	24101	400	1.857	10.384	0.002038210
heldout	Direct-L-BFGS-B	24101	114	0.538	10.542	0.001923288
heldout	Neural-Adam	24101	400	2.002	10.367	0.001925327
heldout	Neural-L-BFGS-B	24101	87	0.437	10.580	0.009608170
heldout	Tangent-Adam	24101	400	1.862	10.389	0.002084464
heldout	Whitened-Adam	24101	400	1.918	10.584	0.01874227
heldout	Diagonal-Adam	24101	400	1.182	10.320	0.002008735
heldout	Direct-Adam	24101	400	1.179	10.325	0.002038249
heldout	Direct-L-BFGS-B	24101	104	0.299	10.435	0.001923296
heldout	Neural-Adam	24101	400	1.305	10.310	0.001925340
heldout	Neural-L-BFGS-B	24101	102	0.322	10.452	0.009608291
heldout	Tangent-Adam	24101	400	1.173	10.328	0.002084499
heldout	Whitened-Adam	24101	400	1.172	10.412	0.01874203
heldout	Diagonal-Adam	24102	400	1.851	10.310	0.002002481
heldout	Direct-Adam	24102	400	1.861	10.321	0.002039189
heldout	Direct-L-BFGS-B	24102	110	0.502	10.371	0.001923288
heldout	Neural-Adam	24102	400	1.991	10.374	0.001925705

Dataset	Arm/rate	Seed	Calls	Gen. s	Check s	Candidate bound
heldout	Neural-L-BFGS-B	24102	400	1.930	10.378	0.001923323
heldout	Tangent-Adam	24102	400	1.831	10.365	0.002157182
heldout	Whitened-Adam	24102	400	1.838	10.375	0.01753457
heldout	Diagonal-Adam	24102	400	1.176	10.341	0.002002522
heldout	Direct-Adam	24102	400	1.182	10.500	0.002039228
heldout	Direct-L-BFGS-B	24102	119	0.343	10.365	0.001923296
heldout	Neural-Adam	24102	400	1.300	10.448	0.001925713
heldout	Neural-L-BFGS-B	24102	400	1.262	10.505	0.001923374
heldout	Tangent-Adam	24102	400	1.177	10.339	0.002157232
heldout	Whitened-Adam	24102	400	1.178	10.343	0.01753422
multicell	Direct-Adam	24300	400	1.171	2.071	0.002708729
multicell	Direct-L-BFGS-B	24300	46	0.133	2.069	0.002587160
multicell	Neural-Adam	24300	400	1.310	2.066	0.002589305
multicell	Tangent-Adam	24300	400	1.181	2.073	0.002692616
multicell	Direct-Adam	24301	400	1.168	2.074	0.002613148
multicell	Direct-L-BFGS-B	24301	161	0.463	2.072	0.002488112
multicell	Neural-Adam	24301	400	1.312	2.077	0.002491675
multicell	Tangent-Adam	24301	400	1.169	2.070	0.002676525
multicell	Direct-Adam	24302	400	1.168	2.088	0.004398790
multicell	Direct-L-BFGS-B	24302	181	0.519	2.105	0.004274863
multicell	Neural-Adam	24302	400	1.299	2.076	0.004280526
multicell	Tangent-Adam	24302	400	1.164	2.074	0.004508256
multicell	Direct-Adam	24303	400	1.170	2.083	0.002379978
multicell	Direct-L-BFGS-B	24303	84	0.243	2.073	0.002267456
multicell	Neural-Adam	24303	400	1.308	2.119	0.002270649
multicell	Tangent-Adam	24303	400	1.168	2.071	0.002440554
multicell	Direct-Adam	24304	400	1.176	2.066	0.002145888
multicell	Direct-L-BFGS-B	24304	202	0.579	2.072	0.002030298
multicell	Neural-Adam	24304	400	1.298	2.061	0.002036150
multicell	Tangent-Adam	24304	400	1.176	2.075	0.002185028
multicell	Direct-Adam	24305	400	1.178	2.057	0.003810485
multicell	Direct-L-BFGS-B	24305	167	0.478	2.067	0.003696931
multicell	Neural-Adam	24305	400	1.308	2.105	0.003698780
multicell	Tangent-Adam	24305	400	1.166	2.065	0.003846639
multicell	Direct-Adam	24306	400	1.179	2.063	0.002767600
multicell	Direct-L-BFGS-B	24306	81	0.235	2.086	0.002664646
multicell	Neural-Adam	24306	400	1.310	2.062	0.002667582
multicell	Tangent-Adam	24306	400	1.178	2.077	0.002799149
multicell	Direct-Adam	24307	400	1.173	2.075	0.002395412
multicell	Direct-L-BFGS-B	24307	169	0.486	2.061	0.002288317
multicell	Neural-Adam	24307	400	1.304	2.062	0.002291286
multicell	Tangent-Adam	24307	400	1.177	2.086	0.002425920
multicell	Direct-Adam	24308	400	1.169	2.062	0.003940201
multicell	Direct-L-BFGS-B	24308	159	0.456	2.089	0.003834396
multicell	Neural-Adam	24308	400	1.307	2.059	0.003837366
multicell	Tangent-Adam	24308	400	1.170	2.069	0.003973639
reference	Diagonal-Adam	24501	400	0.196	0.066	0.00005270608
reference	Direct-Adam	24501	400	0.198	0.066	0.00005026021
reference	Direct-L-BFGS-B	24501	14	0.007	0.067	3.392842×10^{-13}
reference	Neural-Adam	24501	400	0.321	0.066	0.002771190
reference	Tangent-Adam	24501	400	0.197	0.067	0.00001955686

Dataset	Arm/rate	Seed	Calls	Gen. s	Check s	Candidate bound
reference	Whitened–Adam	24501	400	0.196	0.066	0.5345057
reference	Diagonal–Adam	24502	400	0.198	0.066	0.00006450277
reference	Direct–Adam	24502	400	0.199	0.067	0.00005492341
reference	Direct–L-BFGS-B	24502	14	0.007	0.066	9.192647×10^{-14}
reference	Neural–Adam	24502	400	0.321	0.066	3.620557×10^{-8}
reference	Tangent–Adam	24502	400	0.197	0.067	8.427595×10^{-6}
reference	Whitened–Adam	24502	400	0.196	0.073	0.5646302
failure	direct restart	23203	4	0.013	10.704	0.01050307
failure	regenerated	23203	92	0.299	10.934	0.01050307
failure	restart tighter	23203	43	0.141	10.772	0.01050307
failure	scaled	23203	400	1.423	10.800	0.002554827

5. ALL 140 HELD-OUT CHECKPOINTS

Seed/rule	Arm	Cap	Calls	Deployed bound	Gen. s	Cum. check s
24101/12	Diagonal–Adam	25	25	0.007060108	0.120	2.052
24101/12	Diagonal–Adam	50	50	0.003272799	0.241	4.128
24101/12	Diagonal–Adam	100	100	0.002453783	0.473	6.216
24101/12	Diagonal–Adam	200	200	0.002144890	0.937	8.287
24101/12	Diagonal–Adam	400	400	0.002008701	1.855	10.365
24101/12	Direct–Adam	25	25	0.009302293	0.119	2.070
24101/12	Direct–Adam	50	50	0.003987056	0.235	4.132
24101/12	Direct–Adam	100	100	0.002676558	0.466	6.212
24101/12	Direct–Adam	200	200	0.002225434	0.927	8.287
24101/12	Direct–Adam	400	400	0.002038210	1.857	10.384
24101/12	Direct–L-BFGS-B	25	25	0.001930155	0.118	2.109
24101/12	Direct–L-BFGS-B	50	50	0.001923471	0.238	4.220
24101/12	Direct–L-BFGS-B	100	100	0.001923471	0.473	6.314
24101/12	Direct–L-BFGS-B	200	114	0.001923471	0.538	8.433
24101/12	Direct–L-BFGS-B	400	114	0.001923471	0.538	10.542
24101/12	Neural–Adam	25	25	0.002001374	0.133	2.075
24101/12	Neural–Adam	50	50	0.001949498	0.262	4.156
24101/12	Neural–Adam	100	100	0.001934889	0.508	6.231
24101/12	Neural–Adam	200	200	0.001928613	1.012	8.297
24101/12	Neural–Adam	400	400	0.001925327	2.002	10.367
24101/12	Neural–L-BFGS-B	25	25	0.009609658	0.122	2.133
24101/12	Neural–L-BFGS-B	50	50	0.009608181	0.248	4.235
24101/12	Neural–L-BFGS-B	100	87	0.009608181	0.437	6.349
24101/12	Neural–L-BFGS-B	200	87	0.009608181	0.437	8.463
24101/12	Neural–L-BFGS-B	400	87	0.009608181	0.437	10.580
24101/12	Tangent–Adam	25	25	0.004484666	0.119	2.067
24101/12	Tangent–Adam	50	50	0.002390348	0.234	4.141
24101/12	Tangent–Adam	100	100	0.002139910	0.466	6.210
24101/12	Tangent–Adam	200	200	0.002095942	0.927	8.278
24101/12	Tangent–Adam	400	400	0.002084464	1.862	10.389
24101/12	Whitened–Adam	25	25	0.03015331	0.123	2.120
24101/12	Whitened–Adam	50	50	0.02360656	0.247	4.234
24101/12	Whitened–Adam	100	100	0.02025888	0.482	6.351

Seed/rule	Arm	Cap	Calls	Deployed bound	Gen. s	Cum. check s
24101/12	Whitened-Adam	200	200	0.01892330	0.957	8.464
24101/12	Whitened-Adam	400	400	0.01874227	1.918	10.584
24101/8	Diagonal-Adam	25	25	0.007060108	0.075	2.057
24101/8	Diagonal-Adam	50	50	0.003272823	0.148	4.117
24101/8	Diagonal-Adam	100	100	0.002453835	0.296	6.185
24101/8	Diagonal-Adam	200	200	0.002144940	0.597	8.255
24101/8	Diagonal-Adam	400	400	0.002008735	1.182	10.320
24101/8	Direct-Adam	25	25	0.009302295	0.076	2.065
24101/8	Direct-Adam	50	50	0.003987081	0.150	4.137
24101/8	Direct-Adam	100	100	0.002676606	0.297	6.202
24101/8	Direct-Adam	200	200	0.002225488	0.591	8.259
24101/8	Direct-Adam	400	400	0.002038249	1.179	10.325
24101/8	Direct-L-BFGS-B	25	25	0.001930173	0.072	2.085
24101/8	Direct-L-BFGS-B	50	50	0.001923443	0.144	4.174
24101/8	Direct-L-BFGS-B	100	100	0.001923443	0.287	6.254
24101/8	Direct-L-BFGS-B	200	104	0.001923443	0.299	8.332
24101/8	Direct-L-BFGS-B	400	104	0.001923443	0.299	10.435
24101/8	Neural-Adam	25	25	0.002001375	0.085	2.059
24101/8	Neural-Adam	50	50	0.001949502	0.166	4.113
24101/8	Neural-Adam	100	100	0.001934911	0.330	6.177
24101/8	Neural-Adam	200	200	0.001928632	0.656	8.244
24101/8	Neural-Adam	400	400	0.001925340	1.305	10.310
24101/8	Neural-L-BFGS-B	25	25	0.009609672	0.079	2.102
24101/8	Neural-L-BFGS-B	50	50	0.009608327	0.159	4.190
24101/8	Neural-L-BFGS-B	100	100	0.009608327	0.315	6.278
24101/8	Neural-L-BFGS-B	200	102	0.009608327	0.322	8.364
24101/8	Neural-L-BFGS-B	400	102	0.009608327	0.322	10.452
24101/8	Tangent-Adam	25	25	0.004484680	0.075	2.058
24101/8	Tangent-Adam	50	50	0.002390433	0.148	4.124
24101/8	Tangent-Adam	100	100	0.002139947	0.293	6.187
24101/8	Tangent-Adam	200	200	0.002095993	0.585	8.248
24101/8	Tangent-Adam	400	400	0.002084499	1.172	10.328
24101/8	Whitened-Adam	25	25	0.03015330	0.074	2.062
24101/8	Whitened-Adam	50	50	0.02360656	0.148	4.158
24101/8	Whitened-Adam	100	100	0.02025888	0.295	6.220
24101/8	Whitened-Adam	200	200	0.01892319	0.586	8.318
24101/8	Whitened-Adam	400	400	0.01874203	1.172	10.412
24102/12	Diagonal-Adam	25	25	0.006559432	0.117	2.053
24102/12	Diagonal-Adam	50	50	0.003128239	0.231	4.106
24102/12	Diagonal-Adam	100	100	0.002407914	0.462	6.181
24102/12	Diagonal-Adam	200	200	0.002127916	0.922	8.241
24102/12	Diagonal-Adam	400	400	0.002002481	1.850	10.310
24102/12	Direct-Adam	25	25	0.009526626	0.119	2.058
24102/12	Direct-Adam	50	50	0.004024528	0.237	4.122
24102/12	Direct-Adam	100	100	0.002684475	0.466	6.187
24102/12	Direct-Adam	200	200	0.002228160	0.935	8.249
24102/12	Direct-Adam	400	400	0.002039189	1.861	10.321
24102/12	Direct-L-BFGS-B	25	25	0.001930533	0.116	2.077
24102/12	Direct-L-BFGS-B	50	50	0.001923667	0.229	4.156
24102/12	Direct-L-BFGS-B	100	100	0.001923667	0.456	6.226
24102/12	Direct-L-BFGS-B	200	110	0.001923667	0.501	8.300

Seed/rule	Arm	Cap	Calls	Deployed bound	Gen. s	Cum. check s
24102/12	Direct-L-BFGS-B	400	110	0.001923667	0.501	10.371
24102/12	Neural-Adam	25	25	0.002027236	0.131	2.081
24102/12	Neural-Adam	50	50	0.001945647	0.254	4.154
24102/12	Neural-Adam	100	100	0.001933843	0.504	6.227
24102/12	Neural-Adam	200	200	0.001928228	1.001	8.295
24102/12	Neural-Adam	400	400	0.001925705	1.991	10.374
24102/12	Neural-L-BFGS-B	25	25	0.001941987	0.122	2.075
24102/12	Neural-L-BFGS-B	50	50	0.001932375	0.244	4.145
24102/12	Neural-L-BFGS-B	100	100	0.001926151	0.488	6.220
24102/12	Neural-L-BFGS-B	200	200	0.001923514	0.967	8.295
24102/12	Neural-L-BFGS-B	400	400	0.001923514	1.929	10.378
24102/12	Tangent-Adam	25	25	0.007618758	0.114	2.059
24102/12	Tangent-Adam	50	50	0.003729532	0.227	4.138
24102/12	Tangent-Adam	100	100	0.002399877	0.456	6.219
24102/12	Tangent-Adam	200	200	0.002191585	0.918	8.298
24102/12	Tangent-Adam	400	400	0.002157182	1.831	10.365
24102/12	Whitened-Adam	25	25	0.02925194	0.116	2.075
24102/12	Whitened-Adam	50	50	0.02502311	0.230	4.150
24102/12	Whitened-Adam	100	100	0.02137677	0.464	6.230
24102/12	Whitened-Adam	200	200	0.02103722	0.924	8.299
24102/12	Whitened-Adam	400	400	0.01753457	1.838	10.375
24102/8	Diagonal-Adam	25	25	0.006559436	0.073	2.047
24102/8	Diagonal-Adam	50	50	0.003128267	0.148	4.124
24102/8	Diagonal-Adam	100	100	0.002407976	0.295	6.212
24102/8	Diagonal-Adam	200	200	0.002127976	0.588	8.278
24102/8	Diagonal-Adam	400	400	0.002002522	1.176	10.341
24102/8	Direct-Adam	25	25	0.009526628	0.074	2.101
24102/8	Direct-Adam	50	50	0.004024553	0.147	4.203
24102/8	Direct-Adam	100	100	0.002684523	0.296	6.318
24102/8	Direct-Adam	200	200	0.002228214	0.590	8.427
24102/8	Direct-Adam	400	400	0.002039228	1.182	10.500
24102/8	Direct-L-BFGS-B	25	25	0.001930543	0.073	2.075
24102/8	Direct-L-BFGS-B	50	50	0.001923689	0.144	4.143
24102/8	Direct-L-BFGS-B	100	100	0.001923297	0.288	6.217
24102/8	Direct-L-BFGS-B	200	119	0.001923297	0.343	8.288
24102/8	Direct-L-BFGS-B	400	119	0.001923297	0.343	10.365
24102/8	Neural-Adam	25	25	0.002027461	0.085	2.110
24102/8	Neural-Adam	50	50	0.001945621	0.167	4.203
24102/8	Neural-Adam	100	100	0.001933842	0.329	6.291
24102/8	Neural-Adam	200	200	0.001928233	0.651	8.372
24102/8	Neural-Adam	400	400	0.001925713	1.300	10.448
24102/8	Neural-L-BFGS-B	25	25	0.001942199	0.079	2.070
24102/8	Neural-L-BFGS-B	50	50	0.001932827	0.160	4.161
24102/8	Neural-L-BFGS-B	100	100	0.001924547	0.317	6.314
24102/8	Neural-L-BFGS-B	200	200	0.001923533	0.631	8.425
24102/8	Neural-L-BFGS-B	400	400	0.001923533	1.261	10.505
24102/8	Tangent-Adam	25	25	0.007618825	0.074	2.069
24102/8	Tangent-Adam	50	50	0.003729551	0.148	4.136
24102/8	Tangent-Adam	100	100	0.002399927	0.295	6.205
24102/8	Tangent-Adam	200	200	0.002191637	0.588	8.263
24102/8	Tangent-Adam	400	400	0.002157232	1.177	10.339

Seed/rule	Arm	Cap	Calls	Deployed bound	Gen. s	Cum. check s
24102/8	Whitened-Adam	25	25	0.02925194	0.073	2.051
24102/8	Whitened-Adam	50	50	0.02502295	0.148	4.109
24102/8	Whitened-Adam	100	100	0.02137650	0.294	6.174
24102/8	Whitened-Adam	200	200	0.02103689	0.590	8.242
24102/8	Whitened-Adam	400	400	0.01753422	1.178	10.343

6. MACHINE-READABLE EVIDENCE AND REPRODUCTION

`results/COMPLETE_LEDGER.json` and its CSV version enumerate all 110 trajectories. `results/CHECKPOINT_LEDGER.json` and CSV enumerate all 286 checkpoints, including acceptance flags omitted from the narrower printed table. Per-run records retain raw policies, dyadic coefficients, initial and deployed certificates, call counts, objective/gradient histories, termination, generation and checking times. Held-out checkpoints additionally include both proposal gradients, pricing offsets and actual Jacobian singular values.

`results/DEPENDENCY_LOCK.json` identifies runtime-imported repository modules and inherited numerical data with SHA-256 hashes. `PUBLICATION_MANIFEST.json` additionally locks transitive TeX inputs, all R24 evidence and the final PDFs. `results/PRIMARY_EVIDENCE_LOCK.json` verifies the recovered primary bytes, and `results/PRESERVATION_AUDIT.json` verifies every file inherited from the reviewed branch. The publication manifest and enclosing source archive are excluded from their own hash lists to avoid self-reference.

`results/full_state_canonical.json` is the current schema-version-two interpretation of the full-state record. It identifies the discounted all-start-time bound 7.241462443133396 and preserves the old undiscounted quantity as historical. No old summary is overwritten, and no new full-state policy improvement is recorded.

From the repository root, run `replication/assemble_publication.py` within the R24 revision folder, then build `ECTA_R24.tex`, `SUPP_R24.tex` and `RESPONSE_R24.tex`. This regenerates publication tables from frozen evidence, not new optimization. Fresh scientific regeneration uses a disposable checkout, retains the original results elsewhere, and runs `study.py--phaseall` in the locked numerical environment. The complete commands in `REPRODUCE.md` and the manual reproduction workflow keep fresh results separate from the published evidence.

7. PRESERVED R23 AND EARLIER SUPPLEMENTS

The following historical components are unchanged; R24 text governs the new study.

8. CURRENT EVIDENCE, DEPENDENCY MAP, AND REPRODUCIBILITY

The present supplement accompanies the current manuscript, not an alternative numerical claim. R23 is a held-out extension of a previously executed crossed study. The original full-state problem and .01 objective remain unchanged. All scientific source, fixed-policy records, rejected proposals, and earlier proof files are retained. The review entry identifies the exact source, result, and publication commits.

8.1. *Proof-critical dependencies and diagnostic checks*

The proof architecture separates exact mathematical statements, directed bounds for frozen objects, and ordinary numerical diagnostics. The map below uses paths relative to `revisions/2026-09-23-r23/` unless an earlier revision is named. An independent arithmetic implementation does not independently derive the model, the stopping correction, or the quadrature remainder.

TABLE V: Proof dependencies, executable objects, and arithmetic roles

Assertion	Theorem and assumptions	Script and result	Role
Financed quotient, continuous bracket and conditioning	Main Theorem 9; finite positive pricing measure, interior budget, controlled tail mass.	<code>replication/continuous_budget_brackets.py</code> , <code>bracket/main</code> ; <code>results/continuous_budget_brackets.json</code> .	MPFR directed enclosures; proof-critical instantiation for 72 frozen policies.
Finite-node class equality	Main Proposition 8; distinct nodes, finite real weights, sigmoid interiors.	<code>replication/interpolation_certificate.py</code> , <code>main</code> ; <code>results/interpolation_certificate.json</code> .	MPFR interval elimination; proof-critical nonsingularity of one explicit construction.
Uniform payoff and regret	Main Theorem 8; independent fixed-policy intervals, Hessian bounds, pair moments, stopping correction.	<code>replication/continuum_audit.py</code> , <code>main</code> ; <code>results/continuum.json</code> ; <code>R22 moment_jensen.py</code> , <code>moment_jensen</code> ; <code>R20 certify_stochastic.py</code> , <code>certify</code> .	Directed rational/Taylor binary64; proof-critical continuum transfer. Regret additionally uses the inherited dual.
Vertex payoff acceptance	Fixed-policy lower/upper separation; unchanged continuous stopped-model checker.	<code>replication/run_study.py</code> , <code>main</code> ; <code>R22 crossed.py</code> , <code>check</code> ; <code>every results/crossed/**/*.record.json</code> .	Directed payoff certificate; proof-critical acceptance. Proposal objective is not the certificate.
Global all-start-time comparison	Main Proposition 6; admissible current-state actor, trace-compatible witness, residual cover.	<code>R22 time_envelope.py</code> , <code>main</code> ; <code>results/full_state/time_envelope.json</code> and cell certificates.	MPFR cell cover and directed discounted suffix; proof-critical. Not identified actor improvement.

Assertion	Theorem and assumptions	Script and result	Role
Root derivative and quotient regression	First-derivative identity in the budget theorem; finite positive proposal rule.	<code>replication/tests.py</code> , <code>main</code> ; <code>results/root_tests.json</code> .	Ordinary binary64 finite differences and shifts; diagnostic code tests, not economic proof.
Restoration and recovery	Fixed-policy gate plus exact rollback; supervisor changes proposals outside restored state.	<code>R22 results/stress/summary.json</code> and <code>results/rejection_recovery/summary.json</code> .	State hashes and directed payoff intervals; implementation evidence, not convergence theorem.
Sufficient re-budgeted increment	Main Proposition 9; same shapes/reserve, nonexit. Earlier numerical intervals remain separate.	<code>R21 results/continuum_audit.json</code> and frozen compensation checks.	Analytic monotonicity and separately directed welfare comparison; not minimum or unchanged-map variation.
Cover and deployment costs	Main Proposition 7; stated regularity and verified cover assumptions.	<code>R22 results/runtime_cost.json</code> ; <code>R23 results/environment.json</code> and ledgers.	Analytic complexity plus ordinary timing; latency tests do not certify online quadrature error.

The mathematical financing equation and the finite proposal quadrature are different price maps. The latter is differentiated through its implicit equation and is used only to propose policies. The original directed checker computes price and reserve enclosures for the delivered coefficients without assuming that the proposed terminal reserve was attained exactly. The continuous-bracket audit concerns the exact pricing measure and includes its unbounded Gaussian tail. It proves finite existence and a positive derivative enclosure, not a sharp error estimate for every proposal root. In particular, the smallest derivative lower bound in this audit is extremely conservative.

The interpolation certificate is for a fixed rational hidden-layer construction. Interval Gaussian elimination encloses all pivots away from zero. This proves invertibility of the exact matrix, not accuracy of a rounded output-layer solve or successful optimization of that solve. The numerical root regressions test analytic first derivatives against centered finite differences and test common offsets up to magnitude 10^6 . Floating-point translations need not agree bitwise. Invalid logits, negative weights, and endpoint budgets are explicitly rejected.

8.2. Complete held-out ledger and accounting conventions

Each line below is one prescribed final configuration, not a selected best seed. N, D, Q denote neural, direct, and direct-quotient coordinates, with 355, 48, and 47 parameters. All 72 proposals were accepted after an independent payoff comparison. “Extra” is the recorded proxy $\max(0, N_f - 1 - N_{\text{iter}})$ for L-BFGS-B, not an instrumented count of rejected line-search trials. Objective and gradient calls are equal in this implementation. Adam and L-BFGS-B iteration numbers are not treated as interchangeable work units. A budget interruption restores the last accepted L-BFGS-B iterate.

TABLE VI: All 72 held-out final vertex configurations

Ensemble	v	Rep.	Opt.	$N_f = N_g$	Extra	Regret upper	Gen. sec.	Check sec.
23000	0	D	A	400	0	0.01155779	1.301	2.913
23000	0	D	L	75	6	0.00258714	0.239	2.904
23000	0	N	A	400	0	0.00269410	1.531	2.638
23000	0	N	L	400	46	0.00258716	1.586	2.993
23000	0	Q	A	400	0	0.01155440	1.349	2.929
23000	0	Q	L	143	8	0.00258714	0.452	2.932
23000	1	D	A	400	0	0.01206722	1.152	2.588
23000	1	D	L	161	7	0.00283871	0.454	2.587
23000	1	N	A	400	0	0.00302112	1.627	2.939
23000	1	N	L	400	57	0.00283890	1.528	2.822
23000	1	Q	A	400	0	0.01208949	1.188	2.591
23000	1	Q	L	178	10	0.00283872	0.517	2.570
23000	2	D	A	400	0	0.01029960	1.140	2.543
23000	2	D	L	83	9	0.00266465	0.238	2.565
23000	2	N	A	400	0	0.00277932	1.419	2.522
23000	2	N	L	400	48	0.00266467	1.351	2.581
23000	2	Q	A	400	0	0.01029610	1.186	2.553
23000	2	Q	L	60	6	0.00266465	0.179	2.623
23000	3	D	A	400	0	0.01032708	1.144	2.535
23000	3	D	L	86	4	0.00255442	0.246	2.556
23000	3	N	A	400	0	0.00271696	1.418	2.563
23000	3	N	L	400	55	0.00255444	1.341	2.552
23000	3	Q	A	400	0	0.01034883	1.187	2.552
23000	3	Q	L	115	7	0.00255442	0.332	2.561
23100	0	D	A	400	0	0.01152615	1.144	2.562
23100	0	D	L	103	6	0.00258714	0.291	2.530
23100	0	N	A	400	0	0.00270757	1.412	2.597
23100	0	N	L	400	52	0.00258716	1.338	2.594
23100	0	Q	A	400	0	0.01152281	1.190	2.536
23100	0	Q	L	128	6	0.00258714	0.371	2.557
23100	1	D	A	400	0	0.01196019	1.132	2.586
23100	1	D	L	155	8	0.00283872	0.434	2.586
23100	1	N	A	400	0	0.00300570	1.393	2.538
23100	1	N	L	400	63	0.00283873	1.338	2.570
23100	1	Q	A	400	0	0.01198223	1.182	2.551
23100	1	Q	L	122	5	0.00283871	0.352	2.585
23100	2	D	A	400	0	0.01024377	1.139	2.623
23100	2	D	L	73	4	0.00266465	0.210	2.641
23100	2	N	A	400	0	0.00275971	1.419	2.597
23100	2	N	L	400	49	0.00266470	1.332	2.603
23100	2	Q	A	400	0	0.01024030	1.194	2.580
23100	2	Q	L	55	5	0.00266465	0.162	2.614
23100	3	D	A	400	0	0.01039185	1.158	2.591
23100	3	D	L	121	9	0.00255442	0.341	2.571
23100	3	N	A	400	0	0.00269964	1.421	2.596
23100	3	N	L	400	56	0.00255447	1.341	2.628
23100	3	Q	A	400	0	0.01041317	1.182	2.586
23100	3	Q	L	123	9	0.00255442	0.355	2.581
23200	0	D	A	400	0	0.01159225	1.141	2.551
23200	0	D	L	78	6	0.00258714	0.223	2.575

Ensemble	v	Rep.	Opt.	$N_f = N_g$	Extra	Regret upper	Gen. sec.	Check sec.
23200	0	N	A	400	0	0.00270149	1.410	2.595
23200	0	N	L	400	42	0.00258715	1.345	2.596
23200	0	Q	A	400	0	0.01158892	1.187	2.567
23200	0	Q	L	134	11	0.00258714	0.385	2.562
23200	1	D	A	400	0	0.01215505	1.138	2.641
23200	1	D	L	124	7	0.00283871	0.346	2.627
23200	1	N	A	400	0	0.00301599	1.427	2.597
23200	1	N	L	400	56	0.00283875	1.329	2.584
23200	1	Q	A	400	0	0.01217718	1.197	2.540
23200	1	Q	L	191	11	0.00283871	0.545	2.558
23200	2	D	A	400	0	0.01021031	1.141	2.583
23200	2	D	L	62	8	0.00266465	0.174	2.601
23200	2	N	A	400	0	0.00277109	1.418	2.550
23200	2	N	L	400	43	0.00266497	1.334	2.579
23200	2	Q	A	400	0	0.01020688	1.187	2.584
23200	2	Q	L	62	9	0.00266465	0.179	2.574
23200	3	D	A	400	0	0.01032906	1.141	2.609
23200	3	D	L	98	6	0.00255442	0.286	2.585
23200	3	N	A	400	0	0.00270706	1.421	2.563
23200	3	N	L	100	12	0.01050307	0.337	2.590
23200	3	Q	A	400	0	0.01035072	1.186	2.557
23200	3	Q	L	111	6	0.00255442	0.321	2.552

A denotes Adam; L denotes L-BFGS-B. The final-check column excludes initial certification and the shared dual. Full root and memory ledgers are in each record.

The aggregate 290.946 seconds includes initialization checks and overhead; the 68.729 generation and 188.034 final-check seconds should not be expected to sum to the full elapsed time. The peak resident-set measurement covers a shared sequential process and is not an isolated method-specific peak. Python 3.13.15, NumPy 2.3.5, SciPy 1.17.0, and PyTorch 2.10.0+cpu were used on the recorded Linux environment, with OpenBLAS, OpenMP, and MKL thread limits set to one. The same dual library is common to every method; its 415.789-second historical construction cost is reported separately. No new dual solve or new randomized MPFR training run is claimed. The existing MPFR payoff replay and its costs remain in the historical ledger; the new MPFR bracket/rank checks have their own execution records.

8.3. Chronology and nonselection

The R20 seeds and work levels precede its confirmatory execution. R21 mixture, price-transport, and re-budgeting results were certified after those policies were trained. The R22 crossed and current-state protocols precede their respective primary executions; its root-repair and rejection-supervisor passes are separately labeled post-observation diagnostics. They do not replace failed primary runs.

The R23 protocol was committed as `f0ab87bec39e4ba96a49bd01ba5608a7de739bed` after inspecting all available R22 results but before the 72 new configurations. It fixed three held-out seed ensembles, the 400-call cap, the two optimizers, the extra 47-coordinate control, and all continuum comparisons. Scientific source `ba2263e6e8e48abf0c25d5c895df8ca38444f1be` was executed in run 35823417685; the complete primary outputs were committed as `7384d6bc43774ed91738db1c86ec6f21a1e6504e`. The continuous-price and interpolation certificates were subsequently computed for those frozen objects and are

labeled a posteriori proof instantiations. No seed or configuration is discarded, and no new root bracket changes a delivered policy.

8.4. *Publication integrity*

The preceding R22 scientific execution completed before its publication failed. The first failure concerned regeneration of a tracked historical bytecode cache; a subsequent recovery failed because it assumed an absent `run_id` field. Neither is concealed or reinterpreted as a failed scientific experiment. The recovered artifact is hash-pinned; all 360 inherited scientific result files remain byte-identical. New compilation metadata lives under R23 rather than overwriting the original R22 build record. A bitwise preservation check covers every inherited Git blob, with the old root navigation index retained verbatim before its update. The bundle contains the manuscript, technical supplement, response, source, logs, result manifests, and explicit reproduction commands.

9. EARLIER CROSSED EXECUTION AND TECHNICAL DEVELOPMENT

The following material preserves the complete R22 and earlier technical record. Labels such as “current” inside a dated historical subsection refer to its original revision. The current manuscript governs interpretation. In particular, an old end-to-end neural–Adam/direct–L-BFGS-B table is not called a clean hidden-layer ablation, and a re-budgeted increment is not an unchanged-policy compensating variation.

10. NEW EXECUTION AND PROOF-DEPENDENCY RECORD

The prospective crossed experiment is fixed in commit `082a3fdcf97d426df65858702b506bcdd7384ee3`. The separate full-state refinement is fixed in `a1e73f4262a232db147b245521af6af429e8f501`. The root and recovery diagnostics are fixed after observing the primary outcomes, in `98df6e83a2d0ccfae0aab65d5750344e8b629414`. Pair-moment analysis is a posteriori certification of the complete frozen experiment. Historical R20 and R21 inputs are not retrained or overwritten.

The binary64 and MPFR paths share economic assumptions and, where reused, the same stopping and quadrature mathematics. Arithmetic redundancy does not establish mathematical independence. The new pair-moment theorem is proved in the main appendix; an arithmetic overlap or a state-hash check cannot replace it.

11. COMPLETE CONFIGURATION-LEVEL ACCOUNTING

Each row below is one final vertex proposal, including the rejected financing proposal. “Delivered regret” is the certified regret of the delivered policy, which is the initial incumbent when a proposal is rejected. Function and gradient counts are equal in this design; L-BFGS-B line-search counts mean calls beyond its initial call and one call per accepted iterate. There is one shared initial check per block and one final check per configuration. Generation and checker clocks exclude interpreter startup, imports, installation, and the historical dual construction; those exclusions are explicit rather than silently treated as free end-to-end work.

TABLE VII
PROOF-CRITICAL DEPENDENCIES AND DIAGNOSTIC CHECKS

Claim	Analytic and executable dependency	Result and status
Expert payoff and finance	Original decoder and stopped Gaussian reduction; <code>R20 certify_stochastic.py:certify</code>	<code>results/crossed/**/*/candidate_certificate.json</code> ; directed binary64, proof-critical.
Unequal-initialization Jensen upper	Main Theorem 8; absolute Hessian cover; <code>moment_jensen.py:pair_moments</code> and <code>moment_jensen</code>	<code>results/moments/</code> ; directed Cauchy/Taylor quadrature and Gaussian tails, proof-critical.
Uniform payoff and regret	Fixed nonnegative mixture weights, common-event allowance, convex unrestricted upper; <code>moment_jensen.py:main</code>	<code>results/moment_continuum.json</code> ; outward endpoint algebra, proof-critical.
Range-based cross-check	Main Appendix L; <code>continuum.py:general_jensen</code>	<code>results/continuum.json</code> ; separate conservative analytic bound, not independent economic mathematics.
Complex quadrature constants	Main Appendix L.2; <code>proof_checks.py</code>	<code>results/proof_checks.json</code> ; interval inequalities, proof-critical constants.
Full-state global witness	Accessible-face trace theorem and continuous-action oracle; <code>R16 accessibility_certificate.py:audit</code>	<code>results/full_state/</code> ; MPFR, 1,024 cells, zero skipped cells, proof-critical.
Every starting time	Main equation 38; <code>time_envelope.py</code>	<code>results/full_state/time_envelope.json</code> ; directed MPFR recursion, proof-critical.
Matched initialization and counts	<code>crossed.py</code> ; exact delivered-policy equality	<code>matching.json</code> and <code>record.json</code> ; execution invariants, not a value theorem.
Rollback and recovery	<code>crossed.py:run_stress</code> ; <code>replay.py:rejection_recovery</code>	<code>results/stress/</code> and <code>results/rejection_recovery/</code> ; state integrity and finite execution evidence.
Current-state dependence	Actor derivatives at four specified states	<code>state_dependence_diagnostic.json</code> ; diagnostic only, not a coverage proof.
Online cost	Main Proposition 7; <code>runtime_cost.py</code>	<code>results/runtime_cost.json</code> ; ordinary-float timing only, no on-line error certificate.

TABLE VIII: All 48 final vertex proposals

Seed	v	Rep.	Opt.	Grad.	Search	Delivered regret	Check sec.
22000	0	D	Adam	400	0	0.01146888	2.075
22000	0	D	LBFGS	130	5	0.00258714	2.085
22000	0	N	Adam	400	0	0.00270264	2.055
22000	0	N	LBFGS	400	52	0.00258719	2.070
22000	1	D	Adam	400	0	0.01203417	2.064
22000	1	D	LBFGS	122	9	0.00283871	2.083
22000	1	N	Adam	400	0	0.00300464	2.075

Seed	v	Rep.	Opt.	Grad.	Search	Delivered regret	Check sec.
22000	1	N [†]	LBFGS	72	45	0.03770374	0.382
22000	2	D	Adam	400	0	0.01025327	2.036
22000	2	D	LBFGS	84	9	0.00266465	2.056
22000	2	N	Adam	400	0	0.00277380	2.085
22000	2	N	LBFGS	400	60	0.00266476	2.126
22000	3	D	Adam	400	0	0.01024570	2.034
22000	3	D	LBFGS	94	8	0.00255443	2.041
22000	3	N	Adam	400	0	0.00271138	2.031
22000	3	N	LBFGS	400	65	0.00255445	2.037
22100	0	D	Adam	400	0	0.01164557	2.033
22100	0	D	LBFGS	85	6	0.00258714	2.054
22100	0	N	Adam	400	0	0.00271702	2.073
22100	0	N	LBFGS	400	56	0.00258716	2.034
22100	1	D	Adam	400	0	0.01209862	2.056
22100	1	D	LBFGS	121	11	0.00283871	2.077
22100	1	N	Adam	400	0	0.00300355	2.065
22100	1	N	LBFGS	29	10	0.00700843	2.087
22100	2	D	Adam	400	0	0.01026790	2.054
22100	2	D	LBFGS	91	10	0.00266465	2.055
22100	2	N	Adam	400	0	0.00277035	2.084
22100	2	N	LBFGS	400	58	0.00266470	2.087
22100	3	D	Adam	400	0	0.01046757	2.025
22100	3	D	LBFGS	137	8	0.00255442	2.042
22100	3	N	Adam	400	0	0.00271332	2.023
22100	3	N	LBFGS	297	87	0.00255467	2.036
22200	0	D	Adam	400	0	0.01166847	2.048
22200	0	D	LBFGS	103	7	0.00258714	2.047
22200	0	N	Adam	400	0	0.00270142	2.069
22200	0	N	LBFGS	400	61	0.00258718	2.074
22200	1	D	Adam	400	0	0.01192986	2.062
22200	1	D	LBFGS	113	4	0.00283871	2.081
22200	1	N	Adam	400	0	0.00301293	2.053
22200	1	N	LBFGS	292	80	0.00283910	2.059
22200	2	D	Adam	400	0	0.01025820	2.059
22200	2	D	LBFGS	61	7	0.00266465	2.047
22200	2	N	Adam	400	0	0.00277126	2.029
22200	2	N	LBFGS	400	61	0.00266469	2.065
22200	3	D	Adam	400	0	0.01043466	2.016
22200	3	D	LBFGS	156	11	0.00255442	2.042
22200	3	N	Adam	400	0	0.00269921	2.035
22200	3	N	LBFGS	400	57	0.00255447	2.028

†: rejected proposal; the initial incumbent is delivered. Search counts are additional objective calls as defined in the text.

12. POST-OBSERVATION REPAIRS AND RUNTIME MEASUREMENTS

The four root-safe diagnostic configurations are all reported below. They concern only the selected failed block and are not a new confirmatory four-vertex ensemble. Re-budgeting frozen raw parameters changes the delivered financing offset; it is not an unchanged-policy welfare

calculation. The runtime experiment duplicates coefficient arrays to vary the number of priced

TABLE IX

SELECTED-BLOCK ROOT-SAFE DIAGNOSTIC; NOT A REPLACEMENT PRIMARY EXPERIMENT

Configuration	Gradients	Regret upper	Gen. sec.
Neural-Adam	400	0.00300464	2.409
Neural-L-BFGS-B	96	0.01017478	0.415
Direct-Adam	400	0.01203417	1.639
Direct-L-BFGS-B	122	0.00283871	0.494

TABLE X

CONDITIONAL-PRICE TIMING DIAGNOSTIC

Experts	Max. nodes	Coeff. bytes	Median ms	90th pct. ms
4	8192	1536	0.159	0.209
8	16384	3072	0.225	0.331
16	32768	6144	0.354	0.890
32	65536	12288	1.102	2.211

experts. It does not train additional independent experts, certify a larger domain, or establish learned state generalization. Reserve midpoints and ordinary quadrature are used for timing; the mathematical deployment theorem concerns the exact price-defined reserve and hedge.

13. REVIEW SCOPE AND PRESERVED MATERIAL

The complete R21 final report is retained at `reviews/2026-09-23-econometrica-r21-final/referee_report.md`. The response addresses all twelve scientific findings and all seven technical comments individually. The new current-state certificate, local optimizer-conditional advantage, and general Jensen theorem are distinguished from the still unmet global .01 goal and unrestricted neural frontier claim. No certification failure is relabeled as a successful proposal.

All earlier scientific sources and numerical outputs remain in place. The R21 entry index is copied byte-for-byte to the revision archive before the root index is updated. Earlier introduction and conclusion text is reproduced below as historical exposition, so the revised main narrative does not delete it.

14. R21 EVIDENCE IDENTITIES AND REPRODUCTION

The initial referee input is the R18 report at review commit 074849b9aad1812b59e25e1d3833383ed11aa401. The manuscript inherits the R20 source head 6916f250ca2115399dccf2df125c0b13f304547b. The R20 confirmatory protocol was committed before execution at 238ecc0052dc418e1fe6fd06c3daa4a93da1a335; its successful scientific run is 35802793602 at source b1a2ea39d895b330e5a11b4e55d86b7a4985b101. These identifiers distinguish protocol, training, prior paper, new analysis, and publication. The publication manifest does not attempt to contain its own commit hash.

R21 adds continuum proofs and independently executes frozen-policy checks. It preserves each R20 actor, neural-weight file, arithmetic enclosure, acceptance decision and resource

record without alteration. Source and result SHA256 identities are recorded separately. No new training randomness is claimed. The final payoff interval is the hull of the inherited and replayed enclosures, including both arithmetic implementations; taking a more favorable intersection is unnecessary.

14.1. Complete accuracy, price, and cost ledgers

The machine-readable file `results/continuum_audit.json` contains every seed and retained level, the Jensen upper bound, uniform payoff gain, true-regret contraction, directed Hessian bounds, analytic admissibility constants, and every target hit or miss. The file `results/replay_summary.json` identifies every new execution and interval-overlap test. Overlap is an arithmetic consistency test; validity also depends on the shared analytic proof in the main appendix.

TABLE XI

FIRST RETAINED .005 HIT ON K . GENERATION AND CHECKING ARE MEASURED SECONDS; SHARED-UPPER AND ADDITIONAL PUBLICATION AUDITS ARE CHARGED SEPARATELY IN THE TEXT.

Method	Budget	Generation	Checking	Sum	RSS (KiB)
Seed 20100	100	1.107	15.036	16.143	427908
Seed 20101	100	1.096	15.024	16.119	427916
Seed 20102	100	1.099	15.083	16.181	427940
Seed 20103	100	1.084	15.023	16.107	428008
Seed 20104	100	1.074	15.060	16.133	429192
Direct	10	0.153	15.360	15.512	n.r.

TABLE XII

FIXED-POLICY PRICE TRANSPORT: ENDPOINT BOUNDS UNIFORM OVER ALL FIVE NEURAL MIXTURES AND ALL INITIAL STATES IN K . ENDPOINTS BOUND EACH ENTIRE INTERVENING SEGMENT.

Adjustment price k	Regret upper bound
1.5	0.00332694
1.7490234375	0.00298659
2.0	0.00290848
2.0830078125	0.00288265
2.423828125	0.00313176
2.5	0.00326455

At tolerance .002, every seed and the direct stochastic comparator miss the reported target. At .01 and .005, the first retained hitting levels coincide: 100 updates per neural expert and 10 iterations per direct expert. A final-study wall clock includes work after the first hit; it is not substituted for time to a declared tolerance. Conversely, a sum of measured generation and checking times is identified as that sum, rather than mislabeled process wall time. Full R20 resource files retain the actual wall clocks. The shared twenty-node dual-library construction costs 415.789 seconds after upward rounding. Cold-start comparisons add it to every method or state explicitly that it is reused. Final MPFR and R21 publication checks are separately charged audits. Peak memory absent from the original direct run is explicitly missing, not estimated from parameter counts.

14.2. *What is reproduced and what is preserved*

The new replay consists of 80 neural expert checks, 20 final-expert MPFR checks, 16 direct-stochastic checks, and 20 compensated-policy checks: 136 executions. The first group includes initialization and all three retained proposals for each of twenty experts. The MPFR processes are isolated from the binary processes to avoid sharing the loaded interval module. Every reported actor hash is checked. The script fails on a non-overlapping price, reserve, portfolio, or payoff interval. Generated tables are derived from directed calculations and round in the conservative direction.

The inherited record retains the all-ten-seed actor-critic factorial, 30 trace checkpoints, signed classical decompositions, continuation-lift interventions, and high-dimensional learned-gradient/correction study. These are not counted among the new 136 replays. The original all-domain neural error of approximately 7.3, the width-dependent architecture reversal, and accelerated-gradient advantages at stringent nonlinear-control tolerances remain visible. The corrected localized success does not remove any of those observations.

14.3. *Execution commands*

From the repository root, run the new checker and continuum calculation, then build the three documents:

```
R21=revisions/2026-09-23-r21/replication
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  python "$R21/replay.py"
python "$R21/audit.py"
python "$R21/representation_audit.py"
python "$R21/build.py"
```

The original R20 training commands and frozen protocol remain in their original directory. Re-running them is a fresh execution, not a prerequisite for reading the already serialized policies. Their outputs should be directed to a separate copy when preserving the publication snapshot.

15. READING THE RETAINED DEVELOPMENT

The remaining supplement and the retained sections of the main paper reproduce earlier substantive mathematical and numerical material. References to “new” runs within those retained sections mean new in their own R14–R19 revision, not additional R21 training. The historical introductions and conclusions are reproduced verbatim below so that revised exposition does not silently delete prior material. Current headline results and scope are those of the R21 introduction, stochastic-policy section, and conclusion.

16. FURTHER INTERMEDIATE-SNAPSHOT REPORT AND REPRESENTATION AUDIT

The additional report at commit bc118f20f6361cdeae668141e45f64c436ee6a2a reviewed the earlier source-archive snapshot, not the present R21 proofs. The combined response addresses both reports. The new main-text restricted-Markov proposition identifies the exact feedback implemented by the decoder. The representation audit verifies all twenty adverse vertex payoff comparisons and equality of initial parameter dictionaries across vertices within each of five seeds. The old root index is preserved in the history directory; every other inherited file remains at its original path.

17. R19 EXECUTION, SCOPE, AND COMPLETE DIAGNOSTICS

The R19 protocol was committed before the new actor training and the new finite-query study. The actor checkpoints are 0, 50, 200, 800 for seeds 19100–19104. All checkpoint networks, their exact compiled dyadic policies, payoff intervals, acceptance records, and resource logs are retained. The initial and final policies enter the coupled-payoff proof; its deterministic state-extension and wealth-compensation calculations are subsequent analytical consequences, not additional randomized experiments. The attribution study reuses historical frozen networks and is expressly not relabeled as new training.

The primary reproduction evaluates the independent payoff gate before the next Adam update. The optimizer state and network are restored on rejection. An acceptance requires a strictly positive lower endpoint for new payoff minus incumbent payoff; a smaller optimality bound alone is insufficient. All 15 noninitial proposals pass. A separate audit verifies the observable progress test at $\kappa = .02$ for every retained transition, $\kappa = .75$ for each reference-state initial-to-final pair, and the final-to-initial true-regret ratio below .42 on K . These are executed-trajectory statements, not an assumption of global optimizer convergence.

17.1. *All upper-face trajectories*

Tables XIII and XIV report every accessible historical checkpoint, including the signed positive comparison term and the fixed-witness lower floor. Their differences are not treated as actor-value differences. All ten actor–critic interchanges are in the main text. The positive/negative decomposition remains part of the numerical claim, rather than an optional diagnostic hidden behind a total bound.

TABLE XIII
CONTINUATION TRACE TRAJECTORIES, PART 1

Seed	Update	Trace lower	Trace upper	Floor lower	B^+
17100	0	6.679868	6.679869	0.000000	16.983516
17100	100	1.755992	1.755993	4.616272	9.746178
17100	400	0.094970	0.094971	6.277294	7.339625
17101	0	6.661007	6.661008	0.000000	17.020635
17101	100	0.245042	0.245043	6.127222	7.547735
17101	400	0.053474	0.053475	6.318790	7.281386
17102	0	6.686390	6.686391	0.000000	16.975928
17102	100	1.820291	1.820292	4.551973	9.840285
17102	400	0.097382	0.097383	6.274882	7.343312
17103	0	6.648314	6.648315	0.000000	17.006121
17103	100	0.235585	0.235586	6.136679	7.535185
17103	400	0.052760	0.052761	6.319505	7.280303
17104	0	6.694775	6.694776	0.000000	17.003002
17104	100	1.964101	1.964102	4.408163	10.044977
17104	400	0.093338	0.093339	6.278926	7.336434

All 30 accessible checkpoints are included. A fixed-witness floor is not a lower bound on actual policy regret.

17.2. *Classical feedback decomposition*

The finest-grid deterioration is associated with a positive policy-side component, absent at the intermediate grids. The arithmetic decomposition identifies where the reported upper

TABLE XIV
CONTINUATION TRACE TRAJECTORIES, PART 2

Seed	Update	Trace lower	Trace upper	Floor lower	B^+
17105	0	6.675560	6.675561	0.000000	17.035163
17105	100	0.252011	0.252012	6.120253	7.555938
17105	400	0.052751	0.052752	6.319513	7.280018
17106	0	6.666272	6.666273	0.000000	16.981173
17106	100	1.565681	1.565682	4.806583	9.477079
17106	400	0.097374	0.097375	6.274890	7.343666
17107	0	6.682934	6.682935	0.000000	17.099704
17107	100	0.286118	0.286119	6.086146	7.604406
17107	400	0.051775	0.051776	6.320489	7.278320
17108	0	6.653260	6.653261	0.000000	17.019389
17108	100	1.587040	1.587041	4.785224	9.503279
17108	400	0.093895	0.093896	6.278369	7.338342
17109	0	6.694382	6.694383	0.000000	17.076539
17109	100	0.250573	0.250574	6.121691	7.553445
17109	400	0.052498	0.052499	6.319766	7.279140

All 30 accessible checkpoints are included. A fixed-witness floor is not a lower bound on actual policy regret.

TABLE XV
SIGNED DECOMPOSITION OF UNRESTRICTED CLASSICAL FEEDBACK CANDIDATES

Candidate	B^+	B^-	Total
MC-17	7.314510	0.000000	7.314510
MC-25	7.282120	0.481344	7.763463
MC-9	7.339816	0.000000	7.339816
SL-17	7.314559	0.000000	7.314559
SL-25	7.286608	0.487515	7.774123
SL-9	7.342251	0.000000	7.342251

The medium-to-fine deterioration coincides with a positive policy-side term. This separates the terms in the reported bound, but by itself does not identify interpolation error, witness fit, or true policy deterioration.

bound changes. It does not establish that true policy value worsens, and it does not separate interpolation error from witness mismatch without an additional fixed-policy solve. The new SLSQP frontier is a time-control comparison, not a substitute label for these feedback methods.

17.3. All learned warm-start frontiers

The following tables retain all three neural seeds and four tolerances for each dimension. The paired classical method uses the same query vectors. Solver time includes neural inference where applicable; check time is the final MPFR certificate. No finite accounting break-even is returned when the measured saving is nonpositive. Offline training costs are the recorded historical costs, so break-even counts are transferred-cost estimates rather than a matched-hardware retraining study. Negative or absent amortization outcomes are retained. The learned-gradient budgets use a directed maximum over each finite query set in the main paper’s learned-quality theorem. They do not certify the entire cube. The stored floating-point plans are separately checked at the exact stored queries, so an exact-real contraction theorem is not silently used as a floating-point

TABLE XVI
ONLINE WORK AND CONDITIONAL AMORTIZATION AT DIMENSION 8

Seed	Target	NN ms	AG ms	Q_{solve}	Q_{all}	GD bound N/Z
17400	10^{-2}	1.696	3.918	2531	1960	4/9
17401	10^{-2}	1.651	3.918	2596	2361	4/9
17402	10^{-2}	1.725	3.918	3043	4194	4/9
17400	10^{-3}	2.757	4.940	2576	2255	7/12
17401	10^{-3}	2.660	4.940	2580	2586	7/12
17402	10^{-3}	2.729	4.940	3018	5234	7/12
17400	10^{-4}	4.274	6.420	2620	2535	10/15
17401	10^{-4}	4.145	6.420	2586	2814	10/15
17402	10^{-4}	4.297	6.420	3144	5174	10/15
17400	10^{-6}	7.652	9.510	3027	2665	16/21
17401	10^{-6}	7.654	9.510	3170	3414	16/21
17402	10^{-6}	7.631	9.510	3553	8837	17/21

Times are mean five-repeat per-query medians. Break-even counts combine the recorded historical training time with this query execution; they are transferred-cost estimates, not matched-hardware or certified speedups. A dash means no positive measured saving. The all-cost count includes final directed checking. The last column is a proved exact-real GD correction budget using the maximum independently enclosed initial gradient over this finite query set; N and Z denote neural and zero initialization.

TABLE XVII
ONLINE WORK AND CONDITIONAL AMORTIZATION AT DIMENSION 32

Seed	Target	NN ms	AG ms	Q_{solve}	Q_{all}	GD bound N/Z
17400	10^{-2}	2.479	4.977	3037	2537	5/10
17401	10^{-2}	2.343	4.977	3105	2206	5/10
17402	10^{-2}	2.504	4.977	2689	1664	5/10
17400	10^{-3}	3.639	5.903	3351	3269	8/13
17401	10^{-3}	3.613	5.903	3569	2610	8/13
17402	10^{-3}	3.580	5.903	2863	3978	8/13
17400	10^{-4}	5.313	7.489	3485	3010	11/17
17401	10^{-4}	5.282	7.489	3704	2332	11/17
17402	10^{-4}	5.313	7.489	3056	1847	11/17
17400	10^{-6}	9.034	10.796	4305	6124	18/23
17401	10^{-6}	8.903	10.796	4319	2704	17/23
17402	10^{-6}	8.996	10.796	3694	2277	18/23

Times are mean five-repeat per-query medians. Break-even counts combine the recorded historical training time with this query execution; they are transferred-cost estimates, not matched-hardware or certified speedups. A dash means no positive measured saving. The all-cost count includes final directed checking. The last column is a proved exact-real GD correction budget using the maximum independently enclosed initial gradient over this finite query set; N and Z denote neural and zero initialization.

error bound. The complete files contain 1,920 target-check records, including 1,152 neural and 768 classical records; every declared target is attained within the 64-correction cap.

17.4. Reproduction and cost accounting

The entry point is `revisions/2026-09-23-r19/replication/run_all.py`. It runs policy generation and inline acceptance, all-seed attribution, the matched-tolerance query

TABLE XVIII
ONLINE WORK AND CONDITIONAL AMORTIZATION AT DIMENSION 128

Seed	Target	NN ms	AG ms	Q_{solve}	Q_{all}	GD bound N/Z
17400	10^{-2}	5.748	6.577	13535	7216	11/12
17401	10^{-2}	5.748	6.577	16056	8768	11/12
17402	10^{-2}	5.743	6.577	16313	8124	11/12
17400	10^{-3}	7.545	7.518	—	—	14/15
17401	10^{-3}	7.533	7.518	—	239322	14/15
17402	10^{-3}	7.518	7.518	41296137	138022	14/15
17400	10^{-4}	9.411	9.383	—	46790	17/18
17401	10^{-4}	9.436	9.383	—	62773	17/18
17402	10^{-4}	9.464	9.383	—	42578	17/18
17400	10^{-6}	13.985	13.122	—	—	23/25
17401	10^{-6}	13.997	13.122	—	—	23/25
17402	10^{-6}	13.992	13.122	—	—	23/25

Times are mean five-repeat per-query medians. Break-even counts combine the recorded historical training time with this query execution; they are transferred-cost estimates, not matched-hardware or certified speedups. A dash means no positive measured saving. The all-cost count includes final directed checking. The last column is a proved exact-real GD correction budget using the maximum independently enclosed initial gradient over this finite query set; N and Z denote neural and zero initialization.

study, deterministic state and welfare extensions, table generation, and validation. The paper build is separate from numerical execution, so a LaTeX success is not reported as scientific gate closure. The publication workflow records immutable input, protocol, source, and publication objects separately.

Timing records identify generation, policy-value evaluation, query correction, final gradient checking, shared optimal-upper generation/verification, and peak process memory. Shared upper-comparator costs are retained from their historical execution and are not called zero-cost or newly timed. End-to-end neural first-attainment timing includes the earlier retained policy-value checks. Peak memory is process-wide and includes the runtime and loaded libraries; it is not falsely presented as network parameter memory.

All prior tracked files are checked for preservation against the R18 review commit. New arithmetic checks validate the MPFR primitive layer, directed policy ordering, work-count consistency, and stored-plan hashes. They do not constitute an independent formal proof of the model, quadrature, stopping, or comparison theorems. The deployment contract concerns the serialized dyadic controls and checked plans; it does not cover an undocumented online activation implementation or simulator.

18. RETAINED R18 OBJECTIVE AND NUMERICAL DEVELOPMENT

19. CURRENT REVIEW OBJECT AND REPRODUCTION

The current manuscript is `ECTA_R18.tex` with its compiled PDF. This supplement is `SUPP_R18.tex`. The point-by-point response is supplied in both `Mark-down` and `RESPONSE_R18.pdf`. The latest referee input is the 914-line report under `reviews/2026-09-23-econometrica-r16/`. Its immutable review commit is `82af00bc296d`. The reviewed R16 publication commit is `2932b74dad6d`; the R17 scientific source and result commits are `ab79b9db4e7c` and `982326994c85`, respectively.

R18 materializes a new manuscript from those objects and adds the quantitative objective theorem, the finite-horizon defect allocation, the fixed-witness result, the nonlinear-control

proof, and the budget-compensation proof. It does not relabel R17 training as newly executed R18 training. The R18 execution checks all 589 immutable scientific files, re-evaluates all 60 original-economy proposal objectives, replays two complete MPFR cylinder certificates, and independently rechecks all 18 final nonlinear neural query plans. It also checks whole-domain control ranges for six unrestricted classical policies and every one of the twenty fresh-library actors, and runs 450 exact-rational arithmetic tests, 80 exact-rational finite-MDP tests, and derivative/Hessian regression diagnostics. Sampling diagnostics are never substituted for the analytical global results.

From the repository root, the new mathematical and numerical checks are reproduced by

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
python revisions/2026-09-23-r18/replication/reproduce.py
pdflatex -halt-on-error ECTA_R18.tex
pdflatex -halt-on-error ECTA_R18.tex
pdflatex -halt-on-error SUPP_R18.tex
pdflatex -halt-on-error SUPP_R18.tex
pdflatex -halt-on-error RESPONSE_R18.tex
pdflatex -halt-on-error RESPONSE_R18.tex
```

The R17 protocols and execution scripts retain the commands for regenerating the underlying neural experiments, unrestricted baselines, nonlinear plans, wealth compensation, and scalar Riccati policies from their declared inputs. The R18 entry point is a theorem-and-evidence audit, not a replacement for those longer scientific reproduction commands. Both scopes are separately identified in the publication manifest. The publication workflow builds all PDFs, preserves the build logs, verifies the new artifacts, and checks that no historical scientific file has changed.

19.1. Numerical environments and interpretation

The canonical R17 science ran with Python 3.11.16, NumPy 2.3.5, SciPy 1.17.0, and Torch 2.10.0 on the recorded Linux runner. The new local R18 check used Python 3.13.5, the same NumPy/Torch versions, and MPFR 4.2.2. The publication workflow records its own environment independently. Timings are neither pooled across environments nor treated as a neural speed advantage when error levels differ. Stored network weights are exact dyadic inputs; model constants are exact rationals where specified. The directed arithmetic layer does not formally verify the C interface, control derivation, or shared stopping and quadrature arguments.

All numerical table entries are rounded for presentation unless explicitly identified as outward endpoints. Machine-readable JSON retains the exact stored directed endpoint. A printed rounded number should not replace that endpoint in a new certification calculation. The unchanged original-economy target is .01; no R18 audit treats software test success as attainment of that accuracy target.

20. COMPLETE ORIGINAL-ECONOMY NEURAL DECOMPOSITIONS

Tables below retain the signed components and work for every prescribed original-economy checkpoint. Positive and negative entries are the discounted slab maxima entering the certificate. The all-face ablation multiplies the accessibility-aware critic by the additional factor $1 - e^{4(x-2)}$ in its positive lifting term; actor architecture, initialization logic, optimization settings, action comparison, and verification cover are matched. Seeds of even parity use width 16; odd parity uses width 32. The exact cell record is retained for every entry.

TABLE XIX
COMPLETE ACCESSIBILITY-AWARE RECORD

Seed	Updates	Positive	Negative	Total	Train sec.	Audit sec.
17100	0	16.983515	9.316999	26.300514	0.00	12.15
17100	100	9.746177	0.358090	10.104267	17.64	23.76
17100	400	7.339624	0.000000	7.339624	68.49	35.05
17101	0	17.020635	9.479196	26.499830	0.00	31.72
17101	100	7.547734	0.000000	7.547734	26.90	64.64
17101	400	7.281385	0.000000	7.281385	102.01	90.84
17102	0	16.975928	9.350446	26.326374	0.00	10.43
17102	100	9.840284	0.397151	10.237435	18.51	21.81
17102	400	7.343312	0.000000	7.343312	64.26	31.43
17103	0	17.006120	9.505096	26.511217	0.00	27.48
17103	100	7.535185	0.000000	7.535185	21.55	47.47
17103	400	7.280303	0.000000	7.280303	77.80	65.27
17104	0	17.003002	9.364486	26.367488	0.00	7.70
17104	100	10.044976	0.546666	10.591643	12.83	14.51
17104	400	7.336433	0.000000	7.336433	49.19	21.77
17105	0	17.035163	9.433132	26.468295	0.00	21.48
17105	100	7.555938	0.000000	7.555938	14.43	43.38
17105	400	7.280018	0.000000	7.280018	54.39	59.74
17106	0	16.981173	9.371735	26.352908	0.00	5.52
17106	100	9.477079	0.224054	9.701133	8.35	11.19
17106	400	7.343665	0.000000	7.343665	34.17	16.81
17107	0	17.099704	9.567024	26.666728	0.00	16.74
17107	100	7.604406	0.000000	7.604406	12.32	33.22
17107	400	7.278319	0.000000	7.278319	47.39	49.57
17108	0	17.019388	9.320582	26.339971	0.00	5.66
17108	100	9.503279	0.242000	9.745279	7.21	11.41
17108	400	7.338341	0.000000	7.338341	31.20	17.13
17109	0	17.076539	9.450934	26.527472	0.00	17.16
17109	100	7.553444	0.000000	7.553444	13.17	33.16
17109	400	7.279140	0.000000	7.279140	36.65	42.10

Audit seconds are cumulative through the stated checkpoint. All entries have zero skipped or failed cells and zero signed trace budget. These are raw proposals, all retained.

TABLE XX
COMPLETE ALL-FACE RECORD

Seed	Updates	Positive	Negative	Total	Train sec.	Audit sec.
17100	0	21.032537	9.283830	30.316368	0.00	12.18
17100	100	10.730580	0.341597	11.072177	18.50	23.70
17100	400	7.345561	0.000000	7.345561	71.28	34.53
17101	0	21.069026	9.445637	30.514662	0.00	34.68
17101	100	7.637592	0.000000	7.637592	24.99	67.45
17101	400	7.263538	0.000000	7.263538	103.12	92.60
17102	0	21.023728	9.317186	30.340914	0.00	11.29
17102	100	10.732082	0.334124	11.066206	18.02	23.40
17102	400	7.351242	0.000000	7.351242	60.06	30.16
17103	0	21.046882	9.471462	30.518344	0.00	27.69
17103	100	7.610538	0.000000	7.610538	20.08	49.96
17103	400	7.263378	0.000000	7.263378	70.77	73.88
17104	0	21.059772	9.331193	30.390965	0.00	7.60
17104	100	10.592155	0.303616	10.895771	12.29	15.33
17104	400	7.345015	0.000000	7.345015	41.50	20.87
17105	0	21.091783	9.399684	30.491466	0.00	22.13
17105	100	7.676889	0.000000	7.676889	15.77	39.53
17105	400	7.261518	0.000000	7.261518	56.39	55.78
17106	0	21.022430	9.338410	30.360841	0.00	5.86
17106	100	10.897741	0.398215	11.295957	8.40	11.60
17106	400	7.349999	0.000000	7.349999	34.32	17.05
17107	0	21.172118	9.533247	30.705365	0.00	16.84
17107	100	7.678567	0.000000	7.678567	12.97	33.23
17107	400	7.262408	0.000000	7.262408	49.32	49.61
17108	0	21.053287	9.287400	30.340687	0.00	5.68
17108	100	10.054138	0.108039	10.162177	7.91	11.55
17108	400	7.345879	0.000000	7.345879	31.51	17.36
17109	0	21.152839	9.417439	30.570278	0.00	17.41
17109	100	7.640730	0.000000	7.640730	11.99	30.66
17109	400	7.261511	0.000000	7.261511	35.44	39.74

Audit seconds are cumulative through the stated checkpoint. All entries have zero skipped or failed cells and zero signed trace budget. These are raw proposals, all retained.

21. THE OBJECTIVE AUDIT AND NUMERICAL DISCREPANCY

The new checker rejects any missing, duplicated, reordered, or unaccounted-for cell and any mismatch between a network and the hash in its certificate. It uses the immutable MPFR cell endpoints as inputs to the objective-bridge check, rather than claiming to regenerate all 60 interval certificates. The two designated full-cylinder replays are identified separately. For every object, its proposal and certified logits are retained in a compressed numerical array with its own hash.

The smooth-maximum computation subtracts the largest stored logit before exponentiation and includes one exact zero logit. Its temperature is the exact rational $3/20$. Directed discounted masses and pairwise reductions enclose every summed quantity. The correction in Theorem 13 is computed against the clipped proposal logits because both maxima already include zero. This avoids assigning a spurious discrepancy to a negative residual whose positive part is exactly zero. The ordinary proposal graph is still not itself a proof of enclosure.

Across all objects, the integrated correction ranges from approximately 1.11149×10^{-14} to 1.19078×10^{-13} . The worst-case soft-maximum allowance is at most 1.631868. All 60 directed reconstructions satisfy the envelope implication. This is a check of the sufficient inequality, not proof of a global minimum of the nonconvex training objective. The current witness floors are orders of magnitude larger than the numerical correction.

22. FINITE-BELLMAN CONDITIONING

The stationary auxiliary recurrence has evaluation amplification $2q/(1-q)^2$ and greedy amplification $(1-q)^{-1}$. At the original discount and representative time steps, the values are:

Time steps	Evaluation multiplier	Greedy multiplier	Evaluation defect for .01 term
64	5.12000×10^6	1600.5001	1.95313×10^{-9}
128	2.04800×10^7	3200.5001	4.88282×10^{-10}
256	8.19200×10^7	6400.5001	1.22071×10^{-10}

The last column controls only that additive evaluation term in one use of the recurrence, not its infinite-iteration accumulated error. These numbers explain why the stationary result is not used to certify the implemented continuous-time neural algorithm. Proposition 17 instead sums time-local defects over the actual finite horizon. The 80 exact-rational tests enumerate small finite Bellman problems with policy-specific evaluation and greedy defects; they are regression tests of that statement, not tests of an unproved discretization transfer.

23. COMPLETE CONTROL-RANGE DIAGNOSTICS

The following are full-domain enclosures of the final accessible mathematical actor, including the declared output-neighborhood convention. A control-range bound does not identify the optimal control or establish that a change improves payoff.

24. NONLINEAR CONTROL: ALL DIMENSIONS AND RETAINED BUDGETS

The nonlinear experiment has three dimensions, three neural seeds per dimension, two fixed query vectors per network, and five correction budgets. Its 90 neural query objects are distinct from the 15 analytic state-uniform bounds. Each dimension also has ten zero-start-gradient query objects, ten accelerated-gradient query objects, and two L-BFGS query objects, for 66 classical query objects overall. No two-query arithmetic certificate is called a uniform cube certificate.

TABLE XXI
ORIGINAL NEURAL CONTROL RANGES AND CHANGE BOUNDS

Seed	Consumption	Adjustment	Portfolio	Δc	$\Delta\theta$	Δp
17100	[0.5310, 0.7494]	[0.0940, 0.1133]	[0, 0.0874]	0.2187	0.0192	0.1007
17101	[0.5701, 0.7781]	[0.0564, 0.1372]	[-0.0063, 0.1459]	0.1859	0.0538	0.1807
17102	[0.5248, 0.7726]	[0.0844, 0.1196]	[0, 0.0948]	0.2258	0.0222	0.1341
17103	[0.5823, 0.7835]	[0.0613, 0.1315]	[-0.0063, 0.1617]	0.1742	0.0461	0.1928
17104	[0.5237, 0.7292]	[0.0935, 0.1097]	[0, 0.0883]	0.2283	0.0180	0.1126
17105	[0.5754, 0.7749]	[0.0811, 0.1233]	[-0.0015, 0.1457]	0.1788	0.0286	0.1748
17106	[0.5233, 0.7479]	[0.0889, 0.1189]	[0, 0.1019]	0.2308	0.0224	0.1202
17107	[0.5819, 0.7819]	[0.0412, 0.1208]	[-0.0074, 0.1351]	0.1733	0.0643	0.1739
17108	[0.5364, 0.7447]	[0.0924, 0.1194]	[0, 0.0912]	0.2165	0.0245	0.1073
17109	[0.5864, 0.7688]	[0.0792, 0.1362]	[-0.0002, 0.1523]	0.1685	0.0440	0.1865

TABLE XXII
NONLINEAR STORED-PLAN ACCURACY AT ALL RETAINED BUDGETS

Dimension	Corrections	Neural + gradient	Zero + gradient	Accelerated
8	0	1.161	9.03	9.03
8	8	1.652×10^{-5}	8.292×10^{-5}	6.636×10^{-5}
8	16	5.848×10^{-9}	1.485×10^{-8}	5.665×10^{-9}
8	24	3.18×10^{-12}	3.837×10^{-12}	2.795×10^{-13}
8	32	2.089×10^{-15}	1.612×10^{-15}	1.544×10^{-17}
32	0	3.973	36.29	36.29
32	8	5.7×10^{-5}	0.000345	0.0002687
32	16	1.505×10^{-8}	6.218×10^{-8}	2.346×10^{-8}
32	24	5.932×10^{-12}	1.607×10^{-11}	1.143×10^{-12}
32	32	2.917×10^{-15}	5.997×10^{-15}	6.451×10^{-17}
128	0	39.74	144.6	144.6
128	8	0.0004278	0.001383	0.001075
128	16	8.924×10^{-8}	2.487×10^{-7}	9.381×10^{-8}
128	24	2.92×10^{-11}	6.433×10^{-11}	4.567×10^{-12}
128	32	1.626×10^{-14}	2.057×10^{-14}	2.6×10^{-16}

Query-specific total-cost bounds, not per-coordinate losses and not cube-uniform bounds.

For each dimension and correction budget, the next table gives the largest directed stored-plan loss over all retained query objects of each method. The neural maximum is over all six seed/query objects; the other maxima are over the same two initial query vectors. L-BFGS is reported separately because its retained stopping iteration differs from the fixed correction budgets.

Every query certificate includes its objective interval and a rigorously evaluated gradient norm. When the resulting optimal-cost lower bound is positive, the record also reports relative excess cost; otherwise the relative field is absent rather than normalized by an arbitrary positive constant. The complete timing record separates one-time neural training, accumulated online correction time, and independent validation of each retained plan. A retained certificate near machine precision does not prove floating-point convergence at all states.

The nonlinear global proof uses neither a training-set generalization argument nor the two-mode Riccati identity. It bounds the full control-plan Hessian by $4I$ and $(177/8)I$, then contracts the gradient using the exact rational step $16/209$. Its operation count is conditional on the stated

strongly convex deterministic model. A stochastic nonlinear model without that property would require a different argument.

25. WELFARE AND ARITHMETIC QUALIFICATIONS

The wealth-compensation certificate checks all 20 fresh price-library actors, their original hashes, the exact original consumption cap, and both wealth faces. The added initial wealth is $39/5000$, and the constant consumption increment exactly exhausts that addition by the horizon under the original riskless drift. Preference adjustment, portfolio, price, and stopping settlement remain unchanged. The proof gives sufficient compensation for the original optimal value with an explicitly changed feasible policy, not the minimum wealth equivalent of an unchanged policy and not a neural welfare claim.

The full historical 18-node MPFR comparison and the fresh 20-node library are distinct records. Independent arithmetic cannot validate shared analytic premises merely by numerical agreement. The original rational quadrature, normal-tail, stopping, and price-envelope arguments therefore remain in the retained supplement below. The exact-real neural interpretation and the requirement that a portfolio perturbation precede the distance factor likewise remain part of the mathematical contract.

26. ORIGINAL-ECONOMY CLASSICAL AND PRICE-LIBRARY CONTROL RANGES

The certified classical control is the original-action clipping of the exact-real bilinear state interpolant of the stored tensor, held constant within a time step. At the upper wealth column its portfolio is exactly zero. Clipping corrects only the action admissibility convention; it does not multiply interior portfolios by a wealth-distance factor. A bilinear interpolant is a convex combination of four nodal values. Taking their extrema over all time slices and applying monotone action clipping therefore gives a whole-domain enclosure. The new range audit verifies all six tensors and their upper-face portfolios.

The fresh-library selector chooses one of twenty completely enumerated actors. Taking extrema over every stored consumption and preference-adjustment slab therefore also bounds every policy the selector returns for any price in $[\cdot 5, 8]$. This statement concerns the chosen policy's controls, not the unknown optimal controls.

TABLE XXIII
CONTROL RANGES IN THE UNCHANGED STOCHASTIC ECONOMY

Method	Nodes per axis	Consumption	Adjustment	Portfolio
Markov chain	9	$[\cdot 425, \cdot 8]$	$[-\cdot 2, \cdot 2]$	$[0, \cdot 8]$
Markov chain	17	$[\cdot 2375, \cdot 8]$	$[-\cdot 2, \cdot 2]$	$[0, \cdot 8]$
Markov chain	25	$[\cdot 175, \cdot 8]$	$[-\cdot 2, \cdot 2]$	$[0, \cdot 8]$
Semi-Lagrangian	9	$[\cdot 425, \cdot 8]$	$[-\cdot 2, \cdot 2]$	$[0, \cdot 3666667]$
Semi-Lagrangian	17	$[\cdot 2375, \cdot 8]$	$[-\cdot 2, \cdot 2]$	$[0, \cdot 8]$
Semi-Lagrangian	25	$[\cdot 175, \cdot 8]$	$[-\cdot 2, \cdot 2]$	$[0, \cdot 8]$
Fresh library	20 prices	$[\cdot 7618225, \cdot 7721747]$	$[\cdot 0031022, \cdot 2]$	$\{0\}$

Classical ranges hold on the complete state–time cylinder. Library ranges hold for all selected prices and all times before the original stopping time. Decimal endpoints are widened where needed. The JSON record contains the exact stored endpoints and source SHA-256 hashes.

27. PRESERVED TECHNICAL DEVELOPMENT FROM THE PRECEDING REVISION

The following R16 and R14 sections retain their historical scientific content. Statements about their experiment counts and timings describe those revisions, not additional R18 runs. Their existing source and results remain unchanged in the repository. The current results and claims are identified by the R18 entry point and the new sections above.

28. R16 EXPERIMENT IDENTITY AND PROOF DEPENDENCIES

The canonical manuscript is ECTA_R16; this is its technical supplement. The immutable input review is the R14 report dated September 22, 2026. R16 retains the original nonlinear economy, author attribution, preceding proofs, comparison data, unsuccessful experiments, and technical supplements. The new inventory model is explicitly a separate application. The repository's R16_REVIEW.md identifies the current manuscript, response, exact source/result hashes, and canonical build evidence. A staged source transport or a branch name alone is not the reviewed scientific object.

Every numerical bound is attached to a stored mathematical object. The original neural policies use the accessibility-preserving architecture in the main paper. The time-control policies use deterministic consumption and preference actions with zero portfolio until the original

stopping time. The inventory policies use trained time-dependent gains in a coupled linear diffusion. The MPFR audit begins from frozen historical policy and dual inputs; the fresh price experiment does not. These distinctions are maintained even when two experiments reach numerically similar utility tolerances.

28.1. *Original-economy training and complete certification*

Seeds 16100–16109 and both checkpoints are retained. Widths alternate between 16 and 32; both have two hidden tanh layers. Every actor update follows three critic residual updates. Critic and actor learning rates are .0005 and .0002, with 128 samples, double precision, and one Torch thread. The random sampling mixture includes a 25 percent boundary-layer component. The optimizer is a proposal procedure and has no certified convergence claim. The main paper reports the complete twenty-checkpoint table, including all target misses.

The MPFR certifier evaluates every interior box. There are no skipped or failed cells in the retained valid records. The four state boundaries and terminal trace are handled by the algebraic identities and inaccessibility proof, not by omitting an inconvenient face. At upper wealth the witness dominates settlement for the original-control upper comparison. Only the candidate policy’s probability of hitting that face is zero. For a head perturbation, consumption and preference remain clipped to their bounds and the portfolio perturbation precedes the wealth-distance multiplication. An additive portfolio error after that multiplication is outside the certified contract.

A neural-jet diagnostic compares the independently coded value, time derivative, state gradient, and three state Hessian components with automatic differentiation at 24 test states, and checks 72 actor components. It is an implementation diagnostic; the whole-cell MPFR enclosures and the analytic boundary proof establish the certificate. The JSON records retain, separately, positive optimal residuals, negative policy residuals, unsigned evaluation bounds, action-gap allowances, and time-slab integration. They do not identify the unknown true residual maximum or the exact amount of interval wrapping.

TABLE XXIV
FIXED-OBJECT COVER REFINEMENT AND SEPARATE ARCHITECTURE ABLATION

Object	Updates	Cells	Signed budget	Unsigned budget	Seconds
Accessible, coarse	2400	1024	54.004046	84.409078	14.16
Accessible, refined	2400	8192	32.157321	55.142057	37.43
All-face ablation	800	1024	28.195472	52.129689	4.75

Signed means the integrated positive optimal residual plus negative policy residual. Unsigned is the separately recorded $2e + q$ budget. Every trace budget is zero. The ablation is a different 800-update network, not a cover refinement of the 2,400-update candidate. Differences between enclosures are not certified estimates of the exact policy loss.

The all-face hard-trace network has the same invariant actor, seed, width, and 800-update budget specified in the ablation. It is not substituted for an unfavorable primary checkpoint. A smaller finite-resolution bound in that ablation would not refute the proved trace discontinuity: the incompatibility concerns a sequence of continuous all-face witnesses with vanishing global positive residual. Conversely, eliminating the trace budget does not prove that the resulting optimizer attains a small residual.

28.2. *Two same-model state-space methods*

The controlled Markov-chain generator uses upwind drift and two wide-jump diffusion directions with positive weights. The correlation enters those directions, and the stay probability is checked. The semi-Lagrangian generator uses a four-point weak Euler shock rule and bilinear state interpolation. Both apply the original stopping payoff when an approximate jump first leaves the domain, using the straight segment to locate that crossing. Neither treats this discrete boundary approximation as an exact continuous-time stopping simulation.

The three refinements use 9, 17, and 25 state nodes per coordinate and 64, 128, and 256 time steps. Their consumption, preference, and portfolio-head counts are (3, 3, 4), (5, 5, 6), and (7, 7, 8). Bounded bilinear state heads, held piecewise constant in time, define the transferred policies. These policies use the same invariant portfolio-distance factor as the neural class. The separate original-economy verifier encloses every interpolation cell and time-piece range against the continuous control maximum. Its common critic is fixed by the protocol, not selected after seeing the baseline outcomes. Discrete central values are stored as diagnostics and are never used as verified policy values.

The two implementations provide distinct state-space candidate generators. They are not claimed to be optimally tuned implementations or a complete comparison against the strongest possible classical approaches. The common critic is a limitation of the current matched verification exercise; all six resulting bounds miss .01. The paper therefore reports generation and verification costs without claiming a matched-accuracy winner.

28.3. *Established arithmetic and its remaining trusted components*

The R16 arithmetic kernel calls MPFR with 128-bit precision. Lower and upper endpoints use round-toward-minus-infinity and round-toward-plus-infinity for basic operations and elementary functions, including conversion to binary64. Scalar products and reductions propagate intervals instead of treating round-to-nearest BLAS output as exact. Rational constants are enclosed before use. The library rejects invalid logarithms, square roots, nonfinite endpoints, and division through zero. The remote build uses the installed MPFR headers; the source also contains a documented LP64 ABI fallback for the local development environment.

The complete eighteen-node audit retains every primal value interval, upper dual bound, expenditure interval, source-box count, and exact envelope breakpoint. Its C arithmetic and elementary functions differ materially from the retained rational–Taylor kernel. Its analytic conjugate decomposition, rational Gauss roots, derivative remainder argument, stopping corrections, and price-envelope mathematics are shared. Agreement therefore reduces arithmetic uncertainty but does not independently prove those mathematical premises. The 450 rational arithmetic checks, the neural automatic-differentiation comparisons, and the historical high-precision point comparisons are diagnostics, not formal interval proofs.

28.4. *Fresh model-to-price-envelope reproduction*

The fresh calculation initializes only the two endpoint prices. At every generated node, deterministic initializations produce a new feasible 16-slab policy and dual pilot; exact-decimal model constants are unchanged. The full original stopped payoff and adapted-control upper comparison are re-evaluated. The next node is the worst current exact envelope breakpoint rounded to spacing $1/1024$; a duplicate invokes the widest-interval midpoint. The declared termination rule is a proved uniform gap below .01 or thirty generated nodes. All intermediate node additions, optimizer statuses, costs, and certificates are retained, not only the final successful library.

TABLE XXV
COMPLETE MPFR RE-EVALUATION OF ALL EIGHTEEN HISTORICAL PRICE NODES

Price	Policy lower	Optimal upper	Node gap
0.5	-1.2726630012	-1.2628181763	0.0098448248
0.875	-1.2789733032	-1.2692555100	0.0097177931
1.25	-1.2847932336	-1.2751091803	0.0096840532
1.625	-1.2901339789	-1.2804567231	0.0096772557
2	-1.2950178727	-1.2853301725	0.0096877002
2.5	-1.3008396015	-1.2911285459	0.0097110555
3	-1.3059053778	-1.2961649251	0.0097404526
3.5	-1.3102486540	-1.3004746771	0.0097739768
4	-1.3138977523	-1.3040895090	0.0098082433
4.25	-1.3154696749	-1.3056468359	0.0098228390
4.5	-1.3168890699	-1.3070533282	0.0098357416
5	-1.3193518240	-1.3094933092	0.0098585147
5.5	-1.3214155553	-1.3115375893	0.0098779659
6	-1.3231704727	-1.3132756877	0.0098947850
6.5	-1.3246813844	-1.3147719178	0.0099094666
7	-1.3259960813	-1.3160736824	0.0099223988
7.5	-1.3271505957	-1.3172167379	0.0099338578
8	-1.3281726192	-1.3182285110	0.0099441081

Lower endpoints are rounded downward and upper bounds upward. Differences are computed from exact dyadic endpoint fractions, not from the displayed decimals. Every original adapted control is covered by the upper bound; every lower bound belongs to its identified stored time-control policy. The exact all-price envelope is computed from unrounded endpoints and expenditure intervals.

The canonical result has 20 newly generated nodes. It uses the rational–Taylor arithmetic inherited as an algorithm, not historical policy data or preselected dual pilots. The separate historical MPFR result must not be reassigned to this fresh object. Neither price-node refinement nor a denser quadrature eliminates the current dual relaxation’s positive allowances; the retained 16/32/64-slab experiment records the corresponding floor.

28.5. *Coupled-state neural control*

The inventory model, neural parameterization, Riccati residual, and global proof are given in the main paper. Seeds 16200–16209 alternate between widths 8 and 16, with two hidden tanh layers. Adam uses learning rate .005, batch 128, double precision, and one thread. Both 250- and 1,000-update checkpoints are retained. Each is certified on 512 complete time cells. One pair of gain functions is reused at dimensions 4, 8, 16, 32, 64, and 128. The proof covers unbounded state dynamics and every initial state with mean squared inventory at most one; no simulated tail correction or finite state cover is used.

The resource record also includes a structure-aware classical solution of the two scalar Riccati equations using adaptive floating-point RK45. This is a diagnostic reference and is not an interval certificate, a neural training label, or a claimed matched-verified-accuracy benchmark. The theorem compares with the exact Riccati optimum through a differential error inequality, not by treating that numerical reference as ground truth. The per-coordinate and total bounds are both stored; multiplying by dimension is not evidence of dimension-independent total welfare error.

TABLE XXVI
FIXED NEURAL GAIN FUNCTIONS: COMPLETE TIME-COVER REFINEMENT

Time cells	Orthogonal residual	Mean residual	Loss/coordinate
128	0.02124605	0.03542657	0.000411639
512	0.01378842	0.01707312	0.000095606
2048	0.01193411	0.01477912	0.000071640

The same seed 16200 final gain pair is used in every row. This isolates enclosure refinement from training-budget changes. All state dimensions share the same two-mode proof; total loss is dimension times the per-coordinate bound.

TABLE XXVII
INVENTORY TOTAL POLICY-LOSS BOUNDS ACROSS STATE DIMENSION

States	Smallest final bound	Largest final bound
4	0.00010119	0.00038243
8	0.00020237	0.00076485
16	0.00040473	0.00152969
32	0.00080945	0.00305938
64	0.00161889	0.00611876
128	0.00323777	0.01223752

The range is over all ten prescribed final checkpoints, not a selected training result. These are total loss bounds. The initial region scales as $|x_0|^2/d \leq 1$; the full diffusion is state-global. Every dimension reuses the same gain functions and invariant-mode proof, so these rows are not six independent training/verification cost measurements.

28.6. Economic interpretation and reproducibility

The consumption-top-up calculation keeps the preference, adjustment-cost, wealth, and stopping paths fixed while providing externally financed consumption goods. It does not assert budget-feasible wealth compensation or empirical calibration. The main paper gives the marginal-utility inequality, discounted alive-time lower bound, and scale qualification. The original strict-loss and relative-resolution counterfactual diagnostics are preserved below. Policy-control accuracy is not inferred solely from a small value-loss bound; the selected time-control ranges remain distinct from the unknown optimal feedback.

TABLE XXVIII
MEASURED EXPERIMENT RESOURCES

Experiment	Wall seconds	Reported peak RSS (MiB)
Original neural	1093.74	—
Inventory neural	75.98	333.73
Fresh price library	415.79	497.42
MPFR historical nodes	1065.89	276.26
Six classical candidates	95.85	—

Per-experiment timings exclude package installation. Neural seeds and arithmetic nodes may run concurrently; peaks are process-local where stated in JSON and must not be summed as simultaneous memory. A dash means no aggregate peak was measured; seed/process peaks remain in their individual records. The root pipeline receipt reports end-to-end wall time and the execution environment identifies hardware and package versions.

From a clean checkout, the command

`python revisions/2026-09-23-r16/replication/run_all.py` executes the new scientific pipeline. Existing experiment output directories are refused rather than overwritten. The run records the source commit, source hashes, package versions, processor/platform information, per-process resources, and every attempt. The paper table generator reads the resulting JSON and rounds all bound decimals upward. The publication validator compiles both documents, checks internal references, verifies complete counts and target labels, and hashes the materialized scientific objects. The public review receipt identifies the scientific source and result commits separately from any subsequent publication metadata.

The following sections preserve the R14 supplement in full. They describe the retained R14 implementation and historical objects, except where an R16 replacement has been explicitly identified above. Their existence does not make those historical neural policies newly successful or convert a diagnostic comparison into a rigorous proof.

29. RETAINED R14 TECHNICAL DEVELOPMENT

30. SCOPE, NOTATION, AND THE CURRENT EVIDENCE

This supplement supplies details required to verify the R14 manuscript. Time, domain, control bounds, utility, and first-exit settlement are exactly those in the main paper. In particular, the lower calculations do not replace stopped CRRA utility by an unstopped singular expectation, and the upper calculations do not restrict the original adapted portfolio control to zero. The deterministic policies are lower-bound objects; their upper comparison bounds optimize over the original economic opportunities.

Three numerical objects have distinct meanings. The state-feedback tanh actor/critic has a full-domain residual certificate whose current bound is large. The independently verified time-control library has a sharp central-state price-continuum certificate and a new initial-state rectangle certificate. The freshly generated 16-, 32-, and 64-interval policies provide a policy-richness and resource experiment without inherited optimizer inputs. These objects have separate files, hashes, and accuracy predicates.

The main paper contains full proofs of its residual theorem, coefficient transport, and four-corner price construction. Sections below give the analytic economic dual and the independently implemented enclosure lemmas used as inputs to the computer-assisted economic propositions. The shared analytic premise is explicit: two numerical implementations of a false analytic inequality would not establish that inequality.

31. THE AFFINE-PREFERENCE DUAL IN THE ORIGINAL STOPPED ECONOMY

31.1. *Consumption conjugacy and its supporting plane*

Write $r = u - 1$, $l = \log y$. On the rectangle relevant to the witness, the lower consumption cap .05 never binds. The consumption conjugate is

$$\Psi(u, e^l) = \begin{cases} -\exp(-r \log .8)/r - .8e^l, & l \leq -u \log .8, \\ -u \exp(rl/u)/r, & l \geq -u \log .8. \end{cases} \quad (1)$$

Let $S_0 = \Psi + .01e^l - .004 \log(.5)$ and $\beta = \partial_u \Psi$. Both branches match in value and first derivatives at the consumption switch. Their one-sided Hessians need not match; all interval boxes crossing the switch use their hull.

The desired supporting inequality is

$$S_0(u, y) \leq S_0(m, y) + \beta(m, y)(u - m), \quad u \in [1.5, 2.5], \quad m \in [2, 2.2], \quad y \in [.2, 12]. \quad (2)$$

The following derivative facts reduce its proof to one-dimensional covers. On an interior consumption branch, putting $L = l/u$ and $E = \exp((1 - 1/u)l)$, direct differentiation gives

$$\partial_l S_{0,uu} = \frac{EL(2-L)}{u^2}, \quad \partial_l \beta = -\frac{El}{u^2}.$$

The first derivative is nonnegative on the present box: $0 < L < 2$ whenever the interior branch applies. On the capped branch $S_{0,uu}$ is independent of l . Thus its maximum over the y rectangle is controlled at $y = 12$.

A complete interval cover of $u \in [1.5, 2.3]$ at $y = 12$, consisting of 1,600 intervals, gives $S_{0,uu} \leq -.0671742195$. Hence β is decreasing over this interval. On $u \in [2.3, 2.5]$, a separate 2,000-interval cover gives

$$\beta(u, 12) \leq -.9675376119 < -.9572021815 \leq \beta(2.2, 12).$$

To extend this comparison to smaller y , compare $\beta(u, e^l) - \beta(m, e^l)$ for $u > m$. In the interior branch, E/u^2 decreases in u because $\partial_u \log(E/u^2) = l/u^2 - 2/u < 0$ on the box. Therefore $\partial_l \beta(u, e^l) - \partial_l \beta(m, e^l) \geq 0$ whenever both are interior. When the larger u is capped and the smaller is interior, this difference is again nonnegative; when both are capped it is zero. The difference is consequently largest at $y = 12$. Integrating the resulting derivative order from m to u proves the upper-side supporting inequality. Integrating the decreasing derivative on the lower-side interval proves the other side. This proves (2) without asserting global concavity of S_0 .

Additional complete covers of $m \in [2, 2.2]$ verify

$$-7.1 \leq S_0(m, y) \leq 0, \quad |\beta(m, y)| \leq 1.1 \quad (y \in [.2, 12]). \quad (3)$$

Here S_0 is decreasing in y , since its derivative is $-c^* + .01 < 0$, and β is nonincreasing in l where its derivative is nonzero. Checking the two y endpoints therefore suffices. The recorded lower bound is above -6.989553453 , and the recorded absolute coefficient bound is below $.971182123$. All finite covers use the new interval implementation and the formulas (1) with automatic differentiation.

31.2. The auxiliary adapted-control problem

Let Y follow $dY = .02Y dt - .3Y dZ^x$, let $\hat{Y}$ be its clipping to $[\cdot, 12]$, and fix a deterministic reference drift $\bar{\theta} \in [0, .2]$. Put

$$m(t) = 2 + \int_0^t \bar{\theta}_s ds, \quad \phi(u) = -.02(u - 2)^2,$$

$$Q(u) = \phi(m(1)) + \phi'(m(1))(u - m(1)), \quad b_T = \phi'(m(1)).$$

Define the affine source

$$\hat{S}_t(u, y) = S_0(m(t), \hat{y}) + \beta(m(t), \hat{y})(u - m(t))$$

and the full-line auxiliary value

$$A_k(t, u, y) = \sup_{|\theta| \leq .2} \mathbb{E}_{t,u,y} \left[\int_t^1 e^{-\rho(s-t)} \{ \hat{S}_s(U_s, Y_s) - k\theta_s^2/2 \} ds + e^{-\rho(1-t)} Q(U_1) \right], \quad (4)$$

where $dU_s = \theta_s ds + .05dZ_s^u$. The preference action is still adapted. Because Y is exogenous and the coefficients of U and its terminal value are affine, the conditional marginal value is

$$B_s = \mathbb{E} \left[\int_s^1 e^{-\rho(t-s)} \beta(m(t), \hat{Y}_t) dt + e^{-\rho(1-s)} b_T \middle| \mathcal{F}_s \right].$$

Conditional Fubini, followed by pointwise maximization of the strictly concave quadratic in θ_s , gives the exact improvement formula relative to θ :

$$A_k(0, 2, y_0) = \bar{A} + \int_0^1 e^{-\rho s} \mathbb{E} g_k(B_s, \bar{\theta}_s) ds, \quad (5)$$

where $g_k(q, a) = H_k(q) - qa + ka^2/2$ and

$$H_k(q) = \max_{|a| \leq .2} \{qa - ka^2/2\}.$$

The reference expectation is

$$\bar{A} = e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \{\mathbb{E} S_0(m(t), \hat{Y}_t) - k\bar{\theta}_t^2/2\} dt + C, \quad (6)$$

$$C = .00375 \int_0^1 t e^{-\rho t} \mathbb{E} [\partial_t \beta(m(t), \hat{Y}_t)] dt. \quad (7)$$

To obtain the covariance term, use the reference decomposition $U_t = m(t) + .05Z_t^u$. The log-deflator is $l_t = \log y_0 - .025t - .3Z_t^x$. Gaussian integration by parts gives $\mathbb{E}[Z_t^u \beta(l_t)] = .075t \mathbb{E} \beta_l(l_t)$, because the correlation is $-.25$. Multiplication by $.05$ produces $.00375$. Clipping is included in the almost-everywhere derivative. Since $\beta_l \leq 0$, the correction is nonpositive; it is evaluated rather than dropped.

31.3. A quantitative allowance for the random marginal value

The derivative H'_k is $1/k$ -Lipschitz, including across its saturation points. For $b(s) = \mathbb{E} B_s$ this yields

$$\mathbb{E} g_k(B_s, a) \leq g_k(b(s), a) + \frac{\text{Var}(B_s)}{2k}.$$

A Gaussian Poincaré inequality for the past Brownian coordinate and conditional Jensen give

$$\text{Var}(B_s) \leq .09s \left\{ \int_s^1 e^{-\rho(t-s)} \sqrt{\mathbb{E} |\partial_t \beta(m(t), \hat{Y}_t)|^2} dt \right\}^2.$$

For $m \in [2, 2.2]$, the nonzero derivative has $l > 0$ and

$$|\partial_l \beta|^2 \leq \frac{1}{16} e^{(12/11)l} l^2.$$

The derivative vanishes in the capped-consumption and clipped-deflator regions. For a Gaussian l with mean μ and variance v ,

$$\mathbb{E}[e^{al} l^2] = e^{a\mu + a^2 v/2} \{(\mu + av)^2 + v\}.$$

Using $a = 12/11$, $M = \log y_0$, $\mu \leq M$, $v \leq .09$, and the positive mean range $y_0 \in [1.2, 2.4]$, define

$$D^2 = \frac{e^{aM + .045a^2}}{16} \{(M + .09a)^2 + .09\}. \quad (8)$$

The preceding expressions are increasing in the relevant μ, v ranges. Since $\int_0^1 s(1-s)^2 ds = 1/12$ and $C_\rho(1-s) \leq 1-s$, equations (5)–(8) imply

$$A_k(0, 2, y_0) \leq \bar{A} + \int_0^1 e^{-\rho s} g_k(b(s), \bar{\theta}_s) ds + \frac{.09D^2}{24k}. \quad (9)$$

This positive variance allowance is retained at every mesh. An unconditional reference control is not silently treated as the optimal adapted auxiliary control.

31.4. *Returning to the first-exit contract*

Set $h = 1 - t$ and

$$W(t, u, x, y) = y(x - .5) + .1 \log(.5) + A_k(t, u, y).$$

The product identity for Yx cancels every adapted portfolio control. Consumption conjugacy and the supporting plane imply that discounted W plus the original running payoff is a supermartingale up to the relevant state or localization face. This can be proved directly by conditional Fubini and the adapted selector in (5). Alternatively, smoothing the bounded Lipschitz source with a uniform error correction gives a classical verification argument and then the limit.

The affine tangent satisfies $Q \geq \phi$. To control continuation, use the feasible auxiliary action $\theta = 0$. For an initial $u \in [1.5, 2.5]$,

$$\mathbb{E}|U_r - m(r)| \leq .7 + .05 = .75, \quad \mathbb{E}|Q(U_r)| \leq .0008 + .008(.75) = .0068.$$

Applying Itô to Q and using (3) shows

$$A_k(t, u, y) - Q(u) \geq \{-7.1 - 1.1(.75) - .04(.0068)\}h = -7.925272h > -8h. \quad (10)$$

For $x \in [.5, 2]$, $y \geq .2$, $y(x - .5) + .1 \log(.5) - .1 \log x \geq 0$, by differentiation in x and equality at $.5$. Thus W dominates the original settlement on actual wealth exits and at the horizon.

For completeness, G is a global upper comparison for the original economy. The utility is increasing in consumption, and even its largest value on the original state/action box is less than $-.82$. The positive parts of θG_u , the wealth drift term, and $-\rho G$ together are less than $.017$, while the remaining diffusion and adjustment terms are nonpositive or can be bounded within this margin. Hence $\sup_a \mathcal{R}_a G < 0$. Also $G \geq g$ on the stopping trace. Stopped verification therefore gives $V_k \leq G$ for $k \geq .5$.

The potentials

$$Z_u = 8h\{e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h}\},$$

$$Z_y = 8h\{e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h}\}$$

are nonnegative supermartingale potentials. Indeed, $.2(120) + .00125(120)^2 = 42$ bounds the preference exponential generator, while $.025(22) + .045(22)^2 = 22.33$ bounds either sign of the log-deflator generator. Differentiating the factor h makes the killed generator nonpositive.

Each contributes at least $8h$ at its artificial face. Adding them to W , using (10) and then the fallback $V \leq G$, proves the original-economy upper bound in the main paper. Artificial clipping and full-line continuation are therefore proof devices with explicit allowances, not changes to the economic contract.

In the stress calculation the corresponding continuation floor is $-7.11 - 1.11(.75) - .04(.0068) = -7.942772 > -8$. Witness perturbations remain in the same verified reference class, so the same analytic argument applies.

32. INDEPENDENT ARITHMETIC AND QUADRATURE

32.1. *Trusted arithmetic model*

The interval core assumes IEEE 754 binary64 addition, subtraction, multiplication, and division in round-to-nearest arithmetic; correctly directed neighboring numbers from `nextafter`; exact binary decomposition/scaling through `frexp` and `ldexp` where applicable; and gradual underflow rather than flush-to-zero. Each elementary result is expanded outward. Reductions use explicit outward pairwise operations. Neural affine layers do not invoke an unaccounted BLAS dot-product error model. Overflow, nonfinite endpoints, or division through zero cause failure rather than cell deletion.

Every decimal economic constant is first interpreted as a rational. Every stored binary64 parameter is its exact dyadic rational value. A rational is converted to an interval only after checking its position against the candidate floating value. Intervals can overestimate because of dependency, but cannot be narrowed by an optimizer's successful termination flag.

For $|x| \leq 1/8$, the exponential uses its degree-18 Taylor polynomial with the uniform absolute remainder

$$\left| e^x - \sum_{j=0}^{18} \frac{x^j}{j!} \right| \leq \frac{8}{7} \frac{(1/8)^{19}}{19!}.$$

Larger arguments are divided by powers of two until the small-argument condition holds, and enclosed squaring restores the result. For a positive argument, an exact binary exponent and a mantissa in $[1, 2)$ reduce its logarithm to

$$\log m = 2 \sum_{j=0}^{39} \frac{y^{2j+1}}{2j+1} + R, \quad y = \frac{m-1}{m+1}, \quad |R| \leq \frac{2(1/3)^{81}}{81(1-1/9)}.$$

The same construction encloses $\log 2$. Tanh is evaluated from the enclosed exponential and monotonicity, with exact range restriction to $[-1, 1]$. No assumed ulp accuracy of a platform transcendental function enters these proof computations. Separate tests compare 807 elementary enclosures with 100-digit evaluations; these tests check implementation behavior but do not replace the remainder inequalities.

32.2. *Rationally certified Gauss rules*

The eight Gauss–Legendre roots are first proposed by an ordinary floating-point routine. Each root is then isolated in a disjoint rational bracket by exact sign evaluations of the degree-eight Legendre polynomial and 80 bisections. Eight disjoint sign-changing brackets account for all roots. The positive weights are enclosed using

$$w_i = \frac{2}{(1-x_i^2)[P'_8(x_i)]^2}.$$

The program checks positivity and all moments through degree 15. The Gauss rule's exactness through degree $2n - 1$ and its positive weights are the standard construction described in NIST (2026); its nodes and weights are not assumed to equal their floating-point proposals.

Square-root endpoint proposals are validated by exact rational square comparisons and expanded until they bracket the true root. The constant π is enclosed using Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, with 60 alternating-series terms and the first omitted term as a remainder bound.

For an analytic function bounded by M on a complex disk of radius R around the center of a real integration interval of half-width $h < R$, the degree-15 Taylor remainder on that interval is at most $M(h/R)^{16}/(1 - h/R)$. Both integration and the positive Gauss rule have norm equal to the interval length. Subtracting their exact actions on the degree-15 polynomial gives the rigorous remainder

$$|I(f) - Q_8(f)| \leq 2 \text{length} M \frac{(h/R)^{16}}{1 - h/R}. \quad (11)$$

This inequality, interval root/weight errors, and outward summation are used together. Agreement between two mesh levels is not an error bound.

32.3. The original-policy evaluator

For a stored time policy and an initial state in K , let μ_j be its cumulative preference drift at the start of interval j . After $t = v^2$ the covered preference is

$$U = u_0 + \mu_j + \theta_j(v^2 - j/n) + .05vz, \quad z \in [-8, 8].$$

The integrand contains the original utility $-\exp[-(U - 1) \log c_j]/(U - 1)$, the Gaussian density, the Jacobian $2v$, and discount $e^{-.04v^2}$. It is not a polynomial utility surrogate. A tensor Gauss rule uses eight subdivisions in v per policy interval, 32 normal cells, and eight Gauss nodes on each axis.

The complex-neighborhood constants used in (11) are global on the state rectangle and the feasible control ranges. For the v disk use radius $1/8$ with real $z \in [-8, 8]$. One has $|v| \leq 1.125$, and the real part of $U - 1$ stays between .476875 and 1.723125. There is no CRRA pole in the disk. The utility, Gaussian density, discount, and Jacobian together have modulus less than five. For the z disk use radius two with real $v \in [0, 1]$; the real part of $U - 1$ stays between .48 and 1.72, while the Gaussian density's complex factor is at most $e^2/\sqrt{2\pi}$. The complete modulus is below 32. These values give $M_v = 5$, $M_z = 32$. Tensor error follows by writing $I_v I_z - Q_v Q_z = (I_v - Q_v)I_z + Q_v(I_z - Q_z)$ and using positivity and the interval lengths. All remainder terms are summed outward.

To bound the omitted normal tails, extend the running utility by clipping its preference argument to $[1.2, 2.8]$ outside the covered range. On the feasible consumption interval this extension lies in $[-6, 0]$. Its omitted expected contribution lies in $[-12e^{-32}, 0]$, by the Gaussian Chernoff bound. The terminal preference quadratic is integrated analytically without clipping, and terminal wealth is exactly enclosed from the budget. The auxiliary full-horizon functional is connected to the original stopped payoff by a separate estimate; it is not substituted for that payoff without correction.

For all initial states in K and all stored or fresh feasible drifts in $[0, .2]$, either preference exit requires a Brownian excursion of at least .58 after subtracting the largest possible drift. Therefore

$$p_{\text{exit}} \leq 2 \exp\{-.58^2/(2 \cdot .05^2)\}.$$

Also

$$\mathbb{E}(U_1 - 2)^4 \leq .22^4 + 6(.22)^2(.0025) + 3(.0025)^2.$$

Bounding the running-payoff, fee, and bounded terminal differences on the exit event, and using Cauchy–Schwarz for the unbounded terminal quadratic, gives the uniform allowance

$$\epsilon_{\text{stop}} = 15p_{\text{exit}} + .02\sqrt{\{.22^4 + 6(.22)^2(.0025) + 3(.0025)^2\}p_{\text{exit}}} < 3.87 \times 10^{-18}. \quad (12)$$

The constant 15 covers all $k \leq 8$. The wealth path is strictly decreasing for the verified controls, with $x_1 > .5$, so there is no wealth exit before time one. The final policy interval includes the symmetric quadrature error, the one-sided missing utility tail, and the symmetric stopping correction (12).

The central-state stopped expenditure is independently enclosed by its exact deterministic full-horizon value minus at most $.02p_{\text{exit}}$. The state–price concavity construction instead uses the full-horizon expenditure of its covered lower functional; its uniform stopping error has already been paid. Both constructions are therefore valid, but their expenditure labels must not be interchanged.

32.4. The independent dual evaluator

The dual changes variables to $t = v^2$, with 32 subdivisions per reference time interval and 512 normal cells on $[-6, 6]$. The source and its first and second derivatives are generated from (1) by an independent two-variable interval differentiation class. Chain rules map them to (v, z) . Every source box is enclosed, and every possible consumption-switch branch is included. The program verifies that the normal cover does not cross a deflator clipping face; the omitted tails cover the remaining probability.

Write $w_v = 2ve^{-.04v^2}$ and $w_z = (2\pi)^{-1/2}e^{-z^2/2}$. Zeroth, first-centered, and second-centered moments of these weights are integrated by the validated Gauss rule. For their Cauchy bounds the disks have radii $1/8$ and two and base modulus bounds three. Centered moment factors use $(R + 1/1024)^p$, $p = 0, 1, 2$, with an explicit check that midpoint representation errors are below $1/1024$. This accounts for a floating-point center that is not an exact interval midpoint.

Constant and linear source terms retain the signed first-centered moments. The source Hessian on the complete box bounds the remainder by

$$\frac{1}{2}\|S_{vv}\|_{\infty}M_{v,2}M_{z,0} + \|S_{vz}\|_{\infty}r_v r_z M_{v,0}M_{z,0} + \frac{1}{2}\|S_{zz}\|_{\infty}M_{v,0}M_{z,2}.$$

The source is C^1 across the action switch with bounded one-sided Hessians, so the integral Taylor formula with the Hessian hull is valid there. Weighted first moments are not discarded on the false premise that a weighted box is symmetric.

The Gaussian tail probability is at most $2e^{-18}$. The source tail is enclosed between -7.1 times its discounted mass and zero; the coefficient tail has absolute bound 1.1 times its mass. Since $\beta_l \leq 0$ and $|\beta_l| \leq 2.5$ on its nonzero clipped region, the covariance tail has the signed enclosure used in the code. These corrections cover the entire normal distribution.

To enclose $b(s)$, coefficient integrals are accumulated backward in time. The incomplete current cell is bounded by its entire discounted mass and its coefficient interval. The convex function $g_k(q, \theta)$ attains its maximum on the resulting q interval at an endpoint. Its scalar action maximum is evaluated with the clipped maximizer. Terminal, source, covariance, control-gap, variance, localization, and arithmetic terms are finally summed outward. The stored upper endpoint is a bound on the original optimum, not on the fitted pilot objective.

33. INDEPENDENT HIGH-PRECISION CROSS-CHECKS

A third program evaluates the Gaussian expectations by closed truncated-lognormal moments rather than the production or second interval quadrature. If $L \sim N(\mu, s^2)$, define

$$M_q(a, b) = e^{q\mu + q^2 s^2 / 2} \left\{ \Phi\left(\frac{b - \mu - qs^2}{s}\right) - \Phi\left(\frac{a - \mu - qs^2}{s}\right) \right\}.$$

The first weighted moment is

$$M_{1,q}(a, b) = (\mu + qs^2)M_q(a, b) + se^{q\mu + q^2 s^2 / 2} \{\varphi(z_a) - \varphi(z_b)\}.$$

The capped region, interior consumption region, and two deflator-clipping tails are integrated separately using these formulas. In particular, the interior coefficient derivative integrates as $-M_{1,q}/m^2$, not by blindly applying a smooth quadrature rule across its switch. Remaining time integrals are evaluated at 40 and 70 decimal digits.

The source, initial affine coefficient, and covariance at prices .5, 4.25, and 8 all fall inside the independently certified intervals at both precisions. All six runs and their timings are stored. This test is numerically independent in decomposition and precision, but its time quadrature is not asserted to produce an interval proof. The decisive bounds remain those from the complete rational-root/interval execution.

34. NEURAL DERIVATIVES, ACTIONS, AND THE OUTPUT-ERROR CONTRACT

For a scalar composition $w = \tanh z$, the chain rule is

$$w_i = (1 - w^2)z_i, \quad w_{ij} = -2w(1 - w^2)z_i z_j + (1 - w^2)z_{ij}.$$

The interval implementation propagates value, t, u, x first derivatives and uu, ux, xx second derivatives through each affine layer and $\tanh$ layer. For the critic $v = G + (1 - t)N$, this gives, for example,

$$v_t = -N + (1 - t)N_t, \quad v_{ux} = (1 - t)N_{ux},$$

$$v_{uu} = -.04 + (1 - t)N_{uu}, \quad v_{xx} = -.1/x^2 + (1 - t)N_{xx}.$$

Every multiplication and layer reduction is outward. The full cell image is used, not a derivative evaluated only at its center.

The action oracle uses the scalar formulas in the main paper. If the marginal-wealth interval crosses zero, the consumption optimum is enclosed over the whole feasible interval. If the portfolio-curvature interval crosses zero, the complete portfolio range is retained in the upper evaluation. These cases can make the bound wide, but preserve validity for all continuous actions. The candidate action ranges additionally include the 2^{-24} output allowance and are clipped to the exact economic action bounds.

The residual and action records for every interior cell are stored in compressed JSON. The finest cover has 16,384 such cells; each of the four first-exit faces has 512 cells. The terminal face is exact by the critic construction. The per-time-slab maxima are assembled with an enclosed discount integral. A separate undiscounted integral over the full horizon supplies a safe all-initial-times bound. The model-transport modulus encloses drift, variance, cross-covariance, discount, and running-cost changes simultaneously; the correlation remains fixed at $-.25$.

The mathematical feedback uses exact stored weights and real $\tanh$. Its output-neighborhood certificate covers any admissible implementation satisfying the stated action-error contract.

No automatic guarantee for a hardware tanh implementation, a stochastic time integrator, or a floating-point budget reconstruction follows from this contract without a separate error analysis. The network’s fine-domain certificate is deliberately not replaced by the much sharper bound belonging to a different time-control policy.

35. POLICY INTERPRETATION AND ECONOMIC-RESOLUTION RECORDS

TABLE XXIX
CENTRAL-STATE SELECTED POLICY REGIONS AND CONTROL SUMMARIES

Policy k_j	Left	Right	Width	Consumption range	Mean θ
0.5	0.50000	0.67789	0.17789	0.76182–0.77217	0.18363
0.875	0.67789	1.06576	0.38787	0.76197–0.77151	0.17275
1.25	1.06576	1.43620	0.37044	0.76217–0.77092	0.16279
1.625	1.43620	1.81288	0.37668	0.76243–0.77039	0.15319
2	1.81288	2.24707	0.43419	0.76273–0.76995	0.14390
2.5	2.24707	2.74540	0.49833	0.76320–0.76944	0.13209
3	2.74540	3.24895	0.50355	0.76374–0.76903	0.12073
3.5	3.24895	3.74918	0.50023	0.76435–0.76872	0.10983
4	3.74918	4.12169	0.37251	0.76499–0.76849	0.09950
4.25	4.12169	4.37187	0.25017	0.76529–0.76840	0.09446
4.5	4.37187	4.73837	0.36650	0.76557–0.76833	0.08992
5	4.73837	5.23938	0.50102	0.76605–0.76822	0.08205
5.5	5.23938	5.74023	0.50085	0.76646–0.76815	0.07546
6	5.74023	6.24095	0.50072	0.76632–0.76812	0.06985
6.5	6.24095	6.74158	0.50062	0.76616–0.76811	0.06503
7	6.74158	7.24212	0.50054	0.76602–0.76811	0.06083
7.5	7.24212	7.74259	0.50047	0.76590–0.76814	0.05715
8	7.74259	8.00000	0.25741	0.76579–0.76818	0.05389

Endpoints and widths are descriptive rounded values; exact rational endpoints determine the selector. Consumption and average preference adjustment refer to the selected stored policy, with portfolio zero. The state-uniform selector is separately recorded because it minimizes the worst corner gap rather than only the central-state gap.

The exact price program records strict-loss and relative-resolution regions for four anchor prices. Weak monotonicity holds in all cases from the model. A missing strict-resolution region means that the certified interval still includes zero, not that a price increase benefits the decision maker. The relative criterion for complete regions uses the untruncated interval difference and is conservative; selected pairwise tables also use the universal expenditure bound to tighten the upper loss endpoint.

36. REPLICATION, SCOPE OF TIMING, AND HISTORICAL PRESERVATION

The replication directory contains separate entry points for neural generation, neural verification, independent policy evaluation, independent dual evaluation, exact price postprocessing, state–price certification, full-library stress, high-precision diagnostics, and the fresh classical frontier. Executed results are written to new directories rather than overwriting earlier attempts. The build and manifest checks distinguish a numerical failure to meet a requested tolerance from a malformed or missing certificate.

The neural run is a full fresh run after interpreter entry, including all critic and actor updates. Shell timing separately includes imports. The classical frontier separately measures policy generation, dual fitting, primal verification, and dual verification, starts from fixed initial guesses in every case, and retains optimizer reports. The old library audit starts from frozen inputs and is labeled verification only. Historical interrupted work not recorded in the archive is unknown, not zero. Resource clocks from overlapping runs cannot establish a matched-hardware speed ranking.

A source-frozen clean replay reexecutes all 18 second-implementation node calculations and compares the decisive lower endpoints, upper endpoints, and newly computed expenditure intervals exactly. The original development ledger is preserved, including the fact that expenditure fields were subsequently replaced by independent calculations. The clean replay checks the final source rather than silently assigning its hash to an earlier execution. Content hashes identify every included source and result file. A local reconstruction of the R12 archive is distinguished from its actual remote Git commit; publication metadata does not claim a remote write when none has occurred.

Historical preservation is handled at the file level, not by requiring every previous sentence to remain in the current main text. A byte-for-byte audit compared all 1,830 files of the downloaded R12 repository archive with the initial working copy and found no mismatch. Earlier manuscripts, proofs, data, failure records, and code remain unchanged. The old navigation index is archived before its replacement with a single R14 entry point. The current supplement supplies the proofs needed for the current manuscript; it does not concatenate 148 pages of superseded manuscripts.

TABLE XXX
PRESERVATION MAP FOR SUBSTANTIVE INHERITED MATERIAL

Material	Unchanged location and interpretation
Original model and early theoretical development	Root <code>ECTA.tex</code> , <code>ECTA_R2.tex</code> , and earlier source directories; retained research history.
Recursive utility, sophisticated selves, games, stochastic trace identities	Earlier manuscripts and <code>SUPP_R12.tex/.pdf</code> ; retained extensions, not new full-domain numerical claims.
Original continuous-policy evaluation and cover/oracle theory	R9 replication and proof directories; comparison baseline for the independently implemented R14 objects.
Affine-preference dual and earlier loose witnesses	R10 proof, result, and retained-construction files; analytic history and failed/loose certificates remain visible.
External holdout and method-specific tuning	R9–R11 result and paper directories; unfavorable panels and unstable seed orderings preserved.
Exact-reference gain-policy study	R11 accuracy source and results; not recast as the R14 tanh architecture’s accuracy.
Structured 128-dimensional action oracle	R10–R12 retained action-oracle evidence; frozen-jet action optimization, not 128-state dynamic control.
Eighteen-price continuation and historical refinement	R12 source, results, receipts, and refinement history; frozen inputs and their original timing scope preserved.

The response document maps every R13 finding, the inherited R12 issues, and the current evidence paths. It explicitly distinguishes completed calculations from unachieved neural accuracy, absent matched-accuracy classical comparisons, and remote publication status. This preserves the research objective without certifying a claim that the current computation has not established.

37. HISTORICAL R18 EXPOSITION, PRESERVED VERBATIM

The next two sections reproduce the preceding revision’s introduction and conclusion without deletion. Their descriptions of the then-primary method and then-available evidence are historical. The new policy-value mechanism and new evidence are stated in the R19 main text; this annex is not an alternative current claim of numerical performance.

38. INTRODUCTION

Neural approximations to dynamic economic decisions require a criterion that is linked to the value of the policy actually delivered. Average equation residuals need not provide that link. A small error at frequently visited training states can coexist with a large error near a stopping face, and improving a value witness need not improve the policy with which it is paired. These distinctions become operational when an algorithm must return a complete error bound rather than a sampled diagnostic.

This paper develops certificate-aligned neural Bellman operators. The primary algorithm minimizes a differentiable envelope of two signed Bellman residuals on a complete state–time cover. An independent directed-arithmetic verifier checks the proposed policy and witness against the original economic model. Every prescribed proposal is retained; an incumbent is replaced only when its complete certificate decreases. A quantitative theorem connects the trainable envelope to the rigorous policy-loss bound, separating the smoothing allowance from a checkable numerical correction. The theorem concerns the loss actually used in the computation, rather than an unexecuted policy-iteration idealization.

Stopping geometry is part of this construction. In our consumption–investment economy with costly preference adjustment, continuation value has a strict jump exceeding 6.37 utility units at an inward upper-wealth face. A critic forced to equal immediate settlement on every face therefore cannot acquire a vanishing uniform positive residual. A signed accessibility-weighted comparison permits the critic to dominate settlement at a face that the candidate cannot reach, while retaining every originally admissible action in the optimal upper comparison. This corrects an approximation-space inconsistency without modifying preferences, controls, or the stopping contract.

The original-economy experiment compares two generations of training on that unchanged problem. Under the historical sampled objective, extending training from 800 to 2,400 updates worsens all ten certificates, even though sampled residual mean squared error falls for nine seeds. Under the complete-cover objective, all ten accessibility-aware trajectories have decreasing raw certificates at 0, 100, and 400 updates. Their final bounds range from 7.27832 to 7.34367, compared with initial bounds from 26.3005 to 26.6668. A paired ten-seed all-face experiment, unrestricted state-space comparators with method-specific witnesses, and a fixed-network refinement study identify what causes this improvement and what it does not establish. In particular, the .01 original-economy target is unchanged and is not attained. A certified fixed-witness floor and actor–critic interchange show why neither further arithmetic precision nor a monotone incumbent alone establishes accurate policy learning.

A second, nonlinear application supplies an explicit accuracy–work result on a genuinely vector-valued state. A heterogeneous multi-product inventory planner has 12 decision dates, coupled nonquadratic costs, and up to 128 state coordinates. The neural map takes the full initial state and returns a control plan, without optimal-value or optimal-control labels. Strong convexity yields a state-uniform certificate for exact-real gradient corrections to that plan. At 128 coordinates, the total-cost bound falls from 0.124820 after 16 corrections to 0.000359609 after 24 and below 1.037×10^{-6} after 32. This is an unknown-value nonlinear planning problem, not the two-mode Riccati example retained for validation. Zero-start gradient, accelerated-gradient,

and quasi-Newton alternatives are evaluated with the same independent cost-loss certificate. They remain strong competitors; the experiment establishes an accurate neural-initialized solver, not a demonstrated neural speed advantage.

Economic interpretation also requires keeping numerical objects distinct. The sharp deterministic time-control/dual library in the original economy is not a neural result. Its price-uniform regret below 0.009982 can be offset by a rigorously checked, budget-financed initial-wealth increment of .0078 at initial wealth 1.25, using an explicitly adjusted feasible consumption policy. This sufficient compensation measure respects the original consumption cap and stopping contract. The earlier externally financed top-up remains a secondary normalization. Neither measure converts the much larger original neural certificates into sharp welfare claims.

38.1. *Contribution and relation to the literature*

Stopped verification, Bellman contraction, strong-convexity error bounds, and gradient contraction are classical ingredients. Judd (1998) and Kushner and Dupuis (2001) develop numerical dynamic programming and controlled Markov-chain approximation. Barles and Souganidis (1991) establishes a fundamental monotone-approximation framework, and Brumm and Scheidegger (2017) develops adaptive sparse-grid methods for computational economics. Information and control relaxations provide dual comparisons (Brown, Smith, and Sun, 2010, Brown and Smith, 2011). Neural dynamic-control and differential-equation methods include Han and E (2016), Han, Jentzen, and E (2018); the historical external comparison uses Cai, Fang, and Zhou (2025).

Our contribution is the combination of an identified stopping-trace inconsistency, an executed complete-cover neural objective with a checkable policy-loss implication, and a nonlinear control specialization with an explicit state-uniform work-accuracy bound. The new objective theorem does not claim global convergence of Adam, and the nonlinear specialization does not claim to remove the difficulty of arbitrary stochastic high-dimensional Bellman equations. The original-economy upper comparison always retains the full control opportunity. The arithmetic implementation uses MPFR (Fousse et al., 2007); shared analytic formulas, stopping arguments, and program logic remain part of its trusted mathematical base.

The paper presents the model and verification foundations first, followed by the stopping correction, the certificate-aligned algorithm, the original-economy evidence, and the nonlinear control application. Historical coefficient transport, state-price continuation, external holdouts, adverse training outcomes, and specialized dual calculations remain available in the main text and supplement. All new numerical statements are tied to immutable source and result objects; all 60 prescribed original-economy checkpoints are included in the new objective audit.

39. CONCLUSION

The objective of neural Bellman operators is an accurate state-contingent solution of the original economic decision problem. The method developed here makes the connection between training and that objective explicit. A complete-cover signed residual envelope guides neural proposals, independent directed arithmetic determines acceptance, and a quantitative envelope theorem states what the accepted object certifies. Stopping accessibility corrects a genuine inconsistency of all-face hard-settlement critics without changing the economy.

The revised evidence separates numerical progress from statements that the data cannot support. The complete-cover objective produces decreasing raw certificates at all prescribed budgets in ten original-economy trajectories, reversing the earlier sampled-objective anti-alignment. Matched boundary ablations, unrestricted classical candidates, method-specific witnesses, and

a computable fixed-witness floor identify the remaining obstruction. The original .01 neural target is still not met, and monotone certificate descent is not presented as monotone true policy improvement.

The nonlinear multi-product application supplies a different, positive accuracy–work result: a full-state neural initialization with a provably contracting correction attains a state-uniform total-cost bound below 1.037×10^{-6} at 128 coordinates. Strong classical alternatives remain competitive under the same certificates. A separately identified original-economy primal/dual library admits explicit budget-financed wealth compensation. These results preserve the neural-solver objective while making every theorem, numerical claim, comparison, and welfare interpretation refer to the exact object for which it is established.

40. HISTORICAL R19 EXPOSITION, PRESERVED VERBATIM

The following two sections retain the preceding manuscript's exposition. They are historical and do not override the R22 result domains.

41. INTRODUCTION

A numerical policy and an error bound for that policy are different economic objects. An approximation to a value function can improve while the associated consumption, investment, or adjustment decision remains unchanged. This distinction matters for neural dynamic programming: a training criterion that reduces an upper bound on regret need not increase the payoff of the delivered policy. A useful numerical method must identify both changes rather than treating either one as evidence for the other.

We develop Neural Bellman Operators with value-separated policy acceptance. A neural network proposes controls. A policy evaluator supplies a lower and an upper bound for the payoff of those controls under the original stopping contract. An optimal-comparison witness supplies a separate upper bound on the value of the unrestricted economic problem. A policy update is accepted only when the new payoff lower bound exceeds the incumbent payoff upper bound on the declared comparison set. Updating the optimal-comparison witness does not, by itself, change the incumbent policy. The resulting distinction is operational: the acceptance test concerns an actual policy-value difference, whereas the optimality test concerns the distance from that policy to the original optimum.

Our main new analytical result makes policy-sensitive comparison useful beyond a single initial condition. For budget-feasible time controls in the original stochastic economy, a coupled-payoff calculation bounds the variation of a policy-value difference across an initial-state rectangle. It uses differences between the two policies, rather than adding separate, much larger value-function Lipschitz constants. Original CRRA utility, common Gaussian shocks, exact terminal-wealth accounting, and explicit stopping corrections enter the calculation. An independently enclosed value difference at the reference state therefore yields a uniform policy-improvement certificate on a continuum of initial states. A related, observable progress inequality certifies contraction of true policy regret without substituting a critic loss for that regret.

The numerical implementation trains new neural control generators from model-only random starts. No inherited optimal policy, dual witness, or solution label enters their training objective. Five seeds, four prescribed work budgets, and the independent payoff acceptance rule are fixed before execution. All fifteen retained transitions increase certified policy values at the reference state. Initial-to-final improvement is uniform on $K = [1.98, 2.02] \times [1.24, 1.26]$: the gain is at least 0.016513 utility units for every reported seed. Combining this gain with the independently certified final optimality bound proves that true policy regret decreases by at least 58 percent throughout K . At the reference state it decreases by at least 75 percent. These are statements about the economic payoff, not about changes in a residual envelope.

The final reference-state optimality bounds lie between 0.009756577 and 0.009774061, below the unchanged .01 target. The corresponding initial-state-uniform bound is 0.011816625. Each result concerns newly generated neural time controls, held until the original first exit, with a budget-preserving dependence on initial wealth. The optimal upper comparison retains all originally admissible adapted controls. Neither this policy class nor its initial-state rectangle is identified with the full-state feedback architecture. The latter's best retained complete-domain certificate remains approximately 7.27832. The original objective of an accurate solution over the entire state-time domain is preserved; a localized certificate is not presented as its completion.

Continuation values are the second substantive issue. The full-state feedback experiments learn critics whose upper-wealth traces remain close to immediate settlement, even though

continuation is valuable when that face is inaccessible to the candidate. We evaluate the new policies' one-sided continuation values independently. Their certified excess over immediate settlement exceeds 6.708771 uniformly on the tested preference segment. These evaluations provide policy-induced targets for a continuation trace without using the unknown optimal value as a label. A separate, complete-cover intervention raises the old critic's trace while holding its actor fixed. Its global residual bound worsens. The distinction is informative: removing a necessary trace obstruction is not sufficient to construct a globally accurate Bellman upper witness.

The paper also quantifies the contribution of learned initialization in the nonlinear multi-product control problem. On a new, predeclared finite query distribution, neural, zero-start, and accelerated starts are compared at the same directed loss tolerances, rather than after an arbitrary common number of corrections. The learned map reduces gradient work in several regimes, particularly at 8 and 32 state coordinates. At 128 coordinates its advantage is confined to a coarser target; acceleration wins at the strictest target. A learned-gradient-dependent theorem explains the relevant correction budget. The existing cube-uniform guarantee remains explicitly initialization-agnostic, and the new finite-query results are not represented as a cover of that cube.

Finally, economic interpretation is attached to the newly certified neural policies themselves. A wealth increase of .0092, financed through an explicit feasible change in consumption, compensates their uniform bound on K . It is 0.736 percent of reference wealth. This result is distinct from the retained compensation result for the non-neural price library and does not normalize the much larger full-domain feedback certificate.

41.1. *Contribution and relation to the literature*

Stopped verification, Bellman contraction, strong-convexity bounds, and interval ordering are classical ingredients. Judd (1998) and Kushner and Dupuis (2001) develop numerical dynamic programming and controlled Markov-chain approximation. Barles and Souganidis (1991) establishes a monotone-approximation framework, and Brumm and Scheidegger (2017) develops adaptive sparse-grid methods for computational economics. Information and control relaxations provide dual comparisons (Brown, Smith, and Sun, 2010, Brown and Smith, 2011). Neural dynamic-control and differential-equation methods include Han and E (2016), Han, Jentzen, and E (2018); the retained external comparison uses Cai, Fang, and Zhou (2025). Directed arithmetic uses MPFR (Fousse et al., 2007).

Our contribution is the connection of an implemented neural policy-value gate to a coupled-payoff continuum certificate, constructive continuation-trace evaluation, and matched-tolerance accounting for learned representations. The five new actors are not inherited deterministic solutions relabeled as neural networks. Conversely, the faster classical transcription is not hidden, and neither interval ordering nor strong convexity alone is claimed as a new general convergence theorem. The complete-cover objective, accessibility theorem, state-price continuation, dual construction, external holdouts, and adverse historical results are retained. Their roles are separated from the new policy-sensitive results throughout.

42. CONCLUSION

Neural Bellman Operators require a separation between learning a policy, evaluating that policy, and bounding the unrestricted optimum. The value-separated update studied here supplies that separation in executable form. New neural actors improve independently certified original-economy payoffs, and a coupled-payoff theorem promotes the improvement to a continuum

of initial states. Their reference-state certificates meet the unchanged .01 target; their state-rectangle certificates admit explicit feasible wealth compensation. These achievements concern the newly generated policies themselves, rather than an inherited non-neural library.

The same separation clarifies continuation learning and computational work. Feasible-policy evaluation supplies verified one-sided continuation targets, but a scalar critic lift is not a global Bellman solution. Learned initialization reduces matched-tolerance gradient work in identified nonlinear-control regimes, but the cube-uniform correction theorem remains available without training and classical acceleration remains a strong comparator. Complete feedback certificates, every historical failure, and all matched classical results are retained. The objective remains an accurate, competitive solution of the original economic problem over its full state–time domain. The new policy-sensitive results advance that objective without substituting a change of witness, a restricted initial set, or a different numerical method for the result being claimed.

43. HISTORICAL R21 EXPOSITION, PRESERVED VERBATIM

44. INTRODUCTION

A useful numerical solution of a dynamic economic model must improve decisions, not merely the function used to evaluate them. This distinction is particularly important when a neural policy and a value witness are fitted jointly. A smaller upper bound on policy loss need not mean that the policy has improved: the same policy may receive a better witness. We therefore separate the generation of a policy, the verification of its own payoff, and comparison with the unrestricted optimum.

This paper develops Neural Bellman Operators with that separation. The original consumption–portfolio economy permits costly adjustment of preferences, correlated preference and financial shocks, and stopping at the boundary of the economic state space. Its primitives, admissible controls, stopping rule, and accuracy objective are unchanged. The new solver generates genuinely stochastic consumption and a nonzero portfolio. Neural coefficients determine an adapted consumption rule; an explicit contingent-claim price finances the rule and determines its portfolio. Independently enclosed stopped payoffs, rather than a changing critic, decide whether a proposed policy is accepted.

There are three substantive additions. First, we give a continuous-time self-financing and admissibility theorem for the executed neural policies. The decoder is not a discretized wealth recursion: its conditional-price identity holds between decision dates and for every realization of the traded factor. Second, we prove a continuum transfer theorem for four expert policies. Concavity of running utility alone is not enough to order two mixtures, so we also bound the old mixture’s Jensen gain. This supplies an independently verified improvement in actual policy value at every initial state in a rectangle. Third, we combine policy-specific lower bounds with the previously verified unrestricted dual upper bound. This proves useful neural accuracy without requiring the inaccurate historical feedback critic to serve as the evaluator.

Five confirmatory seeds, four experts per seed, and four retained work levels were fixed before the R20 execution. All sixty noninitial expert proposals pass a policy-specific strict acceptance test before further training. The present revision supplies their missing manuscript, the continuum arguments, a price-sensitivity theorem, and independent re-execution of the frozen-policy checks. It does not relabel those R20 optimization runs as new random R21 experiments. At time zero and adjustment price $k = 2$, every final neural mixture has regret below .002909 uniformly on

$$K = [1.98, 2.02] \times [1.24, 1.26].$$

The final policies improve actual payoff over their corresponding initial mixtures by at least 0.01762319 throughout K . Consequently, true regret is reduced by more than 85.8 percent at every point of this rectangle. A further, explicitly post-training calculation keeps the policies fixed and proves a regret bound of 0.00332694 throughout $K \times [1.5, 2.5]$.

The comparison is not made easy for the neural method. A direct, non-neural stochastic-control transcription is evaluated with the same four-state checker, unrestricted optimal upper bound, and continuum theorem. It reaches the same .005 tolerance and is faster in the recorded generation-plus-verification ledger. The five neural policies nevertheless have payoff lower bounds strictly above the policy-specific upper bounds of each of three named deterministic time-control transcriptions, uniformly on K . This identifies an economically consequential stochastic policy improvement without asserting superiority over all conventional solvers. A budget-financed wealth increment of .002 suffices for each learned mixture to attain at least the original optimum’s upper bound, at the uncompensated initial wealth, throughout K .

The domain of each assertion matters. A theorem uniform on K at time zero is not a theorem for all initial states of D and all starting times. The outstanding all-domain feedback target remains .01. Neither the localized dual comparison nor the new pricing decoder is presented as a globally sharp continuation supersolution. The earlier continuation-trace results, complete-domain failures, signed residual decompositions, actor–critic interventions, and nonlinear high-dimensional experiments remain in the paper and supplement. They are indispensable controls on the interpretation of the new results, rather than observations to remove after a successful localized experiment.

45. CONCLUSION

The new stochastic policies establish three different facts with different tests: strict expert-payoff improvement, uniform initial-to-final mixture improvement, and useful optimality bounds on a continuum of initial states and adjustment prices. The Jensen-gap argument is needed for the second fact; it cannot be replaced by four successful expert updates. The martingale-price decoder is needed for admissibility; it cannot be replaced by a sampled wealth trajectory. The unrestricted dual is needed for optimality; it cannot be replaced by policy comparisons alone.

These additions advance the original neural computation program without discarding its difficult evidence. They concern newly trained, genuinely stochastic neural policy generators, not a renamed deterministic library. They also retain the strong direct stochastic comparator, which supplies similarly sharp local accuracy without hidden layers and without a demonstrated neural speed disadvantage being concealed. The older complete-domain feedback certificates, continuation-trace obstruction, and high-dimensional correction results keep their stated scopes.

The all-domain .01 feedback objective is unchanged. Extending the present domain requires an admissible policy system and a sharp comparison object covering additional initial states and starting times, together with a matched global classical frontier. The present manuscript supplies complete, reproducible regional theorems and calculations for that program; it does not replace the global objective with an assertion that it has already been attained.

46. HISTORICAL R22 EXPOSITION, PRESERVED VERBATIM

The following exposition retains the original R22 scope and all qualifications; the R23 main text supplies the current findings.

47. INTRODUCTION

A numerical policy for a dynamic economic problem must earn its accuracy in two different comparisons. It must improve the payoff of the policy it replaces, and its payoff must be close to the unrestricted optimum. A neural representation can help generate a candidate without supplying either comparison. Conversely, a better Bellman witness can improve an error bound without improving the underlying policy. We develop Neural Bellman Operators that keep these tasks separate and test the contribution of the representation rather than identifying it with the performance of a particular optimizer.

Our application is the original stopped consumption–portfolio economy with costly preference adjustment. We retain its continuous action set, correlated shocks, stopping payoff, and full-domain accuracy objective of .01. The central new experiment crosses neural and direct representations with Adam and L-BFGS-B, starting each pair from exactly the same delivered policy. Each of three ensembles has independently initialized corner experts. This design distinguishes an optimizer-conditional representation comparison from the earlier comparison of two end-to-end configurations.

The main mathematical contribution is a continuum policy-comparison theorem based on verified pair moments. The theorem bounds the Jensen benefit of the old mixture without requiring equal expert slopes, equal preference drifts, or synchronized random initialization. The variance identity converts the unknown mixture benefit into six Gaussian integrals per time slab. Directed quadrature and a Hessian bound then control the entire initial-state rectangle, not a sampled state grid. Applied to all three new neural–Adam ensembles, the theorem proves initial-to-final payoff improvement of at least 0.026111 and regret below 0.003013. Under the same Adam optimizer and gradient-call cap, it also proves a uniform payoff advantage of at least 0.003112 over the corresponding direct-parameter mixture. This is a conditional advantage; direct L-BFGS-B remains the stronger end-to-end configuration in these experiments.

Two further executions address operational and global-state questions. A predeclared high-learning-rate test exercises actual rejection and exposes deterministic repetition after exact rollback. A separately recorded supervisory repair restores the complete state before changing the learning rate and resumes accepted improvement. In addition, a current-state network and witness, both taking (t, u, x) , are jointly refined from an explicitly identified historical warm start. An independent complete-domain MPFR audit tightens their certificate to 7.241462444. This is progress on the original witness, not a relabeling of the regional policy theorem and not closure of the .01 objective.

The paper states a constructive residual-correction program that does not require a replicable consumption stream, and quantifies both the local and global costs of a simplicial expert cover. These results separate the general verification mechanism from the financial decoder used by the successful regional policies. We preserve the earlier results, proofs, adverse comparisons, and continuation-trace analysis. The technical supplement records their historical status and provides a proof-to-code map for the new results.

48. CONCLUSION

The new evidence separates what the parameterization contributes from what the optimizer and certifier contribute. Under matched initialization and Adam gradient budgets, neural parameterization produces a uniformly better regional policy than direct coefficients. The pair-moment

argument proves this ordering and initial-to-final improvement for independently initialized experts. Direct L-BFGS-B nevertheless remains the stronger overall regional configuration. These findings replace an unidentified representation claim with an identified conditional comparison without suppressing the adverse benchmark.

The acceptance gate now has executed rejection, complete restoration, a diagnosed financing-root failure, and a documented supervisory recovery. The global current-state actor and witness also have a new, independently checked complete-domain refinement. None of these facts turns the regional stochastic mixture into a globally accurate learned state map. The full-domain .01 objective, continuation-witness problem, and matched global classical frontier remain the paper's original targets. The correction-cone formulation and cover accounting specify mathematical requirements for advancing them; they do not assert that an unexecuted high-dimensional or incomplete-market result has already been obtained.

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NEURAL BELLMAN OPERATORS: RESPONSE TO THE R24 REFEREE REPORT

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We respond to the report dated September 23, 2026 on manuscript commit 14ce582. The revision preserves the original economy, title, target, and historical derivations. It incorporates the already completed R25 evidence, executes new online rejection and recovery tests, strengthens fixed-policy stochastic verification, and supplies a conditional finite-work numerical theorem. Each response distinguishes the objection addressed from any remaining scientific requirement.

1. OVERVIEW OF THE REVISION

We thank the referee for identifying the numerical questions that determine whether the proposed operator solves the stated economic problem. We have not replaced that problem by a smaller one, changed its uniform .01 target, or removed adverse findings. The revised article retains the title *Neural Bellman Operators*, the original stopped consumption–portfolio economy, and the current-state full-domain objective. The complete reviewed article and supplement are reproduced in the new supplement, with all historical scientific source files retained.

The R25 branch already contained a completed scientific execution when we examined the repository. We use it at frozen commit c78d934c0b3b8d346a40c7ddcaf415e167462265; we do not attribute those executions to R26. It supplies expanded rate calibration, a moving historical-transport control, a stronger stochastic benchmark, current-state continuation, and certified secants. R26 adds complete optimizer-state rejection tests with shadow continuations, second-order interval verification of the unchanged stochastic neural policies, exact rational Bernstein verification of the unchanged classical policy, and a finite-work payoff-polling theorem with explicit hypotheses and oracle cost.

These changes establish several new results, but not every requested conclusion. The original full-state bound is now 7.181834580823298 rather than 7.241462443133396; it does not meet .01. No separate original-actor payoff improvement is certified. The continuous stopped-gradient bridge is not instantiated, and direct L-BFGS-B remains the strongest tested central-state method. We do not describe those points as resolved by better exposition or by auxiliary-model certificates. The responses below state the evidence and the remaining requirement together.

2. RESPONSES TO THE BLOCKING FINDINGS

2.1. *R24-F1: Original full-state objective*

Response. The original numerical objective is retained in the abstract, model section, original-results subsection, and conclusion. The imported current-state continuation uses 85 gradient calls across four blocks. Its all-start-time certificate is 7.181834580823298, a reduction of about .059628, or .823%, from the reviewed value. The improvement is in the certified upper bound; the record explicitly does not establish a separate policy-payoff improvement. The target is still missed by about a factor of 718.

The regional financed controller and the auxiliary stochastic model are now identified in the same paragraph or table note as their small bounds. Neither is allowed to stand in for the original

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actor. This finding remains a substantive research requirement. The revision provides new mechanisms and verification tools with which to pursue it, not a declaration that the requested original-state result has already been obtained.

2.2. R24-F2: Unbracketed Adam calibration

Response. The expanded, prospectively declared R25 rate grid is

$$\{.005, .02, .08, .32, 1.28, 5.12, 20.48\}.$$

It is applied to six first-order representations at two disjoint tuning seeds with 200 calls per rate. Selection is based on certified endpoint payoff lower bounds. Neural and whitening Adam select .08; direct, diagonal, static tangent, and moving transport select .32. Each selection is interior and has a worse larger-rate trial. No selected rate is the upper endpoint.

The selected configurations are then evaluated without retuning at two held-out seeds and two proposal quadrature rules. The article reports the frontier within that declared family; the supplement retains every tuning mean and the underlying per-seed files. This addresses the specific grid-censoring objection. It does not claim that a seven-rate family is an exhaustive optimizer of all possible schedules or architectures.

2.3. R24-F3: Direct L-BFGS-B remains strongest

Response. We retain and elevate this result rather than remove its competitor. In the coarse-rule held-out comparison, direct L-BFGS-B has worst regret .0019235324, about 123 mean gradient calls, and about .307 seconds of mean generation time. Neural Adam has .0019263761, 400 calls, and about 1.144 seconds. Matched payoff-difference intervals are strictly negative for neural Adam minus direct L-BFGS-B at both seeds and both rules.

The abstract leads with the classical comparison within the tested class. The moving-transport control does not overturn it. The stochastic direct approximation is also much more accurate than the neural approximation once both are verified rigorously. Accordingly, a neural accuracy/work improvement over the classical frontier remains unestablished. The new claims concern verified online decisions and stronger whole-domain verification, not an unsupported efficiency ranking.

2.4. R24-F4: Static geometry is not the neural Adam mechanism

Response. The new moving arm maintains an explicit neural chart carrier, recomputes its quotient Jacobian at every step, pulls back the current economic gradient, maintains parameter-space Adam first and second moments, and pushes the update into the quotient. It therefore retains historical Jacobian contributions through the moment recursion and uses the current adaptive preconditioner. Its regularizer 10^{-6} and output step cap .5 were fixed before evaluation.

The direct economic coordinates are advanced linearly, so the finite-step nonlinear output remainder is deliberately omitted. The paper states this difference and charges the 355-parameter carrier, its two moment arrays, and the required differentiation. The control is not a cheap autonomous 47-memory-coordinate solver or an exact neural replica. It meets the requested moving/historical-transport comparison while preserving the distinction needed to interpret any remaining performance difference.

2.5. R24-F5: Stopped-gradient error remains uncertified

Response. The revised perturbation proposition explicitly bounds both exact and numerical derivative factors and includes root conditioning and offset error. Its proof is given in the article.

We have not obtained verified numerical constants for every term along the original stopped optimizer trajectories, so the report’s central gradient objection is not marked closed.

There are two constructive additions. First, the inherited stopped-payoff checker supplies rigorous finite secants for five interpolated policies; only the first adjacent secant interval is separated from zero. Second, the new payoff-polling theorem gives a derivative-free route to a stationarity bound under explicit smoothness and shrinking-interval assumptions, including the cost of obtaining the required intervals. Neither addition is called a continuous stopped-gradient certificate. The executed online gate needs only validity of its policy intervals, while the stronger stationarity theorem requires additional hypotheses not verified for every original-economy policy.

2.6. R24-F6: Gate replay versus an actual proposal operator

Response. We inspected the R25 online record rather than relying on its commit message: all 24 blocks were accepted, so that execution did not exercise rejection. R26 runs an explicit three-block stress schedule in six trajectories. Of 18 decisions, 13 accept and five reject. The gate is called synchronously before generating the next block. Each rejection restores economic coordinates, neural/carrier parameters, optimizer moments, and the counter, verified by exact equality of canonical state records and matching digests.

Four rejected aggressive states are continued separately under the same recovery rate and call count as their actual counterpart. All four final states differ from the rolled-back trajectory. These shadow runs are isolated from the production incumbent and timed separately. This closes the implementation distinction between replay and actual coupling for the executed stress mechanism. We do not infer an Adam convergence theorem or an efficiency advantage from this stress design; the finite-work theorem is for the explicitly specified payoff poll.

2.7. R24-F7: Augmented memory does not solve the restart problem

Response. The financed implementation remains explicitly augmented by (Y, λ) . The unrestricted-value equality is an admissible-class argument under the original filtration and is not used to certify arbitrary restricted-library restarts or off-manifold states. The original-model arbitrary-restart accuracy requirement remains unmet.

The added stochastic benchmark has no initialization memory and no conditional-price decoder. Its neural actors use (t, x, z) alone, and the complete interval cover plus stopped Hamiltonian identity prove a bound at every start time and state of that auxiliary economy. The selected restart table merely evaluates that all-time bound. Thus a real current-state restart experiment is supplied without misrepresenting it as a solution to the original financed-library restart problem.

2.8. R24-F8: Small favorable regional cover

Response. The nine-expert, four-cell R24 regional construction and its dimensions are preserved. It is still labeled a $t = 0$ initialized augmented-state patch, not high-dimensional scaling evidence. The new stochastic verification covers the full three-dimensional state–time box $[0, 1] \times [.5, 2] \times [-1, 1]$, with independent unspanned productivity risk and no exact conditional-price financing decoder.

For two unchanged neural policies, the new verifier meets tolerance .001 on 32,768 cells instead of the inherited 262,144. All attempted coarse covers are retained. This supplies a materially different stochastic/restart test and a verified cell-count improvement. It is not evidence of general high-dimensional scaling or original-domain success; those broader requirements remain distinct.

2.9. R24-F9: The deterministic reference is too simple

Response. The stronger reference has two Brownian shocks, a nontrivial risky portfolio, an unspanned factor, first-exit and terminal stopping, and a state–time policy. An independently supplied smooth witness and an explicit transfer yield an exact Hamiltonian completion. The proof identifies the unrestricted value and the exact regret integral; the numerical computation encloses the action errors over the complete domain.

This directly tests continuous stopped stochastic verification without the favorable financed decoder. The reference is nevertheless manufactured: the witness is known and used in training. It is not represented as an independently calibrated second economic application or as learning an unknown Bellman value. The earlier deterministic benchmark remains in the historical annex for its narrower purpose.

2.10. R24-F10: Reference evidence does not show neural efficiency

Response. We agree with the interpretation and keep the comparison strong. The new stochastic neural bounds are .000834337 and .000788415; the exact rational direct-policy bound is approximately 1.298×10^{-7} . The direct policy uses its known structural advantage explicitly. The paper therefore makes no neural efficiency claim from either reference.

The positive numerical result is sharper verification of the same neural weights: the .001 target is established on eight times fewer cells. The bound is not described as better optimization or a runtime speedup across different hardware. The exact rational direct calculation also repairs an arithmetic-enclosure defect in the inherited derivative-coefficient path without silently rewriting that historical code.

2.11. R24-F11: Equivalent-chart fragility

Response. The earlier rescaling outcomes remain in full. The new moving control narrows the mechanism question by reproducing historical pullback and pushforward structure, but does not make neural Adam or L-BFGS-B invariant under equivalent parameter charts. No such empirical robustness conclusion is claimed.

The new quotient poll and its payoff acceptance rule depend on economically identical output policies rather than neural weight coordinates. This gives chart-independent decisions for that component under exact policy identification and valid certificates. Its cost and regularity assumptions are stated, and it has not been run as a 47-dimensional original-economy solver. Thus the paper adds a principled chart-independent safeguard while leaving general neural-optimizer robustness open.

2.12. R24-F12: Theory should yield a numerical-method statement

Response. The new finite-work theorem is more than an explanatory transport identity. At an unsuccessful complete poll of scale h it bounds the quotient gradient by $(\sigma + 2\kappa + L/2)h/\mu$, assuming valid intervals of width at most κh^2 , feasible segments, and Lipschitz gradient. It also bounds the number of payoff evaluations before such a poll by a constant times h^{-2} and retains the oracle cost. The proof states precisely why interval uncertainty contributes 2κ .

The second-order verification theorem and exact rational polynomial lemma are instantiated on saved stochastic policies. The online-improvement theorem is instantiated on the new accepted decisions. The original stopped-gradient and full-state objectives are not inferred from these different results. This adds constructive theory and executed evidence while keeping its conditions visible to the referee.

3. RESPONSES TO TECHNICAL FINDINGS

3.1. *R24-T1: Elevate rate censoring*

The R24 endpoint result remains historical and is not called a tuned frontier. The new article instead uses the expanded, interior-selected R25 experiment, gives the actual grid and selection rule, and includes all rate-level means in the supplement. This is a design change, not merely softer wording.

3.2. *R24-T2: Both numerical and exact derivative factors*

The theorem now states $\max\{|H_a|, |H_{N,a}|\} \leq M_B$ and $\max\{\|P_y\|, \|P_{N,y}\|\} \leq M_C$. All errors are at the respective roots and include their offset-shift allowances. The proof identifies where each factor enters, with no implicit appeal to a bound available only for the exact version.

3.3. *R24-T3: Interpretation of static tangent controls*

The geometry subsection explicitly says that static tangent, diagonal, and whitening charts do not reproduce Adam’s historical adaptive transport. The moving arm is described separately, including its omitted nonlinear remainder and charged carrier. No static-control failure is used to exhaust the space of direct preconditioners.

3.4. *R24-T4: Trajectory counts*

The paper separates 84 calibration trajectories, 32 held-out trajectories, the previous online study, the new stress study, and the fixed stochastic policies. Aggregate counts are not described as independent replications of one scientific effect. The two new stress seeds are not substituted for held-out evidence.

3.5. *R24-T5: Stale revision index*

The root index now names R26 and points to its article, supplement, response, and review entry. The previous index is archived verbatim. The fixed remote base is recorded, and a local branch or generated file is not called remotely published unless a push has succeeded. The new revision adds no changes to the default or existing review branches.

3.6. *R24-T6: End-to-end costs*

The tables separate generation, checking, and measured coupled time. Common dual construction and initial setup are identified where excluded; shadow costs are separate. The supplement gives the batch cost decomposition and states when a shared dual may be amortized. The theorem retains the cost of requesting narrow payoff intervals, rather than counting each certificate as a unit-cost operation. Different CI and local timings are not combined into a hardware speedup.

4. DISPOSITION OF THE REQUESTED NEXT STEPS

4.1. *N1: Bracket first-order performance*

Executed in the frozen R25 experiment and fully incorporated: every first-order selected rate is interior with a worse larger-rate trial, followed by disjoint held-out evaluation. This resolves censoring within the declared finite family.

4.2. *N2: Follow measured neural output geometry*

Executed with a current-Jacobian carrier, historical first moments, adaptive second moments, declared regularization, and an output cap. Nonlinear output curvature is intentionally not reproduced; carrier work and memory are charged.

4.3. *N3: Certify or tightly control the stopped gradient*

The exact/numerical derivative-factor ambiguity is corrected. Rigorous secants and a conditional derivative-free poll are supplied. A complete instantiated original stopped-gradient bound is not supplied and remains a substantive requirement.

4.4. *N4: Successful original current-state full domain*

The original continuation bound improves to 7.181834580823298 but remains far above .01, and no separate payoff improvement is proved. This requested success is not claimed.

4.5. *N5: Nonreplicable stochastic execution*

A manufactured model with unspanned Brownian productivity risk is executed without the conditional-price decoder. Complete state–time neural certificates are obtained. Its known witness and separate economic scope are explicit.

4.6. *N6: Restart and state–time coverage*

The auxiliary current-state policies are certified for all start times and the whole box. This supplies a genuine restart experiment, while original-model global restart accuracy remains unresolved.

4.7. *N7: Stronger known-solution benchmark*

The stochastic stopping benchmark, its exact Hamiltonian completion, and its all-face settlement conditions are provided. Neural and direct policies are independently verified, including an exact rational direct certificate.

4.8. *N8: Decide the paper’s objective*

We retain the original Neural Bellman Operator objective and do not reframe the paper into a claim that the specialized financed construction has solved the original model. The article distinguishes the achieved algorithmic/verification results from the remaining original-domain and efficiency requirements. All earlier scientific content remains available in the historical annex and unchanged source tree.

5. WHAT THE REFEREE CAN VERIFY DIRECTLY

The main article gives the economic model, operator, assumptions, complete proofs, and tables generated from frozen JSON results. The supplement contains every new online decision, all rate-level tuning means, all-time restart bounds, arithmetic details, cost definitions, and the complete reviewed R24 article and supplement. The new replication directory contains the actual stress and verification implementations, exact rational coefficients, every cell file, state snapshots, shadow runs, and the test suite. The response and machine-readable finding map cover every numbered item above without declaring unachieved original-economy results closed.